Effects of Noise on Optimization, Statistics, and Simulation of Quantum Systems

by

Matthew Duschenes

A thesis
presented to the University of Waterloo
in fulfilment of the
thesis requirement for the degree of
Doctor of Philosophy
in
Physics - Quantum Information

Waterloo, Ontario, Canada, 2026

Examining Committee Membership

The following served on the Examining Committee for this thesis. The decision of the Examining Committee is by majority vote.

Supervisors:	Roger Melko Professor, Dept. of Physics & Astronomy, University of Waterloo
	Juan Carrasquilla Associate Professor, Dept. of Physics, ETH Zürich and Adjunct Faculty, Dept. of Physics & Astronomy, University of Waterloo
External Examiner:	William Fefferman Associate Professor, Dept. of Computer Science, University of Chicago
Internal-External Member:	Graeme Smith Associate Professor, Dept. of Applied Mathematics, University of Waterloo
Internal Member:	Kevin Resch Professor, Dept. of Physics & Astronomy, University of Waterloo
Committee Member:	Lauren Hayward Adjunct Faculty, Dept. of Physics & Astronomy, University of Waterloo

Author's Declaration

This thesis consists of material all of which I authored or co-authored: see Statement of Contributions included in the thesis. This is a true copy of the thesis, including any required final revisions, as accepted by my examiners.

I understand that my thesis may be made electronically available to the public.

Statement of Contributions

This thesis consists of material, all of which I am the primary author. I am the sole author of the introduction, preliminaries, and discussion in Chapters 1, 2 and 7. Work presented in this thesis in Chapters 3 to 6 comes from several collaborations, and consists of consolidated ideas from published manuscripts, posted preprints, and unpublished work. My specific contributions to each chapter are as follows.

Chapter 3

I am the primary author, and performed all calculations, numerical simulations, and manuscript writing for this project. Project ideas were conceived by, and manuscript edits were contributed to, by Prof. Juan Carrasquilla and Prof. Raymond Laflamme at the University of Waterloo (UW). This chapter is based on the published manuscript,

- **M. Duschenes**, J. Carrasquilla, and R. Laflamme, Characterization of overparameterization in the simulation of realistic quantum systems, Physical Review A **109**, 062607 (2024). https://doi.org/10.1103/PhysRevA.109.062607 [1]

and data for this chapter are available at,

- **M. Duschenes**, J. Carrasquilla, and R. Laflamme, *Data: Characterization of overparameterization in the simulation of realistic quantum systems*, Zenodo (2024). https://doi.org/10.5281/zenodo.10295792 [2]

Chapter 4

I am the primary author, contributed to project ideas, conceived of all proofs, and performed all calculations and manuscript writing for this project. Project ideas, and manuscript edits were contributed to, by Dr. Marco Cerezo and Dr. Diego García-Martín at Los Alamos National Laboratory (LANL), and Prof. Zoë Holmes at École Polytechnique Fédérale de Lausanne (EPFL). This chapter is based on the preprint manuscript,

- **M. Duschenes**, D. García-Martín, Z. Holmes, and M. Cerezo, Moments of quantum channel ensembles, arXiv preprint arXiv:2511.12700 (2025). https://doi.org/10.48550/arXiv.2511.12700 [3]

with LANL report number LA-UR-25-31295, and is under review for journal publication.

Chapter 5

I am the primary author, conceived the project ideas, and performed all calculations, numerical simulations, and manuscript writing for this project. Manuscript edits were contributed to, by Prof. Roger Melko and Prof. Juan Carrasquilla, with initial project ideas from Prof. Raymond Laflamme, at the University of Waterloo (UW). This chapter is based on the preprint manuscript,

- **M. Duschenes**, R. Melko, J. Carrasquilla, and R. Laflamme, Distributions of Noisy Expectation Values, arXiv preprint arXiv preprint arXiv:2604.05973 (2026). https://doi.org/10.48550/arXiv.2604.05973 [4]

is under review for journal publication, and data for this chapter are available at,

- **M. Duschenes**, R. Melko, J. Carrasquilla, and R. Laflamme, *Data: Distributions of Noisy Expectation Values*, Zenodo (2026). https://doi.org/10.5281/zenodo.18671768 [5]

Chapter 6

I am the sole author, conceived the project ideas, and performed all calculations, numerical simulations, and manuscript writing for this project. Project ideas were contributed to by Prof. Juan Carrasquilla at Eidgenössische Technische Hochschule Zürich (ETH). This chapter is based on unpublished work, and is in preparation for journal submission.

Numerical Simulations

I am the sole author of all numerical simulation code in this thesis, which is available at,

- **M. Duschenes** , *simulation: JAX based Simulator for Quantum Systems* (2022). https://github.com/mduschenes/simulation [6]

Abstract

Understanding interactions between a system and its environment has consistently been at the centre of scientific studies. Indeed, environmental effects are far reaching and rich in behaviour, from heat baths leading to heat engines in thermodynamics, to non-conservative forces leading to dissipation in classical mechanics, to many-body interactions leading to emergent phenomena in statistical mechanics. As we transition from technologies based on classical phenomena, to technologies based on quantum phenomena, understanding system-environment interactions is the central challenge within quantum information sciences.

These interactions arise frequently in quantum settings, due to intentional non-unitary dynamics as part of quantum algorithms, inherent random dynamics within chaotic systems, or unintentional environmental noise in the absence of error mitigation. It is thus essential to derive any underlying structure from these interactions, and to determine their implications on the viability of emerging quantum technologies.

In this thesis, we conduct systematic analytical and numerical analyses of non-unitary dynamics. First, we study the practical effects of noise and experimental constraints on variational quantum algorithms. We find that objectives are initially robust to noise, and decrease exponentially with increased evolution time, before increasing polynomially with evolution time, due to an accumulation of errors. Second, we develop analytical tools to exactly compute statistics of ensembles of random quantum channels. Such formalisms allow us to derive hierarchies between ensembles, to define channel t-designs, and to show that generalized channel-design-induced concentration phenomena can occur. Third, we study distributions of probabilities of generalized measurement outcomes, given simulated noisy random quantum circuits. We develop an accurate and interpretable effective global noise model for these locally noisy distributions. Notably, we show that non-symmetric measurement distributions are multi-modal, whereas symmetric measurement distributions are uni-modal. Fourth, we propose and benchmark a classical simulation method, where measurement probabilities of states are represented by stochastic tensor networks, and non-unitary dynamics are represented by non-negative matrix factorizations.

We conclude this thesis with a discussion of implications of the rich structures underlying non-unitary dynamics. We first interpret and provide examples and counter-examples of the utility of ensembles within channel-centric quantum algorithms. We proceed to discuss long term objectives posed by our studies, regarding constructing phase diagrams of optimization success, across ansatz expressiveness, noise scales, and system sizes. We also propose less passive applications of noise towards steering ensembles towards concentrated or non-concentrated behaviours. Finally, we raise questions of simulability of multi-modal distributions in the search for quantum versus classical advantage.

Acknowledgements

I have been incredibly fortunate to have been mentored by, and to have gotten to contribute to, an exceptional group of people throughout my studies, from Waterloo, to Kingston, to Toronto, to Ann Arbor, to Zurich, to Los Alamos.

First, I would like to truly thank my kind, selfless, and brilliant advisors, Raymond Laflamme, Juan Carrasquilla, and Roger Melko, who nurtured me as a person, and let me develop into a scientist. A few rough quotes of theirs will always stick with me.

My late advisor Raymond, while I was figuring out how to pivot to new graduate studies, once told me, "Just go all in on something, see what happens!". Raymond was a true friend. Whenever something magical happens in physics (and chances are it is from one of his breakthroughs) or it is a nice sunny or snowy day, I will always think about what his reaction would be, and the first thing he would say. No one will ever ask such interesting, insightful questions, that really show how engaged and exceptional he was.

My advisor Juan, while I was visiting his new home of Zurich, once told me, "Be kind to yourself!". Juan has always made sure I was never lost in the shuffle, a difficult task given the many changes over the years. Juan has also always chosen such perfect research questions for me, showing he really cared to make sure that they would resonate. My advisor Roger, while I was struggling to see the bigger picture of my work, once told me, "That's science, you never know how things are going to turn out, or what's going to catch on!". Roger, who I was lucky enough to be advised by in both my Master's and my PhD, has always been my biggest supporter, and the way he is truly present with everyone is something to really strive for. It is hard to put in to words how to thank Roger and Juan for immediately helping me navigate everything upon Raymond's passing.

Second, I would like to sincerely thank all of my committee members, mentors, teachers, and collaborators across UW, IQC, PI, and LANL. Lauren Hayward, Maïté Dupuis, and Aggie Branczyk, who I luckily have known since the beginning of my Master's many years ago now, have all given me countless hours of their time, on communicating, succeeding, and simply being happy within physics. Graeme Smith, Kevin Resch, and Bill Fefferman have also inspired me in their research, and have dedicated a lot of their time to further my academic journey. Finally, Marco Cerezo, Diego García-Martín, and Zoë Holmes, were exciting, motivating collaborators, who taught me how to write a great paper, and how to do impactful science. I also feel incredibly fortunate to have been fully funded by the Natural Sciences and Engineering Research Council of Canada, and to have been funded by the University of Waterloo, the Institute for Quantum Computing, the Perimeter Institute, the Vector Institute, Los Alamos National Laboratory, and ETH Zurich.

Third, I have been exceptionally lucky to have been part of a wide array of groups throughout my studies, and all of my friends there have made the experience. To Ben MacLellan, Andrew Jreissaty, Schuyler Moss, Roeland Wiersema, Zachary Mann, Ningping Cao, Nick Joslyn, and many others, thank you so much for helping me through all of the struggles and successes we have all shared in over the years. It is an irreplaceable time we had together, and I cannot wait to see what we all get up to next.

Finally, none of this would be possible without my sister Sarah, my mom Kathy, and my dad Michael. I am not sure how to thank you for your love through all of these years of endless focus. I am so proud to be your brother and son, and of what you have persevered through yourselves. You have made all of my pursuits meaningful. Love you!

Dedication

This thesis is dedicated to my family.

Table of Contents

List of Figures

List of Tables

List of Abbreviations

NISQ	Noisy Intermediate Scale Quantum devices
NMR	Nuclear Magnetic Resonance
RF	Radio Frequency pulses
Haar	Unitary Haar measure
cHaar	Channel Haar measure
Depolarize	Depolarization measure
CPTP	Completely-Positive Trace-Preserving map
CPTNI	Completely-Positive Trace-Non-Increasing map
POVM	Positive Operator-Valued Measure
PVM	Projection-Valued Measure
SIC	Symmetric Informationally Complete
MPS	Matrix Product State
pMPS	Stochastic Matrix Product State
MPO	Matrix Product Operator
LPDO	Locally Purified Density Operator
SVD	Singular Value Decomposition
NMF	Non-negative Matrix Factorization

List of Symbols

$\mathcal{H}$	System Hilbert space
$\mathcal{E}$	Environment Hilbert space
$\mathcal{O}$	Operator Hilbert space
$\mathcal{B}[\mathcal{H}]$	Bounded linear operators on space
$\mathcal{C},\mathcal{U},\mathcal{N}$	Ensemble of quantum channels, unitaries, noise
$\mathcal{P}$	Set of operators
$[n]$	Set of n indices $\{0, 1, \ldots, n-1\}$
Γ	Subset of n indices $\Gamma \subseteq [n]$
$\mathcal{S}_t$	t-order permutations $\sigma, \pi \in \mathcal{S}_t$
ψ	Pure quantum state $\psi = \|\psi\rangle\langle\psi\|$
ρ	Mixed quantum state
Π, Ξ, Γ	Operator
d	Dimension of system Hilbert space
s	Dimension of environment Hilbert space
l	Dimension of subset of space
q	Dimension of local space
n	Number of local spaces
t	Number of copies of spaces
k	Number of concatenations
m	Number of samples
θ	Parameters
γ	Noise scale
χ	Bond dimension

Chapter 1

Introduction

1.1 Information Sciences

Understanding interactions between ourselves and our world around us has always been at the heart of scientific progress. Striving to understand these interactions has allowed us to advance from simply qualifying, to rigorously quantifying, to ultimately applying the mechanisms behind them to further our technological abilities. At their core, such studies revolve around formalizing notions of the flow of information back and forth between us and our environment. As such, the field of information sciences has steadily evolved, with many so-called classical and quantum fronts running almost in parallel.

Initially, up until the 1800's, we have uncovered classical mechanics involving Newton's laws of motion, Maxwell's equations of electromagnetism, and Boltzmann's statistical mechanics [7, 8], revealing precise ways of describing interactions that we see every day. In the early 1900's, we then have proceeded to realize there exists quantum mechanics involving quantization and incompatibility of different measurements from the Stern-Gerlach experiments [9], resulting in subtle correlations within our world [10]. As our physical intuition has developed, so has the complexity of the mathematics involved to describe these insights. Even more crucially, increasingly complicated and time consuming algorithms have become necessary to process all of this accumulated knowledge and data. Fundamentally different approaches have thus been deemed necessary to progress further. Such realizations inspired brilliant insights by Turing [11], and von-Neumann among many others [12] that algorithms can be implemented, not solely by hand, but by using physical, autonomous computation. It will always be astonishing that we can intentionally harness interactions between materials around us, in order to learn about our world itself.

These insights have given rise to so-called classical computation, that takes advantage of classical interactions. Given this computing paradigm, reasonable questions are thus, what can be computed using these interactions, and crucially, how much robust control over them is necessary? In fact, a critical barrier to implementations of computers is the presence and accumulation of errors and noise that degrade accuracies of computations. To resolve these issues, in the mid 1900's, Shannon and others proved that these new computers can be made robust to errors by strategically encoding and decoding information within larger systems, highlighting the value of information theoretic methods [13]. Only by judiciously applying the most appropriate methods, have we been able to fundamentally understand these sources of error, and their effects on the capabilities and robustness of computers. Classical computing has subsequently progressed exponentially, from theorems, to algorithms, to implementations, to applications, to worldwide adoption. Personal computing, brilliant algorithms, and in particular incredibly efficient and compact hardware have transformed our world, and have primed us for future breakthroughs [14, 15].

Even as classical computing methods have been advancing, throughout the 1900's, it has become clear that alternative, more efficient computing paradigms are necessary to solve the most subtle or perplexing problems. Deutsch, Berstein, Vazirani, and Feynman, among others, subsequently provided or even proved strong statements about universal computation [16–18]. Given the physical world around us obeys quantum mechanical principles, those same quantum phenomena should, and in fact must be taken advantage of when developing computing technologies. Quantum-based technologies are thus likely necessary if we want to efficiently simulate our world. So began the exciting progression of so-called quantum computing using quantum phenomena, from theorems, to algorithms, to implementations, and hopefully someday, applications and worldwide adoption [19, 20].

Quantum algorithms rely on carefully orchestrated combinations of states, that are descriptions of quantum systems using a probabilistic formalism [21]. These states are capable of interfering with each other and of being correlated in such a way that their measurement outcomes yield solutions to computations. Ideally such algorithms also require significantly less computational resources than other methods [22], by storing information in a space that scales exponentially, not polynomially with system size [23, 24]. However, if systems are not isolated from their environment, then their carefully prepared structure may be fundamentally altered. For example, if subsystems become maximally correlated with their environment, due to so-called monogamy of entanglement [13] they cannot be correlated with other subsystems, and are limited in which states they can represent [25]. Similarly, if noise causes systems to become a statistical mixture of states, as opposed to a coherent superposition, then their lack of precise interference will obscure any measurements.

Given the inherently fragile nature of quantum states, the crux of implementing quantum technologies is understanding their more nuanced sources of noise and errors. Fortunately, quantum information can also be protected via encoding and decoding quantum information. Laflamme, Knill, Shor, and others have been able to prove that a sufficient number of layers of error correcting algorithms, with an error rate below a problem-dependent threshold, can lead to robust, so-called fault tolerant universal quantum computation [26–28]. However, such methods require significant insight and overhead of resources to achieve in practice [29, 30]. The probabilistic nature of quantum theory [31], the uncertainty in measurements due to their non-commuting nature [32], the inability to arbitrarily copy quantum information, and the distinct correlations that arise throughout their evolution [33], lead to fundamental difficulties with correcting any errors [16, 34]. Identical to in the development of classical computing, thoroughly understanding the effects of imperfections on the capabilities of quantum computers, is essential to guarantee their viability, and to even take advantage of these phenomena [35].

1.2 Quantum Implementations

Before studying properties of quantum systems, it is essential to understand their current implementations, strengths and weaknesses. As succinctly developed by DiVincenzo [36], any computing system, classical or quantum, should obey the following principles:

i). Scalable (Systems contain well-defined components representing information)

ii). Initializable (Systems have definitive, repeatable initial states)

iii). Universal (Systems are capable of performing arbitrary algorithms)

iv). Measurable (Systems can be interacted with to access information)

v). Robust (Systems perform algorithms with bounded errors)

Each of these criteria arise when assessing whether noisy outputs from a system are close to noiseless outputs [37]. Given an initial state ψ, evolved by an algorithm $\mathcal{U}$, followed by noise $\mathcal{N}$, and corrections Λ [26], and given a property being measured Π [38], it is important to assess whether the noisy and noiseless outputs $\langle\psi\rangle_\Pi$ differ by $\epsilon > 0$,

$$\left|\langle\mathcal{U}(\psi)\rangle_\Pi \;-\; \langle\Lambda(\mathcal{N}(\mathcal{U}(\psi)))\rangle_\Pi\right| < \epsilon \;. \tag{1.1}$$

Such differences ideally are bounded, do not diverge with system size or length of computation, and instead decrease as the necessary corrections are informed by insight into the noise and its interplay with the involved states, algorithms, and measurements.

Regarding specific implementations, given their current successes, let us review nuclear magnetic resonance, optics, trapped ion, superconducting qubit, and neutral atom approaches. Summaries of their properties are shown in Table 1.1. Each approach is shown to have many advantages and disadvantages in terms of their ease of implementation, accuracy, number of quantum qubits of information, speed and number of operations, sources of error, use of existing technologies, and overall viability as a candidate for large-scale quantum computation [46, 54, 60, 61]. From these properties, such implementations are evidently in their infancy with respect to the computing criteria. Current devices can represent around $100-1000$ quantum bits of information, can perform around $100-1000$ universal operations with fidelity around 10^{-3}, and are classified as being noisy-intermediate-scale quantum devices (NISQ) [18]. Throughout their development, NISQ devices have been applied towards several targeted proof of principle problems that lend themselves to quantum settings [24]. Such quantum methods can subsequently be benchmarked against existing, and established classical methods [62–66].

Example current applications include, direct simulation of quantum many-body phenomena [52, 62, 67, 68], quantum chemistry of ground states of specific molecules [69, 70], quantum metrology and sensing [71–74], optimization of classical objective functions [75–77], random quantum circuit models of computation [78–80], quantum machine learning variants of conventional deep learning tasks [81–84], and other hybrid classical-quantum approaches [62, 85, 86]. Such applications have revealed exciting capabilities and potential for quantum approaches to be useful at larger scales. However, the most appropriate applications and problem scales where quantum technologies will be definitively superior to existing classical technologies, are still up for debate [19]. From these NISQ experiments, there is also not one technology that will necessarily solely lead to a useful quantum computer. Instead, the accumulation of techniques learned from each approach are contributing to incrementally understanding how to best harness quantum phenomena.

1.3 Open Quantum Systems

Within the NISQ context, it is thus crucial to thoroughly assess the final computing criteria, that of robustness and capabilities of our existing implementations. Whereas ideal implementations can be modelled as closed quantum systems, that are isolated from their environment, and undergo ideal, reversible unitary dynamics, NISQ implementations must

Table 1.1: **NISQ properties**. Comparison of nuclear magnetic resonance, optics, trapped ion, superconducting qubit, and neutral atom implementations, using generally experimental apparatus and algorithm specific estimates from the literature. Tabulated values are, type and number of qubits n, type and number of operations, or gates k, one U_1 and two U_2 qubit gates and gate times $T_{U_{1,2}}$, measurement procedures and measurement times T_M, and sources of noise, noise scales γ, and decoherence times T. Gates are typically Pauli-operations along the qubit axes $R_{X,Y,Z}$, with nuclei-nuclei J-coupling $U_{\mathrm{J}} \sim R_{ZZ}^{\pi J}$ gates for nuclear magnetic resonance, beam splitters and phase shifters $U_{\mathrm{BS}} \sim R_Y, \mathrm{PS} \sim R_Z$ for optics, Cirac-Zoller $U_{\mathrm{CZ}} \sim CZ$, or Molmer-Sorensen $U_{\mathrm{MS}} \sim R_{XX}$ gates for trapped ions, i-Swap $\sqrt{iS}$ gates for superconducting qubits, and Jaksch $U_{JK} \sim CZ$ gates for neutral atoms. Noise scales are longitudinal or transversal decoherence times $T_{1,2}$, 1 and 2-qubit gate fidelities $\gamma_{1,2}$, and efficiencies η. Each implementation has evident strengths and weaknesses, and currently neutral atoms have the best immediate scalability, trapped ions have the best infidelities, superconducting qubits and optics have the fastest gates, and nuclear magnetic resonance have the most straightforward operation.

Device	Qubits, n	Gates, k	U_1, T_{U1}	U_2, T_{U2}	Measure, T_M	Noise, γ, T
Nuclear Magnetic Resonance [39–41]	Atomic spins $4-12$	RF pulses 1000	$R_{X,Y,Z}$ 75 µs	R_{ZZ} 5 ms	Induced currents 10^2 ms	Inhomogeneity $T_1 \sim 10^3$ ms $T_2 \sim 10^2$ ms
Optics [42–45]	Photon polarizations $10-100$	Optical elements 100	$R_{Y,Z}$ 1 ns	CZ 1 ns	Photon detectors 1 ns	Photon loss $\eta \sim 0.8$
Trapped Ions [46–48] [49–51]	Electron energy levels $100-1000$	Laser pulses 100	$R_{X,Z}$ 5 µs	CZ, R_{XX} 50 µs	State fluorescence 10 µs	Inhomogeneity $\gamma_1 \sim 10^{-5}$ $\gamma_2 \sim 10^{-4}$ $T_2 \sim 5$ s
Super-conducting [52, 53] [41, 54]	Transmon states $50-150$	Micro-wave pulses 100	$R_{X,Z}$ 10 ns	$\sqrt{iS}, CZ$ 100 ns	Dispersive readout 10 ns	Inhomogeneity $\gamma_1 \sim 10^{-3}$ $\gamma_2 \sim 10^{-3}$ $T_2 \sim 200$ µs
Neutral Atoms [55–57] [58, 59]	Atom energy levels $100-1000$	Laser pulses 500	$R_{X,Z}$ 100 ns	CZ 2 µs	State fluorescence 10 ms	Inhomogeneity $\gamma_1 \sim 10^{-3}$ $\gamma_2 \sim 10^{-3}$ $T_2 \sim 1$ s

be modelled as open quantum systems, that are interacting with their environment, and undergo generalized, irreversible non-unitary dynamics. Such systems are subsequently quantified by the dimension of the system d or system size n, the dimension of the environment s, the length of evolution or circuit depth k, the noise scale γ, the dimension of measurements acting on states l. and the number of copies of the system t. Significant analytical and numerical progress has been made to understand these questions of robustness and capabilities, with several distinct research directions. However, many important, concrete open questions regarding noisy behaviours remain.

First, given noisy quantum devices, it is important to assess our abilities to control them in realistic settings. As such, the field of quantum optimal control seeks to develop efficient methods of controlling the evolution of quantum systems by understanding appropriate effective representations of their dynamics [87, 88]. Here, continuous time dynamics are often approximated as a sequence of controls, with each control being chosen from a set of permitted operations. These ansatze depend on experimental constraints such as what controls can be applied within experiments [89, 90], and on fundamental constraints such as adiabaticity limiting how fast quantum systems can evolve [91, 92].

The sequence of controls and their parameters must then be optimized to accurately implement the target dynamics of the system [93–95], and non-trivial compromises arise during optimizations [80]. From the perspective of controllability, or expressing arbitrary quantum dynamics, strongly constrained controls may impose that states evolve within confined subspaces, and cannot evolve to the target states of interest [88, 96]. Conversely, from the perspective of ease of optimization, totally unconstrained dynamics can lead to exponentially large objective landscapes that are highly concentrated and difficult to traverse [85]. Finally, dynamics with conflicting constraints, can lead to highly non-convex objective landscapes with an exponential number of sub-optimal local minima [97, 98].

Such studies have primarily focused on closed quantum systems, where the control operators are unitaries, typically with nice group structure. However, many open questions remain for open quantum systems, given such dynamics have far less algebraic structure [99]. Concurrently, formal learning phenomena [100, 101], such as overparameterization where optimizations can converge to optimality exponentially with the number of control variables, is complicated in the presence of non-trivial constraints and noise [102]. Understanding the robustness of parameterized systems to be optimized in noisy settings is thus essential to determine the limitations of our control.

Second, given adequate control over quantum systems, it is important to understand their dynamics in general settings. As such, the field of randomness within quantum information, particularly random matrix theory, seeks to understand statistics of ensembles of random states and operators [103, 104]. Here, randomness arises often in quantum systems, separate from the inherent probabilistic nature of quantum theory, given randomness can occur due to intrinsic stochastic dynamics [3, 105, 106], or due to intentional random quantum algorithms [34, 107–109]. Assuming randomness can also simplify analytical studies by representing average behaviours over an ensemble, instead of absolute behaviour for all instances [110, 111]. Explicit statistics of sets of strictly unitary operators [112] and sets of states [104] have been well studied, and their exact t-order moments, corresponding to their average action over t-copies of a system, are known for specific ensembles [113–116].

Ensemble-related derivations are greatly simplified when sets of operators form groups, with attractive algebraic structure, allowing formalisms known as Weingarten calculus to be applied [117–119]. Further, comparisons of unitary ensembles arise frequently when assessing the capabilities of the ensembles, for variational quantum algorithms [106, 120], quantum chaos [121–123], and error mitigation [108, 124–126]. Many metrics to compare ensembles exist, and the most general definition is that of ensemble t-designs, where an ensemble's moments correspond with those of a reference ensemble up to t-order.

Generalizing such studies to the case of non-unitary dynamics is complicated once again by their lack of algebraic structure [104]. Notions of ensembles of random quantum channels are just starting to be defined [127–129], with the objective of understanding differences between the structure of non-unitary versus unitary ensembles [130, 131]. Noisy random unitary dynamics are a specific case of these channel ensembles, and involve a yet-to-be fully understood interplay of unitary effects that coherently spread and correlate information across t-copies of a space, versus noise effects that concentrate or dissipate information. Comparisons of channel ensembles have subsequently primarily been done in asymptotically large system limits [128, 132], or for specific small $t = 1, 2$-order moments for means and variances [127]. Formalisms to define and exactly compute closed-form expressions for channel t-designs are thus necessary to shed light on any structures underlying average behaviours of random quantum channels.

Third, given an understanding of random non-unitary dynamics, it is important to assess their concrete measurement statistics. As such, the field of random circuit sampling seeks to understand the structure of random expectation values of measurements, resulting from random quantum systems [78]. Here, samples of measurements from an evolved quantum system yield an empirical distribution of measurement outcome probabilities. This empirical distribution potentially converges to an underlying distribution, with system size, length of evolution, noise scale, and any sources of randomness. In addition to understanding inherently what measurements arise from quantum systems, analysis of such measurement distributions have in fact been at the forefront of the study of the amount of quantum versus classical resources necessary to perform these sampling tasks [62, 133].

Several metrics to assess the convergence of these high dimensional distributions, and to assess any so-called quantum advantage, have been proposed. Example metrics include, definitions in terms of $t = 2$-designs or collision probabilities related to the variances of the distributions [134], a linear cross-entropy quantity describing the similarity of two distributions in a specific basis [135], bounds on the total variational distance between distributions [136], decay of mutual information [137, 138], or even some general approximate t-design results for all t [114, 139]. This convergence is also referred to as anti-concentration, when converging to non-uniform distributions, such as the famous Porter-Thomas distribution [79, 140] for Haar-random unitary dynamics and projective measurements. Crucially, it has been shown than many of these metrics for circuits with local Haar random nearest-neighbour gates converge to the globally Haar random distribution with depths scaling only linearly or even logarithmically with system size [134, 139, 141, 142].

Such convergence bounds in noiseless settings have also recently been extended to systems with unital and non-unital local noise. Intriguingly, the total variational distance of unital noisy circuits is also shown to converge with logarithmically or linearly scaling depth to the depolarized uniform distribution [136, 143]. Conversely, non-unital noise is shown to drive systems away from uniform distributions, and to inhibit anti-concentration [105, 131]. Alternatively, estimates of moments of such locally noisy distributions have also been studied using tensor networks and noisy Weingarten calculus approaches, and are also shown to converge towards the moments of depolarized distributions [130].

It is important to note, that such noisy convergence bounds have primarily been in strictly unitary plus noise, and projective measurement settings. Subsequently, exact, explicit forms of expectation value distributions in non-unitary settings have yet to be systematically investigated, as generalizations of the Porter-Thomas distributions [144, 145]. Equally as important to still understand, is the behaviour of the explicit distributions when considering more general non-projective measurements, such as positive-operator-valued measurements [31, 38, 144, 146]. Such general measurements, in particular if they are informationally complete [146], are necessary to describe quantum states uniquely using tomographic approaches [86, 147, 148]. Insight into expressions for these distributions of expectation values, with respect to non-unitary dynamics and general measurement operators, as functions of circuit depth, environment size, and noise scale, are thus essential to understand how we may most efficiently describe quantum states.

Fourth, given insights into noisy quantum systems, it is important to quantify our ability to classically simulate their dynamics. As such, the field of classical simulation of non-unitary dynamics seeks to understand the most efficient and appropriate representations of correlations within states evolving while correlated with an environment [149]. Simulation methods can be classified by the structure they impose on states or channels throughout simulations. Dynamics can be assumed to map between operators within a closed group, such as Clifford dynamics that maps the Pauli group to itself [150–152]. Given the mixed state nature of states subject to noise, sampling methods also exist, such as unravelling methods, where sets of pure states are each independently evolved, before being sampled to form a mixed state [153–157]. Assumptions about the locality of evolved operators can also be made, such as Pauli-propagation methods, where contributions to expectation values with high-locality operators are assumed to be negligible and are not tracked throughout the dynamics [66, 158, 159]. Finally, correlations and entanglement between qubits can be assumed to be bounded, allowing for various tensor network methods. Such methods represent states and dynamics of interest using localized tensors, with update schemes that preserve and truncate this structure to be as sparse as possible.

Example tensor networks include matrix product states [160], matrix product density operators [161–163] projector entangled pair states [164, 165], and locally purified density operators [166, 167]. In addition to these successful numerical methods, analytical studies have also been conducted on the existence of classical simulation algorithms that scale polynomially with system size, either due to convergence to Haar randomness and anti-concentration properties [163, 168, 169], or Markov properties and decay of mutual information [137], or even due to specific non-unital noise that dominates at the last layers of circuits [131]. Each of these tensor network methods have distinct strengths and weaknesses in terms of their ease of implementation, and their accuracy at capturing various fidelity and entanglement metrics. Most importantly, given these high-dimensional systems being represented, with their non-trivial constraints to ensure they are physically valid, it is essential to assess their scaling of computational resources with their approximation parameter of the maximum bond dimension within each tensor of the network [170].

Given these challenges of classical simulation, we note the prevalence of measurement probabilities within many of these fields of research, and the success of classical simulation methods that avoid explicitly representing quantum states. It thus stands to reason that efficient representations of informationally complete measurement probabilities, using stochastic tensor network methods, could be effective simulation methods [171, 172]. Such stochastic tensors must be updated with non-negative matrix factorizations over non-negative fields [173, 174], as opposed to standard unitary factorizations for quantum states over complex fields [175]. It is thus important to understand the strengths and weaknesses of such implicit approaches, to compare their fidelities and entanglement measures, and in particular to determine their scaling with system parameters of system size, circuit depth, noise scale, and bond dimension.

Given this progress, in this thesis, we seek to answer the following questions:

i). What are appropriate formalisms to describe general quantum systems, and their properties? (Chapter 2)

As preliminaries, we introduce our notation and formalisms used to describe open and closed quantum systems. First, we review quantum mechanics and quantum information concepts, NISQ implementations, and classical-quantum hybrid approaches. Second, we review metrics for quantifying properties of open quantum systems, such as various fidelity, entanglement entropy, mutual information, and Fisher information quantities and their interpretations. Third, we review formalisms to describe ensembles of random quantum states and random operators, and their statistics. Finally, we review the tensor network diagrammatic language, and associated numerical methods for classical simulation.

ii). When do accumulated noise effects begin to dominate noiseless behaviours in the optimization of parameterized quantum systems? (Chapter 3)

As our first study, we investigate learning phenomena, in particular overparameterization of infidelities of parametrized, experimentally constrained, and noisy quantum systems, for state preparation and unitary compilation tasks. We find that infidelities robustly decrease exponentially with increased circuit depth in a convergent regime, before decreasing polynomially with circuit depth and with noise in a noisy regime. This critical circuit depth where noise has catastrophically accumulated, is shown to be logarithmic in the noise scale.

iii). What are statistics of ensembles of random quantum channels? (Chapter 4)

As our second study, we develop analytical tools to define notions of channel t-designs and statistics of ensembles of random quantum channels. Here, we compute exact t-order statistical moment operators for ensembles of quantum channels, derive a hierarchy between such ensembles, and propose appropriate reference ensembles for random quantum channels. We find such random channels, on average, increase support of operators across t-copies of systems, distinct from unitary ensembles that preserve support of operators, leading to distinct depolarizing and non-unital behaviour that constrain any resulting statistics. We then derive channel-design induced concentration phenomena results, including bounds on the convergence towards depolarization of unitary ensembles subject to unital and non-unital noise.

iv). How do non-unitary dynamics affect distributions of expectation values of general measurement operators? (Chapter 5)

> As our third study, we construct empirical distributions of measurement probabilities from simulated noisy quantum circuits, with respect to sets of symmetric and non-symmetric measurements. From analytical insights, we develop an effective global noise distribution model, with interpretable parameters, that captures the peaks behaviours of local noise distributions well, across circuit depth, noise scale, and system size. We show sets of non-symmetric measurements exhibit distinct multi-modal distributions, relative to uni-modal distributions for symmetric measurements, opening up questions about their simulability.

v). Can noisy quantum dynamics and their induced correlations be simulated by stochastic tensor network methods? (Chapter 6)

> As our fourth study, we propose a classical simulation method for states undergoing non-unitary dynamics, based around stochastic matrix product state tensor networks for their measurement probabilities. We show novel non-negative matrix factorization update schemes are necessary to simulate such dynamics in this formalism. We find many intriguing advantages and disadvantages of simulating these correlated systems, their infidelities, and operator entanglement, using non-negative numbers versus complex numbers.

vi). What are implications of these generalized non-unitary phenomena? (Chapter 7)

> As conclusions, we summarize our findings, and discuss implications of the derived non-unitary phenomena. We conclude with proposals of applications of such phenomena towards noisy variational quantum algorithms, random chaotic dynamics, and classical simulation.

As science has progressed, there has been a consistent increase in the generality and complexity of the systems we are studying, from classical mechanics, to classical information, to quantum mechanics, to quantum information, each bringing richness and insight into novel and previously known phenomena. As will be shown throughout this thesis, systematic studies of these more general noisy settings lead to intimate connections between various fields of study, from learning phenomena, to fundamental bounds on distances between states, to properties of permutations, to generalizations of measurement probabilities, to complications when describing states with real versus complex numbers. The intuitive, subtle, and perplexing structures describing these general dynamics is an exciting and open area of research, that will be scrutinized throughout this thesis.

Chapter 2

Preliminaries

Given our objectives of understanding of noise and non-unitarity in quantum systems, in this chapter we build up formalisms and notation to describe such systems. Our formalisms are consolidated from the formalisms introduced in the works the subsequent chapters are based on, namely, M.Duschenes, J. Carrasquilla, and R. Laflamme, PRA 108 (2024) [1] for Chapter 3, M. Duschenes, D. García-Martín, Z. Holmes, and M. Cerezo, arXiv:2511.12700 (2025) [3] for Chapter 4, M. Duschenes, R. Melko, J. Carrasquilla, and R. Laflamme, arXiv:2604.05973 (2026) [4] for Chapter 5, and the unpublished work for Chapter 6. First, we review key concepts related to quantum mechanics concerning quantum systems, their states, dynamics, and measurements. Second, we discuss specific concepts related to quantum information concerning quantum circuit models, quantum channels and measurement operators, and measures of quantum fidelity and entanglement. Third, we introduce particular concepts of randomness within quantum systems concerning ensembles of quantum states and operators, statistics of such quantities, methods of calculating such statistics, and their interpretations. Finally, we introduce classical simulation methods related to tensor network methods, and describe useful diagrammatic notation, discuss different tensor network architectures, and their properties within quantum information settings. Such preliminaries set us up well for our subsequent studies.

2.1 Quantum Mechanics

In this section, let us introduce useful concepts from quantum mechanics [16, 176].

2.1.1 Hilbert Spaces

Here, let us conduct our analyses in the context of d-dimensional Hilbert spaces $\mathcal{H}$. Within such Hilbert spaces, live states,

$$|\psi\rangle \in \mathcal{H}, \tag{2.1}$$

operators which act on these states as bounded linear operators,

$$\Pi \in \mathcal{B}[\mathcal{H}] \;:\; \mathcal{H} \to \mathcal{H} \qquad , \qquad |\psi\rangle \mapsto \Pi|\psi\rangle \ , \tag{2.2}$$

and maps which act on these operators as bounded linear operators,

$$\Lambda \in \mathcal{B}[\mathcal{B}[\mathcal{H}]] \;:\; \mathcal{B}[\mathcal{H}] \to \mathcal{B}[\mathcal{H}] \qquad , \qquad \Pi \mapsto \Lambda(\Pi) \ . \tag{2.3}$$

Operators and maps may map between different Hilbert spaces $\mathcal{H} \to \mathcal{H}'$, but for simplicity, let us assume they map between identical Hilbert spaces $\mathcal{H} \to \mathcal{H}$. Examples include the d-dimensional identity operator I, zero operator 0, identity map $\mathcal{I}$, and zero map $\emptyset$.

Hilbert spaces $\mathcal{H}, \mathcal{H}'$ with dimensions d, d' can be composed with tensor products $\otimes$ into composite Hilbert spaces with dimension dd',

$$\mathcal{H}, \mathcal{H}' \;\to\; \mathcal{H} \otimes \mathcal{H}' \ . \tag{2.4}$$

$$|\psi\rangle \in \mathcal{H} \quad , \quad |\psi'\rangle \in \mathcal{H}' \qquad \to \qquad |\psi\rangle \otimes |\psi'\rangle \in \mathcal{H} \otimes \mathcal{H}' \tag{2.5}$$

$$\Pi \in \mathcal{B}[\mathcal{H}] \quad , \quad \Pi' \in \mathcal{B}[\mathcal{H}'] \qquad \to \qquad \Pi \otimes \Pi' \in \mathcal{B}[\mathcal{H} \otimes \mathcal{H}'] \tag{2.6}$$

$$\Lambda \in \mathcal{B}[\mathcal{B}[\mathcal{H}]] \quad , \quad \Lambda' \in \mathcal{B}[\mathcal{B}[\mathcal{H}']] \qquad \to \qquad \Lambda \otimes \Lambda' \in \mathcal{B}[\mathcal{B}[\mathcal{H} \otimes \mathcal{H}']] \ , \tag{2.7}$$

and d-dimensional objects $\Pi \in \mathcal{B}[\mathcal{H}]$, $\Lambda \in \mathcal{B}[\mathcal{B}[\mathcal{H}]]$ that act non-trivially on one component of a composite space, can be defined to act within the composite space $\mathcal{H} \otimes \mathcal{H}'$ as,

$$\Pi \equiv \Pi \otimes I' \qquad , \qquad \Lambda \equiv \Lambda \otimes \mathcal{I}' \ . \tag{2.8}$$

In the case of t individual spaces $\{\mathcal{H}_i\}_{i\in[t]}$, we use the notation $\mathcal{H} = \otimes_{i\in[t]}\mathcal{H}_i$, and for t identical copies of spaces, $\mathcal{H}_i = \mathcal{H}$, we use the notation $\mathcal{H}^{\otimes t} = \otimes_{i\in[t]}\mathcal{H}_i$.

Hilbert spaces $\mathcal{H}$ with dimension d can also be decomposed with direct sums $\oplus$ into orthogonal subspace Hilbert spaces $\mathcal{O}, \bar{\mathcal{O}} \subseteq \mathcal{H}$ with dimensions $l, d-l$,

$$\mathcal{H} = \mathcal{O} \oplus \bar{\mathcal{O}} \ . \tag{2.9}$$

$l, d-l$-dimensional objects can thus be composed into composite d-dimensional objects,

$$|\psi\rangle \in \mathcal{H} \quad , \quad |\psi'\rangle \in \mathcal{H}' \qquad \to \qquad |\psi\rangle \oplus |\psi'\rangle \in \mathcal{H} \oplus \mathcal{H}' \tag{2.10}$$
$$\Pi \in \mathcal{B}[\mathcal{H}] \quad , \quad \Pi' \in \mathcal{B}[\mathcal{H}'] \qquad \to \qquad \Pi \oplus \Pi' \in \mathcal{B}[\mathcal{H} \oplus \mathcal{H}'] \tag{2.11}$$
$$\Lambda \in \mathcal{B}[\mathcal{H}] \quad , \quad \Lambda' \in \mathcal{B}[\mathcal{H}'] \qquad \to \qquad \Lambda \oplus \Lambda' \in \mathcal{B}[\mathcal{B}[\mathcal{H} \oplus \mathcal{H}']] \ , \tag{2.12}$$

and l-dimensional objects $|\psi\rangle \in \mathcal{H}$, $\Pi \in \mathcal{B}[\mathcal{H}]$, $\Lambda \in \mathcal{B}[\mathcal{B}[\mathcal{H}]]$ that act non-trivially within one subspace, can be defined to act within the entire space as,

$$|\psi\rangle \equiv |\psi\rangle \oplus 0' \qquad , \qquad \Pi \equiv \Pi \oplus 0' \qquad , \qquad \Lambda \equiv \Lambda \oplus \emptyset' \ . \tag{2.13}$$

In the case of t individual spaces $\{\mathcal{H}_i\}_{i\in[t]}$, we use the notation $\mathcal{H} = \oplus_{i\in[t]}\mathcal{H}_i$, and for t identical copies of subspaces $\mathcal{H}_i = \mathcal{H}$, we use the notation $\mathcal{H}^{\oplus t} = \oplus_{i\in[t]}\mathcal{H}_i$.

Given such decompositions of the spaces, and given states $|\psi\rangle \in \mathcal{H}\otimes\mathcal{H}'$ in the composite space, we refer to $|\psi\rangle$ as being a *product* state if there exists states $|\psi\rangle_{\mathcal{H}} \in \mathcal{H}$, $|\psi\rangle_{\mathcal{H}'} \in \mathcal{H}'$ in the individual spaces that factorize the state, and otherwise refer to $|\psi\rangle$ as being an *entangled* state if no such states or factorizations exist,

$$\text{Product} \ : \ |\psi\rangle = |\psi\rangle_{\mathcal{H}} \otimes |\psi\rangle_{\mathcal{H}'} \qquad \text{versus} \qquad \text{Entangled} \ : \ |\psi\rangle \neq |\psi\rangle_{\mathcal{H}} \otimes |\psi\rangle_{\mathcal{H}'} \ . \tag{2.14}$$

2.1.2 Vector Spaces

Hilbert spaces are in fact isomorphic to d-dimensional complex vector spaces, $\mathcal{H} \cong \mathbb{C}^d$, with dual spaces $\mathcal{H}^* \cong \mathbb{C}^d$, with complex vectors and their duals $|\psi\rangle \in \mathcal{H}$, $\langle\psi| \in \mathcal{H}^*$, and complex scalars $\zeta \in \mathbb{C}$. States and operators with respect to $\mathcal{H}$ can thus be thought of as standard vectors and matrices, and any standard linear algebraic operations can be applied. Inner products, $\chi \ : \ \mathcal{H} \times \mathcal{H} \to \mathbb{C}$, can be defined between states $|\psi\rangle, |\psi'\rangle \in \mathcal{H}$,

$$\chi\left(|\psi'\rangle, |\psi\rangle\right) = \langle\psi'|\psi\rangle \ , \tag{2.15}$$

and scalar multiplication and vector addition transform vectors as,

$$|\psi\rangle, |\psi'\rangle \quad \to \quad \zeta|\psi\rangle \ + \ \zeta'|\psi'\rangle \ . \tag{2.16}$$

Let us define an orthonormal basis $\{|\sigma\rangle : \chi(\sigma',\sigma) = \delta_{\sigma\sigma'}\}_{\sigma\in[d]}$ for $\mathcal{H}$,

$$|\psi\rangle = \sum_{\sigma\in[d]} \psi_\sigma \; |\sigma\rangle \quad : \quad \psi_\sigma = \chi(|\sigma\rangle, |\psi\rangle) \;, \tag{2.17}$$

and let us define basis-independent norms $\|\cdot\| \;:\; \mathcal{H} \to \mathbb{C}$ of states as,

$$\||\psi\rangle\| = \sqrt{\chi(|\psi\rangle, |\psi\rangle)} \;. \tag{2.18}$$

Similarly, the set of linear operators is isomorphic to $d \times d$-dimensional complex vector spaces $\mathcal{B}[\mathcal{H}] \cong \mathcal{H} \otimes \mathcal{H}^*$, with matrices and their adjoints, $\Pi \in \mathcal{H} \otimes \mathcal{H}^*$, $\Pi^\dagger \in \mathcal{H} \otimes \mathcal{H}^*$. Inner products $\chi \;:\; \mathcal{B}[\mathcal{H}] \times \mathcal{B}[\mathcal{H}] \to \mathbb{C}$ can be defined between operators $\Pi, \Pi' \in \mathcal{B}[\mathcal{H}]$,

$$\chi\left(\Pi', \Pi\right) = \mathrm{tr}\left(\Pi'^\dagger \; \Pi\right) \qquad , \qquad \chi\left(\Pi\right) = \mathrm{tr}(\Pi) \;\; , \tag{2.19}$$

and scalar multiplication, operator addition and multiplication, traces $\mathrm{tr}\left(\cdot\right) \;:\; \mathcal{B}[\mathcal{H}] \to \mathbb{C}$, and commutators $[\cdot,\cdot] \;:\; \mathcal{B}[\mathcal{H}] \times \mathcal{B}[\mathcal{H}] \to \mathcal{B}[\mathcal{H}]$, transform operators as,

$$\Pi, \Pi' \to \zeta\Pi + \zeta'\Pi' \qquad , \qquad \Pi, \Pi' \to \Pi\Pi' \tag{2.20}$$

$$\Pi \to \mathrm{tr}(\Pi) = \sum_{\sigma\in[d]} \langle\sigma|\Pi|\sigma\rangle \qquad , \qquad \Pi, \Pi' \to [\Pi, \Pi'] = \Pi\Pi' - \Pi'\Pi \;, \tag{2.21}$$

Let us define an orthonormal basis $\mathcal{P} = \{\Pi : \chi(\Pi', \Pi) = \delta_{\Pi\Pi'}\}$ for $\mathcal{B}[\mathcal{H}]$,

$$\Pi = \sum_{\Pi\in\mathcal{P}} \Pi_\Pi \; \Pi = \sum_{\sigma,\sigma'\in[d]} \Pi_{\sigma\sigma'} |\sigma\rangle\langle\sigma'| \qquad : \qquad \Pi_\Pi = \chi(\Pi, \Pi) \quad , \quad \Pi_{\sigma\sigma'} = \langle\sigma|\Pi|\sigma'\rangle \;, \tag{2.22}$$

and let us define basis-independent norms $\|\cdot\| \;:\; \mathcal{B}[\mathcal{H}] \to \mathbb{C}$ of operators as,

$$\|\Pi\| = \sqrt{\chi(\Pi, \Pi)} \;. \tag{2.23}$$

Finally, the set of linear maps is isomorphic to $(d \times d) \times (d \times d)$-dimensional complex vector spaces $(\mathcal{H} \otimes \mathcal{H}^*) \otimes (\mathcal{H} \otimes \mathcal{H}^*)$, with tensors and their adjoints $\Lambda \in (\mathcal{H} \otimes \mathcal{H}^*) \otimes (\mathcal{H} \otimes \mathcal{H}^*)$, $\Lambda^\dagger \in (\mathcal{H} \otimes \mathcal{H}^*) \otimes (\mathcal{H} \otimes \mathcal{H}^*)$ defined in terms of sets of matrices $\{\Gamma_\Lambda\}$, such that,

$$\Lambda(\Pi) = \sum_{\Gamma_\Lambda, \Gamma'_\Lambda} \Gamma_\Lambda \; \Pi \; \Gamma'^\dagger_\Lambda \qquad , \qquad \Lambda^\dagger(\Pi) = \sum_{\Gamma_\Lambda, \Gamma'_\Lambda} \Gamma^\dagger_\Lambda \; \Pi \; \Gamma'_\Lambda \;, \tag{2.24}$$

and the adjoints are defined such that inner products between operators are consistent,

$$\chi(\Pi', \Lambda(\Pi)) = \chi(\Lambda^\dagger(\Pi'), \Pi) \;. \tag{2.25}$$

2.1.3 Examples

Here, we present examples of states, operators and maps, given a d-dimensional unitary orthogonal operator basis, $\mathcal{P} = \{P \ : \ \chi(P,S) = d\delta_{PS} \ , \ \chi(P) = d\delta_{PI} \ , \ P^\dagger \in \mathcal{P}\} \subseteq \mathcal{B}[\mathcal{H}]$. *Maximally entangled* or *Bell* states $|\Omega\rangle \in \mathcal{H} \otimes \mathcal{H}$ across a doubled space are,

$$|\Omega\rangle = \sum_{\sigma\in[d]} |\sigma\sigma\rangle \qquad \to \qquad \Omega = |\Omega\rangle\langle\Omega| = \frac{1}{d}\sum_{P\in\mathcal{P}} P \otimes P^* \ . \tag{2.26}$$

swap operators $S \in \mathcal{B}[\mathcal{H}\otimes\mathcal{H}]$ swap copies of spaces,

$$S \ |\psi\rangle \otimes |\psi'\rangle = |\psi'\rangle \otimes |\psi\rangle \qquad \to \qquad S = \frac{1}{d}\sum_{P\in\mathcal{P}} P \otimes P^\dagger \ , \tag{2.27}$$

projection operators $\Pi \in \mathcal{B}[\mathcal{H}]$ are idempotent,

$$\Pi^2 = \Pi \ , \tag{2.28}$$

unitary operators $U \in \mathcal{U}[\mathcal{H}] \subset \mathcal{B}[\mathcal{H}]$ have an inverse of their adjoint,

$$U^\dagger U = I \ , \tag{2.29}$$

Hermitian operators $H \in \mathcal{B}[\mathcal{H}]$ are self-adjoint,

$$H^\dagger = H \ , \tag{2.30}$$

and *involutory* operators $U \in \mathcal{B}[\mathcal{H}]$ generate projectors, and for any real powers $\varsigma \geq 0$,

$$U^2 = I \to \left(\frac{I \pm U}{2}\right)^2 = \left(\frac{I \pm U}{2}\right) \ , \ U^\varsigma = e^{-i\varsigma\theta\frac{I-U}{2}} = \frac{1+e^{-i\varsigma\theta}}{2}I + \frac{1-e^{-i\varsigma\theta}}{2}U : \theta = \pi. \tag{2.31}$$

Given involutory $U \in \mathcal{B}[\mathcal{H}]$, $V \in \mathcal{B}[\mathcal{H}']$, *controlled unitaries* $C_V U \in \mathcal{B}[\mathcal{H}' \otimes \mathcal{H}]$ consist of a target unitary U, and depend on the projection defined by the control unitary V,

$$C_V U = \frac{I+V}{2} \otimes I \ + \ \frac{I-V}{2} \otimes U = e^{-i\theta \ \frac{I-V}{2}\otimes\frac{I-U}{2}} \qquad : \qquad \theta = \pi \ . \tag{2.32}$$

Depolarizing maps $\mathcal{D}_\gamma \in \mathcal{B}[\mathcal{B}[\mathcal{H}]]$ map operators to the identity, $I \in \mathcal{B}[\mathcal{H}]$, with probability γ, and more generally, *replacement* maps $\mathcal{N}_\eta \in \mathcal{B}[\mathcal{B}[\mathcal{H}]]$ map operators to operators, $\Delta \in \mathcal{B}[\mathcal{H}]$, with probability η, with both maps being trace-preserving,

$$\mathcal{D}_\gamma(\Pi) = (1-\gamma)\Pi \ + \ \gamma\mathrm{tr}(\Pi)\,\frac{1}{d}I \qquad , \qquad \mathcal{N}_\eta(\Pi) = (1-\eta)\Pi \ + \ \eta\frac{\mathrm{tr}(\Pi)}{\mathrm{tr}(\Delta)}\Delta \ . \tag{2.33}$$

Operators $\Pi \in \mathcal{B}[\mathcal{H}]$ can be decomposed into their *singular value decompositions*,

$$\Pi = \sum_\zeta \zeta \; U_\zeta V_\zeta^\dagger \qquad : \qquad U_{\zeta'}^\dagger U_\zeta = \delta_{\zeta\zeta'} I_\zeta \quad , \quad V_{\zeta'}^\dagger V_\zeta = \delta_{\zeta\zeta'} I_\zeta \quad , \quad \sum_\zeta I_\zeta = I \; , \tag{2.34}$$

where $\{\zeta, U_\zeta, V_\zeta\}$ are their distinct singular values ζ, with right and left d_ζ-dimensional eigenspaces, with associated $d \times d_\zeta$-dimensional isometries U_ζ, V_ζ and projectors I_ζ.

Normal operators $\Pi \in \mathcal{B}[\mathcal{H}]$ satisfying $[\Pi, \Pi^\dagger] = 0$ have *eigenvalue decompositions*,

$$\Pi = \sum_\xi \xi \; I_\xi \qquad : \qquad I_\zeta I_{\zeta'} = \delta_{\zeta\zeta'} I_\zeta \quad , \quad \sum_\xi I_\xi = I \; , \tag{2.35}$$

where $\{\xi, I_\xi\}$ are their # number of distinct eigenvalues ξ, with orthogonal d_ξ-dimensional eigenspaces, with associated d_ξ-dimensional projectors I_ξ.

Positive-semidefinite operators $\Pi \in \mathcal{B}[\mathcal{H}]$ have non-negative spectra $\{\xi\}$,

$$\Pi \geq 0 \qquad \leftrightarrow \qquad \{\xi \geq 0\} \; , \tag{2.36}$$

and *completely-positive, trace-non-increasing* (CPTNI) maps $\Lambda \in \mathcal{B}[\mathcal{B}[\mathcal{H}]]$ satisfy [176],

$$\Lambda(\Pi) = \sum_{\Gamma_\Lambda} \Gamma_\Lambda \Pi \Gamma_\Lambda^\dagger \; , \; \sum_{\Gamma_\Lambda} \Gamma_\Lambda^\dagger \Gamma_\Lambda \leq I \; , \; \Pi \geq 0 \; \rightarrow \; \Lambda(\Pi) \geq 0 \; , \; \mathrm{tr}(\Lambda(\Pi)) \leq \mathrm{tr}(\Pi) \; . \tag{2.37}$$

2.1.4 Partial Traces

Given composite Hilbert spaces $\mathcal{H} \otimes \mathcal{H}'$ and operators $\Pi \in \mathcal{B}[\mathcal{H} \otimes \mathcal{H}']$, the *partial trace*,

$$\mathrm{tr}_{\mathcal{H}'} \quad : \quad \mathcal{B}[\mathcal{H} \otimes \mathcal{H}'] \rightarrow \mathcal{B}[\mathcal{H}] \qquad , \qquad \Pi \mapsto \Pi_\mathcal{H} = \mathrm{tr}_{\mathcal{H}'}(\Pi) \; , \tag{2.38}$$

is indispensable, and almost all operations can be expressed as a partial trace. Reduced operators $\Pi_\mathcal{H} \in \mathcal{B}[\mathcal{H}]$ on $\mathcal{H}$ may also be expressed given a basis for the traced out $\mathcal{H}'$,

$$\Pi_\mathcal{H} = \sum_{\sigma' \in [d']} (I \otimes \langle \sigma' |) \;\; \Pi \;\; (I \otimes | \sigma' \rangle) \; , \tag{2.39}$$

and the partial trace is the unique operation [176] that ensures consistent inner products between inherent $\Pi \in \mathcal{B}[\mathcal{H} \otimes \mathcal{H}']$ and reduced $\Pi'_\mathcal{H} \otimes I_{\mathcal{H}'} \in \mathcal{B}[\mathcal{H}]$ operators,

$$\chi(\Pi, \Pi'_\mathcal{H} \otimes I_{\mathcal{H}'}) = \chi(\Pi_\mathcal{H}, \Pi'_\mathcal{H}) \; . \tag{2.40}$$

As important identities, given t operators $\{\Pi_i\}_{i \in [t]}$, products of their traces, and traces of their products, are traces of their tensor products,

$$\prod_{i \in [t]} \mathrm{tr}(\Pi_i) = \mathrm{tr}\big(\bigotimes_{i \in [t]} \Pi_i \big) \;\; , \;\; \mathrm{tr}\big(\prod_{i \in [t]} \Pi_i \big) = \mathrm{tr}\big(\gamma_t \bigotimes_{i \in [t]} \Pi_i \big) \;\; , \;\; \Pi\Pi' = \big(I \otimes \mathrm{tr} \big) \big(S \; \Pi \otimes \Pi' \big) \, , \tag{2.41}$$

up to the t-order cyclic permutation of the t operators, $\gamma_t \; : \; [t] \rightarrow [t] \; , \; i \mapsto i + 1 \bmod t$.

2.1.5 Norms

When comparing d-dimensional states, operators, and maps, identities concerning their *Schatten p-norms* arise frequently, for $p \geq 1$.

States $|\psi\rangle \in \mathcal{H}$ have p-norms of,

$$\||\psi\rangle\|_p^p = \sum_{\sigma\in[d]} |\psi_\sigma|^p \quad : \quad \psi_\sigma = \langle\sigma|\psi\rangle \ , \tag{2.42}$$

operators $\Pi \in \mathcal{B}[\mathcal{H}]$ have p-norms of,

$$\|\Pi\|_p^p = \mathrm{tr}(|\Pi|^p) \quad : \quad |\Pi| = \sqrt{\Pi^\dagger\Pi} \ , \tag{2.43}$$

and maps $\Lambda \in \mathcal{B}[\mathcal{B}[\mathcal{H}]]$ have p-norms of,

$$\|\Lambda\|_p^p = \max_{\substack{O\in\mathcal{B}[\mathcal{H}] \\ \|\Pi\|_p\neq 0}} \frac{\|\Lambda(\Pi)\|_p^p}{\|\Pi\|_p^p} \ , \tag{2.44}$$

with the 1-norm, or trace norm, of $\|\Pi\|_1 = \mathrm{tr}(|\Pi|)$, and diamond-norm, of $\|\Lambda\|_1 \leq \|\Lambda\|_\diamond$.

p-norms for operators $\Pi, \Pi' \in \mathcal{B}[\mathcal{H}]$ only depend on their singular values $\{\zeta\}$,

$$\|\Pi\|_p^p = \sum_\zeta \zeta^p \ , \tag{2.45}$$

satisfy reflectivity, monotonicity for $q \leq p$, sub-additivity, and sub-multiplicativity,

$$\|\Pi\Pi'\|_p = \|\Pi'\Pi\|_p \ \ , \ \ \|\Pi\|_p \leq \|\Pi\|_q \tag{2.46}$$

$$\|\Pi + \Pi'\|_p \leq \|\Pi\|_p + \|\Pi'\|_p \ \ , \ \ \|\Pi\Pi'\|_p \leq \|\Pi\|_p\|\Pi'\|_p \ , \tag{2.47}$$

Holder's inequalities for $1/p + 1/q = 1/r$,

$$\|\Pi\Pi'\|_r \leq \|\Pi\|_p\|\Pi'\|_q \tag{2.48}$$

and von-Neumann's inequalities for $1/p + 1/q = 1$ [177],

$$|\mathrm{tr}(\Pi\Pi')| \leq \|\Pi\|_p\|\Pi'\|_q \ . \tag{2.49}$$

Finally, completely-positive trace-norm-non-increasing maps $\Lambda \in \mathcal{B}[\mathcal{B}[\mathcal{H}]]$, are trace-norm-non-decreasing, and thus have bounded spectra $\{\lambda\}$ [176],

$$\mathrm{tr}(|\Lambda(\Pi)|) \ \leq \ \mathrm{tr}(|\Pi|) \qquad \rightarrow \qquad \mathrm{tr}(|\Lambda|) \ = \ 1 \qquad , \qquad |\lambda| \ \leq \ 1 \ , \tag{2.50}$$

given operators $\Pi = \Pi_+ - \Pi_-$ can be decomposed into positive components $\Pi_\pm \geq 0$,

$$\begin{aligned}\mathrm{tr}(|\Lambda(\Pi)|) &\leq \mathrm{tr}(|\Lambda(\Pi_+)|) + \mathrm{tr}(|\Lambda(\Pi_-)|) = \mathrm{tr}(\Lambda(\Pi_+)) + \mathrm{tr}(\Lambda(\Pi_-)) \\ &\leq \mathrm{tr}(\Pi_+) + \mathrm{tr}(\Pi_-)) = \mathrm{tr}(|\Pi|) \ .\end{aligned} \tag{2.51}$$

2.1.6 Vectorizations

Regarding isomorphisms between Hilbert spaces and complex vector spaces, more formally, operators $\Pi \in \mathcal{B}[\mathcal{H}]$ can be represented as states $|\Pi\rangle\rangle \in \mathcal{H} \otimes \mathcal{H}$, and maps $\Lambda \in \mathcal{B}[\mathcal{B}[\mathcal{H}]]$, with generating $\{\Gamma_\Lambda\}$, can be represented as operators $\hat{\Lambda} \in \mathcal{B}[\mathcal{H} \otimes \mathcal{H}]$, in a doubled space $\mathcal{H} \to \mathcal{H} \otimes \mathcal{H}$, via *vectorizations* vec : $\mathcal{B}[\mathcal{H}] \to \mathcal{H} \otimes \mathcal{H}$,

$$\text{vec} \quad : \quad \Pi \to |\Pi\rangle\rangle \quad , \quad \Lambda \to \hat{\Lambda} \quad : \quad \hat{\Lambda}|\Pi\rangle\rangle = |\Lambda(\Pi)\rangle\rangle \quad , \quad \chi(\Pi, \Pi') = \langle\langle \Pi|\Pi'\rangle\rangle \ . \tag{2.52}$$

Operators $U, V \in \mathcal{B}[\mathcal{H}]$ also satisfy the frequently arising identity, given vectorizations may be defined in terms of the maximally entangled state $|\Omega\rangle \in \mathcal{H} \otimes \mathcal{H}$,

$$|UV^\dagger\rangle\rangle = U \otimes V^* \ |\Omega\rangle \ . \tag{2.53}$$

Further, the partial transpose Γ, acting on one of the doubled spaces, relates the swap operator $S \in \mathcal{B}[\mathcal{H}]$ to the maximally entangled state $\Omega \in \mathcal{B}[\mathcal{H}]$,

$$\Omega^\Gamma = S \qquad \leftrightarrow \qquad \Omega = S^\Gamma \ , \tag{2.54}$$

which arises from the maximally entangled state and the swap operator both being rearrangements of the elements of the identity operator $I \otimes I$.

Linear maps $\Lambda \in \mathcal{B}[\mathcal{B}[\mathcal{H}]]$ can also be represented via channel-operator or even channel-state dualities, as superoperators $\hat{\Lambda}$, or Choi states Φ_Λ,

$$\hat{\Lambda} = \sum_{\Gamma_\Lambda, \Gamma'_\Lambda} \Gamma_\Lambda \otimes \Gamma'^*_\Lambda \qquad \leftrightarrow \qquad \Phi_\Lambda = \sum_{\Gamma_\Lambda, \Gamma'_\Lambda} |\Gamma_\Lambda\rangle\rangle\langle\langle \Gamma'^*_\Lambda| \ , \tag{2.55}$$

In particular, superoperator norms and traces correspond to the Choi state's purity and entanglement fidelity with respect to Ω [126],

$$\left\|\hat{\Lambda}\right\|^2 = \mathrm{tr}\left(\Phi^\dagger_\Lambda \Phi_\Lambda\right) \quad , \quad \mathrm{tr}\left(\hat{\Lambda}\right) = \mathrm{tr}\left(\Omega\Phi_\Lambda\right) \ . \tag{2.56}$$

Even more generally, intriguingly for maps Λ, Λ', inner products between their superoperators equal inner products between their Choi states,

$$\chi\left(\hat{\Lambda}', \hat{\Lambda}\right) = \mathrm{tr}\left(\hat{\Lambda}'^\dagger \hat{\Lambda}\right) = \mathrm{tr}\left(\Phi^\dagger_{\Lambda'} \Phi_\Lambda\right) = \chi\left(\Phi_{\Lambda'}, \Phi_\Lambda\right) \ . \tag{2.57}$$

Any concepts for typical states $|\psi\rangle$ and operators Π, can be thus equivalently applied to the vectorized operators $|\Pi\rangle\rangle$ and superoperators $\hat{\Lambda}$.

2.1.7 Quantum States

Here, let us focus on specific states and operators of interest, with specific constraints such that they describe physical quantum systems, namely distributions for such systems.

Quantum states can often be represented as normalized vectors, which we refer to as *state vectors* or *pure states*,

$$|\psi\rangle \in \mathcal{H} \quad : \quad \langle\psi|\psi\rangle = 1 \ . \tag{2.58}$$

State vectors can be represented equivalently as operators, namely d-dimensional *projectors* $\psi \in \mathcal{B}[\mathcal{H}]$ onto such states, which are uniquely defined,

$$\psi = |\psi\rangle\langle\psi| \in \mathcal{B}[\mathcal{H}] \ . \tag{2.59}$$

Quantum states can always be represented as positive-semi-definite, unit-trace matrices, which we refer to as *density matrices* or *mixed states*,

$$\rho \in \mathcal{B}[\mathcal{H}] \qquad : \qquad \mathrm{tr}(\rho) = 1 \quad , \quad \rho^\dagger = \rho \quad , \quad \rho \geq 0 \ . \tag{2.60}$$

Density matrices of rank s can be represented equivalently as states, namely ds-dimensional *purified* states $|\psi\rangle \in \mathcal{H} \otimes \mathcal{H}_\rho$, given $\mathcal{H}_\rho$ has dimension $d_\rho \geq s$, which are uniquely defined only up to unitaries $U \in \mathcal{U}[\mathcal{H}_\rho] \ : \ |\psi\rangle \to (I \otimes U)\ |\psi\rangle$,

$$\rho = \mathrm{tr}_{\mathcal{H}_\rho}(\psi) \ . \tag{2.61}$$

Pure state vectors have decompositions in terms of a basis $\{|\sigma\rangle\}$, describing their amplitude in the σ state, and mixed state density matrices have decompositions in terms of eigenvalues and eigenstates $\{\lambda, |\lambda\rangle\}$, describing their probability λ of being in the λ state,

$$|\psi\rangle = \sum_\sigma \psi_\sigma\ |\sigma\rangle \qquad : \qquad \psi_\sigma = \langle\sigma|\psi\rangle \quad , \quad \sum_\sigma |\psi_\sigma|^2 = 1 \quad , \quad 0 \leq |\psi_\sigma| \leq 1 \tag{2.62}$$

$$\rho = \sum_\lambda \lambda\ |\lambda\rangle\langle\lambda| \qquad : \qquad \sum_\lambda |\lambda\rangle\langle\lambda| = I \quad , \quad \sum_\lambda \lambda = 1 \quad , \quad 0 \leq \lambda \leq 1 \ . \tag{2.63}$$

Their t-fold *purity* describes their proximity to being a projector,

$$\frac{1}{d^{t-1}} \leq \mathrm{tr}\big(\rho^t\big) \leq 1 \qquad : \qquad \begin{array}{lll} \text{Pure:} & \lambda \in \{0,1\} & \to \mathrm{tr}(\rho^t) = 1 \\ \text{Maximally Mixed:} & \lambda = 1/d & \to \mathrm{tr}(\rho^t) = 1/d^{t-1} \end{array} \ . \tag{2.64}$$

Mixed states thus live within a convex hull of pure states, with boundaries defined by [31],

$$\mathrm{tr}\big(\rho^3\big) = \mathrm{tr}\big(\rho^2\big) = \mathrm{tr}(\rho) = 1 \ , \tag{2.65}$$

given these cubic conditions of $\sum_\lambda \lambda^2 = 1 \ \ , \ \ \sum_\lambda \lambda^2(1-\lambda) = 0$ impose that $\lambda \in \{0,1\}$. Let us refer to state vectors or pure states as ψ or $|\psi\rangle$ interchangeably, general density matrices or mixed states as ρ, and operators as $\Pi_\mathcal{H}$, for $\mathcal{H} \subseteq \mathcal{H} \otimes \mathcal{H}'$, with subscripts $\mathcal{H}$.

2.1.8 Quantum Dynamics

Here, let us focus on specific operators and maps of interest, with specific constraints such that they describe dynamics of physical quantum systems, namely dynamics that preserve the distribution property of such states.

Valid operators and maps applied to quantum states, must yield quantum states. Therefore the most general linear maps $\Lambda \in \mathcal{B}[\mathcal{B}[\mathcal{H}]]$ are *completely-positive, trace-preserving* (CPTP) maps, also referred to as quantum channels, with Kraus operators $\{\Gamma_\Lambda\}$, such that quantum states $\rho \in \mathcal{B}[\mathcal{H}]$ are transformed as,

$$\rho \quad \to \quad \Lambda(\rho) = \sum_{\Gamma_\Lambda} \Gamma_\Lambda\, \rho\, \Gamma_\Lambda^\dagger \qquad : \qquad \sum_{\Gamma_\Lambda} \Gamma_\Lambda^\dagger \Gamma_\Lambda = I \tag{2.66}$$

$$\mathrm{tr}(\Lambda(\rho)) = \mathrm{tr}(\rho) \quad , \quad \Lambda(\rho)^\dagger = \Lambda(\rho) \quad , \quad \rho \geq 0 \leftrightarrow \Lambda(\rho) \geq 0 \ . \tag{2.67}$$

Given Λ is a summation over the adjoint action of Kraus operators, positivity-preservation is satisfied, and given their summation to the identity, trace-preservation is satisfied.

For example, the most fundamental quantum channels are unitaries $U \in \mathcal{U}[\mathcal{H}] \leftrightarrow \mathcal{U} \in \mathcal{U}[\mathcal{B}[\mathcal{H}]]$, with an action on states $|\psi\rangle$, and an adjoint action on states ρ such that,

$$|\psi\rangle \quad \to \quad U|\psi\rangle \qquad , \qquad \rho \quad \to \quad \mathcal{U}(\rho) = U\,\rho\, U^\dagger \ , \tag{2.68}$$

and from their unitarity, preserve the positivity and normalization of $\rho, |\psi\rangle$.

Such dynamics may be derived from the postulate of quantum states obeying *Schrödingers's equation*, and is generated by a Hermitian operator $H \in \mathcal{B}[\mathcal{H}]$ at time t,

$$\partial_t |\psi\rangle = -iH|\psi\rangle \qquad \leftrightarrow \qquad \partial_t \rho = -i[H,\rho] \ . \tag{2.69}$$

Unitaries $U \in \mathcal{B}[\mathcal{H}]$ are generated by a (time-independent) Hermitian generator $H \in \mathcal{B}[\mathcal{H}]$,

$$|\psi\rangle \quad \to \quad U|\psi\rangle \qquad : \qquad U = e^{-iH} \ . \tag{2.70}$$

Channels $\Lambda \in \mathcal{B}[\mathcal{B}[\mathcal{H}]]$, with rank d_Λ and Kraus operators $\{\Gamma_\mu\}_{\mu \in [d_\Lambda]}$, are generated by unitaries $\mathcal{U} \in \mathcal{U}[\mathcal{B}[\mathcal{H} \otimes \mathcal{H}_\Lambda]]$ acting on a composite system-ancillary space $\mathcal{H} \to \mathcal{H} \otimes \mathcal{H}_\Lambda$, where the state $\rho \to \rho \otimes \nu$ is coupled with a pure ancillary state $|\nu\rangle \in \mathcal{H}_\Lambda$. The d_Λ-dimensional space $\mathcal{H}_\Lambda$ is subsequently traced out, in what is known as *Stinespring dilation*,

$$\rho \quad \to \quad \Lambda(\rho) \qquad : \qquad \Lambda(\rho) = \mathrm{tr}_{\mathcal{H}_\Lambda}(\mathcal{U}(\rho \otimes \nu)) \quad , \quad \Gamma_\mu = (I \otimes \langle\mu|)\ U\ (I \otimes |\nu\rangle) \ . \tag{2.71}$$

2.1.9 Quantum Measurements

Here, let us focus on specific operators and maps of interest, with specific constraints such that they describe measurements of physical quantum systems.

Valid measurements of quantum states $\rho \in \mathcal{B}[\mathcal{H}]$, for our purposes, are expectation values x of Hermitian operators $\Pi \in \mathcal{B}[\mathcal{H}]$. Given the positivity of $\rho \geq 0$ and bounds on the spectrum $\Pi = \sum_\xi \xi I_\xi$, of $\sigma \leq \xi \leq \lambda$, such that expectation values satisfy,

$$x = \mathrm{tr}(\Pi\ \rho) \qquad : \qquad \sigma I \leq \Pi \leq \lambda I \qquad , \qquad \sigma\ \leq\ x\ \leq \lambda\ . \tag{2.72}$$

In the case of positive operators $\Pi = \Gamma^\dagger \Gamma$, states are updated post measurement as,

$$\rho \quad \overset{\Pi=\Gamma^\dagger\Gamma}{\rightarrow} \quad \frac{1}{p}\Gamma\rho\Gamma^\dagger \qquad : \qquad p = \mathrm{tr}(\Pi\rho) = \mathrm{tr}\left(\Gamma\rho\Gamma^\dagger\right)\ . \tag{2.73}$$

In the case of pure states $\rho = \psi$, expectation values reduce to overlaps, and when operators are also projectors $\Pi = \psi'$, expectation values further reduce to *Born's* rule,

$$x = \mathrm{tr}(\Pi\ \rho) \quad \overset{\rho=\psi}{\rightarrow} \quad \langle\psi|\Pi|\psi\rangle \quad \overset{\Pi=\psi'}{\rightarrow} \quad |\langle\psi'|\psi\rangle|^2\ . \tag{2.74}$$

In fact, when such operators $0 \leq \Pi \leq I$ are explicitly positive, bounded measurement operators, their expectation values $0 \leq p \leq 1$ can be interpreted as measurement probabilities, of measuring a state ρ in a setting described by Π.

Suppose a space is bi-partitioned into $\mathcal{X} \otimes \mathcal{Y}$. Then separable measurements $0 \leq X \otimes Y \leq I \otimes I$, induce *joint distributions* in outcomes x, y, for states $\rho_{\mathcal{X}\mathcal{Y}} \in \mathcal{B}[\mathcal{X} \otimes \mathcal{Y}]$,

$$p(x,y) = \mathrm{tr}(X \otimes Y\ \rho_{\mathcal{X}\mathcal{Y}})\ , \tag{2.75}$$

with marginal distributions over $\mathcal{X}$ from joint distributions over $\mathcal{X} \otimes \mathcal{Y}$,

$$p(x) = \sum_y\ p(x,y)\ , \tag{2.76}$$

and we can obtain expressions that only depend on measurements and or states within $\mathcal{X}$,

$$p(x) = \mathrm{tr}_{\mathcal{X}\otimes\mathcal{Y}}(X \otimes I\ \rho_{\mathcal{X}\mathcal{Y}}) = \mathrm{tr}_{\mathcal{X}}(X\,\rho_{\mathcal{X}}) \quad : \quad \rho_{\mathcal{X}} = \mathrm{tr}_{\mathcal{Y}}(\rho_{\mathcal{X}\mathcal{Y}})\ . \tag{2.77}$$

Here, $\rho_{\mathcal{X}} \in \mathcal{B}[\mathcal{X}]$ is the partially traced state over $\mathcal{X} \otimes \mathcal{Y} \to \mathcal{X}$, further confirming density matrices as partially traced states lead to consistent joint and marginal distributions.

Here, we are interested in sets $\mathcal{P}$ of measurement operators, namely positive operator-valued measures (POVM),

$$\mathcal{P} = \{\Pi \in \mathcal{B}[\mathcal{H}] \quad : \quad \Pi \geq 0\} \qquad : \qquad \sum_{\Pi \in \mathcal{P}} \Pi = I \ , \tag{2.78}$$

with associated expectation values as measurement probabilities p, such that,

$$p_\Pi = \mathrm{tr}(\Pi\ \rho) = \chi(\Pi, \rho) \ , \tag{2.79}$$

and overlap matrices of,

$$\chi_{\mathcal{P}} = \ \{\chi_{\Pi\Xi} = \chi(\Pi, \Xi)\}_{\Pi,\Xi \in \mathcal{P}} \ . \tag{2.80}$$

Sets of measurement operators are informationally complete (IC) [38, 148] if measurement probabilities are distinct for distinct states, and their overlap matrices are invertible,

$$\forall\ \rho \neq \rho' \in \mathcal{B}[\mathcal{H}] \ \ \exists \ \ \Pi \in \mathcal{P} \quad : \quad \chi(\Pi, \rho) \neq \chi(\Pi, \rho') \quad \leftrightarrow \quad \exists\ \chi_{\mathcal{P}}^{-1} \quad : \quad \chi_{\mathcal{P}}^{-1} \chi_{\mathcal{P}} = I_{|\mathcal{P}|} \, . \tag{2.81}$$

Informationally complete sets of $|\mathcal{P}| = d^2$ operators, such as symmetric (SIC) sets, in fact span the operators $\mathcal{B}[\mathcal{H}]$ [38, 147]. Therefore operators can be expressed as,

$$|\rho\rangle\!\rangle = \sum_{\Pi,\Xi \in \mathcal{P}} \chi_{\Pi\Xi}^{-1}\ p_\Xi\ |\Pi\rangle\!\rangle \tag{2.82}$$

$$p_\Pi = \chi(\Pi, \rho) \qquad , \qquad \sum_{\Pi \in \mathcal{P}} p_\Pi = 1 \ , \tag{2.83}$$

and quantum channels can be expressed as [178, 179],

$$\hat{\Lambda} = \sum_{\Pi,\Xi,\Theta \in \mathcal{P}} \chi_{\Pi\Theta}^{-1}\ L_{\Theta\Xi}\ |\Pi\rangle\!\rangle\langle\!\langle\Xi| \tag{2.84}$$

$$L_{\Pi\Xi} = \sum_{\Theta \in \mathcal{P}} \chi(\Pi, \Lambda(\Theta))\ \chi^{-1}(\Theta, \Xi) \qquad , \qquad \sum_{\Pi \in \mathcal{P}} L_{\Pi\Xi} = 1_\Xi \ . \tag{2.85}$$

States ρ and dynamics Λ can thus be described by their associated probabilities p and quasi-stochastic matrices L without loss of generality,

$$\rho \quad \leftrightarrow \quad p \qquad , \qquad \Lambda \quad \leftrightarrow \quad L \qquad , \qquad \rho \to \Lambda(\rho) \quad \leftrightarrow \quad p \to L\, p \ . \tag{2.86}$$

Expectation values have a variety of interpretations, such as probabilities of measured states occurring for explicit measurements [38, 146, 178], energies of a system for Hamiltonian dynamics [178, 180], or fidelities for distances between states [110, 181–183].

2.2 Quantum Information

In this section, let us introduce specific useful concepts from quantum information [176].

2.2.1 Qudits and Bases

Within quantum information sciences, we wish to represent, and generalize information theoretic concepts within quantum settings, and generally represent n so-called qudits of information as n component local Hilbert spaces $\mathcal{H}_i$ of dimension q, such that the total Hilbert space $\mathcal{H}$ has dimensions $d = q^n$,

$$\mathcal{H} = \otimes_{i\in[n]}\mathcal{H}_i \ . \tag{2.87}$$

For example $q = 2$ qubits are often used, drawing inspiration from classical binary bits of information. Given such localized structure, objects, such as operators $\Pi \to \otimes_{i\in[n]}\Pi_i$ from a basis $\mathcal{P}_d \to \mathcal{P}_q^{\otimes n}$ may be similarly described with a local structure, with support of $\Gamma \subseteq [n]$ where they act non-trivially.

Regarding appropriate bases, here we consider the d-dimensional Pauli basis $\mathcal{P}_d$ [32], which is generated by unitary phase Z_d and shift X_d operators, phases $\omega_d = \omega_d(\omega)$, and is closed under multiplication and inverses up to d-roots of unity phases ω,

$$\mathcal{P}_d = \{P = P(x,z) = \omega_d^w \ X_d^x \ Z_d^z \quad : \quad p = (x,z) \in [d]^2 \quad , \quad w = w(x,z) \in [d]\} \ . \tag{2.88}$$

The basis operators are generated by powers $x, z \in [d] = \{0, 1, \ldots, d-1\}$ of unitary phase and shift generators,

$$X_d = \sum_{\sigma\in[d]} |\sigma + 1\rangle\langle\sigma| \qquad , \qquad Z_d = \sum_{\sigma\in[d]} \omega^\sigma |\sigma\rangle\langle\sigma| \ . \tag{2.89}$$

The basis operator phases $w = w(x,z) \in [d]$, may be chosen as functions of the x, z parameters, and here we choose definitions such that all basis operators have order d,

$$w \to xz \in [d_d] \qquad : \qquad \omega_d, \ d_d = \begin{cases} \omega, \ d & d \text{ odd} \\ \omega^{1/2}, \ 2d & d \text{ even} \end{cases} \ . \tag{2.90}$$

The basis operators $P, S \in \mathcal{P}_d$ with associated symplectic parameters $p, s \in [d]^2$, satisfy the braiding relations [32, 184], given the symplectic matrix Σ,

$$Z_d \ X_d = \omega \ X_d \ Z_d \quad \to \quad P \ S = \omega^{p\Sigma s} \ S \ P \ . \tag{2.91}$$

and further satisfy the commutation relations, given structure constants λ,

$$[P,S] = \sum_{Q\in\mathcal{P}_d} \lambda_{PSQ}\ Q \qquad , \qquad [P,S]=0 \quad \longleftrightarrow \quad p\Sigma s = 0\ . \tag{2.92}$$

Given our choice of ω_d, d_d, products of $k \in [d]$ basis generators,

$$(\omega_d^{xz}\ X_d^x\ Z_d^z)^k = \omega_d^{kxz}\omega^{\binom{k}{2}xz} X_d^{kx} Z_d^{kz} \quad \overset{k=d}{\rightarrow} \quad (\omega_d\ X_d\ Z_d)^d = I_d\ , \tag{2.93}$$

have order d for all even and odd d. Similarly, inverses of basis operators,

$$(\omega_d^{xz}\ X_d^x\ Z_d^z)^\dagger = \omega^{xz}\ \omega_d^{-2xz}(\omega_d^{xz}\ X_d^{-x}\ Z_d^{-z})\ , \tag{2.94}$$

are basis operators for even d, and only basis operators up to phases for odd d.

In the case of composite prime power dimensions, where $d = q^n$ for prime $q > 1$ and $n > 0$, phases $\omega^d = 1 \to \omega^q = 1$ become q-roots of unity, parameters $x, z \in [d]^2 \to x, z \in [q]^{2n}$ become n-tuples, operator phases $w = xz \to x \cdot z$ become sums, and basis operators become order q such that $\omega_d^{d_d} = 1$ with $d_d = 2q, q$ for even, odd q.

For $q = 2$, the basis corresponds to the canonical qubit Pauli basis, $\mathcal{P}_2 = \{I, X, Y, Z\}$, which are uniquely hermitian and unitary, $P^2 = I\ \forall\ P \in \mathcal{P}_2$. Given the change of basis hermitian and unitary Hadamard operator H, the qubit Pauli basis operators satisfy,

$$[X,Y] = i2Z \quad , \quad [Y,Z] = i2X \quad , \quad [Z,X] = i2Y \quad , \quad HXH = Z \quad , \quad HZH = X\ . \tag{2.95}$$

2.2.2 Parameterized Quantum Systems

States $\rho \to \rho_\theta \in \mathcal{B}[\mathcal{H}]$ can be parameterized with parameters θ, such as via parameterized dynamics $\Lambda_\theta \in \mathcal{B}[\mathcal{B}[\mathcal{H}]]$,

$$\rho \quad \to \quad \rho_\theta = \Lambda_\theta(\rho)\ . \tag{2.96}$$

For example, parameterized unitaries $U_\theta \in \mathcal{U}[\mathcal{H}]$, with fixed Hermitian $H \in \mathcal{B}[\mathcal{H}]$ are,

$$U_\theta = R_H^\theta = e^{-i\theta H}\ . \tag{2.97}$$

Such parameterized quantum systems are often optimized to an optimal parameterization θ^*, using appropriate objective functions $\mathcal{L}$, and gradient based methods,

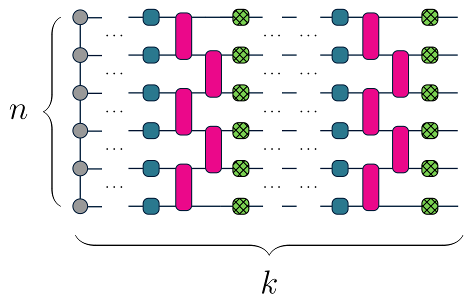
$$\mathcal{L}_\theta = \mathcal{L}(\rho_\theta) \qquad : \qquad \theta \quad \to \quad \theta - \partial_\theta \mathcal{L}_\theta \quad \to \quad \theta^*\ , \tag{2.98}$$

where the optimal states ρ_{θ^*} may only be reached if they are within the span of the ansatz Λ_θ. Gradients of the parameterized states with respect to θ generally must be calculated during optimization, and in the case of parameterized unitaries $\rho_\theta = \mathcal{U}_\theta(\rho)$,

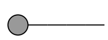
$$\partial_\theta \rho_\theta = -i[H, \rho_\theta]\ . \tag{2.99}$$

2.2.3 Quantum Circuits

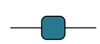
A useful tool for understanding quantum systems and their dynamics is via the quantum circuit diagrammatic language, such as in Fig. 2.1. Here, n qudits are represented by n horizontal lines, in an initial state on the left. Qudits are acted upon from left to right by a sequence of k layers of operators or maps, represented by k vertical boxes connecting the qudit lines where they act. For clarity, let us represent the initial state of qudits as coloured circles, and let us represent operators and maps, also referred to as *gates*, as solid-coloured boxes for unitaries and hatched-coloured boxes for channels. The number of qudits n and the number of layers k are also referred to as the system *size*, and circuit *depth*.



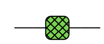
(a) Circuit diagrams $\rho \to \Lambda(\rho)\ :\ \Lambda = \prod_{i\in[k]} \Lambda^{(i)}$



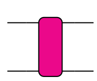
(b) Initial state ρ



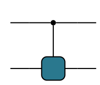
(c) Local unitary U



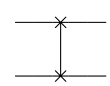
(d) Local channel Λ



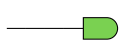
(e) Non-local unitary U



(f) Controlled unitary CU



(g) Swap unitary S



(h) Measurement in basis Π

(i) Partial trace tr

(j) Classical information x

Figure 2.1: **Quantum circuit diagrams**. (a) Quantum circuit with k layers of n qudit local (blue squares) and nearest neighbour (pink rectangles) unitaries, followed by local channels (hatched green squares) after each layer, with an initial state (left-most gray circles). (b-j) Example circuit state, operator, and channel diagrams.

2.2.4 Quantum Channels

Quantum channels $\Lambda \in \mathcal{B}[\mathcal{B}[\mathcal{H}]]$, acting on states $\rho \in \mathcal{B}[\mathcal{H}]$, can be classified as being either *unital* or *non-unital*, depending on their preservation of certain states.

Unital channels preserve maximally mixed states,

$$\Lambda(I) = I \ , \tag{2.100}$$

and typically are entropy-non-decreasing, with fixed points of maximally mixed states. An example is $k \to \infty$ concatenations of the depolarizing channel, for $\gamma > 0$,

$$\mathcal{D}^k_\gamma(\rho) = (1-\gamma)^k \rho \ + \ (1-(1-\gamma)^k)\frac{1}{d}I \qquad : \qquad \lim_{k\to\infty} \mathcal{D}^k_\gamma(\rho) = \frac{1}{d}I \ . \tag{2.101}$$

Non-unital channels do not preserve maximally mixed states,

$$\Lambda(I) = I \ + \ \Delta \ , \tag{2.102}$$

for some traceless operator $\Delta \in \mathcal{B}[\mathcal{H}]$, and are typically entropy-non-increasing, with fixed points of non-maximally mixed states. An example is $k \to \infty$ concatenations of the replacement channel, where the replacement operator is a pure state $|\psi\rangle \in \mathcal{H}$, for $\eta > 0$,

$$\mathcal{N}^k_\eta(\rho) = (1-\eta)^k \rho \ + \ (1-(1-\eta)^k)\psi \qquad : \qquad \lim_{k\to\infty} \mathcal{N}^k_\eta(\rho) = \psi \ . \tag{2.103}$$

2.2.5 Quantum Measurements

Measurement operators $\mathcal{P} = \{\Pi_\mu\}_{\mu\in[|\mathcal{P}|]}$, in particular POVM's, with probabilities $p_\mu = \chi(\Pi_\mu, \rho)$, overlaps $\chi_{\mu\nu} = \chi(\Pi_\mu, \Pi_\nu)$ and structure constants $\Pi_\mu\Pi_\nu = \chi_{\mu\nu\zeta}\Pi_\zeta$, can be classified as being *symmetric*, or *non-symmetric*.

Symmetric informationally complete POVM's (SIC-POVM) consist of pure states, up to a scaling $\lambda = 1/d > 0$, that are equivalent up to unitaries,

$$\mathcal{P} = \{\Pi_\mu = \lambda\psi_\mu\}_{\mu\in[|\mathcal{P}|]} \ , \tag{2.104}$$

and have closed-form expressions for their overlaps [31], given $\xi = d(d+1)$,

$$\chi_{\mu\nu} = (1/\xi)(\delta_{\mu\nu} + (1/d)1_\mu 1_\nu) \quad , \quad \chi^{-1}_{\mu\nu} = \xi\delta_{\mu\nu} - 1_\mu 1_\nu \tag{2.105}$$

$$\chi_{\mu\nu\zeta}1_\zeta = d\chi_{\mu\nu} \quad , \quad \chi_\mu = (1/d)1_\mu \quad , \quad \chi^{-1}_\mu = d\,1_\mu \ . \tag{2.106}$$

Non-symmetric POVM's (NON-SIC-POCM) consist of not-necessarily pure states, that are not equivalent up to unitaries,

$$\mathcal{P} = \{\Pi_\mu \not\propto \psi_\mu\}_{\mu \in [|\mathcal{P}|]} \ , \tag{2.107}$$

and in general do not have closed-form expressions for their overlaps.

$d = q^n$-dimensional measurement operators, informationally complete, symmetric, or otherwise, can also be localized, without loss of generality, as tensor products of n, q-dimensional measurement operators,

$$\mathcal{P} = \{\Pi_\mu\}_{\mu \in [q^2]} \to \mathcal{P}^{\otimes n} = \{\Pi_{\mu\sigma\ldots}\}_{\mu,\sigma,\cdots \in [q^2]} \ , \tag{2.108}$$

with probabilities, overlaps, and structure constants of,

$$p_\mu \to p_{\mu\sigma\ldots} \qquad , \qquad \chi_{\mu\nu} \to \chi_{\mu\nu}\chi_{\sigma\pi}\cdots \qquad , \qquad \chi_{\mu\nu\lambda} \to \chi_{\mu\nu\lambda}\chi_{\sigma\pi\tau}\cdots \ . \tag{2.109}$$

2.2.6 Quantum Metrics

Given quantum states, is important to develop intuitive metrics to assess their similarities, such as their infidelity, their properties, such as their entanglement measures, and their susceptibility to change with parameterizations, such as Fisher information metrics.

Infidelity

When assessing similarity of quantum states, a metric with clear operational meaning is the success probability of distinguishing between two states, occurring with respective probabilities $\lambda, 1-\lambda$. Using the *Holevo-Helstrom* and *Fuchs-van de Graaf* theorems, success probabilities $\mathcal{P}$ can be related to the *infidelity* metric $\mathcal{L}$ of similarity between states [176].

In the classical case, where states are probabilities $p, s \in \mathbb{R}^d_+$, the maximum success probability of distinguishing the states is,

$$\mathcal{P}_{ps} = \frac{1}{2}\left(1 \ + \ \mathrm{tr}(|\lambda p - (1-\lambda)s|)\right) \ , \tag{2.110}$$

and the infidelity between the states is,

$$\mathcal{L}_{ps} = 1 - \sqrt{p} \cdot \sqrt{s} \ . \tag{2.111}$$

In the quantum case, where states are density matrices $\rho, \sigma \in \mathcal{B}[\mathcal{H}]$, the maximum success probability of distinguishing the states is,

$$\mathcal{P}_{\rho\sigma} = \frac{1}{2}\left(1 \;+\; \mathrm{tr}(|\lambda\rho - (1-\lambda)\sigma|)\right) \ , \tag{2.112}$$

and the infidelity between the states is,

$$\mathcal{L}_{\rho\sigma} = 1 - \mathrm{tr}(\sqrt{\rho\sigma}) \ . \tag{2.113}$$

Further, the probabilities, infidelities, and trace distance, are related, for $\lambda = 1/2$,

$$\mathcal{L}_{\rho\sigma} = 1 - \mathrm{tr}(\sqrt{\rho\sigma}) \leq \frac{1}{2}\mathrm{tr}(|\rho - \sigma|) = 2\mathcal{P}_{\rho\sigma} - 1 \ . \tag{2.114}$$

which as per [176], roughly follows, from identities that there exists measurements, such that the quantum infidelity between ρ, σ equals the classical infidelity of their measurement probabilities, p, s, $1 - \mathrm{tr}\big(\sqrt{\rho\sigma}\big) = 1 - \sqrt{p}\cdot\sqrt{s} \leq \mathrm{tr}(|p - s|)$, and that $\mathrm{tr}(|p-s|) \leq \mathrm{tr}(|\rho - \sigma|)$.

In fact, infidelities can be shown [183] to be approximately multiplicative or even additive under a sequence of k approximate operations U, each acting with infidelity γ, resulting in approximate states $|\psi\rangle \to U^k|\psi\rangle \approx |\psi_\chi\rangle$ that have infidelity $\mathcal{L} \approx 1 - (1-\gamma)^k \approx k\gamma$. Given unitaries $\{U^{(i)}\}_{i\in[k]}$, then at iteration k, the exact state $\psi^{(k-1)}$ may be expanded in terms of the approximate state $\psi_\gamma^{(k-1)}$, and its orthogonal state $\phi_\gamma^{(k-1)} \ : \ \langle\phi_\gamma^{(k-1)}|\psi_\gamma^{(k-1)}\rangle = 0$,

$$|\psi^{(k-1)}\rangle \ = \sqrt{1 - \mathcal{L}_\gamma^{(k-1)}}|\psi_\gamma^{(k-1)}\rangle \ + \ \sqrt{\mathcal{L}_\gamma^{(k-1)}}|\phi_\gamma^{(k-1)}\rangle \ , \tag{2.115}$$

and under the application of the unitary $U^{(k)}$ with infidelity $\gamma^{(k)}$, the states have the form,

$$|\psi^{(k)}\rangle = U^{(k)}|\psi^{(k-1)}\rangle \tag{2.116}$$

$$|\psi_\gamma^{(k)}\rangle = \sqrt{1 - \gamma^{(k)}}U^{(k)}|\psi_\gamma^{(k-1)}\rangle \ + \ \sqrt{\gamma^{(k)}}U^{(k)}|\phi_\gamma^{(k-1)}\rangle \ . \tag{2.117}$$

The infidelity between the exact $\psi^{(k)}$ and approximate states $\psi_\gamma^{(k)}$ is thus,

$$1 - \mathcal{L}_\gamma^{(k)} = \left|\langle\psi^{(k)}|\psi_\gamma^{(k)}\rangle\right|^2 = \left(\sqrt{(1-\gamma^{(k)})(1 - \mathcal{L}_\gamma^{(k-1)})} \ + \ \sqrt{\gamma^{(k)}\mathcal{L}_\gamma^{(k-1)}}\right)^2 \ . \tag{2.118}$$

We have the exact bound on the change in fidelity in terms of the operator infidelity $\gamma^{(k)}$,

$$\left|\sqrt{1 - \mathcal{L}_\gamma^{(k)}} - \sqrt{1 - \mathcal{L}_\gamma^{(k-1)}}\right| \leq \sqrt{\gamma^{(k)}} \ . \tag{2.119}$$

and when $\gamma\mathcal{L} \ll (1-\gamma)(1-\mathcal{L})$, infidelities are multiplicative in the operator infidelity γ,

$$1 - \mathcal{L}_\gamma^{(k)} \approx \prod_{i\in[k]}(1 - \gamma^{(i)}) \qquad \overset{\gamma^{(i)}=\gamma}{\rightarrow} \qquad \mathcal{L}_\gamma^{(k)} \approx 1 - (1-\gamma)^k \approx k\gamma \ . \tag{2.120}$$

Entanglement

When assessing correlations within quantum states, we recall that mixed states,

$$\rho_\Gamma = \mathrm{tr}_{\bar{\Gamma}}(\rho) \ , \tag{2.121}$$

given $\rho \in \mathcal{B}[\mathcal{H}]$, arise from pure states due to correlations between subsystems $\Gamma, \bar{\Gamma} \subseteq \mathcal{H}$, or with environments $\mathcal{E}$, as per Fig. 2.2. It is therefore important to develop consistent metrics that distinguish between these types of correlations.

Entanglement entropy $\mathcal{S}_\rho^\Gamma$ captures any correlations of reduced quantum states with other subsystems, with the classical definition, for probabilities p,

$$\mathcal{S}_p^\Gamma = -\frac{1}{\log(d)}\mathrm{tr}(p_\Gamma \log(p_\Gamma)) \ , \tag{2.122}$$

inspiring the quantum definition, for states ρ,

$$\mathcal{S}_\rho^\Gamma = -\frac{1}{\log(d)}\mathrm{tr}(\rho_\Gamma \log(\rho_\Gamma)) \ . \tag{2.123}$$

Operator entanglement $\mathcal{Q}_\rho^\Gamma$ is defined in terms of the entanglement entropy $\mathcal{S}_\rho^\Gamma$ of vectorized states $\rho \to \Psi$, taken to be pure states in a doubled space $\mathcal{H} \to \mathcal{H} \otimes \mathcal{H}$, reduced to the states of subsystems $\Gamma \subseteq [n]$ on both copies of $\mathcal{H}$,

$$\rho \to \Psi = \frac{|\rho\rangle\!\rangle\langle\!\langle\rho|}{\langle\!\langle\rho|\rho\rangle\!\rangle} \to \Psi_\Gamma = \mathrm{tr}_{\bar{\Gamma}\otimes\bar{\Gamma}}(\Psi) \ . \tag{2.124}$$

The operator entanglement then can be defined in terms of ρ and Γ, for the respective d and d^2 dimensional systems, as,

$$\mathcal{Q}_\rho^\Gamma = -\frac{1}{2}\frac{1}{\log(d)}\mathrm{tr}(\Psi_\Gamma \log(\Psi_\Gamma)) = \mathcal{S}_\Psi^\Gamma \ . \tag{2.125}$$

As discussed in [111, 185], whereas entanglement entropy of ρ measures entanglement of a subsystem Γ with its complement $\bar{\Gamma}$ or environment $\mathcal{E}$, the operator entanglement, or the entanglement entropy of the purified states $\rho \to |\rho\rangle\!\rangle$ in the doubled space, emphasizes entanglement of Γ with $\bar{\Gamma}$. In particular, any imposed entanglement between Γ and $\mathcal{E}$, causing impurity in ρ_Γ, does not affect the purity of initially separable $|\rho\rangle\!\rangle = |\rho_\Gamma\rangle\!\rangle \otimes |\rho_{\bar{\Gamma}}\rangle\!\rangle$, and thus does not increase operator entanglement. Further, pure states $\rho = \psi \to |\rho\rangle\!\rangle \cong |\psi\psi\rangle$ have identical normalized entanglement entropy and operator entanglement, making it a consistent entanglement measure.

To motivate our choice of operator entanglement, we note its relationship to entanglement entropy in the following cases.

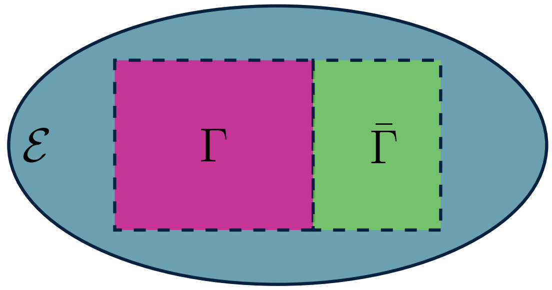


Figure 2.2: **Partitions of Hilbert space**. Partitioning of a space $\mathcal{H} = \Gamma \oplus \bar{\Gamma}$ into disjoint sets $\Gamma, \bar{\Gamma}$, with an environment $\mathcal{E}$ such that the total space is $\mathcal{H} \otimes \mathcal{E} = \Gamma \oplus \bar{\Gamma} \otimes \mathcal{E}$.

- *Pure States*: For pure states $\rho = |\psi\rangle\langle\psi|$,

$$\rho \to |\rho\rangle\rangle = |\psi\psi\rangle\rangle \to |\rho\rangle\rangle\langle\langle\rho| = \rho \otimes \rho \to |\rho\rangle\rangle\langle\langle\rho|_\Gamma = \rho_\Gamma \otimes \rho_\Gamma \tag{2.126}$$

is double of the original state, with identical normalized entanglement measures,

$$\mathcal{Q}^\Gamma_{|\psi\rangle} = \mathcal{S}_{\rho_\Gamma} = \mathcal{S}^\Gamma_{|\psi\rangle} \ . \tag{2.127}$$

- *Mixed States*: For non-traced out states when $\Gamma = [n]$,

$$\rho \to |\rho\rangle\rangle \to |\rho\rangle\rangle\langle\langle\rho| \tag{2.128}$$

is a pure doubled state with zero normalized operator entanglement,

$$\mathcal{Q}_\rho = \mathcal{S}_{\frac{|\rho\rangle\rangle}{\sqrt{\langle\langle\rho|\rho\rangle\rangle}}} = 0 \leq \mathcal{S}_\rho \ . \tag{2.129}$$

- *Product States*: For product states $\rho = \rho_\Gamma \otimes \rho_{\bar{\Gamma}}$,

$$\rho \to |\rho\rangle\rangle = |\rho_\Gamma\rangle\rangle \otimes |\rho_{\bar{\Gamma}}\rangle\rangle \to |\rho\rangle\rangle\langle\langle\rho| \to |\rho_\Gamma\rangle\rangle\langle\langle\rho_\Gamma| \tag{2.130}$$

is a pure doubled state with zero normalized operator entanglement,

$$\mathcal{Q}^\Gamma_{\rho_\Gamma \otimes \rho_{\bar{\Gamma}}} = \mathcal{S}_{\frac{|\rho_\Gamma\rangle\rangle}{\sqrt{\langle\langle\rho_\Gamma|\rho_\Gamma\rangle\rangle}}} = 0 \leq \mathcal{S}_{\rho_\Gamma} \ . \tag{2.131}$$

- *Maximally Mixed States*: For maximally mixed states $\rho = \frac{1}{d}I = \frac{1}{d_\Gamma}I_\Gamma \otimes \frac{1}{d_{\bar{\Gamma}}}I_{\bar{\Gamma}}$,

$$\rho \to |\rho\rangle\rangle = \frac{1}{d}|\Omega\rangle \to |\rho\rangle\rangle\langle\langle\rho| = \frac{1}{d^2}\Omega \to |\rho\rangle\rangle\langle\langle\rho|_\Gamma = \frac{1}{d_\Gamma^2}\Omega_\Gamma \tag{2.132}$$

is a pure maximally entangled state with zero normalized operator entanglement,

$$\mathcal{Q}^\Gamma_I = \mathcal{S}_{\frac{|\Omega_\Gamma\rangle\rangle}{d_\Gamma}} = 0 < \mathcal{S}^\Gamma_I = 1 \ . \tag{2.133}$$

- *Maximally Entangled States*: For maximally entangled states $\rho = \frac{1}{d_\Gamma}\Omega_{\Gamma\bar{\Gamma}}$,

$$\rho \to |\rho\rangle\rangle = \frac{1}{d_\Gamma}|\Omega_{\Gamma\bar{\Gamma}}\Omega_{\Gamma\bar{\Gamma}}\rangle \to |\rho\rangle\rangle\langle\langle\rho| = \frac{1}{d_\Gamma^2}\Omega_{\Gamma\bar{\Gamma}} \otimes \Omega_{\Gamma\bar{\Gamma}} \to |\rho\rangle\rangle\langle\langle\rho|_\Gamma = \frac{d_{\bar{\Gamma}}}{d_\Gamma^2} I_\Gamma \otimes I_\Gamma \quad (2.134)$$

is a maximally mixed state, with maximum normalized operator entanglement,

$$\mathcal{Q}_\Omega^\Gamma = 1 = \mathcal{S}_\Omega^\Gamma \ . \quad (2.135)$$

- *Local Operations and Classical Communication*: For product states mapped to a mixture of separable states, $\sigma = \sigma_\Gamma \otimes \sigma_{\bar{\Gamma}} \overset{LOCC}{\to} \rho_{LOCC} = \sum_\lambda p_\lambda \rho_\Gamma^{(\lambda)} \otimes \rho_{\bar{\Gamma}}^{(\lambda)}$,

$$\rho \to |\rho\rangle\rangle = \sum_\lambda p_\lambda |\rho_\Gamma^{(\lambda)}\rangle\rangle \otimes |\rho_{\bar{\Gamma}}^{(\lambda)}\rangle\rangle \to |\rho\rangle\rangle\langle\langle\rho| = \sum_{\lambda\kappa} p_\lambda p_\kappa |\rho_\Gamma^{(\lambda)}\rangle\rangle\langle\langle\rho_\Gamma^{(\kappa)}| \otimes |\rho_{\bar{\Gamma}}^{(\lambda)}\rangle\rangle\langle\langle\rho_{\bar{\Gamma}}^{(\kappa)}| \quad (2.136)$$

$$\to |\rho\rangle\rangle\langle\langle\rho|_\Gamma = \sum_{\lambda\kappa} p_\lambda \langle\langle\rho_{\bar{\Gamma}}^{(\lambda)}|\rho_{\bar{\Gamma}}^{(\kappa)}\rangle\rangle p_\kappa |\rho_\Gamma^{(\lambda)}\rangle\rangle\langle\langle\rho_\Gamma^{(\kappa)}|$$

is a mixture of vectorized states with non-negative normalized operator entanglement,

$$\mathcal{Q}_{\rho_{LOCC}}^\Gamma \geq 0 \ . \quad (2.137)$$

Finally, the average normalized entanglement entropy $\mathcal{S}_{|\psi\rangle}^l = \left\langle \mathcal{S}_\psi^l \right\rangle_\psi$ over Haar random pure states ψ, for all dimensions d and subsystems of dimension $l \leq d$ [161, 186], is,

$$\mathcal{S}_{|\psi\rangle}^l = \frac{1}{\log(d)} \sum_{l>l}^{d} \frac{1}{l} - \frac{1}{\log(d)} \frac{\binom{l}{2}}{d} \ \leq \ 1 - \frac{\log(l)}{\log(d)} - \frac{\binom{l}{2}}{d} \ . \quad (2.138)$$

Discord

Quantum discord $\mathcal{D}_\rho^{\Gamma|\mathcal{P}}$ is defined as the difference of quantum mutual information definitions, namely the quantum $\mathcal{I}_\rho^{\Gamma}$ and the measurement $\mathcal{I}_\rho^{\Gamma|\mathcal{P}}$ definitions,

$$\mathcal{I}_\rho^{\Gamma} = \mathcal{S}_\rho^{\Gamma} - \mathcal{S}_\rho^{\Gamma|\bar{\Gamma}} \quad : \quad \mathcal{S}_\rho^{\Gamma|\bar{\Gamma}} = \mathcal{S}_\rho - \mathcal{S}_\rho^{\bar{\Gamma}} \quad , \quad p_\Pi = \mathrm{tr}(\Pi\rho) \tag{2.139}$$

$$\mathcal{I}_\rho^{\Gamma|\mathcal{P}} = \mathcal{S}_\rho^{\Gamma} - \mathcal{S}_\rho^{\Gamma|\mathcal{P}} \quad : \quad \mathcal{S}_\rho^{\Gamma|\mathcal{P}} = \sum_{\Pi\in\mathcal{P}} p_\Pi \; \mathcal{S}_{\rho_{\Gamma|\Pi}} \quad , \quad \rho_{\Gamma|\Pi} = \frac{1}{p_\Pi}\mathrm{tr}_{\bar{\Gamma}}(\Pi\rho) \ , \tag{2.140}$$

leading to the definition of quantum discord $\mathcal{D}_\rho^{\Gamma|\mathcal{P}}$, with respect to measurements $\mathcal{P} = \{\Pi\}$,

$$\mathcal{D}_\rho^{\Gamma|\mathcal{P}} = \mathcal{I}_\rho^{\Gamma} \; - \; \mathcal{I}_\rho^{\Gamma|\mathcal{P}} = \left(\mathcal{S}_\rho^{\Gamma} \; + \; \mathcal{S}_\rho^{\bar{\Gamma}} \; - \; \mathcal{S}_\rho\right) \; - \; \left(\mathcal{S}_{\sum_{\Pi\in\mathcal{P}} p_\Pi \rho_{\Gamma|\Pi}} \; - \; \sum_{\Pi\in\mathcal{P}} p_\Pi \; \mathcal{S}_{\rho_{\Gamma|\Pi}}\right) . \tag{2.141}$$

Such differences originate from differences in quantum conditional entropy $\mathcal{S}_\rho^{\Gamma|\bar{\Gamma}}, \mathcal{S}_\rho^{\Gamma|\mathcal{P}}$ definitions, and whether information about a subsystem $\Gamma \subseteq [n]$ is known a priori, or the information is measured using $\Pi \in \mathcal{P}$, yielding marginal states $\rho_{\Gamma|\Pi} = \frac{1}{p_\Pi}\mathrm{tr}_{\bar{\Gamma}}(\Pi \; \rho)$ with probabilities $p_\Pi = \mathrm{tr}(\Pi \; \rho)$ [187, 188]. Discord thus measures the influence of specific measurements on subsystems, and mutual information $\mathcal{I}_\rho^{\Gamma}$, quantum discord $\mathcal{D}_\rho^{\Gamma|\mathcal{P}}$ and the Holevo information $\mathcal{X}_\rho^{\Gamma|\mathcal{P}} = \mathcal{I}_\rho^{\Gamma|\mathcal{P}}$ are related by [189], $\mathcal{I}_\rho^{\Gamma} = \mathcal{D}_\rho^{\Gamma|\mathcal{P}} + \mathcal{X}_\rho^{\Gamma|\mathcal{P}}$, for any $\mathcal{P}$.

Fisher Information

When assessing parameterizations of quantum states, it is important to assess their susceptibility to changes in their parameterization. More concretely, given parameterized states $\rho_\theta \in \mathcal{B}[\mathcal{H}]$, for parameters $\theta = \{\theta_\mu\}_{\mu\in[|\theta|]}$, their infidelity to $\rho \in \mathcal{B}[\mathcal{H}]$ is,

$$\mathcal{L}_{\rho\theta} = 1 - \mathrm{tr}(\sqrt{\rho\rho_\theta}) \ , \tag{2.142}$$

The *Fisher information* $\mathcal{F}_{\rho\theta}$, then may be defined as the leading-order behaviour of the infidelity [182], given a perturbation of parameters $\theta \to \theta + \delta$, evaluated at $\rho = \rho_\theta$,

$$\mathcal{L}_{\rho\theta+\delta} = \mathcal{F}_{\rho\theta\mu\nu}\delta_\mu\delta_\nu \; + \; \mathcal{O}(\delta^3) \ , \tag{2.143}$$

which requires careful derivatives of the infidelity with respect to the parameterized states.

The quantum Fisher information metric for general states, $\rho_\theta = \sum_\lambda \lambda|\lambda\rangle\langle\lambda|$, is,

$$\mathcal{F}_{\rho\theta\mu\nu} = \frac{1}{2} \sum_{\substack{\lambda,\lambda' \\ \lambda+\lambda'>0}} \frac{\langle\lambda|\partial_\mu\rho_\theta|\lambda'\rangle\langle\lambda'|\partial_\nu\rho_\theta|\lambda\rangle}{\lambda+\lambda'} \ , \tag{2.144}$$

and for pure states, $\psi_\theta = |\psi_\theta\rangle\langle\psi_\theta|$, reduces to,

$$\mathcal{F}_{\psi\theta\mu\nu} = \langle\partial_\mu\psi_\theta|\partial_\nu\psi_\theta\rangle \; - \; \langle\partial_\mu\psi_\theta|\psi_\theta\rangle\langle\psi_\theta|\partial_\nu\psi_\theta\rangle \; . \tag{2.145}$$

Such a metric is inspired by the classical Fisher information, where for probabilities p, p_θ from measurements $\mathcal{P} = \{\Pi\}$, the KL-divergence is analogous to the state infidelities,

$$\mathcal{L}_{p\theta} = -\frac{1}{4}\mathrm{tr}\left(p\log\left(\frac{p_\theta}{p}\right)\right) \to \mathcal{F}_{p\theta\mu\nu} = \frac{1}{4}\mathrm{tr}\left(\frac{\partial_\mu p_\theta\,\partial_\nu p_\theta}{p_\theta}\right) = \frac{1}{4}\sum_{\Pi\in\mathcal{P}}\frac{\mathrm{tr}(\Pi\partial_\mu\rho_\theta)\,\mathrm{tr}(\Pi\partial_\nu\rho_\theta)}{\mathrm{tr}(\Pi\rho_\theta)}. \tag{2.146}$$

Further, it can be shown [182], from assumptions of the monotonicity of the infidelities under quantum channels, that for any measurements on states ρ yielding probabilities p, the quantum Fisher information upper bounds the classical Fisher information,

$$\mathcal{F}_{p\theta} \leq \mathcal{F}_{\rho\theta} \; . \tag{2.147}$$

For example, if a pure or mixed state is generated by a Hamiltonian H_θ, with parameterized eigenvalues $\epsilon_{\theta\lambda}$, and constant eigenvectors $|\lambda\rangle$,

$$\begin{array}{lcl} |\psi_\theta\rangle = e^{-iH_\theta}|\psi\rangle & \quad : \quad & H_\theta = \sum_\lambda \epsilon_{\theta\lambda}\;|\lambda\rangle\langle\lambda| \\ \rho_\theta = \frac{1}{Z_\theta}e^{-iH_\theta} & & Z_\theta = \mathrm{tr}(\rho_\theta) \end{array} \quad , \tag{2.148}$$

then the Fisher information is the covariances of the Hamiltonian's gradients,

$$\mathcal{F}_{\psi\theta\mu\nu} \;\propto\; \langle\psi|\partial_\mu H_\theta\partial_\nu H_\theta|\psi\rangle \;\; - \;\; \langle\psi|\partial_\mu H_\theta|\psi\rangle\langle\psi|\partial_\nu H_\theta|\psi\rangle \tag{2.149}$$

$$\mathcal{F}_{\rho\theta\mu\nu} \;\propto\; \mathrm{tr}(\rho_\theta\;\partial_\mu H_\theta\;\partial_\nu H_\theta) \;-\; \mathrm{tr}(\rho_\theta\;\partial_\mu H_\theta)\;\mathrm{tr}(\rho_\theta\;\partial_\nu H_\theta) \; . \tag{2.150}$$

The Fisher information conveys how much information about a state is available, given its parameterization, and indicates in which parameter directions, do distance measures between states change the most. Its classical form of changes in state eigenvalues, versus its quantum form of changes in state eigenvalues and eigenvectors, imply changes in states due to parameterizations are more subtle in quantum settings. Given its connections to parameterization and infidelities, the Fisher information arises within variational quantum algorithms [102], quantum sensing [72, 182], and quantum optimal control [190]. In fact, parameter estimation can be quantified by the Cramer-Rao bound for the covariance $\Sigma_\theta = \mathrm{tr}(\rho_\theta\tilde{\Pi}\tilde{\Pi}) - \mathrm{tr}(\rho_\theta\Pi)\mathrm{tr}(\rho_\theta\Pi)$ of any unbiased estimator $\tilde{\theta} = \mathrm{tr}\left(\tilde{\Pi}\rho\right) \approx \mathrm{tr}(\Pi\rho) = \theta$ [191],

$$\Sigma_\theta \geq \mathcal{F}_\theta^{-1} \; . \tag{2.151}$$

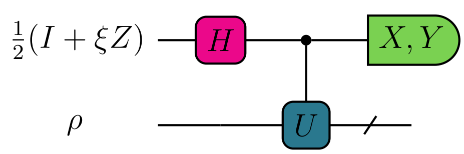
This can be shown using the symmetric-logarithmic derivative L to define state gradients, $2\partial\rho = L\rho + \rho L$, and using the Cauchy-Schwartz inequality for measurements $\mathrm{tr}^2(\Pi\rho L) \leq \mathrm{tr}(\Pi\rho)\,\mathrm{tr}(\Pi\rho L^2)$. Such derivations also show the variance of L is the maximum classical Fisher information, yielding the quantum Fisher information, $\mathcal{F}_p \leq (1/4)\mathrm{tr}(\rho L^2) = \mathcal{F}_\rho$, which is saturated when Π are projectors onto the image of L [192].

2.2.7 Experimental Traces

Suppose we want to calculate the trace of a unitary $U \in \mathcal{U}[\mathcal{H}]$ with a state $\rho \in \mathcal{B}[\mathcal{H}]$,

$$\operatorname{tr}(U\rho) \tag{2.152}$$

but are not able to directly measure the state. Then, using a single ancillary qubit, traces can be computed experimentally with the following *Hadamard test* quantum circuit [193, 194], in terms of I, X, Y, Z, H qubit Pauli and Hadamard operators, with $0 \leq \xi \leq 1$,



. (2.153)

The initial ξ-polarized Z basis qubit state is changed to the X basis (pink), applies a controlled unitary U in the Z basis (blue) to the state ρ, before measurements are performed in the X or Y basis (green) on the qubit, yielding the desired trace,

$$\rho \overset{\otimes}{\rightarrow} \frac{1}{2}(I + \xi Z) \otimes \rho \tag{2.154}$$

$$\overset{H\otimes I}{\rightarrow} \frac{1}{2}(I + \xi X) \otimes \rho \tag{2.155}$$

$$\overset{CU}{\rightarrow} I \otimes \frac{\rho + U\rho U^\dagger}{2} \; + \; Z \otimes \frac{\rho - U\rho U^\dagger}{2} \; + \; X \otimes \xi \frac{U\rho + \rho U^\dagger}{2} \; + \; Y \otimes \xi \frac{U\rho - \rho U^\dagger}{i2} \tag{2.156}$$

$$\overset{X,Y}{\rightarrow} \{\xi \ \mathrm{Re}\,(\operatorname{tr}(U\rho)) \quad , \quad \xi \ \mathrm{Im}\,(\operatorname{tr}(U\rho))\} \ . \tag{2.157}$$

Such steps follow from the action of the controlled unitary $CU = (I - \Pi) \otimes I + \Pi \otimes U$, given the Z-basis projection $\Pi = (I - Z)/2$, with $\Pi X \Pi = 0$, and $\Pi X(I - \Pi) = (X - iY)/2$, plus separate X, Y or total $X + iY$ measurements to obtain $\operatorname{tr}(U\rho)$.

Using Chebyshev's inequality, traces can be computed to within statistical error at most ϵ, with probability $1 - \delta^2$, with a number of measurements,

$$m \geq \frac{2 - |\operatorname{tr}(U\rho)|^2}{(\xi\delta\epsilon)^2} \ . \tag{2.158}$$

Given traces of products of operators can describe almost all operations in quantum settings, such experimental realizations are very general, with wide application. For example, fidelities $\operatorname{tr}(\rho\rho')$ of states ρ, ρ' can be estimated by replacing states with the tensor product, $\rho \to \rho \otimes \rho'$, and the unitary with the swap operator $U \to S$.

2.2.8 Quantum Experiments

Any quantum information protocols must be implemented on devices exhibiting quantum phenomena. Such implementations are generally described by their number of individual q-dimensional qudit components n, algorithms are described by individual gate operators and the number of operations k, measurements arise from various procedures, and noise is characterized by a noise scale γ or decoherence time T. Given their error resistance, but not precise fault tolerance [28], with many initial error correction protocols, such devices are classified as noisy-intermediate-scale quantum devices (NISQ) [18]. Here, given their current strengths and applications, we will review nuclear magnetic resonance, optics, trapped ion, superconducting qubit, and neutral atom approaches.

Nuclear Magnetic Resonance

Nuclear magnetic resonance (NMR) approaches use the nuclear spin of individual atoms of molecules as qubits [195]. Qubits are initialized as an ensemble of atoms within a liquid or solid sample, consisting of identical qubits prepared in pseudo-pure states, meaning spins are weakly polarized along a spin axis. Gates are implemented via radio frequency pulse (RF) magnetic fields, to manipulate the spins, with constant all-to-all 2-qubit gates naturally present from inherent atomic-couplings, and variable individual 1-qubit fields from external pulses [196]. Measurements of spin axes are accomplished via an ensemble measurement of currents induced by the qubit's magnetic fields. Noise and errors occur due to inhomogeneities in any fields, and unintentional spin-spin interactions, with generally stable qubit states. Such noise manifests itself as decay in magnetic signals along the spin axes, which can be accounted for using various procedures like refocusing [195]. The NMR Hamiltonian for n spin qubits can thus be described in terms of the qubit Pauli X, Y, Z operators on $|0\rangle, |1\rangle$, the pulse frequency ω and phase ϕ, the detuning off-resonance frequency Δ from the natural Larmor frequency of the nuclei, the Rabi frequency of oscillations between qubit states Ω, and the nuclei-nuclei J-coupling,

$$H_{\mathrm{NMR}} = \; - \; \frac{1}{2}\sum_{i\in[n]} \Delta_i \, Z_i \; - \; \frac{1}{2}\sum_{i\in[n]} \Omega_i \, \cos(\omega t + \phi) \, X_i \; + \; \frac{1}{2}\sum_{i<j\in[n]} J_{ij} \, Z_i \, Z_j \; . \tag{2.159}$$

Successful implementations include Cory *et al.'s* (Massachusetts Institute of Technology, Los Alamos National Laboratory, 1998) and Zhang *et al.'s* (University of Waterloo, 2012) demonstration of error correction [197, 198], and Negrevergne *et al.'s* (University of Waterloo, 2006) quantum optimal control of $n = 12$ qubits and qutrits [40]. Weaknesses of NMR include its limited scalability to $n \leq \mathcal{O}(10)$ due to qubits being difficult to address individually, and the initialized states being typically exponentially more weakly polarized as system sizes increase, weakening any possible entanglement, and any measurement signals. Strengths of NMR include its intuitive architecture, with stable, well defined, naturally occurring qubits, that allowed for the development of many algorithms such as error correction [197], zero-noise extrapolation [37], dynamical decoupling [199], and refocusing procedures [195], that are the foundation of other quantum devices [195, 200].

Optics

Quantum optics approaches use the state of polarized light, or individual photons as qubits [201]. Qubits are initialized as being encoded into orthogonally polarized states of light, typically within optical fibres, and typically represented as photon mode or coherent or Gaussian states [41]. Gates are implemented via linear optical elements, such as beam splitters and phase shifters for 1-qubit gates, and non-linear, probabilistic, quantum-teleportation protocols for 2-qubit gates [201]. Measurements of photon polarizations are accomplished by photon detectors, which typically are superconducting, and necessitate being substantially cooled. Noise and errors occur due to loss and phase shifts of photons within optical fibres, inefficiencies of photon detectors, and errors within the probabilistic quantum teleportation protocols. The linear optics Hamiltonian for n photon qudits with q modes can thus be described in terms of the photon mode creation and annihilation operators $a^\dagger, a$, with couplings h, such that the number of photons are conserved,

$$H_{\text{Optics}} = \sum_{i\in[n]} \sum_{\mu,\nu\in[q]} h_{i\mu\nu}\, a^\dagger_{i\mu} a_{i\nu} \ . \tag{2.160}$$

Successful implementations include Zhong *et al.'s* (University of Science and Technology of China, 2020), Deng *et al.'s* (University of Science and Technology of China, 2023), and Deshpande *et al.'s* (University of Maryland, Freie Universitat Berlin, Xanadu, 2022) Gaussian boson sampling experiments to compute matrix determinants [43–45], Paparalle *et al.'s* (Sorbonne Université, Instituto de Física Interdisciplinar y Sistemas Complejos, 2026) reservoir computing of correlation functions [202], Main *et al.'s* (University of Oxford, 2025) distributed quantum computing to implement search algorithms [203], and Daley *et al.'s* (University of Waterloo, 2022) verifications of various causal models [204]. Weaknesses of optics include difficulties of correlating and entangling individual pairs of photons, and single photon sources and detectors being exceptionally challenging to engineer. Strengths of optics include their fast gate times, and optics technology being well established, offering both mobile fibre approaches, or manufacturable built-in chip approaches [61].

Trapped Ions

Trapped ion approaches use the electronic energy levels of ions [205]. Qubits are initialized by suspending and separating individual ions in place using oscillating electromagnetic potentials that confine the spatial, or motional modes of the ions. Valence electron energy levels of ions, with appropriate spacing and lifetimes then can be addressed with optical lasers, allowing for qubit states to be encoded, and shuttled between levels [46, 206]. Gates are implemented via ion-laser interactions for both 1 and 2-qubit gates, where the motional and electronic modes become coupled. Depending on the relative frequencies of the modes, Rabi oscillations and so-called sideband or virtual transitions can be induced between the qubit states and the phonon motional modes [41, 48]. Measurements of electronic energy level occupations are accomplished via transferring certain qubit states to auxiliary energy levels, and observing any spontaneous emission, in so-called qubit shelving and state-dependent fluorescence techniques. Noise and errors occur due to fluctuations in lasers and electromagnetic fields, leading to decoherence between qubit motional and electronic states, and due to ions affecting each other, or crosstalk. The trapped ions Hamiltonian for n ion qubits can thus be described in terms of the qubit Pauli X, Y, Z operators on $|0\rangle, |1\rangle$, qubit raising and lowering operators $2S_{\pm} = X \pm iY$, the laser frequency ω and phase ϕ, the detuning off-resonance frequency Δ from the natural ion energy level spacing, the Rabi frequency of oscillations between qubit states Ω, the common motional mode creation and annihilation operators $a^\dagger, a$ and the motional mode frequency χ,

$$H_{\text{Ion}} = -\frac{1}{2}\sum_{i\in[n]} \Delta_i\, Z_i + i\Omega_i\left(e^{i\phi}e^{-i(\Delta_i-\chi)t}a^\dagger S_{i+} - e^{-i\phi}e^{i(\Delta_i-\chi)t}aS_{i-}\right) + \chi\left(a^\dagger a + \frac{1}{2}\right). \quad (2.161)$$

Successful implementations include Low *et al.'s* (University of Waterloo, 2020) generalizations to $q = 7$-dimensional qudits [47], Kotibhaskar *et al.'s* (University of Waterloo, 2024) Ising-like XY gates [67], Dasu *et al.'s* (Quantinuum, 2026) concatenated quantum error correction protocols where 48 logical qubits have better infidelities than 98 physical qubits [49], Kang *et al.'s* (Duke University, 2023) quantum error correction for erasure errors [207], and Hughes *et al.'s* (IonQ, 2025) 2-qubit gate fidelities of 10^{-4} at higher temperatures without significant sideband cooling [50]. Weaknesses of trapped ions include scalability and anticipated difficulties of engineering larger ion traps with lasers for each qubit, and requiring potentially multiple networked traps. Parallel 2-qubit gates are also complicated by having to share a common motional mode, or having to address finely spaced motional modes independently. Strengths of trapped ions include the stable nature of the qubits, being identical atoms, and thus requiring less calibration, with long decoherence times and high gate fidelities. Trapped ions can also naturally be extended to $q \gg 2$ qudit-based architectures, given the many electronic energy levels available [47].

Superconducting Qubits

Superconducting qubit approaches use the quantized energy levels of resonant superconducting circuits [208]. Qubits are initialized by inducing anharmonicity within the energy levels of currents, charges, or phases of ultra-cold circuits, such as so-called transmons for weakly anharmonic charge states. Such circuits involve Josephson junction elements, where pairs of electrons tunnel across junctions with tunable inductances, leading to compromises between charge noise robustness, versus anharmonic and isolated energy levels [53]. Gates are implemented via coupling the circuits to waveguide resonators or adding additional coupling capacitances. For 1-qubit gates, resonators in coherent states mediate Rabi oscillations between transmon states. For 2-qubit gates, coupling capacitances between transmon circuits allow for i-swap or controlled-phase gates. Measurements of transmon states are accomplished via coupling to a resonator, tuned in a dispersive regime to be greatly off resonant from the transmons, and transmon states are inferred from the resonator phase. Noise and errors occur due to imperfections in circuit elements, cooling procedures, and microwave signals, leading to charge noise and fluctuations in the transmon energy level structure [209]. The superconducting qubit Hamiltonian for n transmon qubits can thus be described in terms of the qubit Pauli X, Y, Z operators on $|0\rangle, |1\rangle$, the microwave frequency ω and phase ϕ, the detuning off-resonance frequency Δ from the natural transmon energy level spacing, the Rabi frequency of oscillations between qubit states Ω, and the capacitive coupling between circuits g,

$$H_{\mathrm{SQ}} = \; -\,\frac{1}{2}\sum_{i\in[n]} \Delta_i\, Z_i \;-\; \frac{1}{2}\sum_{i\in[n]} \Omega_i\, \cos(\omega t + \phi)\, X_i \;+\; \sum_{i<j\in[n]} g_{ij}\, Z_i\, X_j \;. \tag{2.162}$$

Successful implementations include Kim *et al.'s* (IBM, 2023) verification of sophisticated error mitigation methods [52, 70, 77], Zhu *et al.'s* (University of Science and Technology, 2022) high fidelity random circuit sampling experiments [210], and Zobrist *et al.'s* (Google, 2024) demonstrations of quantum error correction with increasing code distance and decreasing logical errors [211, 212]. Weaknesses of superconducting qubits are the difficulty of consistently manufacturing the circuits and fine tuning the anharmonicity when creating these artificially isolated q-level systems [41]. It is also difficult to prevent leakage to other energy levels, and to feasibly address each qubit independently with its own unique resonant frequency. Strengths of superconducting qubits are their tunability, and fast gate times due to being artificially constructed, leading to as strong as necessary coupling between qubits. Their use of silicon-based chips also lends itself to being potentially eventually mass-produced using manufacturing processes [54].

Neutral Atoms

Neutral atom approaches use the energy levels of ultra-code neutral atoms as qubits, interacting with power law forces with respect to their positions [59]. Qubits are initialized by using engineered patterns of phases and intensities of lasers, from a spatial light modulator, that form a lattice of individual magneto-optical traps or so-called optical tweezers acting on a cloud of atoms. Each trap consists of gradients in laser intensity that force individual atoms to stochastically occupy specific spatial locations in the lattice, and this probability of single atoms occupying each lattice site can be boosted by layering and repeating multiple traps [58]. Dipole-dipole interactions between atoms then act similar to van der Waals forces, repelling nearby atoms from being simultaneously excited into a third so-called Rydberg state. Gates are implemented via microwave pulses to transition between the atomic energy level states, with interactions mediated by transitions to the Rydberg state [56, 57]. For 1-qubit gates, optimal control pulses sequences are applied to transition between the qubit states. For 2-qubit gates, controlled-phase gates are implemented by coupling one of the control qubit's states to the Rydberg state $|1\rangle \leftrightarrow |r\rangle$, meaning the target qubit cannot also be in the Rydberg state. Applying $\pi, 2\pi, \pi$ Rabi oscillations between $|1\rangle$ and $|r\rangle$ on both target and control qubits then induces a phase unless both qubits are in the $|0\rangle$ state. Measurements of atomic qubit states are accomplished by fluorescence imaging techniques, where specific states are inferred by exciting orthogonal states to auxiliary energy levels, and observing any (lack of) spontaneous emission [213]. Noise and errors occur due to inhomogeneities in lasers, leakage to other energy levels, heating of the apparatus, and general loss of atoms and state preparation errors within the optical traps. The neutral atom qubit Hamiltonian for n atom qubits can thus be described in terms of atom Pauli X, Y, Z operators on $|0\rangle, |1\rangle$, the detuning off-resonance frequency Δ from the natural atomic energy level spacing, the Rabi frequency of oscillations between qubit states Ω, the Rabi frequency of oscillations with the Rydberg state δ, and the oscillation $X_r = |1\rangle\langle r| + |r\rangle\langle 1|$ and occupation $N = |r\rangle\langle r|$ of the Rydberg states $|r\rangle$. Given an inter-atomic power-law potential J/R^α, for atomic-spacings R and powers $\alpha \geq 0$,

$$H_{\text{Atom}} = \frac{1}{2}\sum_{i\in[n]} \Omega_i \, X_i \; + \; \frac{1}{2}\sum_{i\in[n]} \delta_i \, X_{ri} \; - \; \frac{1}{2}\sum_{i\in[n]} \Delta_i \, N_i \; + \sum_{i<j\in[n]} J_{ij} \, N_i \, N_j \tag{2.163}$$
$$J_{ij} = \frac{J}{R_{ij}^\alpha} \; .$$

Successful implementations include Graham *et al.'s* and Radnaev *et al.'s* (Infleqtion, Cold Quanta, University of Wisconsin, 2022) demonstrations of high fidelity quantum algorithms such as phase estimation and entangled state preparation [56, 214]. Bluvstein *et* *al.* and Xu *et al.* (Harvard University, Quera, 2025) have successfully implemented error correction codes with 48 logical qubits and 1225 physical qubits, including teleportation algorithms and mid-circuit measurements via novel atom transport and entropy removal methods [215–217]. Geim *et al.* (Harvard University, 2026) have also simulated quantum spin liquids and Rokhsar-Kivelson many-body states on 271 qubit lattices [68]. Manetsch *et al.* (California Institute of Technology, Oratomic, 2025) have also shown 6000+ neutral atom qubits can be coherently controlled with high fidelity [218], allowing for error correcting codes with potentially highly reduced ancillary qubit overhead [219]. Weaknesses of neutral atoms include their slow state preparation, given atoms must typically be brought close together to perform multi-qubit gates that take advantage of interactions mediated by Rydberg phenomena. Such systems also may have slow measurement times, given atoms must typically be reloaded back into the traps after destructive fluorescence measurements. However, non-destructive measurements are possible by enforcing fluoresced states to transition between known auxiliary states [214]. Strengths of neutral atoms include their inherent scalability and flexibility, given the magneto-optical traps allow for arbitrary configurations of thousands of atoms [215, 220]. Neutral atom qubits are also always identical, and have long coherence times. The Rydberg Hamiltonian also allows for straightforward analogue simulation of various Ising-like models or classical objective functions, without requiring mappings between different qubit representations [59, 213].

Hybrid Quantum-Classical Approaches

Amidst the incredible recent progress in our experimental abilities, it admittedly remains unclear whether NISQ devices have surpassed the capabilities of existing classical technologies at accomplishing their targeted applications [23, 24]. However, there has been a mutually beneficial, back and forth between quantum and classical approaches [62, 63].

For example, approaches from classical machine learning, combined with data and insights about symmetries from experiments, have allowed for highly efficient and interpretable classical ansatze to be developed, with polynomially scaling number of parameters in system size. Such variational ansatze can be optimized to simulate ground state, dynamical, and finite temperature properties of spin, bosonic, and fermionic models with hundreds of qubits [64, 147, 221–225]. Classical tensor network methods have also been shown to be highly versatile at describing many quantum systems of interest by representing correlations and entanglement with carefully chosen tensors and tensor contraction update schemes [65, 226–228]. Direct simulation methods, using sparse, carefully truncated, or constrained representations of quantum states have further shown to be highly effective at capturing specific dynamics [66, 151, 158, 159, 225].

Classical methods have in turn inspired ansatze, initialization procedures, and optimization techniques for quantum methods, giving rise to variational quantum algorithms [81, 83]. Here, data is encoded within quantum states, evolved using parameterized quantum dynamics, and measured to yield objectives of interest. Parameterized quantum systems are then optimized using classical methods [64, 82, 229]. The constrained ansatze permitted in quantum settings also lend themselves to theoretical analyses of optimization phenomena [102, 230, 231], asymptotically vanishing gradient phenomena of barren plateaus [94, 95, 232], and general expressive capabilities of these ansatze [106, 170, 233, 234].

The fruitful combination of data and ansatze from both classical and quantum approaches is promising, however there remain many questions concerning the practical feasibility and scalability of both approaches, given their significant computational resources required during optimization. Quantum approaches are found to be potentially limited by the scaling with system size of the number of samples required to estimate objective functions [24]. Classical approaches are found to be potentially limited by still requiring an exponential numbers of parameters to accurately represent highly entangled quantum states [221]. Benchmarks and datasets must also be carefully chosen to assess which methods are most appropriate [63, 235]. Concurrently, it has also been questioned which classes of applications and systems even necessitate quantum approaches [24, 236]. Systems of interest appear to often fortuitously evolve within small, simulable subspaces [237], with objective functions or properties that can be classically approximated with high accuracy [238] . Overall, it will be important to resolve the correct proportion of classical versus quantum resources necessary to solve our tasks of interest.

2.3 Quantum Ensembles

In this section, let us introduce notions of random quantum states, operators, and maps in quantum settings. In general, states $|\psi\rangle, \rho$ can be generated by operators U and channels Λ from fixed states. Therefore these random objects, and their distributions $P_{|\psi\rangle}, P_\rho, P_U, P_\Lambda$ can take several forms, and be interpreted and derived in several ways [104, 127], depending on which fundamental generating object is taken to be inherently random.

In general, the considered quantities, such as d-dimensional, s-rank states $|\psi\rangle \to \rho$, are within d-dimensional system $\mathcal{H}$ and s-dimensional environment $\mathcal{E}$ spaces,

$$|\psi\rangle \quad \to \quad \rho \qquad : \qquad \mathcal{H}\otimes\mathcal{E} \quad \to \quad \mathcal{H} \ . \tag{2.164}$$

Any $d\times s$-dimensional parameters φ, with measure $d\varphi$, transform under the positive $d\times d$-dimensional linear transformations, $\Pi = \Gamma^\dagger\Gamma$, with determinant factors as [239],

$$\varphi \to \Gamma\varphi \qquad \to \qquad d\varphi \to d\varphi \ \det(\Pi)^s \quad , \quad \delta\big(\Gamma\varphi\varphi^\dagger\Gamma^\dagger\big) \to \det(\Pi)^{-d} \ \delta\big(\varphi\varphi^\dagger\big) \ . \tag{2.165}$$

Here, let us generally consider any distributions, only up to d, s-dependent constants, and study how distributions can become uniform for certain dimensions d, s.

2.3.1 Random Quantum States

Suppose d-dimensional, s-rank quantum states $\rho \in \mathcal{B}[\mathcal{H}]$, are randomly distributed according to a distribution P_ρ, and ds-dimensional pure states $|\psi\rangle \in \mathcal{H}\otimes\mathcal{E}$, are randomly distributed according to a distribution P_φ. Then, given ρ is equivalent to a purification φ, with a reshaped $d\times s$ representation $\varphi \cong |\psi\rangle$, then φ's distribution *induces* ρ's distribution,

$$\begin{matrix} \rho \sim P_\rho(\rho) \\ \varphi \sim P_\varphi(\varphi) \end{matrix} \qquad : \qquad \rho = \varphi\varphi^\dagger \qquad \to \qquad P_\rho(\rho) = \int d\varphi \ P_\varphi(\varphi) \ \delta\big(\rho - \varphi\varphi^\dagger\big) \ . \tag{2.166}$$

To derive the form of the state distribution P_ρ, let us assume the inducing purified distribution P_φ soeley consists of constraints such that any resulting states are physical. If we assume there is soeley the trace constraint on the purified states,

$$P_\varphi(\varphi) \ \propto \ \delta\big(\mathrm{tr}\big(\varphi\varphi^\dagger\big) - 1\big) \ , \tag{2.167}$$

thus, from the transformation $\varphi \to \rho^{-1/2}\varphi$, the distribution of states is uniform when $s = d$,

$$P_\rho(\rho) \ \propto \ \det(\rho)^{s-d} \ \delta(\mathrm{tr}(\rho) - 1) \ . \tag{2.168}$$

The state distribution $P_\rho(\rho) = \int d\varphi \, P_\varphi(\varphi) \, \delta\big(\rho - \varphi\varphi^\dagger/\mathrm{tr}\big(\varphi\varphi^\dagger\big)\big)$ could also explicitly constrain the trace. Such symmetries imply any trace-dependent purified distribution $P_\varphi = P_\varphi(\mathrm{tr}\big(\varphi\varphi^\dagger\big))$ would yield identical expressions, such as the Gaussian, $P_\varphi \propto \exp\big(-\mathrm{tr}\big(\varphi\varphi^\dagger\big)\big)$.

2.3.2 Random Quantum Operators

Given states $\rho \in \mathcal{B}[\mathcal{H}]$ can be generated by a unitary $U \in \mathcal{U}[\mathcal{H} \otimes \mathcal{E}]$ acting on some fixed initial purified state, it turns out to be easiest to conceptualize distributions of random states as being induced by those of random unitaries [127],

$$U \sim P_U(U) \ . \tag{2.169}$$

To derive the form of the unitary distribution P_U, with minimal assumptions concerning its constraints, let us make use of the common distributions of random operators,

$$\textit{Ginibre (Gaussian):}\ \ Z \sim P_Z(Z)\ :\ P_Z(Z) \propto \exp\bigl(-\mathrm{tr}\bigl(ZZ^\dagger\bigr)\bigr) \tag{2.170}$$

$$\textit{Wishart (Hermitian):}\ \ X = ZZ^\dagger \sim P_X(X)\ :\ P_X(X) \propto \int dZ\ P_Z(Z)\delta\bigl(X - ZZ^\dagger\bigr). \tag{2.171}$$

Given $ds \times r$-dimensional Ginibre operators Z, from their elements being independent and identically distributed, partial traces of the ds-dimensional r-rank Wishart operators $X = ZZ^\dagger \to (I \otimes \mathrm{tr})(ZZ^\dagger)$ are also d-dimensional sr-rank Wishart operators. Similarly, other Ginibre-dependent operators, such as unit-trace $ZZ^\dagger/z\ :\ z = \mathrm{tr}\bigl(ZZ^\dagger\bigr)$, or unitary $X^{-1/2}ZZ^\dagger X^{-1/2}\ :\ X = ZZ^\dagger$, given the Wishart matrix is full-rank and invertible almost surely [127], also have distributions induced by the Ginibre distribution.

If we assume the unitary distribution has a unitarity constraint,

$$P_U(U)\ \propto \delta\bigl(U^\dagger U - I\bigr)\ , \tag{2.172}$$

and an assumed *left,right-invariance*, such that for any unitaries $V \in \mathcal{U}[\mathcal{H} \otimes \mathcal{E}]$,

$$P_U(UV) = P_U(VU) = P_U(U)\ , \tag{2.173}$$

then P_U is in fact the unique *Haar* distribution of unitaries [103, 112, 117, 240]. This invariance is almost all that is necessary to define the Haar distribution and its properties.

Therefore, if we find an invariant distribution that satisfies our constraints, and yields unique unitaries in a one-to-one mapping, then we have found the unique Haar distribution. Let us recall that Gaussian-distributed Z generate states, and that any operators are proportional to unitaries Q via a QR decomposition $Z = QR$. Then Haar distributed unitaries U follow from the complex Gaussian Ginibre distributed Z, given a modified QR-decomposition $Z \to QR \to U \propto Q$, to ensure uniqueness [240],

$$\begin{array}{l} Z \sim P_Z(Z) \\ P_Z(Z)\ \propto\ \exp\bigl(-\mathrm{tr}\bigl(Z^\dagger Z\bigr)\bigr) \end{array} \ \rightarrow\ \begin{array}{l} Z = QR \\ T = \mathrm{diag}(R)/|\mathrm{diag}(R)| \end{array} \ \rightarrow U = QT \sim P_U(U)\ . \tag{2.174}$$

When the system and environment are identical, $s = d$, uniform Haar random states are,

$$\rho = ZZ^\dagger/\mathrm{tr}\bigl(ZZ^\dagger\bigr)\ \text{or}\ (I \otimes \mathrm{tr})(U\psi U^\dagger)\ :\ Z \sim P_Z \text{ or } U \sim P_U\ ,\ |\psi\rangle \in \mathcal{H} \otimes \mathcal{E}\ . \tag{2.175}$$

2.3.3 Random Unitary Ensembles

Given relationships between distributions of unitaries, and distributions of states and other operators via purifications and Stinespring dilations, unitary ensembles are evidently central within random matrix theory [241–245]. Here, let us denote arbitrary unitary ensembles as $\mathcal{U}$, with measures dU, which includes implicitly any distribution P_U factors. Finally, let us denote the Haar unitary ensemble as $\mathcal{U}(d)$, with $\mathcal{U} \subseteq \mathcal{U}(d)$.

Ensembles of operators can be described in terms of their t-order statistics over t-copies of the Hilbert space $\mathcal{H} \to \mathcal{H}^{\otimes t}$. In particular, ensembles can be described via the state-dependent average action on operators $X \in \mathcal{B}[\mathcal{H}^{\otimes t}]$ of t-order *twirls*,

$$\mathcal{T}_{\mathcal{U}}^{(t)}(X) = \int_{\mathcal{U}} dU \; U^{\otimes t} \; X \; U^{\otimes t\dagger} \; , \tag{2.176}$$

or via the state-independent superoperators of t-order *moment-operators*,

$$\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)} = \int_{\mathcal{U}} dU \; U^{\otimes t} \otimes U^{\otimes t\,*} \; . \tag{2.177}$$

Spectral properties of moments operators are also essential in relating ensembles,

$$\left\|\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)}\right\| \quad , \quad \mathrm{tr}\left(\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)}\right) \; . \tag{2.178}$$

Ensembles $\mathcal{U}$ are said to form *ϵ-t-designs* with respect ensembles $\mathcal{U}'$, if their t-order statistics, and norms of their moment operators satisfy,

$$\left\|\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)}\right\| \leq \epsilon \; . \tag{2.179}$$

given a task-dependent norm [246, 247], and let us use the Frobenius norm unless specified.

Further, unitary ensemble moment operator norms and traces have an operational interpretation of being the average overlap between unitaries from the ensemble,

$$\left\|\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)}\right\|^2 = \int_{\mathcal{U}\times\mathcal{U}} dU \; dV \; |\langle\!\langle U|V\rangle\!\rangle|^{2t} \quad , \quad \mathrm{tr}\left(\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)}\right) = \int_{\mathcal{U}} dU \; |\mathrm{tr}(U)|^{2t} \; . \tag{2.180}$$

Now let us consider examples of several unitary ensembles.

For the case when $\mathcal{U} = \{U\}$ contains a single unitary U, there is trivially,

$$\widehat{\mathcal{T}}_{U}^{(t)} = U^{\otimes t} \otimes U^{\otimes t\,*} \; , \quad \text{and} \quad \left\|\widehat{\mathcal{T}}_{U}^{(t)}\right\|^2 = d^{2t} \; . \tag{2.181}$$

For the case when $\mathcal{U} = G$ is a group with a Haar measure dU, from the Haar measure invariance, the t-order moment operator is a self-adjoint projector,

$$\widehat{\mathcal{T}}_G^{(t)}\widehat{\mathcal{T}}_G^{(t)} = \widehat{\mathcal{T}}_G^{(t)} \qquad , \qquad \widehat{\mathcal{T}}_G^{(t)\dagger} = \widehat{\mathcal{T}}_G^{(t)} \ , \tag{2.182}$$

and from the structure of the moment operator, projects onto the t-order *commutant* com [118, 248, 249], with dimension dim, of all operators which commute with the ensemble,

$$\mathrm{com}^{(t)}(G) = \{X \in \mathcal{B}[\mathcal{H}^{\otimes t}] \quad : \quad [U^{\otimes t}, X] = 0 \ , \ \forall \ U \in G\} \ . \tag{2.183}$$

The ensemble thus has norm and trace of,

$$\left\|\widehat{\mathcal{T}}_G^{(t)}\right\|^2 = \mathrm{tr}\left(\widehat{\mathcal{T}}_G^{(t)}\right) = \dim(\mathrm{com}^{(t)}(G)) \ . \tag{2.184}$$

Such relationships between moment operator norms and commutant dimensions are useful for studying and bounding approximate ε-t-designs with respect to unitary ensembles $\mathcal{U} \subseteq G$, given Haar-distributed unitary group reference ensembles G. Using soeley the Haar measure invariance of G, for all such ensembles $\mathcal{U} \subseteq G$, their moment operators satisfy,

$$\widehat{\mathcal{T}}_G^{(t)}\widehat{\mathcal{T}}_\mathcal{U}^{(t)} = \widehat{\mathcal{T}}_\mathcal{U}^{(t)}\widehat{\mathcal{T}}_G^{(t)} = \widehat{\mathcal{T}}_G^{(t)} \ , \tag{2.185}$$

and therefore their moment operator norms satisfy,

$$\left\|\widehat{\mathcal{T}}_\mathcal{U}^{(t)} - \widehat{\mathcal{T}}_G^{(t)}\right\|^2 = \left\|\widehat{\mathcal{T}}_\mathcal{U}^{(t)}\right\|^2 - \left\|\widehat{\mathcal{T}}_G^{(t)}\right\|^2 \ , \tag{2.186}$$

and subset ensemble moment operators must explicitly include the group moment operator,

$$\widehat{\mathcal{T}}_\mathcal{U}^{(t)} = \widehat{\mathcal{T}}_G^{(t)} \ + \ \cdots \ . \tag{2.187}$$

These norm relations allow us to establish a partial ordering between the norms of moment operators for ensembles with Haar measures over subgroups of the unitary Haar ensemble $\mathcal{U}(d)$. In particular, given unitary groups $\mathcal{U},\mathcal{U}',\ldots$, such that $\{U\} \subseteq \mathcal{U} \subseteq \mathcal{U}' \subseteq \cdots \subseteq \mathcal{U}(d)$,

$$\left\|\widehat{\mathcal{T}}_{\mathcal{U}(d)}^{(t)}\right\|^2 \ \leq \ \cdots \ \leq \ \left\|\widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)}\right\|^2 \ \leq \ \left\|\widehat{\mathcal{T}}_\mathcal{U}^{(t)}\right\|^2 \ \leq \ \left\|\widehat{\mathcal{T}}_U^{(t)}\right\|^2 \ . \tag{2.188}$$

Such an ordering implies that the more unstructured a group $\mathcal{U}$ is, relative to the group of totally unstructured unitaries $\mathcal{U}(d)$, the smaller the norm of its moment operator. Thus, the Haar ensemble $\mathcal{U}(d)$, at the base of such a hierarchy of unitary ensembles, serves as an appropriate reference ensemble for other subgroups of unitaries.

Let us now focus on the Haar ensemble of unitaries, $\mathcal{U}(d)$, consisting of the group of all unitaries acting on $\mathcal{H}$, distributed according to the Haar measure. Given its group structure, we can use Weingarten calculus [118] to derive its properties. In fact, the commutant of this ensemble, and the space projected on to by its twirls, is spanned by representations of the t-order permutations $\mathcal{S}_t$ [231, 248], which has dimension $t!$ for $t \leq d$.

Given this commutant, for any subgroup of unitaries $\mathcal{U} \subseteq \mathcal{U}(d)$, we have the bounds,

$$t! \;\leq\; \left\| \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}} \right\|^2 \;\leq\; d^{2t} \; . \tag{2.189}$$

In particular, the t-order commutant of $\mathcal{U}(d)$ is given by the system permuting representation $\mathrm{com}^{(t)}(\mathcal{U}(d)) = \mathcal{S}^{(t)}_d$ of $\mathcal{S}_t$, namely all permutations of t copies a space $\mathcal{H}^{\otimes t}$. Further, given a permutation $\sigma \in \mathcal{S}_t$, its representation $V_d \;:\; \mathcal{S}_t \to \mathcal{S}^{(t)}_d \subset \mathcal{B}[\mathcal{H}^{\otimes t}]$ is,

$$V_d(\sigma) = \sum_{\{\mu_i \in [d]\}_{i\in[t]}} \otimes_{i\in[t]} |\mu_{\sigma^{-1}(i)}\rangle\langle\mu_i| \; , \tag{2.190}$$

with associated normalized inner products, or characters,

$$\chi^{(t)}_d(\sigma,\pi) = \frac{\langle\!\langle V_d(\sigma) | V_d(\pi) \rangle\!\rangle}{\langle\!\langle V_d(e) | V_d(e) \rangle\!\rangle} \quad : \quad \chi^{(t)}_d(\sigma) = \chi^{(t)}_d(e,\sigma) \; , \tag{2.191}$$

where e denotes the identity permutation with representation $V_d(e) = I_d^{\otimes t}$. Therefore we can express the Haar ensemble moment operator in terms of the permutations $\mathcal{S}_t$ as,

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)} = \frac{1}{d^t} \sum_{\sigma,\pi\in\mathcal{S}_t} \chi^{(t)-1}_d(\sigma,\pi) \; |V_d(\sigma)\rangle\!\rangle\langle\!\langle V_d(\pi)| \; , \tag{2.192}$$

where the transfer matrix $\chi^{(t)-1}_d$ is the Weingarten matrix, the inverse of the Gram matrix $\chi^{(t)}_d$ over $\mathcal{S}_t$. For $t = 1, 2$, identities $V_d(e) = I_d$ and swaps $V_d(\tau) = S_d$ are mapped between,

$$\widehat{\mathcal{T}}^{(1)}_{\mathcal{U}(d)} = \frac{1}{d} |I_d\rangle\!\rangle\langle\!\langle I_d| \tag{2.193}$$

$$\widehat{\mathcal{T}}^{(2)}_{\mathcal{U}(d)} = \frac{1}{d^2}\frac{d^2}{d^2-1} \left(|I_d^{\otimes 2}\rangle\!\rangle\langle\!\langle I_d^{\otimes 2}| \,+\, |S_d\rangle\!\rangle\langle\!\langle S_d| \right) \,-\, \frac{1}{d^2}\frac{d}{d^2-1} \left(|S_d\rangle\!\rangle\langle\!\langle I_d^{\otimes 2}| \,+\, |I_d^{\otimes 2}\rangle\!\rangle\langle\!\langle S_d| \right) . \tag{2.194}$$

Finally, when acting on t-copies of pure states $\psi \in \mathcal{B}[\mathcal{H}]$, the Haar ensemble t-order twirls project onto the $\binom{d+t-1}{t}$-dimensional symmetric subspace [248] over the t-copies of the d-dimensional space, with the projector $\mathcal{T}^{(t)}_d$,

$$\mathcal{T}^{(t)}_{\mathcal{U}(d)}(\psi) = \int_{\mathcal{U}(d)} dU \; U^{\otimes t} \; \psi^{\otimes t} \; U^{\otimes t\dagger} = \frac{1}{\binom{d+t-1}{t}} \mathcal{T}^{(t)}_d \quad : \quad \mathcal{T}^{(t)}_d = \frac{1}{t!} \sum_{\sigma\in\mathcal{S}_t} V_d(\sigma) \; . \tag{2.195}$$

2.3.4 Permutations

Given their relevance within random matrix theory, let us derive properties of the t-order permutations $\sigma \in \mathcal{S}_t$ [117, 250–252]. Specific permutations include the identity permutation $e = ()$ which permutes no indices, transpositions that swap indices $i \neq j \in [t]$, $\tau = (ij)$, and l-length cycles $\lambda = (012 \cdots l-1)$ which cyclically shift l indices in $[t]$.

A permutation $\sigma \in \mathcal{S}_t$ can be decomposed into a $\#_\sigma$ number of disjoint *cycles*, denoted as $\sigma = \prod_{\lambda \in \sigma} \lambda$, or a $|\sigma|$ number of *transpositions*, denoted as $\sigma = \prod_{\tau \in \sigma} \tau$. All cycles $\lambda \in \sigma$ can be decomposed into $|\lambda| - 1$ number of transpositions, and therefore,

$$\#_\sigma + |\sigma| = t \ . \tag{2.196}$$

Permutations can also be described by the support $\Gamma_\sigma \subseteq [t]$ and locality $l_\sigma = |\Gamma_\sigma| \leq t$ where they act non-trivially, and by the size of their overlap $|\pi^{-1}\sigma|$. Finally, permutations can be described by their *sub-permutations* $\pi \subseteq \sigma$, a partial ordering between permutations [252], that corresponds to factorizations $\sigma = (\pi)(\pi^{-1}\sigma)$ such that,

$$\pi \subseteq \sigma \quad \leftrightarrow \quad \left|\pi^{-1}\sigma\right| = |\sigma| - |\pi| \ . \tag{2.197}$$

Further, the size of permutations $\sigma, \pi, \in \mathcal{S}_t$, also referred to as the *Cayley distance* satisfies a triangle inequality, with upper and lower bounds satisfied by sub-permutations,

$$\big||\sigma| - |\pi|\big| \quad \leq \quad \left|\pi^{-1}\sigma\right| \quad \leq \quad \big||\sigma| + |\pi|\big| \ . \tag{2.198}$$

The representation of the permutations which permutes t copies of a d-dimensional Hilbert space is denoted as $V_d \ : \ \sigma \in \mathcal{S}_t \to V_d(\sigma) \in \mathcal{S}_d^{(t)} \subset \mathcal{B}[\mathcal{H}^{\otimes t}]$. From here, we can define the normalized overlap between the representations of permutations $\sigma, \pi \in \mathcal{S}_t$ as,

$$\chi_d^{(t)}(\sigma, \pi) = \frac{1}{d^{|\sigma^{-1}\pi|}} \ . \tag{2.199}$$

These overlaps can be be used to form a $t! \times t!$ Gram matrix $\chi_d^{(t)}$, with elements, $\chi_d^{(t)}(\sigma, \pi) = \chi_d^{(t)}(\sigma, \pi)$. The inverse Gram matrix $\chi_d^{(t)-1}$, referred to as the Weingarten matrix, which only has known closed-form expressions to leading-order [244], has elements of,

$$\chi_d^{(t)-1}(\sigma, \pi) = c(\sigma^{-1}\pi) \ \frac{1}{d^{|\sigma^{-1}\pi|}} \left[1 + \mathcal{O}\left(\frac{1}{d}\right)\right] \leq \chi_t \ c(\sigma^{-1}\pi) \ \frac{1}{d^{|\sigma^{-1}\pi|}} \ , \tag{2.200}$$

in terms of Möbius functions over a permutation's cycles, $c(\sigma) = \prod_{\lambda \in \sigma} c(\lambda)$, $c(\lambda) = (-1)^{|\lambda|} c_{|\lambda|}$, with Catalan numbers $c_l = \frac{1}{l+1}\binom{2l}{l}$, and t-dependent upper bounds [117] of χ_t.

Here, it is important to note that while the representations of $\sigma \in \mathcal{S}_t$ are linearly independent for $d \geq t$ [248], they are not orthogonal [141]. Hence, it is essential to find an alternative basis to describe permutations, with improved orthogonality properties.

2.4 Tensor Networks

In this section, we present various properties, algorithms, and applications, of so-called tensor network based methods. Given our descriptions of quantum systems, appropriate representations and formalisms are essential, and a tensor diagrammatic language is shown to be exceptionally versatile and suitable to simplify such analyses. After introducing tensor network diagrams [175, 253], we proceed to review fundamental tensor architectures known as matrix product states [160], before discussing their properties [165].

2.4.1 Tensor Diagrams

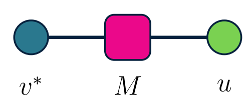
Suppose we have a Hilbert space with tensor product structure $\mathcal{H} = \otimes_{i\in[n]}\mathcal{H}_i$, with n, q-dimensional local component Hilbert spaces $\mathcal{H}_i$. Given this localized structure, n-component quantities $\mathcal{A}$ can be represented with various architectures of tensor networks,

$$\mathcal{A} = \prod_{i\in[n]} A_i \ . \tag{2.201}$$

consisting of products of n tensors $\{A_i\}_{i\in[n]}$. Generic individual tensors are denoted without i labels as A, and may be denoted with their explicit (dummy) indices $\{\alpha,\ldots,\sigma,\ldots,\beta\}$ as $A_{\alpha\ldots\sigma\ldots\beta}$. Tensors, and in particular operations on tensors such as their reshaping and products or contractions, can be described by tensor diagrams, as shown in Fig. 2.3.

For example, a product of matrices M, and vectors u, v, can be represented using coordinate free, explicit summation, Einstein summation, or tensor diagrammatic notations,

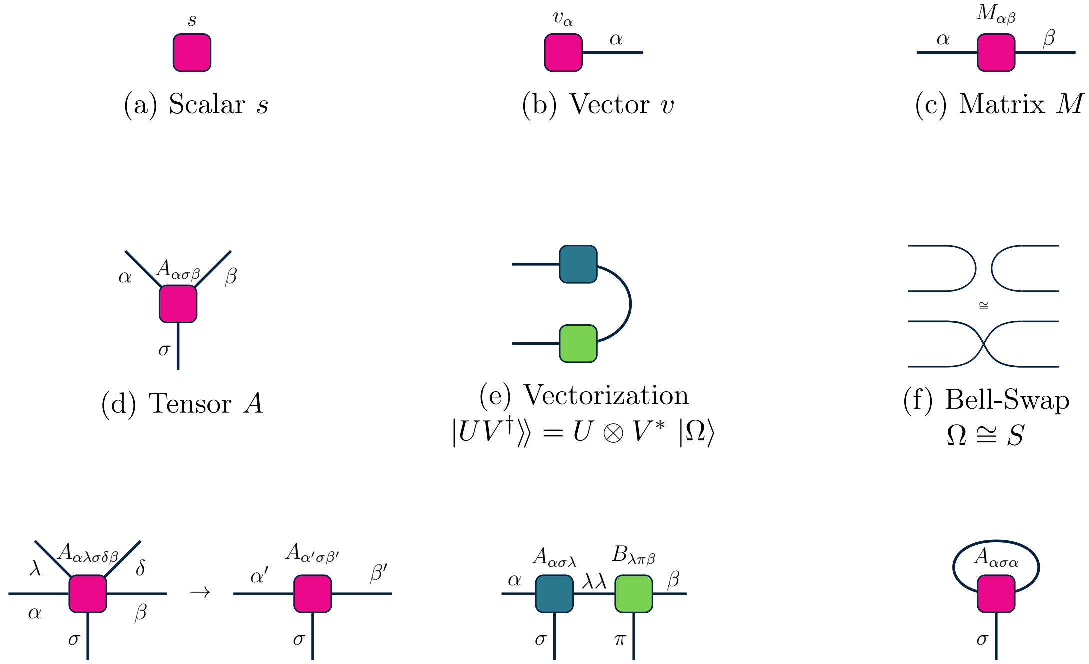
$$v^* M u = \sum_{\mu,\nu} v^*_\mu M_{\mu\nu} u_\nu = v^*_\mu M_{\mu\nu} u_\nu = $$



$$, \tag{2.202}$$

given diagrams for matrices of (pink) squares, and for vectors of (blue,green) circles, with conjugation interpreted by context, usually with tensors pointing in opposite directions.

The localized structure of tensor networks often greatly clarifies analyses. For example, quantum circuits can be analysed by representing states and operators as tensor networks [120, 130, 147, 254]. However, their quintessential use is as a classical simulation method for quantum systems [126, 161, 166, 170, 255], where their sparsity can greatly reduce the number of computations required, relative to naively representing quantities as dense vectors and matrices. Specific tensor network architectures must be chosen such that their structure is easily preserved upon dynamics, particularly for greater spatial dimensions, which involve exponentially more contractions between tensors [164].



(a) Scalar s

(b) Vector v

(c) Matrix M

(d) Tensor A

(e) Vectorization
$|UV^\dagger\rangle\rangle = U \otimes V^* \; |\Omega\rangle$

(f) Bell-Swap
$\Omega \cong S$

(g) Reshape
$A_{\alpha\lambda\sigma\delta\beta} \cong A'_{\alpha'\sigma\beta'} = A_{(\alpha\lambda)\sigma(\delta\beta)}$

(h) Contraction
$C_{\alpha\sigma\pi\beta} = A_{\alpha\sigma\lambda}B_{\lambda\pi\beta}$

(i) Trace
$a_\sigma = A_{\alpha\sigma\alpha}$

Figure 2.3: **Tensor diagrams**. Tensors and their operations can be represented using a diagrammatic language, with boxes representing tensor objects, legs representing tensor indices, and joined legs representing tensor products and summations [253]. Many calculations are instantly clarified when expressed using this language, such as identities of seemingly different quantities, or expressions involving products and tensor products of operators, swapping operators, tracing operators, and rearranging indices across spaces.

2.4.2 Matrix Product States

The canonical tensor network architecture is the *Matrix Product State* (MPS), as shown in Fig. 2.4. MPS states are generally used to represent the complex amplitudes ψ_σ of state vectors, using a product of matrices A_{σ_i} for each component $i \in [n]$,

$$|\psi\rangle = \sum_{\sigma \in [q]^n} \psi_\sigma \; |\sigma\rangle \qquad : \qquad \psi_\sigma = \prod_i A_{\sigma_i} \in \mathbb{C} \tag{2.203}$$

$$A_{\sigma_i} = A_{\alpha_i \sigma_i \alpha_{i+1}} \quad , \quad \sigma_i \in [q] \; , \; \alpha_i \in [\chi] \; . \tag{2.204}$$

This tensor network is thus matrices in a one-dimensional layout, indexed by so-called physical indices $\sigma_i \in [q]$, which correspond to the fixed vector dimensions q, and with so-called virtual indices $\alpha_i \in [\chi]$, which are bounded by the variable *bond dimension* χ.

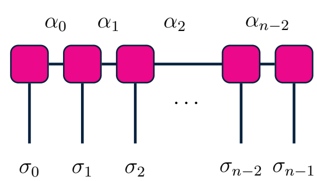
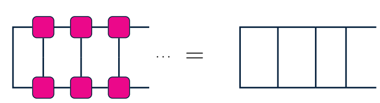


Figure 2.4: **Matrix product states**. Vectors $|\psi\rangle = \prod_{i \in [n]} A_i$ can be represented as products of n tensors, $A_i = A_{\alpha_i \sigma_i \alpha_{i+1}}$, with physical $\sigma \in [q]$, depending on the vector dimension q (non-contracted legs), and virtual $\alpha_i \in [\chi]$, bounded by χ (contracted legs).

MPS must be normalized according to any constraints on ψ, which imposes architecture-defining constraints on the tensors A. Normalized states have the 2-norm constraint,

$$\psi_\sigma = \prod_i A_{\sigma_i} \qquad : \qquad \langle\psi|\psi\rangle = \sum_\sigma |\psi_\sigma|^2 = 1 \tag{2.205}$$



Given their layout, MPS can in fact always be canonically normalized with respect to the 2-norm [164]. As will be derived, by constructing the tensors to be similar to unitaries, and contracting the tensors from left to right (left-normalized, as shown above), or similarly from right to left (right-normalized) the norm is automatically satisfied by contraction with the previous tensor and its adjoint. Here, bare lines indicate identity tensors.

Given quantities are represented by contractions between tensors, the accuracy and efficiency of any tensor network architecture is represented by the largest bond dimension χ of the tensors. Degree-l, r tensor networks, consisting of n tensors with at most l physical indices and r virtual indices, have $\mathcal{O}(nq^l\chi^r)$ elements. Therefore states with naively $\mathcal{O}(d) =$

$\mathcal{O}(q^n)$ elements, can be potentially represented efficiently, depending on the necessary scaling of χ with d, n, and may require exponentially scaling bond dimensions $\chi = \mathcal{O}(d) = \mathcal{O}(q^n)$. It remains to be understood how to construct, factorize, and manipulate such tensor networks, and most importantly, how to appropriately restrict their bond dimension.

2.4.3 Tensor Factorizations and Truncations

To construct matrix product structures from dense tensors, matrix factorizations and truncations of matrices can be used, as per Fig. 2.5. Such algorithms involve reshaping and grouping tensor indices into matrices A, performing a matrix factorization to decompose $A \to UV \approx U_\chi V_\chi$ into local factors U, V, and truncating their virtual indices $U, V \to U_\chi, V_\chi$ to be at most a fixed bond dimension χ. Such procedures can then be iterated, yielding a product of a sequence of n factorized matrices $\prod_{i\in[n]} A_i$. Typically *singular value decomposition* (SVD) matrix factorization algorithms, $A \to USV$ are used, with the diagonal singular values S typically absorbed into the unitary factors U, V. Given SVD factorizations, the 2-norm errors of such methods, $\epsilon_\chi = \|A - U_\chi S_\chi V_\chi\|_2$, when truncating to bond dimension χ, are equal to the discarded singular values of $S \to S_\chi$ [160],

$$\epsilon_\chi^2 = 1 - \mathrm{tr}\left(S_\chi^2\right) \ . \tag{2.206}$$

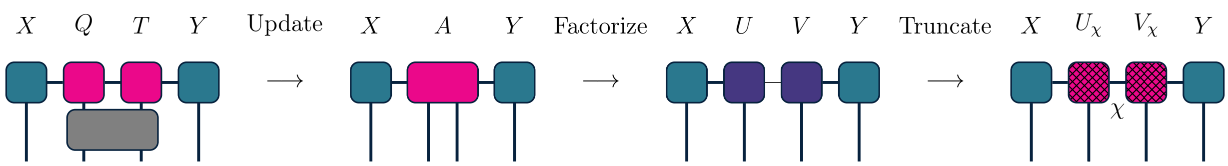


Figure 2.5: **Matrix product state factorization and truncation**. Upon non-local updates to neighbouring tensors Q, R, with boundary tensors X, Y, such that $QR \to A$ loses its local structure, such a tensor can be factorized and truncated, such that $QR \to A \approx UV \approx U_\chi V_\chi$, with bond dimension χ, is a valid matrix product state.

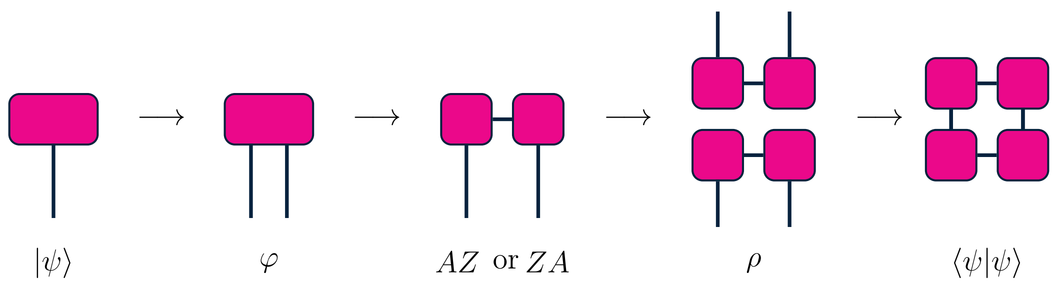
Given such general factorization and truncation procedures, MPS can be in principle, be efficiently implemented as a simulation method of dense tensors undergoing dynamics [160, 171, 172, 175, 253, 254, 256–262]. However, it remains to be understood which systems can be adequately represented by MPS with bond dimensions $\chi \ll q^n$ that scale minimally with system size n. Let us thus derive explicit, and interpretable algorithms to perform factorizations and truncations of MPS. Such algorithms which will reveal capabilities of MPS at representing correlation functions and entanglement within states of interest.

2.4.4 Tensor Network Properties

Tensor networks, in particular matrix product states, must be analysed to determine which states they can efficiently represent. Such analysis may be conducted in terms of their correlations, or entanglement entropy, which gives a direct interpretation of state reshapings, factorizations, and bond dimensions [165, 175].

Suppose we bi-partition a d-dimensional space into $l, d/l$-dimensional components. d-dimensional normalized states $|\psi\rangle$ can be similarly bi-partitioned via a reshaping of its components into a $l \times d/l$-dimensional matrix φ along this bi-partition, with reduced states ρ within this l-dimensional space, and associated tensor diagrams of,

$$|\psi\rangle \;\cong\; \varphi \quad , \quad \rho = \varphi\varphi^\dagger \quad : \quad \langle\psi|\psi\rangle = \langle\langle\varphi|\varphi\rangle\rangle = \mathrm{tr}(\rho) \tag{2.207}$$

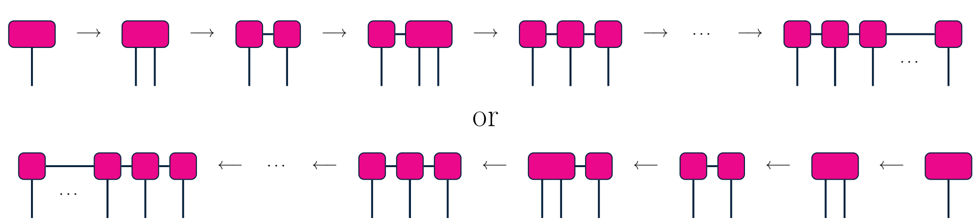


(2.208)

Factorization procedures, such as singular value decompositions, of the bi-partitioned $\varphi = USV^\dagger \cong AZ$ or ZA, yield either a left unitary-like $A \cong U$ and a right matrix $Z \cong SV^\dagger$, or a right unitary-like $A \cong V^\dagger$ and a left matrix $Z \cong US^\dagger$, for left or right normalization. The diagonal singular value $S \to S_\chi$ dimensions can also be truncated to $\chi \leq \min(l, d/l)$.

Given q^n-dimensional states and a fixed χ, this procedure can be iterated to yield $\prod_{i\in[n]} A_i$. At iteration $i \in [n]$, starting from the left or right, $\chi_i q \times q^{n-1-i}$ or $q^{n-1-i} \times \chi_i q$ -dimensional matrices Z_i are factorized, yielding $\chi_i q \times \chi_{i+1}$ or $\chi_{i+1} \times q\chi_i$ -dimensional unitaries $U_{(\alpha_i\sigma_i)\alpha_{i+1}}$ or $V^\dagger_{\alpha_{i+1}(\sigma_i\alpha_i)}$, reshaped into $\chi_i \times q \times \chi_{i+1}$ or $\chi_{i+1} \times q \times \chi_i$ -dimensional tensors $A_i = U_{\alpha_{i+1}\sigma_i\alpha_i}$ or $V^\dagger_{\alpha_{i+1}\sigma_i\alpha_i}$, and $\chi_{i+1}q \times q^{n-1-i-1}$ or $q^{n-1-i-1} \times q\chi_{i+1}$ -dimensional matrices $Z_{i+1} = SV^\dagger_{(\alpha_{i+1}\sigma_{i+1})\pi_{i+1}}$ or $US_{\pi_{i+1}(\sigma_{i+1}\alpha_{i+1})}$ for the next iteration, with,

$$\chi_{i+1} \leq \min(\chi \;,\; \chi_i q \;,\; q^{n-1-i} \;,\; q^{n/2}) \;. \tag{2.209}$$



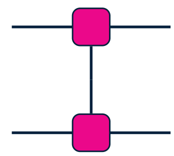
(2.210)

The resulting left or right normalized matrices $\{A_\sigma\}_{\sigma\in[q]}$ at each index $i \in [n]$ satisfy,

$$\sum_\sigma A_\sigma^\dagger A_\sigma = I_\chi \quad \text{or} \quad \sum_\sigma A_\sigma A_\sigma^\dagger = I_\chi \ . \tag{2.211}$$

Such matrices may be interpreted as χ-dimensional *transfer matrices* $\hat{T}$ on the virtual space [160], or even as *completely-positive trace-preserving maps*, with Kraus operators A_σ,

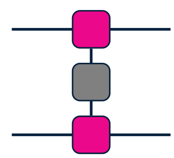
$$\hat{T} \quad = \quad \sum_\sigma A_\sigma \otimes A_\sigma^* \quad =$$



$$. \tag{2.212}$$

which can be generalized to linear maps $\hat{T}_O$, mediated by the physical operator O,

$$\hat{T}_O \quad = \quad \sum_{\sigma,\pi} O_{\sigma\pi} \ A_\sigma \otimes A_\pi^* \quad =$$



$$. \tag{2.213}$$

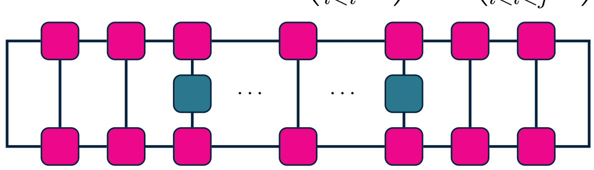
Left or right normalized transfer matrices $\hat{T}$ have maximally entangled left or right eigenvectors $|\Omega_\chi\rangle$, with non-trivial other right or left eigenvectors $|\lambda_\chi\rangle$ given their non-Hermiticity. Their spectra λ_χ are bounded, given they are completely-positive and trace-non-increasing,

$$\langle\lambda_\chi|\hat{T} = \lambda_\chi\langle\lambda_\chi| \quad : \quad \langle\Omega_\chi|\hat{T} = \langle\Omega_\chi| \qquad \text{or} \qquad \hat{T}|\lambda_\chi\rangle = \lambda_\chi|\lambda_\chi\rangle \quad : \quad \hat{T}|\Omega_\chi\rangle = |\Omega_\chi\rangle \tag{2.214}$$

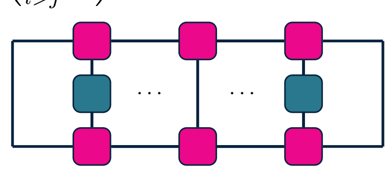
$$|\lambda_\chi| \leq 1 \ . \tag{2.215}$$

Regarding correlation functions, suppose an operator $O^{(k)} = O_i \otimes O_j$ acts on ψ_χ at indices $i, j \in [n]$ separated by $k = |j - i - 1|$. Assuming left-right normalizations, then the matrices to the left of i and to right of j may be trivially contracted, and the expectation value can be interpreted in terms of transfer matrices in the virtual space,

$$\langle\psi_\chi|O_iO_j|\psi_\chi\rangle = \langle\Omega_\chi|\left(\prod_{l<i}\hat{T}_l\right)\hat{T}_{O_i}\left(\prod_{i<l<j}\hat{T}_l\right)\hat{T}_{O_j}\left(\prod_{l>j}\hat{T}_l\right)|\Omega_\chi\rangle \quad \to \quad \langle\Omega_\chi|\hat{T}_{O_i}\ \hat{T}^k\ \hat{T}_{O_j}|\Omega_\chi\rangle$$



$$=$$



$$. \tag{2.216}$$

Finally, given translation invariance in the $n \to \infty$ limit, correlation functions may be expressed in terms of correlation lengths $\xi_\chi = -1/\log(\lambda_\chi)$ for the non-unit spectra $\lambda_\chi < 1$ of the transfer matrices, and matrix elements $o_\chi^{(k)} = \langle \Omega_\chi | \hat{T}_{O_i} | \lambda_\chi \rangle \langle \lambda_\chi | \hat{T}_{O_j} | \Omega_\chi \rangle$,

$$\langle \psi_\chi | O^{(k)} | \psi_\chi \rangle \ \sim \ o_\chi^{(k)} \ e^{-k/\xi_\chi} \ . \tag{2.217}$$

Regarding entanglement, suppose we bi-partition a $d = q^n$-dimensional space into q^l, q^{n-l}-dimensional components, with $l \leq n/2$. d-dimensional states $|\psi\rangle$, bi-partitioned via a reshaping into $q^l \times q^{n-l}$-dimensional parameters φ, and l-dimensional reduced states ρ, have singular value decompositions across the q^l-dimensional bi-partition of,

$$|\psi\rangle \ \cong \ \varphi \qquad , \qquad \rho = \varphi\varphi^\dagger \ . \tag{2.218}$$

$$\varphi = USV^\dagger \qquad \to \qquad \rho = US^2U^\dagger \ . \tag{2.219}$$

Entanglement entropy along this bi-partition strictly depends on these spectra [160],

$$\mathcal{S}_\rho^l = \ - \frac{\mathrm{tr}(S^2 \log(S^2))}{\log(\mathrm{tr}(I))} \ . \tag{2.220}$$

and is bounded when truncating $S \to S_\chi$ with $\chi \leq q^l$ across the bi-partition,

$$\mathcal{S}_{\rho\chi}^l \ \leq \ \frac{\log(\chi)}{\log(d)} \ \leq \ \frac{l}{n} \ . \tag{2.221}$$

Similarly, when considering states ψ in d-dimensional systems and s-dimensional environments $\mathcal{H} \to \mathcal{H} \otimes \mathcal{E}$, then the bond dimension, interpreted as the rank of the resulting reduced states ρ in $\mathcal{H}$, will be generally bounded by $\chi \leq \min(d, s)$. Such a partitioning is also referred to as a *Schmidt decomposition* [175], and the maximum necessary bond dimension, or number of non-trivial spectra, is the Schmidt rank.

Ultimately, although the exact bond-dimension χ-dependence on the resulting transfer matrices and spectra is highly non-trivial, MPS matrix factorizations immediately yield both virtual completely-positive trace-preserving maps that offer insight into correlation functions, and truncatable singular values that offer insight into entanglement entropy bounds. Such quantities offer interpretable insights into the capabilities of MPS at representing systems of interest. Other tensor network architectures for higher-spatial-dimensional systems [65, 165, 228] or systems exhibiting different types of correlations [161, 162, 263, 264], generally cannot be canonically normalized, and thus must be interpreted using more advanced techniques from random matrix theory.

Chapter 3

Overparameterization of Quantum Systems

Given exceptional advances in quantum computing experiments, in this chapter, based on the published manuscript, M. Duschenes, J. Carrasquilla, and R. Laflamme, PRA 108 (2024) [1], we are compelled to explore our current abilities to control and optimize quantum systems to perform tasks of interest. In particular, quantum computing devices require exceptional control of their experimental parameters to prepare quantum states and simulate other quantum systems. Classical optimization procedures used to find such optimal control parameters, have further been shown in idealized settings to exhibit different regimes of learning. Of interest here is the overparameterization regime, where for systems with a sufficient number of parameters, global optima for prepared state and compiled unitary fidelities may potentially be reached exponentially quickly. Here, we study the robustness of overparameterization phenomena in the presence of experimental constraints on the controls, such as bounding or sharing parameters across operators, as well as in the presence of noise inherent to experimental setups. We observe that overparameterization phenomena are resilient in these realistic settings at short times, however fidelities decay to zero past a critical simulation duration due to accumulation of either quantum or classical noise. This critical depth is found to be logarithmic in the scale of noise, and optimal fidelities initially increase exponentially with depth, before decreasing polynomially with depth, and with noise. Our results demonstrate that parameterized ansatze can mitigate entropic effects from their environment, offering tantalizing opportunities for their application and experimental realization in near-term quantum devices.

3.1 Introduction

There exist many active avenues and experimental platforms for the quantum information community to explore and advance quantum technologies, including trapped ions [46, 47], superconducting qubits [52, 53, 209], neutral atoms [55, 57, 90, 215], nuclear magnetic resonance [39, 265], and several other intriguing approaches. To harness these technologies' full potential for tasks of interest in quantum information [18, 85, 109], in particular state preparation or unitary compilation [266], it is imperative to exercise precise control through the manipulation of experimental parameters. This complex, high-dimensional quantum control problem arises in numerous applications [80, 200, 267–269] and is addressed through classical simulation and parameter optimization. Insight into such procedures, in experimentally relevant quantum settings, is therefore necessary to further our ability to rigorously control such systems.

Quantum control shares striking similarities with the field of classical deep learning, where large parameterized models are used to discover and represent complex patterns in large amounts of data. Beyond the resemblance of being variational algorithms concerned with optimizing high dimensional systems for technological advancement, a series of observations [82, 83, 101] has lead to direct equivalencies in learning phenomena between variational quantum algorithms and deep learning. A striking example is overparameterization and lazy training [270, 271]. In classical systems, excessively parameterized models can learn efficiently with negligible adjustments to their parameters, leading to improved generalization performance and training efficiency. A similar phenomenon has been anticipated in ideal settings for noise-free variational quantum algorithms [100, 230, 272]. It is observed that in the overparameterized regime, the optimization landscape becomes almost free of sub-optimal minima and optimization may converge exponentially quickly.

Here we explore overparameterization via the classical simulation of quantum systems, within experimentally feasible settings. We investigate these phenomena within the quantum optimal control paradigm, where systems evolve under continuous time evolution [39, 80, 267], and within the variational quantum algorithms paradigm, where systems evolve under discretized sets of operations [83, 266, 273].

We find that overparameterization phenomena are robust under realistic settings, including constrained parametrizations, and imposing noisy non-unitary ansatz. For a given periodic ansatz, where quantum circuit depth dictates the number of model parameters, we find that inclusion of parameter constraints shifts the overparameterization depth boundary by a system size dependent factor. However, the dominant overparameterization phenomenon, of exponential convergence of optimization routines with depth, persists.

In noisy settings, we observe that there are different regimes of optimization convergence. For depths beyond the overparameterization depth, but before a noise-induced critical depth, exponential convergence of optimization routines with depth, still occurs. However, beyond this critical depth, an excessive amount of noise accumulates, and the optimization diverges polynomially with depth, and with noise. To complement these numerical findings, analytical investigations into the noise and depth scaling of the infidelity objectives, and other metrics including the entropy and purity of the parameterized states, provide an explanation for these behaviours. Overall, these findings suggest overparameterization is resilient when imposing experimental feasibility, offering opportunities for this phenomenon to be exploited in future simulated and existing quantum experiments.

The chapter is structured as follows. In Section 3.2, we define general parameterized quantum channels, and objective tasks of interest, namely noisy infidelities for state preparation and unitary compilation. We proceed to interpret the form of noisy parameterized channels as expectation values of s-error channels, and we perform an analysis on the scaling of infidelity objectives with respect to noise and depth. In Section 3.3, we demonstrate overparameterization and other learning phenomena in constrained and noisy parameterized quantum systems. From numerical studies, we quantify the relationships between noise and depth at optimality. Finally in Section 3.4, we discuss the implications of these results and the resulting compromises that occur between numerical and experimental feasibility.

3.2 Preliminaries

Here, we aim to understand the abilities of parameterized quantum systems in realistic, experimentally feasible settings. Critically, noise, resulting from systems interacting with their environment, is well known to be detrimental to quantum computation [124, 266, 274]. Example effects include noise-induced symmetry breaking [275], fundamental differences in sampling and annealing trajectories [76], and noise-induced barren plateaus [276]. Initial numerical investigations by Fontana *et al.* [277] demonstrate that noise, with scale γ, accumulates with depth k. There are also well known compromises between expressiveness, how much of the desired space of solutions can be represented by an ansatz via increased depth [106], and trainability, the ability of an ansatz to be optimized. However, these precise relationships in noisy settings have yet to be confirmed quantitatively.

3.2.1 Parameterized Quantum Systems

The parameterized systems of interest simulated here are representative of various noisy intermediate scale quantum devices (NISQ) [52, 90, 209]. These systems consist of n qubits, each with local space dimension $q = 2$, and total space dimension $d = q^n$. These qubits may have fixed inter-qubit couplings, however they can be manipulated with external, time-dependent fields over a time T, or equivalently depth of simulation k. Please refer to Appendix 3.A for a complete description of the studied parameterized ansatze. Such experimental parameters are generally constrained due to feasibility [269, 274], and these studies seek to quantify the amount with which constraints affect the capabilities of parameterized quantum systems. We assume there are generally on the order of $p = \mathcal{O}(\text{poly}(n)k)$ variable parameters in the system, where generally the system size n dependence is held fixed. Thus changes in parameter counts are reflected in the simulation depth k, and any notion of overparameterization is discussed in the context of depth. Here, we take as an example, nuclear magnetic resonance (NMR) quantum systems where nuclei, acting as n, $q = 2$ level qubits, are manipulated by $\mathcal{O}(p)$ time-dependent magnetic pulses [195, 196, 265, 269].

Underlying this analysis, are principles from learning theory, based on studies of overparameterization in ideal quantum settings, including unitary compilation [100], variational quantum eigensolvers [272, 278], and quantum circuits [230]. These works have subsequently been followed up by initial studies on the effects of noise. Within an information theoretic context, the quantum Fisher information [102] is used as a metric to determine whether a quantum system is overparameterized. Within a general optimization context to complement neural-tangent kernel approaches for asymptotic learning dynamics [101], Riemannian gradient flow dynamics [278] are used to assert the convergence of overparameterized systems with bounded gradient noise. For our purposes, overparameterization refers to when there are an adequate number of parameters $p > \tilde{p}$, or depths $k > \tilde{k}$, such that the full space of g-dimensional solutions to tasks is spanned by the ansatz.

Within this regime, we investigate the resulting overparameterization phenomena of possible exponential convergence of noisy optimization procedures with depth. Imposing constraints on the optimization related to experimental feasibility, including restricting individual control of qubits, is also known to decrease convexity in the objective landscape [85, 279, 280], and it requires many optimization heuristics. Please refer to Appendix 3.B for a more complete description of overparameterization, including quantitative bounds on the overparameterization depth related to the quantum Fisher information.

Based on these studies, we hypothesize that there exists a critical evolution time or depth $k_\gamma > \tilde{k}$, where overparameterization has occurred, however too much noise has also accumulated. This noise is expected to prevent parameterized systems from accomplishing fidelity-based tasks with arbitrary precision. To confirm these predictions, we will consider the average behaviour of infidelities, optimized independently over a distribution of tasks. We therefore conjecture that there are convergent and divergent phases of the optimization

$$\text{Optimization} \sim \begin{cases} \text{Plateau} & k < \tilde{k} \\ \text{Convergent} & \tilde{k} < k < k_\gamma \\ \text{Divergent} & k > k_\gamma \end{cases} . \tag{3.1}$$

These studies aim to confirm these conjectures of depth-dependent regimes of learning phenomenon in non-ideal settings.

To quantify the effects of noise on the evolution and abilities of parameterized quantum systems to perform tasks of interest, we define parameterized quantum channels as,

$$\Lambda_{\theta\gamma} = \mathcal{N}_\gamma \circ \mathcal{U}_\theta \ , \tag{3.2}$$

with a unitary component $\mathcal{U}_\theta$ parameterized by variable parameters θ, and a non-unitary component $\mathcal{N}_\gamma$ parameterized by constant noise scales γ.

For the unitary component, we assume the Hamiltonian driving the evolution is,

$$H_\theta^{(t)} = \sum_\mu H_\mu^{(t)} \ : \ H_\mu^{(t)} = \theta_\mu^{(t)} G_\mu \tag{3.3}$$

at a continuous time $t \in [0, T]$, with time-dependent parameters $\theta_\mu^{(t)}$, and constant generators $\{G_\mu\}$. As per Appendix 3.A, we also assume the evolution is approximately piecewise constant over k time steps $\tau = T/k$, allowing for first-order temporal Trotterization,

$$U_\theta = \mathcal{T} e^{-i \int_0^T dt \ H_\theta^{(t)}} = \prod_\kappa^k U_\theta^{(\kappa)} \ + \ \mathcal{O}(\tau^2) \ . \tag{3.4}$$

Q-order spatial Trotterization across the n qubits at time index κ is also assumed, as a product of lower-order Q-dependent operators, $U_\mu^{(\kappa)}$, indexed by an index μ, such that,

$$U_\theta^{(\kappa)} = e^{-i\tau H_\theta^{(\kappa)}} = \prod_\mu U_\mu^{(\kappa)} \ + \ \mathcal{O}(\tau^{Q+1}) \ . \tag{3.5}$$

The overall first-order temporal and $Q = 2$-order spatial Trotterization is thus,

$$\mathcal{U}_\theta = \ \circ_{\kappa \in [k]} \mathcal{U}_\theta^{(\kappa)} \ : \ \mathcal{U}_\theta^{(\kappa)} = \left[\circ_{\mu \in \{0,\ldots,k-1\}} \mathcal{U}_\mu^{(\kappa)}\right] \circ \left[\circ_{\mu \in \{k-1,\ldots,0\}} \mathcal{U}_\mu^{(\kappa)}\right] \ , \ U_\mu^{(\kappa)} = e^{-i\frac{1}{2}\tau H_\mu^{(\kappa)}} \ . \tag{3.6}$$

For the non-unitary component, we consider temporally and spatially local, independent noise acting on the $s = nk$ sites, indexed by $(\kappa, i) \in [k] \times [n]$, of possible errors

$$\mathcal{N}_\gamma = \circ_{\kappa \in [k]} \mathcal{N}_\gamma^{(\kappa)} \quad : \quad \mathcal{N}_\gamma^{(\kappa)} = \circ_{i \in [n]} \mathcal{N}_{\gamma_i}^{(\kappa)} \ . \tag{3.7}$$

For our purposes, we decompose each local noise channel into a convex combination of an identity component, and what we refer to as a non-identity error component,

$$\mathcal{N}_{\gamma_i}^{(\kappa)} \equiv (1-\gamma)\mathcal{I}_i \ + \ \gamma \mathcal{K}_{\gamma_i}^{(\kappa)} \ . \tag{3.8}$$

The forms of the non-trivial $\mathcal{K}_\gamma$, with noise scale γ, depend on the specific noise model of interest. Noise models considered here include local dephasing, amplitude damping, and depolarization noise. Such local noise models are known to be relevant in several quantum computing implementations [37, 281]. Our analytical and numerical approaches are easily transferable to spatially correlated noise models across multiple qubits, however such studies concerning any non-trivial effect of correlated noise are left for future studies. Please refer to Appendix 3.C for a complete description of each noise channel, and to Appendix 3.D for an analytical treatment of noise-induced effects. Given the temporal and spatial locality of the Trotterized unitary and non-unitary channel, we finally reach the explicit ansatz form of interlaced noise and unitary evolution,

$$\Lambda_{\theta\gamma} = \ \circ_{\kappa \in [k]} \left(\mathcal{N}_\gamma^{(\kappa)} \circ \mathcal{U}_\theta^{(\kappa)} \right) \ , \tag{3.9}$$

leading to our overall quantum channel circuit diagram in Fig. 3.1.

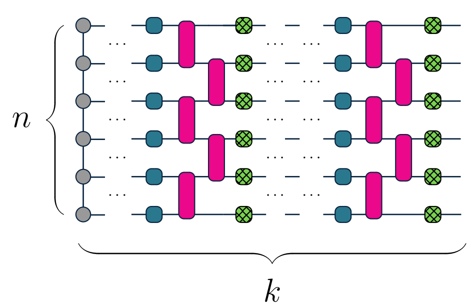


Figure 3.1: **Trotterized parameterized quantum channel**. Quantum channel with k layers of n qubit trotterized local (blue squares) and all-to-all (pink rectangles) unitary operators (not all operators shown), followed by local noise channels (hatched green squares) after each layer, with an initial state (left-most gray circles).

Here, we take as our parameterized unitary ansatz, evolution generated by the nuclear magnetic resonance (NMR) Hamiltonian consisting of $q = 2$ qubit Pauli operators,

$$H_\theta^{(\kappa)} = \sum_i \theta_i^{x(\kappa)} X_i \; + \; \sum_i \theta_i^{y(\kappa)} Y_i \; + \; \sum_i h_i Z_i \; + \; \sum_{i<j} J_{ij} Z_i Z_j \; . \tag{3.10}$$

Generally in such systems, we have control over the variable time-dependent local transverse X and Y fields θ at qubit i, with additional constant time-independent longitudinal local Z fields h at qubit i and non-local ZZ fields J at qubits $i \neq j$. Such Hamiltonians allow for universal control over qubits, with the local and non-local gates allowing entangling gates to be implemented. We use experimentally relevant scales [268] for our ansatz in Table 3.1, and details of the ansatz are discussed in Appendix 3.E.

To compare our choice of NMR ansatz to other implementations, we collect rough estimates [282] from recent literature of the 1-qubit T_{U1} and 2-qubit T_{U2} gate times, and decoherence times T_γ, for NMR [39, 41, 268], trapped ion [41, 46, 47], superconducting qubit [41, 52, 53], and neutral atom [55, 57, 215] quantum computing experiments in Table 3.E.2 of Appendix 3.E. NMR systems are limited by experimental feasibility. Their small non-local coupling constants J limit how much qubits can be correlated at each time step $T_{U1} = \mathcal{O}(\tau)$, increasing significantly the necessary 2-qubit gate times $T_{U2} = \mathcal{O}(1/J)$. This translates into NMR systems having the largest effective depth $T_{U2}/T_{U1} \sim \mathcal{O}(100)$ required for each 2-qubit gate, and having the smallest effective maximum depth $T_\gamma/T_{U2} \sim \mathcal{O}(10^2)$ before decoherence, amongst considered implementations. Therefore any conclusions drawn from these studies regarding the explicit scale of noise or circuit depth where phenomena occurs, are specific to this NMR ansatz. However, we believe other similarly universal ansatz should exhibit comparable behaviour, at their specific ansatz-dependent scales.

The choice of a universal ansatz spanning the full space of $g = \mathcal{O}(d^2)$ unitaries, also simplifies transferability to other universal ansatze, and avoids any bias by restricting evolution to being within a subspace. Techniques used by other implementations [52] generally have origins in NMR techniques and NMR's proof of principle quantum algorithm experiments [195, 200], including the highly relevant zero-noise extrapolation [37], dynamical decoupling [199], and refocusing procedures [195].

Given this parameterized ansatz, we wish to assess its ability to represent targets of interest, such as operator compilation, where sequences of operators are optimized to approach a target operator, and state preparation, where initial states transformed by sequences of operators, are optimized to approach a target state [283]. Such tasks depend crucially on targets being within span of the ansatz.

Table 3.1: **NMR ansatz**. Experimentally relevant NMR parameters [39, 268].

n	Number of qubits $\sim 1-4$
k	Number of time steps $\sim \mathcal{O}(10^0 - 10^4)$
p	Number of parameters $\sim \mathcal{O}(\mathrm{poly}(n)k)$
τ	Trotterization time step $\sim \mathcal{O}(75 - 100\ \mu\mathrm{s})$
T	Evolution time $= k\tau \sim \mathcal{O}(375\ \mu\mathrm{s} - 500\ \mathrm{ms})$
Q	Spatial Trotterization order $= 2$
J	Constant longitudinal coupling $\sim \mathcal{O}(\pi/2 \times 10^2\ \mathrm{Hz})$
h	Constant longitudinal field $\sim \mathcal{O}(\pi/2\ \mathrm{kHz})$
θ	Variable transverse field $\sim \mathcal{O}(\pi/2\ \mathrm{MHz})$
γ	Noise scale $\sim \mathcal{O}(10^{-14} - 10^{-1})$

3.2.2 Quantum Optimal Control Tasks

Here we focus on unitary compilation and pure state preparation, given a parameterized ansatz with universal control over the full space of unitary operators. Given an initial pure state σ, target unitaries U or pure states ρ, and ansatz parameters θ we assess the abilities of the optimized respective parameterized unitaries $U_{\theta\gamma}$ and states $\rho_{\theta\gamma}$,

$$U_{\theta\gamma} \approx U \tag{3.11}$$

$$\rho_{\theta\gamma} = \Lambda_{\theta\gamma}(\sigma) \approx \rho = \mathcal{U}(\sigma)\ . \tag{3.12}$$

Objective metrics of infidelities with respect to the given unitary compilation and state preparation tasks [87] are chosen to quantify these ansatze through optimization,

$$\mathcal{L}^U_{\theta\gamma} = 1 - (1/d^2)\left|\mathrm{tr}\left(U^\dagger U_{\theta\gamma}\right)\right|^2 \tag{3.13}$$

$$\mathcal{L}^\rho_{\theta\gamma} = 1 - \mathrm{tr}(\rho\rho_{\theta\gamma})\ . \tag{3.14}$$

We also define the impurity, von-Neumann entropy, and relative entropy divergence, relative to a pure state ρ as follows, for later interpretations of noise-induced phenomena,

$$\mathcal{I}_{\theta\gamma} = 1 - \mathrm{tr}\left(\rho^2_{\theta\gamma}\right) \tag{3.15}$$

$$\mathcal{S}_{\theta\gamma} = -\,\mathrm{tr}(\rho_{\theta\gamma}\log(\rho_{\theta\gamma}))\,/\log(d) \tag{3.16}$$

$$\mathcal{D}^\rho_{\theta\gamma} = -\,\mathrm{tr}(\rho\log(\rho_{\theta\gamma}))\,/\log(d)\ . \tag{3.17}$$

In general, the optimization of objectives $\mathcal{L}_{\theta\gamma} \to \mathcal{L}^*_{\theta\gamma}$, particularly in noisy settings to determine the optimal parameterization θ^*_γ, has no closed forms [284]. Whether there are similarities between optimal noisy and noiseless quantities, such as infidelities $\mathcal{L}^*_{\theta\gamma} \approx \mathcal{L}^*_{\theta}$, or even more strongly, parameters $\theta^*_{\gamma} \approx \theta^*$, remains an open question [275]. In our subsequent analysis, in addition to numerical studies, we derive analytically the leading-order scalings of quantities with respect to system parameters of circuit depth k and noise scale γ.

We also note that all plotted statistics here reflect average behaviours of the ansatz across independent optimizations with respect to Haar random initial pure states, and Haar random targets. The minimum infidelity reached for each independent optimization for a given fixed depth and noise scale, is used for statistics across samples. Error bars and shaded regions represent one standard deviation from the mean of samples. Lower error bars are plotted equal in length to the upper error bar on a log scale, and non-visible error bars can be considered to be equal in scale to any plot markers. Error bars here generally appear to be relatively orders of magnitude smaller than their corresponding average values, and $m = 50 \leq \mathcal{O}(100)$ samples are deemed adequate to capture all behaviours well in practice for $n \leq 4$ qubits. We note however that optimization hyperparameters, as per Appendix 3.F, in particular the use of a modified conjugate-gradient-based optimizer with a Wolfe condition line search and appropriate learning rates, must be carefully selected. Our initial studies indicated that at larger depths, and larger small noise scales, after an initially smooth convergence in infidelity, optimization routines can oscillate rapidly between local minima, as also observed in previous optimization studies [272].

Quantities computed within a noisy context generally have γ subscripts, such as parameters obtained in a noisy setting θ_{γ}. Otherwise in noiseless settings, any noise subscripts are dropped. All logarithms $\log(\cdot) \equiv \log_e(\cdot)$ are defined to be natural logarithms, unless specified $\log_b$ with an explicit base b. Ranges of numbers up to n are denoted by $[n] = \{0, 1, \ldots, n-1\}$, with subsets $\Gamma \subseteq [n]$, and indicator delta-functions are $\delta_{i\in[n]}$.

3.2.3 Overparameterization Metrics

We now develop formalisms to understand the scaling of channel-dependent quantities with respect to depth and noise scales. We first develop methods for comparing constrained versus unconstrained noiseless ansatze, as detailed in Appendix 3.B. We follow the formalism developed by Larocca *et al.* [230], which sets bounds on the rank $r = r(p) \leq p$ of the quantum Fisher information $\mathcal{F}_{\theta_{\mu\nu}}$ to determine whether a quantum system with p parameters is overparameterized. At an overparameterization limit $\tilde{p} = \mathcal{O}(g)$, defined in terms of the dimension of the space spanned by the ansatz, this metric's rank is shown to transition from being full rank $r = p$ for underparameterized $p < \tilde{p}$, to saturating at this limit $r = \tilde{p}$ for adequately or overparameterized $p \geq \tilde{p}$.

Underlying these definitions of overparameterization is the quantum Fisher information's rank generally reflecting how many directions a parameterized ansatz may span in its space of solutions. We seek to generalize these intuitions from state-dependent ansatze to unitary-dependent ansatze, independent of the initial state being transformed. The conventional state Fisher information $\mathcal{F}^{\rho}_{\theta}$ is defined in terms of states ρ_θ, given a state preparation objective $\mathcal{L}^{\rho}_{\delta} \sim \mathrm{tr}(\rho_\theta \rho_{\theta+\delta})$. We define a generalized unitary Fisher information $\mathcal{F}^{U}_{\theta}$, developed concurrently by [190], given a unitary compilation objective $\mathcal{L}^{U}_{\delta} \sim \mathrm{tr}\left(U^\dagger_\theta U_{\theta+\delta}\right)$,

$$\mathcal{F}^{U}_{\theta_{\mu\nu}} = \frac{1}{d^2}\mathrm{Re}\left(d\ \mathrm{tr}\left(\partial_\mu U^\dagger_\theta \partial_\nu U_\theta\right)\ -\ \mathrm{tr}\left(\partial_\mu U^\dagger_\theta U_\theta\right)\mathrm{tr}\left(U^\dagger_\theta \partial_\nu U_\theta\right)\right)\ ,$$

which corresponds to the leading-order $\delta \to 0$ deviation of the unitary objective, and which reduces to the state Fisher information in the limit of tracing over pure states $\rho_\theta = \psi_\theta$,

$$\mathcal{F}^{\rho}_{\theta_{\mu\nu}} = \mathrm{Re}\left(\langle \partial_\mu \psi_\theta | \partial_\nu \psi_\theta\rangle - \langle \psi_\theta | \partial_\mu \psi_\theta\rangle\langle \partial_\nu \psi_\theta | \psi_\theta\rangle\right)\ . \tag{3.18}$$

The rank of the unitary quantum Fisher information subsequently offers insight into the capabilities of an ansatz to span a set of unitaries, and to potentially become overparameterized for unitary compilation tasks.

We also can derive expressions for noise-dependent quantities that allow for an easier interpretation for noise-induced phenomena. Crucially, we differentiate states by their number of $l \leq s = nk$ errors due to noise,

$$\mathcal{N}_\gamma = \sum_{l}^{s} \sum_{\substack{\Gamma \subseteq [k]\times[n] \\ |\Gamma| = l}} \gamma^l (1-\gamma)^{s-l} \circ_{(\kappa,i)\in\Gamma} \mathcal{K}^{(\kappa)}_{\gamma_i}. \tag{3.19}$$

Here, we refer to an error as any single non-identity noise operation $\mathcal{K}_\gamma$ acting locally at any of $s = nk$ possible spatial and temporal indices, interlaced within a unitary ansatz $\mathcal{U}_\theta$. We then may define l-error channels $\Lambda_{\theta\gamma_l}$, that are convex combinations of channels consisting of all possible locations of $l \leq s$ errors, each with probability γ^l. The channels that generate at most l-error noisy states, can therefore be seen to have an interesting form of an expectation value over a distribution of l-error channels,

$$\Lambda_{\theta\gamma} =\ \langle \Lambda_{\theta\gamma_l}\rangle_{l\sim p_{s\gamma}}\ . \tag{3.20}$$

For noise models defined in terms of strictly identity or non-trivial errors, the exact distribution over the errors is the binomial distribution with mean $s\gamma$,

$$p_{s\gamma}(l) = \tbinom{s}{l}\gamma^l(1-\gamma)^{s-l}\ . \tag{3.21}$$

As discussed in Appendix 3.C, other interpretations for noise can arise due to the binomial distribution being equivalent to other distributions in various limits of γ and s. Now, grouping the l, γ-dependent terms yields the deviation from the noiseless channel,

$$\Lambda_{\theta\gamma} - \Lambda_{\theta} = \sum_{l>0}^{s} \tbinom{s}{l} \gamma^l \Lambda_{\theta\gamma \leq l} \ , \tag{3.22}$$

where l-error operators are the uniform combination of all error locations,

$$\Lambda_{\theta\gamma_l} = \frac{1}{\binom{s}{l}} \sum_{\substack{\Gamma \subseteq [k]\times[n] \\ |\Gamma|=l}} \Lambda_{\theta\gamma\Gamma} \ , \tag{3.23}$$

at-most-l-error operators are the non-unfiorm combination of all $k \leq l$ error locations,

$$\Lambda_{\theta\gamma\leq l} = \sum_{l'}^{l} (-1)^{l-l'} \tbinom{l}{l'} \Lambda_{\theta\gamma_{l'}} \ , \tag{3.24}$$

and specific l-error channels with errors at locations Γ are,

$$\Lambda_{\theta\gamma\Gamma} = \ \circ_{\kappa\in[k]} \ \circ_{i\in[n]} \mathcal{K}_{\gamma_i}^{(\kappa)\delta_{(\kappa,i)\in\Gamma}} \circ \mathcal{U}_{\theta}^{(\kappa)} \ . \tag{3.25}$$

For example, the $l = 1$-order deviation from the noiseless channel

$$\Lambda_{\theta\gamma\leq 1} = \frac{1}{s} \sum_{(\kappa,i)\in[k]\times[n]} \mathcal{U}_{\theta}^{(>\kappa)} \circ (\mathcal{K}_{\gamma_i}^{(\kappa)} - \mathcal{I}_i) \circ \mathcal{U}_{\theta}^{(\leq\kappa)} \tag{3.26}$$

depends on the non-triviality of the s possible local channels $\mathcal{K}_i^{(\kappa)} \neq \mathcal{I}_i$. More generally, the l-order deviation depends on how the $\binom{s}{l}$ possible $l \leq s$-error channels deviate from the trivial noiseless channel. Channels consisting of multiple types of errors may also be written as an expectation over a generalized multinomial distribution of channels.

As derived in Appendix 3.C, gradients along parameter directions μ of objectives, which are linear in the parameterized channels with constant noise, follow parameter shift rules [285], denoted with perturbing parameters φ and coefficients π_φ,

$$\partial_\mu \Lambda_{\theta\gamma} = \sum_{\varphi} \pi_{\varphi}^{\mu} \ \Lambda_{\theta+\varphi \ \gamma} \ . \tag{3.27}$$

These parameter shift rules indicate that the linear nature of the noise interlaced with the parameterized unitary ansatz perturbatively affects the noisy quantities.

Whether classical sources of noise impose similar noise-induced phenomena to the quantum noise sources studied, is a separate important question when determining the robustness of parameterized systems. As developed in Appendix 3.D, we introduce the notion of classical noise due to numerical precision, or floating point error scale ϵ, and we determine its similarity to the quantum noise scale γ. This error affects both the representation and numerical operations of floating point scalars, which we extend to d-dimensional matrices A, and l-fold matrix multiplication, given error matrices Σ, interlaced at locations $\Gamma \subseteq [l]$,

$$\prod_{\mu\in[l]} A_\mu \to \prod_{\mu\in[l]}{}_{\epsilon} A_\mu = \sum_{\Gamma\subseteq[l]} \prod_{\mu\in[l]} A_\mu \Sigma^{\delta_{\mu\in\Gamma}} \ . \tag{3.28}$$

We then relate classical ϵ and quantum γ noise phenomena, given the number of $l = \mathcal{O}(s) = \mathcal{O}(\mathrm{poly}(n)k)$ matrix multiplication operations that are subject to classical error, are related to the number of s possible quantum errors in the ansatz. We are thus able to derive exact bounds on the deviations of noisy infidelities from their noiseless counterparts,

$$|\mathcal{L}_{\theta\epsilon} - \mathcal{L}_\theta| \leq \ \left|1 - (1+\epsilon)^{2l}\right| \sim \mathcal{O}(2l\epsilon) \tag{3.29}$$

$$|\mathcal{L}_{\theta\gamma} - \mathcal{L}_\theta| \leq 2|1 - (1-\gamma)^s| \sim \mathcal{O}(2s\gamma) \ , \tag{3.30}$$

with the use of various properties of norms such as Holder and von-Neumann's inequalities for Schatten matrix norms [286]. Both classical and quantum noisy infidelities scale similarly: polynomially in the noise scale, exponentially in the number of errors, and identically in the limit of small local error scales. However, the constant classical error scales $\epsilon = \epsilon(d) = \|\Sigma\|$ depend on the dimension of the space of interest, and the number of parameters, compared to the assumed constant quantum noise scales $\gamma = \mathcal{O}(\|\mathcal{K}\|)$. Therefore as the dimension of classically simulated systems increases, they potentially accumulate polynomially greater classical error.

This scaling of infidelities analysis is ansatz-independent and provides valuable insight into general deviations of noisy quantities from their noiseless counterparts. However tighter bounds are potentially possible if there are symmetries or relationships between the noise and dynamics, or if the optimization is analytically tractable, and if an exact form for $\mathcal{L}_{\theta^*_\gamma\gamma}$ may be derived. These upper bounds, as will be discussed in Section 3.3, also relate to the critical point at which noise effects begin to dominate for greater numbers of operations, and optimization enters a divergent regime.

3.3 Results

Here, classical simulations and analytical calculations are performed to quantify changes in the behaviour of parameterized systems, with depth and with noise scale. Simulation details are described in Appendices 3.E and 3.F, and all data can be found in [2].

3.3.1 Unconstrained versus Constrained Optimization

We first investigate the effects of constraints on noiseless Haar random unitary compilation, with respect to the number of optimization iterations and various depths k. The controllable transverse fields $\theta_i^{x,y(\kappa)}$ at each time step $\kappa \in [k]$ and qubit $i \in [n]$, are constrained due to it being difficult to exercise individual control over qubits. Therefore in the constrained setting, we impose that all transverse fields are equal across qubits, $\theta_i^{x,y(\kappa)} = \theta^{x,y(\kappa)}$, pulse amplitudes are bounded $|\theta_i^{x,y(\kappa)}| \leq \bar{\theta}$, and pulse amplitudes are generally turned off at start and end times, $\theta_i^{x,y(\kappa=0,k-1)} = 0$.

Given the optimization trends of the $n = 4$ NMR ansatz with Haar random targets, as well as the rank saturation of the unitary Fisher information in Fig. 3.2, we observe exponential infidelity convergence beyond a depth $k > \tilde{k} \approx g \sim \mathcal{O}(256)$. We thus claim numerically that noiseless constrained optimization converges, given some rate α, as,

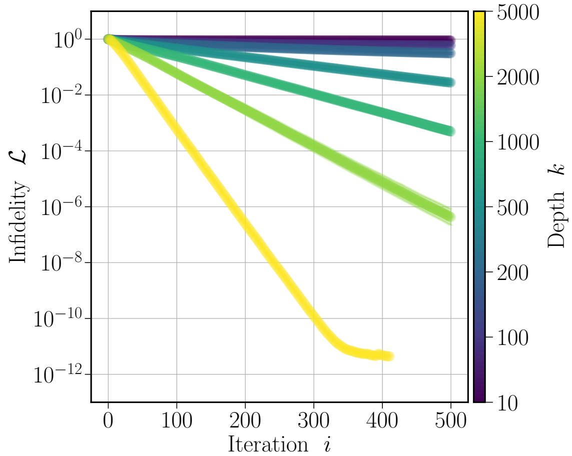
$$\mathcal{L}_\theta \sim e^{-\alpha k} \ : \ k > \tilde{k} \ . \tag{3.31}$$

In practice, $k \approx \mathcal{O}(1000) \gg \tilde{k}$ is necessary for adequate convergence within a feasible number of optimization iterations. We also note that at larger depths, constrained tasks exhibit greater variance relative to unconstrained tasks, however variances are generally low, suggesting an appropriate choice of optimization hyperparameters, as per Appendix 3.F.

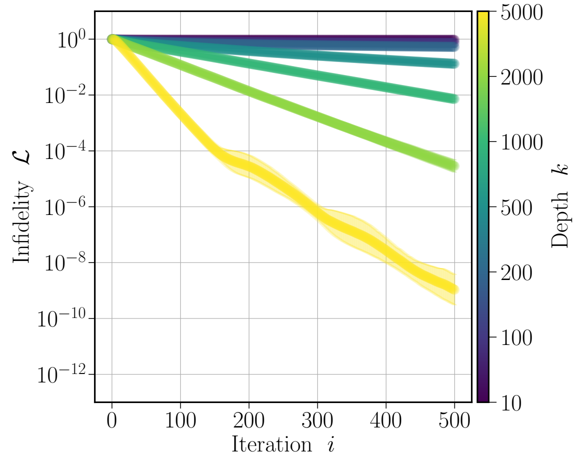
We note that the non-local coupling $J \ll h, \theta$ is much smaller than other scales, and for experimentally realistic time steps, $\tau J \ll 1$. This limits how much of an entangling gate can be implemented in a single time step. From these simulations, we conjecture that these constraints necessitate that the minimum depth where overparameterization can occur, depends on this scale J, and how many poly(n) less local parameters there are,

$$\tilde{k}_{\text{Constrained}} \sim \mathcal{O}(\frac{1}{\tau J}\text{poly}(n))\tilde{k}_{\text{Unconstrained}} \ . \tag{3.32}$$

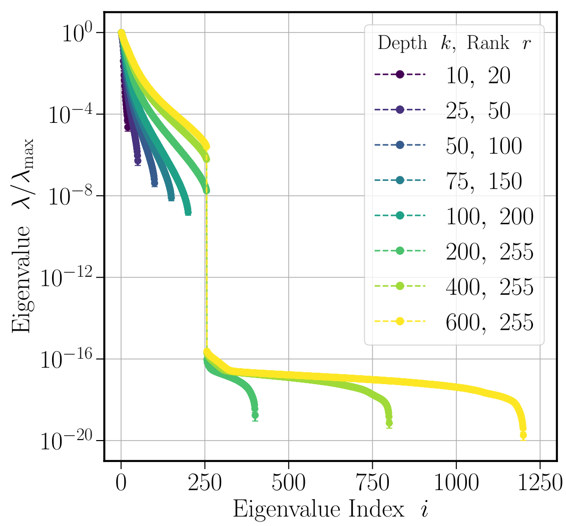
Other than a shift in the overparameterization boundary, the given constraints do not appear to fundamentally affect the exponential infidelity convergence. We also note that even for relatively small system sizes of $n \leq 4$, several orders of magnitude deeper circuits than are typically classically simulated [100, 230, 272], with up to $\mathcal{O}(10^5)$ gates, are necessary.



(a) Unconstrained



(b) Constrained



(c) Unitary Quantum Fisher Information

Figure 3.2: **Unitary compilation**. Convergence of unitary compilation infidelity with respect to optimization iteration and depth k (colours) for the $n = 4$ NMR ansatz. (a) Unconstrained parameterization with independent qubit parameters, and no boundary conditions. (b) Constrained parameterization with shared qubit parameters, and zero-field temporal boundary conditions. Constrained tasks require comparable iterations or depth, to converge comparably to unconstrained tasks, and exhibit exponential convergence beyond $k > \tilde{k} \approx \mathcal{O}(d^2) = \mathcal{O}(256)$. (c) Constrained quantum Fisher information eigenvalue spectrum at optimality. The spectrum is full-rank $r = p$ for $k < \tilde{k}$ before saturating at rank $r = \tilde{p} \sim \mathcal{O}(d^2) \leq p$ for $k \geq \tilde{k}$, indicating overparameterization.

3.3.2 Noisy State Preparation

We now investigate the effects of local noise on Haar random pure state preparation. Here, parameters are unconstrained to ensure all observed phenomena are due to noise. Given our findings on the effects of constraints, to leading-order, imposing constraints should only shift the depth dependent results by a noise-independent factor.

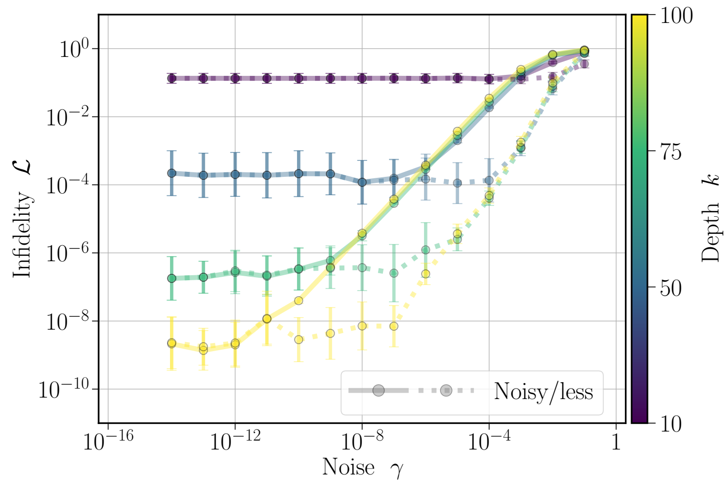
To demonstrate the interplay between depth and noise scales on optimized infidelities, we plot infidelities in Fig. 3.3. Here we display unital dephasing noise, and other unital and non-unital noise models are shown to exhibit similar behaviour in Appendix 3.C. When varying the noise scale in Fig. 3.3a, for small noise scales, the average optimal infidelity (solid lines in Fig. 3.3a) strictly scales as expected with depth. Increasing the noise scale past a depth-dependent critical noise scale γ_k causes infidelities to increase polynomially, between linearly and quadratically, until the infidelities plateau at their maximum value.

We also investigate trends when inserting parameters learned in the noisy setting into an identical, but noiseless unitary ansatz, yielding infidelities (dashed lines in Fig. 3.3a). Such noiseless, tested infidelities are superior than their corresponding trained noisy infidelities, in terms of their greater critical noise scale $\gamma_{k_{\text{Noiseless}}} > \gamma_{k_{\text{Noisy}}}$: the noiseless infidelities increase in the divergent regime at much larger noise scales. This suggests the optimization is learning about the underlying unitary dynamics, and is not just preparing another mixed state that happens to be close to the target pure state.

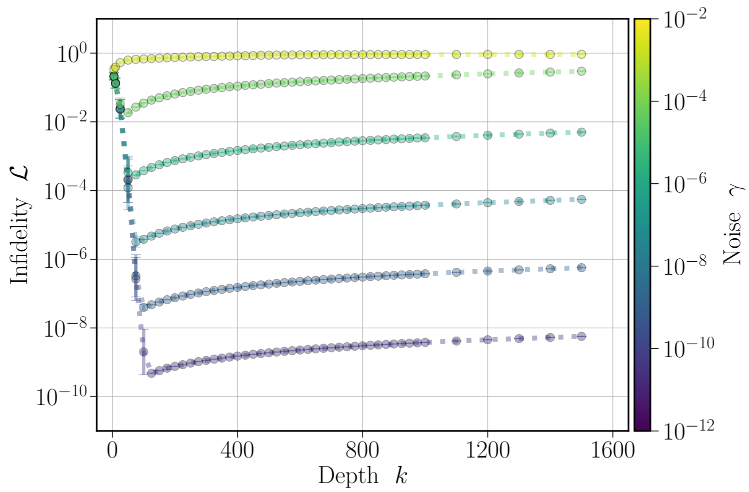
The error bars of both the noisy trained and noiseless tested infidelities are equal in the convergent regime with $\gamma < \gamma_k$ (overlapped markers in the left of Fig. 3.3a), and they are orders of magnitude smaller in the divergent regime with $\gamma > \gamma_k$. Overall, the trends in error suggests the optimization and dynamics are most uncertain within the regime around and past the noise scale γ_k, indicating increased complexity near this transition.

The noiseless behaviour can partly be explained from parameter shift rules for gradients of parameterized channels with constant noise, as derived in Appendix 3.C. The noisy state is a convex combination of parameterized pure states, each with identical gradient directions to the noiseless case, albeit with magnitudes that are scaled by polynomials of the noise scale. Therefore, the trajectory of the gradient-based optimization remains similar at small noise scales in both noisy and noiseless cases.

Here, we observe a critical noise-dependent depth k_γ that occurs in Fig. 3.3b. Previously decreasing infidelities with depth in a convergent regime for $k < k_\gamma$, increase uniformly with depth in a divergent regime for $k > k_\gamma$. Beyond this depth, any increase in expressiveness of the ansatz to prepare arbitrary states from increasing the number of variable parameters, is outweighed by the accumulated noise from increased sources of error.



(a) Infidelity versus noise scale γ, for various depths k.



(b) Infidelity versus depth k, for various noise scales γ.

Figure 3.3: **State preparation**. Behaviour of state preparation infidelity with respect to unital dephasing noise γ and depth k for the $n = 4$ NMR ansatz. (a) Trained noisy infidelity (solid), and tested infidelity of noisy parameters in noiseless ansatz (dashed), with respect to noise scale, for various depths k (colours). Infidelities are depth dependent and noise independent for small noise scales, before universally increasing polynomially with noise. Tested noiseless infidelities indicate that the underlying unitary dynamics are being learned resiliently. (b) Critical depth for noisy infidelity for various noise scales (colours). Infidelities improve exponentially with depth, up until a noise-induced critical depth, where entropic effects worsen infidelities polynomially.

From a trainability standpoint, beyond this critical depth, noise-induced barren plateaus may be occurring, leading to a decrease in trainability where the gradients are unable to find the ideal trajectory to reach optimality. Alternatively, from an expressiveness standpoint, noise potentially increases in an uncontrolled manner the number of directions permitted to be explored in the objective landscape [102]. Future studies should investigate the density of mixed states that are perturbatively away from a given pure state [266, 287]. As discussed below, through analytical calculations, we offer complementary interpretations into exactly how entropic effects begin to dominate infidelity behaviours.

3.3.3 Universal Effects of Classical and Quantum Sources of Noise

Such studies of noise bring to mind the question of whether noise-induced critical depth phenomena can be attributed to strictly quantum, or potentially classical noise phenomena. For very deep, ideal noiseless ansatz with $k > \mathcal{O}(1000)$, infidelities approach machine precision, and start to increase with depth. This suggests floating point errors accumulate for large numbers of simulated operations. As per the noise models derived in Appendix 3.D, we insert artificial classical floating point error of different scales into simulations, as per Fig. 3.4. Adding zero-mean random errors with standard deviation proportional to error scales, to results of floating point operations, shows similar trends in noise-induced convergent and divergent regimes. Since unitary compilation tasks have quadratically more degrees of freedom than state compilation tasks, their infidelities decrease slower with depth. Furthermore, infidelity curves reach their critical depth at earlier depths for smaller system sizes due to the exponentially smaller spaces to search.

Infidelities are shown to increase, and enter a divergent regime when they reach a scale proportional to that of the error scale. This error scale may be approximately upper bounded by the derived deviations of the noisy infidelities from their noiseless counterparts. By decreasing the artificial floating point error, the curves also converge to the supposedly noiseless case, offering an estimate for the machine precision of $\varepsilon \sim \mathcal{O}(10^{-16})$. These trends question the viability of large-scale simulations close to machine precision with finite floating point arithmetic. Can arbitrarily large systems be accurately simulated in principle, without resorting to inefficient arbitrary precision arithmetic, or classical error mitigation, or error correction approaches [37, 109]?

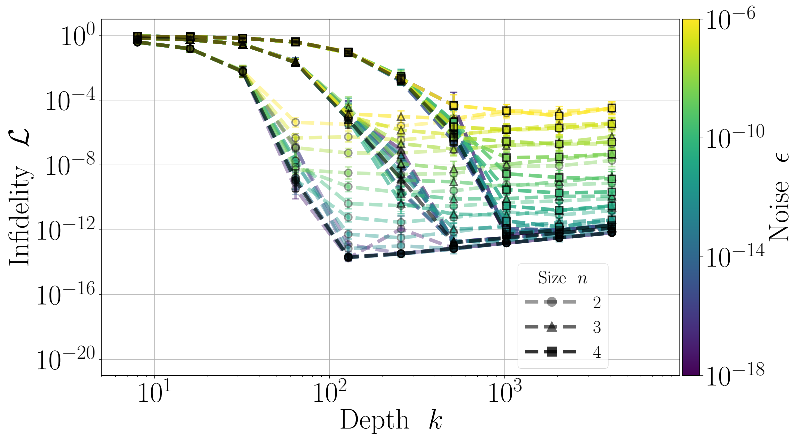


Figure 3.4: **Classical floating point error**. Behaviour of unitary compilation infidelity with respect to depth k and classical floating point error scale ϵ (colours), for various n NMR ansatz, relative to the noiseless case (black). The classical noise model is shown to be suitable, and estimates an architecture dependent machine precision of $\varepsilon \sim \mathcal{O}(10^{-16})$. Classical noise is also shown to exhibit a critical depth k_ϵ and divergent regime.

3.4 Discussion

From numerical studies, we find for a given noise scale γ there is a critical depth k_γ, beyond which optimal infidelities increase due to an accumulation of noise. From fitting procedures in Appendix 3.C, the critical depth is shown to be logarithmic in noise scale,

$$k_\gamma \sim \log(1/\gamma) \ . \tag{3.33}$$

The optimal infidelity is therefore approximately, with derived scaling $1 \leq \alpha \leq 2$,

$$\mathcal{L}_{\theta^*\gamma|k_\gamma} \sim e^{-\alpha k_\gamma} \sim \gamma^\alpha \ , \tag{3.34}$$

confirming previous conjectures of linear, or quadratic scaling of infidelity with noise [277].

The interpretation of a noisy channel being a binomial distribution of s-error channels also suggests that parameterized quantum channels can mitigate approximately

$$\bar{s}_\gamma \sim \gamma \log(1/\gamma) \ \text{ errors} \ . \tag{3.35}$$

Determining whether the optimization is finding a parameterization that explicitly performs error mitigation [37, 274, 288], or even error correction through a parameterized encoding [289], would constitute important future contributions [109, 290].

The presence of a noise-induced depth is also reminiscent of measurement-induced phase transitions [291]. However, these noise-induced effects should occur at all system sizes, and do not seem to be related to typical indicators of criticality such as scale invariance.

As derived in Appendix 3.C, the Bloch representation of quantum states allows explicit leading-order scaling of quantities with respect to depth and noise to be derived. Here, we represent states as $\rho = (1/d)(I + \lambda \cdot \omega)$, with Bloch coefficients λ associated with a set of $d^2 - 1$ non-identity, trace orthogonal basis operators ω. Channels Λ may then be represented as affine transformations $\lambda \to \Upsilon\lambda + \upsilon$. Parameterized noisy channels $\Lambda_{\theta\gamma}$ with s possible errors may be decomposed into strictly unitary u_θ, and unital $u_{\theta\gamma}$ and non-unital $\eta_{\theta\gamma}$ noise-dependent components, such that,

$$\Upsilon_{\theta\gamma} = (1-\gamma)^s u_\theta + (1-(1-\gamma)^s) u_{\theta\gamma} \tag{3.36}$$

$$\upsilon_{\theta\gamma} = (1-(1-\gamma)^s)\eta_{\theta\gamma} \ . \tag{3.37}$$

In this Bloch representation, we may then express a parameterized noisy state as,

$$\rho_{\theta\gamma} = (1-\gamma)^s \rho_\theta + (1-(1-\gamma)^s)\epsilon_{\theta\gamma} + \Delta_{\theta\gamma} \ . \tag{3.38}$$

This decomposition expresses the interplay of the parameterized unitary and noise-induced non-unitary components of the channel. The unitary component rotates what we refer to as the pure component of the state. This component consists of a superposition of the pure state ρ_θ, with coefficients λ_θ, and an orthogonal pure state with orthogonal coefficients $\zeta_\theta \perp \lambda_\theta$, represented within the traceless deviation term $\Delta_{\theta\gamma}$. The noise component of the channel scales the pure component of the state with the noise scale $1-\gamma$, plus it shifts the state by what we refer to as the mixed component of the state $\epsilon_{\theta\gamma}$, with coefficients $\varepsilon_{\theta\gamma}$.

In the limit of the optimization reaching optimality in the noisy setting $\theta \to \theta^*_\gamma$, the pure component of the state approaches the pure target state $\lambda_\theta \to \lambda$, the deviation term $\Delta_{\theta^*_\gamma\gamma} \to 0$ approaches zero, and there only remains an inherent noise-dependent mixed component $\epsilon_{\theta^*_\gamma\gamma}$. We may then assume that channel-dependent quantities may be expanded in the number of errors s, and noise scale γ.

At optimality, our metrics analytically scale similarly to leading-order in s, γ,

$$\mathcal{L}^\rho_{\theta\gamma} \sim s\gamma \frac{d-1}{d}\left(1 - \frac{\lambda \cdot \varepsilon_{\theta\gamma}}{\lambda^2}\right) \ + \ \mathcal{O}(\tbinom{s}{2}\gamma^2) \ , \tag{3.39}$$

$$\mathcal{I}_{\theta\gamma} \sim 2s\gamma \frac{d-1}{d}\left(1 - \frac{\lambda \cdot \varepsilon_{\theta\gamma}}{\lambda^2}\right) \ + \ \mathcal{O}(\tbinom{s}{2}\gamma^2) \ , \tag{3.40}$$

$$\mathcal{S}_{\theta\gamma} \sim \mathcal{O}(s\gamma) \quad , \quad \mathcal{D}^\rho_{\theta\gamma} \sim \mathcal{O}(s\gamma) \ . \tag{3.41}$$

As derived in Appendix 3.C, the Bloch representation allows exact leading-order terms to be derived for quantities that are strictly polynomial functions of the Bloch coefficients. Quantities that are more complex, for example logarithmic, functions of the Bloch coefficients require knowledge of the algebras governing the specific choice of basis ω, yielding in principle calculable [292], but unintuitive forms.

Critically, when the mixed components $\varepsilon_{\theta\gamma} \parallel \lambda$ become pure, or aligned with the pure target coefficients, quantities differ from their noiseless values by strictly higher-order terms. This purification of the mixed component could be due to the lack of noise, or the specific parameterization forcing the system towards a pure state, and it describes general error mitigation. The use of Bloch representations thus allows simplified, and occasionally ansatz-independent depictions of noise-induced phenomena.

Beyond the critical depth, infidelities appear both analytically and numerically to be linear functions of entropy and impurity, and they scale with the overlap of the pure target state with the mixed component. In particular, at low noise scales in the divergent noisy regime, analytical predictions and numerical simulations of the discussed quantities all correspond precisely, as demonstrated in Fig. 3.C.4 in Appendix 3.C.

Further, from Fig. 3.5, all quantities appear to collapse together with increasing noise scales. Reasoning that noise phenomena start to dominate at $k = k_\gamma$, when the scale of the optimization-driven decreasing optimized infidelity $\mathcal{L}^{\rho}_{\theta^*_\gamma\gamma}$, reaches the scale of the entropic-driven increasing analytical infidelity $\mathcal{L}^{\rho}_{\theta^*\gamma}$, thus the following infidelity descriptions are equal at this transition point,

$$\mathcal{L}^{\rho}_{\theta^*_\gamma\gamma} \sim e^{-\alpha k}|_{k_\gamma} \approx \mathcal{L}^{\rho}_{\theta^*\gamma} \sim nk\gamma|_{k_\gamma} \ . \tag{3.42}$$

From such a relation, we recover our numerically predicted noise-induced critical depth that is logarithmic in noise scale,

$$k_\gamma \sim \log(1/\gamma) \ . \tag{3.43}$$

We note that bi-partite entanglement entropy, between a system and its environment, is generally bounded strictly by the system size $n = s/k$. However, noisy quantities are polynomials of l-error factors $p_{s\gamma}(l) \sim \mathcal{O}(s^l\gamma^l)$. The additional degree of freedom representing the strength of system-environment interactions γ, appears to suppress the higher-order polynomial factors.

Universal behaviours in the divergent regime can be attributed to entropy-increasing phenomena, once the parameterized channel has rotated the state to within a depth-dependent distance from the target state. However, at large noise scales, there are important distinctions between unital versus non-unital types of noise.

For unital noise, such as dephasing noise in Fig. 3.3, a potential explanation is that the combination of the potentially close to Haar random parameterized unitaries, and the accumulated noise, induces depolarization. Entropy is shown to increase linearly with depth, at practically all noise scales, and dominates the infidelity behaviours.

For non-unital noise, such as amplitude damping noise in Fig. 3.5, at small noise scales, the behaviour appears qualitatively similar to unital noise, of linearly increasing quantities with noise scale. However, non-unital noise appears to have fundamentally different, non-universal behaviour at large noise scales. Non-unital noise forces the state into a specific (pure) state, which appears to improve the infidelities. Unlike in the overparameterized regime however, infidelities appear to decrease polynomially with depth. Adequately parameterized unitaries at depths beyond the typical overparameterization bound appear necessary to rotate some components of this forced state towards the target state.

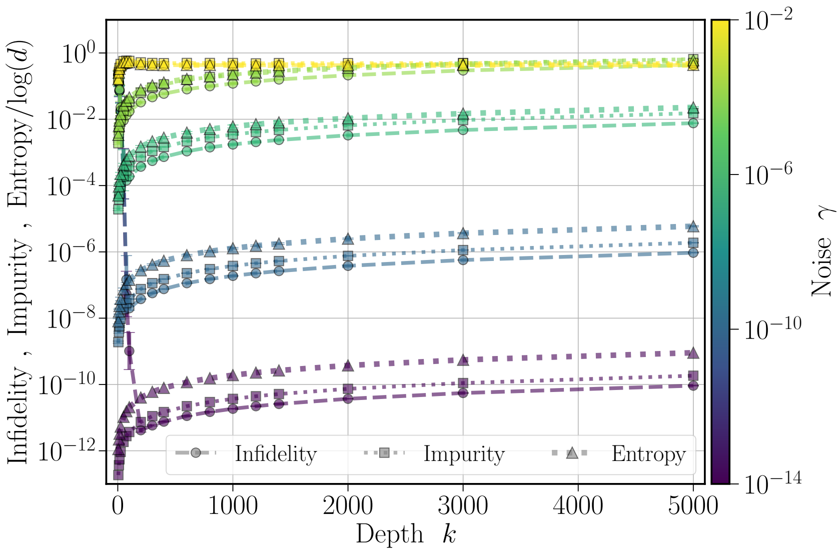


Figure 3.5: **Noisy infidelity, entropy, and impurity**. Behaviour of infidelity, entropy, and impurity with respect to depth k and non-unital amplitude damping noise γ (colours), for the $n = 4$ NMR ansatz. At small noise scales, quantities scale identically linearly with depth and with noise in the divergent, entropic driven regime. At large noise scales, unlike unital noise, non-unital noise infidelities decrease polynomially with depth, once the parameterized unitary aligns the state towards the target pure state.

Ultimately, optimization routines in a noisy setting are shown to be capable of rotating the pure components of states towards target pure states, and they exhibit overparameterization phenomena. Once objectives approach a noise-induced entropy-dependent scale, entropic effects then dominate objective behaviour with increasing depth. Finally, fundamental differences between unital and non-unital noise at large depths and large noise scales are relevant when considering NISQ applications. Through our classical simulation and analytical treatments, overparameterization phenomena for quantum systems are shown to be robust under realistic settings. Infidelities decrease exponentially with depth in the convergent regime, before increasing polynomially with depth and with noise in the divergent regime. These scalings provide essential data for the experimental design of variational quantum algorithms.

When assessing the relevance of these studies, given we simulate specifically NMR systems with depths $k \in [10, 5000]$, and noise scales $\gamma \in [10^{-18}, 10^{-2}]$, we must assess the depth and noise scales of other implementations, such as trapped ions, superconducting qubits, and neutral atoms in Table 3.E.2.

Given our derived logarithmic dependence with noise scales of the noise-induced critical depth, we conjecture that NMR systems' robustness deteriorates at depths $k_\gamma \sim \mathcal{O}(200)$, and noise scales of $\gamma_k < \mathcal{O}(10^{-3})$ if we require infidelities $\mathcal{L} < \mathcal{O}(10^{-4})$. Although the full span of considered depths and noise scales exceeds currently experimentally feasible regimes for NISQ implementations, the derived limits where robustness deteriorates are still approaching currently feasible scales. Other implementations are expected to show identical phenomena, with some ansatz-dependent shift in these depth and noise scales.

It should also be noted that non-unital noise appears to allow for re-improved infidelities in the divergent regime at exceptionally large depths of $k \sim \mathcal{O}(5000)$ and large noise $\gamma \sim \mathcal{O}(10^{-2})$. There may be intriguing non-trivial, noise-type induced emergent phenomena at these large depths, even if these regimes are currently impractical experimentally. The conclusions drawn, therefore, support the necessity of quantum error correction, and challenge aspirations [24, 75] of existing NISQ applications being scaled to thousands of qubits and gates.

Finally, we remark that entropic effects appear to dominate only beyond a critical number of errors in the system. General parameterized systems are thus shown to be capable of suppressing entropic behaviour imposed by their environment. By locating the noise-induced critical depth, problems can be optimized to their best-case objectives across all depths, for example in coveted quantum control problems [80]. This opens up intriguing applications [293] for variational ansatze, both classical and quantum, and we are excited about their potential.

Chapter 4

Moments of Random Quantum Channels

Given the empirical effects of noise in variational settings, in this chapter, based on the preprint manuscript, M. Duschenes, D. García-Martín, Z. Holmes, and M. Cerezo, arXiv:2511.12700 (2025) [3], we are motivated to understand noise-induced phenomena more rigorously, and develop analytical tools to calculate exact statistics of systems undergoing general, random non-unitary dynamics. In particular, moments of ensembles of unitaries have played a central role in quantum information theory as they capture the statistical properties of dynamics of systems with some form of randomness. Indeed, concepts such as approximate t-designs arise when comparing how close an associated moment operator of a given unitary ensemble is to that of another, reference ensemble. Despite the importance of moment operators, their properties have not been as explored for quantum channels. Here, we develop a theoretical framework to compute moment operators for ensembles of quantum channels, for all moment orders t, with a special focus on determining ensembles that can be used as points of reference. By deriving hierarchies between ensembles, via inequalities of their moment operator norms, we give them operational meaning, and define useful concepts such as that of channel t-designs. Finally, we show that different types of noise can decrease the norm of the moment operators, as well as increase it, and generalize noise-induced concentration phenomena to channel-design-induced phenomena. Along the way, we find a block-orthogonal basis for permutations, which greatly simplifies our analyses, and may be of independent interest. Ultimately, such quantum channel statistics are shown to exhibit rich, distinct behaviour from unitary statistics, and suggest several applications in randomized benchmarking experiments, error mitigation, and connections between quantum information sciences and representation theory.

4.1 Introduction

Quantum experiments, whether intrinsically or by design, often involve randomness. Such randomness occurs often in quantum information science, including within quantum-supremacy experiments [84, 125, 134, 212, 294, 295], quantum chaos [296–299], variational quantum algorithms [76, 95, 106, 232, 276, 300–304], concentration phenomena [76, 95, 106, 276, 300, 301], randomized benchmarking [86, 305–309], and error mitigation [310, 311]. The statistical properties of such experiments are described by so-called moment operators, superoperators given by the ensemble-averaged action of operators from an ensemble. Moment operators can be though of as Markov transfer matrices [113, 114, 121–123, 139, 226, 232, 247, 297, 298, 312–315], and whose operational meaning will be elucidated as we study its properties. Ultimately, by focusing on the moment operator, we can study ensembles independently of initial states and final measurements.

Given an ensemble, it is often useful to quantify its similarity to a reference ensemble, using a distance measure between moment operators. This problem has been extensively studied for ensembles of unitaries, where an ensemble forms a t-design with respect to a reference if their moment operators are equal up to the t-order [113, 114, 116, 139, 247, 314–323]. Computing and comparing moment operators are often intractable tasks. However, such tasks simplify if the reference ensemble is a group, as its moment operators are projectors onto the group's commutant, namely all operators that commute with the group's operators, and its properties can be derived using Weingarten calculus [118, 244, 248]. Of particular relevance is the ensemble of the unitary group with a Haar measure. Not only is this a natural reference ensemble, it also has desirable extremal properties: there is a hierarchy between unitary ensembles arising from inequalities between moment-operator norms, with the Haar ensemble at the base of the hierarchy [324].

These previous studies, however, have been almost entirely restricted to ensembles of strictly unitaries. In contrast, ensembles of completely-positive trace-preserving maps, or quantum channels, have been largely unexplored [119, 127, 128, 132, 325, 326]. For instance, it is still unknown whether hierarchies and desirable properties for reference ensembles of more general operators can be found. Ultimately, exact forms of moment operators are necessary to determine the most appropriate reference ensemble. Indeed, there are many important open questions concerning ensembles that must be addressed, such as: *What is the most natural generalization of the uniform (Haar) distribution of unitaries to channels?*, or, *Is there a reference ensemble of channels that plays a privileged role over other ensembles?*

In this chapter, we contribute to answering the previous questions by presenting a framework to study moments of ensembles of quantum channels. In Section 4.2, we introduce our formalism and notation, which include as a special case, ensembles of strictly unitaries. In Section 4.3, we start by generalizing the canonical reference ensemble of the Haar measure over the unitary group, which corresponds to sampling uniformly from the set of all unitaries, and we consider the ensemble of the Lebesgue measure [326] over quantum channels, which corresponds to sampling uniformly from the set of all quantum channels. This study prompted us to consider a parametrized family of reference channel ensembles [127] which we dub the "channel Haar" (cHaar) ensemble. The cHaar ensemble is shown to interpolate between two reference ensembles of interest, the Haar reference ensemble of strictly unitaries, and what we dub as the Depolarize reference ensemble, where the maximally depolarizing channel is applied with unit probability.

By deriving the exact forms and spectral properties for the associated moment operators, we compute deviations of the cHaar ensemble from the Haar and Depolarize ensembles. Such calculations indicate that random quantum channels are on average, inherently depolarizing, with perturbative non-unital behaviour, and interpolate with environment dimension between unitary and depolarizing behaviour. Finally, given these insights, we deduce a hierarchy between these ensembles, with the Depolarize ensemble found to be at the base of the hierarchy. Thus, we argue that the Depolarize ensemble is the most appropriate reference ensemble for quantum channels.

Expanding on these studies, we explicitly calculate the extent to which quantum circuits consisting of random unitaries, and fixed unital or non-unital noise, form approximate Depolarize t-designs. Here, we observe that certain types of noise can decrease the norm of the moment operator, or can increase it. Finally, we study concentration of expectation values, and find that what has previously been attributed to be noise-induced concentration phenomena [276], can be more generally thought of as channel-design-induced concentration phenomena. Finally, in Section 4.4, we interpret the derived average behaviours of random quantum channels, and discuss several future applications of our formalisms.

4.2 Preliminaries

In this section we present the notation and formalisms used throughout these studies, with additional details in Appendix 4.A.

4.2.1 Moments of Ensembles of Channels

Here, we denote d-dimensional quantum Hilbert spaces as $\mathcal{H} = \mathbb{C}^d$, the space of linear operators acting on $\mathcal{H}$ as $\rho, \Pi \in \mathcal{B}[\mathcal{H}]$, the space of linear maps, acting on $\mathcal{B}[\mathcal{H}]$ as $\Lambda \in \mathcal{B}[\mathcal{B}[\mathcal{H}]]$, and sets of t indices as $[t] = \{0, 1, \ldots, t-1\}$. Here, we also make use of normalized Hilbert-Schmidt inner products between $X, Y \in \mathcal{B}[\mathcal{H}]$, in terms of vectorizations,

$$\chi_d(X, Y) = \frac{\langle\langle X|Y\rangle\rangle}{\langle\langle I_d|I_d\rangle\rangle} \quad , \quad \chi_d(X) = \chi_d(I_d, X) \ , \tag{4.1}$$

given the d-dimensional identity operator I_d.

Let us also make use of an orthogonal basis for $\mathcal{B}[\mathcal{H}]$,

$$\mathcal{P}_d = \{P, S \in \mathcal{B}[\mathcal{H}] \ : \ \chi_d(P, S) \propto \delta_{PS}\} \ , \tag{4.2}$$

with $I_d \in \mathcal{P}_d$, and an extension to t copies for $\mathcal{B}[\mathcal{H}^{\otimes t}]$ of,

$$\mathcal{P}_d^t = \mathcal{P}_d^{\otimes t} = \{P = \otimes_{i \in \Gamma_P} P_i \ : \ P_i \in \mathcal{P}_d\} \tag{4.3}$$

with operators $P \in \mathcal{P}_d^t$ of support $\Gamma_P \subseteq [t]$ and locality $|P| = |\Gamma_P|$.

Here, we will analyse ensembles of quantum channels $\mathcal{C} \subseteq \mathcal{B}[\mathcal{B}[\mathcal{H}]]$, namely sets of channels $\{\Lambda \in \mathcal{B}[\mathcal{B}[\mathcal{H}]]\}$ accompanied with a probability distribution $d\Lambda$, and ensemble averages of channel-dependent quantities $\mathcal{F}_\Lambda$,

$$\langle \mathcal{F}_\Lambda \rangle_{\mathcal{C}} = \int_{\mathcal{C}} d\Lambda \ \mathcal{F}_\Lambda \ . \tag{4.4}$$

Note that here we have assumed that the ensemble is continuous, leading to an integral over $\mathcal{C}$. The discrete case follows by simply replacing the integral by a summation.

Here, we will consider expectation values of an initial state ρ evolved by a quantum channel Λ, sampled from $\mathcal{C}$, with respect to an operator Π, with t-order moments of,

$$\left\langle \mathrm{tr}(\Lambda(\rho)\Pi)^t \right\rangle_{\mathcal{C}} = \mathrm{tr}\Big(\mathcal{T}_{\mathcal{C}}^{(t)}(\rho^{\otimes t})\Pi^{\otimes t}\Big) \ . \tag{4.5}$$

Such t-order moments can be expressed in terms of t-order twirl operator $\mathcal{T}_{\mathcal{C}}^{(t)} : \mathcal{B}[\mathcal{H}^{\otimes t}] \to \mathcal{B}[\mathcal{H}^{\otimes t}]$, with associated moment operator $\widehat{\mathcal{T}}_{\mathcal{C}}^{(t)}$, whose action on operators $X \in \mathcal{B}[\mathcal{H}^{\otimes t}]$ is,

$$\mathcal{T}_{\mathcal{C}}^{(t)}(X) = \left\langle \Lambda^{\otimes t}(X) \right\rangle_{\mathcal{C}} = \int_{\mathcal{C}} d\Lambda \ \Lambda^{\otimes t}(X) \qquad \to \qquad \widehat{\mathcal{T}}_{\mathcal{C}}^{(t)} = \int_{\mathcal{C}} d\Lambda \ \widehat{\Lambda^{\otimes t}} \ , \tag{4.6}$$

such that expectation values take the form,

$$\left\langle \mathrm{tr}(\Lambda(\rho))^t \right\rangle_{\mathcal{C}} = \langle\!\langle \Pi^{\otimes t} | \widehat{\mathcal{T}}_{\mathcal{C}}^{(t)} | \rho^{\otimes t} \rangle\!\rangle \ . \tag{4.7}$$

Moment operators are central in our formalisms, as they allow us to study the statistical properties of ensembles independently from the initial state and measurement operator. Moreover, we can use them to study experiments involving k concatenations of several random channels Λ, each sampled identically and independently from $\mathcal{C}$ according to $d\Lambda$. In this case, we obtain expectation values $\langle\!\langle \Pi^{\otimes t} | \widehat{\mathcal{T}}_{\mathcal{C}}^{(t)k} | \rho^{\otimes t} \rangle\!\rangle$, with the moment operator replaced by its k-th power,

$$\widehat{\mathcal{T}}_{\mathcal{C}}^{(t)} \rightarrow \widehat{\mathcal{T}}_{\mathcal{C}}^{(t)k} \ , \tag{4.8}$$

and consider ensemble statistics with respect to d, t, k.

Here, we also find it convenient to express moment operators as,

$$\widehat{\mathcal{T}}_{\mathcal{C}}^{(t)} = \widehat{\mathcal{D}}_d^{\otimes t} \ + \ \widehat{\Delta}_{\mathcal{C}}^{(t)} \ , \tag{4.9}$$

where we have defined an explicit trace-preserving term, and extraneous terms mapping to non-identity operators,

$$\widehat{\mathcal{D}}_d^{\otimes t} = \frac{1}{d^t} | I_d^{\otimes t} \rangle\!\rangle\langle\!\langle I_d^{\otimes t} | \quad , \quad \widehat{\Delta}_{\mathcal{C}}^{(t)} = \frac{1}{d^t} \sum_{\substack{P \in \mathcal{S}_{\mathcal{C}}^{(t)} \setminus \{ I_d^{\otimes t} \} \\ S \in \mathcal{S}_{\mathcal{C}}^{(t)}}} \tau_{\mathcal{C}}^{(t)}(P,S) \ | P \rangle\!\rangle\langle\!\langle S | \ , \tag{4.10}$$

in terms of an ensemble-dependent basis $P, S \in \mathcal{S}_{\mathcal{C}}^{(t)}$ and transfer matrix elements, $\tau_{\mathcal{C}}^{(t)}(P,S)$.

Analytically evaluating the moment operator can lead to intractable calculations. In these cases, one can instead opt to study how ϵ-close in norm the moment operator of $\mathcal{C}$ is to that of a reference ensemble $\mathcal{C}'$. This allows us to introduce the notion of an additive ϵ-approximate t-channel design, with respect to $\mathcal{C}'$,

$$\left\| \widehat{\mathcal{T}}_{\mathcal{C}}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{C}'}^{(t)} \right\| \leq \epsilon \ . \tag{4.11}$$

Here we consider ϵ to be additive error, but our definitions can be adapted to relative error. Importantly, often the evaluation of norms of moment operator differences leads to analysis of the moment operator norm $\| \widehat{\mathcal{T}}_{\mathcal{C}}^{(t)} \|$.

The choice of norm $\|\cdot\|$ is task-dependent, and we generally use the Hilbert-Schmidt or 2 norm for its simplicity, given we derive that norms of differences of moment operators bound the diamond norm of differences of twirls, from the operator-state correspondence,

$$\left\| \mathcal{T}_{\mathcal{C}}^{(t)} - \mathcal{T}_{\mathcal{C}'}^{(t)} \right\|_{\diamond} \;\leq\; d^t \left\| \widehat{\mathcal{T}}_{\mathcal{C}}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{C}'}^{(t)} \right\|_{2} \;\leq\; d^t \left\| \widehat{\mathcal{T}}_{\mathcal{C}}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{C}'}^{(t)} \right\|_{1}. \tag{4.12}$$

While deriving relationships between moment operator norms for channel ensembles, let us recall the established relationships between moment operator norms for unitary ensembles $\mathcal{U}$ due their invariance properties. Such relationships allow us to derive a partial ordering between moment operator norms for unitary subgroups with Haar measures, namely $\mathcal{U}, \mathcal{U}', \ldots$, such that $\{U\} \subseteq \mathcal{U} \subseteq \mathcal{U}' \subseteq \cdots \subseteq \mathcal{U}(d)$, leading to the hierarchy,

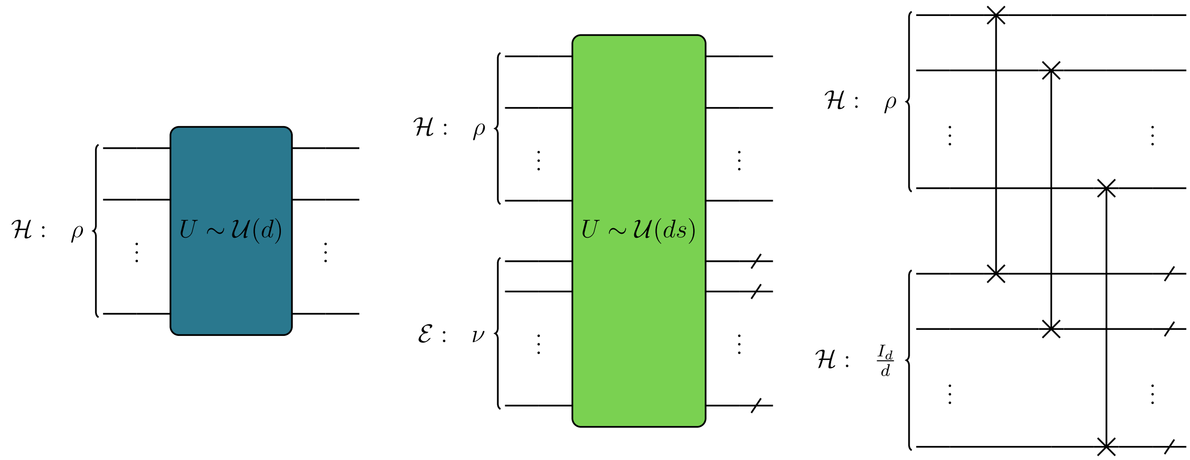
$$\left\| \widehat{\mathcal{T}}_{\mathcal{U}(d)}^{(t)} \right\|^2 \;\leq\; \cdots \;\leq\; \left\| \widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)} \right\|^2 \;\leq\; \left\| \widehat{\mathcal{T}}_{\mathcal{U}}^{(t)} \right\|^2 \;\leq\; \left\| \widehat{\mathcal{T}}_{U}^{(t)} \right\|^2 . \tag{4.13}$$

This hierarchy implies that the more unstructured a group $\mathcal{U}$ is, relative to the group of totally unstructured unitaries $\mathcal{U}(d)$, the smaller the norm of its moment operator. Thus, the Haar ensemble $\mathcal{U}(d)$, at the base of such a hierarchy, serves as an appropriate reference ensemble for other subgroups of unitaries. Moreover, we note that such norm inequalities appear within the context of variational quantum algorithms [37, 81, 83, 327], and as a measure of the expressiveness of parameterized ensembles [114, 132, 231, 234, 300, 315]. In this context, it has been shown that the smaller the norm, the greater the concentration of the resulting variational landscape [95, 106, 233, 234, 328, 329].

However, unlike unitary ensembles $\mathcal{U}$, ensembles of channels $\mathcal{C}$ are rarely invariant, and have less algebraic structure, obfuscating their interpretation, and intuition for which reference ensemble is most appropriate. Channel ensemble moment operator norms and traces still retain an operational meaning, relevant to randomized benchmarking [126], of being the average overlap between Kraus operators from the ensemble,

$$\left\| \widehat{\mathcal{T}}_{\mathcal{C}}^{(t)} \right\|^2 = \int_{\mathcal{C}\times\mathcal{C}} d\Lambda \; d\Upsilon \sum_{\Gamma_\Lambda, \Gamma_\Upsilon} |\langle\!\langle \Gamma_\Lambda | \Gamma_\Upsilon \rangle\!\rangle|^{2t} \quad , \quad \mathrm{tr}\left(\widehat{\mathcal{T}}_{\mathcal{C}}^{(t)} \right) = \int_{\mathcal{C}} d\Lambda \sum_{\Gamma_\Lambda} |\mathrm{tr}(\Gamma_\Lambda)|^{2t} . \tag{4.14}$$

Such interpretations motivate studying spectral properties of moment operators when choosing reference ensembles. Concurrently, design properties of ensembles of unitaries have been attributed to their expressive power within variational quantum algorithms [106, 233, 234, 330]. However it remains to be seen whether similar notions of capability or utility of ensembles of quantum channels directly follow from their design properties, and any potential resulting hierarchies between ensembles. Other questions regarding operational interpretations will arise within these studies once exact forms for moment operators over specific ensembles of quantum channels are derived.



(a) Haar (b) cHaar (c) Depolarize

Figure 4.1: **Circuits implementing reference ensembles**. (a): The Haar ensemble can be implemented by initializing t-copies of a state, $\rho \in \mathcal{B}[\mathcal{H}]$, and applying a unitary from $\mathcal{U}(d)$. (b): The cHaar ensemble can be implemented via initializing t-copies of a composite system state and environment pure state, $\rho \otimes \nu \in \mathcal{B}[\mathcal{H} \otimes \mathcal{E}]$, applying a unitary from $\mathcal{U}(ds)$, and tracing out the environment. (c): The Depolarize ensemble can be implemented by initializing t-copies of a composite system state and maximally mixed ancillary state, $\rho \otimes I_d/d \in \mathcal{B}[\mathcal{H} \otimes \mathcal{H}]$, swapping each system and ancillary copy, and tracing out the ancilla.

4.3 Results

In this section, we present our main results, with additional details in the appendices. Specifically, we define reference ensembles of quantum channels, derive their relationships and properties (Appendix 4.A and Appendix 4.B), calculate moments of specific ensembles of quantum channels (Appendix 4.C), and investigate concentration phenomena arising in the context of noisy variational quantum algorithms (Appendix 4.D).

4.3.1 Reference Ensembles

Here, it is convenient to introduce reference ensembles to define concepts such as approximate t-designs. Here, we consider three candidate reference ensembles for quantum channels, derive their properties, and determine whether a hierarchy exists between these ensembles that suggests the most appropriate reference ensemble.

For our first candidate reference ensemble, we take the Haar ensemble of unitaries, given its place at the base of the unitary ensemble hierarchy. We will denote it as the *Haar* ensemble $\mathcal{U}(d)$, with a t-order twirl of,

$$\mathcal{T}^{(t)}_{\mathcal{U}(d)}(X) = \int_{\mathcal{U}(d)} dU \; U^{\otimes t} \; X \; U^{\otimes t \dagger} \; , \tag{4.15}$$

and an associated moment operator of,

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)} = \int_{\mathcal{U}(d)} dU \; U^{\otimes t} \otimes U^{\otimes t \, *} \; . \tag{4.16}$$

For our second candidate reference ensemble, we recall the definition of the Stinespring dilation [13], which states that any quantum channel $\Lambda : \mathcal{B}[\mathcal{H}] \to \mathcal{B}[\mathcal{H}]$, can expressed as a unitary $U_\Lambda : \mathcal{H} \otimes \mathcal{E} \to \mathcal{H} \otimes \mathcal{E}$ acting on the tensor product of the system Hilbert space $\mathcal{H}$ of dimension d and an environment Hilbert space $\mathcal{E}$ of dimension s, with the environment being in a pure state $\nu = |\nu\rangle\langle\nu| \in \mathcal{B}[\mathcal{E}]$. Thus the action of any quantum channel is

$$\Lambda(X) = \mathrm{tr}_{\mathcal{E}}\Big(U_\Lambda \, (X \otimes \nu) \, U_\Lambda^\dagger \Big) \; , \tag{4.17}$$

where the environment $\mathcal{E}$ is partially traced out. A key advantage of working with the Stinespring dilation is that we can now analyse ensembles of random quantum channels in terms of the random Stinespring unitaries U_Λ that generate such channels. Sampling quantum channels can thus be directly related to sampling unitaries in larger composite spaces. Notably, it has been shown that a consistent measure $d\Lambda$ for quantum channels exists that is equivalent to sampling unitaries U_Λ according to the Haar measure dU over $\mathcal{U}(ds)$ [127]. In fact, when $s = d^2$, this measure $d\Lambda$ is the Lebesgue measure over channels.

Given such Stinespring dilations, we define our second candidate reference ensemble as the environment dimension s-parametrized family of ensembles of channels, which we denote as the channel-Haar or *cHaar* ensemble $\mathcal{C}(d,s)$, with a t-order twirl of,

$$\mathcal{T}^{(t)}_{\mathcal{C}(d,s)}(X) = \int_{\mathcal{U}(ds)} dU \; \mathrm{tr}_{\mathcal{E}^{\otimes t}}\big(U^{\otimes t} (X \otimes \nu^{\otimes t}) U^{\otimes t \dagger} \big) \; , \tag{4.18}$$

and an associated moment operator of,

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)} = \big(I_d^{\otimes 2t} \otimes \langle\!\langle I_s^{\otimes t} | \big) \; \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(ds)} \; \big(I_d^{\otimes 2t} \otimes | \nu^{\otimes t} \rangle\!\rangle \big) \; . \tag{4.19}$$

Finally, we can use expansions for Haar ensemble moment operators in terms of its commutant of permutations $\mathcal{S}_t$ [117], with representations $V_d \,:\, \mathcal{S}_t \to \mathcal{S}_d^{(t)} \subset \mathcal{B}[\mathcal{H}^{\otimes t}]$, and overlap Weingarten matrices $\chi_{ds}(\sigma,\pi) = \langle\!\langle V_{ds}(\sigma)|V_{ds}(\pi)\rangle\!\rangle / \langle\!\langle V_{ds}(e)|V_{ds}(e)\rangle\!\rangle \to \chi_{ds}^{(t)-1}(\sigma,\pi)$ to find,

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)} = \frac{1}{d^t} \sum_{\sigma,\pi\in\mathcal{S}_t} \chi_s^{(t)}(\sigma)\ \chi_{ds}^{(t)-1}(\sigma,\pi)\ |V_d(\sigma)\rangle\!\rangle\langle\!\langle V_d(\pi)|\ . \tag{4.20}$$

Above we have used the fact that under a simple reordering of the Hilbert space [226], $(\mathcal{H}\otimes\mathcal{E})^{\otimes t} \to \mathcal{H}^{\otimes t}\otimes\mathcal{E}^{\otimes t}$, we can express $|V_{ds}(\sigma)\rangle\!\rangle \to |V_d(\sigma)\rangle\!\rangle \otimes |V_s(\sigma)\rangle\!\rangle$, as well as the fact that $\langle\!\langle V_s(\pi)|\nu^{\otimes t}\rangle\!\rangle = 1$, for all permutations $\pi \in \mathcal{S}_t$ and pure states $|\nu\rangle \in \mathcal{E}$.

For our third candidate reference ensemble, we recall the depolarizing channel

$$\mathcal{D}_d(X) = \frac{\mathrm{tr}(X)}{d} I_d\ , \tag{4.21}$$

that has been shown to describe the trace-preservation component in all ensembles, and will be key to our analysis. We thus define our third candidate reference ensemble as the ensemble containing only the depolarizing channel with unit probability, which we denote as the *Depolarize* ensemble $\mathcal{D}(d)$, with a t-order twirl of,

$$\mathcal{T}^{(t)}_{\mathcal{D}(d)}(X) = \mathcal{D}_d^{\otimes t}(X) = \frac{\mathrm{tr}(X)}{d^t} I_d^{\otimes t}\ , \tag{4.22}$$

and an associated moment operator of,

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} = \widehat{\mathcal{D}}_d^{\otimes t} = \frac{1}{d^t} |I_d^{\otimes t}\rangle\!\rangle\langle\!\langle I_d^{\otimes t}|\ . \tag{4.23}$$

Quantum circuits to experimentally implement each of the three candidate reference ensembles are shown in Fig. 4.1.

In the following sections, after deriving explicit moments of each ensemble, we will compare the Haar, cHaar and Depolarize reference ensembles, and deduce their hierarchy.

4.3.2 Properties of Moment Operators

Having defined the Haar, cHaar, and Depolarize ensembles, we now proceed to analyse their moment operators. First, simply from their definitions, we can immediately prove the following properties regarding their relationships:

Proposition 1 (Properties of Reference Ensembles)**.** *Let* $\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)}$, $\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)}$ *and* $\widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}$ *respectively denote the moment operators for the Haar, cHaar and Depolarize ensembles. Then, for* $t=1$ *and any* s*, the ensembles are identical,*

$$\widehat{\mathcal{T}}^{(1)}_{\mathcal{U}(d)} = \widehat{\mathcal{T}}^{(1)}_{\mathcal{C}(d,s)} = \widehat{\mathcal{T}}^{(1)}_{\mathcal{D}(d)} \ , \tag{4.24}$$

with such equalities not holding for $t>1$*. While* $\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)}$ *and* $\widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}$ *are self-adjoint projectors,*

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)}\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)} \quad , \quad \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}\widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} \ , \tag{4.25}$$

$\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)}$ *is not a self-adjoint projector for* $t>1$ *and* $s>1$*,*

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)}\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)} \neq \widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)} \quad , \quad \widehat{\mathcal{T}}^{(t)\dagger}_{\mathcal{C}(d,s)} \neq \widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)} \ . \tag{4.26}$$

Moreover, the cHaar ensemble is invariant under unitary ensembles $\mathcal{U}$*,*

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)}\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}}\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)} \ . \tag{4.27}$$

The Depolarize ensemble is right-invariant under any ensemble of channels $\mathcal{C}$ *and is left-invariant under the ensemble of their adjoints,*

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} \quad , \quad \widehat{\mathcal{T}}^{(t)\dagger}_{\mathcal{C}}\widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} \ , \tag{4.28}$$

and is left-invariant under unital ensembles of channels $\mathcal{C}_{\mathcal{U}}$*,*

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}_{\mathcal{U}}}\widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} \ . \tag{4.29}$$

Therefore Hilbert-Schmidt norms of differences of moment operators relative to $\mathcal{D}(d)$ *are,*

$$\left\|\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}\right\|^2 = \left\|\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}}\right\|^2 - \left\|\widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}\right\|^2 \ . \tag{4.30}$$

These relationships between ensembles already suggest that the Depolarize ensemble, although not corresponding to uniformly sampling all quantum channels, is potentially an appropriate reference ensemble. The relevance of this Depolarize ensemble follows partially from its right, and often left invariance under concatenation with other ensembles. It is also important to note that any moment operator of any trace-preserving ensemble involves a Depolarize moment operator term (which is reminiscent of how the moment operator for any unitary ensembles contain the term $\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)}$). In contrast, the cHaar ensemble is found to not be left- or right- invariant under concatenations with other ensembles.

It is perhaps surprising that the cHaar ensemble, given its lack of invariance properties, does not appear to be a natural generalization of the Haar ensemble, and does not immediately appear to be the most appropriate reference ensemble for quantum channels. After all, the Depolarize ensemble only contains a single channel! To fully support this proposal, we must derive exact properties of the cHaar ensemble and deduce its place in the hierarchy of ensembles, as a function of environment dimension s, order t, and concatenations k.

Theorem 1 (Hierarchy of moment operator norms)**.** *The following dimension-dependent relationships between the Depolarize, cHaar and Haar ensembles hold:*

1) The cHaar ensemble, and its k-concatenations, interpolates with system and environment dimension between the Haar and Depolarize ensembles. That is, for all $k \geq 0$:

$$\lim_{ds\to d} \widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)} \quad , \quad \lim_{ds\to\infty} \widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} \ . \tag{4.31}$$

In fact, the cHaar ensemble moment operator has an exact closed form, consisting of trace-preserving, non-unital, and unital terms, expressed for simplicity, in terms of some t, k-dependent traceless operators $T_d^{(t,k)}, T_d^{(t,k)\prime} \in \mathcal{B}[\mathcal{H}^{\otimes t}]$, with asymptotic scalings of transfer matrix elements $\tau^{(t)}_{\mathcal{C}(d,s)}$ in the asymptotic dimension $d, s \to \infty$ limit,

$$\begin{aligned} \lim_{ds\to\infty} \widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)} = & \ \frac{1}{d^t} |I_d^{\otimes t}\rangle\!\rangle\langle\!\langle I_d^{\otimes t}| \\ & + \ \frac{1}{d^t}\mathcal{O}\left(\frac{1}{ds}\right) \ |T_d^{(t,k)}\rangle\!\rangle\langle\!\langle I_d^{\otimes t}| \\ & + \ \frac{1}{d^t}\mathcal{O}\left(\frac{1}{d^2 s^k}\right) |T_d^{(t,k)}\rangle\!\rangle\langle\!\langle T_d^{(t,k)\prime}| \ . \end{aligned} \tag{4.32}$$

2) The cHaar ensemble forms an ϵ-approximate Depolarize t-design, for $\epsilon \in \mathcal{O}\big(\frac{1}{s}\big)$, in the fixed t and the asymptotic $ds \to \infty$ limit.

3) In the asymptotic $ds \to \infty$ limit, there exists a strict hierarchy between the k-concatenated t-order ensembles, via their moment operator norms, with $k \geq k'$,

$$\left\|\widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}\right\|^2 \ \leq \ \left\|\widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)}\right\|^2 \ \leq \ \left\|\widehat{\mathcal{T}}^{(t)k'}_{\mathcal{C}(d,s)}\right\|^2 \ \leq \ \left\|\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)}\right\|^2 \ . \tag{4.33}$$

As per part 1) and 2) of Theorem 1, in the appropriate limits, the cHaar ensemble reduces to either the Depolarize ensemble or the Haar ensemble, and can be thought as interpolating with dimension between these ensembles.

For large enough systems and environments such that $ds \to \infty$ (for example if s is a function of d, such as the maximal $s = d^2$ necessary to implement any quantum channel) the cHaar ensemble approaches the Depolarize ensemble. Moreover, even with finite-sized systems and environments, the perturbative next-leading-order non-unital contributions to the moment operator vanish inversely with system and environment dimensions, and are intriguingly independent of concatenations. The exact forms for cHaar moment operators generalize previous derivations of random quantum channels being asymptotically depolarizing [132], and offer a complete depiction of their average behaviour.

For trivial environments such that $s \to 1$, the cHaar ensemble reduces to the Haar ensemble. The hierarchy between ensembles in part 3) of Theorem 1 then implies that concatenating cHaar ensembles decreases their norm, tending towards the base of the hierarchy set by the Depolarize ensemble. Thus, as a consequence of the dissipative nature of tracing out the environment, whereas cHaar ensembles with a finite environment are not maximally dissipative, their behaviour can be made more so via concatenation. (We note that while the hierarchy between ensembles in Theorem 1 holds asymptotically when $ds \to \infty$, we have explicitly verified it for $t = 2, 3, 4$, $k = 1, 3$, and various d, s in Fig. 4.B.5 of Appendix 4.B, and we conjecture that it holds for all t, k, and d, s.)

Theorem 1 is derived explicitly in Appendix 4.B by studying the spectrum of the cHaar moment operator. Namely, since the cHaar moment operator fails to be a projector, unlike the Haar and Depolarize moment operators, its eigenvalues indicate the inherent average dissipation of information of random quantum channels. This realization, as well as the derived concatenation-independent behaviour of its next-leading-order non-unital contributions, prompted us to study the spectral properties of $\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)}$.

Theorem 2 (Spectrum of cHaar Moment Operator)**.** *The t-order cHaar moment operator has the exact leading term of a projector,*

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)} \;=\; \frac{|\psi\rangle\!\rangle\langle\!\langle\varphi|}{\langle\!\langle\varphi|\psi\rangle\!\rangle} \;+\; \cdots \;, \tag{4.34}$$

which implies that it has a leading eigenvalue $\lambda = 1$ with right- and left-eigenvectors,

$$|\psi\rangle\!\rangle = \sum_{\sigma\in\mathcal{S}_t} \chi^{(t)}_{ds}(\sigma)\ |V_d(\sigma)\rangle\!\rangle \quad , \quad |\varphi\rangle\!\rangle = |V_d(e)\rangle\!\rangle \;, \tag{4.35}$$

and norms and traces, up to t, k-dependent constants, of,

$$\left\|\widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)}\right\|^2 \;\leq\; 1 \;+\; \frac{1}{s^2}\; C^{(t,k)}_{\|\tau\|} \qquad , \qquad \mathrm{tr}\Big(\widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)}\Big) \;\leq\; 1 \;+\; \frac{1}{s^k}\; C^{(t,k)}_{\mathrm{Tr}[\tau]} \;. \tag{4.36}$$

In the fixed t, asymptotic $ds \to \infty$ limits, the t-order cHaar moment operator has a single leading eigenvalue $\lambda = 1$, and all non-leading eigenvalues are $\lambda \in \mathcal{O}\big(\frac{1}{s}\big) < 1$.

We note that a direct consequence of the previous theorem is that k-concatenations of the cHaar ensemble will have moment operators with the fixed points,

$$\widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)} = \frac{|\psi\rangle\!\rangle\langle\!\langle\varphi|}{\langle\!\langle\varphi|\psi\rangle\!\rangle} + \cdots \ . \tag{4.37}$$

Moreover, Theorem 2 implies that the cHaar ensemble moment operator's only fixed point is its eigenvector $|\psi\rangle\!\rangle$. In turn, this leading eigenvector converges with dimension to the Depolarize channel's single fixed point

$$\lim_{ds\to\infty} |\psi\rangle\!\rangle, |\varphi\rangle\!\rangle \ \to \ |I_d^{\otimes t}\rangle\!\rangle, |I_d^{\otimes t}\rangle\!\rangle \qquad \leftrightarrow \qquad \lim_{ds\to\infty} \frac{|\psi\rangle\!\rangle\langle\!\langle\varphi|}{\langle\!\langle\varphi|\psi\rangle\!\rangle} \ \to \ \widehat{\mathcal{D}}_d^{\otimes t} \ . \tag{4.38}$$

This observation regarding the eigenvectors and the single, depolarization fixed point of the cHaar ensemble further underlies the hierarchy between ensembles in Theorem 1.

These theorems have several important implications, that culminate in the Depolarize ensemble being found to be a natural reference ensemble for ensembles of quantum channels.

Claim 1. *The Depolarize ensemble $\mathcal{D}(d)$ is a natural generalization of the Haar ensemble. The invariance properties of $\mathcal{D}(d)$ impose that its moment operator is present in the moment operators of all quantum channel ensembles, and place the Depolarize ensemble at the bottom of the moment operator norm hierarchy of all quantum channel ensembles. Therefore, any ensemble of channels $\mathcal{C}$ satisfies*

$$\left\|\widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}\right\|^2 \leq \left\|\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}}\right\|^2 \ . \tag{4.39}$$

Claim 2. *The Depolarize ensemble therefore serves as a natural reference ensemble for quantum channels for defining t-designs. Concretely, an ensemble $\mathcal{C}$ is defined to be an ϵ-approximate Depolarize t-design, if and only if*

$$\left\|\mathcal{T}^{(t)}_{\mathcal{C}} - \mathcal{T}^{(t)}_{\mathcal{D}(d)}\right\|_\diamond \leq \epsilon \ . \tag{4.40}$$

It admittedly remains surprising that from our criteria of positions within hierarchies of moment operator norms, the most appropriate reference ensemble for quantum channels is found to be the Depolarize ensemble and not the cHaar ensemble. To explain this outcome, we recall that only when $s = d^2$, does the cHaar ensemble have a uniform Lebesgue measure over quantum channels, and only when $s = d$, is there a resulting uniform cHaar distribution of quantum states [127]. However, even at these dimensions, the cHaar ensemble still does not share the defining characteristics of the Haar or Depolarize ensembles that make them the most appropriate reference ensembles, namely invariance under concatenations. In fact, cHaar ensembles for sufficiently large environment dimension s, with non-leading eigenvalues $|\lambda| < 1$, will dissipate information upon concatenation. It is therefore proposed that criteria of ensemble hierarchies, as opposed to (the admittedly desirable property of) uniformity of measure, are used when considering reference ensembles.

A possible geometric explanation for the Depolarize ensemble being at the base of the norm hierarchy for quantum channels is in terms of its distribution of Kraus operators. In particular, the norm of the t-order moment operator can be interpreted as the t-order moment of the overlaps between its Kraus operators, or frame potentials [234],

$$\left\| \widehat{\mathcal{T}}_{\mathcal{C}}^{(t)} \right\|^2 = \int_{\mathcal{C}\times\mathcal{C}} d\Lambda \; d\Upsilon \sum_{\Gamma_\Lambda, \Gamma_\Upsilon} |\langle\!\langle \Gamma_\Upsilon | \Gamma_\Lambda \rangle\!\rangle|^{2t} \;, \tag{4.41}$$

For unitaries, the Haar reference ensemble minimizes the frame potentials, and hence maximizes the spread of the corresponding (unitary) Kraus operators. For quantum channels, it is precisely the Depolarize ensemble that maximizes the spread of the Kraus operators.

Our derivations of the properties of the cHaar moment operator are made possible via an analysis of the transfer matrices $\tau_{\mathcal{C}}^{(t)}$ of the Haar and cHaar moment operator in the permutation basis, namely the $t! \times t!$ matrices that map permutations to permutations. In Appendices 4.A and 4.B we derive a crucial change of basis, from non-orthogonal permutations [141], to so-called localized permutations. This novel localized basis consists of operators, each with support over a strict subset of copies of the Hilbert space, that are orthogonal to any operators with non-identical support. Due to its well-defined support and block-orthogonality, this basis block diagonalizes the Haar transfer matrix and block lower-triangularizes the cHaar transfer matrix.

We note that this $t! \times t!$-dimensional change of basis is not the $d^t \times d^t$-dimensional Schur transformation [331]. This localized basis is not soeley a technical convenience, but the main tool that makes our results accessible, and we expect it to be useful in other permutation-based calculations.

The resulting transfer matrix elements are depicted in Fig. 4.2, for $t = 4$ and $k = 1, 3$ concatenations, obtained from closed-form expressions and symbolic methods [332]. Shown is the leading order leading-order scaling l of matrix elements with d. The Haar ensemble transfer matrix is block-diagonal, with elements that are invariant under concatenations. Such structure reflects that its moment operator is a projector, and is support-preserving. The cHaar ensemble transfer matrix is block-lower-triangular, with elements that decay with concatenations, except for the first column which corresponds to the fixed-point leading eigenvector. Such structure reflects that its moment operator is non-unital, and is support-non-decreasing, given it is generated by a support-preserving Haar ensemble, with its output environment traced over. Given the exact forms of the transfer matrices, in particular the k-concatenated transfer matrices $\tau_{\mathcal{C}}^{(t,k)}$, the spectral properties, norms and traces of the associated moment operators may be derived.

In fact, we can derive a general result concerning the support of operators mapped between by moment operators.

Theorem 3 (Support Preservation of Quantum Channel Ensembles Induced by Unitary Ensembles)**.** *Let a quantum channel ensemble $\mathcal{C}$ over a d-dimensional space $\mathcal{H}$ be induced via Stinespring dilation by a unitary ensemble $\mathcal{U}$ with projector moment operators over a composite ds-dimensional space $\mathcal{H} \otimes \mathcal{E}$, with an environment state $\nu_{\mathcal{E}} \in \mathcal{B}[\mathcal{E}]$. Such an ensemble's moment operators are support-preserving for trivial $s = 1$ environments, and support-non-decreasing for non-trivial $s > 1$ environments, given an orthogonal basis $\mathcal{P}_d^t$,*

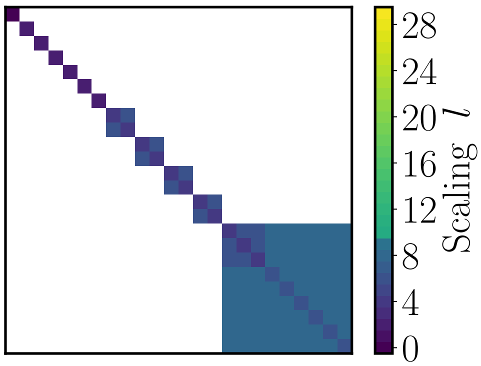
$$\widehat{\mathcal{T}}_{\mathcal{C}}^{(t)} = \frac{1}{d^t} \sum_{\substack{P,S \in \mathcal{P}_d^t \\ \Gamma_P \supseteq \Gamma_S}} \tau_{\mathcal{C}}^{(t)}(P,S)\ |P\rangle\!\rangle\langle\!\langle S| \tag{4.42}$$

$$\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)} = \sum_{\Gamma \subseteq [t]} \widehat{\mathcal{T}}_{\mathcal{U}}^{(\Gamma)} \quad : \quad \widehat{\mathcal{T}}_{\mathcal{U}}^{(\Gamma)} = \frac{1}{d^t} \sum_{\substack{P,S \in \mathcal{P}_d^t \otimes \mathcal{P}_s^t \\ \Gamma_P = \Gamma_S = \Gamma}} \tau_{\mathcal{U}}^{(|S|)}(P,S) |P\rangle\!\rangle\langle\!\langle S| \ , \tag{4.43}$$

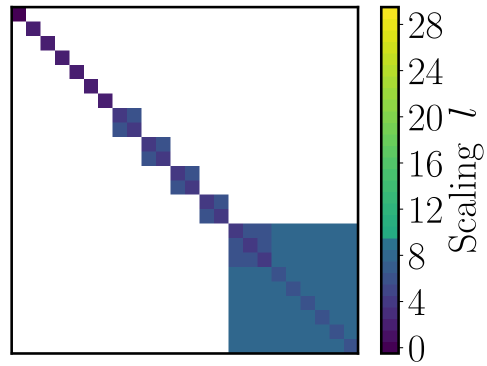
with transfer matrices in terms of the generating ensemble, and the environment state,

$$\tau_{\mathcal{C}}^{(t)}(P,S) = \sum_{\substack{T_{\mathcal{E}} \in \mathcal{P}_s^t \\ \Gamma_{P_{\mathcal{H}}} = \Gamma_{S_{\mathcal{H}}} \cup \Gamma_{T_{\mathcal{E}}}}} \langle\!\langle T_{\mathcal{E}} | \nu_{\mathcal{E}}^{\otimes t} \rangle\!\rangle\ \tau_{\mathcal{U}}^{(t)}(P_{\mathcal{H}} \otimes I_s^{\otimes t}, S_{\mathcal{H}} \otimes T_{\mathcal{E}}) \ . \tag{4.44}$$

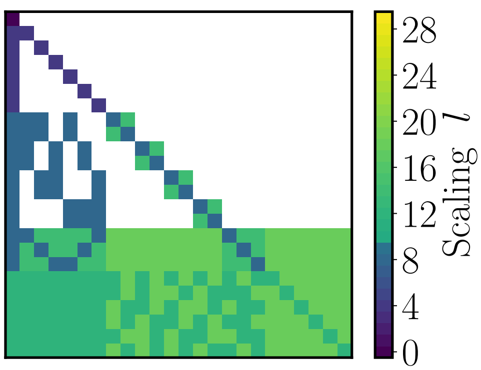
From this general notion of channel ensemble moment operators being support-non-decreasing, in principle, their moment operators can be computed, given sufficient insight into the generating unitary ensemble, and the environment state. In the cHaar ensemble case, given the well-understood generating Haar ensemble commutant of permutations, and our derived localized basis, we can compute its exact, support-non-decreasing transfer matrix. Further, from the invariance of copies of pure states under permutations, the cHaar ensemble is independent of any pure environment state. For quantum channel ensembles induced by more structured unitary ensembles, such as the Clifford group [243], it may prove more difficult to find such an orthogonal basis [115, 333], and such ensembles may depend on the environment state, such as its purity or magic [334].



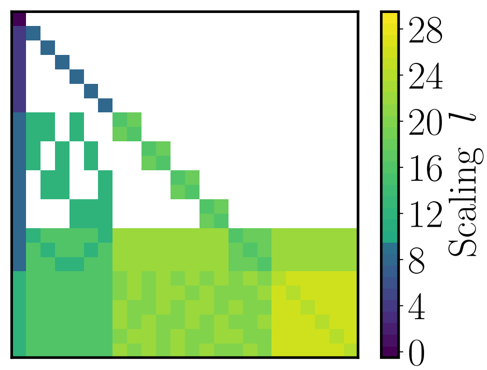
(a) $\widehat{\mathcal{T}}^{(t)k}_{\mathcal{U}(d)}$ for $k = 1$



(b) $\widehat{\mathcal{T}}^{(t)k}_{\mathcal{U}(d)}$ for $k = 3$



(c) $\widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)}$ for $k = 1$



(d) $\widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)}$ for $k = 3$

Figure 4.2: **Scaling of Haar and cHaar transfer matrices**. We show the scaling of the k-concatenated t-order moment operators for the Haar (top) and cHaar (bottom) ensembles for $s = d^2$, $k = 1$ (left) , $k = 3$ (right), and $t = 4$. In this basis, the $t!$ elements are sorted by their support, from the smallest identity (top-left corner), to the largest cycle (bottom-right corner). Colours represent the leading-order scaling l with $1/d$ of elements, such that, $\tau^{(t,k)}_{\mathcal{C}} \sim \mathcal{O}(1/d^l)$.

4.3.3 Moment Operators of Noisy Quantum Circuits

Here, we use our formalisms to study the moment operators of noisy quantum circuits. As per Fig. 4.3, we consider k-layers of random unitaries sampled from a unitary ensemble $\mathcal{U}$, followed by a fixed noise channel $\mathcal{N}$.

The t-order moment operator for this evolution is $(\widehat{\mathcal{N}}^{\otimes t}\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)})^k$. Here, we choose an orthogonal basis for the noise model, $\mathcal{S}_{\mathcal{N}} = \mathcal{P}_d = \{I_d, P, \ldots, S\}$ acting on $\mathcal{H}$, with a generalization to $\mathcal{S}_{\mathcal{N}}^{(t)} = \mathcal{P}_d^t = \{P = \otimes_{i\in\Gamma_P} P_i : P_i \in \mathcal{P}_d, \Gamma_P \subseteq [t]\}$ acting on $\mathcal{H}^{\otimes t}$. We thus consider a constant noise model acting on $\mathcal{H}$ in the basis $P, S \in \mathcal{S}_{\mathcal{N}_{\gamma\eta}} = \mathcal{P}_d$ of,

$$\widehat{\mathcal{N}}_{\gamma\eta} = \frac{1}{d}\begin{bmatrix} 1 & 0 & \cdots & 0 \\ \eta_d(P) & 1-\gamma_d(P) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \eta_d(S) & 0 & \cdots & 1-\gamma_d(S) \end{bmatrix} , \tag{4.45}$$

where $\gamma \leq \gamma_d(P) \leq 1$, $\eta_d(P) \leq \eta \leq 1$, $\forall\ P \in \mathcal{P}_d$, with appropriate constraints on γ_d, η_d such that $\mathcal{N}_{\gamma\eta}$ is completely positive [335]. This noise model is composed of a strictly diagonal part which reduces the magnitude of operators, and a non-unital part that maps identity to non-identity operators, encompasses both unital Pauli-like and non-unital noise [13, 266], and allows us to understand how noise propagates across copies of Hilbert spaces. This model represents effective global noise behaviours of $d = q^n$ qudit quantum circuits, given local qudit noise interlaced with unitaries has exactly diagonal and non-unital components in the case of $q = 2$ qubits [131]. The strictly diagonal and non-unital components of the n-fold tensor products of this local noise reflects leading-order independent unital or non-unital effects, justifying our choice of model. The t-fold superoperator for the noise component, in the basis $P, S \in \mathcal{S}_{\mathcal{N}_{\gamma\eta}}^{(t)} = \mathcal{P}_d^t$, is thus,

$$\widehat{\mathcal{N}}_{\gamma\eta}^{\otimes t} = \frac{1}{d^t} \sum_{P,S\in\mathcal{S}_{\mathcal{N}_{\gamma\eta}}^{(t)}} \tau_{\mathcal{N}_{\gamma\eta}}^{(t)}(P,S)\ |P\rangle\!\rangle\langle\!\langle S| , \tag{4.46}$$

with transfer matrix elements that scale with γ, η as,

$$\tau_{\mathcal{N}_{\gamma\eta}}^{(t)}(P,S) \in \mathcal{O}\big((1-\gamma)^{|S|}\ \eta^{|P|-|S|}\big)\ \delta_{\Gamma_P \supseteq \Gamma_S} . \tag{4.47}$$

Finally, the following theorem holds, on the effect of noise on channel design properties.

Theorem 4 (Noise-Induced Depolarize t-Designs)**.** *For k-concatenated t-order unitary ensembles, diagonal unital noise decreases, and non-unital noise increases moment operator norms. The respective ensembles approach and deviate from being Depolarize t-designs.*

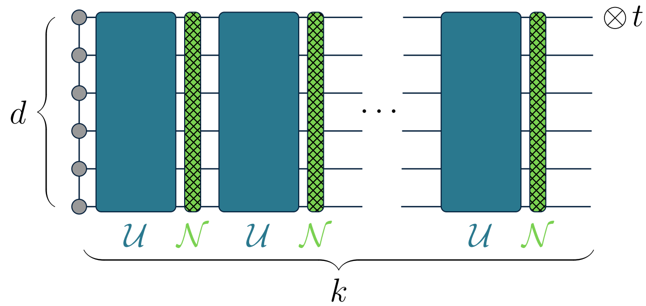


Figure 4.3: **Concatenation of random quantum channels**. t-copies of k-concatenated quantum circuits with random unitaries $\mathcal{U}$ (blue) and fixed noise $\mathcal{N}$ (green).

More explicitly, we can compute the scaling of composite moment operator norms. Suppose a unitary ensemble $\mathcal{U}$ is an ϵ-t-design with respect to a reference Haar-distributed unitary ensemble $\mathcal{U}'$, with t-order commutant elements $\mathcal{S}_{\mathcal{U}'}^{(t)}$ with locality-$\geq l$, and rank-r moment operators. Given strictly diagonal unital, and non-unital noise $\widehat{\mathcal{N}}_{\gamma\eta}^{(t)}$, with bounded norms, $\|\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)}\|_\infty \leq \epsilon$, $\|\widehat{\mathcal{N}}_{\gamma}^{(t)}\|_\infty \leq 1-\gamma$, $\|\widehat{\mathcal{N}}_{\eta}^{(t)}\|_\infty \leq \eta$, we have the exact bound,

$$\left\| \left(\mathcal{N}_{\gamma\eta}^{\otimes t}\mathcal{T}_{\mathcal{U}}^{(t)}\right)^k - \mathcal{D}_d^{\otimes t} \right\|_\diamond \leq d^t \frac{(r-1+\epsilon)^k\,(1-\gamma)^k - 1}{(r-1+\epsilon)\,(1-\gamma)-1}\ \eta \ + \ d^t\,(r-1+\epsilon)^k\,(1-\gamma)^k \ . \tag{4.48}$$

Suppose the unitary ensemble $\mathcal{U}$ is an exact Haar-distributed group t-design, or consists of randomly parametrized unitaries $\mathcal{G}_\Theta = \{U_\theta^G = e^{-i\theta G}, G \in \mathcal{P}_d, \theta \sim \Theta\}$ with parameters $\theta \sim \Theta$, and a single involutory G with commutant $\mathcal{S}_G$. Given strictly diagonal unital $\widehat{\mathcal{N}}_{\gamma}^{(t)}$ and non-unital $\widehat{\mathcal{N}}_{\eta}^{(t)}$ noise, we have the leading-order scaling,

$$\left\| \left(\widehat{\mathcal{N}}_{\gamma}^{\otimes t}\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)}\right)^k - \widehat{\mathcal{D}}_d^{\otimes t} \right\|_1 = (r-1)\ \mathcal{O}\big((1-\gamma)^{lk}\big) \tag{4.49}$$

$$\left\| \left(\widehat{\mathcal{N}}_{\gamma\eta}^{\otimes t}\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)}\right)^k - \widehat{\mathcal{D}}_d^{\otimes t} \right\|_2^2 = t\ |\mathcal{P}_d \backslash \{I_d\}|\ \mathcal{O}\big(\eta^2\big) \tag{4.50}$$

$$\left\| \left(\widehat{\mathcal{N}}_{\gamma}^{\otimes t}\widehat{\mathcal{T}}_{\mathcal{G}_\Theta}^{(t)}\right)^k - \widehat{\mathcal{D}}_d^{\otimes t} \right\|_2^2 = t\ |\mathcal{S}_G \backslash \{I_d\}|\ \mathcal{O}\big((1-\gamma)^{2k}\big) \ . \tag{4.51}$$

Here, in the $\gamma, \eta \to 0$ limit, independent of d, t, the leading-order $\mathcal{O}(\gamma, \eta)$ behaviour is due to k concatenations of the noise component acting on any unitary component operators of minimum locality l. The leading-order scaling is thus only dependent on the locality and rank of unitary operators, and is otherwise independent of the specific details of the unitary ensemble. Unital noise is inherently depolarizing, decreasing moment operator norms exponentially with concatenations and converges towards a Depolarize t-design.

Such noise-induced behaviour can be explained, given our derivations that any Haar-distributed unitary ensemble moment operators are support-preserving and consist of locality-l, rank-$r_l \leq r$ projectors. Unital noise predominantly scales such projectors by $\sim (1-\gamma)^{lk}$ throughout the k-concatenations, leading to an effect proportional to the total amount of noise, $\sim 1 - lk\gamma$, independent of the number of operators scaled.

Conversely, non-unital noise predominantly occurs at the last concatenation, independent of k, scaling $\mathcal{P}_d \backslash \{I_d\}$ number of non-identity basis operators mapped to from the identity. Any intermediate non-unital noise that increases support of operators at concatenation $s < k$ would contribute next-leading-order scaling $\sim \eta(1-\gamma)^{l(k-s)}$, leading to a mixture of unital and non-unital effects as unitary and noise component basis operators are mapped between each other.

4.3.4 Channel Designs and Expectation Value Concentration

In this section, given moment operator norms for ensembles of unitaries have been given operational meaning in the context of variational algorithms [106], we show that similar phenomena occur for ensembles of quantum channels, and relate concentration of expectation values and their gradients to channel design properties.

In particular, let us define the expectation value, and its gradient at index μ

$$\mathcal{L}_\Lambda(\rho, \Pi) = \mathrm{tr}(\Lambda(\rho)) \quad \rightarrow \quad \partial_\mu \mathcal{L}_\Lambda(\rho, \Pi) \ , \tag{4.52}$$

where ρ is a quantum state, Π an observable, $\Lambda = \circ_\mu \Lambda_\mu$ is a channel of concatenated channels indexed by μ, collectively from an ensemble $\mathcal{C}$, and ensembles of channels before and after index μ are denoted respectively as $\mathcal{C}_{\mu_R}, \mathcal{C}_{\mu_L}$. We can bound the probability of $\mathcal{L}_\Lambda$ and its gradient deviating from its mean $\mu_{\mathcal{L}|\mathcal{C}} = \langle \mathrm{tr}(\Lambda(\rho)\Pi) \rangle_{\mathcal{C}}$ by $\varepsilon > 0$, in terms of its variance $\sigma^2_{\mathcal{C}} = \langle \mathrm{tr}(\Lambda(\rho)\Pi)^2 \rangle_{\mathcal{C}} - \langle \mathrm{tr}(\Lambda(\rho)\Pi) \rangle^2_{\mathcal{C}}$, via Chebyshev's inequality

$$\text{Probability}\left[\left| \mathcal{L}_\Lambda - \mu_{\mathcal{L}|\mathcal{C}} \right| \geq \varepsilon \right] \ \leq \ \frac{\sigma^2_{\mathcal{L}|\mathcal{C}}}{\varepsilon^2} \ . \tag{4.53}$$

Given computing variances $\sigma^2_{\mathcal{L}|\mathcal{C}}$ for arbitrary ensembles of quantum channels $\mathcal{C}$ may be difficult, we can instead study concentration phenomena by comparing the proximity of ensembles $\mathcal{C}, \mathcal{C}'$. In particular, one can show that the following theorems hold, regarding bounds on the variance of expectation values and their gradients.

Theorem 5 (Variance Bounds in terms of Reference Ensemble)**.** *Let $\mathcal{L}_\Lambda = \mathcal{L}_\Lambda(\rho, \Pi)$ be an expectation value , $\mathcal{C}, \mathcal{C}'$ be ensembles of quantum channels, with states ρ and observables Π. If $\mathcal{C}$ is an ϵ-t-design with respect to $\mathcal{C}'$, the expectation value variance,*

$$\sigma^2_{\mathcal{C}} \;=\; \sigma^2_{\mathcal{C}'} \;+\; \delta^{(2)}_{\mathcal{C}\mathcal{C}'} \;, \tag{4.54}$$

with deviations from the reference ensemble variance, upper bounded by terms,

$$\begin{aligned}\delta^{(2)}_{\mathcal{C}\mathcal{C}'} \leq \quad & \min_{1/p+1/q=1} \|\rho\|_p^2 \;\|\Pi\|_q^2 \;\|\mathcal{T}^{(2)}_{\mathcal{C}} - \mathcal{T}^{(2)}_{\mathcal{C}'}\|_p \\ + \quad & \min_{1/p+1/q=1} 2 \left|\mu_{\mathcal{L}|\mathcal{C}'}\right| \;\|\rho\|_p \|\Pi\|_q \;\|\mathcal{T}^{(1)}_{\mathcal{C}} - \mathcal{T}^{(1)}_{\mathcal{C}'}\|_p \;.\end{aligned} \tag{4.55}$$

In particular, when ρ is a quantum state, the observable Π has trace $|\mathrm{tr}(\Pi)| \leq 1$ (such as a projector or Pauli operator), the reference ensemble is $\mathcal{C}' \in \{\mathcal{C}(d,s), \mathcal{D}(d)\}$, and the norms are diamond $\|\cdot\|_\diamond$ or operator $\|\cdot\|_{\mathrm{op}}$ norms, the expectation value variance is,

$$\sigma^2_{\mathcal{C}'} = \begin{cases} \mathcal{O}\big(\frac{1}{d^2 s}\big) \;\; \|\rho\|_2^2 \;\|\Pi\|_2^2 & \mathcal{C}' = \mathcal{C}(d,s) \\ 0 & \mathcal{C}' = \mathcal{D}(d) \end{cases} \quad . \tag{4.56}$$

$$\begin{gathered}\delta^{(2)}_{\mathcal{C}\mathcal{C}'} \leq \\ \|\Pi\|_\infty^2 \|\mathcal{T}^{(2)}_{\mathcal{C}} - \mathcal{T}^{(2)}_{\mathcal{C}'}\|_\diamond + \; 2\frac{1}{d}\|\Pi\|_\infty \|\mathcal{T}^{(1)}_{\mathcal{C}} - \mathcal{T}^{(1)}_{\mathcal{C}'}\|_\diamond \;, \\ \text{or} \\ \|\rho\|_2^2 \|\Pi\|_2^2 \|\mathcal{T}^{(2)}_{\mathcal{C}} - \mathcal{T}^{(2)}_{\mathcal{C}'}\|_{\mathrm{op}} \;+\; 2\frac{1}{d}\|\rho\|_2 \|\Pi\|_2 \|\mathcal{T}^{(1)}_{\mathcal{C}} - \mathcal{T}^{(1)}_{\mathcal{C}'}\|_{\mathrm{op}} \;.\end{gathered} \tag{4.57}$$

Theorem 6 (Gradient Variance Bounds in terms of Reference Ensembles)**.** *Let $\mathcal{L}_\Lambda = \mathcal{L}_\Lambda(\rho, \Pi)$ be an expectation value, with gradients at index μ, $\partial_\mu \mathcal{L}_\Lambda$, and $\mathcal{C} = \mathcal{C}_{\mu_L}\mathcal{C}_{\mu_R}, \mathcal{C}' = \mathcal{C}'_{\mu_L}\mathcal{C}'_{\mu_R}$ be ensembles of quantum channels, with states ρ and observables Π. If $\mathcal{C}_{\mu_R}, \mathcal{C}_{\mu_L}$, are ϵ-t-designs with respect to $\mathcal{C}'_{\mu_R}, \mathcal{C}'_{\mu_L}$, the gradient variance is bounded by,*

$$\sigma^2_{\partial_\mu \mathcal{L}|\mathcal{C}_{\mu_{RL}}}(\rho, \Pi) \leq \begin{cases} \left[\begin{array}{l} \mathcal{O}\big(\frac{1}{d^2 s^2}\big) \;\; \|\rho\|_2^2 \;\|\Pi\|_2^2 \\ \quad + \mathcal{O}\big(\frac{1}{d^2 s}\big) \;\; \|\rho\|_1^2 \;\|\Pi\|_2^2 \;\|\mathcal{T}^{(2)}_{\mathcal{C}_{\mu_R}} - \mathcal{T}^{(2)}_{\mathcal{C}'_{\mu_R}}\|_\diamond \\ \quad + \mathcal{O}\big(\frac{1}{d^2 s}\big) \;\; \|\rho\|_2^2 \;\|\Pi\|_1^2 \;\|\mathcal{T}^{(2)\dagger}_{\mathcal{C}_{\mu_L}} - \mathcal{T}^{(2)\dagger}_{\mathcal{C}'_{\mu_L}}\|_1 \\ \quad + 4 \;\|\rho\|_1^2 \;\|\Pi\|_1^2 \;\|\mathcal{T}^{(2)}_{\mathcal{C}_{\mu_R}} - \mathcal{T}^{(2)}_{\mathcal{C}'_{\mu_R}}\|_\diamond \|\mathcal{T}^{(2)\dagger}_{\mathcal{C}_{\mu_L}} - \mathcal{T}^{(2)\dagger}_{\mathcal{C}'_{\mu_L}}\|_1 \end{array}\right] & \mathcal{C}' = \mathcal{C}(d,s) \\ \\ 4 \;\|\rho\|_1^2 \;\|\Pi\|_1^2 \;\|\mathcal{T}^{(2)}_{\mathcal{C}_{\mu_R}} - \mathcal{T}^{(2)}_{\mathcal{C}'_{\mu_R}}\|_\diamond \;\|\mathcal{T}^{(2)\dagger}_{\mathcal{C}_{\mu_L}} - \mathcal{T}^{(2)\dagger}_{\mathcal{C}'_{\mu_L}}\|_1 & \mathcal{C}' = \mathcal{D}(d) \end{cases} \tag{4.58}$$

The exact variance and gradient variance bounds must be considered case-by-case, depending on which norms of the states, observables, and moment operator differences are optimal. These theorems encompass the results for unitary ensembles of [106], and allows us to study non-unitary concentration phenomena. For instance, noise-induced barren plateaus described in [276], can now be more generally thought of as arising from noisy circuits becoming a t-design with respect to the Depolarize ensemble.

4.4 Discussion

Here, we introduce a framework for studying moment operators of ensembles of random quantum channels. A key contribution is the proposal of three reference ensembles: Haar, cHaar, and Depolarize, for which we derive exact expressions for their spectra, traces, and norms, and obtain a hierarchy between these ensembles. In particular, we show that the cHaar ensemble interpolate, via the environment dimension, between the unitary and depolarizing limits. Indeed, we find that its dominant fixed point is predominantly depolarizing with a perturbatively non-unital component, highlighting persistent non-unital effects and the intrinsic dissipative nature of quantum channels [105, 131, 336, 337]. Finally, instead of uniformity of measure, derived hierarchies between ensembles, generally arising from invariance properties, are shown to be more versatile criteria to compare ensembles in these more general settings. Based on these criteria, we show that the Depolarize ensemble is the most natural reference ensemble for ensembles of quantum channels.

The norm of the ensemble moment operator plays a central role in describing relationships between ensembles. It is natural to ask for further operational interpretation of this quantity, particularly to describe the capabilities of random quantum channels. In the case of ensembles of random unitaries, this quantity is interpreted as the expressive power of the ensemble [234]. However, in the case of ensembles of random quantum channels, given their inherent depolarizing and therefore not necessarily useful behaviour, this feels less applicable to describe channel-centric algorithms [35]. The Depolarize or asymptotic dimension cHaar reference ensembles do capture in some sense the uniformity or symmetry of an ensemble, in terms of its ability to dissipate information evenly in all directions. However, neither term is quite as satisfactory as expressivity for unitary ensembles.

Applying the formalism to noisy parametrized circuits with diagonal unital and non-unital Pauli-like noise, we observe distinct trends for moment-operator norms. Unital noise typically decreases such norms, while non-unital amplitude damping increases such norms. Circuit depth drives ensembles toward the appropriate reference designs, and concentration of observables emerges. Such analysis ultimately offers a channel-design-centric view on noisy barren-plateau–type phenomena [95, 105, 106, 248, 276, 336]. Crucially, this perspective of channel-design induced concentration suggests that noise is not solely a passive perturbation. Instead, by driving channel ensembles toward (unital) or away (non-unital) from the Depolarize reference ensemble, noise can itself become the mechanism governing concentration phenomena.

By establishing the previous clear connection to variational algorithms, we highlight that our framework can also be directly relevant to more general hybrid classical–quantum protocols that rely on sampling and inversion, such as randomized benchmarking, classical shadows, and error-mitigation schemes. Known design parameters and exact moment forms can potentially tighten sample-complexity bounds and guide experimental choices, particularly when device noise models are difficult to characterize [305, 338–342].

Furthermore, it has recently been shown that random measurement probabilities p_γ from local noise circuits, converge to measurement probabilities from globally depolarized noisy circuits p, in average total variational distance $\|p_\gamma - p\|_1$ [136, 143]. Given our derivation that the moment operator norm bounds the twirl diamond norm, which in turn bounds the total variational distance, our noisy moment operator norms $\|\widehat{\mathcal{T}} - \widehat{\mathcal{T}}_\gamma\|_1$ are shown to be interpretable, scaling similarly, and potentially placing loose bounds on these variational distance results [136, 143],

$$\begin{aligned} &\big\langle \|p_\gamma - p\|_1 \big\rangle_{\mathcal{U}\mathcal{N}_\gamma} \;\sim\; \mathcal{O}\big(e^{-2k\gamma}\big) \;\leq\; \big\langle \|\mathcal{N}_\gamma \mathcal{U} - \mathcal{D}_d\|_1 \big\rangle_{\mathcal{U}\mathcal{N}_\gamma} \;\leq\; \big\langle \|\mathcal{N}_\gamma \mathcal{U} - \mathcal{D}_d\|_\diamond \big\rangle_{\mathcal{U}\mathcal{N}_\gamma} \\ &\qquad\qquad\qquad\qquad \overset{?}{\leftrightarrow} \\ &\left\| \mathcal{T}^{(t)}_{\mathcal{U}\mathcal{N}_\gamma} - \mathcal{D}_d^{\otimes t} \right\|_\diamond \;\leq\; d^t \left\| \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}\mathcal{N}_\gamma} - \widehat{\mathcal{D}}_d^{\otimes t} \right\|_1 \;\sim\; \mathcal{O}\big(d^t \; t! \; (1-\gamma)^{2k}\big) \;. \end{aligned} \tag{4.59}$$

Our bounds on the twirl diamond norm suggest at least depth,

$$k = \Omega\left(t \; \frac{\log(d) + \log(t)}{\log\left(\frac{1}{1-\gamma}\right)} \right) \;, \tag{4.60}$$

that is linear in system size, $n = \log(d)$, is sufficient for t-design Haar unitary ensembles plus unital noise to converge to Depolarize t-designs. Such loosely bounded linear scaling depths with n for all t is comparable to previous derivations of $k = \Omega(n \log(n))$ number of channels, or $\Omega(\log(n))$ depth for ϵ-$t = 2$-design Haar unitary ensembles with local noise and caveats on the noise scaling [136, 143]. Our bounds can potentially be tightened if focusing specifically on the $t = 2$ case, given closed-form expressions for such moment operators.

In conclusion, these studies opens up several future research directions. First, non-asymptotic spectral analyses and representation-theoretic proofs for the cHaar–Haar hierarchy and related block structures, with connections to Weingarten calculus and partition algebras [117, 132, 141] are likely possible. Second, characterization of more specific, physically motivated, and experimentally relevant ensembles of channels, such as Stinespring unitaries beyond Haar, structured entanglement, composite ansatze, and principled criteria for channel designs [231, 304, 343–346] would be beneficial to understand random non-unitary phenomena more completely. Third, recent work has shown that design-like phenomena can emerge dynamically in chaotic many-body systems, even without explicit sampling from an ensemble. In particular, projected or subsystem ensembles generated by chaotic dynamics can realize exact or approximate quantum state-design behaviours [299, 347–349]. Our formalisms suggest a natural generalization of these settings to the case of the statistics of channels induced by chaotic dynamics acting on a system coupled to an environment. Finally, specific experiments should be performed that probe higher-order moments, benchmark predicted concentration trends, and test their implications for noise-assisted or active-error-mitigation strategies [131, 199, 350–352].

Chapter 5

Distributions of Expectation Values

Given the distinct statistics that arise when considering ensembles of quantum channels, in this chapter, based on the preprint manuscript, M. Duschenes, R. Melko, J. Carrasquilla, and R. Laflamme, arXiv:2604.05973 (2026) [4], we are inspired to understand such statistics within the tangible setting of distributions of expectation values. In particular, expectation values of measurement operators, interpreted as measurement probabilities, arise frequently throughout quantum algorithms. As such, when quantum states are randomly distributed, any expectation values are also randomly distributed. Here, we generalize previous derivations for distributions of expectation values (Campos Venuti and Zanardi, Physics Letters A (377), 2013) to the case of sets of measurement operators and random mixed quantum states within variable sized environments. Using combinatorics approaches, we derive novel expressions for their moments. We proceed to construct empirical distributions of simulated Haar random brickwork quantum circuits with local depolarizing noise, and compare their form to a proposed effective global-depolarizing-like model with variable effective noise scales and environment dimensions. The fitted effective distributions reproduce peak behaviour across circuit depths, noise scales, and system sizes, while deviations in the distribution tails arise from local noise effects. The fit effective model parameters are also shown to vary smoothly and consistently with circuit depth and noise scale. Finally, sets of non-symmetric measurement operators are shown to exhibit distinct multi-modal distributions relative to uni-modal distributions for symmetric measurement operators, opening up questions about their simulability.

5.1 Introduction

Expectation values of systems described by quantum states, given quantum theory is inherently probabilistic [21, 353], are at the heart of quantum information. Experimental probes, numerical simulations, or analytical studies thus involve measurement procedures that allow us to interact with and extract information from a system of interest. Expectation values can have a variety of interpretations. Expectation values of measurements specify the probabilities of measurement outcomes [38, 146, 178], of Hamiltonians specify average energies [178, 180], and fidelities quantify distances between states [110, 181–183].

Randomness in quantum states [181, 354] arises due to intrinsic stochastic dynamics [3, 105, 106], algorithmic design [34, 107–109], and chaos and thermalization in quantum many-body systems [355]. Randomness is also a key concept in random circuit sampling experiments to compare classical and quantum resources [23, 52, 78, 105, 155, 210, 212, 294, 356, 357], tomography to efficiently learn system properties [86, 147, 255, 358, 359], concentration phenomena within variational algorithms [3, 95, 106], optimal control [1, 29, 29, 85, 360], and error mitigation [34, 108, 109, 145, 361, 362]. Random quantum circuits consisting of layers of unitaries followed by measurements, give rise to measurement induced phase transitions [78]. As the probability of local measurements is increased, states can transition from being volume-law entangled, to being area-law entangled, with well-defined unitary ensemble-dependent critical exponents and universality classes [291, 363].

More concretely, given states that are randomly distributed from a known distribution, their properties will also be randomly distributed, from a yet-to-be-determined distribution. Expectation values are also always calculated with respect to specific states and operators, and of particular relevance to quantum computing applications are measurement operators [21, 353, 364]. It is thus essential to understand these expectation value distributions, given specific operators, and ensembles of states.

When sets of measurement operators are so-called informationally complete [38, 146, 148], meaning their measurement probabilities for distinct quantum states are uniquely defined, then consistent tomography can be performed [147, 255, 365–368]. Such measurements have far-reaching applications, such as classical shadow tomography [86, 148, 358, 369] and measurement-induced phase transitions [291, 363]. By posing such applications in terms of full distributions rather than expectation values alone, it is possible to obtain a more complete characterization of how observables depend on system parameters such as system size or evolution time. Further, when considering distributions over sets of operators [364], total distributions of expectation values become sums over conditional distributions, conditioned on the choice of operator. Such total distributions, and their dependence on any defining properties of these sets of operators, have yet to be systematically explored.

Significant progress has been made in understanding distributions of expectation values of projective measurements for pure, uniformly distributed states. Such settings give rise to the famous Porter-Thomas distribution [79, 140], and mixed state variants of this distribution are explored throughout this chapter. Here, notions of uniformity imply dynamics are distributed according to the unitarily-invariant Haar measure [118, 127]. Crucially, a quantum circuit's architecture that dictates these dynamics, and any global versus local effects, greatly affect ease of analysis [113, 134, 136, 370]. In fact, closed-form distribution expressions have been derived for Hermitian operators and Haar random pure states [144, 371–373]. Recently, more experimentally realistic [29, 52, 357] random quantum circuits with brickwork layouts [113, 120, 320], and projective measurements, have also been investigated. In such analyses, it is often intractable to compute expressions for distributions, and instead moments or other statistics are computed [33, 120, 130, 374] using Weingarten calculus [117, 118, 244]. Subsequently, distributions, approximated using these moments, are shown to be in good agreement with empirical histograms. However, without closed form expressions, quantitatively interpreting such distributions remains difficult.

Further complicating these studies is when systems undergo noisy dynamics [1, 37, 105, 138, 155, 356, 369], or directly interact in open settings with their environment [3, 375, 376]. The subtle interplay between the effects of noise with specific noise scales [149, 155] or environment dimensions [3, 127, 132, 145], versus the effects of entangling unitary dynamics [78], is just beginning to be studied, using insights from random matrix theory [104, 119, 130] and variational quantum algorithms [3, 106, 128].

In this chapter, we seek to answer, *How do parameters of system size, circuit depth, and noise scale affect distributions of expectation values of sets of general measurement operators?*, and *Given insight into distributions for globally noisy random systems, are there effective analytical models that capture behaviours of locally noisy random systems?*

In Section 5.2, we introduce our formalisms to describe states, operators, and their statistics. In Section 5.3, we extend the previously derived distributions of expectation values to the case of Haar random mixed states with non-trivial environments, and by using combinatorics approaches, we derive expressions for their moments. We proceed to form empirical distributions of symmetric and non-symmetric positive-operator-valued (POVM) measurement probabilities from simulated noisy brickwork circuits, and propose an effective analytical model with interpretable parameters. By using metrics to distinguish continuous variable distributions, we demonstrate the suitability of these effective models at describing peak versus tails behaviours. We find that these distributions develop sharp peaks with increased circuit depth and noise scale, become uni-modal for symmetric measurements, and multi-modal for non-symmetric measurements. Finally in Section 5.4, we show how the fit effective model parameters depend on system size, circuit depth, and noise scale.

5.2 Preliminaries

Here, we introduce formalisms to study expectation values of Hermitian operators, with respect to quantum states within d-dimensional spaces $\mathcal{H}$, and s-dimensional environments $\mathcal{E}$. After introducing our notation, we will introduce distributions of various quantities, before discussing metrics for numerically quantifying differences between such distributions.

5.2.1 Quantum States, Operators, and Expectation Values

Regarding states, we consider s-rank d-dimensional quantum states $\rho \in \mathcal{B}[\mathcal{H}]$, which can be parameterized by $d \times s$-dimensional parameters $\varphi \in \mathcal{B}[\mathcal{H} \otimes \mathcal{E}]$,

$$\rho = \varphi\varphi^\dagger \ . \tag{5.1}$$

Regarding dynamics, we consider trace-preserving completely positive maps, or quantum channels $\Lambda \in \mathcal{B}[\mathcal{B}[\mathcal{H}]] \ : \ \mathcal{B}[\mathcal{H}] \to \mathcal{B}[\mathcal{H}]$, composed into k channels, such that,

$$\Lambda = \prod_{i \in [k]} \Lambda^{(i)} \quad : \quad \rho \to \Lambda(\rho) \ , \tag{5.2}$$

where the k-layers are indexed by $[k] = \{0, 1, \ldots, k-1\}$.

Regarding operators, we consider Hermitian l-rank operators $\Pi \in \mathcal{B}[\mathcal{H}]$, with $\#$ distinct real eigenvalues $\{\sigma \leq \xi \leq \lambda\}$, such that,

$$\Pi = \sum_\xi \xi \ I_\xi \quad : \quad \sum_\xi I_\xi = I \ , \tag{5.3}$$

with d_ξ-dimensional eigenspaces, with projectors I_ξ.

Given states, dynamics, and operators, here we consider expectation values x of operators Π, with respect to states ρ, defined via the linear maps,

$$\rho \to \tau_\Pi(\rho) = \mathrm{tr}(\Pi \ \rho) \ \equiv \ x \ . \tag{5.4}$$

From the positivity and unit-trace properties of quantum states, operators and their expectation values are bounded by their maximum λ and minimum σ eigenvalues,

$$\sigma I \leq \Pi \leq \lambda I \qquad , \qquad \sigma \leq x \leq \lambda \ . \tag{5.5}$$

We find it convenient to normalize operators,

$$\Pi \to \frac{\Pi - \sigma I}{\lambda - \sigma} \quad , \quad x \to \frac{x - \sigma}{\lambda - \sigma} \ , \tag{5.6}$$

which are bounded by their eigenvalues,

$$0 \leq \Pi \leq I \qquad , \qquad 0 \leq x \leq 1 \ . \tag{5.7}$$

The normalized $\Pi \equiv \Gamma^\dagger \Gamma \geq 0$ are positive, rank $0 \leq l \leq d$, with # number of distinct non-negative eigenvalues. The image of Π induces $l, d-l$-dimensional subspaces with associated $l, d-l$-rank projectors $I_\Pi, I - I_\Pi$, which are often useful in bi-partitioned spaces.

States ρ, operators Π, and expectation values x can be jointly transformed by hermiticity-preserving trace-preserving maps, such as quantum channels Λ and their associated adjoints $\Lambda^\dagger$, with the collective mappings of these quantities of,

$$\rho \to \rho_\Lambda = \Lambda(\rho) \quad \leftrightarrow \quad \Pi \to \Pi_{\Lambda^\dagger} = \Lambda^\dagger(\Pi) \tag{5.8}$$

$$x \to x_\Lambda = \tau_\Pi\left(\rho_\Lambda\right) = \tau_{\Pi_{\Lambda^\dagger}}(\rho) \ . \tag{5.9}$$

5.2.2 Distributions of Expectation Values of Operators

Given randomly distributed quantum states, here we build up intuition for our choice of distributions, as detailed in Appendix 5.A. Properties of these randomly distributed states, such as expectation values, are thus also randomly distributed, and we can derive a useful invariance property of this distribution.

More concretely, given fixed operators Π, and randomly distributed states $\rho \sim P_\rho$ or parameters $\varphi \sim P_\varphi$, such distributions are related by,

$$P_\rho(\rho) = \int d\varphi \ P_\varphi(\varphi) \ \delta\big(\rho - \varphi\varphi^\dagger\big) \ , \tag{5.10}$$

and any corresponding expectation values with respect to such randomly distributed states, given fixed operators, $x \sim P_\Pi$ are thus also randomly distributed, with distributions,

$$P_\Pi(x) = \int d\rho \ P_\rho(\rho) \ \delta(x - \tau_\Pi(\rho)) \ . \tag{5.11}$$

We now choose to make some assumptions about the distribution of states and parameters, with the objective of understanding expectation values with respect to approximately uniformly random states [103, 104, 119, 181, 354]. Unlike in the pure state case, the general mixed state case has additional freedom in choosing distributions for both the state eigenvectors and the eigenvalues independently. In particular, care must be taken when discussing mixed states as being uniformly random, given the induced d-dimensional distributions with s-dimensional environments, depend on factors proportional to $\sim \delta(\mathrm{tr}(\rho) - 1)\ \det(\rho)^{d-s}$, and only reduce to being uniform when identically $d = s$ [104].

Here we will assume general unitary invariance, such that our distribution of mixed states is induced by a uniform distribution of pure states $\psi \in \mathcal{B}[\mathcal{H} \otimes \mathcal{E}]$ in the composite system and environment space. Given the unitary-invariance of the parameter measure $d\varphi$, we will assume the parameter distribution $P_\varphi = P_\varphi(\|\varphi\|)$ is also unitarily-invariant, and is isomorphic to the Haar measure of uniformly random ds-dimensional pure states $d\psi$, with solely the norm constraint, given $2ds$-dimensional areas, Ω_{2ds},

$$P_\varphi(\varphi)\ d\varphi \quad = \quad \frac{1}{\frac{1}{2}\Omega_{2ds}} \delta\big(\|\varphi\|^2 - 1\big)\, d\varphi \quad \cong \quad d\psi\ . \tag{5.12}$$

Given such unitary invariance and constraints, we derive that expectation value distributions have shift and scale invariance, and thus unnormalized and normalized operator distributions are related by the equation,

$$P_\Pi(x) = \frac{1}{\lambda - \sigma} P_{\frac{\Pi - \sigma I}{\lambda - \sigma}} \left(\frac{x - \sigma}{\lambda - \sigma}\right). \tag{5.13}$$

For example, suppose we have a global depolarizing channel with noise scale γ, then states, operators, and distributions are shifted and scaled as,

$$\rho \to \rho_\gamma = (1-\gamma)\rho + \gamma \frac{\mathrm{tr}(\rho)}{d} I \tag{5.14}$$

$$\leftrightarrow$$

$$\Pi \to \Pi_\gamma = (1-\gamma)\Pi + \gamma \frac{\mathrm{tr}(\Pi)}{d} I \tag{5.15}$$

$$x \to x_\gamma = (1-\gamma)x + \gamma \frac{\mathrm{tr}(\Pi)}{d} \tag{5.16}$$

$$P_{\Pi_\gamma}(x_\gamma) = \frac{1}{1-\gamma}\ P_\Pi \left(\frac{x_\gamma - \gamma \mathrm{tr}(\Pi)/d}{1-\gamma}\right). \tag{5.17}$$

We seek to understand such distributions of expectation values, namely their analytical form, and their similarity to empirical distributions from simulated systems.

5.2.3 Metrics Quantifying Differences in Distributions

Given an analytical distribution P or cumulative distribution F, m independent and identically distributed samples $\{x_i \sim P\}_{i\in[m]}$ form an empirical distribution,

$$\tilde{F}(x) = \frac{1}{m}\sum_{i\in[m]} \delta(x \geq x_i) \;\approx\; F(x) = \int^{x} dz\ P(z)\ , \tag{5.18}$$

with the m-dependent means and variances in terms of the true distribution F,

$$\mu_{\tilde{F}}(x) = F(x) \quad , \quad \Sigma_{\tilde{F}}(x) = \frac{1}{m}F(x)(1-F(x))\ . \tag{5.19}$$

As detailed in Appendix 5.B, a metric $\mathcal{L} = \mathcal{L}(\tilde{F}, F)$ must be chosen to determine how the empirical distribution converges with system parameters to a known reference distribution F. The empirical distribution may be biased and converge to a different distribution, $\tilde{F} \to F' \neq F$, such as in noisy, finite system size or finite depth simulation settings.

Potential metrics vary in their relevance to discrete versus continuous variable distributions, their use of empirical versus analytic expressions, their computational complexity, and ultimately their sample complexity, such that the metrics accurately reflect differences in the distributions [377]. The Kullback–Leibler divergence, being a statistical distance, is an attractive choice of metric [147] in particular in discrete variable settings. However, given it involves calculating the entropy of the true distribution, such a metric cannot usually be efficiently estimated from samples in continuous variable settings [378]. We therefore choose a statistical test that is better suited for empirical distributions.

Due to its intuitive form, here we use the Kolmogorov–Smirnov [377, 379] metric, of the maximum difference between cumulative distributions,

$$\mathcal{L} = \max_{x} \left|\tilde{F}(x) - F(x)\right|\ . \tag{5.20}$$

Given the monotonicity of cumulative distributions, such a maximization over the entire domain x can be made numerically tractable by being upper-bounded by an empirical metric $\mathcal{L} \leq \tilde{\mathcal{L}}$ over the m samples as,

$$\tilde{\mathcal{L}} = \max_{i\in[m]} \left|\tilde{F}(x_i) - F(x_i)\right| + \left|F(x_{i+1}) - F(x_i)\right|\ . \tag{5.21}$$

Finally, if we assume $\tilde{F} \to F$ is unbiased, and converges with $m \to \infty$ samples, given $F = 1/2$ maximizes the sample variance, we can bound the metric sample complexity using Chebyshev's inequality for the probability at most $\delta^2 > 0$ of quantities differing from their mean by an amount $\epsilon > 0$,

$$P\left(\left|\tilde{F} - \mu_{\tilde{F}}\right| \geq \epsilon\right) \leq \frac{\Sigma_{\tilde{F}}}{\epsilon^2} \leq \delta^2 \quad \to \quad m \geq \left(\frac{1}{2\delta\epsilon}\right)^2. \tag{5.22}$$

Given such definitions and assumptions, our goal is to derive expressions for the distribution of expectation values correspond to particular random quantum systems. Given analytical and empirical distributions, we can then use our metrics to assess whether the distributions converge as a function of various system parameters.

5.3 Results

Here, we show derived expressions for distributions of expectation values of general Hermitian operators, and discuss their properties. In Appendix 5.A we extend previous derivations for pure states that used contour integration [144, 145, 371], to derive the following properties of distributions of mixed states using intuitive combinatorics and geometry. We then perform numerical studies of empirical distributions of measurement probabilities, given simulated noisy quantum circuits and measurement operators.

5.3.1 Projection Operators

First, let us study properties for projectors,

$$\Pi = I_\Pi \ , \tag{5.23}$$

with $0 \leq l \leq d$-rank, and expectation values $0 \leq x \leq 1$ that follow the elegant l, s-dependent Beta distributions, given the Gamma functions $\Gamma_d = (d-1)!$,

$$P_\Pi(x) = \frac{\Gamma_{ds}}{\Gamma_{ls}\ \Gamma_{(d-l)s}}\ x^{ls-1}\ (1-x)^{(d-l)s-1} \ , \tag{5.24}$$

with the following properties,

$$\begin{aligned}
\text{Family:} \quad & f(x) = \mathrm{Beta}_{ls,(d-l)s}(x) && \rightarrow \begin{cases} x^{ls}e^{-dsx} & d\to\infty\ ,\ s\to\infty \\ e^{-dx} & s\to 1\ ,\ l\to 1 \end{cases} \\
\text{Moments:} \quad & x_t = \frac{\Gamma_{ds}}{\Gamma_{ls}}\frac{\Gamma_{(d-l)s+t}}{\Gamma_{ds+t}} && \rightarrow \begin{cases} (l/d)^t & d\to\infty\ ,\ s\to\infty \\ t!/d^t & s\to 1\ ,\ l\to 1 \end{cases} \\
\text{Optima:} \quad & x_* \in \left\{0, \frac{ls-1}{ds-2}, 1\right\} && \rightarrow \begin{cases} l/d & d\to\infty\ ,\ s\to\infty \\ 0 & s\to 1\ ,\ l\to 1 \end{cases} \\
\text{Roots:} \quad & z \in \left\{0, 1\right\} \ .
\end{aligned} \tag{5.25}$$

In this projector case, the operator rank l and environment dimension s greatly affect the form of the resulting distributions, as shown in Fig. 5.1. For $l = 1$-rank projectors, trivial $s = 1$ environments lead to distributions $P(x) \sim (1-x)^{d-2}$ that are monotonically decreasing with x and are peaked at the boundary $x = 0$. Conversely, non-trivial $s > 1$ environments lead to distributions $P(x) \sim x^{s-1}(1-x)^{(d-1)s-1}$, that are not monotonic with x and are peaked at $0 < (s-1)/(ds-2) < 1$. Further, when the dimensions d or s are large, these distributions are sharply peaked at l/d, with moments as powers of this ratio, representing the proportion of the space that is measured by the l-rank operator. Finally, given the shift and scale invariance of the distributions, let us arbitrarily define the k-dependent noise scales $1 - \gamma^{(k)} = (1-\gamma)^k$ and environment dimensions $s^{(k)} = k+1$. Operators become $\Pi \rightarrow \Pi_\gamma^{(k)} = (1-\gamma^{(k)})\Pi + \gamma^{(k)}(\mathrm{tr}(\Pi)/d)I$, yielding effective global depolarizing-like noisy distributions $P_\Pi \rightarrow P_{\Pi_\gamma}^{(k)}$. Evidently, the rank l, dimensions d, s, number of distinct eigenvalues $\#$, and their relative eigenspace dimensions $\{\xi, d_\xi\}$, crucially affect the form of these distributions.

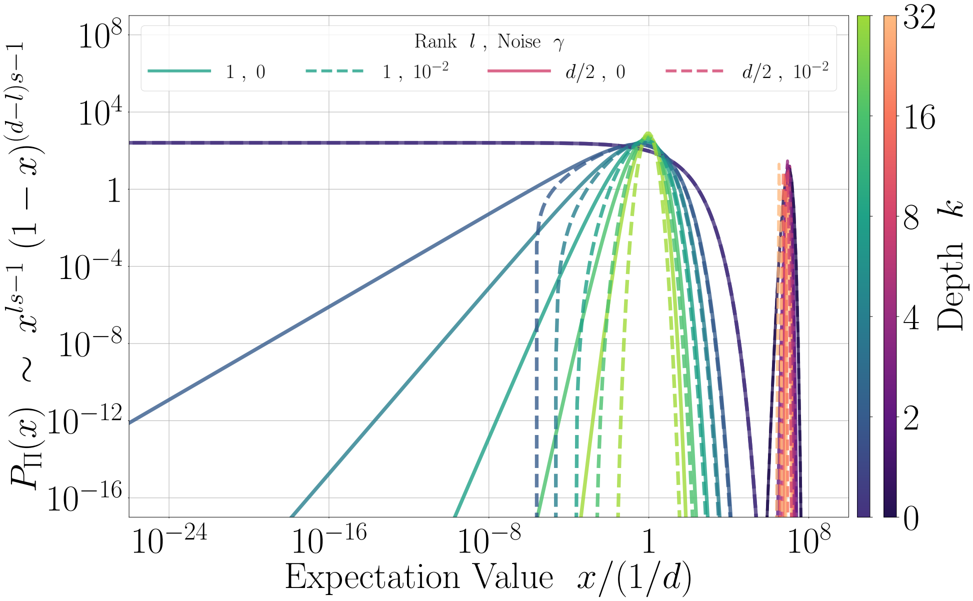


Figure 5.1: **Generalized noisy Porter-Thomas distributions**. Analytical distributions for l-rank, $d = 2^8$-dimensional projectors Π, with global noise γ, $P_\Pi(x) \to P_{\Pi\gamma}^{(k)}(x) = (1/(1-\gamma^{(k)}))\, P_\Pi((x - \gamma^{(k)} l/d)/(1-\gamma^{(k)}))$, with depth-$k$-dependent noise scales $1-\gamma^{(k)} = (1-\gamma)^k$ and environment dimensions $s^{(k)} = k+1$ (colours for different k). Distributions change from having no peak at $ls = 0$, to becoming sharply peaked around $x = l/d$ as rank and environments $ls \to \infty$ (green/blue curves for $l = 1$, red/purple curves for $l = d/2$), with shifts due to noise $\gamma \geq 0$ (solid curves for $\gamma = 0$, dashed curves for $\gamma = 10^{-2}$).

5.3.2 Hermitian Operators

Second, let us study properties for general operators,

$$\Pi = \sum_{\xi} \xi I_{\xi} \quad : \quad \sigma \leq \xi \leq \lambda \ , \tag{5.26}$$

with # distinct eigenvalues $\{\xi, d_{\xi}\}$, and expectation values $\sigma \leq x \leq \lambda$, that follow the polynomial distributions,

$$P_{\Pi}(x) = \begin{cases} \sum\limits_{\substack{\xi \\ l_{\xi} \in [d_{\xi} s]}} \pi_{l_{\xi}} \ \mathrm{sign}(\xi - x) \ (\xi - x)^{(d-d_{\xi})s + l_{\xi} - 1} & \# > 1 \\ \dfrac{1}{\lambda - \sigma} \dfrac{\Gamma_{ds}}{\Gamma_{d_{\sigma} s} \Gamma_{d_{\lambda} s}} \ \left(\dfrac{x - \sigma}{\lambda - \sigma}\right)^{d_{\lambda} s - 1} \left(\dfrac{\lambda - x}{\lambda - \sigma}\right)^{d_{\sigma} s - 1} & \# = 2 \\ \delta(x - \lambda) & \# = 1 \end{cases} \tag{5.27}$$

with t-order moments for this distribution of,

$$x_t = \sum_{\substack{\xi \\ l_{\xi} \in [d_{\xi} s]}} \chi_{l_{\xi}, t} \ \ \xi^t \tag{5.28}$$

which we show are the complete homogenous symmetric polynomials in the spectra $\{\xi, d_{\xi}\}$ [380]. The polynomial coefficients for the respective distributions and moments have remarkably similar forms, that do not appear to immediately follow from each other,

$$\pi_{l_{\xi}} = \frac{1}{2} \ (-1)^{l_{\xi}} \ \frac{\Gamma_{ds}}{\Gamma_{d_{\xi} s - l_{\xi}} \Gamma_{(d-d_{\xi})s + l_{\xi}}} \ \sum_{\sum_{\zeta \neq \xi} l_{\zeta} = l_{\xi}} \prod_{\zeta \neq \xi} \binom{d_{\zeta} s + l_{\zeta} - 1}{l_{\zeta}} \frac{1}{(\xi - \zeta)^{d_{\zeta} s + l_{\zeta}}} \tag{5.29}$$

$$\chi_{l_{\xi}, t} = \quad (-1)^{l_{\xi}} \ \ \frac{\Gamma_{ds}}{\Gamma_{d_{\xi} s - l_{\xi}}} \frac{\Gamma_{d_{\xi} s - l_{\xi} + t}}{\Gamma_{ds + t}} \ \ \sum_{\sum_{\zeta \neq \xi} l_{\zeta} = l_{\xi}} \prod_{\zeta \neq \xi} \binom{d_{\zeta} s + l_{\zeta} - 1}{l_{\zeta}} \frac{\xi^{d_{\zeta} s} \ \zeta^{l_{\zeta}}}{(\xi - \zeta)^{d_{\zeta} s + l_{\zeta}}} \ . \tag{5.30}$$

5.3.3 Empirical Distributions of Measurement Probabilities

We now seek to understand the behaviour of empirical distributions of expectation values from simulated noisy random quantum circuits, as detailed in Appendix 5.B, using the library [6], with data available at [5].

Here we sample random quantum channels $\Lambda_\gamma^{(k)}$, as per Fig. 5.2, with k layers of one-dimensional brickwork layouts of n qubit, $d = q^n$, $q = 2$, of Haar random nearest-neighbour unitaries $\mathcal{U}$, interspersed with fixed local depolarizing noise channels $\mathcal{N}_\gamma$ with noise scale $0 \leq \gamma \leq q^2/(q^2-1)$, that are defined as,

$$\mathcal{U} = \prod_{i,j \in [\frac{n}{\mathrm{brick}}]} U_{ij} \ : \ U_{ij} \ \sim \ \mathcal{U}(q^2) \tag{5.31}$$
$$\mathcal{N}_\gamma = \otimes_{i\in[n]} \mathcal{N}_{\gamma i} \ : \mathcal{N}_{\gamma i}(\rho) = (1-\gamma)\rho + \gamma \frac{\mathrm{tr}_i(\rho)}{q} \otimes I_i \ .$$

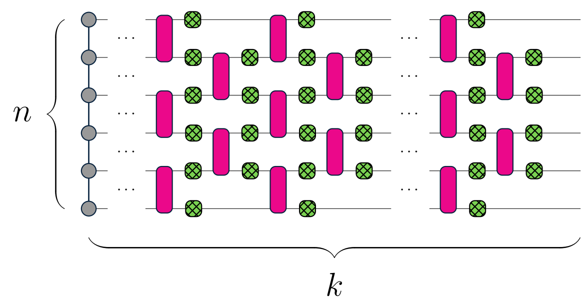


Figure 5.2: **Brickwork quantum circuits**. k layers of n component brickwork layouts of unitaries (pink) and local noise (green).

Regarding the operators associated with our expectation values of interest, here we study sets of positive-operator-valued (POVM) [38, 146, 178] measurement operators,

$$\mathcal{P} = \{0 \leq \Pi \leq I\} \quad : \quad \sum_{\Pi \in \mathcal{P}} \Pi = I \ . \tag{5.32}$$

Expectation values $0 \leq p \leq 1$ can thus be interpreted as measurement probabilities, with distributions $P_{\mathcal{P}}(p)$ derived from conditional $P_\Pi(p)$, given $\Pi \sim P_{\Pi|\mathcal{P}}(\Pi)$,

$$P_{\mathcal{P}}(p) = \sum_{\Pi \in \mathcal{P}} P_{\Pi|\mathcal{P}}(\Pi) \ P_\Pi(p) \ . \tag{5.33}$$

Examples of such sets of measurement operators include,

$$\begin{array}{ll} \text{PVM (Projector)} & \Pi = I_\Pi \\ \text{SIC-POVM (Quasi-Projector)} & \Pi = \lambda I_\Pi \\ \text{NON-SIC-POVM (Non-Projector)} & \Pi = \sum_\xi \xi \ I_\xi \ . \end{array} \tag{5.34}$$

As shown in Fig. 5.3, we simulate circuits with depths $k \in \{0, 2, 4, 8, 16, 32\}$, noise scales $\gamma \in \{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$, system sizes $n \in \{4, 6, 8, 10\}$, in $q = 2$ qubit initial product pure states, and $m \leq 128$ samples. Operators in $\mathcal{P}$ are uniformly, deterministically measured, $P_{\Pi|\mathcal{P}}(\Pi) = 1/|\mathcal{P}|$, yielding $m|\mathcal{P}|$ samples of measurement probabilities $\{p_i\}_{i\in[m|\mathcal{P}|]} \to \{p'_i\}_{i\in[m']}$, which are further uniformly logarithmically binned into $m' = 10^4$ samples $p'_i \in [10^{-20}, 1]$, resulting in the binned empirical total distributions,

$$\tilde{P}_{\mathcal{P}}(p) = \sum_{\Pi\in\mathcal{P}} P_{\Pi|\mathcal{P}}(\Pi)\tilde{P}_{\Pi}(p) \quad \overset{\text{bin}}{\to} \quad \frac{1}{m'}\sum_{i\in[m']} \tilde{P}'_{\mathcal{P}}(p|p'_i) \ . \tag{5.35}$$

As we vary the system parameters k, γ, n, m, the empirical distribution $\tilde{P}^{(k)}_{\mathcal{P}\gamma}$ of samples $p = \mathrm{tr}\left(\Pi\ \Lambda^{(k)}_{\gamma}(\rho)\right)$, will converge towards a potentially biased distribution $\hat{P}_{\mathcal{P}} \approx P_{\mathcal{P}}$,

$$\tilde{P}^{(k)}_{\mathcal{P}\gamma} \ \to \ \hat{P}_{\mathcal{P}} \ \approx \ P_{\mathcal{P}} \quad \leftrightarrow \quad \mathcal{L}^{(k)}_{\mathcal{P}\gamma} \ = \ \mathcal{L}\left(\tilde{F}^{(k)}_{\mathcal{P}\gamma}, \hat{F}_{\mathcal{P}}\right) \ \to \ 0 \ . \tag{5.36}$$

In fact, local noisy brickwork systems do not yield exactly globally Haar random mixed states in finite settings [113], $\tilde{P}^{(k)}_{\mathcal{P}\gamma} \to \hat{P}_{\mathcal{P}} \neq P_{\mathcal{P}}$, and $\hat{P}_{\mathcal{P}}$ is unknown relative to the known Haar distribution $P_{\mathcal{P}}$. However, such systems are appropriate benchmarks given their experimentally relevant structure [52, 357] and recent analysis [33, 113, 120, 130, 374, 381].

To partially account for these biases, we propose an effective distribution $P^{(k)}_{\mathcal{P}\gamma} \approx \hat{P}_{\mathcal{P}}$ of global depolarizing-like noise, $p \to (1-\tilde{\gamma})p + \tilde{\gamma}\mathrm{tr}(\Pi)/d$, with variable effective noise scales $\tilde{\gamma}$ and environment dimensions $\tilde{s}$,

$$P^{(k)}_{\Pi\gamma}(p) \equiv \frac{1}{1-\tilde{\gamma}} P_{\Pi}\left(\frac{p - \tilde{\gamma}\mathrm{tr}(\Pi)/d}{1-\tilde{\gamma}}\right) \qquad : \quad 0 \leq \tilde{\gamma} \leq 1 \quad , \quad \tilde{s} \geq 1 \ . \tag{5.37}$$

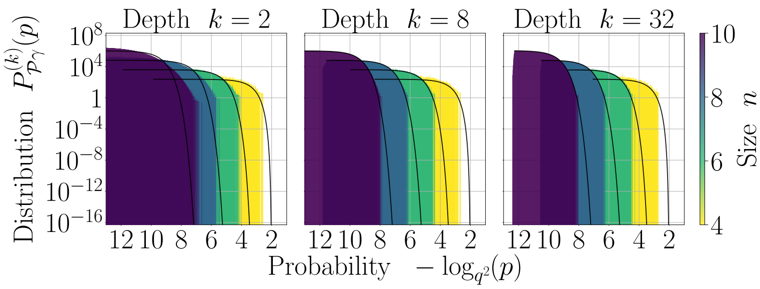
Given the shift and scale invariance of the distributions for Π, we are free to shift operators by arbitrary scales $\tilde{\gamma}$, reflecting global depolarization, with arbitrary environment dimensions $\tilde{s}$, reflecting global entanglement with the environment. Such interpretable parameters thus represent effective changes to distributions, given actual local noise. Such a model is inspired by recent bounds on the convergence of locally noisy to globally noisy distributions [136, 143]. This effective model can be fit using empirical data, via minimization of the mean-squared-error between $P^{(k)}_{\Pi\gamma}(p), \tilde{P}^{(k)}_{\Pi\gamma}(p)$, before comparing $F^{(k)}_{\Pi\gamma}(p), \tilde{F}^{(k)}_{\Pi\gamma}(p)$ with the empirical Kolmogorov–Smirnov metric $\tilde{\mathcal{L}}^{(k)}_{\mathcal{P}\gamma} \approx \mathcal{L}^{(k)}_{\mathcal{P}\gamma}$.

First, let us simulate SIC-POVM operators, namely $l = 1$-rank quasi-projectors with scalings $\lambda = 1/d$, as per Figs. 5.3a and 5.3b. The distributions correspond to those of pure states at low depths or noise scales, and those of depolarized states at large depths and noise scales, with respective smooth decays or sharp peaks at $\sim (l/d)\lambda$.

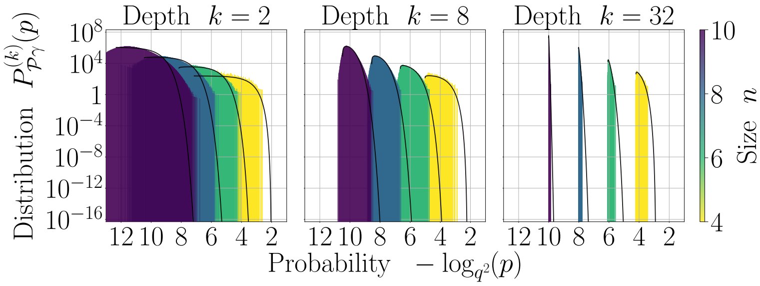
Second, let us simulate NON-SIC-POVM operators, namely $l \leq d$-rank non-projectors, as per Figs. 5.3c and 5.3d. At low depths or noise scales, the distributions are more uniformly spread out than the SIC distributions. At large depths or noise, the distributions become sharply multiply-peaked, with a number of peaks that scales with system size, due to the corresponding $|\mathcal{P}| = d^2$ non-symmetric measurement operators in the set, each contributing shifted and peaked conditional distributions. Unlike in the symmetric case where each of the identical contributing conditional distributions have a single common peak, in the non-symmetric case many of the distinct contributing conditional distributions have distinct peaks. Any differences in spectra of operators across a set directly contribute to any multi-modality in distributions.

To quantify these behaviours, we calculate empirical Kolmogorov–Smirnov metrics, as per Fig. 5.4. Such metrics decrease with depth until plateauing for small noise, or increasing for large noise. Such behaviours are primarily attributed to inherent biases in the model at capturing peak versus tails behaviours. There is also potential overfitting of the minimization of probability density differences, which does not necessarily generalize to minimization of cumulative distribution differences.

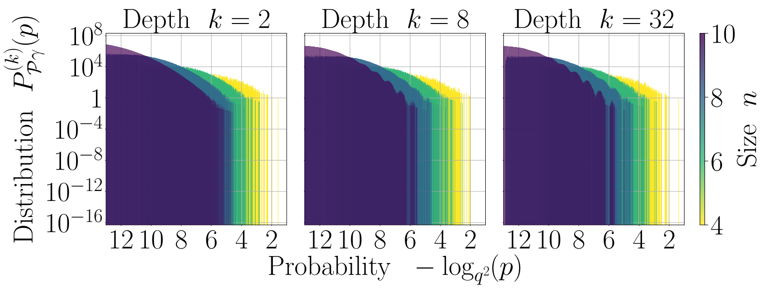
Regarding finite-size and finite-sample effects, the metrics converge to smaller values with increased system size n, and behave similarly across number of samples m. The upper bounded empirical metrics are also non-zero at finite binning m',
$\tilde{\mathcal{L}} \sim \mathcal{O}(\max_{x,y} |F(x) - F(y)|/m') > 0$, leading to inherent bias. Such biases may mask some sample-complexity-dependencies, but do not prevent peaks behaviours from being well described by the effective model. Finally, independent of binning, we may require an exponentially greater number of samples with system size to capture tails information about this distribution on the logarithmic scale. Subsequently, the precise sample complexity and accuracy of the effective model may be masked by being in this relatively low sample regime.



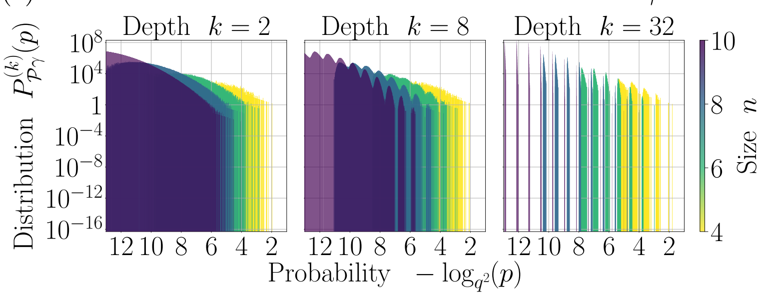
(a) SIC-POVM distribution with noise scale $\gamma = 10^{-4}$.



(b) SIC-POVM distribution with noise scale $\gamma = 10^{-2}$.



(c) NON-SIC-POVM distribution with noise scale $\gamma = 10^{-4}$.



(d) NON-SIC-POVM distribution with noise scale $\gamma = 10^{-2}$.

Figure 5.3: **SIC and NON-SIC POVM distributions**. Empirical SIC-POVM (a,b) and NON-SIC (c,d) histograms $\tilde{P}^{(k)}_{\mathcal{P}_\gamma}(p)$ for depths k, noise scales γ, system sizes n, and $m = 128$ samples, for Haar random brickwork and local depolarization circuits. Solid lines indicate effective SIC-POVM distributions $P^{(k)}_{\mathcal{P}_\gamma}$. Additional plots of distributions for noise scales, $\gamma \in \{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$, are shown in Figs. 5.B.1 and 5.B.2 in Appendix 5.B.

5.4 Discussion

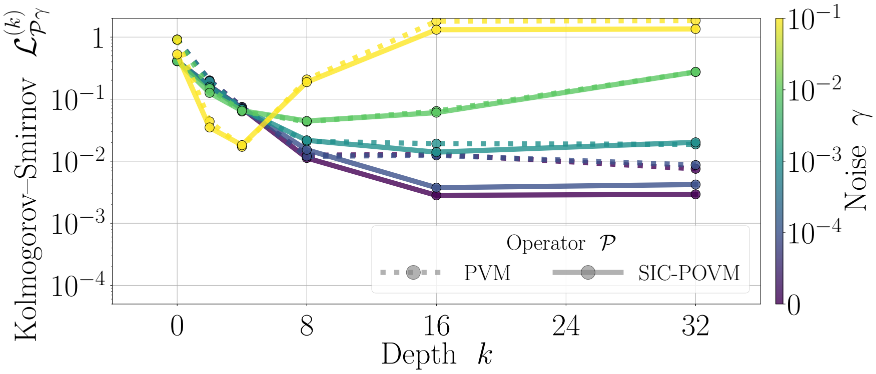
Here, we have investigated analytical and empirical distributions of expectation values with respect to Haar random states, with an environment traced out. Distributions over sets of non-projective and not-necessarily-symmetric Hermitian operators are studied in depth, demonstrating important differences in their properties to previously studied projectors [120, 144, 371].

Using a combinatorics approach, with derivations shown in Appendix 5.A, we extend expressions for expectation value distributions [144], to the case of mixed states with non-trivial environments, and provide new expressions for their moments. Whereas systems with trivial environments or with unit-rank operators have only peaks at the domain boundaries, systems with non-trivial environments or higher-rank operators become sharply peaked at interior domain points. The forms of moments are shown to have intimate connections to symmetric polynomials [380], hypergeometric functions [382], and the symmetric subspaces [248] of each of the eigenspaces of the operator.

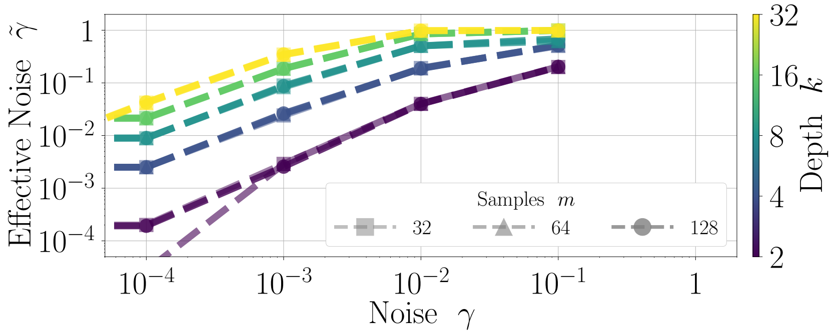
By sampling measurement probabilities given a set of measurement operators from instances of noisy quantum circuits, we confirm the predicted distribution behaviours. Our analytical noiseless expressions subsequently inspire a proposed effective model, with variable effective global noise scale and environment dimension parameters. The effective models are shown to be capable of describing peaks, and less capable of describing tails of the distributions. These behaviours are attributed to a combination of limited samples in these tails on the logarithmic scale, due to inherent biases in the effective global noise model at capturing local noise behaviour. In particular, an exponential $\mathcal{O}(d^2) \gg m = 128$ number of samples per operator are potentially necessary to obtain tails information. The effective model, with its explicit Beta functional form, also potentially requires significantly more variable parameters to fit the local noise-induced tails. Potentially obscuring the analysis is also the approximations made involving Kolmogorov–Smirnov upper bounds and binning procedures, which overestimate model errors around sharp peaks in distributions.

As discussed in Appendix 5.B, it is important to assess the validity and interpretability of these effective models. We find, from Figs. 5.4 and 5.B.4 the effective noise $\tilde{\gamma}$ and environment dimension $\tilde{s}$ vary with noise scale γ and circuit depth k as the approximately polynomial functions, given some threshold γ^*,

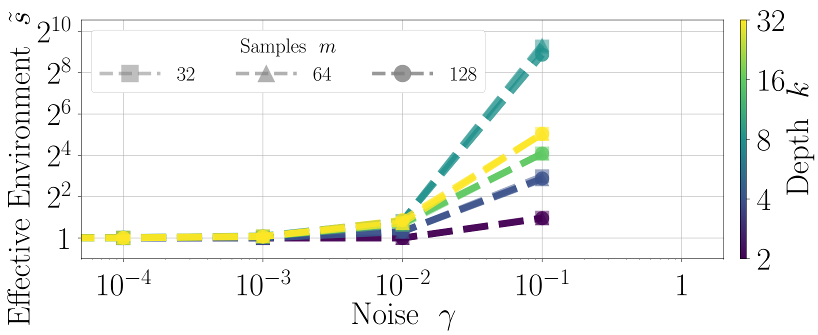
$$\begin{matrix} \tilde{\gamma} \sim \mathcal{O}(\mathrm{poly}(\gamma)) \\ \tilde{s} \sim 1 \;+\; \mathcal{O}(\mathrm{poly}(\gamma)) \;\; \delta_{\gamma > \gamma^*} \end{matrix} \quad : \quad \lim_{\gamma \to 0} \begin{matrix} \tilde{\gamma} \to 0 \\ \tilde{s} \to 1 \end{matrix} \;, \tag{5.38}$$



(a) Empirical metric $\tilde{\mathcal{L}}^{(k)}_{\mathcal{P}_\gamma}$ for size $n = 10$.



(b) Effective noise $\tilde{\gamma}$ versus noise scale γ.



(c) Effective environment $\tilde{s}$ versus noise scale γ.

Figure 5.4: **Kolmogorov–Smirnov metrics**. (a): Empirical Kolmogorov–Smirnov metric $\tilde{\mathcal{L}}^{(k)}_{\mathcal{P}_\gamma}$ of upper-bounded maximum difference between empirical and analytical cumulative distributions for SIC-POVM and PVM distributions, as a function of depth k, for noise scales γ (colours), system size $n = 10$, and $m = 128$ samples, for Haar random brickwork and local depolarization circuits. As depth increases, the metrics converge towards zero, before plateauing due to inherent biases in the models. (b,c): Optimized effective noise $\tilde{\gamma}$ and effective environment dimension $\tilde{s}$ as a function of noise scale γ and depth k (colours). Additional plots of empirical Kolmogorov–Smirnov metrics and effective model parameters, for system sizes $n \in \{4, 6, 8, 10\}$, and samples $m \in \{32, 64, 128\}$, are shown in Figs. 5.B.3 and 5.B.4 in Appendix 5.B.

Curiously, the effective noise scale varies more smoothly with noise than the effective environment dimension. The effective environment is also largest at intermediate depths (blue curve in Fig. 5.4c), potentially indicative of where mixed states are maximally Haar random. Although it is not proven that such trends are representative of global optima of the effective model, these smooth relationships indicate the optimization is within a physically valid local minimum.

Regarding the universality of our results, other operators could equally be studied using the derived analytical expressions, replacing operators and their spectra $\Pi \to \Lambda^\dagger(\Pi)$ with those corresponding with the adjoint action of a channel Λ. Depending on the fixed points of the channel [3], such distributions may be multiply-peaked, and spread out over the logarithmic scale. Further, any non-isotropic noise models may not be able to use the derived shift and scale invariance of the distributions, requiring more thought into appropriate forms of effective models. In particular, the effects of non-unital noise, which generally drive states towards non-maximally mixed states, possibly result in significantly less anti-concentrated, or less uniformly spread out distributions than we observe for our unital depolarizing noise for both SIC and NON-SIC measurement operators [131, 333].

We also note that analytical forms of the NON-SIC-POVM measurement operator sets also follow from the above derivations. However, such computations scale at least thrice-exponentially in system size (these computations involve summations over $\mathcal{O}(\mathrm{poly}(d^2))$ operators $\times$ $\mathcal{O}(\mathrm{poly}(d))$ eigenvalues $\times$ $\mathcal{O}(\mathrm{poly}(l^l))$ terms per multiplicity-l eigenvalue, per individual evaluation of $P_{\mathcal{P}}(x)$), and involve exponentially small and large numbers, making them so far numerically infeasible to compare to simulated empirical distributions.

This work sets the stage for further investigations into noisy expectation values. Our effective models could be supplemented with recent estimates of moments of local noisy systems [130], and offer insights into classical shadow techniques [86, 362, 369]. Moments of such expectation values also appear to have deep connections to algebraic combinatorics [248, 249]. Further, in principle, cross entropy benchmarks for our multi-model distributions, or moments with respect to subgroups of unitaries [249, 302], can be computed.

Finally, given the recent back and forth between quantum and classical methods on simulating noisy random quantum circuits [23, 52, 133, 169, 210, 212, 294, 383, 383–385], it should be emphasized that those results are for strictly projective measurement operators, and are typically assumed to converge to uni-modal distributions. It will thus be important to assess the computational and sample complexities of our multi-modal distributions. For which sets of operators and distributions of quantum states, does any claimed advantage of quantum versus classical methods hold?

Chapter 6

Stochastic Tensor Network Methods

Given the rich behaviours that non-unitarity has been shown to bring to dynamics of quantum systems, in this chapter, based on unpublished work, we are drawn to understand new ways of classically simulating such dynamics. In particular, different types of correlations based on quantum and classical phenomena arise in such open systems, opening up several questions of what are the most appropriate representations of such correlations. Here, we propose a classical simulation method for non-unitary dynamics of quantum systems, entitled POVM-MPS that represents informationally complete probability distributions undergoing quasi-stochastic transformations as stochastic matrix product states (pMPS). To update such pMPS under application of quantum channels, we propose generalized variants of non-negative matrix factorization (NMF) that require local and global information to appropriately update and factorize distributions. Such NMF factorization algorithms are intuitive, but are shown to be difficult to implement in practice. In their place, we use conventional singular value decomposition (SVD) algorithms. We subsequently study the robustness of the method for simulations across system sizes, circuit depths, noise scales, and tensor bond dimensions. Operator entanglement is shown to increase smoothly with increasing bond dimension and with decreasing noise scale, and corresponds well with previously developed simulation methods, and with analytical noiseless average entanglement. Quantum discord is shown to be overestimated by an amount that scales logarithmically with sub-maximum bond dimension, and inversely with noise scale. Finally, negative spectra in quantum states, due to the approximate SVD algorithms, are shown to appear at small noise scales, but disappear at large circuit depths and large noise scales. We conclude this chapter with a discussion of the advantages and disadvantages of POVM-MPS methods, and of future research directions related to understanding differences in tensor network methods over complex versus non-negative fields.

6.1 Introduction

As quantum technologies develop, particularly while such technologies are in their infancy and implement noisy, imperfect operations, it is important to understand methods of efficiently classically simulating such systems. In fact, significant numerical and analytical progress has been recently made on development of methods to simulate general states or their properties in these non-unitary settings [137, 138, 154, 155, 161, 162, 166, 386, 387]. As will be discussed, each method makes distinct assumptions on the nature of correlations that arise throughout dynamics, and such choices have distinct effects on various results. In particular, within a d-dimensional Hilbert space $\mathcal{H}$, simulating $d \times d$-dimensional mixed state matrices ρ and their evolution due to quantum channels Λ, is notably different than simulating d-dimensional pure state vectors $|\psi\rangle$ and their evolution due to unitaries U,

$$U|\psi\rangle \in \mathbb{C}^{d} \quad \rightarrow \quad \Lambda(\rho) \in \mathbb{C}^{d\times d} \; . \tag{6.1}$$

Here, we are not just simulating what is equivalently up to twice as large a system $d = q^n \rightarrow d^2 = q^{2n}$, but unlike the vectors $|\psi\rangle$ that are only constrained to be normalized $|\langle\psi|\psi\rangle|^2 = 1$, the matrices ρ are non-trivially constrained to have unit-trace $\mathrm{tr}(\rho) = 1$, and to be positive-semidefinite $\rho \geq 0$. Such constraints must be maintained throughout dynamics, however due to their assumptions, many existing simulation methods struggle satisfying these constraints at long times, and this is in fact an open problem [388, 389].

More concretely, states ρ with spectral decompositions $\{\xi, \psi_\xi\}$, quantum channels Λ with Kraus decompositions $\{K\}$, and expectation values with respect to operators Π are,

$$\rho = \sum_{\xi} \xi \; \psi_\xi \qquad , \qquad \Lambda(\rho) = \sum_{K} K\rho K^\dagger \qquad \rightarrow \qquad \mathrm{tr}(\Lambda(\rho)\;\Pi) = \mathrm{tr}\big(\rho \; \Lambda^\dagger(\Pi)\big) \;\; , \tag{6.2}$$

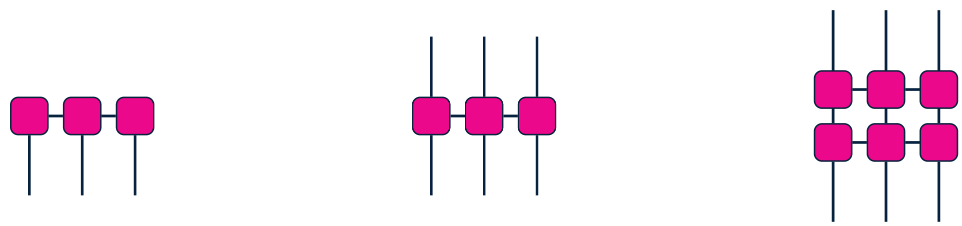
and methods must be developed to efficiently represent such quantities.

From a numerical perspective, simulation methods can be categorized by the structure they impose on states or channels throughout simulations. As a first example, dynamics can be assumed to map between operators within a closed group, such as stabilizer state simulations, where states are acted on by the Clifford group that maps the Pauli group to itself [150–152]. As a second example, sampling methods exist, such as unravelling methods, sets of pure states are each independently evolved, before being sampled to form a mixed state [153–157]. As a third example, assumptions about the locality can be made, in so-called Pauli-propagation methods, where contributions to expectation values with high-locality operators are assumed to be small and are discarded [66, 158, 159].

Finally, as the most relevant to these studies, n-component quantities $\mathcal{A}$ can be represented with various architectures of tensor networks,

$$\mathcal{A} = \prod_{i \in [n]} A^{(i)} \ . \tag{6.3}$$

consisting of products of n tensors $\{A^{(i)}\}_{i\in[n]}$. Generic tensors are denoted without i labels as A, and may be denoted with their explicit (dummy) indices $\{\alpha, \ldots, \sigma, \ldots, \beta\}$ as $A_{\alpha\ldots\sigma\ldots\beta}$. Example architectures include Matrix Product States (MPS), Matrix Product Operators (MPO), and Locally Purified Density Operators (LPDO) are shown in Fig. 6.1.



(a) MPS $A_{\alpha\sigma\beta}$ (b) MPO $A_{\alpha\sigma\pi\beta}$ (c) LPDO $A_{\alpha\sigma\zeta\beta}$

Figure 6.1: **Tensor networks**. Various architectures exist to model quantities $\prod_{i\in[n]} A^{(i)}$, with advantages and disadvantages in their ease of canonical updates, and satisfaction of constraints. Tensors $A^{(i)} = A^{(i)}_{\alpha\sigma\zeta\pi\beta}$ have physical dimensions $\sigma, \pi \in [q]$ and virtual dimensions $\alpha, \beta \in [\chi]$, $\zeta \in [\kappa]$ bounded by bond dimensions χ, κ.

The localized n-component structure of tensor networks as a product of n local tensors, is what contributes to their efficient manipulation, and thus specific architectures must be chosen such that their updates throughout dynamics preserve this structure. In general, these architectures are derived from principles based on MPS architectures for quantum states $\rho = \varphi\varphi^\dagger$ such that the 2-norm $\|\varphi\|_2 = 1$ is satisfied. As we will discuss, if tensor networks are implemented for other quantities obeying other constraints, such that for example the p-norm $\|\varphi\|_p = 1$ is satisfied, then other architectures must be chosen.

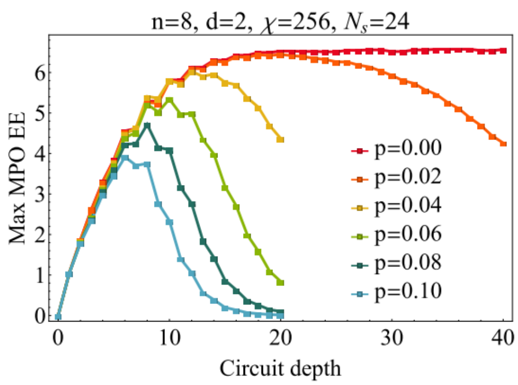
MPS architectures for vectors $|\psi\rangle = \prod_{i\in[n]} A^{(i)}$, with $A^{(i)} = A_{\alpha\sigma\beta}$, such as for pure quantum states, have one physical index $\sigma \in [q]$, and virtual indices for neighbouring bonds $\alpha, \beta \in [\chi]$. MPS's, given their one-dimensional layout, have efficient updates throughout dynamics [160, 175, 253, 254, 256–262]. MPO architectures for matrices $\rho = \prod_{i\in[n]} A^{(i)}$, with $A^{(i)} = A_{\alpha\sigma\pi\beta}$, such as for mixed quantum states, have two physical indices $\sigma, \pi \in [q]$, and virtual indices for neighbouring bonds $\alpha, \beta \in [\chi]$. MPO's, as a generalization of MPS, have potentially efficient updates throughout dynamics, but are not necessarily guaranteed to remain positive semi-definite [161, 163–165, 258]. LPDO architectures for partially traced matrices $\rho = \prod_{i\in[n]} A^{(i)}A'^{(i)\dagger}$, with $A^{(i)} = A_{\alpha\sigma\zeta\beta}$, $A'^{(i)} = A_{\alpha'\pi\zeta\beta'}$, such as for mixed quantum states within environments, have one physical index $\sigma \in [q]$, and virtual indices for neighbouring bond and noise components $\alpha, \beta, \alpha', \beta' \in [\chi]$, $\zeta \in [\kappa]$. LPDO's, as partially traced pure MPS, have complicated updates throughout dynamics, however are guaranteed to be positive semi-definite [162, 166, 167, 170, 390, 391]. Finally, there exist variants of MPS for probability distributions, namely stochastic or probabilistic MPS (pMPS) [171, 174] and tensor trains [172, 264, 392–394], and due to representing probability densities and not probability amplitudes, require distinct algorithms [173, 395–400].

Given quantities are represented by contractions between tensors, the accuracy and efficiency of any tensor network architecture is represented by the largest bond dimension χ, κ of the tensors. Degree-l, r tensor networks, consisting of n tensors with at most l physical indices and r virtual indices, have $\mathcal{O}\big(nq^l\chi^r\big)$ elements. Therefore states with naively $\mathcal{O}(d) = \mathcal{O}(q^n)$ elements, can be potentially represented efficiently, depending on the necessary scaling of χ with d, n. To represent quantities and calculations exactly thus requires up to exponentially scaling bond dimensions $\chi = \mathcal{O}(d) = \mathcal{O}(q^n)$.

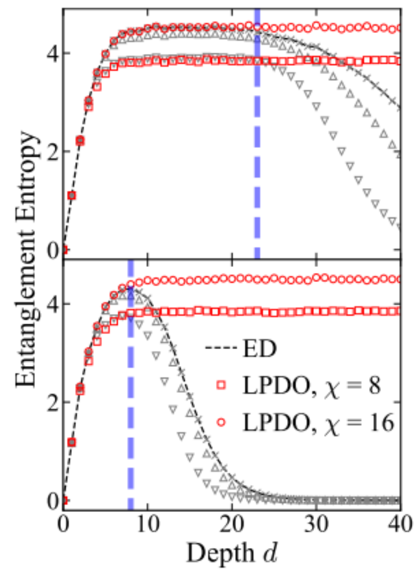
From an analytical perspective, various random matrix theoretic approaches have been applied to study non-unitary dynamics, in particular related to entanglement measures in mixed state settings. Measures of entangling power or operator entanglement [111, 185], dynamics of entanglement [296, 298, 401, 402], and conformal theoretic analysis [403] probe the regime where correlations within noisy systems have accumulated due to quantum phenomena, namely entanglement via entangling unitaries and have not yet decayed or been masked by accumulation of correlations due to classical phenomena, namely noise.

As per Fig. 6.2, within noisily evolving systems there exists an initial unitary regime of increasing entanglement, followed by a noisy regime of decreasing entanglement [161, 401]. Entanglement measures subsequently assess the accuracy of simulation methods at representing such correlations faithfully. MPO-based methods, as per [161] and Fig. 6.2a, although requiring greater resources via larger bond dimensions, accurately represent the unitary and noisy regimes as circuit depth and noise scales are increased. In contrast, LPDO-based methods, as per [166] and Fig. 6.2b, although requiring less resources via smaller bond dimensions, only accurately represent the initial unitary regime, before saturating instead of decreasing in the noisy regime.

Here, we propose a simulation method for noisy quantum systems based around a distinct tensor network architecture. To motivate our method, suppose we model dynamics in terms of measurement probabilities [21, 31, 353], as opposed to explicit quantum states. Such probabilistic methods are known in the classical stochastic simulation community



(a) MPO operator entanglement for noise scales $p \in \{0, 0.02, \ldots, 0.10\}$ (colours), from [161].



(b) LPDO operator entanglement for noise scales $\epsilon = 0.01$ (top), 0.03 (bottom), from [166].

Figure 6.2: **Operator entanglement for MPO and LPDO methods**. Operator entanglement as a function of circuit depth for $n = 8$ qubit noisy brickwork circuits.

[172, 264, 392, 393], and have recently been used for quantum state tomography [174], however have yet to be applied to non-unitary dynamics.

Given such methods, we are interested in answering the questions, *Do stochastic tensor network methods circumvent previously observed challenges of maintaining physical, accurate representations of evolved quantum systems?* and *If not, where does such complexity appear within these stochastic tensor network methods [171, 398, 404]?*

In Section 6.2 we outline the formalisms used to describe our method. We then introduce our specific tensor network architecture, its approximations, and how this architecture can be implemented in practice. In Section 6.3, we discuss commonly used noisy quantum circuit systems in which to simulate, and demonstrate the accuracy of the method for various system parameters of system size, noise scale, depth of quantum circuit, and bond dimension. Finally, in Section 6.4, we discuss the advantages and disadvantages of the proposed methods, in terms of their ease of implementations, accuracy, and appropriateness for simulating noisy quantum systems.

6.2 Preliminaries

Here, we introduce formalisms for representing quantities in terms of measurement probabilities, before describing implementation details of simulating such probabilities.

6.2.1 States, Operators, and Probabilities

Within a d-dimensional Hilbert space be $\mathcal{H}$, let d-dimensional quantum states be $\rho \in \mathcal{B}[\mathcal{H}]$, operators $X, Y \in \mathcal{B}[\mathcal{H}]$, with inner products $\chi_d(X, Y) = \mathrm{tr}\big(X^\dagger Y\big) = \langle\!\langle X|Y\rangle\!\rangle$, and d-dimensional maps between states be quantum channels,

$$\Lambda \quad : \quad \mathcal{B}[\mathcal{H}] \to \mathcal{B}[\mathcal{H}] \ . \tag{6.4}$$

Let a set of measurement operators $\Pi \in \mathcal{B}[\mathcal{H}]$ be denoted as $\mathcal{P} = \{\Pi\}$, with associated expectation values, interpreted as measurement probabilities p, for states $\rho \in \mathcal{B}[\mathcal{H}]$, of,

$$p_\Pi = \mathrm{tr}(\Pi\ \rho) = \langle\!\langle \Pi|\rho\rangle\!\rangle \ , \tag{6.5}$$

and overlap matrices of,

$$\chi_\mathcal{P} = \ \{\chi_{\Pi\Xi} = \chi_d(\Pi, \Xi)\}_{\Pi,\Xi\in\mathcal{P}} \ . \tag{6.6}$$

Sets of measurement operators are informationally complete [38, 148] if measurement probabilities are distinct for distinct states, or equivalently their overlap matrices are invertible,

$$\sum_{\Pi\in\mathcal{P}} \Pi = I \quad , \quad \forall\ \rho \neq \sigma\ \exists\ \Pi \in \mathcal{P}\ :\ \langle\!\langle \Pi|\rho\rangle\!\rangle \neq \langle\!\langle \Pi|\sigma\rangle\!\rangle \quad \leftrightarrow \quad \exists\ \chi_\mathcal{P}^{-1}\ :\ \chi_\mathcal{P}^{-1}\chi_\mathcal{P} = I_{|\mathcal{P}|} \ . \tag{6.7}$$

Informationally complete sets of $|\mathcal{P}| = d^2$ operators, such as symmetric (SIC) sets, in fact span the operators $\mathcal{B}[\mathcal{H}]$ [38, 147]. Therefore d-dimensional operators can be expressed as,

$$|\rho\rangle\!\rangle = \sum_{\Pi,\Xi\in\mathcal{P}} p_\Xi\ \chi_{\Xi\Pi}^{-1}\ |\Pi\rangle\!\rangle \quad : \quad p_\Pi = \langle\!\langle \Pi|\rho\rangle\!\rangle \qquad , \qquad \sum_{\Pi\in\mathcal{P}} p_\Pi = 1 \ , \tag{6.8}$$

and d-dimensional quantum channels can be expressed as [178, 179],

$$\hat{\Lambda} = \sum_{\Pi,\Xi,\Theta\in\mathcal{P}} L_{\Theta\Xi}\ \chi_{\Pi\Theta}^{-1}\ |\Pi\rangle\!\rangle\langle\!\langle\Xi| \quad : \quad L_{\Pi\Xi} = \sum_{\Theta\in\mathcal{P}} \langle\!\langle \Pi|\Lambda(\Theta)\rangle\!\rangle\ \chi_{\Theta\Xi}^{-1} \quad , \quad \sum_{\Pi\in\mathcal{P}} L_{\Pi\Xi} = 1_\Xi \ . \tag{6.9}$$

States ρ and dynamics Λ can thus be described by their associated probabilities p and quasi-stochastic matrices L without loss of generality,

$$\rho \ \leftrightarrow\ p \qquad , \qquad \Lambda \ \leftrightarrow\ L \qquad , \qquad \rho \to \Lambda(\rho) \ \leftrightarrow\ p \to L\,p \ . \tag{6.10}$$

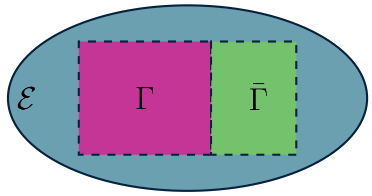


Figure 6.3: **Partitions of Hilbert space**. Partitioning of a space $\mathcal{H} = \Gamma \oplus \bar{\Gamma}$ into disjoint sets $\Gamma, \bar{\Gamma}$, with an environment $\mathcal{E}$ such that the total space is $\mathcal{H} \otimes \mathcal{E} = \Gamma \oplus \bar{\Gamma} \otimes \mathcal{E}$.

6.2.2 Entanglement Measures

Here, we describe measures of entanglement for noisy mixed quantum states, namely measures of operator entanglement [111], quantum mutual information and quantum discord [188], with additional details in Appendix 6.A. We seek to assess capabilities of ansatze at capturing correlations within subsystems $\Gamma, \bar{\Gamma} \subseteq \mathcal{H}$, and not correlations with an environment $\mathcal{E}$, as per Fig. 6.3.

Operator entanglement $\mathcal{Q}_\rho^\Gamma$ is defined [111, 185] in terms of the entanglement entropy $\mathcal{S}_\rho^\Gamma$ of vectorized states $\rho \to \Psi$, taken to be pure states in a doubled space $\mathcal{H} \to \mathcal{H} \otimes \mathcal{H}$, reduced to the states of the $l = |\Gamma|$ component subsystems $\Gamma \subseteq [n]$ on both copies of $\mathcal{H}$,

$$\rho \to \Psi = \frac{|\rho\rangle\!\rangle\langle\!\langle\rho|}{\langle\!\langle\rho|\rho\rangle\!\rangle} \to \Psi_\Gamma = \mathrm{tr}_{\bar{\Gamma}\otimes\bar{\Gamma}}(\Psi) \ . \tag{6.11}$$

The normalized entanglement entropy and operator entanglement then can be defined in terms of ρ and Γ, for the respective d and d^2 dimensional systems, as,

$$\mathcal{S}_\rho^\Gamma = -\frac{1}{\log(d)}\mathrm{tr}(\rho_\Gamma \log(\rho_\Gamma)) \qquad , \qquad \mathcal{Q}_\rho^\Gamma = -\frac{1}{2}\frac{1}{\log(d)}\mathrm{tr}(\Psi_\Gamma \log(\Psi_\Gamma)) = \mathcal{S}_\Psi^\Gamma \ . \tag{6.12}$$

Quantum discord $\mathcal{D}_\rho^{\Gamma|\mathcal{P}}$ is defined as the difference of quantum mutual information definitions [187, 189], namely the quantum $\mathcal{I}_\rho^\Gamma$ and the measurement $\mathcal{I}_\rho^{\Gamma|\mathcal{P}}$ definitions,

$$\mathcal{I}_\rho^\Gamma = \mathcal{S}_\rho^\Gamma - \mathcal{S}_\rho^{\Gamma|\bar{\Gamma}} \quad : \quad \mathcal{S}_\rho^{\Gamma|\bar{\Gamma}} = \mathcal{S}_\rho - \mathcal{S}_\rho^{\bar{\Gamma}} \quad , \quad p_\Pi = \mathrm{tr}(\Pi\rho) \tag{6.13}$$

$$\mathcal{I}_\rho^{\Gamma|\mathcal{P}} = \mathcal{S}_\rho^\Gamma - \mathcal{S}_\rho^{\Gamma|\mathcal{P}} \quad : \quad \mathcal{S}_\rho^{\Gamma|\mathcal{P}} = \sum_{\Pi\in\mathcal{P}} p_\Pi \ \mathcal{S}_{\rho_{\Gamma|\Pi}} \quad , \quad \rho_{\Gamma|\Pi} = \frac{1}{p_\Pi}\mathrm{tr}_{\bar{\Gamma}}(\Pi\rho) \ , \tag{6.14}$$

leading to the definition of quantum discord $\mathcal{D}_\rho^{\Gamma|\mathcal{P}}$, with respect to measurements $\mathcal{P} = \{\Pi\}$,

$$\mathcal{D}_\rho^{\Gamma|\mathcal{P}} = \mathcal{I}_\rho^\Gamma - \mathcal{I}_\rho^{\Gamma|\mathcal{P}} = \left(\mathcal{S}_\rho^\Gamma + \mathcal{S}_\rho^{\bar{\Gamma}} - \mathcal{S}_\rho\right) - \left(\mathcal{S}_{\sum_{\Pi\in\mathcal{P}} p_\Pi \rho_{\Gamma|\Pi}} - \sum_{\Pi\in\mathcal{P}} p_\Pi \ \mathcal{S}_{\rho_{\Gamma|\Pi}}\right) \ . \tag{6.15}$$

6.2.3 Stochastic Matrix Product States

For numerical tractability, but without loss of generality, let us impose a local structure, of n, q-dimensional components, on operators $\mathcal{P}$ and probabilities p, as per Fig. 6.4.

First, let us assume the $d = q^n$-dimensional measurement operators $\Pi \in \mathcal{P}$ have a tensor-product structure over n, local q-dimensional informationally complete operators,

$$\mathcal{P} \to \mathcal{P}^{\otimes n} \qquad \leftrightarrow \qquad \Pi \to \otimes_{i\in[n]}\Pi_i \quad : \quad \sum_{\Pi_i \in \mathcal{P}_i} \Pi_i = I_i \ , \ \ \exists \ \chi_{\mathcal{P}_i}^{-1} \ \forall \ i \in [n] \ . \tag{6.16}$$

Therefore many operator-dependent quantities, such as $\chi_{\mathcal{P}} = \oplus_{i\in[n]}\chi_{\mathcal{P}_i}$ factorize.

Second, let us assume that the probabilities p also have a local structure in terms of stochastic matrix product states (pMPS), with bond dimension χ,

$$p \to p_\chi = \prod_{i\in[n]} A^{(i)} \quad : \quad A^{(i)} = A^{(i)}_{\alpha_i\mu_i\alpha_{i+1}} \quad , \quad \mu_i \in [q^2] \ , \ \alpha_i \in [\chi] \ . \tag{6.17}$$

Therefore states ρ can be represented as tensor networks, which we denote as POVM-MPS,

Third, let us assume that the pMPS $p_\chi \to p'_\chi$ have an efficient update structure when the corresponding states or probabilities are transformed by quantum channels L,

$$p_\chi \to L\, p_\chi \approx p'_\chi \ . \tag{6.18}$$

Given efficient updates, arbitrary dynamics $\rho \to \Lambda(\rho)$ can be simulated efficiently.

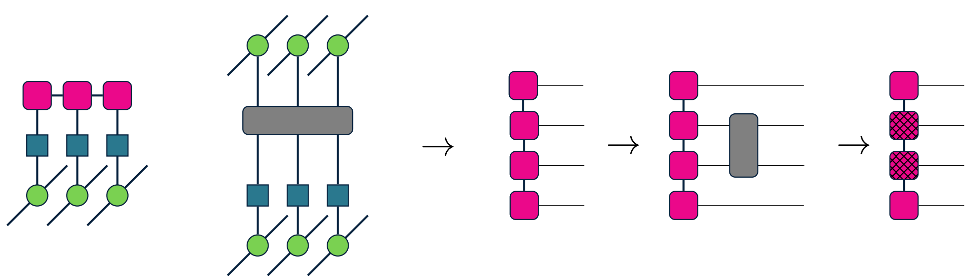


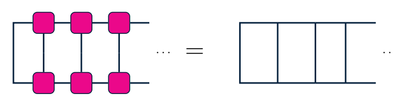
Figure 6.4: **POVM-MPS architecture**. States ρ (left) and channels Λ (centre) can be expanded in terms of local operators $\Pi \in \mathcal{P}$ (green circles), with inverse overlap matrices $\chi_{\mathcal{P}}^{-1}$ (blue squares), and probabilities p (pink squares). Probabilities can be represented as stochastic matrix product states (right), which can be updated to $p \to p'$ (pink-hatched squares) with quasi-stochastic matrices L (gray rectangles) associated with Λ.

6.2.4 Non-Negative Matrix Factorizations

Given our pMPS ansatz models probability densities [171], as opposed to probability amplitudes [160], pMPS versus MPS algorithms have many key similarities and differences. Here, we consider tensors $A_{\alpha\mu\beta}$, with physical $\mu \in [p]$ and bond $\alpha, \beta \in [\chi]$ dimensions.

MPS states ψ must be normalized according to the 2-norm,

$$\psi_\sigma = \prod_i A_{\sigma_i} \in \mathbb{C} \quad : \quad |\psi|^2 = \sum_\sigma |\psi_\sigma|^2 = 1$$

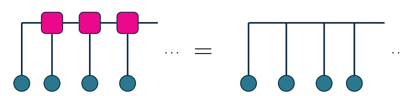


(6.19)

and given their one-dimensional layout, can be canonically normalized [164] by contracting the tensors from left to right such that the 2-norm is automatically satisfied by contraction with the previous tensor and its adjoint. Here, bare lines indicate identity tensors.

pMPS probabilities p must be normalized according to the 1-norm,

$$p_\mu = \prod_i A_{\mu_i} \in \mathbb{R}_+ \quad : \quad |p| = \sum_\mu p_\mu = 1$$



. (6.20)

and given their one-dimensional layout, can be canonically normalized [174] by contracting the tensors from left to right such that the 1-norm is automatically satisfied by summing with the previous tensor. Here, bare lines and blue circles indicate tensors of ones.

To construct matrix product structures from tensors ψ, p, matrix factorizations and truncations of $A = USV \approx U_\chi V_\chi$ can be used, as per Fig. 6.5, with p-norm errors $\epsilon_{\chi,p} = \|A - U_\chi S_\chi V_\chi\|_p$ in terms of the retained singular-like values ξ of the diagonal S [160, 171],

$$\epsilon^2_{\chi\mathrm{MPS},2} = 1 - \sum_{\zeta<\chi} \xi^2_\zeta \qquad , \qquad \epsilon_{\chi\mathrm{pMPS},1} = 1 - \sum_{\zeta<\chi} \xi_\zeta \ . \tag{6.21}$$

However, although such procedures exist in principle for pMPS similar to MPS, unlike MPS which can be decomposed using singular value decompositions (SVD) [165, 175, 258], pMPS must be decomposed with non-negative factorizations (NMF) [173, 392, 395–398]. SVD factorizations for MPS over the complex numbers $\mathbb{C}$ exist for all χ, are unique up to a unitary gauge invariance, are convergent, and always yields a valid MPS [175, 400]. NMF factorizations for pMPS over the non-negative numbers $\mathbb{R}_+$ are not guaranteed to exist for all χ, are unique up to an inverse gauge invariance, there is currently no consensus on a convergent algorithm, and do not always yield a valid pMPS [173, 395].

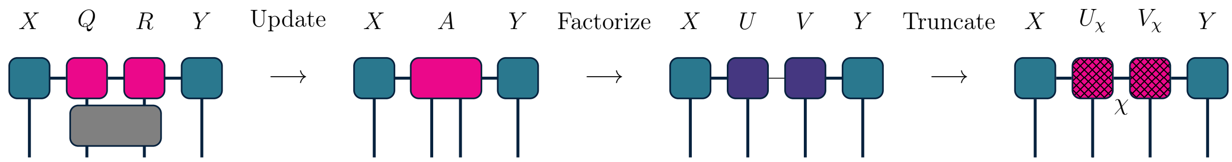


Figure 6.5: **Matrix product factorization and truncation**. Upon 2-local updates to neighbouring tensors Q, R, with boundary tensors X, Y, such that $QR \to A$ loses its local structure, such a tensor can be factorized and truncated, $A \approx UV \approx U_\chi V_\chi$ with bond dimension χ, such that $XAY \approx XU_\chi V_\chi Y$ is a valid matrix product tensor network.

Further, as per Fig. 6.5, a crucial aspect to pMPS NMF that has yet to be considered, given they have only been applied to classical dynamics [172, 174, 264], is that we must consider the boundary tensors X, Y at indices $i, i+1 \in [n]$, with physical dimensions p,

$$\text{MPS-SVD} : XAY \approx XUVY \in \mathbb{C}^{p^i \times p \times p \times p^{n-i-2}} \ : \ U, V \in \mathbb{C}^{\chi \times p \times \chi} \ , \ A \in \mathbb{C}^{\chi \times p \times p \times \chi} \quad (6.22)$$

$$\text{pMPS-NMF} : XAY \approx XUVY \in \mathbb{R}_+^{p^i \times p \times p \times p^{n-i-2}} \ : \ U, V \in \mathbb{R}_+^{\chi \times p \times \chi} \ , \ A \notin \mathbb{R}_+^{\chi \times p \times p \times \chi} . \quad (6.23)$$

For MPS or classical pMPS, upon application of updates with unitaries or stochastic matrices to neighbouring tensors Q, R to obtain $QR \to A$, A remains a complex or non-negative-valued tensor that is a valid complex MPS amplitude or pMPS probability, regardless of any contraction with its boundaries X, Y. As such, any SVD or NMF factorizations $A \approx UV$ only require the local information in A to yield a valid factorization. For quantum pMPS, upon the application of updates as quasi-stochastic matrices to neighbouring tensors Q, R to obtain $QR \to A$, A does not necessarily remain a non-negative-valued tensor that is a valid non-negative pMPS probability, unless it is contracted with its boundaries X, Y. As such, as per Fig. 6.6, any NMF factorizations $XAY \approx XUVY$ requires the local information in A and the non-local information in X, Y to yield a valid pMPS.

Given the non-local information in the boundaries must be considered for valid pMPS updates, if only approximately valid pMPS tensor networks are desired after NMF algorithms, the form of the boundaries is found to greatly impact the accuracy and the efficiency of the overall algorithm. As per Fig. 6.6, given a dense tensor A to be factorized, the boundaries X, Y can either take into account all information about the joint distribution of the involved tensors being updated and their neighbours (Joint NMF), or only some of the information (Marginal NMF), or naively none of the information (Naive NMF). Although guaranteed to yield a valid pMPS in principle, Joint NMF involves NMF routines with up to p^{n-2} larger tensors, relative to Marginal or Naive NMF, and given the lack of NMF routines with guaranteed convergence for larger problems [397, 404], there are evidently significant compromises between efficiency and validity of the overall algorithm.

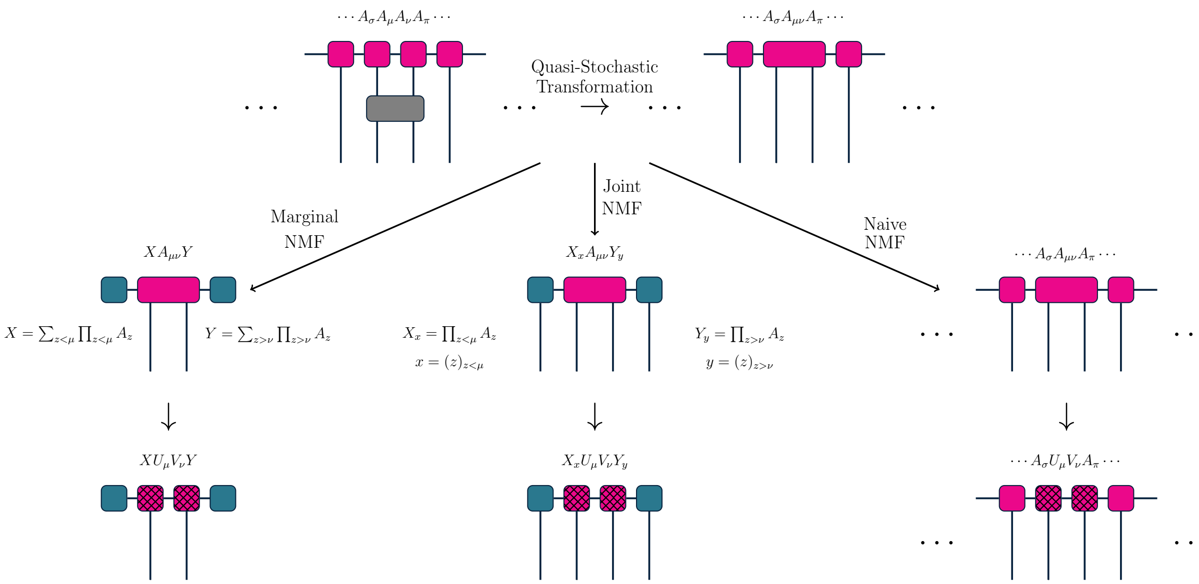


Figure 6.6: **pMPS NMF algorithms**. Matrix products under quasi-stochastic transformations $\cdots A_\sigma A_\mu A_\nu A_\pi \cdots \to \cdots A_\sigma A_{\mu\nu} A_\pi \cdots \approx \cdots A_\sigma U_\mu V_\nu A_\pi \cdots$ can be updated with factorization algorithms for $A_{\mu\nu} \approx U_\mu V_\nu$ that are NMF algorithms once taking into account boundaries X_x, Y_y, with physical indices $x = (\ldots, \sigma)$, $y = (\pi, \ldots)$, to yield valid or approximately valid pMPS $X_x A_{\mu\nu} Y_y \approx X_x U_\mu V_\nu Y_y$. Joint NMF (centre): Boundaries X_x, Y_y are contractions of all tensors to the left or right of $A_{\mu\nu}$, excluding contracting the physical indices x, y, such that the resulting joint distribution tensor and its NMF $X_x A_{\mu\nu} Y_y \approx X_x U_\mu V_\nu Y_y$ are both valid, non-negative joint distributions for x, μ, ν, y. Marginal NMF (left): Boundaries X, Y are contractions of all tensors to the left or right of $A_{\mu\nu}$, including contracting the physical indices x, y, such that the resulting marginal distribution tensor and its NMF $XA_{\mu\nu}Y \approx XU_\mu V_\nu Y$ are only approximately valid, non-negative joint distributions for μ, ν. Naive NMF (right): Boundaries X_x, Y_y are not used, and the resulting tensor and its NMF $A_{\mu\nu} \approx U_\mu V_\nu$ are not necessarily valid, non-negative joint or marginal distributions for μ, ν. Unlike in MPS, the global tensor must be considered and contracted during updates for pMPS. There are significant compromises in improved NMF algorithm efficiency and convergence for the potentially invalid pMPS for simpler Naive or Marginal algorithms involving smaller tensors, versus valid pMPS for the more complex Joint algorithm involving larger tensors.

Finally, given our pMPS algorithms with boundaries, we must choose a specific NMF routine for A, X, Y and error E, such that,

$$Z = Z_{x\mu\nu y} \ = \ X_{x\alpha} A_{\alpha\mu\nu\beta} Y_{\beta z} \ \approx \ X_{x\alpha} U_{\alpha\mu\lambda} V_{\lambda\nu\beta} Y_{\beta z} \ + \ E_{x\mu\nu y} \ = \ XUVY \ + \ E \ , \quad (6.24)$$

where the products,

$$XUVY = \sum_i U_i (XVY)_i = \sum_i (XUY)_i V_i \qquad , \qquad E = \sum_i E_i \qquad (6.25)$$

can be decomposed into independent factors, indexed by a collection of tensor indices denoted by i. Such an NMF routine involves a specific NMF factorization algorithm, an objective function $\mathcal{L}(A, UV)$, and an initialization procedure for the initial U, V [399, 400].

Here, after considering many combinations of NMF algorithms [173, 396, 397], we consider NMF factorization algorithms $A = USV$ with direct interpretations for $A = A_{\mu\nu}$ as a joint distribution $P_{\mu\nu}$ in μ, ν, and $U = U_{\mu\lambda}$, $V = V_{\lambda\nu}$ as conditionally independent distributions $P_{\mu\lambda}$, $P_{\nu\lambda}$ for $\mu|\lambda$, $\nu|\lambda$, given latent variables λ with distribution $P_\lambda = S_\lambda$,

$$P_{\mu\nu} = \sum_\lambda P_{\mu\lambda} \ P_\lambda \ P_{\nu\lambda} \qquad : \qquad \begin{matrix} P_{\mu\nu} = A_{\mu\nu} & P_{\mu\lambda} = U_{\mu\lambda} U_\lambda^{-1} & P_{\nu\lambda} = V_\lambda^{-1} V_{\lambda\nu} \\ P_\lambda = U_\lambda V_\lambda & U_\lambda = \sum_\mu U_{\mu\lambda} & V_\lambda = \sum_\nu V_{\lambda\nu} \end{matrix} \quad . \qquad (6.26)$$

Given initial U, V, and A, X, Y, we propose the following iterative NMF updates for $U, V \to U, V - \varsigma_{U,V} \partial_{U,V} \mathcal{L}$, generalized to the case of boundaries.

Frobenius Norm - Multiplicative Updates (MU):

$$\mathcal{L}_{\text{MU}} = \frac{1}{2} \| Z - XUVY \|^2 \quad \to \quad \begin{cases} U \to \frac{(Z)(XVY)}{(XUVY)(XVY)} \cdot U \\ V \to \frac{(Z)(XUY)}{(XUVY)(XUY)} \cdot V \end{cases} \qquad (6.27)$$

KL Divergence - Multiplicative Updates (KL):

$$\mathcal{L}_{\text{KL}} = -Z \log\left(\frac{XUVY}{Z}\right) - 1_Z Z + 1_{XUVY}(XUVY) \quad \to \quad \begin{cases} U \to \frac{\frac{(Z)}{(XUVY)}(XVY)}{1_{XVY}(XVY)} \cdot U \\ V \to \frac{\frac{(Z)}{(XUVY)}(XUY)}{1_{XUY}(XUY)} \cdot V \end{cases} \qquad (6.28)$$

Hierarchical Alternating Least Squares - Local Updates (HALS):

$$\mathcal{L}_{\text{HALS}} = \frac{1}{2} \| Z - XUVY + (XUVY)_i - (XUVY)_i \|^2 \to \begin{cases} U_i \to \left[\frac{E_i (X_i V_i Y_i)}{(X_i V_i Y_i)(X_i V_i Y_i)} \right]_{+\epsilon} \\ V_i \to \left[\frac{E_i (X_i U_i Y_i)}{(X_i U_i Y_i)(X_i U_i Y_i)} \right]_{+\epsilon} \end{cases} \qquad (6.29)$$

Such updates follow from direct computation of gradients $\partial\mathcal{L}$, and choosing learning rates ς such that $U, V - \varsigma_{U,V} \cdot \partial_{U,V}\mathcal{L} > 0$ to guarantee non-negativity of U, V. Further, in the MU,KL cases, denoting the positive and negative components of gradients as $\partial\mathcal{L} = \partial\mathcal{L}_+ - \partial\mathcal{L}_-$, letting $\varsigma_{U,V} = U, V / \partial\mathcal{L}_{U,V+}$ and $U, V \to \frac{\partial\mathcal{L}_{U,V-}}{\partial\mathcal{L}_{U,V+}} \cdot U, V$ imposes that positive or negative gradients respectively decrease or increase U, V [173, 405]. Here, products of tensors denotes contractions, 1_A denotes elementwise summations over indices unique to A, log, $\cdot$ and / denote elementwise logarithms, multiplication, and division, and $[A]_{+\epsilon}$ denotes elementwise projection to the nearest positive elements $> \epsilon$.

6.3 Results

Here, we perform numerical studies of our proposed methods, including studies of pMPS NMF convergence, and simulations of noisy quantum circuits using POVM-MPS, using the simulation library [6].

6.3.1 pMPS NMF Algorithms

Here, we study the convergence of pMPS NMF. In Fig. 6.7, we show the convergence of various NMF updates (MU,KL,HALS) for the proposed NMF algorithms (Marginal,Joint) as bond dimension χ increases, for a single stochastic update to tensors Q, R with boundaries X, Y within a pMPS, such that, $XQRY \to XAY \approx XUVY$, with fixed dimensions p, location of the updates $i = n/2$, and number of tensors n. We also choose random non-negative SVD initializations of U, V [400], and norm (MU,HALS) and divergence (KL) objectives $\mathcal{L} < \epsilon$ to assess convergence criteria of machine-precision ϵ [173].

There are compromises in practice between the efficiency and overall capabilities of pMPS NMF at representing dense tensors as non-negative factors $A \approx UV$. The Marginal versus Joint algorithms converge similarly for a given update scheme, with the Joint algorithms (purple-red colours), involving p^{n-2} times more matrix elements, generally converging to larger objectives than the Marginal algorithms (blue-green colours).

Regarding the effect of different update schemes, the MU (circle markers) and KL (square markers) updates plateau at non-machine-precision objective values. However, given their global updates to whole U, V tensors, their wall-time to converge remains constant as bond dimension increases.

In contrast, the HALS updates and Joint algorithms (purple-red coloured X markers) plateau at small bond dimensions, and only transition to being capable of representing U, V and converging exponentially with iterations towards machine-precision as $\chi \to p^{n/2}$. However, these capabilities come at the cost of potentially orders of magnitude greater wall-time. HALS updates each of the $p\chi^2$ elements of U, V locally and sequentially per iteration, and each update involves contractions over up to $p^n\chi^3$ elements to obtain the necessary boundary information to yield the joint distribution represented by X, U, V, Y.

As discussed in [398, 405], unlike for SVD algorithms with rigorous dependence on bond dimension χ, NMF algorithms have non-trivial dependence on bond dimension and on the necessary rank of the factors. Given our previous interpretations that χ of pMPS corresponds to the number of latent variables $S = P_\lambda$ that cause $U = P_{\mu|\lambda}, V = P_{\nu|\lambda}$ to be conditionally independent, further analysis and interpretation of the correlations between U, V, given various quasi-stochastic updates $Q, R \to A \approx UV$ are warranted.

These convergence results, for the factorization of a single pMPS transformation, do not provide evidence of its validity in quasi-stochastic settings. In particular, larger numbers of tensors $n > 6$, bond dimensions $\chi \sim p^n$, and $k > 1$ transformations appear to be currently infeasible. Several combinations of NMF update schemes have been considered [394, 396, 399], and can continue to be explored. Algorithm implementations can also be improved from their current JIT-compiled JAX-based implementations [6]. Further, NMF can be NP-hard [404] and potentially will always require $\mathcal{O}(\mathrm{poly}(p^n, \chi))$ resources per iteration to represent joint distributions. BLAS routines for MPS SVD algorithms have also been highly optimized over decades [406, 407], and similarly optimized generalized NMF routines could address these technical challenges. Improvement of pMPS NMF is thus an important future contribution to assess the practicality of these methods.

Further, even sufficiently converged pMPS for any bond dimension may be outside the quantum probability simplex of valid probabilities [21, 31, 353]. Tensor products of n, q-dimensional SIC-POVM's $\mathcal{P} = \{\Pi_\mu\}_{\mu\in[q^2]} \to \mathcal{P}^{\otimes n} = \{\Pi_{\mu\sigma\dots}\}_{\mu,\sigma,\dots\in[q^2]}$, have probabilities $p_\mu \to p_{\mu\sigma\dots}$, overlaps $\chi_{\mu\nu} \to \chi_{\mu\nu}\chi_{\sigma\pi}\dots$, and structure constants $\chi_{\mu\nu\lambda} \to \chi_{\mu\nu\lambda}\chi_{\sigma\pi\tau}\cdots$, of,

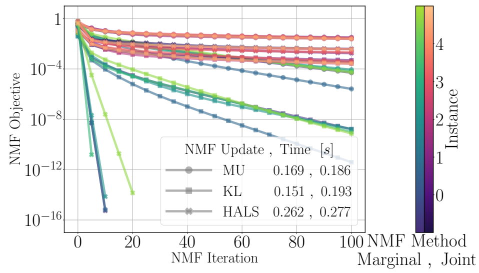
$$p_\mu = \langle\!\langle \Pi_\mu | \rho \rangle\!\rangle \quad , \quad \chi_{\mu\nu} = \langle\!\langle \Pi_\mu | \Pi_\nu \rangle\!\rangle \quad , \quad \xi = q(q+1) \tag{6.30}$$

$$\chi_{\mu\nu} = (1/\xi)(\delta_{\mu\nu} + (1/q)1_\mu 1_\nu) \quad \to \quad \chi^{-1}_{\mu\nu} = \xi\delta_{\mu\nu} - 1_\mu 1_\nu \quad , \quad \chi_\mu = (1/q)1_\mu \tag{6.31}$$

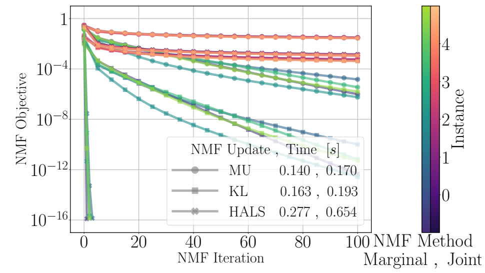
$$\Pi_\mu \Pi_\nu = \chi_{\mu\nu\lambda}\Pi_\lambda \quad \to \quad \chi_{\mu\nu\lambda}1_\lambda = q\chi_{\mu\nu} \quad , \quad \chi^{-1}_\mu = q\,1_\mu \ , \tag{6.32}$$

Valid probabilities are within the simplex formed by the constraints defining pure states,

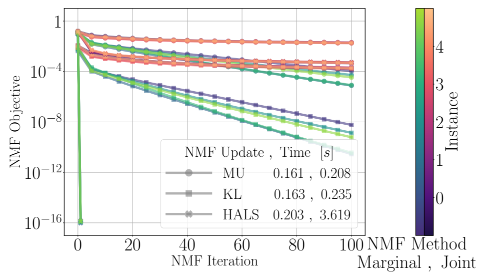
$$0 \leq \mathrm{tr}\big(\rho^3\big) = \mathrm{tr}\big(\rho^2\big) = \mathrm{tr}(\rho) = 1 \leq 1 \ . \tag{6.33}$$



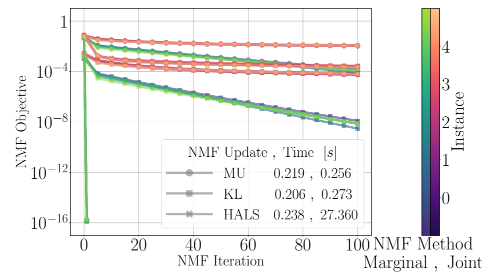
(a) Bond dimension $\chi = 2$



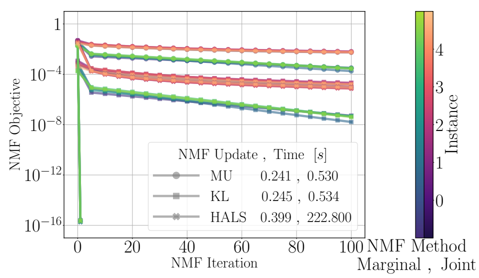
(b) Bond dimension $\chi = 4$



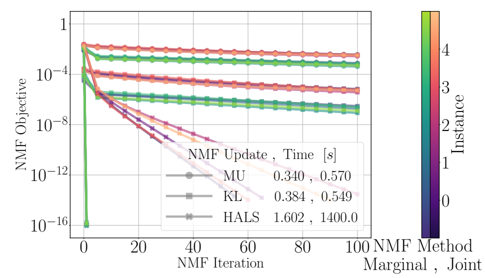
(c) Bond dimension $\chi = 8$



(d) Bond dimension $\chi = 16$



(e) Bond dimension $\chi = 32$



(f) Bond dimension $\chi = 64$

Figure 6.7: **pMPS NMF convergence**. Convergence of pMPS NMF objectives over NMF iterations, for random tensors $XQRY \to XAY \approx XUVY$, with random stochastic transforms $L : QR \to A$, for $Q, R, U, V \in \mathbb{R}^{\chi\times p\times\chi}$, $X \in \mathbb{R}^{x\times\chi}$, $Y \in \mathbb{R}^{\chi\times y}$, $A \in \mathbb{R}^{\chi\times p\times p\times\chi}$, $L \in \mathbb{R}^{(p\times p)\times(p\times p)}$, for $p = 2^2$, $x = p^i$, $y = p^{n-i-2}$, $i = n/2-1$, $p = 2^2$, $n = 6$. Different bond-dimension $\chi \leq p^{n/2}$ (a-f), MU,KL,HALS update schemes (markers), NMF Marginal (blue-green) and Joint (purple-red) algorithms, and 5 random instances (colours) are compared. Average CPU wall-times for instances to converge, for Marginal and Joint methods, are shown in the legend. MU and KL updates plateau above, and HALS updates converge exponentially with iteration towards machine-precision objectives, and decrease with bond dimension. Wall-times for MU and KL updates, being global updates, are approximately constant with bond dimension. Wall-times for HALS updates, being local updates, increase by orders of magnitude with bond dimension.

For $n = 1$, we see that q-dimensional states ρ and probabilities p_μ must satisfy [31, 353],

$$\rho = \xi p_\mu \Pi_\mu - I \tag{6.34}$$
$$1_\mu p_\mu = 1 \tag{6.35}$$
$$1 \leq \xi p_\mu p_\mu \leq 2 \tag{6.36}$$
$$1 \leq 2\xi p_\mu p_\mu - 1 \leq \xi^2 \chi_{\mu\nu\lambda} p_\mu p_\nu p_\lambda \leq 2\xi p_\mu p_\mu \leq 4 \ . \tag{6.37}$$

For $n = 2$, we derive that q^2-dimensional states ρ and probabilities $p_{\mu\sigma}$ must satisfy,

$$\rho = \xi^2 p_{\mu\sigma} \Pi_{\mu\sigma} - \xi p_\mu \Pi_\mu - \xi p_\sigma \Pi_\sigma + I \tag{6.38}$$
$$1_{\mu\sigma} p_{\mu\sigma} = 1 \tag{6.39}$$
$$1 \leq \xi(p_\mu p_\mu + p_\sigma p_\sigma) - 1 \leq \xi^2 p_{\mu\sigma} p_{\mu\sigma} \leq \xi(p_\mu p_\mu + p_\sigma p_\sigma) \leq 4 \tag{6.40}$$
$$\begin{aligned} -13 - 2\xi^2 p_\mu p_\sigma p_{\mu\sigma} &\leq \xi^4 \chi_{\mu\nu\lambda} \chi_{\sigma\pi\tau} p_{\mu\sigma} p_{\nu\pi} p_{\lambda\tau} - 2\xi^3 (\chi_{\mu\nu\lambda} p_{\mu\sigma} p_\nu p_{\lambda\sigma} + \chi_{\sigma\pi\tau} p_{\mu\sigma} p_\pi p_{\mu\tau}) \\ 4 - 2\xi^2 p_\mu p_\sigma p_{\mu\sigma} &\geq \ . \end{aligned} \tag{6.41}$$

Such non-trivial cubic ξ-dependent constraints appear difficult to impose within NMF algorithms, further obscuring the robustness of these methods.

6.3.2 POVM-MPS Simulations

Here, we study the capabilities of POVM-MPS methods at simulating noisy dynamics.

In particular, as per Fig. 6.8, we study random quantum channels $\Lambda_\gamma^{(k)}$, consisting of k layers of one-dimensional brickwork layouts of Haar random nearest-neighbour unitaries $\mathcal{U}$, with local depolarizing noise $\mathcal{N}_\gamma$ with noise scales γ, and evolve initial Haar random product pure states $\psi = \otimes_{i \in [n]} \psi_i$ with such quantum channels, where,

$$\begin{aligned} \mathcal{U} &= \textstyle\prod_{i,j \in [\substack{n \\ \text{brick}}]} \ U_{ij} \ : \ U_{ij} \ \sim \ \mathcal{U}(q^2) \\ \mathcal{N}_\gamma &= \otimes_{i \in [n]} \ \mathcal{N}_{\gamma i} \ : \mathcal{N}_{\gamma i}(\rho) = (1-\gamma)\rho + \gamma \frac{\mathrm{tr}_i(\rho)}{q} \otimes I_i \ . \end{aligned} \tag{6.42}$$

Here, we simulate $k \in \{0, \ldots, 50\}$ layers, noise scales $\gamma \in \{10^{-4}, \ldots, 5 \cdot 10^{-1}\}$, system sizes $n \in \{4, 6, 8\}$, $q = 2$-dimensional qubits, bond dimensions $\chi \leq q^{2n/2}$, subsystems Γ of size $n/2$. Quantities are plotted as averages and standard deviations over $m \leq 64$ circuit instances. We also use local Tetrad SIC-POVM measurement operators $\mathcal{P} = \{\Pi\}$ for the POVM-MPS architecture [146, 147].

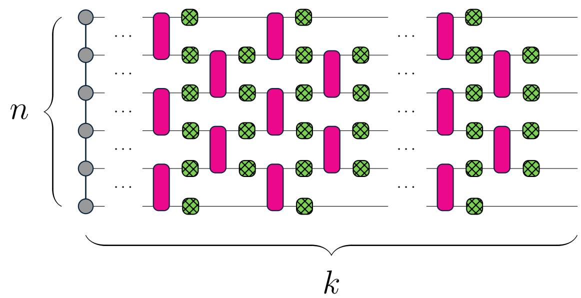


Figure 6.8: **Brickwork quantum circuits**. k layers of n component brickwork layouts of unitaries (pink) and local noise (green).

Given these noisy quantum systems, let us assess the robustness of our POVM-MPS methods at accurately representing various fidelity and entanglement measures, as a function of system parameters of bond dimension χ, noise scale γ, circuit depth k, and system size n. Given initial states ψ and subsystems Γ, simulations yield the evolved states,

$$\psi \quad \rightarrow \quad \rho_{\gamma}^{(k)} = \Lambda_{\gamma}^{(k)}(\psi) \quad \rightarrow \quad \rho_{\gamma\chi}^{(k)} = \Lambda_{\gamma\chi}^{(k)}(\psi) \tag{6.43}$$

$$\Psi_{\gamma\chi}^{(k)} = \frac{|\rho_{\gamma\chi}^{(k)}\rangle\!\rangle\langle\!\langle\rho_{\gamma\chi}^{(k)}|}{\langle\!\langle\rho_{\gamma\chi}^{(k)}|\rho_{\gamma\chi}^{(k)}\rangle\!\rangle} \quad \rightarrow \quad \Psi_{\gamma\chi\Gamma}^{(k)} = \mathrm{tr}_{\bar{\Gamma}}\big(\Psi_{\gamma\chi}^{(k)}\big) \ . \tag{6.44}$$

Simulating random quantum channels using the POVM-MPS formalism, we find in fact pMPS NMF methods (NMF POVM-MPS) are numerically unstable and do not converge, or require infeasible computation per channel, even after a single sweep of $\leq n$ individual channels in one layer of a circuit. Although local channels act with no overheard, only changing local tensors without changing bond dimensions, nearest-neighbour channels require factorizations and increased bond dimension to restore the localized tensor structure. Even at maximum bond dimension, these studies indicate our pMPS NMF methods in these settings can accumulate non-trivial errors, resulting in invalid non-positive probabilities.

Therefore, given the lack of convergence of pMPS NMF algorithms, let us use standard pMPS SVD algorithms, which decompose nearest-neighbour tensors $QR \rightarrow A \approx UV$ using SVD factorizations and without use of boundary tensors [164]. Within such approximations, the absolute value of spectra of any states $\rho \rightarrow |\rho|$, $\Psi \rightarrow |\Psi|$ are used in calculations, such as entropies, which require positive spectra. As such, let us turn to assessing the robustness of SVD-based POVM-MPS (SVD POVM-MPS) methods at representing various quantities of interest, even if such methods also yield potentially unphysical states.

First, let us investigate classical fidelity measures of evolved probabilities $p^{(k)}_{\gamma\chi}$,

$$\mathcal{L}^{(k)}_{\gamma\chi} = 1 - \sqrt{p^{(k)}_{\gamma\chi}} \cdot \sqrt{p^{(k)}_{\gamma}} \ . \tag{6.45}$$

As per Fig. 6.9, the infidelities of SVD POVM-MPS increase sharply, then plateau with circuit depth k, and the saturated values decrease polynomially with bond dimension χ, and with noise scale γ. Non-machine-precision infidelities at close to maximum bond dimension $\chi < d$, that sharply improve only at $\chi = d$ or $\gamma \to 1$, suggest SVD based methods are inherently more robust at capturing noisy dynamics than random unitary dynamics.

Second, let us investigate operator entanglement measures within subsystems Γ,

$$\mathcal{Q}^{(k)\Gamma}_{\gamma\chi} = \ -\frac{1}{2}\frac{1}{\log(d)}\mathrm{tr}\left(\Psi^{(k)}_{\gamma\chi\Gamma}\log\left(\Psi^{(k)}_{\gamma\chi\Gamma}\right)\right) \ . \tag{6.46}$$

As per Fig. 6.10, operator entanglement of SVD POVM-MPS, for subsystems of size $n/2$, is shown to exhibit the expected trends [401] of smoothly increasing then decreasing over circuit depth k. Maximum entanglement occurs at intermediate circuit depths, and is shown to increase logarithmically with bond dimension χ, to decrease logarithmically with noise scale γ, and to increase with system size n. Such SVD POVM-MPS entanglement scaling is identical to that of MPO methods [161, 162], and unlike LPDO methods [166, 170], decreases instead of plateaus in the noisy regime. Although SVD POVM-MPS states are potentially non-physical, with non-positive probabilities, and exponentially worse fidelity with circuit depth, they appear capable of representing entanglement behaviours.

To investigate SVD POVM-MPS entanglement more thoroughly, in Fig. 6.11, we show the maximum operator entanglement over circuit depth as a function of bond dimension χ or noise scale γ. Such plots indicate intuitive, smooth scaling with these parameters, and confirm the logarithmic scaling relationships, suggesting SVD POVM-MPS can represent one-dimensional area-law entangled states [261]. Further, as per Fig. 6.11a, at bond dimension $\log(\chi) = n$, entanglement curves saturate at a maximum value. Within errorbars, this value is the analytical average entanglement entropy of Haar random pure states [186],

$$\lim_{\chi\to d}\mathcal{Q}^{(k)l}_{\gamma\chi} \ \to \ \mathcal{S}^{l}_{|\psi\rangle} \ = \ \frac{1}{\log(d)}\sum_{i>l}^{d}\frac{1}{i} - \frac{1}{\log(d)}\frac{\binom{l}{2}}{d} \ \leq \ 1 - \frac{\log(l)}{\log(d)} - \frac{\binom{l}{2}}{d} \ . \tag{6.47}$$

Finally, as per Fig. 6.11b, at large noise scales $\gamma \to 1$, entanglement curves shift from being independent of noise scale, with offsets by bond dimension, to decreasing and collapsing to a single curve. Such collapse suggests universal noisy behaviour as the subsystems become less entangled with each other, and more entangled with their environment.

Third, let us investigate quantum discord measures with respect to measurements $\mathcal{P}$,

$$\mathcal{D}^{(k)\Gamma|\mathcal{P}}_{\gamma\chi} = \left| \mathcal{I}^{\Gamma}_{\rho^{(k)}_{\gamma\chi}} - \mathcal{I}^{\Gamma|\mathcal{P}}_{\rho^{(k)}_{\gamma\chi}} \right| . \tag{6.48}$$

As per Fig. 6.12, quantum discord of SVD POVM-MPS, for subsystems of size $n/2$, and using the same measurement set $\mathcal{P}$ as is used to represent the states, is shown to smoothly increase then decrease over circuit depth k. Such behaviour of quantum discord, being a measure of the effects of partial information about entangled subsystems, appears similar to the behaviour of operator entanglement. However, as depicted by the maximum quantum discord curves in Fig. 6.13, quantum discord is significantly overestimated at sub-maximum bond dimensions $\chi < d$. Any non-positive spectra in SVD POVM-MPS appear to have a greater effect on the more intricate quantum discord than on operator entanglement.

Finally, let us investigate the ratio of negative to positive spectra $\{\lambda^{(k)}_{\gamma\chi}\}$ of states, to provide intuition for any invalid behaviours due to unphysical SVD POVM-MPS updates,

$$\mathcal{Z}^{(k)}_{\gamma\chi} = \left| \frac{\sum_{\lambda<0} \lambda^{(k)}_{\gamma\chi}}{\sum_{\lambda\geq 0} \lambda^{(k)}_{\gamma\chi}} \right| \quad : \quad \rho^{(k)}_{\gamma\chi} = \sum_{\lambda} \lambda^{(k)}_{\gamma\chi} \; \psi_{\lambda^{(k)}_{\gamma\chi}} . \tag{6.49}$$

As per Fig. 6.14, for sub-maximal bond dimensions and all noise scales, negative spectra immediately appear at small circuit depths, and saturate to be almost equal in proportion to positive spectra. As circuit depth increases, such large negative spectra ratios remain for small noise scales, but decrease to zero for large noise scales.

Ultimately, SVD POVM-MPS methods with sub-maximum bond dimension, given the current infeasibility of NMF POVM-MPS, are shown to be capable of representing some measures more accurately than others, even with non-zero infidelity. The immediate presence of negative spectra at small circuit depths across all noise scales, that decay to zero for large enough circuit depths and noise scales, contribute to infidelities similarly increasing immediately at small circuit depths, and improving with increased noise scale. The use of absolute values of negative spectra for total entanglement entropies $\mathcal{S}_{\rho^{(k)}_{\gamma\chi}}$ in quantum mutual information calculations, likely significantly contribute to the quantum discord being greatly overestimated at large depths and small noise scales. Intriguingly, the presence of negative spectra appear to affect operator entanglement entropies $\mathcal{S}^{n/2}_{\Psi^{(k)}_{\gamma\chi}}$, less than total entanglement entropies $\mathcal{S}_{\rho^{(k)}_{\gamma\chi}}$, possibly from fortuitous cancellations occurring.

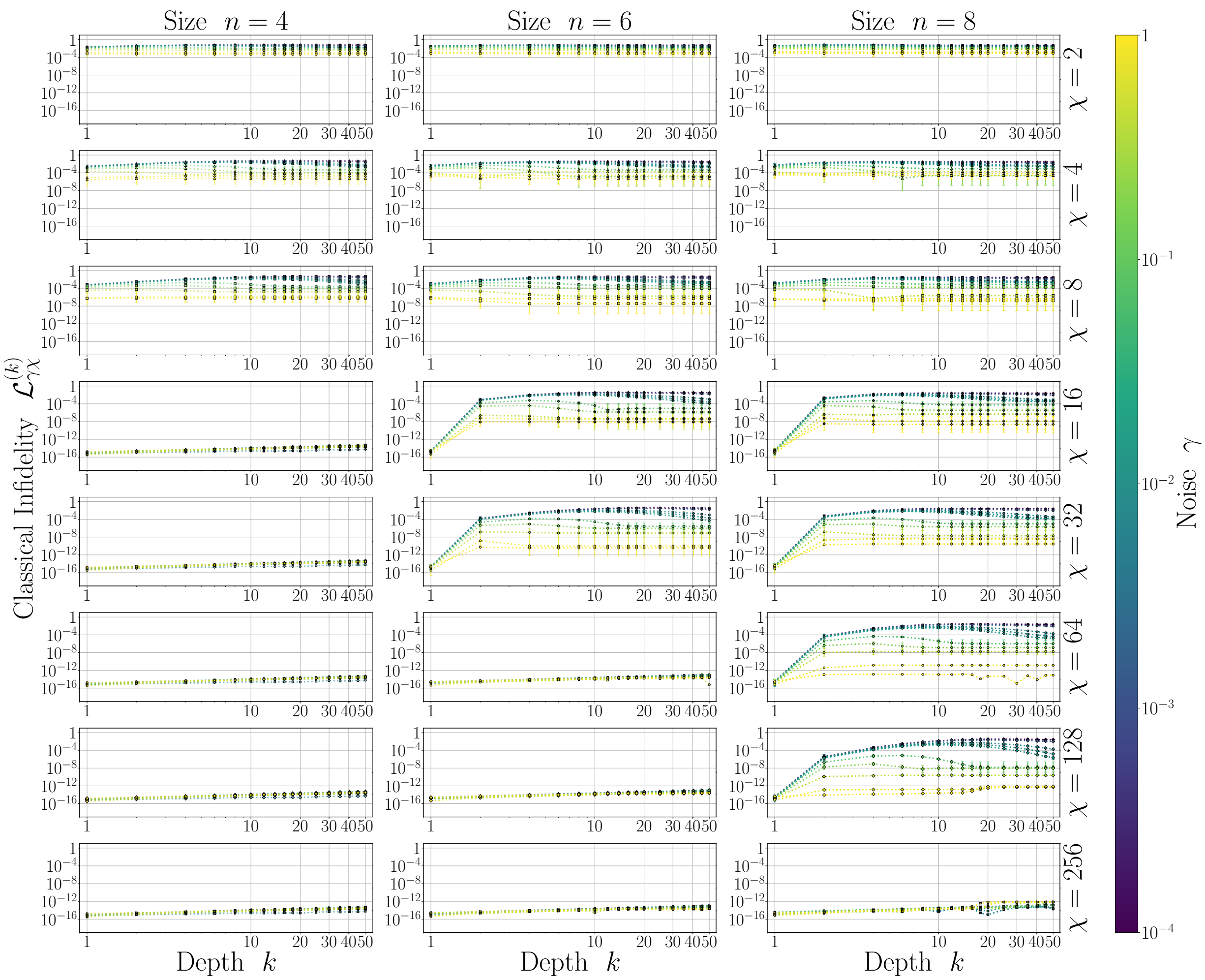


Figure 6.9: **SVD POVM-MPS classical infidelity**. Classical infidelity of $p^{(k)}_{\gamma\chi}$ versus $p^{(k)}_{\gamma}$ for Haar random brickwork and local depolarization noise circuits, as a function of circuit depth k, for various noise scales $\gamma \in \{10^{-4}, \ldots, 5 \cdot 10^{-1}\}$ (colours), bond dimensions $\chi \in \{2, 4, 8, 16, 32, 64, 128, 256\}$ (rows, top to bottom), system sizes $n \in \{4, 6, 8\}$ (columns), and over $m = 64$ instances. Infidelities decrease polynomially with bond dimension and noise scale, and saturate at intermediate circuit depth. SVD POVM-MPS are shown to be more capable of representing noisy dynamics than unitary dynamics.

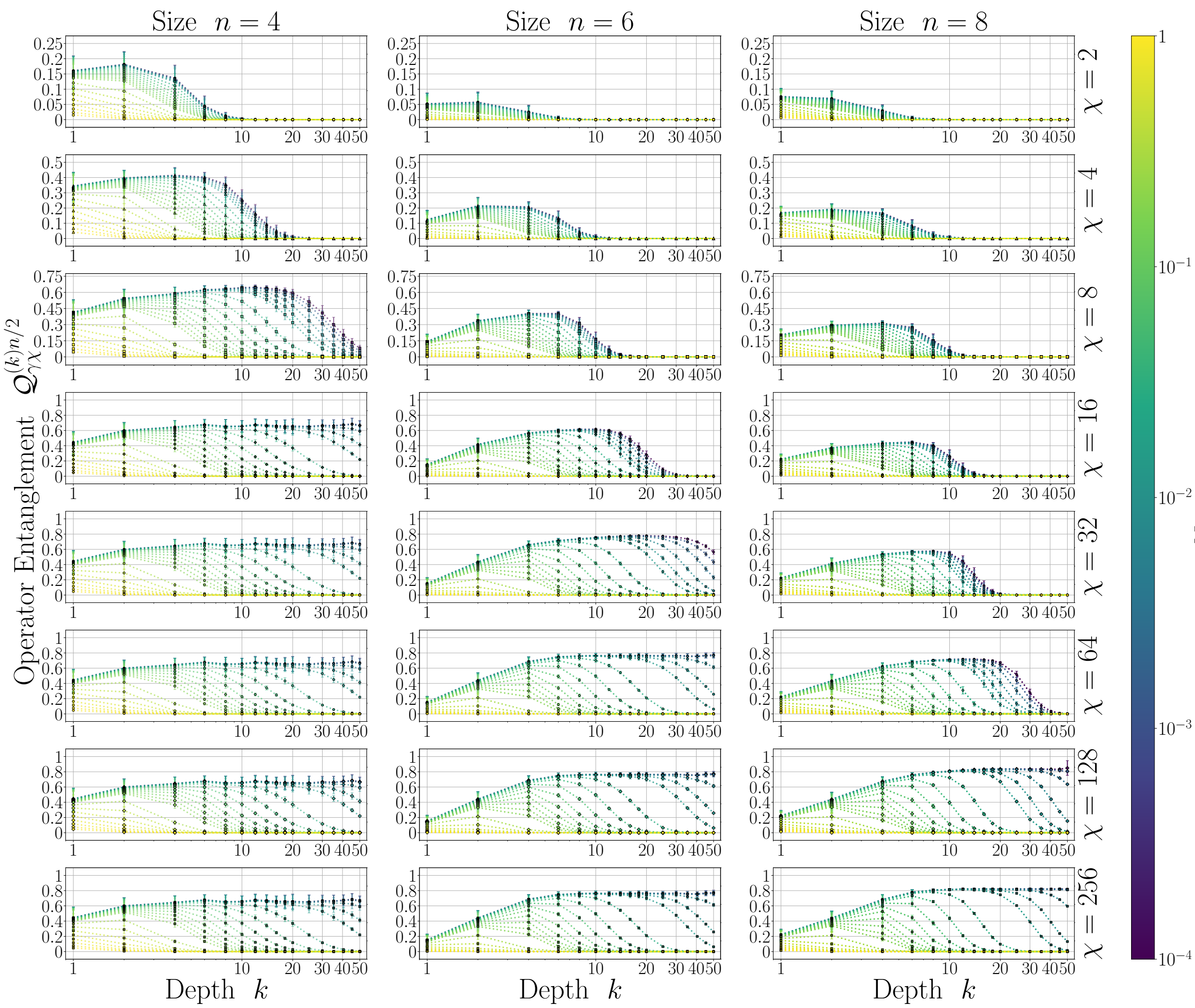
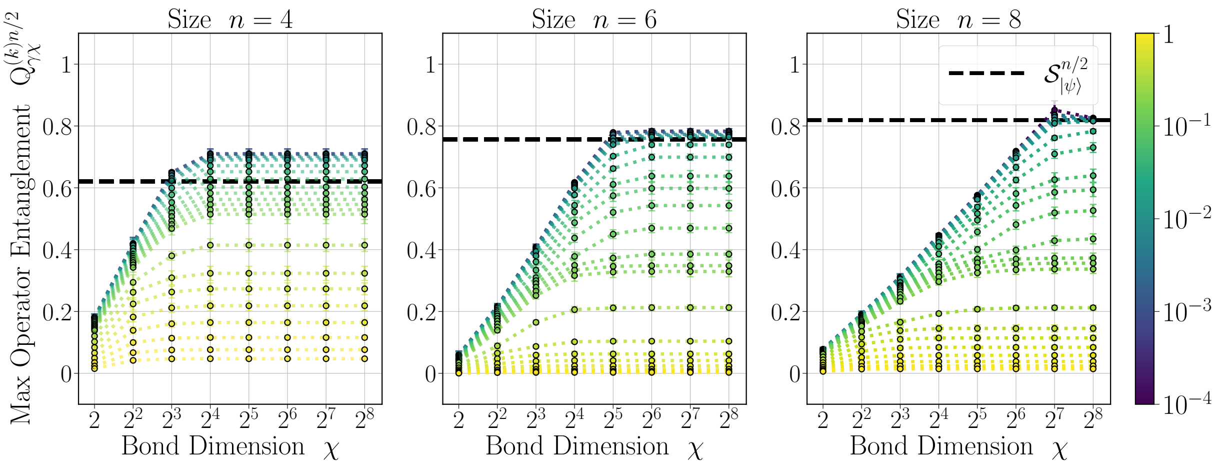
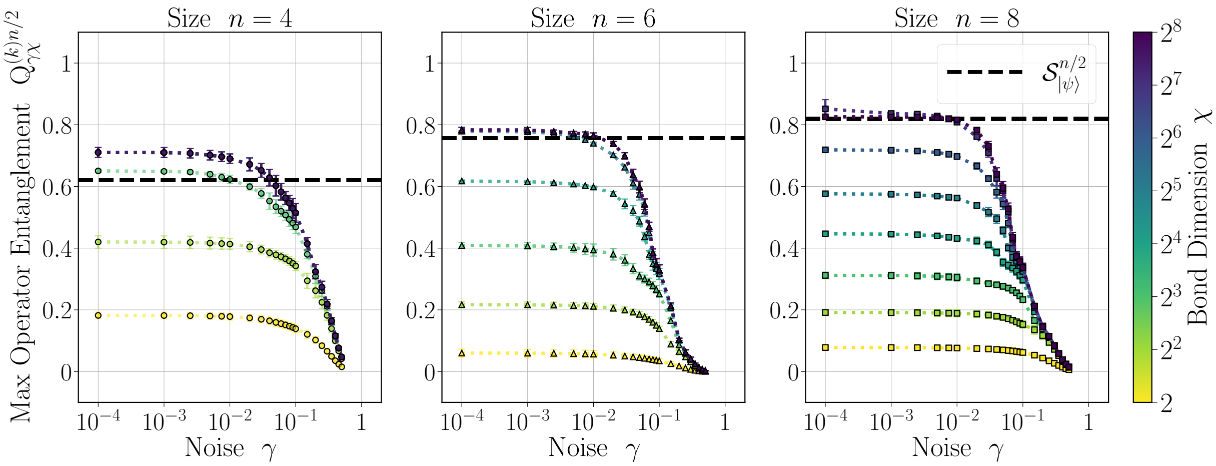


Figure 6.10: **SVD POVM-MPS operator entanglement**. Operator entanglement of $\rho_{\gamma\chi}^{(k)}$ for a subsystem of size $n/2$ for Haar random brickwork and local depolarization noise circuits, as a function of circuit depth k, for various noise scales $\gamma \in \{10^{-4}, \ldots, 5 \cdot 10^{-1}\}$ (colours), bond dimensions $\chi \in \{2, 4, 8, 16, 32, 64, 128, 256\}$ (rows, top to bottom), and system sizes $n \in \{4, 6, 8\}$ (columns), and over $m = 64$ instances. Entanglement smoothly rises then falls with circuit depth in respective unitary and noisy regimes, with peak entanglement increasing logarithmically with increasing bond dimension and with decreasing noise scale. POVM-MPS are shown to be capable of representing qualitative behaviour of entanglement with potentially non-physical SVD updates.



(a) Maximum operator entanglement versus bond dimension χ, for various noise scales γ.



(b) Maximum operator entanglement versus noise scales γ, for various bond dimensions χ.

Figure 6.11: **SVD POVM-MPS maximum operator entanglement**. Maximum operator entanglement of $\rho^{(k)}_{\gamma\chi}$ for a subsystem of size $n/2$ for Haar random brickwork and local depolarization noise circuits, over circuit depth k, as a function of bond dimension χ (a) and noise scale γ (b) for system sizes $n \in \{4, 6, 8\}$ (columns), and over $m = 64$ instances. (a): Maximum entanglement increases smoothly and logarithmically with bond dimension, before saturating at $\chi = d$, with plateau locations and values decreasing with noise scale. (b): Maximum entanglement collapses to a single curve that decreases logarithmically with noise scale. Dashed black line indicates analytical average entanglement entropy $\mathcal{S}^{l}_{|\psi\rangle}$ of subsystems of size $n/2$ for Haar random pure states, and corresponds with peak entanglement of noiseless states.

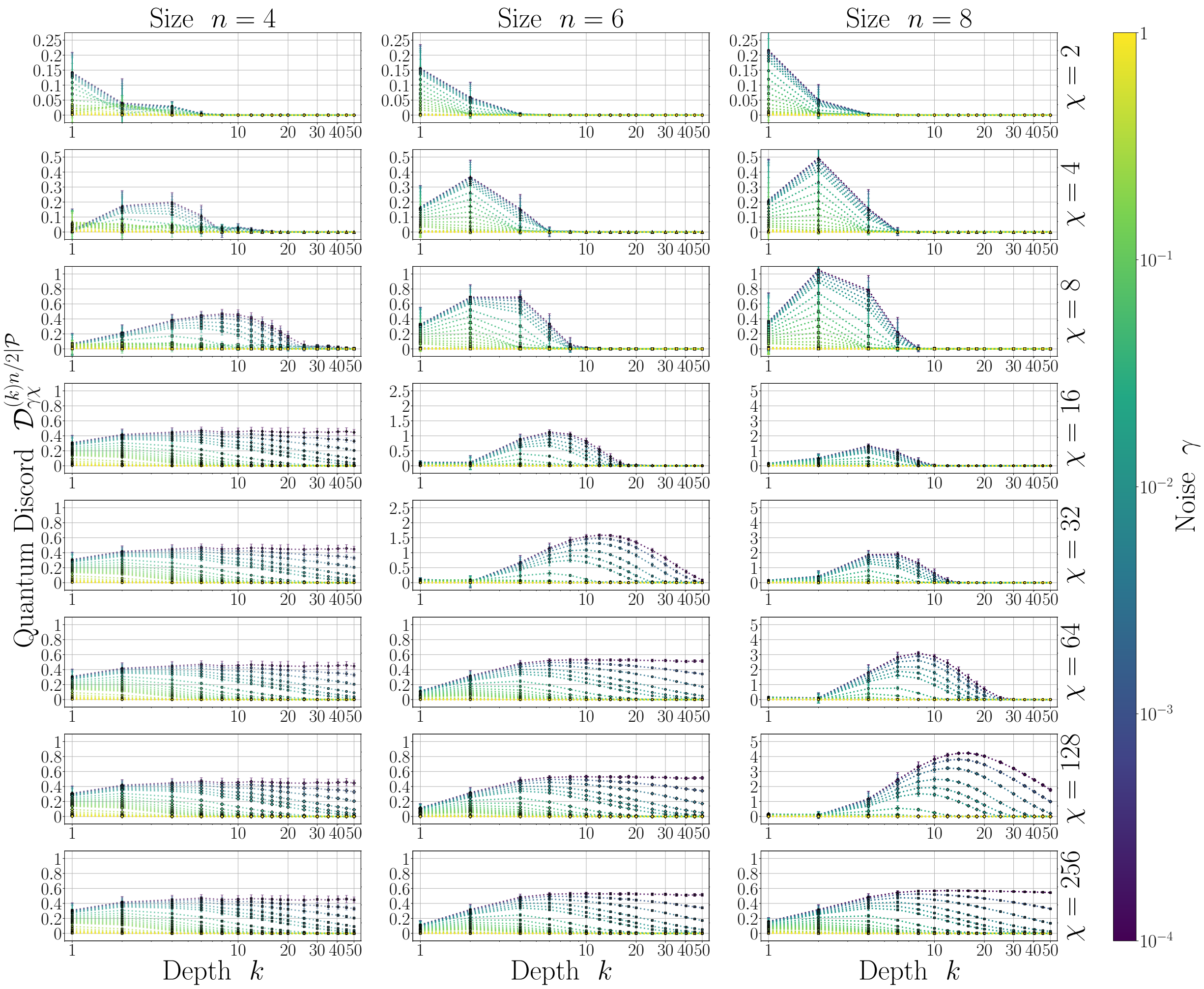
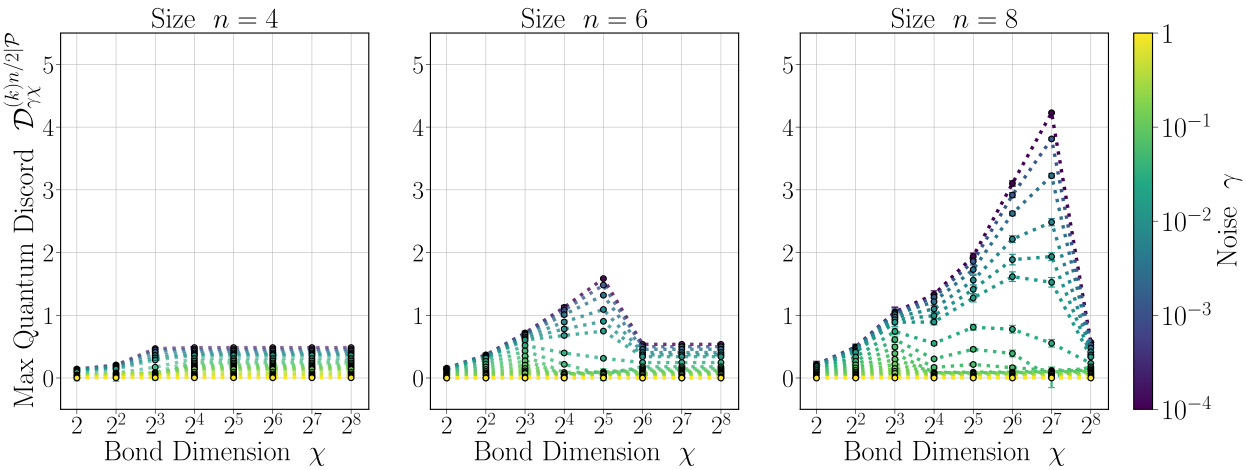
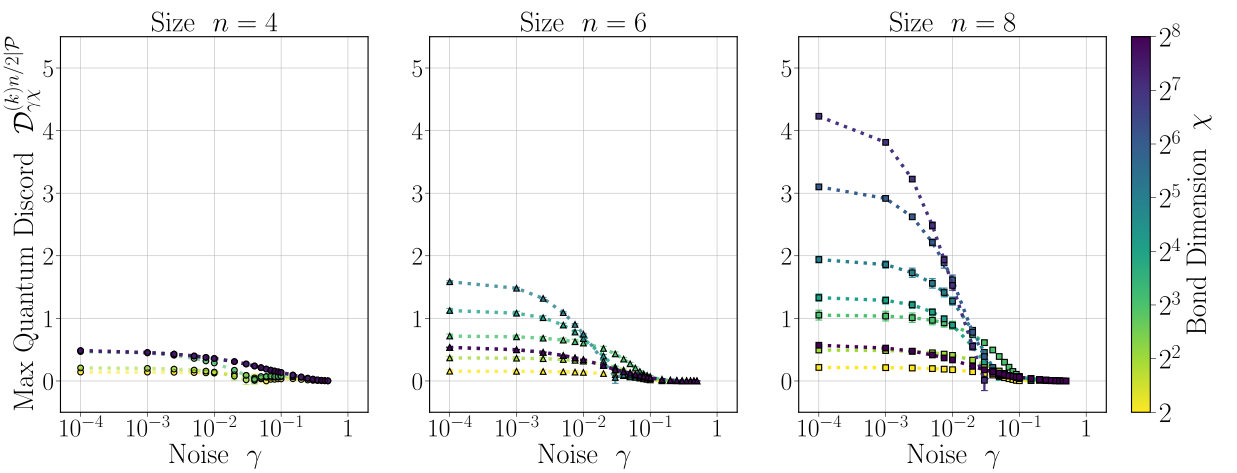


Figure 6.12: **SVD POVM-MPS quantum discord**. Quantum discord of $\rho_{\gamma\chi}^{(k)}$ for a subsystem of size $n/2$ for Haar random brickwork and local depolarization noise circuits, as a function of circuit depth k, for various noise scales $\gamma \in \{10^{-4}, \ldots, 5 \cdot 10^{-1}\}$ (colours), bond dimensions $\chi \in \{2, 4, 8, 16, 32, 64, 128, 256\}$ (rows, top to bottom), and system sizes $n \in \{4, 6, 8\}$ (columns), and over $m = 64$ instances. Similar to operator entanglement, discord smoothly rises then falls with circuit depth in respective unitary and noisy regimes, with peak values increasing logarithmically with bond dimension and with noise scale. Given the exact behaviour at maximum bond dimension $\chi = d$, SVD POVM-MPS are shown to overestimate discord at sub-maximum bond dimensions $\chi < d$.



(a) Maximum quantum discord versus bond dimension χ, for various noise scales γ.



(b) Maximum quantum discord versus noise scales γ, for various bond dimensions χ.

Figure 6.13: **SVD POVM-MPS maximum quantum discord**. Maximum quantum discord of $\rho^{(k)}_{\gamma\chi}$ for a subsystem of size $n/2$ for Haar random brickwork and local depolarization noise circuits, over circuit depth k, as a function of bond dimension χ (a) and noise scale γ (b) for system sizes $n \in \{4, 6, 8\}$ (columns), and over $m = 64$ instances. (a): Maximum discord increases monotonically and logarithmically with bond dimension, before decreasing to exact values at $\chi = d$, with unphysical peaks that decrease with noise scale. (b): Maximum discord collapses to a single curve that decreases logarithmically with noise scale, with exact curves at $\chi = d$ (dark purple) shown to be overestimated by large, sub-maximum bond dimensions $0 \ll \chi < d$.

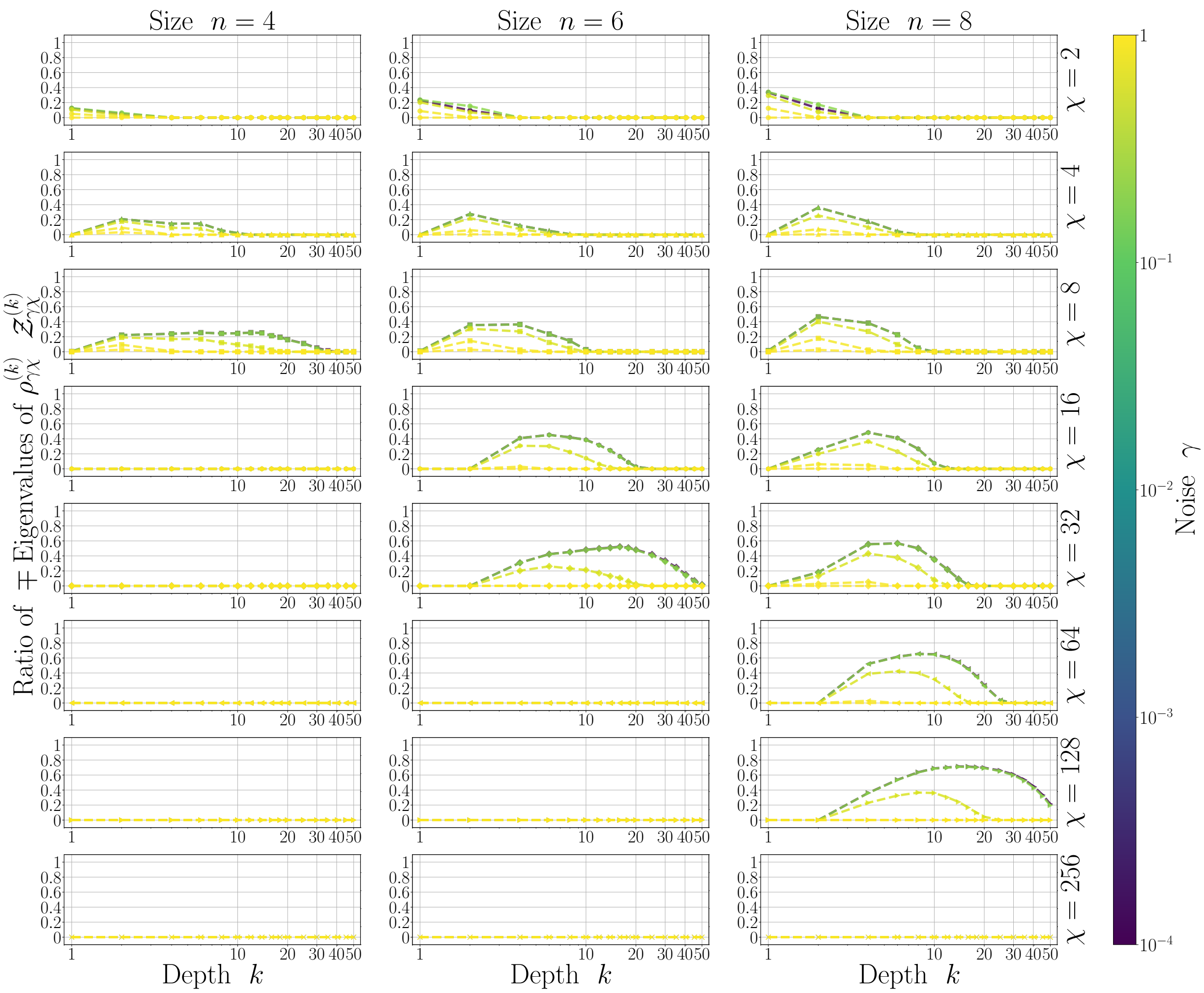


Figure 6.14: **SVD POVM-MPS spectrum ratio**. Ratio of negative to positive spectra of $\rho_{\gamma\chi}^{(k)}$ for Haar random brickwork and local depolarization noise circuits, as a function of circuit depth k, for various noise scales $\gamma \in \{10^{-18}, 10^{-4}, 10^{-2}, 5 \cdot 10^{-2}, 10^{-2}\}$ (colours), bond dimensions $\chi \in \{2, 4, 8, 16, 32, 64, 128, 256\}$ (rows, top to bottom), and system sizes $n \in \{4, 6, 8\}$ (columns), and over $m = 32$ instances. At small noise scales, negative spectra ratios appear at small circuit depths, and remain almost equal in proportion to positive spectra as depth increases. At large noise scales, negative spectra ratios remain small, but increase with depth before decreasing to zero. The non-physicality of SVD POVM-MPS is thus shown to decrease with noise, suggesting a noisy regime exists where SVD updates form correct non-negative factorizations, resulting in physical quantum states.

6.4 Discussion

Here, we propose a classical simulation method (POVM-MPS) that uses informationally complete measurement probabilities and stochastic matrix product states (pMPS) to represent quantum systems undergoing non-unitary dynamics. Such POVM-MPS methods, used to represent probabilities undergoing quasi-stochastic transformations, require distinct algorithms from conventional tensor network-based simulation methods, used to explicitly represent quantum states. Here, we propose generalized variants of non-negative matrix factorization (NMF) that require local and global information to appropriately update and factorize distributions upon undergoing dynamics.

These NMF factorization algorithms are intuitive and promising, but are shown to be difficult to implement in practice, and require further development to fully understand the potential of the POVM-MPS method. In their place, we use conventional singular value decomposition (SVD) algorithms as local update schemes. We subsequently study their robustness at simulating various probability infidelity, operator entanglement, and quantum discord measures of interest, for brickwork quantum circuits with depolarizing noise, across system sizes, circuit depths, noise scales, and tensor bond dimensions.

Infidelities of inexact simulations are shown to immediately increase with circuit depth, but improve with increased noise scale. Operator entanglement is shown to increase smoothly with increasing bond dimension and with decreasing noise scale, and corresponds with analytical average entanglement entropy results in the noiseless case. Quantum discord is shown to be overestimated by an amount that scales logarithmically with sub-maximum bond dimension, and inversely with noise scale. Finally, negative spectra in quantum states, due to the approximate SVD algorithms, are shown to explicitly increase in proportion to positive spectra at small noise scales, with the negative spectra disappearing at large circuit depths and large noise scales. Such results suggest the viability and robustness of SVD POVM-MPS as efficient simulators in noisy settings, with anticipated greater robustness across noise scales, given improved factorization implementations.

Overall, we find the performance of SVD POVM-MPS method to be in good agreement with existing noisy quantum circuit simulation methods [161, 162, 166, 170], in particular with respect to operator entanglement scaling smoothly and logarithmically with bond dimension and with noise scale. While comparing methods, one unanswered question is how to precisely interpret the POVM-MPS bond dimension between components of the joint measurement distribution, and its ability to represent both unitary and noise effects. Such a bond dimension differs from that of matrix product operator (MPO) methods [161], with their single bond dimension representing correlations between components of density matrices, and locally purified density operator (LPDO) methods [166], with their two bond dimensions representing correlations with neighbouring components and with environments. Similarly, although during NMF factorizations of pMPS we have access to yet-to-be-interpreted distributions of latent variables [173], as the equivalent of singular values in SVD factorizations of MPS, their utility is not immediately apparent. Having access to singular values greatly simplifies calculations such as for entanglement entropies [160], thus it is important to understand whether knowledge of latent variables can also simplify important calculations.

Further, many classical simulation methods, such as sampling methods [153–157], are instantly complicated or their accuracy is reduced in the presence of even local noise. In contrast, POVM-MPS methods, given its tensor network architecture, have the distinct advantage of being able to simulate arbitrary local quantum channels without any overhead or increase in bond dimension. At the same time, such a tensor network architecture requires potentially infeasible global information about the corresponding distribution to simulate even nearest-neighbour quantum channels. Ultimately, it remains an open question of why simulating tensor networks over complex fields [408, 409], with limited constraints [259], allowing for SVD factorization algorithms, appears to be significantly more stable and convergent in practice than simulating tensor networks over non-negative fields with probability simplex constraints [31], requiring non-negative factorization algorithms.

In conclusion, our proposed methods and the current questions surrounding their viability as an efficient classical simulator of open quantum systems, have opened up several future directions of study. In addition to explicitly improving NMF implementations to the level of BLAS SVD routines, it will be important to understand whether other factorization routines exist for distributions, given quasi-stochastic dynamics [179]. Further, it is seemingly necessary to have global information when performing local updates to distributions, which appears to be unique to the non-negative probability setting in contrast to the complex amplitude setting. This suggests it is essential to assess how much information about joint distributions is necessary to update marginal distributions, which is reminiscent of the quantum marginal problem [410].

Finally, it remains perplexing that non-physical states, with their negative spectra that result from inexact SVD POVM-MPS updates, still seemingly describe operator entanglement behaviours well when the spectra are made positive, with smooth monotonic behaviour across bond dimension and noise scale. Interpretations of such non-physical states, their causes, their relation to quasi-probability distributions [179, 334] and conditional mutual information within systems [138], and their properties, all warrant future study. POVM-MPS, and implications of its successful implementation on classical simulability of noisy quantum systems are captivating prospects, that we are excited to pursue.

Chapter 7

Discussion

7.1 Summary

In this thesis, we present systematic analytical and numerical analyses of the effects of non-unitary dynamics on properties of quantum systems. By focusing on learning phenomena during optimization, generalized statistics of ensembles of random quantum channels, distributions of measurement probabilities, and classical simulation of non-unitarity, we provide rigorous and empirical studies of these phenomena.

First, we have learned about practical implications of noise and non-unitarity in variational quantum algorithms. By investigating universal effects of noise throughout optimization of unitary compilation and state preparation tasks in finite dimensional Hilbert spaces, we have shown that overparameterization phenomena are robust to experimental constraints. Given convergent optimization, infidelities decrease exponentially with circuit depth, up until a noise-induced critical depth that scales logarithmically with noise scale. General parameterized systems are thus shown to be capable of suppressing entropic behaviour imposed by their environment [102, 277]. Generic problems may be prevented by noise from achieving arbitrary accuracy, but fortunately are capable of being optimized to a non-trivial best-case objective across all depths and noise scales.

Unfortunately, in asymptotically large Hilbert spaces, or spaces coupled to large environments, or significantly noisy settings, we have shown various forms of concentration phenomena occur, potentially masking such robustness. These derivations explain and generalize what was previously attributed as strictly unitary or noise-induced phenomena, to be general channel-design-induced phenomena. Averages over a sufficient number of concatenations of ensembles of quantum channels, or ensembles of unitaries and unital noise, are shown to converge in various norm distances to the globally depolarizing channel, with some small non-unital perturbations. Such convergence is shown to occur when circuit depths scale linearly in system size, logarithmically in noise scale, and linearly in the order of ensemble moments being considered. Such convergence further imposes that channel-design-induced barren plateaus in objective landscapes occur with high probability.

Concurrently, we have shown empirically that distributions of expectation values of sets of symmetric measurement operators in local unital noisy settings, converge towards a depolarized distribution. In the case of non-symmetric measurement operators, such as in the case of noisy measurements, concentration also occurs, but towards multi-modal distributions, with sharp peaks that scale in number and in sharpness with system size. This multi-modality brings in to question whether various concentration and complexity results related to cross-entropy [23, 62, 133] and total variational distance benchmarks [105, 136, 143], where uni-modality is often implied, still hold in these more general settings.

More positively, in contrast to these unital noise-induced concentration results, we have shown even local non-unital noise prevents such inherent concentration. This non-unital impedance supplements previous results regarding noise-induced barren plateaus [276, 336], random circuit sampling [105], and ideas of non-unital noise allowing quantum simulation via effective low-depth circuits [131]. Relevant experimental demonstrations have also shown non-unitality can allow for open quantum simulation [411, 412]. Noise is thus not an entirely undesirable or passive phenomena, but can drive systems towards or away from concentration properties. Overall, our findings quantify detrimental effects of non-unitarity on variational algorithms. Any non-unitary should be used judicially.

Second, we have developed general tools to exactly compute statistics of ensembles of random quantum channels. Such formalisms allow hierarchies between channel ensembles to be derived, based on invariance properties rather than uniformity of measure. Notably, an ensemble of solely the depolarizing channel is found to be at the base of this hierarchy. By using concatenations of moment operator formalisms, independent of initial states or measurements, it becomes readily apparent how average channels spread information across copies of Hilbert spaces. Subsequently, these insights allow us to evaluate distributions of expectation values, with respect to states evolved by random quantum channels. Underlying these analyses are insights into the Haar ensemble of unitaries, that are anticipated to be useful for the quantum information community. We derive an elegant block-orthogonal basis for permutations, that reveals their intimate structure, and greatly simplifies derivations that are otherwise obfuscated by their non-orthogonality [117, 132, 141]. The techniques used to derive this basis can also likely be applied towards finding orthogonal bases for representations of other groups, and for deriving tighter bounds on various norms.

Third, we have refined our understanding of measurements within non-unitary settings. Informationally complete measurement probabilities are exceptionally succinct and intuitive formalisms for describing quantum systems. By deriving generalized forms of their distributions using combinatoric techniques, and proposing accurate effective models for these distributions, we are able to better understand how local noise phenomena accumulate into global noise phenomena. However, these formalisms come with many challenges. Obtaining the tails of such distributions of probabilities necessitates potentially exponentially scaling sample complexity with system size. Further, by imposing stochastic matrix product structure on these probabilities, observing scaling of their corresponding state's operator entanglement, and revealing numerical instability of their non-negative matrix factorizations, we demonstrate fundamental challenges of their use in practice. Ultimately, the inherent difficulties, of maintaining physically valid descriptions of entangled systems undergoing non-unitary dynamics throughout simulations [166, 170, 365], are not completely alleviated by using non-negative numbers versus complex numbers.

7.2 Outlook

Our studies of non-unitary dynamics of quantum systems have revealed their rich structure, and often times subtle, and occasionally uncooperative behaviours. Such analytical and numerical findings suggest several implications of noisy behaviours, and lay groundwork for future studies and applications.

As a direct follow up, our optimization findings should be extended to error-mitigated settings, or even non-Markovian settings, given recent derivations of bounds on system-environment couplings in control settings [199, 413], and studies of error mitigated quantum algorithms in Markovian [109, 124, 287, 341, 362] and non-Markovian settings [74, 414, 415]. Supplementing our ansatze and objectives with error mitigation terms [109], and explicit environment details, would lead to competition between potential increased robustness of the ansatze, versus increased non-convexity or concentration of the objectives, with increased computational and sample complexity [341]. Ultimately, there likely exists several regimes of trainability, over the noise scale, circuit depth, system size, environment size, and ansatz expressivity axes. These axes have yet to be systematically studied numerically in quantum settings, and would extend recent studies of the effect of Gaussian measurement noise, system size, and sample size on optimization success probabilities [416]. Overall, an ambitious long term objective would be deriving phase diagrams where optimization robustness, versus noise effects, versus dimensional effects, accumulate and dominate.

Intriguingly, depolarization has been highlighted throughout our studies as a central reference, and is an extremal, inherent phenomena within quantum information [136, 143, 417–419]. However, it remains to be seen how to best interpret the implications of such manifestations of depolarization. Regarding ensemble behaviours, a primary motivation for developing notions of hierarchies between ensembles is to obtain notions of an ensemble's utility, expressive power, and capabilities of representing the statistics of a reference ensemble. In the case of unitary ensembles, even for totally unstructured, uniformly distributed ensembles of unitaries, there are many generic and appropriate definitions of ensemble expressivity, that all directly correspond to an ensemble's usefulness [106, 233, 420]. Conversely for channel ensembles, totally unstructured, uniformly distributed ensembles of quantum channels are shown to not be useful for non-depolarizing-related tasks. As such, notions of expressivity and usefulness are more subtle in the channel case, perhaps requiring definitions that depend on structures of specific channel ensembles, and that are tailored to specific algorithms. Whether there are more general principles that dictate whether an ensemble forms a channel design [345], such as has been established in elegant ways for unitaries [346], would also reveal properties about the underlying algebraic structures of quantum channels [304]. It is therefore important to continue development of channel-centric quantum algorithms, and to pair these algorithms with specific sets of quantum channels, whose average behaviours are derived to accomplish desired tasks.

Although explicit definitions of expressivity of ensembles of quantum channels remain elusive, these studies bring to attention many relationships between structures of ensembles and their ability to transmit quantum information. Given twirling over unitaries has been used to unify various scrambling and out of time correlator quantities for closed quantum systems [421], our expressions for twirls over quantum channels can be used to assess general notions of non-commutativity of quantum channels at different times [422]. Further, recalling that the norm and trace of a channel's super-operator corresponds to the purity and entanglement fidelity of its corresponding Choi state, we note several relationships between Choi state properties and the difficulty of recovering quantum information. First, bounds on norms of differences of twirls can be expressed in terms of norms of differences of moment operators, due to these state-operator correspondences. Second, the sample complexity of many error mitigation protocols has been shown to be inversely proportional to the purity of a depolarization noise channel's Choi state [108]. Third, it has been shown that the rank of a channel's Choi state is directly related to the success of the channel in state discrimination and exclusion tasks [423]. Therefore, there are intimate relationships between the structure of the average behaviours of an ensemble, and its ability to distinguish states, and its susceptibility to error mitigation.

Finally, these findings open up several future directions of study concerning hybrid classical-quantum algorithms [236] that involve sampling, such as randomized benchmarking [305, 338], classical shadow tomography [339], and error mitigation [340–342]. In the worst case, sampling resources for such methods scale exponentially with system size when not accounting for algorithm-specific details [108], or specific symmetries [275, 424]. Alternatively, randomized benchmarking and probabilistic error cancellation strategies based around ensembles of quantum channels instead of unitaries, may account for more general symmetries and more complicated experiments to be probed. It is reasonable to expect that insights from our explicit expressions for channel statistics, spectral properties of ensemble moment operators, and forms of noisy distributions, could improve sample complexities of these algorithms. Ultimately, by using these more general quantum channel-centric approaches, we hope that it will become easier to calibrate quantum devices with unknown noise models, and that we will be able to study broader classes of errors.

References

[1] M. Duschenes, J. Carrasquilla, and R. Laflamme, Characterization of over-parametrization in the simulation of realistic quantum systems, Physical Review A **109**, 062607 (2024).

[2] M. Duschenes, J. Carrasquilla, and R. Laflamme, Data: Characterization of Overparameterization in Simulation of Realistic Quantum Systems, Zenodo (2024).

[3] M. Duschenes, D. García-Martín, Z. Holmes, and M. Cerezo, Moments of quantum channel ensembles, arXiv preprint arXiv:2511.12700 (2025).

[4] M. Duschenes, R. Melko, J. Carrasquilla, and R. Laflamme, Distributions of Noisy Expectation Values over Sets of Measurement Operators, arXiv preprint arXiv:2604.05973 (2026).

[5] M. Duschenes, R. Melko, J. Carrasquilla, and R. Laflamme, Data: Distributions of Noisy Expectation Values over Sets of Measurement Operators, Zenodo (2026).

[6] M. Duschenes, simulation: JAX based Simulator for Quantum Systems (2022), https://github.com/mduschenes/simulation.

[7] L. D. Landau and E. M. Lifshitz, *Mechanics: Volume 1*, 3rd ed. (Butterworth-Heinemann, 1976).

[8] J. D. Jackson, *Classical Electrodynamics*, 3rd ed. (Wiley, 1999).

[9] H. Schmidt-Böcking, L. Schmidt, H. J. Lüdde, W. Trageser, and T. Sauer, The Stern-Gerlach Experiment Revisited, European Physical Journal H **41**, 327 (2016).

[10] D. Home and F. Selleri, Bell's theorem and the EPR paradox, La Rivista del Nuovo Cimento **14**, 1 (1991).

[11] J. v. L. S. Barry Cooper, *Alan Turing: His Work and Impact* (Elsevier Science, 2013).

[12] H. H. Goldstine, *The Computer from Pascal to von Neumann*, Princeton Paperbacks (Princeton University Press, 1993).

[13] M. M. Wilde, *Quantum information theory* (Cambridge University Press, 2013).

[14] S. Zhu, T. Yu, T. Xu, H. Chen, S. Dustdar, S. Gigan, D. Gunduz, E. Hossain, Y. Jin, F. Lin, B. Liu, Z. Wan, J. Zhang, Z. Zhao, W. Zhu, Z. Chen, T. S. Durrani, H. Wang, J. Wu, T. Zhang, and Y. Pan, Intelligent Computing: The Latest Advances, Challenges, and Future, Intelligent Computing **2**, 45 (2023).

[15] D. A. Roberts, Why is AI hard and Physics simple?, arXiv preprint arXiv:2104.00008 (2021).

[16] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information* (Cambridge University Press, 2000).

[17] E. Grumbling and M. Horowitz, *Quantum Computing: Progress and Prospects* (National Academies Press, Washington, DC, 2019).

[18] J. Preskill, Quantum Computing in the NISQ era and beyond, Quantum **2**, 79 (2018).

[19] J. Eisert and J. Preskill, Mind the gaps: The fraught road to quantum advantage, arXiv preprint arXiv:2510.19928 (2025).

[20] J. Preskill, Quantum Computing 40 Years Later, in *Feynman Lectures on Computation* (CRC Press, 2023) 2nd ed., pp. 193–243.

[21] M. Appleby, C. A. Fuchs, B. C. Stacey, and H. Zhu, Introducing the Qplex: A Novel Arena for Quantum Theory, European Physical Journal D **71**, 197 (2016).

[22] P. W. Shor, Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer, SIAM Journal on Computing **26**, 1484 (1997).

[23] R. LaRose, A brief history of quantum vs classical computational advantage, arXiv preprint arXiv:2412.14703 (2024).

[24] M. Schuld and N. Killoran, Is Quantum Advantage the Right Goal for Quantum Machine Learning?, PRX Quantum **3**, 030101 (2022).

[25] W. G. Ritter, Quantum Channels and Representation Theory, Journal of Mathematical Physics **46**, 082103 (2005).

[26] E. Knill and R. Laflamme, A Theory of Quantum Error-Correcting Codes, arXiv preprint arXiv:quant-ph/9604034 (1996).

[27] E. Knill, R. Laflamme, and L. Viola, Theory of Quantum Error Correction for General Noise, Physical Review Letters **84**, 2525 (2000).

[28] D. Gottesman, An Introduction to Quantum Error Correction and Fault-Tolerant Quantum Computation, arXiv preprint arXiv:0904.2557 (2009).

[29] V. V. Sivak, A. Eickbusch, B. Royer, S. Singh, I. Tsioutsios, S. Ganjam, A. Miano, B. L. Brock, A. Z. Ding, L. Frunzio, S. M. Girvin, R. J. Schoelkopf, and M. H. Devoret, Real-time quantum error correction beyond break-even, Nature **616**, 50 (2023).

[30] A. L. Grimsmo and S. Puri, Quantum Error Correction with the Gottesman-Kitaev-Preskill Code, PRX Quantum **2**, 020101 (2021).

[31] C. A. Fuchs and R. Schack, Quantum-Bayesian Coherence: The No-Nonsense Version, Reviews of Modern Physics **85**, 1693 (2013).

[32] J. M. Farinholt, An Ideal Characterization of the Clifford Operators, Journal of Physics A **47**, 305303 (2014).

[33] H. Lóio, G. Cecile, S. Gopalakrishnan, G. Lami, and J. De Nardis, Correlations, spectra, and entanglement transitions in ensembles of matrix product states, Physical Review B **112**, 035127 (2025).

[34] J. Emerson, M. Silva, O. Moussa, C. Ryan, M. Laforest, J. Baugh, D. G. Cory, and R. Laflamme, Symmetrized characterization of noisy quantum processes, Science **317**, 1893 (2007).

[35] S. Lohani, J. M. Lukens, D. E. Jones, T. A. Searles, R. T. Glasser, and B. T. Kirby, Improving application performance with biased distributions of quantum states, Physical Review Research **3**, 043145 (2021).

[36] D. P. DiVincenzo and D. Loss, Quantum information is physical, Superlattices and Microstructures **23**, 419 (1998).

[37] K. Bharti, A. Cervera-Lierta, T. H. Kyaw, T. Haug, S. Alperin-Lea, A. Anand, M. Degroote, H. Heimonen, J. S. Kottmann, T. Menke, *et al.*, Noisy intermediate-scale quantum algorithms, Reviews of Modern Physics **94**, 015004 (2022).

[38] A. J. Scott, Tight informationally complete quantum measurements, Journal of Physics A: Mathematical and General **39**, 13507 (2006).

[39] J. Peterson, R. S. Sarthour, and R. Laflamme, Enhancing Quantum Control by Improving Shaped-Pulse Generation, Physical Review Appl. **13**, 054060 (2020).

[40] C. Negrevergne, T. S. Mahesh, C. A. Ryan, M. Ditty, F. Cyr-Racine, W. Power, N. Boulant, T. Havel, D. G. Cory, and R. Laflamme, Benchmarking Quantum Control Methods on a 12-Qubit System, Physical Review Letters **96**, 170501 (2006).

[41] S. Majidy, C. Wilson, and R. Laflamme, *Building Quantum Computers: A Practical Introduction* (Cambridge University Press, 2024).

[42] X. Ding, Y. P. Guo, M. C. Xu, R. Z. Liu, G. Y. Zou, J. Y. Zhao, Z. X. Ge, Q. H. Zhang, H. L. Liu, L. J. Wang, M. C. Chen, H. Wang, Y. M. He, Y. H. Huo, C. Y. Lu, and J. W. Pan, High-efficiency single-photon source above the loss-tolerant threshold for efficient linear optical quantum computing, Nature Photonics **19**, 387 (2025).

[43] Y. H. Deng, Y. C. Gu, H. L. Liu, S. Q. Gong, H. Su, Z. J. Zhang, H. Y. Tang, M. H. Jia, J. M. Xu, M. C. Chen, J. Qin, L. C. Peng, J. Yan, Y. Hu, J. Huang, H. Li, Y. Li, Y. Chen, X. Jiang, L. Gan, G. Yang, L. You, L. Li, H. S. Zhong, H. Wang, N. L. Liu, J. J. Renema, C. Y. Lu, and J. W. Pan, Gaussian Boson Sampling with Pseudo-Photon-Number-Resolving Detectors and Quantum Computational Advantage, Physical Review Letters **131**, 150601 (2023).

[44] A. Deshpande, A. Mehta, T. Vincent, N. Quesada, M. Hinsche, M. Ioannou, L. Madsen, J. Lavoie, H. Qi, J. Eisert, D. Hangleiter, B. Fefferman, and I. Dhand, Quantum computational advantage via high-dimensional Gaussian boson sampling, Science Advances **8**, 7894 (2022).

[45] H. S. Zhong, H. Wang, Y. H. Deng, M. C. Chen, L. C. Peng, Y. H. Luo, J. Qin, D. Wu, X. Ding, Y. Hu, P. Hu, X. Y. Yang, W. J. Zhang, H. Li, Y. Li, X. Jiang, L. Gan, G. Yang, L. You, Z. Wang, L. Li, N. L. Liu, C. Y. Lu, and J. W. Pan, Quantum computational advantage using photons, Science **370**, 1460 (2020).

[46] C. D. Bruzewicz, J. Chiaverini, R. McConnell, and J. M. Sage, Trapped-Ion Quantum Computing: Progress and Challenges, Applied Physics Reviews **6**, 021314 (2019).

[47] P. J. Low, B. M. White, A. A. Cox, M. L. Day, and C. Senko, Practical trapped-ion protocols for universal qudit-based quantum computing, Physical Review Research **2**, 033128 (2020).

[48] C. Monroe, W. C. Campbell, L. M. Duan, Z. X. Gong, A. V. Gorshkov, P. Hess, R. Islam, K. Kim, N. Linke, G. Pagano, P. Richerme, C. Senko, and N. Y. Yao, Programmable Quantum Simulations of Spin Systems with Trapped Ions, Reviews of Modern Physics **93**, 025001 (2021).

[49] S. Dasu, M. DeCross, A. Y. Guo, A. Lavasani, J. Behrends, A. Benhemou, Y.-H. Chen, K. Mayer, C. N. Self, S. Simsek, B. Srivastava, M. S. Allman, J. Arkinstall, J. G. Bohnet, N. Q. Burdick, J. P. Campora, A. Chernoguzov, S. F. Cooper, R. D. Delaney, J. M. Dreiling, B. Estey, C. Figgatt, C. Foltz, J. P. Gaebler, A. Hall, C. A. Holliman, A. A. Husain, A. Isanaka, C. J. Kennedy, Y. Kodama, N. Kotibhaskar, N. K. Lysne, I. S. Madjarov, M. Mills, A. R. Milne, B. Neyenhuis, A. J. Park, A. Ransford, A. P. Reed, S. J. Sanders, C. H. Baldwin, D. Hayes, B. Criger, A. C. Potter, and D. Amaro, Computing with many encoded logical qubits beyond break-even, arXiv preprint arXiv:2602.22211 (2026).

[50] A. C. Hughes, R. Srinivas, C. M. Löschnauer, H. M. Knaack, R. Matt, C. J. Ballance, M. Malinowski, T. P. Harty, and R. T. Sutherland, Trapped-ion two-qubit gates with $> 99.99\%$ fidelity without ground-state cooling, arXiv preprint arXiv:2510.17286 (2025).

[51] M. Liu, R. Shaydulin, P. Niroula, M. DeCross, S. H. Hung, W. Y. Kon, E. Cervero-Martín, K. Chakraborty, O. Amer, S. Aaronson, A. Acharya, Y. Alexeev, K. J. Berg, S. Chakrabarti, F. J. Curchod, J. M. Dreiling, N. Erickson, C. Foltz, M. Foss-Feig, D. Hayes, T. S. Humble, N. Kumar, J. Larson, D. Lykov, M. Mills, S. A. Moses, B. Neyenhuis, S. Eloul, P. Siegfried, J. Walker, C. Lim, and M. Pistoia, Certified randomness using a trapped-ion quantum processor, Nature **640**, 343 (2025).

[52] Y. Kim, A. Eddins, S. Anand, K. X. Wei, E. van den Berg, S. Rosenblatt, H. Nayfeh, Y. Wu, M. Zaletel, K. Temme, and A. Kandala, Evidence for the utility of quantum computing before fault tolerance, Nature **618**, 500 (2023).

[53] M. Werninghaus, D. J. Egger, F. Roy, S. Machnes, F. K. Wilhelm, and S. Filipp, Leakage reduction in fast superconducting qubit gates via optimal control, npj Quantum Information **7**, 1 (2021).

[54] M. AbuGhanem, Superconducting Quantum Computers: Who Is Leading the Future?, EPJ Quantum Technology **12**, 102 (2025).

[55] K. Wintersperger, F. Dommert, T. Ehmer, A. Hoursanov, J. Klepsch, W. Mauerer, G. Reuber, T. Strohm, M. Yin, and S. Luber, Neutral Atom Quantum Computing

Hardware: Performance and End-User Perspective, EPJ Quantum Technology **10**, 32 (2023).

[56] T. M. Graham, Y. Song, J. Scott, C. Poole, L. Phuttitarn, K. Jooya, P. Eichler, X. Jiang, A. Marra, B. Grinkemeyer, M. Kwon, M. Ebert, J. Cherek, M. T. Lichtman, M. Gillette, J. Gilbert, D. Bowman, T. Ballance, C. Campbell, E. D. Dahl, O. Crawford, N. S. Blunt, B. Rogers, T. Noel, and M. Saffman, Multi-qubit entanglement and algorithms on a neutral-atom quantum computer, Nature **604**, 457 (2022).

[57] D. Bluvstein, H. Levine, G. Semeghini, T. T. Wang, S. Ebadi, M. Kalinowski, A. Keesling, N. Maskara, H. Pichler, M. Greiner, V. Vuletić, and M. D. Lukin, A quantum processor based on coherent transport of entangled atom arrays, Nature **604**, 451 (2022).

[58] N. C. Chiu, E. C. Trapp, J. Guo, M. H. Abobeih, L. M. Stewart, S. Hollerith, P. L. Stroganov, M. Kalinowski, A. A. Geim, S. J. Evered, S. H. Li, X. Lyu, L. M. Peters, D. Bluvstein, T. T. Wang, M. Greiner, V. Vuletić, and M. D. Lukin, Continuous operation of a coherent 3,000-qubit system, Nature **646**, 1075 (2025).

[59] A. Browaeys and T. Lahaye, Many-body physics with individually controlled Rydberg atoms, Nature Physics **16**, 132 (2020).

[60] F. Tacchino, A. Chiesa, S. Carretta, and D. Gerace, Quantum Computers as Universal Quantum Simulators: State-of-the-Art and Perspectives, Advanced Quantum Technologies **3**, 1900052 (2020).

[61] N. P. de Leon, K. M. Itoh, D. Kim, K. K. Mehta, T. E. Northup, H. Paik, B. S. Palmer, N. Samarth, S. Sangtawesin, and D. W. Steuerman, Materials challenges and opportunities for quantum computing hardware, Science **372**, eabb2823 (2021).

[62] A. J. Daley, I. Bloch, C. Kokail, S. Flannigan, N. Pearson, M. Troyer, and P. Zoller, Practical quantum advantage in quantum simulation, Nature **607**, 667 (2022).

[63] J. Bowles, S. Ahmed, and M. Schuld, Better than classical? The subtle art of benchmarking quantum machine learning models, arXiv preprint arXiv:2403.07059 (2024).

[64] J. Carrasquilla and R. G. Melko, Machine learning phases of matter, Nature Physics **13**, 431 (2017).

[65] J. Tindall and M. Fishman, Gauging tensor networks with belief propagation, SciPost Physics **15**, 222 (2023).

[66] P. Rall, D. Liang, J. Cook, and W. Kretschmer, Simulation of qubit quantum circuits via Pauli propagation, Physical Review A **99**, 062337 (2019).

[67] N. Kotibhaskar, C. Y. Shih, S. Motlakunta, A. Vogliano, L. Hahn, Y. T. Chen, and R. Islam, Programmable XY-type couplings through parallel spin-dependent forces on the same trapped ion motional modes, Physical Review Research **6**, 033038 (2024).

[68] A. A. Geim, N. U. Koyluoglu, S. J. Evered, R. Sahay, S. H. Li, M. Xu, D. Bluvstein, N. O. Gjonbalaj, N. Maskara, M. Kalinowski, T. Manovitz, R. Verresen, S. F. Yelin, J. Feldmeier, M. Greiner, V. Vuletic, and M. D. Lukin, Engineering quantum criticality and dynamics on an analog-digital simulator, arXiv preprint arXiv:2602.18555 (2026).

[69] L. Zhao, J. Goings, K. Shin, W. Kyoung, J. I. Fuks, J. K. Kevin Rhee, Y. M. Rhee, K. Wright, J. Nguyen, J. Kim, and S. Johri, Orbital-optimized pair-correlated electron simulations on trapped-ion quantum computers, npj Quantum Information **9**, 1 (2023).

[70] J. Robledo-Moreno, M. Motta, H. Haas, A. Javadi-Abhari, P. Jurcevic, W. Kirby, S. Martiel, K. Sharma, S. Sharma, T. Shirakawa, I. Sitdikov, R. Y. Sun, K. J. Sung, M. Takita, M. C. Tran, S. Yunoki, and A. Mezzacapo, Chemistry beyond the scale of exact diagonalization on a quantum-centric supercomputer, Science Advances **11**, 9991 (2025).

[71] C. L. Degen, F. Reinhard, and P. Cappellaro, Quantum sensing, Reviews of Modern Physics **89**, 035002 (2017).

[72] J. J. Meyer, J. Borregaard, and J. Eisert, A variational toolbox for quantum multiparameter estimation, npj Quantum Information **7**, 89 (2020).

[73] B. MacLellan, P. Roztocki, S. Czischek, and R. G. Melko, End-to-end variational quantum sensing, npj Quantum Information **10**, 118 (2024).

[74] Z. Mann, N. Cao, R. Laflamme, and S. Zhou, Quantum Error-Corrected Non-Markovian Metrology, PRX Quantum **6**, 030321 (2025).

[75] A. Abbas, A. Ambainis, B. Augustino, A. Bärtschi, H. Buhrman, C. Coffrin, G. Cortiana, V. Dunjko, D. J. Egger, B. G. Elmegreen, N. Franco, F. Fratini, B. Fuller,

J. Gacon, C. Gonciulea, S. Gribling, S. Gupta, S. Hadfield, R. Heese, G. Kircher, T. Kleinert, T. Koch, G. Korpas, S. Lenk, J. Marecek, V. Markov, G. Mazzola, S. Mensa, N. Mohseni, G. Nannicini, C. O'Meara, E. P. Tapia, S. Pokutta, M. Proissl, P. Rebentrost, E. Sahin, B. C. Symons, S. Tornow, V. Valls, S. Woerner, M. L. Wolf-Bauwens, J. Yard, S. Yarkoni, D. Zechiel, S. Zhuk, and C. Zoufal, Challenges and opportunities in quantum optimization, Nature Reviews Physics **6**, 718 (2024).

[76] D. S. Franca and R. Garcia-Patron, Limitations of optimization algorithms on noisy quantum devices, Nature Physics **17**, 1221 (2020).

[77] J. Yu, J. R. Moreno, J. T. Iosue, L. Bertels, D. Claudino, B. Fuller, P. Groszkowski, T. S. Humble, P. Jurcevic, W. Kirby, T. A. Maier, M. Motta, B. Pokharel, A. Seif, A. Shehata, K. J. Sung, M. C. Tran, V. Tripathi, A. Mezzacapo, and K. Sharma, Quantum-Centric Algorithm for Sample-Based Krylov Diagonalization, arXiv preprint arXiv:2501.09702 (2025).

[78] M. P. Fisher, V. Khemani, A. Nahum, and S. Vijay, Random Quantum Circuits, Annual Review of Condensed Matter Physics **14**, 335 (2023).

[79] S. Mullane, Sampling random quantum circuits: a pedestrian's guide, arXiv preprint arXiv:2007.07872 (2020).

[80] A. B. Magann, C. Arenz, M. D. Grace, T. S. Ho, R. L. Kosut, J. R. McClean, H. A. Rabitz, and M. Sarovar, From Pulses to Circuits and Back Again: A Quantum Optimal Control Perspective on Variational Quantum Algorithms, PRX Quantum **2**, 010101 (2021).

[81] Y. Gujju, A. Matsuo, and R. Raymond, Quantum machine learning on near-term quantum devices: Current state of supervised and unsupervised techniques for real-world applications, Physical Review Applied **21**, 067001 (2024).

[82] M. Schuld, I. Sinayskiy, and F. Petruccione, An introduction to quantum machine learning, Contemp. Phys. **56**, 172 (2015).

[83] M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, J. R. McClean, K. Mitarai, X. Yuan, L. Cincio, and P. J. Coles, Variational quantum algorithms, Nat. Rev. Phys. **3**, 625 (2021).

[84] H.-Y. Huang, R. Kueng, G. Torlai, V. V. Albert, and J. Preskill, Provably efficient machine learning for quantum many-body problems, Science **377**, eabk3333 (2022).

[85] X. Ge, R. B. Wu, and H. Rabitz, The optimization landscape of hybrid quantum–classical algorithms: From quantum control to NISQ applications, Annu. Rev. Control **54**, 314 (2022).

[86] H.-Y. Huang, R. Kueng, and J. Preskill, Predicting many properties of a quantum system from very few measurements, Nature Physics **16**, 1050 (2020).

[87] S. J. Glaser, T. Schulte-Herbrüggen, M. Sieveking O, Schedletzky, N. C. Nielsen, O. W. Sørensen, and C. Griesinger, Unitary control in quantum ensembles: Maximizing signal intensity in coherent spectroscopy, Science **280**, 421 (1998).

[88] C. Altafini, Controllability properties for finite dimensional quantum Markovian master equations, Journal of Mathematical Physics **44**, 2357 (2004).

[89] R. Gautier, É. Genois, and A. Blais, Optimal Control in Large Open Quantum Systems: The Case of Transmon Readout and Reset, Physical Review Letters **134**, 070802 (2025).

[90] H. Levine, A. Keesling, A. Omran, H. Bernien, S. Schwartz, A. S. Zibrov, M. Endres, M. Greiner, V. Vuletić, and M. D. Lukin, High-Fidelity Control and Entanglement of Rydberg-Atom Qubits, Physical Review Lett. **121**, 123603 (2018).

[91] T. Caneva, M. Murphy, T. Calarco, R. Fazio, S. Montangero, V. Giovannetti, and G. E. Santoro, Optimal Control at the Quantum Speed Limit, Physical Review Letters **103**, 240501 (2009).

[92] S. Deffner and S. Campbell, Quantum speed limits: from Heisenberg's uncertainty principle to optimal quantum control, Journal of Physics A **50**, 453001 (2017).

[93] M. Larocca, P. M. Poggi, and D. A. Wisniacki, Quantum control landscape for a two-level system near the quantum speed limit, Journal of Physics A **51**, 385305 (2018).

[94] M. Larocca, P. Czarnik, K. Sharma, G. Muraleedharan, P. J. Coles, and M. Cerezo, Diagnosing Barren Plateaus with Tools from Quantum Optimal Control, Quantum **6**, 824 (2022).

[95] M. Larocca, S. Thanasilp, S. Wang, K. Sharma, J. Biamonte, P. J. Coles, L. Cincio, J. R. McClean, Z. Holmes, and M. Cerezo, Barren plateaus in variational quantum computing, Nature Reviews Physics **7**, 174–189 (2025).

[96] C. P. Koch, Controlling open quantum systems: Tools, achievements, and limitations, Journal of Physics: Condensed Matter **28**, 213001 (2016).

[97] E. R. Anschuetz and B. T. Kiani, Quantum variational algorithms are swamped with traps, Nature Communications **13**, 1 (2022).

[98] G. Riviello, K. M. Tibbetts, C. Brif, R. Long, R. B. Wu, T. S. Ho, and H. Rabitz, Searching for quantum optimal controls under severe constraints, Physical Review A **91**, 043401 (2015).

[99] G. Dirr, U. Helmke, I. Kurniawan, and T. Schulte-Herbrueggen, Lie-Semigroup Structures for Reachability and Control of Open Quantum Systems: Viewing Markovian Quantum Channels as Lie Semigroups and GKS-Lindblad Generators as Lie Wedge, Reports on Mathematical Physics **64**, 93 (2009).

[100] B. T. Kiani, S. Lloyd, and R. Maity, Learning Unitaries by Gradient Descent, arXiv preprint arXiv:2001.11897 (2020).

[101] J. Liu, F. Tacchino, J. R. Glick, L. Jiang, and A. Mezzacapo, Representation Learning via Quantum Neural Tangent Kernels, Physical Review X Quantum **3**, 030323 (2022).

[102] D. García-Martín, M. Larocca, and M. Cerezo, Effects of noise on the overparametrization of quantum neural networks, Physical Review Research **6**, 013295 (2024).

[103] K. Zyczkowski, K. A. Penson, I. Nechita, and B. Collins, Generating random density matrices, Journal of Mathematical Physics **52**, 062201 (2011).

[104] K. Zyczkowski and H.-J. Sommers, Induced measures in the space of mixed quantum states, Journal of Physics A: Mathematical and General **34**, 7111 (2001).

[105] B. Fefferman, S. Ghosh, M. Gullans, K. Kuroiwa, and K. Sharma, Effect of Nonunital Noise on Random-Circuit Sampling, PRX Quantum **5**, 030317 (2024).

[106] Z. Holmes, K. Sharma, M. Cerezo, and P. J. Coles, Connecting ansatz expressibility to gradient magnitudes and barren plateaus, PRX Quantum **3**, 010313 (2022).

[107] J. Emerson, R. Alicki, and K. Zyczkowski, Scalable Noise Estimation with Random Unitary Operators, Journal of Optics B **7** (2005).

[108] Y. Quek, D. Stilck França, S. Khatri, J. J. Meyer, and J. Eisert, Exponentially tighter bounds on limitations of quantum error mitigation, Nature Physics **20**, 1648 (2024).

[109] S. Wang, P. Czarnik, A. Arrasmith, M. Cerezo, L. Cincio, and P. J. Coles, Can Error Mitigation Improve Trainability of Noisy Variational Quantum Algorithms?, Quantum **8**, 1287 (2024).

[110] K. Zyczkowski and H.-J. Sommers, Average fidelity between random quantum states, Physical Review A **71**, 032313 (2003).

[111] P. Zanardi, Entanglement of Quantum Evolutions, Physical Review A **63**, 1 (2001).

[112] K. Zyczkowski and M. Kus, Random unitary matrices, Journal of Physics A: Mathematical and General **27**, 4235 (1994).

[113] D. Belkin, J. Allen, S. Ghosh, C. Kang, S. Lin, J. Sud, F. T. Chong, B. Fefferman, and B. K. Clark, Approximate t-Designs in Generic Circuit Architectures, PRX Quantum **5**, 040344 (2024).

[114] A. W. Harrow and S. Mehraban, Approximate unitary t-designs by short random quantum circuits using nearest-neighbor and long-range gates, Communications in Mathematical Physics **401**, 1531 (2023).

[115] L. Bittel, J. Eisert, L. Leone, A. A. Mele, and S. F. E. Oliviero, A complete theory of the Clifford commutant, arXiv preprint arXiv:2504.12263 (2025).

[116] A. W. Harrow and R. A. Low, Random quantum circuits are approximate 2-designs, Communications in Mathematical Physics **291**, 257 (2009).

[117] B. Collins and S. Matsumoto, Weingarten calculus via orthogonality relations: new applications, arXiv preprint arXiv:1701.04493 (2017).

[118] B. Collins, S. Matsumoto, and J. Novak, The Weingarten Calculus, Notices Of The American Mathematical Society **69**, 734 (2022).

[119] B. Collins and I. Nechita, Random matrix techniques in quantum information theory, Journal of Mathematical Physics **57**, 015215 (2016).

[120] A. Sauliere, B. Magni, G. Lami, X. Turkeshi, and J. D. Nardis, Universality in the anticoncentration of chaotic quantum circuits, Physical Review B **112**, 134312 (2025).

[121] W. Brown and O. Fawzi, Scrambling speed of random quantum circuits, arXiv preprint arXiv:1210.6644 (2012).

[122] Y. Sekino and L. Susskind, Fast scramblers, Journal of High Energy Physics **2008**, 065 (2008).

[123] P. Hosur, X.-L. Qi, D. A. Roberts, and B. Yoshida, Chaos in quantum channels, Journal of High Energy Physics **2016**, 1 (2016).

[124] L. Botelho, A. Glos, A. Kundu, J. A. Miszczak, Ö. Salehi, and Z. Zimborás, Error mitigation for variational quantum algorithms through mid-circuit measurements, Physical Review A **105**, 022441 (2022).

[125] E. Knill, D. Leibfried, R. Reichle, J. Britton, R. B. Blakestad, J. D. Jost, C. Langer, R. Ozeri, S. Seidelin, and D. J. Wineland, Randomized benchmarking of quantum gates, Physical Review A **77**, 012307 (2008).

[126] C. J. Wood, J. D. Biamonte, and D. G. Cory, Tensor networks and graphical calculus for open quantum systems, arXiv preprint arXiv:1111.6950 (2011).

[127] R. Kukulski, I. Nechita, L. Pawela, Z. Puchala, and K. Zyczkowski, Generating random quantum channels, Journal of Mathematical Physics **62**, 062201 (2021).

[128] B. Collins and I. Nechita, Random quantum channels I: graphical calculus and the Bell state phenomenon, Communications in Mathematical Physics **297**, 345 (2009).

[129] I. Nechita and S. J. Park, Random Covariant Quantum Channels, Annales Henri Poincare **27**, 847–907 (2026).

[130] A. Sauliere, G. Lami, C. Boyer, J. De Nardis, and A. De Luca, Universality in the Anticoncentration of Noisy Quantum Circuits at Finite Depths, arXiv preprint arXiv:2508.14975 (2025).

[131] A. A. Mele, A. Angrisani, S. Ghosh, S. Khatri, J. Eisert, D. Stilck França, and Y. Quek, Noise-induced shallow circuits and absence of barren plateaus, arXiv preprint arXiv:2403.13927 (2024).

[132] J. Bai, J. Wang, and Z. Yin, Primitivity for random quantum channels, Quantum Information Processing **23**, 1 (2024).

[133] A. Bouland, B. Fefferman, C. Nirkhe, and U. Vazirani, On the complexity and verification of quantum random circuit sampling, Nature Physics **15**, 159 (2018).

[134] A. M. Dalzell, N. Hunter-Jones, and F. G. S. L. Brandão, Random Quantum Circuits Anticoncentrate in Log Depth, PRX Quantum **3**, 010333 (2022).

[135] X. Gao, M. Kalinowski, C. N. Chou, M. D. Lukin, B. Barak, and S. Choi, Limitations of Linear Cross-Entropy as a Measure for Quantum Advantage, PRX Quantum **5**, 010334 (2024).

[136] A. Deshpande, P. Niroula, O. Shtanko, A. V. Gorshkov, B. Fefferman, and M. J. Gullans, Tight bounds on the convergence of noisy random circuits to the uniform distribution, PRX Quantum **3**, 040329 (2022).

[137] S.-u. Lee, S. Ghosh, C. Oh, K. Noh, B. Fefferman, and L. Jiang, Classical simulation of noisy random circuits from exponential decay of correlation, arXiv preprint arXiv:2510.06328 (2025).

[138] S. Lee, C. Oh, Y. Wong, S. Chen, and L. Jiang, Universal Spreading of Conditional Mutual Information in Noisy Random Circuits, Physical Review Letters **133**, 200402 (2024).

[139] T. Schuster, J. Haferkamp, and H.-Y. Huang, Random unitaries in extremely low depth, Science **389**, 92 (2025).

[140] P. W. Claeys and G. De Tomasi, Fock-space delocalization and the emergence of the Porter-Thomas distribution from dual-unitary dynamics, Physical Review Letters **134**, 050405 (2025).

[141] A. W. Harrow, Approximate orthogonality of permutation operators, with application to quantum information, Letters in Mathematical Physics **114**, 1 (2024).

[142] B. Fefferman, S. Ghosh, and W. Zhan, Anti-Concentration for the Unitary Haar Measure and Applications to Random Quantum Circuits, arXiv preprint arXiv:2407.19561 (2024).

[143] A. M. Dalzell, N. Hunter-Jones, and F. G. S. L. Brandão, Random quantum circuits transform local noise into global white noise, Communications in Mathematical Physics **405**, 78 (2021).

[144] L. Campos Venuti and P. Zanardi, Probability density of quantum expectation values, Physics Letters A **377**, 1854 (2013).

[145] V. B. Bulchandani, S. L. Sondhi, and J. T. Chalker, Random-matrix models of monitored quantum circuits, Journal of Statistical Physics **191** (2024).

[146] J. M. Renes, R. Blume-Kohout, A. J. Scott, and C. M. Caves, Symmetric Informationally Complete Quantum Measurements, Journal of Mathematical Physics **45**, 2171 (2003).

[147] J. Carrasquilla, G. Torlai, R. G. Melko, and L. Aolita, Reconstructing quantum states with generative models, Nature Machine Intelligence **1**, 155 (2019).

[148] A. Acharya, S. Saha, and A. M. Sengupta, Informationally complete POVM-based shadow tomography, arXiv preprint arXiv:2105.05992 (2021).

[149] J. Denzler, J. Carrasco, J. Eisert, and T. Guaita, Simulation of noisy quantum circuits using frame representations, arXiv preprint arXiv:2601.05131 (2026).

[150] S. Aaronson and D. Gottesman, Improved Simulation of Stabilizer Circuits, Physical Review A **70**, 052328 (2004).

[151] C. Gidney, Stim: a fast stabilizer circuit simulator, Quantum **5**, 497 (2021).

[152] S. Masot-Llima and A. Garcia-Saez, Stabilizer Tensor Networks: Universal Quantum Simulator on a Basis of Stabilizer States, Physical Review Letters **133**, 230601 (2024).

[153] T. Vovk and H. Pichler, Entanglement-Optimal Trajectories of Many-Body Quantum Markov Processes, Physical Review Letters **128**, 243601 (2022).

[154] S. Cichy, P. K. Faehrmann, L. Bittel, J. Eisert, H. Pashayan, and H. Hai, Classical simulation of noisy quantum circuits via locally entanglement-optimal unravelings, arXiv preprint arXiv:2508.05745 (2025).

[155] Z. Cheng and M. Ippoliti, Efficient Sampling of Noisy Shallow Circuits Via Monitored Unraveling, PRX Quantum **4**, 040326 (2023).

[156] R. Daraban, F. Salas-Ramírez, and J. Schachenmayer, Non-unitarity maximizing unraveling of open quantum dynamics, SciPost Physics **18**, 048 (2025).

[157] Z. Chen, Y. Bao, and S. Choi, Optimized trajectory unraveling for classical simulation of noisy quantum dynamics, Physical Review Letters **133**, 230403 (2024).

[158] V. Martinez, A. Angrisani, E. Pankovets, O. Fawzi, and D. S. França, Efficient simulation of parametrized quantum circuits under non-unital noise through Pauli backpropagation, Physical Review Letters **134**, 250602 (2025).

[159] M. S. Rudolph, T. Jones, Y. Teng, A. Angrisani, and Z. Holmes, Pauli Propagation: A Computational Framework for Simulating Quantum Systems, arXiv preprint arXiv:2505.21606 (2025).

[160] U. Schollwock, The density-matrix renormalization group in the age of matrix product states, Annals of Physics **326**, 96 (2011).

[161] K. Noh, L. Jiang, and B. Fefferman, Efficient classical simulation of noisy random quantum circuits in one dimension, Quantum **4**, 318 (2020).

[162] S. Cheng, C. Cao, C. Zhang, Y. Liu, S.-Y. Hou, P. Xu, and B. Zeng, Simulating noisy quantum circuits with matrix product density operators, Physical Review Res. **3**, 023005 (2021).

[163] Z.-Y. Wei, J. Rajakumar, J. Nelson, D. Malz, M. J. Gullans, and A. V. Gorshkov, Noise-induced contraction of MPO truncation errors in noisy random circuits and Lindbladian dynamics, arXiv preprint arXiv:2603.20400 (2026).

[164] I. Cirac, D. Perez-Garcia, N. Schuch, and F. Verstraete, Matrix Product States and Projected Entangled Pair States: Concepts, Symmetries, and Theorems, Reviews of Modern Physics **93**, 045003 (2020).

[165] R. Orús, A practical introduction to tensor networks: Matrix product states and projected entangled pair states, Annals of Physics **349**, 117 (2014).

[166] Y. Guo and S. Yang, Locally purified density operators for noisy quantum circuits, arXiv preprint arXiv:2312.02854 (2023).

[167] Y. Guo, J. H. Zhang, H. R. Zhang, S. Yang, and Z. Bi, Locally Purified Density Operators for Symmetry-Protected Topological Phases in Mixed States, Physical Review X **15**, 021060 (2025).

[168] T. Schuster, C. Yin, X. Gao, and N. Y. Yao, A polynomial-time classical algorithm for noisy quantum circuits, Physical Review X **15**, 041018 (2025).

[169] D. Aharonov, X. Gao, Z. Landau, Y. Liu, and U. Vazirani, A Polynomial-Time Classical Algorithm for Noisy Random Circuit Sampling, in *Proceedings of the 55th Annual ACM Symposium on Theory of Computing*, STOC 2023 (2023) p. 945–957.

[170] I. Glasser, R. Sweke, N. Pancotti, J. Eisert, and J. I. Cirac, Expressive power of tensor-network factorizations for probabilistic modeling, with applications from hidden Markov models to quantum machine learning, arXiv preprint arXiv:1907.03741 (2019).

[171] K. Temme and F. Verstraete, Stochastic Matrix Product States, Physical Review Letters **104**, 210502 (2010).

[172] T. H. Johnson, S. R. Clark, and D. Jaksch, Dynamical simulations of classical stochastic systems using matrix product states, Physical Review E **82**, 036702 (2010).

[173] D. Lee and H. S. Seung, Algorithms for non-negative matrix factorization, in *Advances in Neural Information Processing Systems*, Vol. 13, edited by T. Leen, T. Dietterich, and V. Tresp (MIT Press, 2000).

[174] D. Han, C. Guo, and X. Wang, Density matrix reconstruction using non-negative matrix product states, Physical Review A **106**, 042435 (2022).

[175] J. Biamonte, Lectures on Quantum Tensor Networks a pathway to modern diagrammatic reasoning, arXiv preprint arXiv:1912.10049 (2019).

[176] J. Watrous, *The Theory of Quantum Information* (Cambridge University Press, 2018).

[177] B. Baumgartner, An inequality for the trace of matrix products, using absolute values, arXiv preprint arXiv:1106.6189 (2011).

[178] V. I. Yashin, E. O. Kiktenko, A. S. Mastiukova, and A. K. Fedorov, Minimal informationally complete measurements for probability representation of quantum dynamics, New Journal of Physics **22**, 103026 (2020).

[179] E. O. Kiktenko, A. O. Malyshev, A. S. Mastiukova, V. I. Man'ko, A. K. Fedorov, and D. Chruściński, Probability representation of quantum dynamics using pseudostochastic maps, Physical Review A **101**, 052320 (2020).

[180] H. J. D. Miller, Statistical Mechanics of Random Mixed State Ensembles with Fixed Energy, arXiv preprint arXiv:2508.00809 (2025).

[181] H. Sommers and K. Zyczkowski, Bures volume of the set of mixed quantum states, Journal of Physics A: Mathematical and General **36**, 10083 (2005).

[182] J. J. Meyer, Fisher Information in Noisy Intermediate-Scale Quantum Applications, Quantum **5**, 539 (2021).

[183] Y. Zhou, M. Stoudenmire, and X. Waintal, What Limits the Simulation of Quantum Computers?, Physical Review X **10**, 041038 (2020).

[184] N. Rengaswamy, R. Calderbank, H. D. Pfister, and S. Kadhe, Synthesis of logical clifford operators via symplectic geometry, in *2018 IEEE International Symposium on Information Theory (ISIT)* (2018) pp. 791–795.

[185] P. Zanardi, C. Zalka, and L. Faoro, On the Entangling Power of Quantum Evolutions, Physical Review A **62**, 4 (2000).

[186] S. Sen, Average Entropy of a Quantum Subsystem, Physical Review Letters **77**, 1 (1996).

[187] H. Ollivier and W. H. Zurek, Quantum Discord: A Measure of the Quantumness of Correlations, Physical Review Letters **88**, 017901 (2001).

[188] W. H. Zurek, Quantum Discord and Maxwell's Demons, Physical Review A **67**, 6 (2003).

[189] M. Zwolak and W. H. Zurek, Complementarity of quantum discord and classically accessible information, Scientific Reports **3**, 1729 (2013).

[190] T. Haug and M. S. Kim, Generalization of quantum machine learning models using quantum fisher information metric, Physical Review Lett. **133**, 050603 (2024).

[191] C. W. Helstrom, Minimum mean-squared error of estimates in quantum statistics, Physics Letters A **25**, 101 (1967).

[192] M. G. Paris, Quantum Estimation for Quantum Technology, International Journal of Quantum Information **7**, 125 (2011).

[193] E. Knill and R. Laflamme, On the Power of One Bit of Quantum Information, Physical Review Lett. **81**, 5672 (1998).

[194] Y. Quek, E. Kaur, and M. M. Wilde, Multivariate trace estimation in constant quantum depth, Quantum **8**, 1220 (2024).

[195] D. G. Cory, R. Laflamme, E. Knill, L. Viola, T. F. Havel, N. Boulant, G. Boutis, E. Fortunato, S. Lloyd, R. Martinez, C. Negrevergne, M. Pravia, Y. Sharf, G. Teklemariam, Y. S. Weinstein, and W. H. Zurek, NMR Based Quantum Information Processing: Achievements and Prospects, Fortschritte der Phys. **48**, 875 (2000).

[196] R. Laflamme, E. Knill, D. G. Cory, E. M. Fortunato, T. Havel, C. Miquel, R. Martinez, C. Negrevergne, G. Ortiz, M. A. Pravia, Y. Sharf, S. Sinha, R. Somma, and L. Viola, Introduction to NMR Quantum Information Processing, arXiv preprint arXiv:0207172 (2002).

[197] D. G. Cory, M. D. Price, W. Maas, E. Knill, R. Laflamme, W. H. Zurek, T. F. Havel, and S. S. Somaroo, Experimental Quantum Error Correction, Physical Review Letters **81**, 2152 (1998).

[198] J. Zhang, R. Laflamme, and D. Suter, Experimental Implementation of Encoded Logical Qubit Operations in a Perfect Quantum Error Correcting Code, Physical Review Letters **109**, 100503 (2012).

[199] D. A. Lidar, Review of Decoherence-Free Subspaces, Noiseless Subsystems, and Dynamical Decoupling, Adv. Chem. Phys. **154**, 295 (2014).

[200] G. Feng, F. H. Cho, H. Katiyar, J. Li, D. Lu, J. Baugh, and R. Laflamme, Gradient-based closed-loop quantum optimal control in a solid-state two-qubit system, Physical Review A **98**, 052341 (2018).

[201] E. Knill, R. Laflamme, and G. J. Milburn, A scheme for efficient quantum computation with linear optics, Nature **409**, 46 (2001).

[202] I. Paparelle, J. Henaff, J. García-Beni, É. Gillet, D. Montesinos, G. L. Giorgi, M. C. Soriano, R. Zambrini, and V. Parigi, Experimental memory control in continuous-variable optical quantum reservoir computing, Nature Photonics **20**, 413–420 (2026).

[203] D. Main, P. Drmota, D. P. Nadlinger, E. M. Ainley, A. Agrawal, B. C. Nichol, R. Srinivas, G. Araneda, and D. M. Lucas, Distributed quantum computing across an optical network link, Nature **638**, 383 (2025).

[204] P. J. Daley, K. J. Resch, and R. W. Spekkens, Experimentally adjudicating between different causal accounts of Bell-inequality violations via statistical model selection, Physical Review A **105**, 042220 (2022).

[205] J. I. Cirac and P. Zoller, Quantum Computations with Cold Trapped Ions, Physical Review Letters **74**, 4091 (1995).

[206] J. S. Chen, E. Nielsen, M. Ebert, V. Inlek, K. Wright, V. Chaplin, A. Maksymov, E. Páez, A. Poudel, P. Maunz, and J. Gamble, Benchmarking a trapped-ion quantum computer with 30 qubits, Quantum **8**, 1516 (2024).

[207] M. Kang, W. C. Campbell, and K. R. Brown, Quantum Error Correction with Metastable States of Trapped Ions Using Erasure Conversion, PRX Quantum **4**, 020358 (2023).

[208] Y. Nakamura, Y. A. Pashkin, and J. S. Tsai, Coherent control of macroscopic quantum states in a single-Cooper-pair box, Nature **398**, 786 (1999).

[209] G. Wendin, Quantum information processing with superconducting circuits: a review, Reports on Progress in Physics **80**, 106001 (2017).

[210] Q. Zhu, S. Cao, F. Chen, M. C. Chen, X. Chen, T. H. Chung, H. Deng, Y. Du, D. Fan, M. Gong, C. Guo, C. Guo, S. Guo, L. Han, L. Hong, H. L. Huang, Y. H. Huo, L. Li, N. Li, S. Li, Y. Li, F. Liang, C. Lin, J. Lin, H. Qian, D. Qiao, H. Rong, H. Su, L. Sun, L. Wang, S. Wang, D. Wu, Y. Wu, Y. Xu, K. Yan, W. Yang, Y. Yang, Y. Ye, J. Yin, C. Ying, J. Yu, C. Zha, C. Zhang, H. Zhang, K. Zhang, Y. Zhang, H. Zhao, Y. Zhao, L. Zhou, C. Y. Lu, C. Z. Peng, X. Zhu, and J. W. Pan, Quantum computational advantage via 60-qubit 24-cycle random circuit sampling, Science Bulletin **67**, 240 (2022).

[211] N. Zobrist, N. Zhu, Y. Zhang, A. Zalcman, G. Young, N. Yosri, J. Yoo, B. Ying, P. Yeh, Z. J. Yao, C. Xing, B. W. Woo, K. Wong, T. White, T. Weidel, K. Weber, B. Ware, S. X. Wang, S. Waltman, C. V. Heidweiller, B. Villalonga, G. Vidal, S. Vdovichev, J. Vargas, A. Vaishnav, M. M. Torunbalci, A. Torres, D. Thor, A. Sztein, A. Szasz, J. Suchard, D. Strain, G. Sterling, S. Springer, R. D. Somma, W. C. Smith, V. Smelyanskiy, S. Small, J. Skruzny, V. Sivak, S. Singh, V. Shvarts, N. Shutty, A. Shorter, M. J. Shearn, A. W. Senior, C. Schuster, H. F. Schurkus, K. J. Satzinger, K. Sankaragomathi, D. Sank, N. Saei, N. C. Rubin, P. Roushan, E. Rosenfeld, E. Rosenberg, G. Roberts, D. M. Rhodes, J. Redding, M. J. Reagor, G. Ramachandran, C. Quintana, L. P. Pryadko, O. Pritchard, R. Potter, J. Platt, A. Pizzuto, A. Petukhov, K. Ottosson, A. Opremcak, W. D. Oliver, T. E. O'Brien, M. Y. Niu, C. H. Ni, M. Nguyen, A. Nguyen, J. H. Ng, M. Newman, H. Neven, A. Nersisyan, C. Neill, M. Neeley, O. Naaman, W. Mruczkiewicz, R. Movassagh, A. Morvan, S. Montazeri, S. Molina, R. Molavi, A. Mieszala, K. C. Miao, X. Mi,

A. Megrant, S. Meeks, M. McEwen, J. R. McClean, C. Maxfield, S. Martin, O. Martin, L. S. Martin, J. Manyika, S. Mandrà, A. Maloney, F. D. Malone, S. Madhuk, A. Lunt, D. Lundahl, E. Lucero, A. Locharla, W. P. Livingston, W. Liu, A. T. Lill, Y. Li, W. Y. Li, B. J. Lester, S. Leon, Y. D. Lensky, K. Lee, J. Lee, J. Ledford, L. Le Guevel, K. M. Lau, P. Laptev, B. W. Langley, T. Lange-Dei, D. Landhuis, N. Lacroix, V. D. Kurilovich, J. M. Kreikebaum, B. Kozlovskii, R. Kothari, F. Kostritsa, A. N. Korotkov, P. Kohli, B. Kobrin, A. R. Klots, P. V. Klimov, S. Kim, M. Khezri, T. Khattar, T. Khaire, J. Kelly, K. Kechedzhi, A. H. Karamlou, H. Kang, D. Kafri, P. Juhas, C. Joshi, S. Jordan, C. Jones, Z. Jiang, E. Jeffrey, J. Iveland, S. V. Isakov, L. B. Ioffe, W. J. Huggins, A. Huff, H. Y. Huang, S. Hong, G. Holland, J. Hilton, G. Hill, O. Higgott, P. Heu, S. Heslin, F. J. Heras, S. D. Harrington, M. P. Harrigan, M. Hansen, M. C. Hamilton, J. Hall, S. Habegger, J. A. Gross, A. Greene, D. Graumann, A. G. Dau, R. Gosula, D. Gilboa, C. Gidney, W. Giang, É. Genois, R. Gasca, G. Garcia, S. Ganjam, B. Foxen, A. G. Fowler, E. Forati, L. F. Burgos, V. S. Ferreira, E. Farhi, L. Faoro, C. Erickson, M. Elzouka, A. M. Elbag, A. Eickbusch, T. Edlich, C. Earle, A. Dunsworth, I. Drozdov, P. Donohoe, A. Di Paolo, M. Devoret, S. Demura, D. M. Debroy, L. De Lorenzo, A. Davies, S. Das, B. Curtin, A. L. Crook, W. Courtney, P. Conner, R. Collins, J. Cogan, A. Y. Cleland, J. Claes, C. Chou, D. Chik, B. Chiaro, Z. Chen, Y. Chen, H. S. Chang, J. Campero, A. Cabrera, N. Bushnell, B. Burkett, T. Burger, D. A. Buell, B. B. Buckley, B. Buchea, D. A. Browne, M. Broughton, L. Brill, J. Bovaird, A. Bourassa, G. Bortoli, S. Boixo, S. Blackwell, A. Bilmes, A. Bengtsson, J. Bausch, J. C. Bardin, B. Ballard, D. Bacon, R. Babbush, J. Atalaya, N. Astrakhantsev, A. Asfaw, K. Arya, F. Arute, M. Ansmann, T. I. Andersen, I. Aleiner, L. Aghababaie-Beni, D. A. Abanin, and R. Acharya, Quantum error correction below the surface code threshold, Nature **638**, 920 (2024).

[212] F. Arute, K. Arya, R. Babbush, D. Bacon, J. C. Bardin, R. Barends, R. Biswas, S. Boixo, F. G. S. L. Brandao, D. A. Buell, B. Burkett, Y. Chen, Z. Chen, B. Chiaro, R. Collins, W. Courtney, A. Dunsworth, E. Farhi, B. Foxen, A. Fowler, C. Gidney, M. Giustina, R. Graff, K. Guerin, S. Habegger, M. P. Harrigan, M. J. Hartmann, A. Ho, M. Hoffmann, T. Huang, T. S. Humble, S. V. Isakov, E. Jeffrey, Z. Jiang, D. Kafri, K. Kechedzhi, J. Kelly, P. V. Klimov, S. Knysh, A. Korotkov, F. Kostritsa, D. Landhuis, M. Lindmark, E. Lucero, D. Lyakh, S. Mandrà, J. R. McClean, M. McEwen, A. Megrant, X. Mi, K. Michielsen, M. Mohseni, J. Mutus, O. Naaman, M. Neeley, C. Neill, M. Y. Niu, E. Ostby, A. Petukhov, J. C. Platt, C. Quintana, E. G. Rieffel, P. Roushan, N. C. Rubin, D. Sank, K. J. Satzinger, V. Smelyanskiy, K. J. Sung, M. D. Trevithick, A. Vainsencher, B. Villalonga, T. White, Z. J. Yao, P. Yeh, A. Zalcman, H. Neven, and J. M. Martinis, Quantum supremacy using a

programmable superconducting processor, Nature **574**, 505 (2019).

[213] M. Saffman, Quantum computing with atomic qubits and Rydberg interactions: progress and challenges, Journal of Physics B **49**, 202001 (2016).

[214] A. G. Radnaev, W. C. Chung, D. C. Cole, D. Mason, T. G. Ballance, M. J. Bedalov, D. A. Belknap, M. R. Berman, M. Blakely, I. L. Bloomfield, P. D. Buttler, C. Campbell, A. Chopinaud, E. C. Copenhaver, M. K. Dawes, S. Y. Eubanks, A. J. Friss, D. M. Garcia, J. Gilbert, M. Gillette, P. Goiporia, P. Gokhale, J. Goldwin, D. Goodwin, T. M. Graham, C. J. Guttormsson, G. T. Hickman, L. Hurtley, M. Iliev, E. B. Jones, R. A. Jones, K. W. Kuper, T. B. Lewis, M. T. Lichtman, F. M. Majdeteimouri, J. J. Mason, J. K. McMaster, J. A. Miles, P. T. Mitchell, J. D. Murphree, N. A. Neff-Mallon, T. O. Oh, V. Omole, C. Parlo Simon, N. Pederson, M. A. Perlin, A. Reiter, R. Rines, P. R. Romlow, A. M. Scott, D. Stiefvater, J. R. Tanner, A. K. Tucker, I. V. Vinogradov, M. L. Warter, M. Yeo, M. S. Saffman, and T. W. Noel, Universal Neutral-Atom Quantum Computer with Individual Optical Addressing and Nondestructive Readout, PRX Quantum **6**, 030334 (2025).

[215] D. Bluvstein, S. J. Evered, A. A. Geim, S. H. Li, H. Zhou, T. Manovitz, S. Ebadi, M. Cain, M. Kalinowski, D. Hangleiter, J. P. B. Ataides, N. Maskara, I. Cong, X. Gao, P. S. Rodriguez, T. Karolyshyn, G. Semeghini, M. J. Gullans, M. Greiner, V. Vuletic, and M. D. Lukin, Logical quantum processor based on reconfigurable atom arrays, Nature **626**, 58 (2023).

[216] D. Bluvstein, A. A. Geim, S. H. Li, S. J. Evered, J. P. Bonilla Ataides, G. Baranes, A. Gu, T. Manovitz, M. Xu, M. Kalinowski, S. Majidy, C. Kokail, N. Maskara, E. C. Trapp, L. M. Stewart, S. Hollerith, H. Zhou, M. J. Gullans, S. F. Yelin, M. Greiner, V. Vuletić, M. Cain, and M. D. Lukin, A fault-tolerant neutral-atom architecture for universal quantum computation, Nature **649**, 39 (2025).

[217] Q. Xu, J. P. Bonilla Ataides, C. A. Pattison, N. Raveendran, D. Bluvstein, J. Wurtz, B. Vasić, M. D. Lukin, L. Jiang, and H. Zhou, Constant-overhead fault-tolerant quantum computation with reconfigurable atom arrays, Nature Physics **20**, 1084 (2024).

[218] H. J. Manetsch, G. Nomura, E. Bataille, X. Lv, K. H. Leung, and M. Endres, A tweezer array with 6,100 highly coherent atomic qubits, Nature **647**, 60 (2025).

[219] M. Cain, Q. Xu, R. King, L. R. B. Picard, H. Levine, M. Endres, J. Preskill, H.-Y. Huang, and D. Bluvstein, Shor's algorithm is possible with as few as 10,000 reconfigurable atomic qubits, arXiv preprint arXiv:2603.28627 (2026).

[220] S. J. Evered, D. Bluvstein, M. Kalinowski, S. Ebadi, T. Manovitz, H. Zhou, S. H. Li, A. A. Geim, T. T. Wang, N. Maskara, H. Levine, G. Semeghini, M. Greiner, V. Vuletić, and M. D. Lukin, High-fidelity parallel entangling gates on a neutral-atom quantum computer, Nature **622**, 268 (2023).

[221] H. Lange, A. Van de Walle, A. Abedinnia, and A. Bohrdt, From architectures to applications: a review of neural quantum states, Quantum Science and Technology **9**, 040501 (2024).

[222] M. Medvidović and J. R. Moreno, Neural-network quantum states for many-body physics, The European Physical Journal **139**, 631 (2024).

[223] M. S. Moss, R. Wiersema, M. Hibat-Allah, J. Carrasquilla, and R. G. Melko, Leveraging recurrence in neural network wavefunctions for large-scale simulations of Heisenberg antiferromagnets on the square lattice, Physical Review B **112**, 134450 (2025).

[224] G. Torlai, J. Carrasquilla, M. T. Fishman, R. G. Melko, and M. P. A. Fisher, Wavefunction positivization via automatic differentiation, Physical Review Research **2**, 032060(R) (2019).

[225] A. Jreissaty, H. Zhang, J. C. Quijano, J. Carrasquilla, and R. Wiersema, Entanglement and optimization within autoregressive neural quantum states, Physical Review Research **8**, 013147 (2026).

[226] P. Braccia, P. Bermejo, L. Cincio, and M. Cerezo, Computing exact moments of local random quantum circuits via tensor networks, Quantum Machine Intelligence **6**, 54 (2024).

[227] A. Berezutskii, M. Liu, A. Acharya, R. Ellerbrock, J. Gray, R. Haghshenas, Z. He, A. Khan, V. Kuzmin, D. Lyakh, D. Lykov, S. Mandrà, C. Mansell, A. Melnikov, A. Melnikov, V. Mironov, D. Morozov, F. Neukart, A. Nocera, M. A. Perlin, M. Perelshtein, M. Steinberg, R. Shaydulin, B. Villalonga, M. Pflitsch, M. Pistoia, V. Vinokur, and Y. Alexeev, Tensor networks for quantum computing, Nature Reviews Physics **7**, 581 (2025).

[228] L. Ma, M. T. Fishman, E. M. Stoudenmire, and E. Solomonik, Approximate Contraction of Arbitrary Tensor Networks with a Flexible and Efficient Density Matrix Algorithm, Quantum **8**, 1580 (2024).

[229] Y. Alexeev, M. H. Farag, T. L. Patti, M. E. Wolf, N. Ares, A. Aspuru-Guzik, S. C. Benjamin, Z. Cai, S. Cao, C. Chamberland, Z. Chandani, F. Fedele, I. Hamamura,

N. Harrigan, J. S. Kim, E. Kyoseva, J. G. Lietz, T. Lubowe, A. McCaskey, R. G. Melko, K. Nakaji, A. Peruzzo, P. Rao, B. Schmitt, S. Stanwyck, N. M. Tubman, H. Wang, and T. Costa, Artificial intelligence for quantum computing, Nature Communications **16**, 10829 (2025).

[230] M. Larocca, N. Ju, D. García-Martín, P. J. Coles, and M. Cerezo, Theory of overparametrization in quantum neural networks, Nat. Comput. Sci. **3**, 542 (2021).

[231] D. García-Martín, M. Larocca, and M. Cerezo, Quantum neural networks form Gaussian processes, Nature Physics **21**, 1153 (2025).

[232] J. Napp, Quantifying the barren plateau phenomenon for a model of unstructured variational ansätze, arXiv preprint arXiv:2203.06174 (2022).

[233] Y. Du, M. H. Hsieh, T. Liu, and D. Tao, Expressive power of parametrized quantum circuits, Physical Review Research **2**, 033125 (2020).

[234] S. Sim, P. D. Johnson, and A. Aspuru-Guzik, Expressibility and Entangling Capability of Parameterized Quantum Circuits for Hybrid Quantum-Classical Algorithms, Advanced Quantum Technologies **2**, 1900070 (2019).

[235] W. Huggins, P. Patil, B. Mitchell, K. Birgitta Whaley, and M. Stoudenmire, Towards quantum machine learning with tensor networks, Quantum Science Technology **4**, 024001 (2019).

[236] M. Cerezo, M. Larocca, D. García-Martín, N. L. Diaz, P. Braccia, E. Fontana, M. S. Rudolph, P. Bermejo, A. Ijaz, S. Thanasilp, E. R. Anschuetz, and Z. Holmes, Does provable absence of barren plateaus imply classical simulability?, Nature Communications **16**, 7907 (2025).

[237] M. L. Goh, M. Larocca, L. Cincio, M. Cerezo, and F. Sauvage, Lie-algebraic classical simulations for quantum computing, Physical Review Research **7**, 033266 (2025).

[238] H. Mhiri, R. Puig, S. Lerch, M. S. Rudolph, T. Chotibut, S. Thanasilp, and Z. Holmes, A unifying account of warm start guarantees for patches of quantum landscapes, arXiv preprint arXiv:2502.07889 (2025).

[239] L. Zhang, Dirac Delta Function of Matrix Argument, International Journal of Theoretical Physics **60**, 2445 (2020).

[240] F. Mezzadri, How to generate random matrices from the classical compact groups, arXiv preprint arXiv:math-ph/0609050 (2006).

[241] N. LaRacuente and F. Leditzky, Approximate Unitary k-Designs from Shallow, Low-Communication Circuits, arXiv preprint arXiv:2407.07876 (2024).

[242] L. Grevink, J. Haferkamp, M. Heinrich, J. Helsen, M. Hinsche, T. Schuster, and Z. Zimborás, Will it glue? On short-depth designs beyond the unitary group, arXiv preprint arXiv:2506.23925 (2025).

[243] H. Zhu, R. Kueng, M. Grassl, and D. Gross, The Clifford group fails gracefully to be a unitary 4-design, arXiv preprint arXiv:1609.08172 (2016).

[244] B. Collins and P. Śniady, Integration with respect to the Haar measure on unitary, orthogonal and symplectic group, Communications in Mathematical Physics **264**, 773 (2006).

[245] D. Petz and J. Réffy, On asymptotics of large Haar distributed unitary matrices, Periodica Mathematica Hungarica **49**, 103 (2004).

[246] S. N. Hearth, M. O. Flynn, A. Chandran, and C. R. Laumann, Unitary k-designs from random number-conserving quantum circuits, Physical Review X **15**, 021022 (2025).

[247] N. Hunter-Jones, Unitary designs from statistical mechanics in random quantum circuits, arXiv preprint arXiv:1905.12053 (2019).

[248] A. A. Mele, Introduction to Haar Measure Tools in Quantum Information: A Beginner's Tutorial, Quantum **8**, 1340 (2024).

[249] M. Ragone, Q. T. Nguyen, L. Schatzki, P. Braccia, M. Larocca, F. Sauvage, P. J. Coles, and M. Cerezo, Representation Theory for Geometric Quantum Machine Learning, arXiv preprint arXiv:2210.07980 (2022).

[250] P. Zinn-Justin, Jucys-Murphy elements and Weingarten matrices, Lett. Math. Phys. **91**, 119 (2009).

[251] M. Schroeder, Permutations Cycles and Derangements, in *Number Theory in Science and Communication: With Applications in Cryptography, Physics, Digital Information, Computing, and Self-Similarity* (Springer Berlin Heidelberg, 2009) pp. 147–158.

[252] T. Brady, A Partial Order on the Symmetric Group and New K(π,1)'s for the Braid Groups, Advances in Mathematics **161**, 20 (2001).

[253] J. Bridgeman and C. Chubb, Hand-waving and interpretive dance: an introductory course on tensor networks, Journal of Physics A **50**, 223001 (2017).

[254] M. S. Rudolph, J. Chen, J. Miller, A. Acharya, and A. Perdomo-Ortiz, Decomposition of Matrix Product States into Shallow Quantum Circuits, Quantum Sci. Technol **9**, 015012 (2024).

[255] G. Torlai, C. J. Wood, A. Acharya, G. Carleo, J. Carrasquilla, and L. Aolita, Quantum process tomography with unsupervised learning and tensor networks, Nature Communications **14**, 2858 (2023).

[256] F. Verstraete and I. Cirac, Matrix product states represent ground states faithfully, Physical Review B **73**, 094423 (2006).

[257] S. Paeckel, T. Köhler, A. Swoboda, S. R. Manmana, U. Schollwöck, and C. Hubig, Time-evolution methods for matrix-product states, Annals of Physics **411**, 167998 (2019).

[258] S.-J. Ran, E. Tirrito, C. Peng, X. Chen, L. Tagliacozzo, G. Su, and M. Lewenstein, Lecture Notes of Tensor Network Contractions, in *Tensor Network Contractions* (2017) 1st ed., p. 150.

[259] A. J. Ferris and G. Vidal, Perfect Sampling with Unitary Tensor Networks, Physical Review B **85**, 165146 (2012).

[260] G. Vidal, Efficient simulation of one-dimensional quantum many-body systems, Physical Review Lett. **93**, 040502 (2004).

[261] M. Hastings, An Area Law for One Dimensional Quantum Systems, Journal of Statistical Mechanics **8**, P08024.

[262] J. Eisert, M. Cramer, and M. B. Plenio, Colloquium: Area laws for the entanglement entropy, Reviews of Modern Physics **82**, 277 (2010).

[263] J. Tindall, M. Fishman, E. Miles Stoudenmire, and D. Sels, Efficient tensor network simulation of IBM's Eagle kicked Ising experiment, PRX Quantum **5**, 010308 (2024).

[264] T. H. Johnson, T. J. Elliott, S. R. Clark, and D. Jaksch, Capturing exponential variance using polynomial resources: applying tensor networks to non-equilibrium stochastic processes, Physical Review Letters **114**, 090602 (2015).

[265] K. R. Rao, H. Katiyar, T. S. Mahesh, A. SenDe, U. Sen, and A. Kumar, Multipartite quantum correlations reveal frustration in a quantum Ising spin system, Physical Review A **88**, 022312 (2013).

[266] K. Sharma, S. Khatri, M. Cerezo, and P. J. Coles, Noise resilience of variational quantum compiling, New Journal of Physics **22**, 043006 (2020).

[267] M. Dalgaard and F. Motzoi, Fast, high precision dynamics in quantum optimal control theory, J. Phys. B At. Mol. Opt. Phys. **55**, 085501 (2022).

[268] J. P. Peterson, H. Katiyar, and R. Laflamme, Fast Simulation of Magnetic Field Gradients for Optimization of Pulse Sequences, arXiv preprint arXiv:2006.10133 (2020).

[269] J. Pan, Y. Cao, X. Yao, Z. Li, C. Ju, H. Chen, X. Peng, S. Kais, and J. Du, Experimental realization of quantum algorithm for solving linear systems of equations, Physical Review A **89**, 022313 (2014).

[270] Y. Bahri, J. Kadmon, J. Pennington, S. S. Schoenholz, J. Sohl-Dickstein, and S. Ganguli, Statistical Mechanics of Deep Learning, Annu. Rev. Condens. Matter **11**, 501 (2020).

[271] L. Chizat, E. Oyallon, and F. Bach, On Lazy Training in Differentiable Programming, Adv. Neural Inf. Process. Syst. **32** (2018).

[272] R. Wiersema, C. Zhou, Y. De Sereville, J. F. Carrasquilla, Y. B. Kim, and H. Yuen, Exploring Entanglement and Optimization within the Hamiltonian Variational Ansatz, Physical Review X Quantum **1**, 020319 (2020).

[273] M. Benedetti, M. Fiorentini, and M. Lubasch, Hardware-efficient variational quantum algorithms for time evolution, Physical Review Res. **3**, 033083 (2021).

[274] G. S. Ravi, K. Smith, J. M. Baker, T. Kannan, N. Earnest, A. Javadi-Abhari, H. Hoffmann, and F. T. Chong, Navigating the dynamic noise landscape of variational quantum algorithms with qismet, in *Proceedings of the 28th ACM International Conference on Architectural Support for Programming Languages and Operating Systems*, ASPLOS 2023, Vol. 2 (Association for Computing Machinery, 2023) pp. 515–529.

[275] E. Fontana, M. Cerezo, A. Arrasmith, I. Rungger, and P. J. Coles, Non-trivial symmetries in quantum landscapes and their resilience to quantum noise, Quantum **6**, 804 (2020).

[276] S. Wang, E. Fontana, M. Cerezo, K. Sharma, A. Sone, L. Cincio, and P. J. Coles, Noise-Induced Barren Plateaus in Variational Quantum Algorithms, Nat. Commun. **12**, 6961 (2020).

[277] E. Fontana, N. Fitzpatrick, D. M. Ramo, R. Duncan, and I. Rungger, Evaluating the noise resilience of variational quantum algorithms, Physical Review A **104**, 022403 (2021).

[278] X. You, S. Chakrabarti, and X. Wu, A Convergence Theory for Over-parameterized Variational Quantum Eigensolvers, arXiv preprint arXiv:2205.12481 (2022).

[279] K. W. Moore and H. Rabitz, Exploring constrained quantum control landscapes, J. Chem. Phys. **137**, 134113 (2012).

[280] Y. Song, J. Li, Y. J. Hai, Q. Guo, and X. H. Deng, Optimizing quantum control pulses with complex constraints and few variables through autodifferentiation, Physical Review A **105**, 012616 (2022).

[281] K. Georgopoulos, C. Emary, and P. Zuliani, Modeling and simulating the noisy behavior of near-term quantum computers, Physical Review A **104**, 062432 (2021).

[282] N. M. Linke, D. Maslov, M. Roetteler, S. Debnath, C. Figgatt, K. A. Landsman, K. Wright, and C. Monroe, Experimental Comparison of Two Quantum Computing Architectures, Proceedings of the National Academy of Sciences of the United States of America **114**, 3305 (2017).

[283] S. Ashhab, N. Yamamoto, F. Yoshihara, and K. Semba, Numerical analysis of quantum circuits for state preparation and unitary operator synthesis, Physical Review A **106**, 022426 (2022).

[284] J. Nocedal and S. J. Wright, *Numerical Optimization*, 2nd ed. (Springer, 2006) p. 664.

[285] D. Wierichs, J. Izaac, C. Wang, and C. Y. Y. Lin, General parameter-shift rules for quantum gradients, Quantum **6**, 677 (2022).

[286] L. Mirsky, A trace inequality of John von Neumann, Monatshefte fur Math. **79**, 303 (1975).

[287] J. Kattemolle and G. Burkard, Ability of error correlations to improve the performance of variational quantum algorithms, Physical Review A **107**, 042426 (2023).

[288] Z. Cai, X. Xu, and S. C. Benjamin, Mitigating coherent noise using Pauli conjugation, Nature Quantum Information **6**, 1 (2020).

[289] P. D. Johnson, J. Romero, J. Olson, Y. Cao, and A. Aspuru-Guzik, QVECTOR: an algorithm for device-tailored quantum error correction, arXiv preprint arXiv:1711.02249 (2017).

[290] P. Niroula, S. Gopalakrishnan, and M. J. Gullans, Error Mitigation Thresholds in Noisy Quantum Circuits, arXiv preprint arXiv:2302.04278 (2023).

[291] B. Skinner, J. Ruhman, and A. Nahum, Measurement-Induced Phase Transitions in the Dynamics of Entanglement, Physical Review X **9**, 031009 (2019).

[292] S. Sarkar, On the trace of the product of Pauli and Dirac matrices, Acta Phys. Acad. Sci. Hungaricae **30**, 351 (1971).

[293] S. Endo, S. C. Benjamin, and Y. Li, Practical Quantum Error Mitigation for Near-Future Applications, Physical Review X **8**, 031027 (2018).

[294] S. Boixo, S. V. Isakov, V. N. Smelyanskiy, R. Babbush, N. Ding, Z. Jiang, M. J. Bremner, J. M. Martinis, and H. Neven, Characterizing quantum supremacy in near-term devices, Nature Physics **14**, 595 (2018).

[295] Y. Wu, W.-S. Bao, S. Cao, F. Chen, M.-C. Chen, X. Chen, T.-H. Chung, H. Deng, Y. Du, D. Fan, *et al.*, Strong quantum computational advantage using a superconducting quantum processor, Physical Review Letters **127**, 180501 (2021).

[296] A. Nahum, J. Ruhman, S. Vijay, and J. Haah, Quantum entanglement growth under random unitary dynamics, Physical Review X **7**, 031016 (2017).

[297] C. W. von Keyserlingk, T. Rakovszky, F. Pollmann, and S. L. Sondhi, Operator hydrodynamics, OTOCs, and entanglement growth in systems without conservation laws, Physical Review X **8**, 021013 (2018).

[298] A. Nahum, S. Vijay, and J. Haah, Operator spreading in random unitary circuits, Physical Review X **8**, 021014 (2018).

[299] W. W. Ho and S. Choi, Exact emergent quantum state designs from quantum chaotic dynamics, Physical Review Letters **128**, 060601 (2022).

[300] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven, Barren plateaus in quantum neural network training landscapes, Nature Communications **9**, 1 (2018).

[301] M. Cerezo, A. Sone, T. Volkoff, L. Cincio, and P. J. Coles, Cost function dependent barren plateaus in shallow parametrized quantum circuits, Nature Communications **12**, 1 (2021).

[302] M. Ragone, B. N. Bakalov, F. Sauvage, A. F. Kemper, C. Ortiz Marrero, M. Larocca, and M. Cerezo, A Lie algebraic theory of barren plateaus for deep parameterized quantum circuits, Nature Communications **15**, 7172 (2024).

[303] E. Fontana, D. Herman, S. Chakrabarti, N. Kumar, R. Yalovetzky, J. Heredge, S. H. Sureshbabu, and M. Pistoia, Characterizing barren plateaus in quantum ansätze with the adjoint representation, Nature Communications **15**, 7171 (2024).

[304] N. L. Diaz, D. García-Martín, S. Kazi, M. Larocca, and M. Cerezo, Showcasing a Barren Plateau Theory Beyond the Dynamical Lie Algebra, arXiv preprint arXiv:2310.11505 (2023).

[305] A. Elben, S. T. Flammia, H.-Y. Huang, R. Kueng, J. Preskill, B. Vermersch, and P. Zoller, The randomized measurement toolbox, Nature Review Physics **5**, 9 (2023).

[306] A. Zhao, N. C. Rubin, and A. Miyake, Fermionic partial tomography via classical shadows, Physical Review Letters **127**, 110504 (2021).

[307] K. H. Wan, O. Dahlsten, H. Kristjánsson, R. Gardner, and M. Kim, Quantum generalisation of feedforward neural networks, npj Quantum information **3**, 1 (2017).

[308] F. Sauvage and M. Larocca, Classical shadows with symmetries, arXiv preprint arXiv:2408.05279 (2024).

[309] M. West, A. A. Mele, M. Larocca, and M. Cerezo, Real classical shadows, arXiv preprint arXiv:2410.23481 (2024).

[310] H.-Y. Hu, A. Gu, S. Majumder, H. Ren, Y. Zhang, D. S. Wang, Y.-Z. You, Z. Minev, S. F. Yelin, and A. Seif, Demonstration of robust and efficient quantum property learning with shallow shadows, Nature Communications **16**, 2943 (2025).

[311] Z. Cai, R. Babbush, S. C. Benjamin, S. Endo, W. J. Huggins, Y. Li, J. R. McClean, and T. E. O'Brien, Quantum error mitigation, Reviews of Modern Physics **95**, 045005 (2023).

[312] B. Barak, C.-N. Chou, and X. Gao, Spoofing linear cross-entropy benchmarking in shallow quantum circuits, arXiv preprint arXiv:2005.02421 (2020).

[313] A. Letcher, S. Woerner, and C. Zoufal, Tight and efficient gradient bounds for parameterized quantum circuits, Quantum **8**, 1484 (2024).

[314] S. Mittal and N. Hunter-Jones, Local random quantum circuits form approximate designs on arbitrary architectures, arXiv preprint arXiv:2310.19355 (2023).

[315] A. E. Deneris, P. Bermejo, P. Braccia, L. Cincio, and M. Cerezo, Exact spectral gaps of random one-dimensional quantum circuits, Physical Review A **112**, 062619 (2025).

[316] C. Dankert, R. Cleve, J. Emerson, and E. Livine, Exact and approximate unitary 2-designs and their application to fidelity estimation, Physical Review A **80**, 012304 (2009).

[317] F. G. Brandao, A. W. Harrow, and M. Horodecki, Local random quantum circuits are approximate polynomial-designs, Communications in Mathematical Physics **346**, 397 (2016).

[318] J. Haferkamp, Random quantum circuits are approximate unitary t-designs in depth $O\left(nt^{5+o(1)}\right)$, Quantum **6**, 795 (2022).

[319] J. Haferkamp and N. Hunter-Jones, Improved spectral gaps for random quantum circuits: Large local dimensions and all-to-all interactions, Physical Review A **104**, 022417 (2021).

[320] W. G. Brown and L. Viola, Convergence Rates for Arbitrary Statistical Moments of Random Quantum Circuits, Physical Review Lett. **104**, 250501 (2010).

[321] C.-F. Chen, J. Docter, M. Xu, A. Bouland, F. G. Brandão, and P. Hayden, Efficient unitary designs from random sums and permutations, in *2024 IEEE 65th Annual Symposium on Foundations of Computer Science (FOCS)* (IEEE, 2024) pp. 476–484.

[322] C.-F. Chen, J. Haah, J. Haferkamp, Y. Liu, T. Metger, and X. Tan, Incompressibility and spectral gaps of random circuits, arXiv preprint arXiv:2406.07478 (2024).

[323] T. Yada, R. Suzuki, Y. Mitsuhashi, and N. Yoshioka, Non-Haar random circuits form unitary designs as fast as Haar random circuits, arXiv preprint arXiv:2504.07390 (2025).

[324] D. Gross, K. Audenaert, and J. Eisert, Evenly distributed unitaries: On the structure of unitary designs, Journal of mathematical physics **48**, 052104 (2007).

[325] I. Bengtsson and K. Zyczkowski, *Geometry of Quantum States: An Introduction to Quantum Entanglement* (Cambridge University Press, 2006).

[326] W. Bruzda, V. Cappellini, H.-J. Sommers, and K. Życzkowski, Random quantum operations, Physics Letters A **373**, 320 (2009).

[327] Y. Wang and J. Liu, A comprehensive review of quantum machine learning: from nisq to fault tolerance, Reports on Progress in Physics **87**, 116402 (2024).

[328] K. Nakaji and N. Yamamoto, Expressibility of the alternating layered ansatz for quantum computation, Quantum **5**, 434 (2021).

[329] R. J. Garcia, Y. Zhou, and A. Jaffe, Quantum scrambling with classical shadows, Physical Review Research **3**, 033155 (2021).

[330] L.-W. Yu, W. Li, Q. Ye, Z. Lu, Z. Han, and D.-L. Deng, Expressibility-induced Concentration of Quantum Neural Tangent Kernels, arXiv preprint arXiv:2311.04965 (2023).

[331] A. W. Harrow, Applications of coherent classical communication and the Schur transform to quantum information theory, arXiv preprint quant-ph/0512255 10.48550/arXiv.0512255 (2005).

[332] Y. Cardin, H. de Guise, and N. Quesada, Haarpy, a Python library for the symbolic calculation of Weingarten functions (2024).

[333] B. Magni, A. Christopoulos, A. De Luca, and X. Turkeshi, Anticoncentration in Clifford Circuits and Beyond: From Random Tensor Networks to Pseudo-Magic States, Physical Review X **15**, 031071 (2025).

[334] P. Sonya Tarabunga, M. Frau, T. Haug, X. Wang, M. M. Wilde, and Y. Su, Quantifying the magic of quantum channels, New Journal of Physics **21**, 103002 (2019).

[335] D. Greenbaum, Introduction to quantum gate set tomography, arXiv preprint arXiv:1509.02921 (2015).

[336] P. Singkanipa and D. A. Lidar, Beyond unital noise in variational quantum algorithms: noise-induced barren plateaus and limit sets, Quantum **9**, 1617 (2025).

[337] C. Hirche, C. Rouzé, and D. Stilck França, On contraction coefficients, partial orders and approximation of capacities for quantum channels, Quantum **6**, 862 (2022).

[338] E. Magesan, J. M. Gambetta, and J. Emerson, Scalable and robust randomized benchmarking of quantum processes, Physical Review Letters **106**, 180504 (2011).

[339] J. Kunjummen, M. C. Tran, D. Carney, and J. M. Taylor, Shadow process tomography of quantum channels, Physical Review A **107**, 042403 (2023).

[340] Y. Hama and H. Nishi, Quantum-Error-Mitigation Circuit Groups for Noisy Quantum Metrology, arXiv preprint arXiv:2303.01820 (2023).

[341] A. Ijaz, C. H. Alderete, F. Sauvage, L. Cincio, M. Cerezo, and M. L. Goh, More buck-per-shot: Why learning trumps mitigation in noisy quantum sensing, Materials Today Quantum **6**, 100042 (2025).

[342] E. Van Den Berg, Z. K. Minev, A. Kandala, and K. Temme, Probabilistic error cancellation with sparse Pauli–Lindblad models on noisy quantum processors, Nature Physics **19**, 1 (2023).

[343] D. García-Martín, P. Braccia, and M. Cerezo, Architectures and random properties of symplectic quantum circuits, Quantum Science and Technology **11**, 015012 (2025).

[344] Y. Ge and J. Eisert, Area laws and efficient descriptions of quantum many-body states, New Journal of Physics **18**, 083026 (2016).

[345] J. Czartowski and K. Zyczkowski, Quantum pushforward designs, Physical Review A **111**, 032433 (2025).

[346] Z. Webb, The Clifford group forms a unitary 3-design, Quantum Information and Computation **16**, 1379 (2016).

[347] J. S. Cotler, D. K. Mark, H.-Y. Huang, F. Hernandez, J. Choi, A. L. Shaw, M. Endres, and S. Choi, Emergent quantum state designs from individual many-body wave functions, PRX quantum **4**, 010311 (2023).

[348] P. W. Claeys and A. Lamacraft, Emergent quantum state designs and biunitarity in dual-unitary circuit dynamics, Quantum **6**, 738 (2022).

[349] M. Ippoliti and W. W. Ho, Dynamical purification and the emergence of quantum state designs from the projected ensemble, PRX Quantum **4**, 030322 (2023).

[350] J. D. Guimaraes, J. Lim, M. I. Vasilevskiy, S. F. Huelga, and M. B. Plenio, Noise-Assisted Digital Quantum Simulation of Open Systems Using Partial Probabilistic Error Cancellation, PRX Quantum **4**, 040329 (2023).

[351] M.-J. So and M.-S. Choi, Symmetry and Liouville Space Formulation of Decoherence-Free Subsystems, arXiv preprint arXiv:2507.15506 (2025).

[352] C. Strydom and M. Tame, Implementation of single-qubit measurement-based t-designs using IBM processors, Scientific Reports **12**, 1 (2022).

[353] D. M. Appleby, Å. Ericsson, and C. A. Fuchs, Properties of QBist State Spaces, Foundations of Physics **41**, 564 (2009).

[354] H.-J. Sommers and K. Zyczkowski, Statistical properties of random density matrices, Journal of Physics A: Mathematical and General **37**, 8457 (2004).

[355] M. Srednicki, Chaos and quantum thermalization, Physical Review E **50**, 888 (1994).

[356] M. Zhang, C. Wang, and Y. Han, Noisy Random Quantum Circuit Sampling and its Classical Simulation, Advanced Quantum Technologies **6**, 2300030 (2023).

[357] S. Czischek, G. Torlai, S. Ray, R. Islam, and R. G. Melko, Simulating a measurement-induced phase transition for trapped ion circuits, Physical Review A **104**, 062405 (2021).

[358] H. Hu, S. Choi, and Y. You, Classical shadow tomography with locally scrambled quantum dynamics, Physical Review Research **5**, 023027 (2023).

[359] F. Hu, G. Angelatos, S. A. Khan, M. Vives, E. Türeci, L. Bello, G. E. Rowlands, G. J. Ribeill, and H. E. Türeci, Tackling Sampling Noise in Physical Systems for Machine Learning Applications: Fundamental Limits and Eigentasks, Physical Review X **13**, 041020 (2023).

[360] K. Georgopoulos, C. Emary, and P. Zuliani, Modelling and Simulating the Noisy Behaviour of Near-term Quantum Computers, Physical Review A **104**, 062432 (2021).

[361] K. Tsubouchi, Y. Mitsuhashi, K. Sharma, and N. Yoshioka, Symmetric Clifford twirling for cost-optimal quantum error mitigation in early FTQC regime, npj Quantum Information **11**, 104 (2025).

[362] H. Jnane, J. Steinberg, Z. Cai, H. C. Nguyen, and B. Koczor, Quantum Error Mitigated Classical Shadows, PRX Quantum **5**, 010324 (2024).

[363] Y. Li, X. Chen, and M. P. Fisher, Quantum Zeno effect and the many-body entanglement transition, Physical Review B **98**, 205136 (2018).

[364] T. Heinosaari, M. A. Jivulescu, and I. Nechita, Random positive operator valued measures, Journal of Mathematical Physics **61**, 042202 (2019).

[365] A. Anshu and S. Arunachalam, A survey on the complexity of learning quantum states, Nature Reviews Physics **6**, 59 (2023).

[366] R. T. Thew, K. Nemoto, A. G. White, and W. J. Munro, Qudit quantum-state tomography, Physical Review A **66**, 012303 (2002).

[367] D. Gross, Y. K. Liu, S. T. Flammia, S. Becker, and J. Eisert, Quantum state tomography via compressed sensing, Physical Review Letters **105**, 150401 (2010).

[368] M. Cramer, M. B. Plenio, S. T. Flammia, R. Somma, D. Gross, S. D. Bartlett, O. Landon-Cardinal, D. Poulin, and Y. K. Liu, Efficient quantum state tomography, Nature Communications **1**, 149 (2010).

[369] D. E. Koh and S. Grewal, Classical Shadows With Noise, Quantum **6**, 776 (2020).

[370] M. Kus, J. Mostowski, and F. Haake, Universality of eigenvector statistics of kicked tops of different symmetries, Journal of Physics A: Mathematical and General **21**, L1073 (1988).

[371] C. F. Dunkl, P. Gawron, J. A. Holbrook, J. A. Miszczak, Z. Puchała, and K. Życzkowski, Numerical shadow and geometry of quantum states, Journal of Physics A **44**, 335301 (2011).

[372] E. Gutkin and K. Zyczkowski, Joint numerical ranges, quantum maps, and joint numerical shadows, Linear Algebra and its Applications **438**, 2394 (2013).

[373] L. Zhang, S. Luo, S.-M. Fei, and J. Wu, Uncertainty regions of observables and state-independent uncertainty relations, Quantum Information Processing **20**, 357 (2021).

[374] A. De Luca, C. Liu, A. Nahum, and T. Zhou, Universality Classes for Purification in Nonunitary Quantum Processes, Physical Review X **15**, 041024 (2025).

[375] S. Sang, Y. Zou, and T. H. Hsieh, Mixed-state Quantum Phases: Renormalization and Quantum Error Correction, Physical Review X **14**, 031044 (2023).

[376] M. Zhang, C. Wang, S. Dong, H. Zhang, Y. Han, and L. He, Entanglement entropy scaling of noisy random quantum circuits in two dimensions, Physical Review A **106**, 052430 (2022).

[377] S. G. Bobkov and F. Götze, Concentration of empirical distribution functions with applications to non-i.i.d. models, Bernoulli **16**, 1385 (2010).

[378] F. Perez-Cruz, Kullback-Leibler divergence estimation of continuous distributions, in *2008 IEEE International Symposium on Information Theory* (2008) pp. 1666–1670.

[379] D. S. Dimitrova, V. K. Kaishev, and S. Tan, Computing the Kolmogorov-Smirnov Distribution When the Underlying CDF is Purely Discrete, Mixed, or Continuous, Journal of Statistical Software **95**, 1 (2020).

[380] S. G. Krantz and H. R. Parks, *A Primer of Real Analytic Functions*, 1st ed. (Birkhäuser Basel, 1992) pp. 1–184.

[381] M. Arienzo, M. Heinrich, I. Roth, and M. Kliesch, Closed-form analytic expressions for shadow estimation with brickwork circuits, Quantum Information and Computation **23**, 961 (2023).

[382] M. J. Schlosser, Multiple Hypergeometric Series: Appell Series and Beyond, in *Computer Algebra in Quantum Field Theory: Integration, Summation and Special Functions*, edited by C. Schneider and J. Blümlein (Springer Vienna, 2013) pp. 305–324.

[383] D. Hangleiter and J. Eisert, Computational advantage of quantum random sampling, Rev. Mod. Phys. **95**, 035001 (2023).

[384] A. Zlokapa, B. Villalonga, S. Boixo, and D. A. Lidar, Boundaries of quantum supremacy via random circuit sampling, npj Quantum Information **9**, 36 (2023).

[385] F. Pan, K. Chen, and P. Zhang, Solving the Sampling Problem of the Sycamore Quantum Circuits, Physical Review Lett. **129**, 090502 (2022).

[386] M. Myers II, M. H. Teo, R. Mishra, J. H. Chai, and H. K. Ng, Simulating general noise nearly as cheaply as Pauli noise, arXiv preprint arXiv:2512.07304 (2025).

[387] A. Sander, M. Fröhlich, M. Ali, M. Eigel, J. Eisert, M. Hintermüller, C. B. Mendl, R. M. Milbradt, and R. Wille, Quantum circuit simulation with a local time-dependent variational principle, arXiv preprint arXiv:2508.10096 (2025).

[388] M. S. Byrd and N. Khaneja, Characterization of the positivity of the density matrix in terms of the coherence vector representation, Physical Review A **68**, 062322 (2003).

[389] A. Liniov, I. Meyerov, E. Kozinov, V. Volokitin, I. Yusipov, M. Ivanchenko, and S. Denisov, Unfolding quantum master equation into a system of real-valued equations: computationally effective expansion over the basis of $SU(N)$ generators, Physical Review E **100**, 053305 (2019).

[390] A. H. Werner, D. Jaschke, P. Silvi, M. Kliesch, T. Calarco, J. Eisert, and S. Montangero, Positive Tensor Network Approach for Simulating Open Quantum Many-Body Systems, Physical Review Letters **116**, 237201 (2016).

[391] Y. Guo and S. Yang, Quantum state tomography with locally purified density operators and local measurements, Communications Physics **7**, 1 (2024).

[392] N. Lee, A.-H. Phan, F. Cong, and A. Cichocki, Nonnegative Tensor Train Decompositions for Multi-domain Feature Extraction and Clustering, in *Neural Information Processing*, edited by A. Hirose, S. Ozawa, K. Doya, K. Ikeda, M. Lee, and D. Liu (Springer International Publishing, 2016) pp. 87–95.

[393] E. M. Shcherbakova, Nonnegative Tensor Train Factorization with DMRG Technique, Lobachevskii Journal of Mathematics **40**, 1863 (2019).

[394] E. M. Shcherbakova and E. E. Tyrtyshnikov, Fast Nonnegative Tensor Factorizations with Tensor Train Model, Lobachevskii Journal of Mathematics **43**, 882 (2022).

[395] E. Gaussier and C. Goutte, Relation between plsa and nmf and implications, in *Proceedings of the 28th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR '05 (Association for Computing Machinery, 2005) p. 601–602.

[396] A. Cichocki, R. Zdunek, A. H. Phan, and S. I. Amari, *Nonnegative Matrix and Tensor Factorizations: Applications to Exploratory Multi-Way Data Analysis and Blind Source Separation* (John Wiley and Sons, 2009) pp. 1–477.

[397] X. Fu, K. Huang, N. D. Sidiropoulos, and W. K. Ma, Nonnegative Matrix Factorization for Signal and Data Analytics: Identifiability, Algorithms, and Applications, IEEE Signal Processing Magazine **36**, 59 (2018).

[398] Y. T. Guo, Q. Q. Li, and C. S. Liang, The rise of nonnegative matrix factorization: Algorithms and applications, Information Systems **123**, 102379 (2024).

[399] T. Sano, T. Migita, and N. Takahashi, A novel update rule of HALS algorithm for nonnegative matrix factorization and Zangwill's global convergence, Journal of Global Optimization **84**, 755 (2022).

[400] C. Boutsidis and E. Gallopoulos, SVD based initialization: A head start for nonnegative matrix factorization, Pattern Recognition **41**, 1350 (2008).

[401] D. Wellnitz, G. Preisser, V. Alba, J. Dubail, and J. Schachenmayer, Rise and fall, and slow rise again, of operator entanglement under dephasing, Physical Review Letters **129**, 170401 (2022).

[402] Z. Li, S. Sang, and T. H. Hsieh, Entanglement Dynamics of Noisy Random Circuits, Physical Review B **107**, 014307 (2023).

[403] J. Dubail, Entanglement scaling of operators: a conformal field theory approach, with a glimpse of simulability of long-time dynamics in 1+1d, Journal of Physics A **50**, 234001 (2017).

[404] S. A. Vavasis, On the Complexity of Nonnegative Matrix Factorization, SIAM Journal on Optimization **20**, 1364 (2009).

[405] N. Gillis, The Why and How of Nonnegative Matrix Factorization, in *Regularization, Optimization, Kernels, and Support Vector Machines* (CRC Press, 2014) pp. 257–291.

[406] J. Kossaifi, Y. Panagakis, A. Anandkumar, and M. Pantic, TensorLy: Tensor Learning in Python, Journal of Machine Learning Research **20**, 1 (2019).

[407] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, *et al.*, SciPy 1.0: fundamental algorithms for scientific computing in Python, Nature methods **17**, 261 (2020).

[408] Y. Yīng, M. C. Alañón, D. Centeno, J. Surace, M. M. Ansanelli, R. Liu, D. Schmid, and R. W. Spekkens, On whether quantum theory needs complex numbers: the foil theories perspective, arXiv preprint arXiv:2506.08091 (2025).

[409] M. O. Renou, D. Trillo, M. Weilenmann, T. P. Le, A. Tavakoli, N. Gisin, A. Acín, and M. Navascués, Quantum theory based on real numbers can be experimentally falsified, Nature **600**, 625 (2021).

[410] I. H. Kim, Entropy Scaling Law and the Quantum Marginal Problem, Physical Review X **11**, 021039 (2021).

[411] S. Schmitt, L. Ekstrøm, A. Bottarelli, and X. Bonet-Monroig, Experimental demonstration of the absence of noise-induced barren plateaus using information content landscape analysis, arXiv preprint arXiv:2602.22851 (2026).

[412] M. Rose, S. Kais, and D. A. Mazziotti, Open-quantum-system simulation through exploiting noise on quantum computers, Physical Review A **111**, 052612 (2025).

[413] L. Ding, J. Fan, and X. Qiu, Universally robust control of open quantum systems, npj Quantum Information **12**, 22 (2025).

[414] K. Wang and X. Li, Non-Markovian noise mitigation: Practical implementation, error analysis, and the role of spectral properties of the environment, Physical Review A **112**, 012406 (2025).

[415] Z. Liu, Y. Xiao, and Z. Cai, Non-Markovian Noise Suppression Simplified through Channel Representation, arXiv preprint arXiv:2412.11220 (2024).

[416] A. Bärligea, B. Poggel, and J. M. Lorenz, Scalability challenges in variational quantum optimization under stochastic noise, Physical Review A **112**, 032407 (2025).

[417] G. Styliaris, N. Anand, and P. Zanardi, Information scrambling over bipartitions: Equilibration, entropy production, and typicality, Physical Review Letters **126**, 030601 (2020).

[418] Y. Xiang and S. J. Xiong, Entropy exchange, coherent information, and concurrence, Physical Review A **76**, 014306 (2007).

[419] C. King, The capacity of the quantum depolarizing channel, IEEE Transactions on Information Theory **49**, 221 (2002).

[420] Y. Du, Z. Tu, X. Yuan, and D. Tao, Efficient measure for the expressivity of variational quantum algorithms, Physical Review Letters **128**, 080506 (2022).

[421] L. Leone, S. F. Oliviero, and A. Hamma, Isospectral Twirling and Quantum Chaos, Entropy **23**, 1073 (2021).

[422] P. Zanardi and N. Anand, Information Scrambling and Chaos in Open Quantum Systems, Physical Review A **103**, 062214 (2021).

[423] B. Stratton, C.-Y. Hsieh, and P. Skrzypczyk, Operational interpretation of the choi rank through exclusion tasks, Physical Review A **110**, L050601 (2024).

[424] T. Scheiber, P. Haubenwallner, and M. Heller, Reduced Sampling Overhead for Probabilistic Error Cancellation by Pauli Error Propagation, Quantum **9**, 1840 (2025).

[425] X. Yang, X. Nie, Y. Ji, T. Xin, D. Lu, and J. Li, Improved quantum computing with higher-order trotter decomposition, Physical Review A **106**, 042401 (2022).

[426] D. J. Egger, C. Capecci, B. Pokharel, P. K. Barkoutsos, L. E. Fischer, L. Guidoni, and I. Tavernelli, Pulse variational quantum eigensolver on cross-resonance-based hardware, Physical Review Res. **5**, 033159 (2023).

[427] R. Wiersema, E. Kökcü, A. F. Kemper, and B. N. Bakalov, Classification of dynamical Lie algebras of 2-local spin systems on linear, circular and fully connected topologies, npj Quantum Information **10**, 110 (2024).

[428] J. Haah, R. Kothari, R. O'Donnell, and E. Tang, Query-optimal estimation of unitary channels in diamond distance, in *2023 IEEE 64th Annual Symposium on Foundations of Computer Science (FOCS)* (2023) pp. 363–390.

[429] H. Fischer, *A History of the Central Limit Theorem: From Classical to Modern Probability Theory*, edited by J. Buchwald (Springer New York, 2010) pp. 1–415.

[430] T. Crowder, A linearization of quantum channels, Journal of Geometry and Physics **92**, 157 (2015).

[431] G. Kimura, The Bloch Vector for N-Level Systems, Phys. Lett. A **314**, 339 (2003).

[432] T. Nuradha and M. M. Wilde, Fidelity-Based Smooth Min-Relative Entropy: Properties and Applications, IEEE Transactions on Information Theory **70**, 4170 (2024).

[433] J. Wilkinson, *The Algebraic Eigenvalue Problem*, 1st ed. (Oxford University Press, 1965) pp. 112–116.

[434] N. Shirai, K. Kubo, K. Mitarai, and K. Fujii, Quantum tangent kernel, Physical Review Research **6**, 033179 (2024).

[435] G. Kreweras, Sur les partitions non croisees d'un cycle, Discrete Mathematics **1**, 333 (1972).

[436] A. Nica and R. Speicher, *Lectures on the Combinatorics of Free Probability*, London Mathematical Society Lecture Note Series (Cambridge University Press, 2006).

[437] K. Ebrahimi-Fard, L. Foissy, J. Kock, and F. Patras, Noncrossing arithmetic, arXiv preprint arXiv:2407.17660 (2024).

[438] S. Linton, N. Ruškuc, and V. Vatter, *Permutation Patterns*, London Mathematical Society Lecture Note Series (Cambridge University Press, 2010).

[439] R. Rama, Inclusion Exclusion Principle, in *Topics in Combinatorics and Graph Theory* (2025) pp. 111–129.

[440] D. M. Topkis, The structure of sublattices of the product of n lattices, Pacific Journal of Mathematics **65**, 525 (1976).

[441] R. A. Horn and C. R. Johnson, *Matrix analysis* (Cambridge university press, 2012).

[442] D. Chruściński, Dynamical maps beyond Markovian regime, Physics Reports **992**, 1 (2022).

[443] Y. Virkar and A. Clauset, Power-law distributions in binned empirical data, Annals of Applied Statistics **8**, 89 (2014).

Appendices

Chapter 3 Appendices

3.A Parameterized Quantum Systems

In these appendixes, we elaborate on the specific parameterized quantum channel ansatze studied. We discuss the unitary and non-unitary components of the channel, and their approximations and derivations via Trotterization. These Trotterized forms correspond to quantum circuit models of continuous evolution of quantum systems up to a specified order of precision, and they are used for classical simulation.

3.A.1 Unitary Evolution

The unitary evolution takes the form of the time ordered exponential of a Hamiltonian,

$$U_\theta = \mathcal{T}e^{-i\int_0^T dt\ H_\theta^{(t)}} \ . \tag{3.A.1}$$

Here the time-dependent Hamiltonian driving the evolution takes the parameterized form,

$$H_\theta^{(t)} = \sum_\mu H_\mu^{(t)} \qquad : \qquad H_\mu^{(t)} = \theta_\mu^{(t)} G_\mu \ , \tag{3.A.2}$$

given a set of poly(n) variable, time-dependent parameters $\theta^{(t)}$, and a set of fixed, time-independent generators $\mathcal{G} = \{G_\mu\}$, at each time $t \in [0, T]$.

This continuous evolution generated by exponential maps of Hamiltonians must be discretized temporally and spatially across the space of subsystems for feasible classical simulation. This discretization further allows for comparison with the variational quantum circuit paradigm. Depending on the control problem of interest, there are several choices for the specific discretization scheme. Further, given constraints placed on the parameters, the explicitly optimized parameters may take various functional forms.

3.A.2 Trotterization

To classically simulate such time-dependent systems, unitaries are trotterized, both temporally and spatially across the space. To first-order in time,

$$U_\theta \approx \prod_{\kappa}^{k} U_\theta^{(\kappa)} \ + \ \mathcal{O}(\tau^2) \qquad : \qquad U_\theta^{(\kappa)} = e^{-i\tau H_\theta^{(\kappa)}} \ . \tag{3.A.3}$$

where the time has been discretized into k time steps of size $\tau = T/k$.

Further, depending on the commutation relations between terms in the Hamiltonian, to Q-order in space across the qubits, at a given instance κ in time,

$$U_\theta^{(\kappa)} \approx \prod_{\mu} U_\mu^{(\kappa)} \ + \ \mathcal{O}(\big[H_\mu^{(\kappa)}, H_\nu^{(\kappa)}\big]_{Q+1} \tau^{Q+1}) \ . \tag{3.A.4}$$

The product is over some function of lower-order Trotterizations, denoted by $U_\mu^{(\kappa)}$, with the error being proportional to the $Q+1$-deep nested commutator of Hamiltonian terms at time κ [425]. For $Q = 2$, the product is equal to the forward and backward ordering of first-order Trotterized operators,

$$U_\theta^{(\kappa)} \overset{Q=2}{=} \overrightarrow{\prod_{\mu}} U_\mu^{(\kappa)/Q} \overleftarrow{\prod_{\mu}} U_\mu^{(\kappa)/Q} \ + \ \mathcal{O}(\tau^3) \ . \tag{3.A.5}$$

The resulting evolution can be directly described by a parameterized quantum circuit, consisting of operators $U_\mu^{(\kappa)}$ with locality of the corresponding Hamiltonian generator G_μ. Therefore the following results, up to the precision of these discretizations, are relevant both in the discrete gate-based circuit model, and the continuous time evolution model, more generally seen in pulse-level and quantum control problems [80, 426].

3.A.3 Parameterized Quantum Channels

To understand the effects of noise on the evolution and abilities of such a system to represent various targets, we describe the evolution with parameterized quantum channels,

$$\Lambda_{\theta\gamma} = \mathcal{N}_\gamma \circ \mathcal{U}_\theta \tag{3.A.6}$$

such that the evolved parameterized state from an initial state σ is

$$\rho_{\theta\gamma} = \Lambda_{\theta\gamma}(\sigma) \ . \tag{3.A.7}$$

The channel is composed of a noiseless unitary channel with variable parameters θ,

$$\mathcal{U}_{\theta}(\cdot) = U_{\theta} \cdot U_{\theta}^{\dagger} \ , \tag{3.A.8}$$

and a noisy non-unitary Kraus operator channel with fixed noise parameters γ,

$$\mathcal{N}_{\gamma}(\cdot) = \sum_{K_{\gamma}} K_{\gamma} \cdot K_{\gamma}^{\dagger} \ . \tag{3.A.9}$$

Given the Trotterization of the continuous time evolution, we define the channel as a composition of k layers of channels at each time step,

$$\Lambda_{\theta\gamma} = \ \circ_{\kappa\in[k]} \left(\mathcal{N}_{\gamma}^{(\kappa)} \circ \mathcal{U}_{\theta}^{(\kappa)}\right) \ . \tag{3.A.10}$$

Let us denote composition of channels before or after an index κ as,

$$\Lambda_{\theta\gamma}^{\lesseqgtr\kappa} = \ \circ_{\eta\lesseqgtr\kappa} \left(\mathcal{N}_{\gamma}^{(\eta)} \circ \mathcal{U}_{\theta}^{(\eta)}\right) \ . \tag{3.A.11}$$

We also write the Q-order spatial Trotterization of the unitary part of the channel as a composition over the Trotterized unitaries, with identical notation to the products of unitaries when deriving their Trotterization,

$$\mathcal{U}_{\theta}^{(\kappa)} = \ \circ_{\mu} \, \mathcal{U}_{\mu}^{(\kappa)} \ . \tag{3.A.12}$$

Let us also denote composition of unitaries before or after an index μ as,

$$\mathcal{U}_{\lesseqgtr\mu}^{(\kappa)} = \ \circ_{\nu\lesseqgtr\mu} \, \mathcal{U}_{\nu}^{(\kappa)} \ , \tag{3.A.13}$$

and finally denote partial channels before or after an index (μ, κ) as,

$$\Lambda_{\theta\gamma} = \Lambda_{\gamma}^{(>\kappa)} \circ \mathcal{N}_{\gamma}^{(\kappa)} \circ \mathcal{U}_{>\mu}^{(\kappa)} \circ \mathcal{U}_{\mu}^{(\kappa)} \circ \mathcal{U}_{<\mu}^{(\kappa)} \circ \Lambda_{\gamma}^{(<\kappa)} \ . \tag{3.A.14}$$

The corresponding partially forward evolved state relative to a state σ from the action of the channel before an index (μ, κ) may then be denoted as

$$\rho_{<\mu\gamma}^{(<\kappa)} = \Lambda_{<\mu\gamma}^{(<\kappa)}(\sigma) \ , \tag{3.A.15}$$

and the corresponding backward evolved operator relative to an input operator Π from the adjoint action of the channel after an index (μ, κ) may also be denoted as

$$\Pi_{>\mu\gamma}^{(>\kappa)} = \Lambda_{>\mu\gamma}^{(>\kappa)^{\dagger}}(\Pi) \ . \tag{3.A.16}$$

This notation is used to understand parameter shift rules for channels, and to derive the scaling of objectives with noise and with depth.

3.B Learning Phenomena

In these appendixes, we study learning phenomena, specifically overparameterization phenomena in quantum settings. We derive a unitary version of the quantum Fisher information, and we show numerically that its rank acts as an indicator of overparameterization identically to the state quantum Fisher information studied in previous works. Finally, we use these results to confirm that overparameterization phenomena occur in realistic settings, as per established definitions.

A key aspect of these studies involves understanding whether in realistic settings, indicators of overparameterization, in particular exponential convergence of the optimization with the number of parameters, still occur. We follow the approaches by Larocca *et al.* [230], later followed up by Garcia-Martin *et al.* [102], which set bounds on the rank of the quantum Fisher information to determine whether a quantum system is overparameterized, both in noiseless and then noisy settings.

Overparameterization in this context, as per the Fisher information definition [230], refers to when there are adequate number p parameters such that the model ansatz can span the space [100, 101] of the dynamical Lie algebra $\mathcal{G}$ formed by its generators $\{G_\mu\}$. This parameterization generally translates to the optimization procedure being able to converge exponentially with the number of optimization iterations. This may occur due to a fundamental change in the objective landscape where it becomes much more convex in this regime. It may also be accompanied by what is known as lazy training, where the optimal parameters are negligibly different from their random initial values.

In the continuous time evolution, or gate based circuit formalisms, these generators are the non-commuting terms in the underlying Hamiltonian that drives the evolution, and the dynamical Lie algebra is formed by the Lie closure

$$\mathcal{G} = \langle \{G_\mu\} \rangle_{\mathrm{Lie}} \ , \tag{3.B.1}$$

of all linearly independent nested commutators, of these generators.

This dynamical Lie algebra span forms a subspace $\mathcal{G} \subseteq \mathcal{H}$ of the full space $\mathcal{H}$ where the operators act, and it has dimension

$$g = |\mathcal{G}| \ . \tag{3.B.2}$$

Within the context of spaces with dimension $d = q^n$, example algebras associated with the Lie closure that arise in quantum control settings include the special unitary algebra $su(q^n)$ with dimension $|su(q^n)| = q^{2n} - 1$, or the symplectic algebra $sp(2n)$ with dimension

$|sp(2n)| = 2n^2 + n$. Generally, the dynamical Lie algebra dimension is either a polynomial or an exponential function of the system size n, and one-dimensional Lie algebras have been classified in [427]. The number of parameters p, for a fixed periodic ansatz repeated for k layers, generally scales as

$$p \sim \mathcal{O}(\mathrm{poly}(n)k) \ , \tag{3.B.3}$$

and the depth itself may depend on the system size. For overparameterization to occur, the number of parameters must be of at least similar order to this dimension,

$$p > \tilde{p} \sim \mathcal{O}(g) \ , \tag{3.B.4}$$

and in ideal settings, overparameterization occurs when exactly $\tilde{p} = g$.

A key indicator [102, 230] of overparameterization is whether the rank of the quantum Fisher information $\mathcal{F}_\theta$ saturates at this dimension,

$$r_\theta^{\mathcal{F}} = \mathrm{rank}(\mathcal{F}_\theta) \leq g \ . \tag{3.B.5}$$

It can be shown that this saturation is independent of where you are in the objective or parameter landscapes, and it does not occur only at optimality.

Similar bounds for the Hessian $\mathcal{H}_\theta$ of the objective, being objective- and target-dependent, only occur at optimality,

$$r_{\theta^*}^{\mathcal{H}} = \mathrm{rank}(\mathcal{H}_\theta) \leq g \ , \tag{3.B.6}$$

and otherwise the Hessian is generally full rank $r_\theta^{\mathcal{H}} = p$.

We should also note recent developments [102] in understanding the effects of noise on variational quantum circuits, in the context of the spectrum of the quantum Fisher information. Small amounts of local noise are shown to allow more directions to be searched [287]. These additional directions in the state space may increase or decrease the parameterized state's purity. However, for increasing noise scales, the system becomes exponentially less sensitive to its parameters, can search fewer and fewer directions. As shown in Section 3.3, we also observe different regimes depending on noise scales. Specifically, we observe quasi-linear scaling of infidelity with noise at the boundary of the convergent and divergent regimes. Finite noise suppresses the system's abilities to achieve perfect infidelity, however not so much as to be confined within the divergent regime.

3.B.1 Unitary Quantum Fisher Information

We now study the form of the quantum Fisher information, which can be shown [182] to be proportional to the second-order correction to the Bures metric, also referred to as the Fubini-Study metric in the case of pure states. This quantity offers insight into which directions of the space can be reached, given the variational ansatz. Here, we generalize the definition of the Fisher information, from states, to a state-independent definition that only depends on the underlying parameterized unitary, similar in form to another recent study of generalized metrics [190].

We now investigate the second-order term, or Fisher-like information of distance measures $\mathcal{L}_\theta^U$ between a parameterized unitary U_θ with parameters $\theta = \{\theta_\mu\}$, and a fixed reference unitary U, over a d-dimensional space. We define $\mathcal{L}_{\theta^*}^U = \mathcal{L}_\theta^U|_{U_\theta=U}$ to be the parameterized unitary evaluated exactly at the reference unitary.

Let a distance $\mathcal{L}_\theta^U$ between unitaries (that is not a proper distance metric as it does not satisfy the triangle inequality), be related to the absolute trace overlap

$$\mathcal{L}_\theta^U = 1 - \frac{1}{d^2}\left|\mathrm{tr}\left(U^\dagger U_\theta\right)\right|^2 , \tag{3.B.7}$$

with derivatives with respect to θ of

$$\begin{aligned}\partial_\mu \mathcal{L}_\theta^U &= -2\frac{1}{d^2}\mathrm{Re}\left(\mathrm{tr}\left(U_\theta^\dagger U\right)\mathrm{tr}\left(U^\dagger \partial_\mu U_\theta\right)\right) &(3.B.8)\\ \partial_{\mu\nu} \mathcal{L}_\theta^U &= -2\frac{1}{d^2}\mathrm{Re}\left(\mathrm{tr}\left(U_\theta^\dagger U\right)\mathrm{tr}\left(U^\dagger \partial_{\mu\nu} U_\theta\right) - \mathrm{tr}\left(\partial_\mu U_\theta^\dagger U\right)\mathrm{tr}\left(U^\dagger \partial_\nu U_\theta\right)\right) . &(3.B.9)\end{aligned}$$

At optimality where $U_\theta = U$,

$$\begin{aligned}\mathcal{L}_{\theta^*}^U = 0 \quad , \quad \partial_\mu \mathcal{L}_{\theta^*}^U = 0 & &(3.B.10)\\ \partial_{\mu\nu}\mathcal{L}_{\theta^*}^U = 2\frac{1}{d^2}\mathrm{Re}\left(d\ \mathrm{tr}\left(\partial_\mu U_\theta^\dagger \partial_\nu U_\theta\right) - \mathrm{tr}\left(\partial_\mu U_\theta^\dagger U_\theta\right)\mathrm{tr}\left(U_\theta^\dagger \partial_\nu U_\theta\right)\right) . & &(3.B.11)\end{aligned}$$

To define the Fisher information metric, we define it as the leading-order behaviour of the objective given a perturbation of parameters $\theta \to \theta + \delta$, evaluated at $U = U_\theta$, yielding

$$\mathcal{L}_{\theta+\delta}^U = \mathcal{F}_{\theta_{\mu\nu}}^U \delta_\mu \delta_\nu \ + \ \mathcal{O}(\delta^3) , \tag{3.B.12}$$

with the Fisher information metric being

$$\mathcal{F}_{\theta_{\mu\nu}}^U = \frac{1}{d^2}\mathrm{Re}\left(d\ \mathrm{tr}\left(\partial_\mu U_\theta^\dagger \partial_\nu U_\theta\right) - \mathrm{tr}\left(\partial_\mu U_\theta^\dagger U_\theta\right)\mathrm{tr}\left(U_\theta^\dagger \partial_\nu U_\theta\right)\right) . \tag{3.B.13}$$

This state-independent quantity, identical to other Fisher information metrics, contains a term that reflects the change in the ansatz, plus a corrective term to ensure gauge invariance, with additional dimensionality d factors to reflect that the action of the ansatz with respect to specific states is not being considered.

We can also use a proper distance metric in terms of the Frobenius norm $\|A\|^2 = \mathrm{tr}\left(A^\dagger A\right)$ [428], such that a proper objective is invariant up to phases between the operators

$$\tilde{\mathcal{L}}_\theta^U = 1 - \sqrt{1 - \mathcal{L}_\theta^U} = \frac{1}{2}\frac{1}{d}\max_\phi \left\|U_\theta - e^{i\phi}U\right\|^2 , \tag{3.B.14}$$

with derivatives with respect to θ of

$$\partial_\mu \tilde{\mathcal{L}}_\theta^U = \frac{1}{2}\frac{1}{1-\tilde{\mathcal{L}}_\theta^U}\partial_\mu \mathcal{L}_\theta^U \tag{3.B.15}$$

$$\partial_{\mu\nu} \tilde{\mathcal{L}}_\theta^U = \frac{1}{2}\frac{1}{(1-\tilde{\mathcal{L}}_\theta^U)^2}\left((1-\tilde{\mathcal{L}}_\theta^U)\partial_{\mu\nu}\mathcal{L}_\theta^U + \partial_\mu \mathcal{L}_\theta^U \partial_\nu \mathcal{L}_\theta^U\right) , \tag{3.B.16}$$

such that the optimal proper and improper quantities are identical up to constant scalings,

$$\tilde{\mathcal{L}}_{\theta^*}^U = 0 \quad , \quad \partial_\mu \tilde{\mathcal{L}}_{\theta^*}^U = 0 \quad , \quad \partial_{\mu\nu}\tilde{\mathcal{L}}_{\theta^*}^U = \frac{1}{2}\partial_{\mu\nu}\mathcal{L}_{\theta^*}^U \ . \tag{3.B.17}$$

and similarly for the proper and improper Fisher information metric definitions,

$$\tilde{\mathcal{F}}_{\theta_{\mu\nu}}^U = \frac{1}{2}\mathcal{F}_{\theta_{\mu\nu}}^U \ . \tag{3.B.18}$$

For example, when the trace is over $d = 1$ states, $|\psi_\theta\rangle = U_\theta|\sigma\rangle \to \rho_\theta = \psi_\theta$ from an initial fixed state $|\sigma\rangle$, or equivalently the unitaries are projected onto a $d = 1$-dimensional subspace, the Fisher information reduces to the standard definition

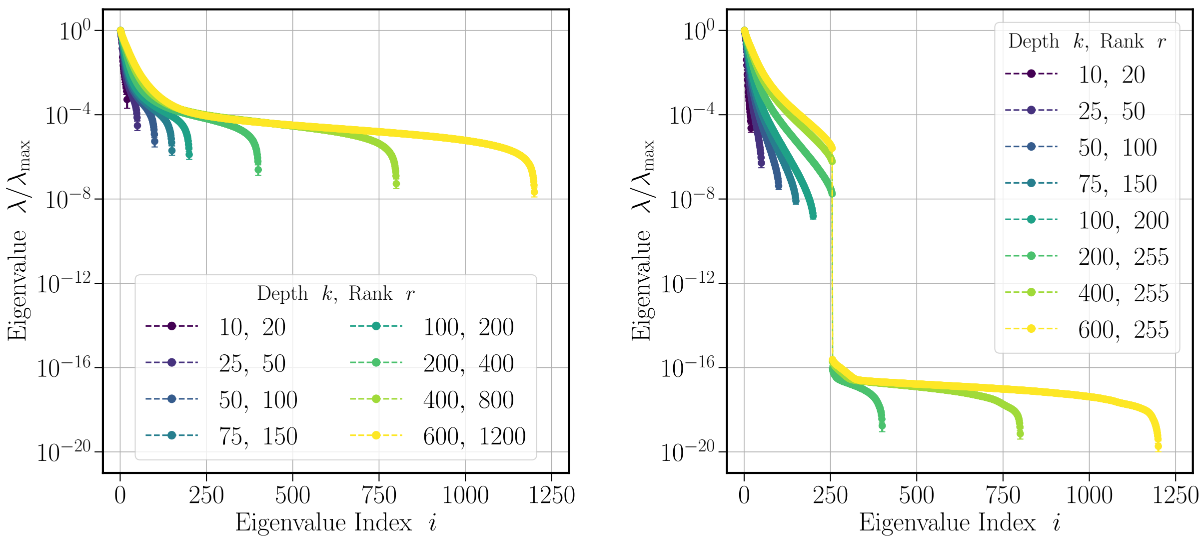
$$\mathcal{F}_{\theta_{\mu\nu}}^\rho = \mathrm{Re}\left(\langle\partial_\mu\psi_\theta|\partial_\nu\psi_\theta\rangle - \langle\psi_\theta|\partial_\mu\psi_\theta\rangle\langle\partial_\nu\psi_\theta|\psi_\theta\rangle\right) \ . \tag{3.B.19}$$

3.B.2 Numerical Overparameterization

We now investigate the effects of overparameterization numerically via the quantum Fisher information $\mathcal{F}_\theta$ and the objective Hessian $\mathcal{H}_\theta$. Here, we optimize $n = 4$ qubit, constrained, noiseless unitary compilation tasks. For the NMR ansatz there are $q^2 - 2 = 2$ variable parameters per time step, meaning there are $p = 2k$ variable parameters per ansatz. Given the universal ansatz, $g = q^n - 1$, we observe in Fig. 3.B.1 that the expected rank saturation occurs at $r = p = g = d^2 - 1 = 255$ for $n = 4$ for the Fisher information, and full rank saturation occurs at likely $r = p \gg g$ for the Hessian.

In agreement with previous studies [230], in Fig. 3.B.1 the Fisher metric exhibits saturation behaviour and indicates overparameterization at any point in the objective or parameter landscape. However, the Hessian rank does not saturate, and remains full rank at this point in the landscape achieved by the optimization. This point in the constrained landscape, even for $p \geq g$ is likely not optimal, and therefore does not saturate the Hessian rank. This non-optimality is attributed to the previous studies indicating that $k \sim \mathcal{O}(1000)$ depth is necessary for this constrained ansatz to achieve infidelities close to machine precision. Due to the quadratic scaling of the memory requirements for computing such metrics with respect to these $p \times p$ dimensional dense matrices, plus determining their spectrum, only $k \sim \mathcal{O}(600)$ are currently feasible to compute in the current implementation.

We should note that for finite machine precision simulations, there is not a definitive method of determining the rank, or number of non-zero eigenvalues of matrices. Here we choose the heuristic when there is an obvious visual distinction between the set of zero and non-zero spectra. In the case in which there is not an obvious cutoff, we choose a relative precision of $\lambda/\lambda_{\max} > p\varepsilon$ for p parameters and machine precision ε. It remains an interesting question whether there is a more principled, and physics-informed approach for determining the rank of matrices in finite precision settings.



(a) Hessian

(b) Fisher Information

Figure 3.B.1: **Overparameterization metrics**. Metrics of overparameterization for constrained unitary compilation of an $n = 4$ qubit NMR ansatz for various depths k (colours). (a) Hessian spectrum of eigenvalues. The Hessian is shown to be full-rank at optimality for depths below the constrained overparameterization depth. Except at optimality in the overparameterized regime, all directions of the ansatz' span are able to be reached during optimization with respect to a target unitary. (b) Quantum Fisher information spectrum of eigenvalues. The quantum Fisher information is shown to be full-rank at optimality for depths below the overparameterization depth. It then saturates at the dynamical Lie algebra dimension $g = d^2 - 1 = 255$, given $p \leq 2k = 1200$ number of parameters, for depths k above the overparameterization depth, indicating the capabilities of the ansatz, independent of the specific target unitary.

3.C Behaviour of Multiple Layer Noise Channels

In these appendixes, we investigate properties of quantum channels consisting of parameterized unitary layers, interlaced with noise. We show that many local noise models form a binomial distribution, over the number of errors, or non-identity noise operations applied throughout the layers. We perform additional numerical studies to show differences in the behaviour of infidelities with depth for unital versus non-unital noise. We extract from piecewise fitting, the scaling of the noise-induced critical depth for infidelities, to be logarithmic in the noise scale. Finally, we perform analytical calculations of the leading-order scaling, with depth and noise scale, of infidelities, impurities, entropies, and relative entropy divergences. These scalings match our numerical studies exactly, and confirm that the divergent regime of optimization is driven by entropic effects.

There are many choices in the exact noise model $\mathcal{N}_\gamma$ used, and whether the noise acts globally or locally, both temporally and spatially. We assume that all noise acts independently in time and locally on each qubit. The total noise channel is chosen to take the form of temporally local noise

$$\Lambda_{\theta\gamma} = \ \circ_{\kappa\in[k]} \, \Lambda^{(\kappa)}_{\theta\gamma} \tag{3.C.1}$$

where the unitary and non-unitary components are separable

$$\Lambda^{(\kappa)}_{\theta\gamma} = \mathcal{N}^{(\kappa)}_{\gamma} \circ \mathcal{U}^{(\kappa)}_{\theta} \ . \tag{3.C.2}$$

Here, we also consider spatially local non-unitary noise models,

$$\mathcal{N}^{(\kappa)}_{\gamma} = \ \circ_{i\in[n]} \, \mathcal{N}^{(\kappa)}_{\gamma_i} \ , \tag{3.C.3}$$

which act identically, and independently, on qubit i at time step κ, with noise scale $\gamma^{(\kappa)}_i = \gamma$.

We will discuss noise models in terms of the number of errors, or non-identity operations that they apply to the evolution. In particular, the considered noise models have,

$$s = nk \tag{3.C.4}$$

independent possible errors at each site in time and in space.

As a key assumption for our analysis, we explicitly assume that the local non-unitary components of the channel can be written as a convex combination of an identity channel and a non-trivial (non-identity) error channel

$$\mathcal{N}_\gamma \equiv (1-\gamma)\mathcal{I} \ + \ \gamma\mathcal{K}_\gamma \ . \tag{3.C.5}$$

Here, there is a scale γ of a non-trivial operation $\mathcal{K}_\gamma$ applied, causing the evolution to deviate from being strictly unitary, resulting in what we refer to as an error. The non-trivial error channel is normalized identically as $\mathcal{N}_\gamma$. If the explicit noise model $\mathcal{N}_\gamma$ does not strictly contain an identity Kraus operator, for example if it is non-unital, then the non-trivial error is defined as

$$\mathcal{K}_\gamma = \frac{1}{\gamma}\mathcal{N}_\gamma - \frac{1-\gamma}{\gamma}\mathcal{I} \ . \tag{3.C.6}$$

This noise model for the non-unitary component of the channel forms a binomial distribution over s possible errors defined by $\mathcal{K}_\gamma$. Errors can occur at spatial and temporal sites $i \in [n]$, and $\kappa \in [k]$, and we use the multi-index $\Gamma \subseteq [k] \times [n]$ for where the $|\Gamma| = l \leq s$ errors occur across the sites. Given this binomial distribution description, and using the binomial expansion of the noise scale factors, the channel may be written as a convex combinations of l-error, or at most l-error channels

$$\begin{aligned}
\Lambda_{\theta\gamma} &= \sum_{l}^{s} \tbinom{s}{l} \gamma^l (1-\gamma)^{s-l} \Lambda_{\theta\gamma_l} && (3.C.7)\\
&= \sum_{l}^{s} \sum_{l'}^{s-l} \tbinom{s}{l} \tbinom{s-l}{l'} (-1)^{l'} \gamma^{l+l'} \Lambda_{\theta\gamma_l} && (3.C.8)\\
&= \sum_{l}^{s} \sum_{l'=l}^{s} \tbinom{s}{l} \tbinom{s-l}{l'-l} (-1)^{l'-l} \gamma^{l'} \Lambda_{\theta\gamma_l} && (3.C.9)\\
&= \sum_{l}^{s} \sum_{l'}^{l} \tbinom{s}{l} \tbinom{l}{l'} (-1)^{l-l'} \gamma^{l} \Lambda_{\theta\gamma_{l'}} && (3.C.10)\\
&= \sum_{l}^{s} \tbinom{s}{l} \gamma^{l} \left[\sum_{l'}^{l} \tbinom{l}{l'} (-1)^{l-l'} \Lambda_{\theta\gamma_{l'}} \right] && (3.C.11)\\
&= \sum_{l}^{s} \tbinom{s}{l} \gamma^{l} \Lambda_{\theta\gamma \leq l} && (3.C.12)
\end{aligned}$$

We denote trace-preserving channels with l non-trivial errors as the uniform convex combination of all possible locations of the $l \leq s$ errors,

$$\Lambda_{\theta\gamma_l} = \frac{1}{\binom{s}{l}} \sum_{\substack{\Gamma \subseteq [k]\times[n] \\ |\Gamma| = l}} \Lambda_{\theta\gamma\Gamma} \ , \tag{3.C.13}$$

and we denote traceless operators with at most l non-trivial errors as the non-uniform combination of all possible locations of the $l' \leq l$ errors,

$$\Lambda_{\theta\gamma \leq l} = \sum_{l'}^{l} (-1)^{l-l} \tbinom{l}{l'} \Lambda_{\theta\gamma_{l'}} \ , \tag{3.C.14}$$

where specific l-error channels with l non-trivial errors at indices Γ are denoted as

$$\Lambda_{\theta\gamma\Gamma} = \ \circ_{\kappa\in[k]} \ \circ_{i\in[n]} \mathcal{K}_{\gamma_i}^{(\kappa)\delta_{(\kappa,i)\in\Gamma}} \circ \mathcal{U}_{\theta}^{(\kappa)} \ . \tag{3.C.15}$$

Therefore, to leading-order, the noisy and noiseless channels differ as per the non-triviality $\mathcal{K}_{\gamma_i}^{(\kappa)} \neq \mathcal{I}_i$ of s possible local errors,

$$\Lambda_{\theta\gamma} - \Lambda_{\theta} = \left(\sum_{\kappa}^{k} \sum_{i}^{n} \ \mathcal{U}_{\theta}^{(>\kappa)} \circ (\mathcal{K}_{\gamma_i}^{(\kappa)} - \mathcal{I}_i) \circ \mathcal{U}_{\theta}^{(\leq\kappa)} \right) \gamma \ + \ \mathcal{O}(\tbinom{s}{2}\gamma^2) \ . \tag{3.C.16}$$

3.C.1 Noise Models

Here, we consider several noise models $\mathcal{N}_\gamma = (1-\gamma)\mathcal{I} \ + \ \gamma\mathcal{K}_\gamma$ that are relevant to existing quantum devices, namely independent, local, dephasing, amplitude damping, and depolarizing noise. These noise models all belong to the class of unital or non-unital Pauli noise, that transform local Pauli operators $\mathcal{P}_q$. For the case of $q = 2$ qubits, $\mathcal{P}_2 = \{I, Z, X, Y\}$. The forms of the models are identified by their non-trivial error component $\mathcal{K}_\gamma$ and are described as follows.

Unital local dephasing noise for all inputs may be written as

$$\mathcal{K}_{\gamma}{}^{\text{dephase}}(\cdot) = Z \cdot Z^\dagger \ , \tag{3.C.17}$$

where the non-trivial channel is unitary.

Unital local depolarizing noise for all inputs may be written as

$$\mathcal{K}_{\gamma}{}^{\text{depolarize}}(\cdot) = \frac{\text{tr}(\cdot)}{q} I \ , \tag{3.C.18}$$

where the non-trivial channel is maximally depolarizing.

Non-unital amplitude damping noise for Pauli inputs $P \in \mathcal{P}_2$ may be written as

$$\mathcal{K}_{\gamma}{}^{\text{amplitude}}(P) = \begin{cases} P + \gamma Z & P \in \{I\} \\ (1-\gamma)P & P \in \{Z\} \\ \sqrt{1-\gamma}P & P \in \mathcal{P}_2 \backslash \{I, Z\} \end{cases} \ . \tag{3.C.19}$$

In the case of the non-trivial components of the noise channel being single unitary operators V such that $\mathcal{K}_\gamma(\cdot) = V \cdot V^\dagger$, each s-error channel is unitary, and the original unitary channel is interlaced with unitaries V at indices Γ where errors occur. In the case of depolarizing noise where $\mathcal{K}_\gamma(\cdot) = (\mathrm{tr}(\cdot)/q)\mathcal{I}$, partial traces remove any information about the state at the local indices in Γ where errors occur.

3.C.2 Probabilistic Interpretation of Noise Channels

Within this formalism, the channel can be represented as an expectation over a distribution of the number of $l \leq s$ possible errors,

$$\Lambda_{\theta\gamma} = \langle \Lambda_{\theta\gamma_l} \rangle_{l \sim p_{s\gamma}} \tag{3.C.20}$$

where for moments of channels with l non-trivial errors,

$$\Lambda_{\theta\gamma_l} = \frac{1}{\binom{s}{l}} \sum_{\substack{\Gamma \subseteq [k]\times[n] \\ |\Gamma|=l}} \Lambda_{\theta\gamma\Gamma} \ . \tag{3.C.21}$$

The exact distribution with this local noise model is the binomial distribution,

$$p_{s\gamma}(l) = p_{s\gamma}^{\mathrm{Binomial}}(l) = \tbinom{s}{l}\gamma^l(1-\gamma)^{s-l} \tag{3.C.22}$$

which has mean and variance

$$\mu_{s\gamma} = s\gamma \quad , \quad \Sigma_{s\gamma} = s\gamma(1-\gamma) \ . \tag{3.C.23}$$

Such interpretations are also used in error mitigation approaches [293].

In the $s \to \infty$ limit, this distribution tends to the Gaussian distribution,

$$p_{s\gamma}(l) \quad \overset{s\to\infty}{\to} \quad p_{s\gamma}^{\mathrm{Gaussian}}(l) = \sqrt{\frac{1}{2\pi\Sigma_{s\gamma}}} e^{-\frac{1}{2}(l-\mu_{s\gamma})\Sigma_{s\gamma}^{-1}(l-\mu_{s\gamma})} \ , \tag{3.C.24}$$

which can be shown using the De Moivre - Laplace theorem [429], a central-limit version of the standardized binomially distributed variables $(l-\mu_{s\gamma})/\sqrt{\Sigma_{s\gamma}}$.

In the $s \to \infty, \gamma \to 0$ limit, with finite $s\gamma \to \lambda_{s\gamma}$, the distribution tends to the Poisson distribution,

$$p_{s\gamma}(l) \quad \overset{s\to\infty,\gamma\to 0}{\to} \quad p_{s\gamma}^{\mathrm{Poisson}}(l) = \frac{1}{l!}(s\gamma)^l e^{-s\gamma} \ , \tag{3.C.25}$$

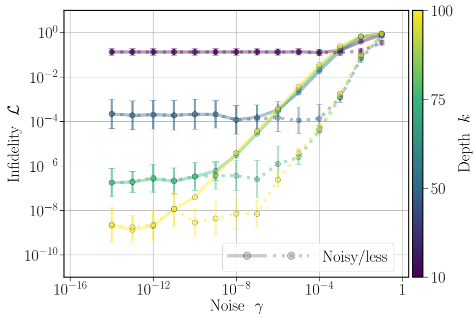
which can be shown using Sterling's approximation for the binomial coefficient, and equating the generating functions for the distributions.

3.C.3 Noisy State Preparation

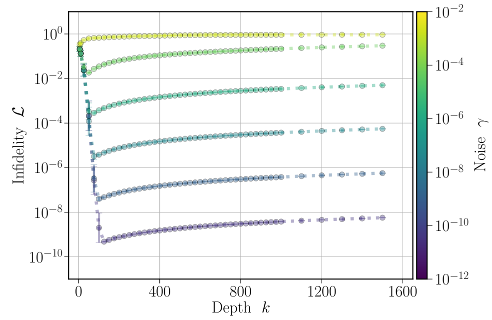
Here, we conduct studies in Fig. 3.C.2 of the noise and depth dependence on the optimal infidelities for each of the unital dephasing and depolarizing, and non-unital amplitude damping noise models. For unital noise, we observe that the critical depth phenomena seem to be consistent across noise models at all noise scales, and entropic effects dominate. For non-unital noise, we observe that the critical depth phenomena seem to be consistent with unital noise models, at small noise scales. However, for large noise scales, non-unital noise forces the state into a specific (pure) state, which dominates over entropic effects. The infidelities then appear to converge to non-unity values or potentially decrease polynomially to zero. The unitary component of the channel appears to be able to slowly rotate this noise-induced pure state, towards the correct target pure state. In both unital and non-unital noise models, inserting parameters learned in the noisy setting, into the corresponding noiseless unitary ansatz indicates that the noiseless infidelities have greater resilience to increasing noise scales. This suggests the underlying unitary transformations are being learned in the noisy settings, however with greater variance.

3.C.4 Noise-induced Critical Depth

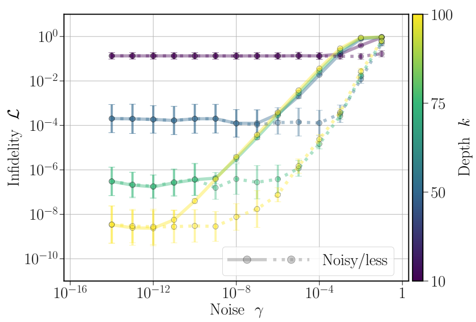
To develop a relationship between noise and an induced critical depth beyond which infidelities no longer converge exponentially, we perform piecewise fits, as per Fig. 3.C.3. For each noise scale, we perform exponential, and then polynomial fits, respectively, before and after an approximate location of the critical depth suggested by the finite amount of data points available. We are then able to approximate where the piece wise curves intersect, indicating the location of the critical depth as a function of noise. Plotting this relationship suggests a logarithmic relationship between noise and critical depth. This leads to the optimal infidelity scaling approximately polynomially, between linearly and quadratically with noise, and in agreement with previous conjectures about these relationships [277].



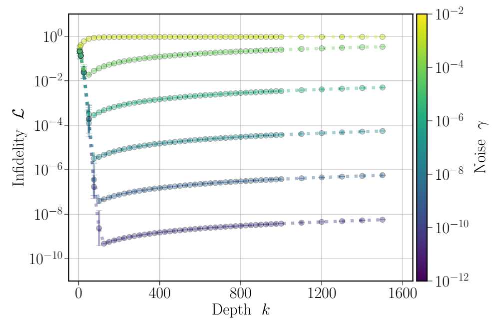
(a) Dephasing



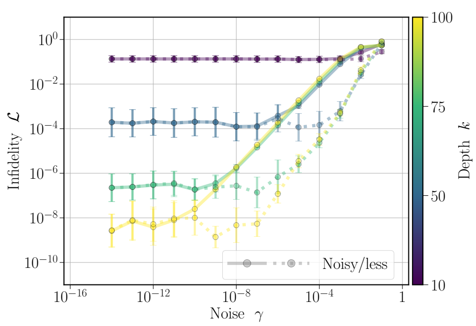
(b) Dephasing



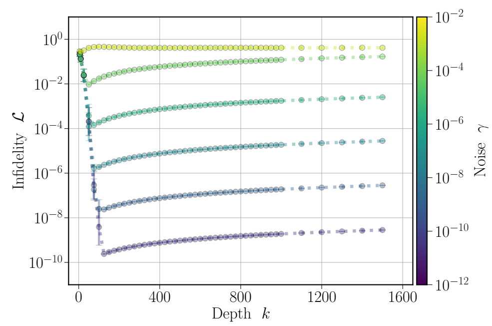
(c) Depolarizing



(d) Depolarizing

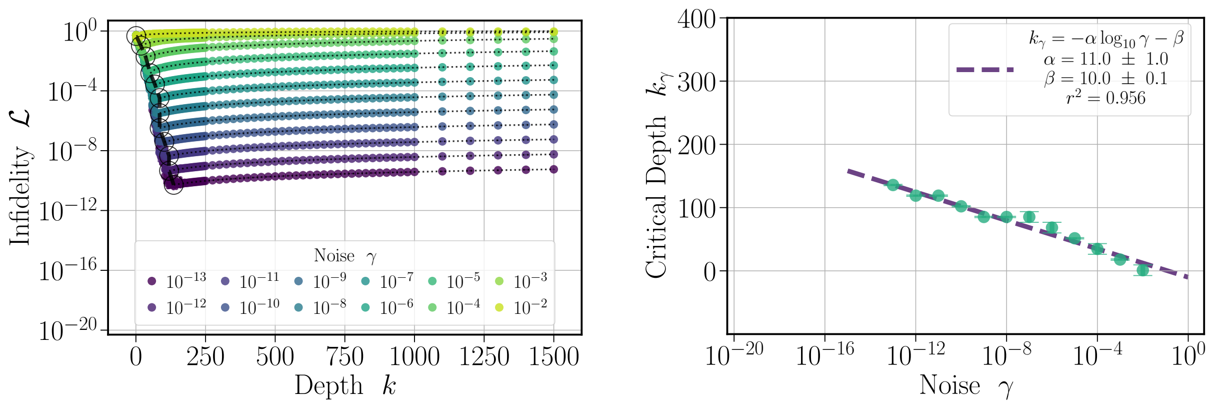


(e) Amplitude Damping



(f) Amplitude Damping

Figure 3.C.2: **Noisy infidelities**. Behaviour of infidelity objectives with respect to noise and depth for the $n = 4$ NMR ansatz. Left) Trained noisy infidelity (solid), and tested infidelity of noisy parameters in noiseless ansatz (dashed) for various depths k (colours). Infidelities of noisy ansatz are depth dependent and noise independent for small noise scales, before universally increasing polynomially with noise. Inserting parameters learned in the noisy setting into an identical noiseless ansatz indicates that the underlying unitary evolution is being learned, and is resilient to noise. Right) Critical depth for noisy infidelity for various noise scales γ (colours). Infidelities improve exponentially with depth, up until a noise-induced critical depth, where entropic effects worsen infidelities polynomially.



(a) Piecewise Fit of Noisy Infidelity

(b) Linear-Log Fit of Critical Depth

Figure 3.C.3: **Noise-induced critical depth**. Fit of noise-induced critical depth. (a) Piecewise fits for various noise scales γ (colours). Objectives decrease exponentially with depth, for depths less than the critical depth (black circles), and objectives increase polynomially with depth for depths greater than the critical depth. (b) Uncertainty propagation is used to estimate the error of the linear-log fit (dashed line) between the estimated critical depths where the piecewise curves intersect, and the associated noise scales.

3.C.5 Parameter Shift Rules for Quantum Channels

Here, we derive parameter shift rules for the gradients of multiple layer quantum channels. This generalizes previous results, and provides some explanation for why the optimization routines in noisy settings generally appear to converge to the noiseless optima at small noise scales. Given our composite channel definition over k layers, written decomposed into channels before and after an index (μ, κ), our state preparation objectives of the trace infidelity with respect to a pure state ρ, and initial state σ may be written as

$$\mathcal{L}_{\theta\gamma} = 1 - \mathrm{tr}\left(\mathcal{U}^{(\kappa)}_{\mu}\left(\sigma^{(<\kappa)}_{<\mu\gamma}\right)\mathcal{U}^{(\kappa)\dagger}_{>\mu}\left(\mathcal{N}^{(\kappa)\dagger}_{\gamma}\left(\rho^{(>\kappa)}_{>\mu\gamma}\right)\right)\right) . \tag{3.C.26}$$

These objectives are crucially linear in the states, and therefore gradients of the objectives are linear functions of the gradients of the states.

Let $\mathcal{U}_\theta$ be a single parameter unitary channel, with the corresponding unitary operator $U_\theta = e^{-i\theta G}$, where the hermitian generator is involutory up to a factor $G^2 = \zeta^2 I$. The gradient of such a unitary channel for an arbitrary input follows the parameter shift rule

$$\partial_\theta \mathcal{U}_\theta = -i[G, \mathcal{U}_\theta] = \zeta(\mathcal{U}_{\theta+\varphi} - \mathcal{U}_{\theta-\varphi}) \qquad : \qquad \varphi = \pi/4\zeta \ , \tag{3.C.27}$$

and other choices of φ are also possible depending on experimental feasibility. This can be extended to the k-order gradient of this channel as the k-order nested commutator

$$\partial_\theta^k \mathcal{U}_\theta = (-i)^k [G, \mathcal{U}_\theta]_k = \zeta^k \sum_l^k (-1)^l \tbinom{k}{l} \mathcal{U}_{\theta+(k-2l)\varphi} \ . \tag{3.C.28}$$

Unitary channels with generators with more complicated spectra can be expressed as a linear combination of unitary channels, with perturbative φ, weighted by coefficients π_φ,

$$\partial_\theta^k \mathcal{U}_\theta = \sum_\varphi \pi_\varphi \mathcal{U}_{\theta+\varphi} \ . \tag{3.C.29}$$

Thus gradients of general parameterized channels for constant noise scales γ have the form,

$$\partial_\mu \Lambda_{\theta\gamma} = \sum_\varphi \pi_{\varphi_\mu} \Lambda_{\theta+\varphi_\mu \ \gamma} \ , \tag{3.C.30}$$

and similarly any linear objectives such as $\partial_\mu^{(\kappa)} \mathcal{L}_{\theta\gamma}$ have an identical form.

3.C.6 Relationship between Infidelity, Impurity and von-Neumann Entropy

To better understand the phenomena dictating the behaviour of infidelities with respect to noise scales and depth, we also investigate the scaling of the impurity and entropy of the noisy parameterized states. For completeness, we restate the normalized impurity, von-Neumann entropy, conditional entropy, and relative entropy divergence, between a d-dimensional state ρ and a pure state ρ', as

$$\mathcal{L}'_\rho = 1 - \operatorname{tr}(\rho'\rho) \tag{3.C.31}$$

$$\mathcal{I}_\rho = 1 - \operatorname{tr}\left(\rho^2\right) \tag{3.C.32}$$

$$\mathcal{S}_\rho = -\operatorname{tr}(\rho \log(\rho)) \,/\, \log(d) \tag{3.C.33}$$

$$\mathcal{S}_\rho^{\rho'} = -\operatorname{tr}(\rho' \log(\rho)) \,/\, \log(d) \tag{3.C.34}$$

$$\mathcal{D}_\rho^{\rho'} = \mathcal{S}_\rho^{\rho'} - \mathcal{S}_{\rho'} \ . \tag{3.C.35}$$

We also note that when we use the definitions of parameterized states $\rho_{\theta\gamma}$ and target reference states ρ in place of ρ and ρ', we instead use the subscripts $\theta\gamma$ and superscripts ρ.

For this analysis, we use the Bloch representation [430], which describes operators in terms of a trace orthogonal basis $\mathcal{P}_d$, such that $\operatorname{tr}\left(S^\dagger P\right) = d\delta_{PS}$ for all operators $P, S \in \mathcal{P}_d$. This basis contains the identity, is of size $|\mathcal{P}_d| = d^2$, with structure constants defining their commutation relations. Let $\omega = \{P \,:\, P \in \mathcal{P}_d \backslash \{I\}\}$ represent the vector of all non-identity operators such that quantum states with unit traces have the form

$$\rho = \frac{I + \lambda \cdot \omega}{d} \ . \tag{3.C.36}$$

Here the $|\mathcal{P}_d| - 1$ Bloch coefficients λ fully describe quantum states. These coefficients are constrained to represent positive-semidefinite operators, and their magnitude is bounded by the pure-state boundary described by

$$\lambda^2 \leq d - 1 \ . \tag{3.C.37}$$

The action of arbitrary trace-preserving channels on quantum states can further be described by the affine linear transformation on the coefficients,

$$\Lambda \ : \ \lambda \to \Upsilon\lambda + \upsilon \ . \tag{3.C.38}$$

This affine linear transformation represents a rotation and scaling of λ by Υ, plus a translation to a different axis by υ, and it can be thought of as a transformation on the generalized d-dimensional Bloch sphere of radius $d-1$. The primary constraints on the transformations are to remain within this boundary such that $(\Upsilon\lambda + \upsilon)^2 \leq d - 1$.

If the channel is unital, then the transformation is strictly linear and $\upsilon = 0$, otherwise the channel is non-unital. Finally, if the channel is unitary, $\Upsilon = u$ is an orthogonal transformation that preserves the length of λ^2.

We also note a useful identity when λ is associated with a pure state with zero entropy, given the expansion for the matrix logarithm

$$\log(I + \lambda \cdot \omega) = \sum_{k>0} \frac{(-1)^{k+1}}{k} (\lambda \cdot \omega)^k \tag{3.C.39}$$

and the tracelessness of $\lambda \cdot \omega$, meaning for pure states, their coefficients λ satisfy,

$$\frac{1}{d\log(d)} \sum_{k>1} \frac{(-1)^k}{k(k-1)} \operatorname{tr}\left((\lambda \cdot \omega)^k\right) = 1 \ . \tag{3.C.40}$$

This identity avoids complicated expressions for powers of Bloch vectors [431].

We may now define quantities in terms of the Bloch coefficients. Similarity between quantum states ρ, ρ' can be described either by their trace inner products

$$\mathrm{tr}(\rho\rho') = \frac{1 + \lambda \cdot \lambda'}{d} = 1 - \mathcal{L}_\rho^{\rho'} \ , \tag{3.C.41}$$

or by their cosine similarity with an angle $\phi_\rho^{\rho'}$ between their coefficients

$$\cos\left(\phi_\rho^{\rho'}\right) = \frac{\lambda \cdot \lambda'}{\sqrt{\lambda^2 \lambda'^2}} = \frac{d(1 - \mathcal{L}_\rho^{\rho'}) - 1}{\sqrt{(d(1 - \mathcal{I}_\rho) - 1)(d(1 - \mathcal{I}_{\rho'}) - 1)}} \ . \tag{3.C.42}$$

Similarly for other functions of interest with respect to a pure state ρ',

$$\mathcal{L}_\rho^{\rho'} = 1 - \frac{1 + \lambda \cdot \lambda'}{d} \tag{3.C.43}$$

$$\mathcal{I}_\rho = 1 - \frac{1 + \lambda^2}{d} \tag{3.C.44}$$

$$\mathcal{S}_\rho = 1 - \frac{1}{d\log(d)} \sum_{k>1} \frac{(-1)^k}{k(k-1)} \mathrm{tr}\big((\lambda \cdot \omega)^k\big) \tag{3.C.45}$$

$$= 1 - \frac{1}{2}\frac{1}{\log(d)}\lambda^2 \ + \ \mathcal{O}(\lambda^3) \tag{3.C.46}$$

$$\mathcal{D}_\rho^{\rho'} = 1 - \frac{1}{d\log(d)} \sum_{k>0} \frac{(-1)^{k+1}}{k} \mathrm{tr}\big((I + \lambda' \cdot \omega)(\lambda \cdot \omega)^k\big) \tag{3.C.47}$$

$$= 1 - \frac{1}{2}\frac{1}{\log(d)} \left(2\frac{\lambda \cdot \lambda'}{\lambda^2} - 1\right) \lambda^2 \ + \ \mathcal{O}(\lambda^3) \ . \tag{3.C.48}$$

We can also define the generalized α-Renyi infidelities, impurities, entropies, and divergences for $\alpha \geq 1/2$,

$$\mathcal{L}_\rho^{\rho'(\alpha)} = \mathrm{tr}\left((\rho'^{\frac{1-\alpha}{2\alpha}} \rho \rho'^{\frac{1-\alpha}{2\alpha}})^\alpha\right) \tag{3.C.49}$$

$$\mathcal{I}_\rho^{(\alpha)} = 1 - \mathrm{tr}(\rho^\alpha) \tag{3.C.50}$$

$$\mathcal{S}_\rho^{(\alpha)} = \frac{1}{\alpha - 1}\frac{1}{\log(d)} \log\big(1 - \mathcal{I}_\rho^{(\alpha)}\big) \tag{3.C.51}$$

$$\mathcal{D}_\rho^{\rho'(\alpha)} = \frac{1}{\alpha - 1}\frac{1}{\log(d)} \log\left(1 - \mathcal{L}_\rho^{\rho'(\alpha)}\right) \ . \tag{3.C.52}$$

Using these definitions we can show [432] that the entropies and divergences are monotonically increasing with decreasing α, and we can identify $\mathcal{D}_\rho^{\rho'}$ with $\alpha \to 1$, and related divergences and infidelities with $\mathcal{D}_{\rho}^{\rho'(1/2)} = -2\log\big(1-\mathcal{L}_{\rho}^{\rho'}\big)/\log(d)$. Based on these relationships, we have the bounds relating the conditional entropy and the infidelity,

$$\mathcal{S}_{\rho}^{\rho'} \geq \frac{2}{\log(d)}\mathcal{L}_{\rho}^{\rho'} \ , \tag{3.C.53}$$

however we have not found any bounds relating entropy and infidelity or impurity.

Given our previous expressions for channels consisting of layers of non-unitary noise with scale γ, interlaced by unitary channels with parameters θ, resulting in a binomial distribution over states with at most l errors, we may represent such parameterized quantum channels in this formalism as

$$\Lambda_{\theta\gamma} \ : \ \Upsilon_{\theta\gamma} = (1-\gamma_s)u_{\theta} + \gamma_s u_{\theta\gamma} \quad , \quad \upsilon_{\theta\gamma} = \gamma_s \eta_{\theta\gamma} \ . \tag{3.C.54}$$

Here, we have explicitly separated the noiseless unitary rotation from the other non-unitary and non-unital transformations, and we defined the error-dependent noise scale as,

$$(1-\gamma_s) = (1-\gamma)^s \ . \tag{3.C.55}$$

Therefore the transformed coefficients, given an initial state σ with coefficients ξ, are transformed by unitary and noise components of the transformation,

$$\lambda_{\theta\gamma} = (1-\gamma_s)u_{\theta}\xi + \gamma_s u_{\theta\gamma}\xi + \gamma_s\eta_{\theta\gamma} \tag{3.C.56}$$
$$= (1-\gamma_s)\lambda_{\theta|\gamma} + \gamma_s\varepsilon_{\theta\gamma} \ . \tag{3.C.57}$$

Here, we decompose the transformed state into what we refer to as the pure and mixed components. The pure component of the state may be written in terms of a reference pure state ρ with coefficients λ, and a pure state with orthogonal coefficients $\zeta \perp \lambda$ as,

$$\lambda_{\theta|\gamma} = u_{\theta}\xi = (1-\alpha_{\theta|\gamma})\lambda + \beta_{\theta|\gamma}\zeta \ , \tag{3.C.58}$$

where the coefficients implicitly contain a dependence on the noise from optimization, and they are constrained such that $(1-\alpha_{\theta|\gamma})^2 + \beta_{\theta|\gamma}^2 = 1$. The mixed components of the state can be written in terms of its unital and non-unital components,

$$\varepsilon_{\theta\gamma} = u_{\theta\gamma}\xi + \eta_{\theta\gamma} \ . \tag{3.C.59}$$

This decomposition allows us to understand how the unitary and noise components transform the state. The unitary component, independent of the noise, rotates the state into the pure components parallel and perpendicular to ρ. The noise component then scales both the pure and mixed components with γ_s, and performs an affine translation with $\eta_{\theta\gamma}$. In the limit that $\gamma \to 0$, and assuming converged optimization $\theta \to \theta^*$ such that $\rho_{\theta^*} \to \rho$, the coefficients should reduce to $\alpha_{\theta^*}, \beta_{\theta^*}, \eta_\theta \to 0$.

Due to the non-trivial optimization, we do not have a general closed-form expression depicting the θ, γ, K dependence of the components $\alpha_{\theta|\gamma}, \beta_{\theta|\gamma}, \varepsilon_{\theta\gamma}$. However, by expanding out the quantities of interest in terms of these expressions for the coefficients $\lambda_{\theta\gamma}$, we can obtain the leading-order scaling of quantities in terms of s, γ.

The inner products of the coefficients are

$$\begin{aligned} \lambda_{\theta\gamma}^2 &= (1-\gamma_s)^2\lambda^2 + \gamma_s^2\varepsilon_{\theta\gamma}^2 + 2\gamma_s(1-\gamma_s)^2((1-\alpha_{\theta|\gamma})\lambda + \beta_{\theta|\gamma}\zeta)\cdot\varepsilon_{\theta\gamma} && (3.C.60)\\ &= \lambda^2 - 2s\gamma\left(1-(1-\alpha_{\theta|\gamma})\frac{\lambda\cdot\varepsilon_{\theta\gamma}}{\lambda^2} - \beta_{\theta|\gamma}\frac{\zeta\cdot\varepsilon_{\theta\gamma}}{\lambda^2}\right)\lambda^2 \;+\; \mathcal{O}(\tbinom{s}{2}\gamma^2) && (3.C.61)\\ \lambda_{\theta\gamma}\cdot\lambda &= (1-\gamma_s)(1-\alpha_{\theta|\gamma})\lambda^2 + \gamma_s\lambda\cdot\varepsilon_{\theta\gamma} && (3.C.62)\\ &= \lambda^2 - \alpha_{\theta|\gamma}\lambda^2 - s\gamma\left(1-\alpha_{\theta|\gamma} - \frac{\lambda\cdot\varepsilon_{\theta\gamma}}{\lambda^2}\right)\lambda^2 \;+\; \mathcal{O}(\tbinom{s}{2}\gamma^2)\;. && (3.C.63) \end{aligned}$$

Powers of Bloch vectors, even when $\alpha_{\theta|\gamma} = \beta_{\theta|\gamma} = 0$, scales with γ as

$$(\lambda_{\theta\gamma}\cdot\omega)^k = (1-\gamma_s)^k(\lambda\cdot\omega)^k \;+\; \mathcal{O}(\gamma)\;. \qquad (3.C.64)$$

All powers of Bloch vectors, such as those that occur in infinite series expansions for logarithmic functions appearing in entropies, therefore contribute leading-order terms in s, γ. It does not then suffice to retain only some $\mathcal{O}(\lambda^k)$-order terms in expansions of such functions, to capture their leading-order behaviour. Obtaining these coefficients, for all dimensions d, is possible, however it requires opaque expressions for products $\mathrm{tr}\big((\lambda\cdot\omega)^k\big)$ in terms of the structure constants, such as those found for the Pauli basis [292].

Therefore, to leading-order in s, γ, the quantities of interest show similar scaling with overlaps of the parameterized state with the target pure state,

$$\begin{aligned} \mathcal{L}_{\theta\gamma}^{\rho} &= \frac{d-1}{d}\alpha_{\theta|\gamma} \;+\; s\gamma\frac{d-1}{d}\left((1-\alpha_{\theta|\gamma}) - \frac{\lambda\cdot\varepsilon_{\theta\gamma}}{\lambda^2}\right) \;+\; \mathcal{O}(\tbinom{s}{2}\gamma^2)\\ \mathcal{I}_{\theta\gamma} &= 2s\gamma\frac{d-1}{d}\left(1-(1-\alpha_{\theta|\gamma})\frac{\lambda\cdot\varepsilon_{\theta\gamma}}{\lambda^2} - \beta_{\theta|\gamma}\frac{\zeta\cdot\varepsilon_{\theta\gamma}}{\lambda^2}\right) \;+\; \mathcal{O}(\tbinom{s}{2}\gamma^2) && (3.C.65)\\ \mathcal{S}_{\theta\gamma} &= s\gamma\frac{d-1}{\log(d)}\left(1-(1-\alpha_{\theta|\gamma})\frac{\lambda\cdot\varepsilon_{\theta\gamma}}{\lambda^2} - \beta_{\theta|\gamma}\frac{\zeta\cdot\varepsilon_{\theta\gamma}}{\lambda^2}\right) \;+\; \mathcal{O}(\lambda_{\theta\gamma}^3) \;\sim\; \mathcal{O}(s\gamma)\\ \mathcal{D}_{\theta\gamma}^{\rho} &= \frac{d-1}{\log(d)}\alpha_{\theta|\gamma} \;+\; s\gamma\frac{d-1}{\log(d)}\left(\alpha_{\theta|\gamma}(1+\frac{\lambda\cdot\varepsilon_{\theta\gamma}}{\lambda^2}) - \beta_{\theta|\gamma}\frac{\zeta\cdot\varepsilon_{\theta\gamma}}{\lambda^2}\right) \;+\; \mathcal{O}(\lambda_{\theta\gamma}^3) \;\sim\; \mathcal{O}(s\gamma)\;. \end{aligned}$$

If the unitary component of the channel transforms the pure component of the state to exactly the target state such that $\alpha_{\theta|\gamma} = \beta_{\theta|\gamma} = 0$, then expressions simplify further. A hierarchy between quantities can be partially observed to leading-order in terms of the overlap $\lambda \cdot \varepsilon_{\theta\gamma}$ between the mixed component of the state and the target state

$$\mathcal{L}^{\rho}_{\theta\gamma} = s\gamma\frac{d-1}{d}\left(1 - \frac{\lambda \cdot \varepsilon_{\theta\gamma}}{\lambda^2}\right) \ + \ \mathcal{O}(\tbinom{s}{2}\gamma^2) \tag{3.C.66}$$

$$\mathcal{I}_{\theta\gamma} = 2s\gamma\frac{d-1}{d}\left(1 - \frac{\lambda \cdot \varepsilon_{\theta\gamma}}{\lambda^2}\right) \ + \ \mathcal{O}(\tbinom{s}{2}\gamma^2) \tag{3.C.67}$$

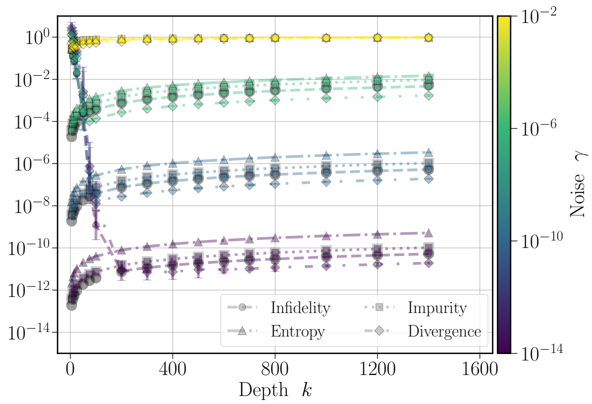
$$\mathcal{S}_{\theta\gamma} = \mathcal{O}(s\gamma) \qquad , \qquad \mathcal{D}^{\rho}_{\theta\gamma} = \mathcal{O}(s\gamma) \ . \tag{3.C.68}$$

Plotting each of these quantities from numerical optimizations in Fig. 3.C.4 for unital dephasing noise, we observe that once the infidelities or divergences are less than the impurities or entropies, there is a transition from the convergent to the divergent regime. This transition occurs at a critical depth, before which increasing depth allows the dominant unitary component of the channel to rotate the state to converge to the target state. Beyond this critical depth, the noise component of the channel dominates, and the parameterized state scales towards the mixed state. This divergence of all quantities is shown numerically at small noise scales to be linear in depth and noise scale. We also plot the derived analytical leading-order scaling of the infidelity and impurities with gray enlarged markers, which are in excellent agreement with the numerical results. In this divergent regime at small noise scales, we also observe the numerical hierarchy of the quantities

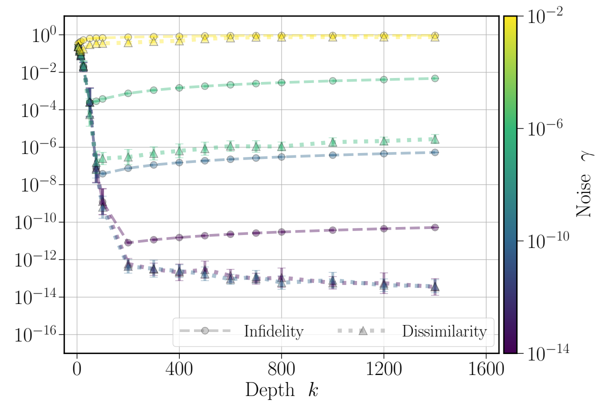
$$\mathcal{D}^{\rho}_{\theta\gamma} \leq \mathcal{L}^{\rho}_{\theta\gamma} \leq \mathcal{I}_{\theta\gamma} \leq \mathcal{S}_{\theta\gamma} \ . \tag{3.C.69}$$

All quantities also numerically appear to converge together as noise increases. In the unital case, all quantities diverge to the worst-case unity values as the noise and depth increases. In non-unital case, quantities potentially converge to optimality polynomially with depth as the noise and depth increases. This contrasting behaviour to the unital noise case suggests the non-unital terms $\eta_{\theta\gamma} \neq 0$ dominate the leading-order scaling in this regime.

We also note that the relative entropy divergences show identical convergent and divergent overparameterized regimes of optimization as the infidelities. The numerically found hierarchy of quantities also suggests that infidelity is potentially lower bounded by the divergence. This appears to be reasonable due to both quantities reflecting distances between distributions corresponding to the quantum states. Given that the divergence is used in classical learning tasks, it is also suitable as an objective function for quantum fidelity-based tasks, although it is more demanding to compute.



(a) Impurity, Entropy, Divergence



(b) Cosine Dissimilarity

Figure 3.C.4: **Noisy infidelity, impurity, entropy, relative entropy divergence, and cosine dissimilarity**. Behaviour of parameterized states relative to pure target states, with respect to dephasing noise γ (colours) and depth for the $n = 4$ NMR ansatz. (a) Impurity, entropy, relative entropy divergence. Once the optimized parameterized ansatz achieves infidelities or relative entropy divergences of the order of the impurity and entropies, the system is dominated by entropic effects. In this divergent regime, all quantities (coloured markers) scale identically, and leading-order infidelity and impurity analytical values (enlarged gray markers) show exact agreement. (b) Cosine dissimilarity. The unitary component of the channel is able to align the parameterized state with the target state. For sufficiently low noise, this dissimilarity remains constant and infidelities increase in the divergent regime due to being scaled by the depth-dependent noise scale.

To assess the validity of this assumption of the unitary component aligning the pure component of the state with the target state, for small noise scales, the cosine dissimilarity

$$1 - \left|\cos\left(\phi^{\rho}_{\theta\gamma}\right)\right| \sim \mathcal{O}\left(\left|\alpha_{\theta|\gamma}\right|\right) \tag{3.C.70}$$

is of order of the alignment of the pure component with the target state. Plots of the cosine dissimilarity in Fig. 3.C.4 indicate that for sufficiently low noise scales, the dissimilarity remains constant, or even decreases to machine precision scales. This suggests that divergences in the infidelities at large depths, are strictly due to the noise component of the channel scaling the pure component by the depth-dependent noise scales. At larger noise scales, the cosine dissimilarity is larger, but still orders of magnitude smaller than the infidelities, and diverges similarly to the infidelities.

3.D Classical and Quantum Error Analysis

In these appendixes, we derive and compare the noise-induced bias of the parameterized noisy states from their noiseless values, for both classical and quantum noise. We show both noise types have biases that scale polynomially with the noise scale, and exponentially with the number of errors induced by the noise. Finally, we discuss the implications of classical floating point error on the viability of large scale simulations without error mitigation.

To bound generally independent errors in simulations, we use Schatten norms $\|A\| = \|A\|_p$, $p \geq 1$ for d-dimensional matrices A, B. Such norms satisfy the convenient properties of monotonicity $\|A\|_p \leq \|A\|_q$, $q \leq p$, sub-additivity $\|A+B\|_p \leq \|A\|_p + \|B\|_p$, sub-multiplicativity $\|AB\|_p \leq \|A\|_p \|B\|_p$, and unitary-invariance $\left\|UAV^\dagger\right\|_p = \|A\|_p$, for unitaries U, V. Unitary matrices U have constant norm $\|U\|_p = d^{1/p}$, and we denote the $\lim_{p\to\infty} \|A\|_p = \lambda(A)$ norm as the largest singular value. Finally, Schatten norms satisfy Holder's $\|AB\|_s \leq \|A\|_p \|B\|_q$, $1/p + 1/q = 1/s$, and von-Neumann $|\mathrm{tr}(AB)| \leq \|A\|_p \|B\|_q$, $1/p + 1/q = 1$ inequalities [286].

3.D.1 Classical Error Analysis

We first investigate the effect of classical floating point errors on scalar operations, generalizing the results of [433] to the case of an arbitrary number of matrix multiplications.

Scalar Floating Point Error

Let us assume that we are performing binary floating point operations, with the exact, ideal operations denoted by $\circ \in \{+, -, \times, /\}$. We denote floating point representations of any exact operations $\circ$ with subscripts $\circ_\epsilon$.

For operations between scalars, and scalar values themselves, we assume they may only be represented with a relative error, upper bounded by ε, sometimes referred to as machine precision. For vectorized operations comprised of vectors of dimension d, let the relative error be upper bounded by $\epsilon = \epsilon(d, \varepsilon)$. This dependence of the error on the number of operations d and the machine precision ε may be polynomial, or even exponential. This general definition includes the case of more sophisticated computer architectures, that perform operations such as fused-addition-multiplications.

The scalar product between scalars x, y, when $d = 1$, has error

$$x \circ_\epsilon y - x \circ y = (x \circ y)\epsilon \ , \tag{3.D.1}$$

and in this scalar case, the error is generally proportional to the machine precision

$$\epsilon = \varepsilon \ . \tag{3.D.2}$$

This error could be considered as deterministic error, where a fixed magnitude error ϵ is associated with operations. Instead, stochastic error could be considered,

$$x \circ_\epsilon y - x \circ y = (x \circ y)\xi \ , \tag{3.D.3}$$

where the error is assumed to be the random variable

$$\xi \sim \Omega_\epsilon \tag{3.D.4}$$

from a distribution Ω_ϵ, with zero mean and a standard deviation that is proportional to ϵ.

For successive additions, the order of the operations affects which terms have greater error, as initial terms accumulate more error over the course of successive additions. The sum of d scalars $\{a_\mu\}$ has error

$$\sum_\mu^d{}_\epsilon\, a_\mu - \sum_\mu^d a_\mu = \sum_\mu^d a_\mu \epsilon_\mu \ , \tag{3.D.5}$$

where the accumulated error from addition is

$$\epsilon_\mu = (1 + \varepsilon)^{d-\mu} - 1 \ . \tag{3.D.6}$$

The error of summations of d scalars $a = \{a_\mu\}$ can be represented as inner products with a ones vectors 1, and error vectors $\epsilon = \{\epsilon_\mu\}$ to account for the accumulated error,

$$1 \cdot_\epsilon a - 1 \cdot a = \epsilon \cdot a \ . \tag{3.D.7}$$

Matrix Floating Point Error

Generalizing beyond scalars, the deterministic error of inner products of d-dimensional matrices A, B can be represented as an inner product with respect to an error matrix

$$AB_\epsilon - AB = A\Sigma B \ . \tag{3.D.8}$$

Similarly, the stochastic error of inner products of d-dimensional matrices A, B can be represented as an inner product with respect to an error matrix,

$$AB_\epsilon - AB = A\Xi B \tag{3.D.9}$$

where the error is assumed to be the random variable

$$\Xi \sim \Omega_\epsilon^{d\times d} \ , \tag{3.D.10}$$

from a distribution Ω_ϵ, with zero mean and a standard deviation that is proportional to ϵ.

The relative error matrix is arbitrary depending on the model of computation, and we choose it to be diagonal,

$$\Sigma = \{\epsilon_\mu = (1+\varepsilon)^{d-\mu} - 1\} \ ., \tag{3.D.11}$$

with norm

$$\|\Sigma\|_p = \left(\sum_\mu^d ((1+\varepsilon)^{d-\mu} - 1)^p\right)^{1/p} \approx \left(\sum_\mu^d (d-\mu)^p\right)^{1/p} \varepsilon \tag{3.D.12}$$

$$\approx \mathcal{O}(d^{\frac{p+1}{p}})\varepsilon \ \sim \ \mathcal{O}(\mathrm{poly}(d))\varepsilon \ \equiv \ \epsilon^{(p)} \ \leq \ \epsilon \ , \tag{3.D.13}$$

which we define to be upper bounded by a p-norm independent error ϵ.

This can be extended to a product of k matrices $\prod_\mu^k A_\mu$. We assume the error acts recursively, propagating from the initial to final matrix multiplications. Therefore. the matrix multiplication operation becomes a sum over all possible locations of the error matrix interlaced within the products, denoted with a multi-index $\Gamma \subseteq [k]$, with the number of errors $|\Gamma| = l \leq k$. We assume that errors also exist on the representation of the matrix elements, yielding an extra error matrix factor at the end of the product, and errors from additions of the error terms themselves are a secondary effect.

The error of the product of k matrices $\prod_\mu^k A_\mu$ therefore takes the form

$$\prod_{\mu}^{k}{}_\epsilon A_\mu - \prod_\mu^k A_\mu = \sum_{l>0}^k \sum_{\substack{\Gamma\subseteq[k] \\ |\Gamma|=l}} \prod_\mu^k A_\mu \Sigma^{\delta_{\mu\in\Gamma}} \ . \tag{3.D.14}$$

We then use norm inequalities, and define an upper bound on the norms values $\|A_\mu\|_p \leq \|A\|_p$, to bound the norm of the absolute matrix multiplicative error,

$$\left\| \prod_{\mu \epsilon}^{k} A_\mu - \prod_{\mu}^{k} A_\mu \right\|_s = \left\| \sum_{l>0}^{k} \sum_{\substack{\Gamma \subseteq [k] \\ |\Gamma| = l}} \prod_{\mu}^{k} A_\mu \Sigma^{\delta_{\mu \in \Gamma}} \right\|_s \tag{3.D.15}$$

$$\leq \sum_{l>0}^{k} \sum_{\substack{\Gamma \subseteq [k] \\ |\Gamma| = l}} \left\| \prod_{\mu}^{k} A_\mu \Sigma^{\delta_{\mu \in \Gamma}} \right\|_s \tag{3.D.16}$$

$$\leq \sum_{l>0}^{k} \sum_{\substack{\Gamma \subseteq [k] \\ |\Gamma| = l}} \min_{\sum_{\mu}^{k} \frac{1}{p_\mu} + \frac{\delta_{\mu \in \Gamma}}{q_\mu} = \frac{1}{s}} \prod_{\mu}^{k} \|A_\mu\|_{p_\mu} \|\Sigma\|_{q_\mu}^{\delta_{\mu \in \Gamma}} \tag{3.D.17}$$

$$\leq \sum_{l>0}^{k} \min_{\frac{k}{p} + \frac{l}{q} = \frac{1}{s}} \binom{k}{l} \|A\|_p^k \|\Sigma\|_q^l \tag{3.D.18}$$

$$\leq \sum_{l>0}^{k} \binom{k}{l} \lambda(A)^k \epsilon^l \tag{3.D.19}$$

$$= \lambda(A)^k ((1+\epsilon)^k - 1) \ . \tag{3.D.20}$$

Therefore, the relative multiplicative error is upper bounded for a general norm by

$$\frac{\left\| \prod_{\epsilon \mu}^{k} A_\mu - \prod_{\mu}^{k} A_\mu \right\|}{\left\| \prod_{\mu}^{k} A_\mu \right\|} \leq \frac{\lambda(A)^k}{\left\| \prod_{\mu}^{k} A_\mu \right\|} \Big((1+\epsilon)^k - 1 \Big) \overset{A_\mu = U_\mu}{\rightarrow} \frac{1}{\mathrm{poly}(d)} \Big((1+\epsilon)^k - 1 \Big) \ , \tag{3.D.21}$$

where in the case of unitary matrices with unit singular values, the bound is simplified.

3.D.2 Classical Error Scaling

Let us define a unitary evolution as a product of k, d-dimensional unitaries $U = \prod_{\mu}^{k} U_\mu$ that transforms an initial state as $\sigma \to \rho = U \sigma U^\dagger$. Suppose the evolution is subject to classical floating point error $U \to U_\epsilon$ and $\sigma \to \rho_\epsilon$. An interesting interpretation is that this noisy evolution with classical floating point error can be represented by unormalized Kraus-like operators, $\{I, \Sigma\}$, where $I + \Sigma^\dagger \Sigma \neq I$, meaning the operation is not trace-preserving.

The composite adjoint action of unitaries with l matrix multiplications with classical error is therefore

$$\rho_\epsilon = U_\epsilon \sigma U_\epsilon^\dagger = \rho + \delta_\epsilon \ , \tag{3.D.22}$$

where the perturbation is

$$\delta_\epsilon = \sum_{l+l'>0}^{k} \sum_{\substack{\Gamma,\Gamma'\subseteq[k] \\ |\Gamma|=l,|\Gamma'|=l'}} \left[\prod_{\mu}^{k} U_\mu \Sigma^{\delta_{\mu\in\Gamma}}\right] \sigma \left[\prod_{\mu'}^{k} \Sigma^{\delta_{\mu'\in\Gamma'}\dagger} U_{\mu'}{}^\dagger\right] \ . \tag{3.D.23}$$

The absolute norm error, given the non-unitary classical noise has norm $\|\Sigma\| = \epsilon$, and the operators are unitaries with $\|U_\mu\| = \mathrm{poly}(d)$, is therefore bounded by

$$\begin{aligned}
\|\rho_\epsilon - \rho\| &\leq \sum_{l+l'>0}^{k} \sum_{\substack{\Gamma,\Gamma'\subseteq[k] \\ |\Gamma|=l,|\Gamma'|=l'}} \left\| \left[\prod_{\mu}^{k} U_\mu \Sigma^{\delta_{\mu\in\Gamma}}\right] \sigma \left[\prod_{\mu'}^{k} \Sigma^{\delta_{\mu'\in\Gamma'}\dagger} U_{\mu'}{}^\dagger\right] \right\| && (3.D.24)\\
&\leq \sum_{l+l'>0}^{k} \sum_{\substack{\Gamma,\Gamma'\subseteq[k] \\ |\Gamma|=l,|\Gamma'|=l'}} \left[\prod_{\mu}^{k} \left\|U_\mu \Sigma^{\delta_{\mu\in\Gamma}}\right\|\right] \|\sigma\| \left[\prod_{\mu'}^{k} \left\|\Sigma^{\delta_{\mu'\in\Gamma'}\dagger} U_{\mu'}^\dagger\right\|\right] && (3.D.25)\\
&= \sum_{l+l'>0}^{k} \sum_{\substack{\Gamma,\Gamma'\subseteq[k] \\ |\Gamma|=l,|\Gamma'|=l'}} \left[\prod_{\mu}^{k} \left\|\Sigma^{\delta_{\mu\in\Gamma}}\right\|\right] \|\sigma\| \left[\prod_{\mu'}^{k} \left\|\Sigma^{\delta_{\mu'\in\Gamma'}\dagger}\right\|\right] && (3.D.26)\\
&\leq \sum_{l+l'>0}^{k} \sum_{\substack{\Gamma,\Gamma'\subseteq[k] \\ |\Gamma|=l,|\Gamma'|=l'}} \|\sigma\| \prod_{\mu,\mu'}^{k} \|\Sigma\|^{\delta_{\mu\in\Gamma}+\delta_{\mu'\in\Gamma'}} && (3.D.27)\\
&= \sum_{l+l'>0}^{k} \tbinom{k}{l}\tbinom{k}{l'} \|\sigma\| \epsilon^{l+l'} && (3.D.28)\\
&= \|\sigma\| \left[\left(\sum_{l}^{k} \tbinom{k}{l} \epsilon^l \right)^2 - 1 \right] && (3.D.29)\\
&= \|\sigma\| \left((1+\epsilon)^{2k} - 1 \right) \ . && (3.D.30)
\end{aligned}$$

Given the norm of pure input and noiseless output states is $\|\rho\| = \|\sigma\| = 1$, the relative error for pure states is

$$\frac{\|\rho_\epsilon - \rho\|}{\|\rho\|} \leq \left|1 - (1+\epsilon)^{2k}\right| \ . \tag{3.D.31}$$

We may then calculate the infidelity with a (pure) state ρ, as

$$\mathcal{L}_\epsilon = 1 - \mathrm{tr}(\rho\rho_\epsilon) = \mathcal{L} - \mathrm{tr}(\rho\delta_\epsilon) \ . \tag{3.D.32}$$

and using the Cauchy-Schwartz inequality,

$$|\mathcal{L}_\epsilon - \mathcal{L}| = |\mathrm{tr}(\rho\delta_\epsilon)| \leq \min_{\frac{1}{p}+\frac{1}{q}=1} \|\rho\|_p \|\delta_\epsilon\|_q \leq \min_{\frac{1}{p}+\frac{1}{q}=1} \|\rho\|_p \|\sigma\|_q \left|1 - (1+\epsilon)^{2k}\right| \ , \tag{3.D.33}$$

and this noisy linear objective deviation, for pure states, scales as

$$|\mathcal{L}_\epsilon - \mathcal{L}| \leq \left|1 - (1+\epsilon)^{2k}\right| \ . \tag{3.D.34}$$

An example simulation of classical floating point error for k successive matrix multiplications, by artificially including noise with different scales ε, is shown in Fig. 3.D.5. To investigate this classical error, we compare analytical results (exact upper bounds on the difference between noisy and noiseless matrix multiplications), probabilistic results (numerically adding zero-mean uniformly random errors $U[-\varepsilon/2, \varepsilon/2]$ to results of floating point operations), and numerical results (numerically performing matrix multiplications without artificial noise for standard datatypes: single point 32-bit precision with $\varepsilon \approx 10^{-7}$, double point 64-bit precision with $\varepsilon \approx 10^{-16}$, and quadruple point 128-bit precision with $\varepsilon \approx 10^{-19} - 10^{-24}$, depending on whether the precision is only simulated to $\sim$ 80-bit for double precision architectures). We observe good agreement across all models, and we note the drift in the probabilistic results at a large number of matrix multiplications k, to an effective ε an order of magnitude smaller than simulated. This suggests the built-in floating point operation algorithms may be cacheing results or eliminating some sources of error. We repeat such experiments across $m = 10$ samples, and we also observe negligible errorbars within plot markers. This classical error also shows similar trends to the quantum noise case of noise-induced convergent and divergent regimes.

Finally, we note that it is difficult to precisely define the machine precision ε, or the number of decimal points of accuracy of a simulation, even for a recognized floating point data-type. Apart from the often non-linear dependence of the true machine precision value on the number of operations and dimensionality of the problem, there are additionally software effects. These effects include how basic floating point operations may be vectorized or fused together. Similarly, there may be hardware effects, such as how the exact memory layout can affect the way in which operations are performed. These studies, in particular the exact agreement between analytical and 32 and 64-bit data-type numerical results, however confirm the appropriateness of our derived models, and they offer valuable insight into upper bound estimates for a range of machine precision values.

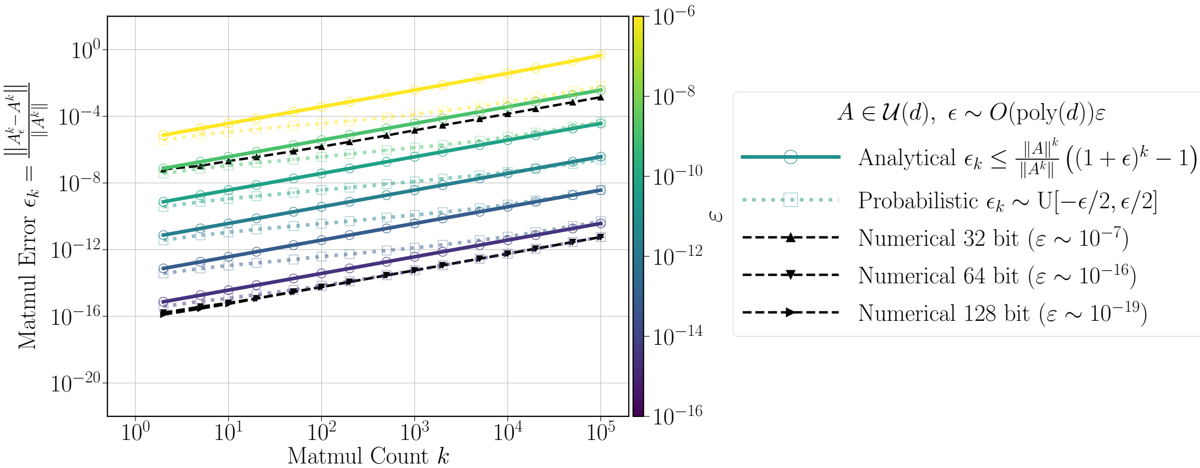


Figure 3.D.5: **Matrix multiplication floating point error**. Error scaling with number of multiplications k, with analytical (circles/solid lines), numerical probabilistic (squares/dotted lines) and numerical data-type (triangles/dashed lines) models scaling for $d = 2^2$-dimensional random unitaries A. Simulated floating point error is shown to increase polynomially with the number of floating point operations. Exact floating point data-types are in exact agreement with the analytical upper bounds, with the exception of simulated 128-bit data-type, whose error appears to be bounded by the 64-bit precision, likely attributed to intermediate backend calculations being cast to 64-bit precision.

The simulation package developed for these studies [6], as discussed in Appendix 3.F, can perform single $\varepsilon \sim \mathcal{O}(10^{-7})$, double $\varepsilon \sim \mathcal{O}(10^{-16})$, or (simulated) quadruple $\varepsilon \sim \mathcal{O}(10^{-19})$ floating point arithmetic, and each data-type is useful in understanding the sources of floating point error. However, quadruple floating point arithmetic is unable to currently be compiled efficiently into computational graphs for automatic differentiation for large systems, in addition to the overhead of the simulated precision. Automatic differentiation computations also empirically have less than double precision accuracy, typically $\varepsilon \sim \mathcal{O}(10^{-12})$. Therefore current large scale simulations involving automatically differentiated workflows are infeasible with this quadruple floating point data-type.

3.D.3 Quantum Error Scaling

Let us define a unitary evolution as a product of k, d-dimensional unitaries $U = \prod_{\mu}^{k} U_\mu$ that transforms an initial state as $\sigma \to \rho = U\sigma U^\dagger$. Suppose the evolution is subject to quantum error $U \to U_\gamma$ and $\sigma \to \rho_\gamma$. This noisy evolution with quantum error can be represented by the normalized Kraus operators $\{\sqrt{1-\gamma}I, \sqrt{\gamma}\Sigma\}$, where $(1-\gamma)I + \gamma\Sigma^\dagger\Sigma = I$, meaning the operation is trace-preserving.

The composite adjoint action of unitaries with k matrix multiplications with quantum error is therefore

$$\rho_\gamma = U_\gamma \sigma U_\gamma^\dagger = (1-\gamma)^k \rho + \delta_\gamma \ , \tag{3.D.35}$$

where the perturbation is

$$\delta_\gamma = \sum_{l>0}^{k} \sum_{\substack{\Gamma \subseteq [k] \\ |\Gamma| = l}} \gamma^l (1-\gamma)^{k-l} \left[\prod_{\mu}^{k} U_\mu \Sigma^{\delta_{\mu\in\Gamma}}\right] \sigma \left[\prod_{\mu}^{k} \Sigma^{\delta_{\mu\in\Gamma}\dagger} {U_\mu}^\dagger\right] \ . \tag{3.D.36}$$

The absolute norm error, given the unitary quantum noise has norm $\|\Sigma\| = \text{poly}(d)$, and the operators are unitaries with $\|U_\mu\| = \text{poly}(d)$, is therefore bounded by

$$\begin{aligned}
\|\rho_\gamma - \rho\| &\leq ((1-\gamma)^k - 1)\big\|U\sigma U^\dagger\big\| \qquad (3.D.37)\\
&\quad + \sum_{l>0}^{k} \sum_{\substack{\Gamma \subseteq [k] \\ |\Gamma| = l}} \gamma^l (1-\gamma)^{k-l} \left\| \left[\prod_{\mu}^{k} U_\mu \Sigma^{\delta_{\mu\in\Gamma}}\right] \sigma \left[\prod_{\mu}^{k} \Sigma^{\delta_{\mu\in\Gamma}\dagger} {U_\mu}^\dagger\right] \right\| \\
&= ((1-\gamma)^k - 1)\|\sigma\| + \sum_{l>0}^{k} \sum_{\substack{\Gamma \subseteq [k] \\ |\Gamma| = l}} \gamma^l (1-\gamma)^{k-l} \|\sigma\| \qquad (3.D.38)\\
&= ((1-\gamma)^k - 1)\|\sigma\| + \sum_{l>0}^{k} \tbinom{k}{l} \gamma^l (1-\gamma)^{k-l} \|\sigma\| \qquad (3.D.39)\\
&= 2\|\sigma\|((1-\gamma)^k - 1) \ . \qquad (3.D.40)
\end{aligned}$$

Given the norm of pure input and noiseless output states is $\|\rho\| = \|\sigma\| = 1$, the relative error for pure states is

$$\frac{\|\rho_\epsilon - \rho\|}{\|\rho\|} \leq 2\big|1 - (1-\gamma)^k\big| \ . \tag{3.D.41}$$

We may then calculate the infidelity with a (pure) state ρ as

$$\mathcal{L}_\gamma = 1 - \mathrm{tr}(\rho\rho_\gamma) = \mathcal{L} - \mathrm{tr}(\rho\delta_\gamma) \ . \tag{3.D.42}$$

and using the Cauchy-Schwartz inequality,

$$|\mathcal{L}_\gamma - \mathcal{L}| = |\mathrm{tr}(\rho\delta_\gamma)| \leq \min_{\frac{1}{p}+\frac{1}{q}=1} \|\rho\|_p \|\delta_\gamma\|_q \leq \min_{\frac{1}{p}+\frac{1}{q}=1} 2\|\rho\|_p \|\sigma\|_q \left|1 - (1-\gamma)^k\right| \ , \tag{3.D.43}$$

and this noisy linear objective deviation, for pure states, scales as

$$|\mathcal{L}_\gamma - \mathcal{L}| \leq 2\left|1 - (1-\gamma)^k\right| \ . \tag{3.D.44}$$

3.D.4 Classical versus Quantum Error Scaling

We may relate the classical and quantum error scales by identifying the dimension dependent classical scale

$$\epsilon = \epsilon(d) \tag{3.D.45}$$

and the number of classical matrix multiplication operations l can also be identified relative to the $s = \mathrm{poly}(n)k$ number of quantum operations, such that,

$$l = \mathcal{O}(s) = \mathcal{O}(\mathrm{poly}(n)k) \ . \tag{3.D.46}$$

The error scaling of classical and quantum noisily evolved states is thus,

$$\|\rho_\epsilon - \rho\| \leq \ \left|1 - (1+\epsilon)^{2l}\right| \tag{3.D.47}$$
$$\|\rho_\gamma - \rho\| \leq 2|1 - (1-\gamma)^s| \ , \tag{3.D.48}$$

and the linear functions of the perturbed states are also perturbed with identical scaling,

$$|\mathcal{L}_\epsilon - \mathcal{L}| \leq \ \left|1 - (1+\epsilon)^{2l}\right| \tag{3.D.49}$$
$$|\mathcal{L}_\gamma - \mathcal{L}| \leq 2|1 - (1-\gamma)^s| \ . \tag{3.D.50}$$

A subtle point to keep in mind is that we are comparing infidelities at fixed parameter θ values. If the parameters have been optimized in a noisy setting, then the associated noiseless infidelity $\mathcal{L}_{\theta_\gamma}$, evaluated with these parameters, likely differs from the true optimal noiseless infidelity $\mathcal{L}_\theta$ with parameters optimized in a noiseless setting. If we happen to be in an overparameterized regime, where the phenomenon of lazy training occurs [434], parameters may change negligibly from their initial values. In this regime, the noiseless infidelities, with noisy and noiseless trained parameters, possibly approximately coincide,

$$\theta^*_\gamma \approx \theta^* \ . \tag{3.D.51}$$

In this setting, functions of the parameters may also approximately coincide,

$$\mathcal{L}_{\theta^*_\gamma} \approx \mathcal{L}_{\theta^*} \ , \tag{3.D.52}$$

and they can be used in place of each other. Bounds such as those above comparing noisy and noiseless fidelities may be relevant, however future studies should investigate the noise-induced bias in optimization.

3.E Nuclear Magnetic Resonance Ansatz

In these appendixes, we include details about the nuclear magnetic resonance (NMR) ansatz. We describe the model Hamiltonian, and we document all experimentally relevant parameter scales,

$$H^{(\kappa)}_\theta = \sum_{i\in[n]} \theta^{x(\kappa)}_i X_i \ + \ \sum_{i\in[n]} \theta^{y(\kappa)}_i Y_i \ + \ \sum_{i\in[n]} h_i Z_i \ + \ \sum_{i<j\in[n]} J_{ij} Z_i Z_j \ . \tag{3.E.1}$$

Here we have control over the variable time-dependent transverse X and Y fields, with constant time-independent longitudinal Z and ZZ fields. In these units, the parameters in Table 3.E.1 are used for this ansatz. We note that the non-local coupling scale is $J \ll h, \theta$, and there are time step sizes such that $\tau J \ll 1$. To generate a finite angle ($\pi/4$ for ZZ) rotation necessary to implement a single entangling gate, depths of order $k \sim 1/\tau J \sim \mathcal{O}(10^4)$ are necessary. Furthermore, the number of entangling gates necessary to compile Haar random unitaries is generally exponential $\mathcal{O}(q^n)$ in the number of qubits [283]. Therefore, these parameter scales, even for relatively small system sizes of $n \leq 4$, indicate that in general an even greater total number, possibly super-exponential $k > \mathcal{O}(q^n)$, of physical gates are necessary in practice for these tasks of interest.

Table 3.E.1: **NMR parameters**. Constrained experimentally relevant parameters for NMR ansatz [39]. Some parameters, in particular the time step of 100 µs (unitary compilation), and 75 µs (state preparation), are chosen to be on the maximum end of what is experimentally feasible, to allow for reasonably efficient simulation.

Parameter	Description	Value
n	Number of qubits	$1 - 4$
k	Number of time steps	$5 - 5000$
p	Number of parameters	$kn(n+5)/2$
τ	Trotterization time step	$75 - 100$ µs
T	Evolution time	375 µs $-$ 500 ms
Q	Spatial Trotterization order	2
$J_{i<j}$	Constant longitudinal coupling	$\pi/2 \left\{ \begin{array}{l} 72.4, -130.0, 50.0, 210.0, \\ 20.0, -190.0, -30.0, 60.0, \\ 90.0, -60.0 \end{array} \right\}$ Hz
h_i	Constant longitudinal field	$\pi/2\ \{-10.0, 0, -1.0, 29.0, -20.0\}$ kHz
$\bar{\theta}$	Variable transverse field	$\pi/2$ MHz
γ	Noise scale	$10^{-14} - 10^{-1}$
$\tilde{\theta}$	Constraints	Constrained and Shared Transverse Fields Zero-Field Dirichlet Boundary Conditions $\theta_i^{x,y\,(\kappa)} = \theta^{x,y\,(\kappa)}\ ,\ \left\lvert\theta^{x,y\,(\kappa)}\right\rvert \leq \bar{\theta}\ ,\ \theta^{x,y\,(0,k)} = 0$

Table 3.E.2: **NISQ parameters**. Comparison of 1-qubit gate times T_{U1}, 2-qubit gate times T_{U2}, decoherence times T_γ, experimentally achieved number of qubits n_U, experimentally achieved depths of 2-qubit gates k_U, ratio of 2-qubit to 1-qubit gate times $k_{U12} = T_{U2}/T_{U1}$, and the ratio of decoherence to 2-qubit gate times $k_{\gamma U2} = T_\gamma/T_{U2}$, for various experimental quantum computing implementations. The ratio of 2 to 1-qubit gate times can be interpreted as the effective depth k_{U12} necessary to implement 2-qubit gates. The ratio of decoherence times T_γ to 2-qubit gate times can be interpreted as the effective maximum depth $k_{\gamma U2}$ before decoherence. For NMR, 1-qubit gate times $T_{U1} = \mathcal{O}(\tau)$ are approximately the pulse time steps, 2-qubit gate times $T_{U2} = \mathcal{O}(1/J)$ are approximately inversely related to the 2-qubit coupling, and decoherence times $T_\gamma = \mathcal{O}(T_2)$ are approximately the dephasing times. Tabulated values are experimental apparatus and algorithm specific [282], taken from a range of values in recent literature.

Implementation	T_{U1}	T_{U2}	T_γ	n_U	k_U	k_{U12}	$k_{\gamma U2}$
NMR [39, 41, 268]	75 µs	5000 µs	10^3 ms	12	1000	70	$2 \cdot 10^2$
Trapped Ions [41, 46–48, 69]	1 µs	50 µs	10^4 ms	12	64	50	$1 \cdot 10^6$
Superconducting [41, 52, 53]	0.02 µs	0.1 µs	1 ms	127	60	5	$1 \cdot 10^4$
Neutral Atoms [55, 57, 215]	2 µs	0.5 µs	10^3 ms	280	516	0.25	$2 \cdot 10^6$

We also compare gate times, decoherence times, experimentally tested depths, and effective depths based on these times for various NISQ implementations in Table 3.E.2. We note that each implementation has better and worse properties. NMR in our settings has the largest 2-qubit gate times $T_{U2} = \mathcal{O}(1/J)$, which translates to NMR having the largest effective depth $T_{U2}/T_{U1} \sim \mathcal{O}(100)$ required for each 2-qubit gate, and having the smallest effective maximum depth $T_\gamma/T_{U2} \sim \mathcal{O}(10^2)$ before coherence. We emphasize, however, that tabulated values taken from a range of values in recent literature are very experimental apparatus- and algorithm- specific [282].

3.F Classical Simulation and Optimization

In these appendixes, we discuss details of our classical simulation and optimization settings. We explain our hyperparameter choices, optimization routines, and we list all optimization settings.

The quantum systems here are simulated and optimized classically using a compiled, automatic-differentiation library using the Python JAX backend, developed as a general differentiable exponentially deep circuit simulator [6]. This library is optimized for noisy density matrix simulations with few qubits $n < 5$, and state-of-the-art large depth circuits with $p \sim \mathcal{O}(10^5)$ gates per circuit instance.

For example, our NMR ansatz, circuits consist of $Q = 2$ order spatially Trotterized evolution, consisting of fully connected two-body gates. Local noise on all qubits after each layer requires up to $k \sim 5 \times 10^3$ circuit layers, and the circuits are simulated roughly 10^9 times throughout the optimization routines and loops over ansatz settings. All optimizer hyperparameters are shown in Table 3.F.3.

When performing gradient-based optimization, the gradients of these objectives can be expressed analytically with parameter shift rules, however for the $n \leq 4$ system sizes considered here, we use automatic differentiation for efficiency. An exception is when computing the Hessian and quantum Fisher information, where computing analytical gradients is most memory-efficient. For small system sizes, the unitary and quantum noise operations, and infidelities, can also be calculated exactly. Sampling methods, which introduces forms of shot-noise, and other quantum algorithms, such as the Hilbert-Schmidt test, and other forms of process tomography, are not necessary to be implemented. The effects of estimating such infidelities with sampling and approximate operators for larger system sizes are important studies to be conducted in future works.

For computing statistics such as the mean and variance across m independent optimizations, we sample any inputs from specific distributions. We generally sample the initial states σ, target unitaries U, and target states ρ according to the Haar measure, to avoid any biases in targeting a specific subspace [106]. Expectation values of parameterized functions of the initial and target states $\mathcal{F}_\theta(\sigma, \rho)$, such as infidelities, may then be computed,

$$\langle \mathcal{F}_\theta \rangle^{(m)} = \frac{1}{m} \sum_{\substack{i \in [m] \\ \tilde{\theta}^{(i)} \sim \text{Uniform} \\ \sigma^{(i)} \sim \text{Haar} \ , \ \rho^{(i)} \sim \text{Haar}}} \mathcal{F}_{\theta^{(i)}}(\sigma^{(i)}, \rho^{(i)}) \ . \tag{3.F.1}$$

For artificially simulating floating point error, we add random matrices to each successive distinct matrix multiplication in a calculation. Plotted quantities are also calculated at the optimal parameterization, which is not necessarily at the last optimization iteration. We also uniformly randomly sample initial parameters $\theta = \tilde{\theta}$, and we then smooth them over the k time steps with cubic interpolation to be experimentally implementable.

Here, we define the parameters as $\theta = \theta(\phi)$, functions of explicit variables ϕ that are explicitly optimized. In the unconstrained case, all parameters for each qubit are independent, and are not constrained in magnitude. However in the constrained case, we impose that the fields are constrained, and coupled to act uniformly across all sites for each operator. Therefore, $\theta_i^{x,y} = \theta^{x,y}$ for each qubit $i \in [n]$. We also bound all transverse field magnitudes $|\theta_i^{x,y}| \leq \bar{\theta}$. We finally impose Dirichlet boundary conditions in time, such that the initial and final fields are approximately zero, $\theta^{x,y(0)} = \theta^{x,y(k)} = 0$.

To perform the classical simulation and optimization of the parameterized channels, we perform gradient based optimization routines to minimize the objectives $\mathcal{L}_{\theta\gamma}$ with respect to the variable parameters θ. We may then compute gradients of objectives $\zeta = \partial\mathcal{L}_{\theta}$.

Due to the high dimensionality of the problem, we choose a variant of the first-order conjugate gradient scheme, where the search direction ξ is updated iteratively [284] at iteration $0 \leq i \leq L$:

$$\theta^{(i+1)} = \theta^{(i)} + \alpha^{(i)}\xi^{(i)} \ , \tag{3.F.2}$$

$$\xi^{(i+1)} = \ -\zeta^{(i+1)} + \beta^{(i)}\xi^{(i)} \ , \tag{3.F.3}$$

given initial conditions of $\theta^{(0)}$ and $\xi^{(0)} = -\zeta^{(0)}$. The learning rates α, β must be chosen to satisfy the Wolfe convergence conditions [284], which guarantee the parameters and search directions are updated such that the objective is monotonically decreasing.

For the parameter learning rate α, a line search is conducted, that involves at most L_α objective calls per iteration l, and it ensures the objective maximally decreases. For the search learning rate β, rates that obey

$$|\beta| \leq \bar{\beta} = \frac{\zeta^{(i+1)} \cdot \zeta^{(i+1)}}{\zeta^{(i)} \cdot \zeta^{(i)}} \tag{3.F.4}$$

ensure convergence [284]. Various forms for this parameter include the standard Fletcher-Reeves rate $\beta_{\mathrm{FR}}^{(i)} = \bar{\beta}^{(i)}$. However, for the range of problems considered here, we find that it and many definitions lead to the optimizer immediately getting stuck in local minima. We find the Hestenes-Stiefel rate, however, to be quite effective,

$$\beta_{\mathrm{HS}}^{(i)} = \frac{\zeta^{(i+1)} \cdot (\zeta^{(i+1)} - \zeta^{(i)})}{\xi^{(i)} \cdot (\zeta^{(i+1)} - \zeta^{(i)})} \ . \tag{3.F.5}$$

The hyperparameters, as per Table 3.F.3, are chosen after heuristic searches for iterations L, learning rates α, β, and other parameters c_i, that indicated stability and adequately fast convergence for noisy and noiseless state preparation and unitary compilation tasks. Additional heuristic stopping conditions ϵ are also chosen to avoid unnecessary iterations.

Table 3.F.3: **Optimization hyperparameters**. All settings are selected from manual parameter searches, and are only heuristically shown to reasonably guarantee optimization convergence across all system settings.

Hyperparameter	Description	Value
Optimizer	Optimizer routine	Conjugate grad
Line Search	Line search routine	Wolf
Conjugate Search	Conjugate search routine	Hestenes-Stiefel
$p_{\tilde{\theta}}$	Initial parameters	Uniform
$p_{U,\rho}$	Objectives distribution	Haar
p_σ	Initial state distribution	Haar
m	Number of sample objectives	50
L	Maximum number of optimization iterations	500
$L_<$	Minimum number of optimization iterations	50
L_α	Maximum number of line search iterations per iteration	2500
$\epsilon_{\mathcal{L}}$	Minimum objective	10^{-16}
$\epsilon_{\Delta\mathcal{L}}$	Minimum absolute difference in objective per iteration	0
$\epsilon_{\delta\mathcal{L}}$	Maximum relative increase in objective per iteration	10^{-3}
$\epsilon_{\partial\mathcal{L}}$	Minimum gradient norm	0
$\epsilon_{\Delta\partial\mathcal{L}}$	Minimum absolute difference in gradient norm per iteration	0
$\epsilon_{\delta\partial\mathcal{L}}$	Maximum relative increase in gradient norm per iteration	∞
α	Initial parameter learning rate	10^{-4}
β	Initial search learning rate	10^{-4}
$\alpha_{\lessgtr}$	Bounds on parameter learning rate before reset to α	$[0, \infty]$
$\beta_{\lessgtr}$	Bounds on search learning rate before reset to β	$[10^{-10}, 10^{10}]$
c_1	Wolf objective parameter	$10^{-5} - 10^{-4}$
c_2	Wolf gradient parameter	$10^{-1} - 9 \times 10^{-1}$

Chapter 4 Appendices

4.A Spaces, Operators, and Permutations

In this appendix, we present useful preliminary definitions, including relevant spaces, operators, norms, and permutation operators.

Let us denote the d-dimensional Hilbert space as $\mathcal{H} = \mathbb{C}^d$, the set of bounded linear operators $\rho, \Pi \in \mathcal{B}[\mathcal{H}]$, the set of bounded linear superoperators acting on operators as $\Lambda \in \mathcal{B}[\mathcal{B}[\mathcal{H}]]$, and a basis of orthogonal operators as $\mathcal{P}_d \subset \mathcal{B}[\mathcal{H}]$. The t-fold tensor product of the Hilbert space is denoted as $\mathcal{H}^{\otimes t}$, and we define the set of t indices as, $[t] = \{0, 1, \ldots, t-1\}$.

Moreover, from $\mathcal{P}_d$ we define a basis for the space of linear operators $\mathcal{B}[\mathcal{H}^{\otimes t}]$, which is simply given by $\mathcal{P}_d^t = \mathcal{P}_d^{\otimes t}$. Given two operators $X, Y \in \mathcal{B}[\mathcal{H}^{\otimes t}]$, their normalized inner product is denoted as $\chi_d^{(t)}(X,Y) = \langle\langle X|Y\rangle\rangle / \langle\langle I_d^{\otimes t}|I_d^{\otimes t}\rangle\rangle$. Here, we indicate t-fold generalizations with t or (t) superscripts.

Given an operator $X = \otimes_{i\in[t]} X_i \in \mathcal{B}[\mathcal{H}^{\otimes t}]$ with tensor product structure, we define its support Γ_X and locality as the indices of non-identity operators over copies of the space,

$$\Gamma_X = \{i \in [t] \ : \ X_i \notin \{I_d\}\} \qquad , \qquad |X| = |\Gamma_X| \in [t+1] \ . \tag{4.A.1}$$

Operators $X = \otimes_{i\in\Gamma_X} X_i$, restricted to a subset of their support $\Gamma \subseteq \Gamma_X$, are denoted as $X_\Gamma = \otimes_{i\in\Gamma} X_i$. Operators $X = \sum_x X_x$ with support $\Gamma_X = \cup_x \Gamma_{X_x}$, are denoted as localized with definitive support, if all operators within their expansion have identical support $\Gamma_{X_x} = \Gamma_X$. Operators $X, Y \in \mathcal{B}[\mathcal{H}^{\otimes t}]$ are denoted as being orthogonal with respect to support, if they are orthogonal due to having non-identical support, $\chi_d^{(t)}(X,Y) \propto \delta_{\Gamma_X\Gamma_Y}$.

To simplify our analysis of permutations $\mathcal{S}_t$, their representations $V_d : \sigma \in \mathcal{S}_t \to V_d(\sigma) \in \mathcal{S}_d^{(t)} \subset \mathcal{B}[\mathcal{H}^{\otimes t}]$, and normalized characters $\chi_d^{(t)}(\sigma,\pi) = \langle\langle V_d(\sigma)|V_d(\pi)\rangle\rangle / \langle\langle V_d(e)|V_d(e)\rangle\rangle$, let us define the set of character-normalized permutations $\hat{\mathcal{S}}_t$,

$$\hat{\mathcal{S}}_t = \{\hat{\sigma} = V_d(\sigma)/\chi_d^{(t)}(\sigma)\}_{\sigma\in\mathcal{S}_t} \qquad : \qquad \chi_d^{(t)}(\hat{\sigma},\hat{\pi}) = \frac{\chi_d^{(t)}(\sigma,\pi)}{\chi_d^{(t)}(\sigma)\ \chi_d^{(t)}(\pi)} \ . \tag{4.A.2}$$

Here, it is important to note that while the representations of elements of $\mathcal{S}_t$ are linearly independent for $d \geq t$ [248], they are not orthogonal [141]. Hence, we find it convenient to derive an alternative basis denoted as the localized permutations. The elements of this basis are in one-to-one correspondence with the original permutations, are localized with definitive support, and crucially are orthogonal with respect to support.

Let us motivate our localized basis by considering a simple example for $t = 3$. First, let us first recall that the normalized operator for a transposition $\tau = (ij)$ is,

$$\hat{\tau} = \sum_{P\in\mathcal{P}_d} P_i \otimes P_j^\dagger \ . \tag{4.A.3}$$

Then, given the $l = 3$-cycle $\lambda = (012)$, we can decompose it as $(012) = (01)(02) = (21)(10)$,

$$\begin{aligned}\hat{\lambda} &= \sum_{P\in\mathcal{P}_d} P_0 \otimes P_1^\dagger \sum_{S\in\mathcal{P}_d} S_0 \otimes S_2^\dagger \ = \ \sum_{P,S\in\mathcal{P}_d} P_0 S_0 \otimes P_1^\dagger \otimes S_2^\dagger \\ &= \underbrace{I_d \otimes I_d \otimes I_d}_{()} \\ &\quad + \sum_{P\in\mathcal{P}_d\backslash\{I_d\}} \underbrace{P \otimes P^\dagger \otimes I_d}_{(01)} + \sum_{P\in\mathcal{P}_d\backslash\{I_d\}} \underbrace{P \otimes I_d \otimes P^\dagger}_{(02)} + \sum_{P\in\mathcal{P}_d\backslash\{I_d\}} \underbrace{I_d \otimes P \otimes P^\dagger}_{(12)} \\ &\quad + \sum_{P\neq S\in\mathcal{P}_d\backslash\{I_d\}} \underbrace{PS \otimes P^\dagger \otimes S^\dagger}_{(012)} \ . \end{aligned} \tag{4.A.4}$$

The brackets in the previous equation showcase that the final decomposition contains grouping of tensor products of operators that match the expansion of λ into its sub-permutations, for example $\{\pi = (), (01), (02), (12), (012) \subseteq (012) = \lambda\}$. Importantly, since the decomposition of a permutation into transpositions is not unique, the previous decomposition is also not unique. However, such expansions suggest there exist general relationships between representations of permutations, and their sub-permutations. We seek to exploit such relationships to define the basis of localized permutations.

Theorem 7 (Localized Basis for Permutations)**.** *Given the character normalized permutations $\hat{\mathcal{S}}_t$, there exists a change of basis, $\phi_t \;:\; \hat{\mathcal{S}}_t \to [\hat{\mathcal{S}}_t]$, to localized permutations $[\hat{\mathcal{S}}_t]$,*

$$\hat{\sigma} \;=\; \sum_{\pi \subseteq \sigma} [\hat{\pi}] \quad \leftrightarrow \quad [\hat{\sigma}] \;=\; \sum_{\pi \subseteq \sigma} \phi_t(\sigma,\pi)\; \hat{\pi} \;, \tag{4.A.5}$$

where the elements of $[\hat{\mathcal{S}}_t] = \{[\hat{\sigma}]\}_{\sigma \in \mathcal{S}_t}$ depend on sub-permutation relationships $\pi \subseteq \sigma$, and obey the following properties:

i). Change of basis transformation: $\phi_t \;:\; \hat{\mathcal{S}}_t \to [\hat{\mathcal{S}}_t]$ is invertible, lower-triangular as per the sub-permutation ordering, is dimension-d-independent, and is solely dependent on the structure of the t-order permutations:

$$\phi_t(\sigma,\pi) \;=\; c(\pi^{-1}\sigma)\; \delta_{\sigma \supseteq \pi} \qquad , \qquad c(\sigma) = \prod_{\lambda \in \sigma} c(\lambda) \tag{4.A.6}$$

$$\phi_t^{-1}(\sigma,\pi) \;=\; \delta_{\sigma \supseteq \pi} \qquad , \qquad c(\lambda) = (-1)^{|\lambda|}\; c_{|\lambda|} \quad , \quad c_l = \tfrac{1}{l+1}\tbinom{2l}{l}\;. \tag{4.A.7}$$

ii). Linearly independent for $d \geq t$, and in one-to-one correspondence operators with permutations: $[\hat{\sigma}] \in [\hat{\mathcal{S}}_t] \leftrightarrow \sigma \in \mathcal{S}_t$

iii). Orthogonal with respect to support, and localized with definitive support $\Gamma_{[\hat{\sigma}]} = \Gamma_\sigma$:

$$\chi_d^{([\hat{\mathcal{S}}_t])} \;=\; \bigoplus_{\Gamma \subseteq [t]} \chi_d^{(\Gamma)} \;:\; \chi_d^{(\Gamma)} = \begin{bmatrix} \chi_d^{(t)}([\hat{\sigma}],[\hat{\sigma}]) & \cdots & \chi_d^{(t)}([\hat{\sigma}],[\hat{\pi}]) \\ \vdots & \ddots & \vdots \\ \chi_d^{(t)}([\hat{\pi}],[\hat{\sigma}]) & \cdots & \chi_d^{(t)}([\hat{\pi}],[\hat{\pi}]) \end{bmatrix}_{\substack{\sigma,\pi \in \mathcal{S}_t \\ \Gamma_\sigma = \Gamma_\pi = \Gamma}} \tag{4.A.8}$$

$$\left|\chi_d^{(t)}([\hat{\sigma}],[\hat{\pi}])\right| \;\leq\; c_\sigma^2\, c_\pi^2\, d^{|\sigma|+|\pi|}\; \delta_{\Gamma_\sigma \Gamma_\pi}$$

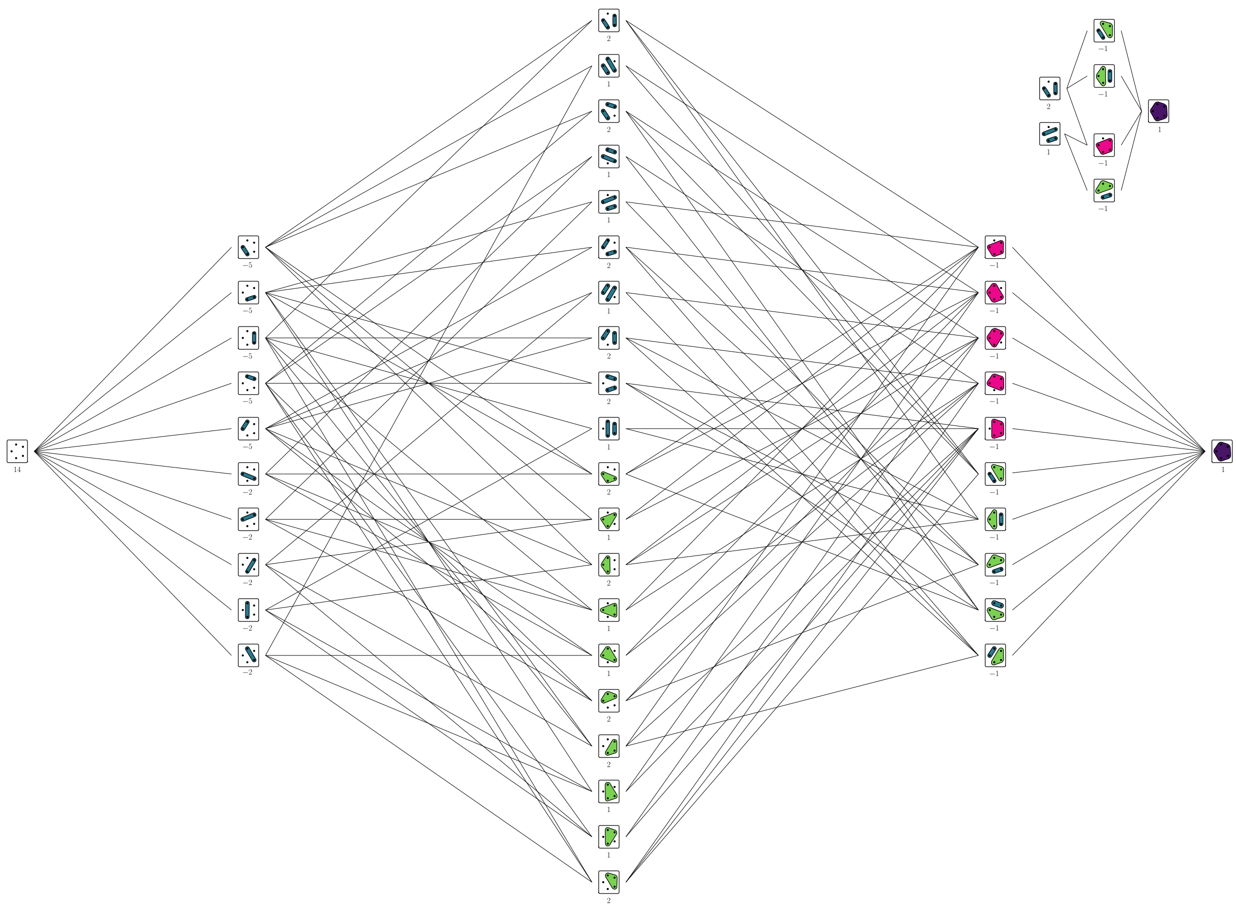


Figure 4.A.1: **Sub-permutation lattice**. All $c_5 = 42$ sub-permutations $e \subseteq \xi \subseteq \sigma$ for an $l = 5$ cycle $\sigma = (01234)$ are shown as nodes on a lattice, each as non-crossing partitions of points, ordered counter-clockwise from the leftmost point on an $l = 5$-polygon. Nodes of the lattice depict individual sub-permutations $e \subseteq \xi \subseteq \sigma$ between minimal e and maximal σ, ordered in l columns, by size $0 \leq |\xi| \leq l - 1$, from left to right. Edges of the lattice depict sub-permutation relationships, from left to right. At a given node ξ, edges and paths to rightward nodes ξ' represent larger sup-permutations $\xi \subset \xi' \subseteq \sigma$, such that $1 \leq |\xi^{-1}\xi'| \leq |\xi^{-1}\sigma| \leq l - 1$. Such sup-permutations $\{\xi \subseteq \xi' \subseteq \sigma\}$ of ξ form a sub-lattice with minimal ξ and maximal σ. Permutations ξ are labelled with their localized permutation change of basis coefficient $\phi_l(\sigma, \xi) = c(\xi^{-1}\sigma)$, such that $\sum_{\pi \subseteq \xi \subseteq \sigma} \phi_l(\sigma, \xi) = \delta_{\sigma\pi}$. Inset (top right) is the sub-lattices forming a set S_σ, with minimal permutations $\pi \in T_\sigma = \{\pi = (01)(23), (03)(12) \subset \sigma = (01234)\}$, with support $\Gamma = \Gamma_\pi = \{0, 1, 2, 3\} \subset \Gamma_\sigma = \{0, 1, 2, 3, 4\}$, and a smallest common sup-permutation $\pi_\sigma = (0123) \subset \sigma = (01234)$, such that $\sum_{\xi \in S_\sigma} \phi_l(\sigma, \xi) = \sum_{\pi \subseteq \xi \subseteq \sigma}^{\pi \in T_\sigma} \phi_l(\sigma, \xi) - \sum_{\pi \subseteq \xi \subseteq \sigma}^{\pi = \pi_\sigma} \phi_l(\sigma, \xi) = 0$.

Proof. To prove properties of the localized permutation basis, we use properties of sub-permutations $\pi \subseteq \sigma$ such that $|\pi^{-1}\sigma| = |\sigma| - |\pi|$, which represent a partial ordering between permutations [252, 435], and is transitive, such that if $\pi \subseteq \xi$, and $\xi \subseteq \sigma$, then $\pi \subseteq \sigma$, and if $\pi \subseteq \xi \subseteq \sigma$, then $\pi^{-1}\xi \subseteq \pi^{-1}\sigma$. Concurrently, the number of sub-permutations of an l-length cycle also corresponds to the number of non-crossing partitions of the set $[l]$ that respect the order of the permuted indices [252]. As depicted in Fig. 4.A.1, the set of sub-permutations $\{\pi \subseteq \xi \subseteq \sigma\} \cong \{e \subseteq \xi \subseteq \pi^{-1}\sigma\}$ forms a lattice containing all paths of sub-permutations ξ between some minimum π and maximum σ. We also note that any sub-permutations $\{\pi \subseteq \lambda\}$ of a cycle λ, with identical support $\Gamma_\pi = \Gamma \subseteq \Gamma_\lambda$ and locality $l_\pi = l \leq l_\lambda$, are within a subset of permutations that are isomorphic to the l-order permutations $\mathcal{S}_l$. Further, any subset $T \subseteq \{\pi \subseteq \lambda\}$ of such Γ-support sub-permutations of λ have a unique smallest common sup-permutation π' (which is referred to as the join of T in the lattice literature [436]). Further, this join π' has Γ-support and is a sub-permutation of the unique Γ-support, l-length cycle $\pi_\lambda \subseteq \lambda$ (joins of T can be seen to preserve common singletons, the elements $[t]\backslash\Gamma$ acted on trivially by all $\pi \in T$ [437]).

Further, many permutation-dependent quantities can be partitioned by the support $\Gamma_\sigma = \Gamma \subseteq [t]$ and locality $l_\sigma = l = |\Gamma| \leq t$ of permutations σ. Properties of sub-permutations also hold independently for disjoint cycles, given permutations $\pi = \prod_{\lambda\in\sigma}\pi_\lambda$ can be decomposed into disjoint sub-permutations $\{\pi_\lambda \subseteq \lambda\}_{\lambda\in\sigma}$ for any sup-permutation $\sigma = \prod_{\lambda\in\sigma}\lambda \supseteq \pi$. It follows that sup-permutation dependent quantities can often be factorized, such as $[\hat{\pi}] = \prod_{\lambda\in\sigma}[\hat{\pi}_\lambda]$, and $\phi_t(\sigma,\pi) = \prod_{\lambda\in\sigma}\phi_{l_\lambda}(\lambda,\pi_\lambda)$.

Finally, we can count the number of sub-permutations and localized permutations. We note that the number of sub-permutations, which equals the number of non-crossing partitions, equals the Catalan numbers, $c_l = \frac{1}{l+1}\binom{2l}{l}$ [438]. The number of sub-permutations between permutations $\pi \subseteq \sigma$ subsequently are,

$$c_{\sigma,\pi} = |\{\xi \in \mathcal{S}_t \ : \ \pi \subseteq \xi \subseteq \sigma\}| = c_{\pi^{-1}\sigma}\ \delta_{\sigma\supseteq\pi} \tag{4.A.9}$$

$$c_\sigma = |\{\xi \in \mathcal{S}_t \ : \ \xi \subseteq \sigma\}| = \prod_{\lambda\in\sigma} c_{l_\lambda}\ . \tag{4.A.10}$$

The number of locality-l blocks of permutations is $\binom{t}{l}$, and the number of permutations non-trivially permuting their l-locality support Γ, is by definition, $!l = \left\lfloor \frac{l!}{e} + \frac{1}{2} \right\rfloor$, the number of length-$l$ derangements [251].

We will prove each property $i)$ to $iii)$ of the mapping between the permutation and localized permutation bases separately.

i) To prove the existence of the localized permutation basis, we will show that an invertible relationship exists, also known as Möbius inversion [436], between $\hat{\mathcal{S}}_t$ and $[\hat{\mathcal{S}}_t]$. Let $\phi_t^{-1} \ : \ [\hat{\mathcal{S}}_t] \to \hat{\mathcal{S}}_t$ be such a change of basis, with matrix elements $\phi_t^{-1}(\sigma,\pi) = \delta_{\sigma\supseteq\pi}$, which is lower-triangular, with unit diagonal. Its sub-permutation dependent structure further impose that its strictly lower-triangular component is t-nilpotent, given the (σ,π) element of its k powers is the number of paths, of sub-permutations in the lattice formed by $\pi \subset \sigma$, of length k, and each path has at most $|\pi^{-1}\sigma| < t$ elements. The change of basis matrix inverse thus exists, $\phi_t \ : \ \hat{\mathcal{S}}_t \to [\hat{\mathcal{S}}_t]$, in fact with a closed form Möbius inverse [436], with identical sub-permutation dependent structure to its inverse, $\phi_t(\sigma,\pi), \phi_t^{-1}(\sigma,\pi) \propto \delta_{\sigma\supseteq\pi}$, that only depends on the overlaps and structure of permutations,

$$\phi_t = \sum_{k\in[t]} (I_{t!} - \phi_t^{-1})^k \ , \ \phi_t(\sigma,\pi) = \phi_t(\pi^{-1}\sigma) \ \delta_{\sigma\supseteq\pi} \quad : \quad \begin{array}{c} \phi_t(\sigma) = c(\sigma) = \prod_{\lambda\in\sigma} c(\lambda) \\ \sum_{\pi\subseteq\sigma} \phi_t(\pi) = \delta_{\sigma e} \end{array} . \tag{4.A.11}$$

Such relationships are independent of the representation dimension d.

ii) From the invertible relationship between the bases, both the permutations and the localized permutations are linearly independent for $d \geq t$ [248], and the localized permutations are in one-to-one correspondence with the permutations.

iii) To prove that localized permutations are localized, we consider the effect of the support of a basis operator string, $P = \otimes_{i\in\Gamma_P} P_i \in \mathcal{P}_d^t$, with support $\Gamma_P \subseteq \Gamma_\sigma \subseteq [t]$, on its overlap with a localized permutation $[\hat{\sigma}] \in [\hat{\mathcal{S}}_t]$, with potentially non-localized support $\Gamma_{[\hat{\sigma}]} = \cup_{\pi\subseteq\sigma}\Gamma_\pi = \Gamma_\sigma$. As per Fig. 4.A.2, the overlap between a permutation and an operator string is the tracefullness of products of operators, in the order of the permuted indices within each cycle of the permutation. Such products of operators are invariant under any sup-permutations, meaning non-zero overlaps of operator strings with all permutations π, π' with a common sup-permutation $\sigma \supseteq \pi, \pi'$, must be identical,

$$\chi_d^{(t)}(\hat{\pi}, P) \ = \ \chi_d^{(t)}(\hat{\pi}', P) \ = \ \chi_d^{(t)}(\hat{\sigma}, P) \ = \ \frac{1}{\chi_d^{(t)}(\sigma)} \prod_{\lambda\in\sigma} \chi_d^{(t)}(\prod_{i\in\Gamma_\lambda} P_i) \tag{4.A.12}$$

$$\chi_d^{(t)}(\hat{\pi}, P) \ = \ \chi_d^{(t)}(\hat{\pi}', P) \ = \ \chi_d^{(t)}(\hat{\sigma}, P) \ \propto \ \delta_{\Gamma_\sigma\supseteq\Gamma_P} \ \delta_{\Gamma_\pi\supseteq\Gamma_P} \ \delta_{\Gamma_{\pi'}\supseteq\Gamma_P} \ . \tag{4.A.13}$$

Therefore overlaps of localized permutations with an operator string reduce to summations of coefficients ϕ_t over the subset of permutations, $S_\sigma = \{\xi \subseteq \sigma \ : \ \Gamma_\xi \supseteq \Gamma_P \ , \ \chi_d^{(t)}(\xi, P) \neq 0\}$ that have non-zero overlap with the operator string,

$$\frac{\chi_d^{(t)}([\hat{\sigma}], P)}{\chi_d^{(t)}(\hat{\sigma}, P)} = \sum_{\xi\subseteq\sigma} \phi_t(\sigma,\xi) \ \frac{\chi_d^{(t)}(\hat{\xi}, P)}{\chi_d^{(t)}(\hat{\sigma}, P)} = \sum_{\xi\in S_\sigma} \phi_t(\sigma,\xi) \ \frac{\chi_d^{(t)}(\hat{\xi}, P)}{\chi_d^{(t)}(\hat{\sigma}, P)} = \sum_{\xi\in S_\sigma} \phi_t(\sigma,\xi) \ . \tag{4.A.14}$$

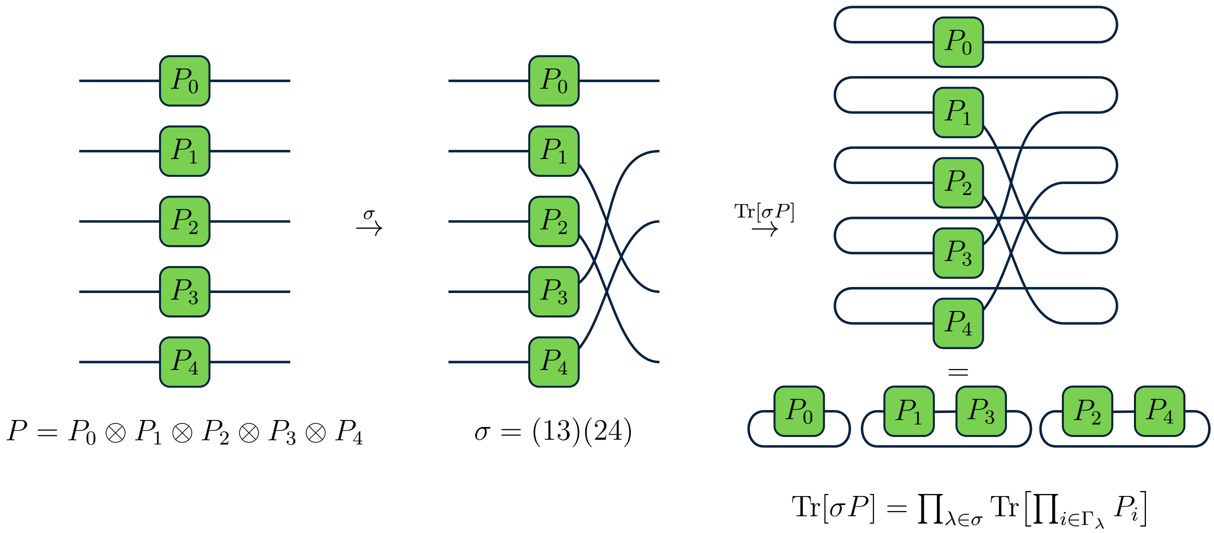


Figure 4.A.2: **Overlap between operators and permutations**. Overlaps between an operator string P, and a permutation σ are equal to the tracefullness of the products of an operator's components $\otimes_{\lambda \in \sigma} \prod_{i \in \Gamma_\lambda} P_i$, in the order of indices $\cup_{\lambda \in \sigma} \Gamma_\lambda$ permuted by each of the permutation's disjoint cycles.

Given summations of the coefficients ϕ_t over lattices $\{\pi \subseteq \xi \subseteq \sigma\}$ are non-zero if and only if $\pi = \sigma$, we seek to express the unknown summations over S_σ in terms of these known summations over lattices, potentially leading to cancellations. The set $S_\sigma = \cup_{\pi \in T_\sigma} S_{\sigma\pi}$, although not a lattice, can be partitioned into lattices, $S_{\sigma\pi} = \{\xi \in S_\sigma \ : \ \pi \subseteq \xi \subseteq \sigma\}$, each defined by a unique minimal permutation, $\pi \in T_\sigma = \{\pi \in S_\sigma \ : \ \pi \not\supset \xi \ \ \forall\, \xi \in S_\sigma\}$, as in the inset of Fig. 4.A.1. As will be crucial, all minimal $\pi \in T_\sigma$ have identical support $\Gamma_\pi = \Gamma_P$, otherwise they would not be minimal permutations in S_σ. As such, as above, any subset $T \subseteq T_\sigma$ of permutations always have smallest common sup-permutations π' with Γ_P-support, and which is itself a sub-permutation $\pi' \subseteq \pi_\sigma$, of the unique where Γ_P-support permutation $\pi_\sigma = \prod_{\lambda \in \sigma} \pi_{\sigma\lambda} \subseteq \sigma$, comprised of unique disjoint cycles $\{\pi_{\sigma\lambda} : \pi_{\sigma\lambda} \subseteq \lambda\}_{\lambda \in \sigma}$. Finally, in the equal support case, $\Gamma_P = \Gamma_\sigma \quad \to \quad \pi_\sigma \subseteq \sigma$, whereas in the strict subset of support case, $\Gamma_P \subset \Gamma_\sigma \ \ \to \ \ \pi_\sigma \subset \sigma$, and thus any smallest common sup-permutations $\pi' \subseteq \pi_\sigma$ of $T \subseteq T_\sigma$, are constrained in size to be at most π_σ.

Given this decomposition of a set of permutations into a union of lattices, we use the inclusion-exclusion principle [439], that relates counting of unions as counting of intersections, less multiply counted elements in multiple intersections,

$$\sum_{\xi \in \cup_{\pi \in T_\sigma} S_{\sigma\pi}} = \sum_{\{\} \subset T \subseteq T_\sigma} (-1)^{|T|-1} \sum_{\xi \in \cap_{\pi \in T} S_{\sigma\pi}} . \tag{4.A.15}$$

As one example, in the equal support case, if there is a single minimal permutation $\pi = \sigma$, forming a single lattice, $S_{\sigma\pi} = S_\sigma = \{\sigma\}$, then trivially $\sum_{\xi\in S_\sigma} \phi_t(\sigma,\xi) = 1$. As another example, in the strict subset of support case, if there is a single minimal permutation $\pi \subset \sigma$, forming a single lattice, $S_{\sigma\pi} = S_\sigma = \{\pi \subseteq \xi \subseteq \sigma\}$, then trivially $\sum_{\xi\in S_\sigma} \phi_t(\sigma,\xi) = 0$. It is therefore essential to understand any cases of σ, P that yield non-trivial intersections of lattices $S_{\sigma\pi}$ associated with minimal permutations $\pi \in T_\sigma$, in order to understand the summations over S_σ using the inclusion-exclusion principle.

Although intersections of lattices, $S'_\sigma = \cap_{\pi\in T} S_{\sigma\pi}$ for some $T \subseteq T_\sigma$, are not necessarily lattices [440], any sup-permutation $\xi'' \supseteq \xi'$ of any permutation in the intersection $\xi' \in S'_\sigma$, is necessarily also in this intersection, $\xi'' \in S'_\sigma$. Therefore, intersections of lattices can be themselves decomposed into new lattices $S'_{\sigma\pi}$, such that $S'_\sigma = \cup_{\pi'\in T'_\sigma} S'_{\sigma\pi'}$, each defined by a unique minimal permutation, $\pi' \in T'_\sigma = \{\pi \in S'_\sigma : \pi \not\supset \xi \ \ \forall\, \xi \in S'_\sigma\}$, and crucially $\pi' \subseteq \pi_\sigma$.

Intersections of lattices can thus be recursively defined as unions of sub-lattices, $S_\sigma \to S'_\sigma \to \cdots \to S^*_\sigma$. Such intersecting sub-lattices are depicted in Fig. 4.A.3, where coloured patches and nodes represent their non-intersecting components, with highlighted minimal elements in one of $T_\sigma \to T'_\sigma \to \cdots \to T^*_\sigma$. Given the unique common sup-permutation π_σ of any subsets of $T_\sigma, T'_\sigma, \ldots, T^*_\sigma$, and that each lattice has paths of at most length $|\sigma| + 1$ such a sequence of unions of sublattices terminates after at most $|\sigma|$ iterations, yielding a single lattice $S^*_\sigma = \{\pi_\sigma \subseteq \xi \subseteq \sigma\}$, with minimal element $T^*_\sigma = \{\pi_\sigma\}$.

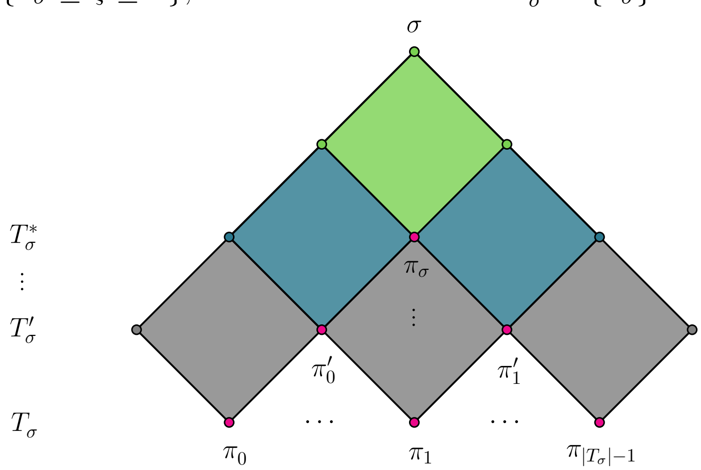


Figure 4.A.3: **Decomposition of sublattices**. Decomposition of $S_\sigma = \cup_{\pi\in\bar{T}_\sigma} S_{\sigma\pi}$ into intersecting sub-lattices $S_{\sigma\pi} = \{\pi \subseteq \xi \subseteq \sigma\}$. Gray, blue, green patches and nodes depict disjoint subsets of each sub-lattice, with pink nodes denoting the minimal permutation of each sublattice $\pi \in \bar{T}_\sigma = \cup\{T_\sigma, T'_\sigma, \ldots, T^*_\sigma\}$ with smallest common minimal permutation π_σ across all sublattices that satisfies $\pi \subseteq \pi_\sigma \subseteq \sigma$, such that $T^*_\sigma = \{\pi_\sigma\}$.

Summations over S_σ therefore take the form of summations over lattices, defined by $\bar{T}_\sigma = \{\xi\ :\ \exists\ \pi \in T_\sigma : \pi \subseteq \xi \subseteq \pi_\sigma\}$, weighted by some non-negative multiplicities and signs $\{n_\pi, k_\pi\}_{\pi\in\bar{T}_\sigma}$. Finally, given summations over $\phi_t(\sigma,\pi)$ are only non-zero when $\pi = \sigma$, summations over S_σ reduce to summations over the lattice $\{\pi_\sigma \subseteq \xi \subseteq \sigma\}$,

$$\sum_{\xi\in S_\sigma} \phi_t(\sigma,\xi) \ = \ \sum_{\pi\in\bar{T}_\sigma} (-1)^{k_\pi}\ n_\pi \sum_{\pi\subseteq\xi\subseteq\sigma} \phi_t(\sigma,\xi) \ = \ (-1)^{k_\sigma}\ n_\sigma\ \delta_{\pi_\sigma\sigma} \ \propto \ \delta_{\Gamma_P\Gamma_\sigma}\ . \tag{4.A.16}$$

Localized permutations $[\hat{\sigma}]$ thus have zero overlap with operator strings P with $\Gamma_P \subset \Gamma_\sigma$, meaning localized permutations are localized, with definitive support, $\Gamma_{[\hat{\sigma}]} = \Gamma_\sigma$,

$$[\hat{\sigma}] \ = \ \sum_{P\in\mathcal{P}_d^t} \chi_d^{(t)}([\hat{\sigma}],P)\ P \quad : \quad \chi_d^{(t)}([\hat{\sigma}],P) \ = \ (-1)^{k_\sigma}\ n_\sigma\ \frac{\chi_d^{(t)}(\sigma,P)}{\chi_d^{(t)}(\sigma)}\delta_{\Gamma_P\Gamma_\sigma} \tag{4.A.17}$$

$$\chi_d^{(t)}([\hat{\sigma}],[\hat{\pi}]) \ \propto \ \delta_{\Gamma_\sigma\Gamma_\pi}\ . \tag{4.A.18}$$

Intriguingly, such arguments could potentially hold for the localization of other change of bases $\{\hat{X}\} \to \{[\hat{X}]\}$ that are in terms of other partial orderings $Y \subseteq X$ between operators, their Möbius inverses ϕ_t, and their summations $\sum_{Y\in S_X} \phi_t(X,Y)$, given symmetries of overlaps with sub and sup-operators $\chi_d^{(t)}(\hat{Y},P) = \chi_d^{(t)}(\hat{X},P)$ for $Y \subseteq X,\ P \in \mathcal{P}_d^t$.

To prove the bounds on the overlap between localized permutations, we recall,

$$\chi_d^{(t)}([\hat{\sigma}],[\hat{\pi}]) \ = \ \sum_{\substack{\eta\subseteq\sigma\\ \kappa\subseteq\pi}} \phi_t(\sigma,\eta)\ \phi_t(\pi,\kappa)\frac{\chi_d^{(t)}(\eta,\kappa)}{\chi_d^{(t)}(\eta)\chi_d^{(t)}(\kappa)} \tag{4.A.19}$$

$$\frac{\chi_d^{(t)}(\sigma,\pi)}{\chi_d^{(t)}(\sigma)\chi_d^{(t)}(\pi)} \ = \ \sum_{\substack{\eta\subseteq\sigma\\ \kappa\subseteq\pi}} \chi_d^{(t)}([\hat{\eta}],[\hat{\kappa}])\ . \tag{4.A.20}$$

Given bounds on permutation sizes, $|\sigma|+|\pi|-|\sigma^{-1}\pi| \ \leq \ |\sigma|+|\pi|$, their overlaps are, $\chi_d^{(t)}(\sigma,\pi)/\chi_d^{(t)}(\sigma)\chi_d^{(t)}(\pi) \leq d^{|\sigma|+|\pi|}$, given the monotonicity of the Catalan numbers $c_l \leq c_{l+1}$, the Möbius functions are, $|c(\pi^{-1}\sigma)| \leq |c(\sigma)| \leq c_\sigma$, and given summations are bounded by the largest summand times the number of summands, $|\chi_d^{(t)}([\hat{\sigma}],[\hat{\pi}])| \leq c_\sigma^2\, c_\pi^2\, d^{|\sigma|+|\pi|}\,\delta_{\Gamma_\sigma\Gamma_\pi}$.

Finally, we note there is a trade-off in this choice of localized basis. The mapping from the localized permutations to the permutations has a simple, attractive closed form, with known enumerations of the expansions in terms of sub-permutations, with unit expansion coefficients. Permutations can equally be expanded in terms of a basis of strings of orthogonal operators, grouped by their support. However such groupings are not enumerated by the sub-permutations, and the associated expansion coefficients have no closed form, requiring the algorithms used in this proof. □

4.B Properties of Moment Operators

In these appendices, we consider properties of t-order moment operators for various ensembles, generalizing previous expressions [106], and adapt approaches for integrating over ensembles of operators to the case of ensembles of quantum channels [118, 231, 244].

4.B.1 Twirls and Moment Operators

Here, we denote the t-order moment operator for an ensemble of channels $\mathcal{C} \subseteq \mathcal{B}[\mathcal{B}[\mathcal{H}]]$ as $\widehat{\mathcal{T}}_{\mathcal{C}}^{(t)} = \int_{\mathcal{C}} d\Lambda\ \widehat{\Lambda^{\otimes t}}$. Then, the moment operator for k-concatenated independent ensembles $\{\mathcal{C}_i\}_{i\in[k]}$ is the product of the moment operators,

$$\widehat{\mathcal{T}}_{\{\mathcal{C}_i\}}^{(t)} = \prod_{i\in[k]} \widehat{\mathcal{T}}_{\mathcal{C}_i}^{(t)} = \prod_{i\in[k]} \int_{\mathcal{C}_i} d\Lambda_i\ \widehat{\Lambda_i^{\otimes t}} = \widehat{\mathcal{D}}_d^{\otimes t} + \widehat{\Delta}_{\{\mathcal{C}_i\}}^{(t)} , \tag{4.B.1}$$

where we defined,

$$\widehat{\mathcal{D}}_d^{\otimes t} = \frac{1}{d^t} |I_d^{\otimes t}\rangle\!\rangle\langle\!\langle I_d^{\otimes t}| \quad \text{and} \quad \widehat{\Delta}_{\{\mathcal{C}_i\}}^{(t)} = \frac{1}{d^t} \sum_{P\in\mathcal{S}_{\mathcal{C}_0}^{(t)}\setminus\{I_d^{\otimes t}\},\ S\in\mathcal{S}_{\mathcal{C}_{k-1}}^{(t)}} \tau_{\{\mathcal{C}_i\}}^{(t)}(P,S) |P\rangle\!\rangle\langle\!\langle S| \ . \tag{4.B.2}$$

Here, $\tau_{\{\mathcal{C}_i\}}^{(t)}$ denote the composite expansion coefficients, also known as the transfer matrix elements. These composite transfer matrices can be expanded in terms of the individual transfer matrices $\{\tau_{\Lambda_i}^{(t)}\}_{i\in[k]}$ for each moment operator, and overlaps $\chi_d^{(t)}$ between the elements of some ensemble-dependent bases $\{\mathcal{S}_{\mathcal{C}_i}^{(t)} \subseteq \mathcal{B}[\mathcal{H}^{\otimes t}]\}$ as,

$$\tau_{\{\mathcal{C}_i\}}^{(t)}(P,S) = \sum_{\substack{\{P_i,S_i\in\mathcal{S}_{\mathcal{C}_i}^{(t)}\}_{i\in[k]} \\ P_0=P,\ S_{k-1}=S}} \prod_{i\in[k]} \tau_{\mathcal{C}_i}^{(t)}(P_i,S_i) \prod_{i\in[k-1]} \chi_d^{(t)}(S_i,P_{i+1}) \ . \tag{4.B.3}$$

In the case of concatenating k identical ensembles, such that $\mathcal{C}_i = \mathcal{C}\ \forall i \in [k]$, we use k-powers, or k-superscripts, such that, $\widehat{\mathcal{T}}_{\{\mathcal{C}_i\}}^{(t)} \to \widehat{\mathcal{T}}_{\mathcal{C}}^{(t)k}$, $\widehat{\Delta}_{\{\mathcal{C}_i\}}^{(t)} \to \widehat{\Delta}_{\mathcal{C}}^{(t,k)}$, and $\tau_{\{\mathcal{C}_i\}}^{(t)} \to \tau_{\mathcal{C}}^{(t,k)}$.

To compare two ensembles $\mathcal{C}, \mathcal{C}'$, let us define the difference of their moment operators,

$$\widehat{\Delta}_{\mathcal{C}\mathcal{C}'}^{(t)} = \widehat{\mathcal{T}}_{\mathcal{C}}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{C}'}^{(t)} \qquad \to \qquad \epsilon_{\mathcal{C}\mathcal{C}'}^{(t)} = \left\| \widehat{\Delta}_{\mathcal{C}\mathcal{C}'}^{(t)} \right\| \quad , \quad \epsilon_{\mathcal{C}}^{(t)} = \left\| \widehat{\Delta}_{\mathcal{C}}^{(t)} \right\| . \tag{4.B.4}$$

Specific norms $\|\cdot\|_*$ correspond to $\epsilon^{(t|*)}$, and unless noted, we use the Hilbert-Schmidt norm.

Our choice of Hilbert-Schmidt norm $\|\cdot\| = \|\cdot\|_2$ for moment operators can be motivated, by its intuitive form that lends itself to simplified calculations, but also due to it bounding the more commonly used diamond norm $\|\cdot\|_\diamond$, such that, diamond norms of twirls can be bounded by Hilbert-Schmidt norms of their moment operators.

Lemma 1 (Moment Operator Hilbert-Schmidt Norms Bound Twirl Operator Diamond Norms)**.** *Given ensembles of quantum channels $\mathcal{C}, \mathcal{C}'$, the completely bounded diamond and Hilbert-Schmidt norm of their twirl operators are bounded by the Hilbert-Schmidt norm of their moment operators, up to a dimension-dependent scaling,*

$$\left\| \mathcal{T}^{(t)}_{\mathcal{C}} - \mathcal{T}^{(t)}_{\mathcal{C}'} \right\|_\diamond \;\leq\; d^t \left\| \widehat{\mathcal{T}}^{(t)}_{\mathcal{C}} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{C}'} \right\|_2 \;\leq\; d^t \left\| \widehat{\mathcal{T}}^{(t)}_{\mathcal{C}} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{C}'} \right\|_1 . \tag{4.B.5}$$

Proof. Given vectorizations, quantum channels $\Lambda \in \mathcal{B}[\mathcal{B}[\mathcal{H}]]$ can also be represented via channel-operator or even channel-state dualities, as superoperators $\hat{\Lambda}$, or Choi states Φ_Λ,

$$\hat{\Lambda} = \sum_{\Gamma_\Lambda} \Gamma_\Lambda \otimes \Gamma^*_\Lambda \qquad \leftrightarrow \qquad \Phi_\Lambda = \sum_{\Gamma_\Lambda} |\Gamma_\Lambda\rangle\!\rangle\langle\!\langle\Gamma^*_\Lambda| \ , \tag{4.B.6}$$

In fact quantum channels $\Lambda, \Lambda' \in \mathcal{B}[\mathcal{B}[\mathcal{H}]]$, inner products between their superoperators equal inner products between their Choi states [126],

$$\chi\left(\hat{\Lambda}', \hat{\Lambda}\right) = \mathrm{tr}\left(\hat{\Lambda}'^\dagger \hat{\Lambda}\right) = \mathrm{tr}\left(\Phi^\dagger_{\Lambda'} \Phi_\Lambda\right) = \chi\left(\Phi_{\Lambda'}, \Phi_\Lambda\right) \ . \tag{4.B.7}$$

Finally, given its convexity, the completely-bounded 1-norm, or diamond $\diamond$ norm,

$$\|\Lambda\|_\diamond = \max_{|\psi\rangle \in \mathcal{H}\otimes\mathcal{H}} \|(\Lambda \otimes I_d)(\psi)\|_1 \ , \tag{4.B.8}$$

is the $p = 1$-norm of the operator acting on some normalized pure state $|\psi\rangle = (1/d) U \otimes I_d |\Omega\rangle$ on the doubled space $\mathcal{H} \otimes \mathcal{H}$ for some unitary $U \in \mathcal{U}(d)$ [176]. Further, the p-norm obeys Holder's inequality of $\|\cdot\cdot\|_r \leq \|\cdot\|_p \|\cdot\|_q$ for $1/p + 1/q = 1/r$, the 2-norm obeys $\|\cdot\|_1 \leq \sqrt{d}\|\cdot\|_2$, and unitaries have ∞-norms of the 1-norm one of their columns u, bounded by $\|U\|_\infty = \|u\|_1 \leq \sqrt{d}$. The completely bounded 1-norm of the difference of quantum channels Λ, Λ' can thus be bounded by the 2-norm of their associated Choi states [176], and thus by the 2 and 1-norms of their superoperators,

$$\begin{aligned}
\|\Lambda - \Lambda'\|_\diamond &= \|(\Lambda \otimes I_d)(\psi) - (\Lambda' \otimes I_d)(\psi)\|_1 && \text{(4.B.9)}\\
&= \frac{1}{d}\ \|(\Lambda \otimes I_d)((\mathcal{U} \otimes I_d)(\Omega)) - (\Lambda' \otimes I_d)((\mathcal{U} \otimes I_d)(\Omega))\|_1 && \text{(4.B.10)}\\
&= \frac{1}{d}\ \left\|(I_d \otimes U^*)\left(\Phi_\Lambda - \Phi_{\Lambda'}\right)\left(I_d \otimes U^T\right)\right\|_1 && \text{(4.B.11)}\\
&\leq \frac{1}{d}\ \|I_d \otimes U^*\|_\infty \left\|I_d \otimes U^T\right\|_\infty \|\Phi_\Lambda - \Phi_{\Lambda'}\|_1 && \text{(4.B.12)}\\
&\leq \frac{1}{d}\ \sqrt{d}\ \sqrt{d}\ \|\Phi_\Lambda - \Phi_{\Lambda'}\|_1 && \text{(4.B.13)}\\
&= \|\Phi_\Lambda - \Phi_{\Lambda'}\|_1 && \text{(4.B.14)}\\
&\leq d\ \|\Phi_\Lambda - \Phi_{\Lambda'}\|_2 && \text{(4.B.15)}\\
&= d\ \left\|\widehat{\Lambda} - \widehat{\Lambda}'\right\|_2 && \text{(4.B.16)}\\
&\leq d\ \left\|\widehat{\Lambda} - \widehat{\Lambda}'\right\|_1\ . && \text{(4.B.17)}
\end{aligned}$$

□

Further, when ensembles contain strictly unital operators, we can prove how the associated moment operators preserve the support of their inputs.

Definition 1 (Projector Ensembles)**.** *Projector ensembles $\mathcal{C}$ have self-adjoint projector moment operators:*

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}}\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{C}} \qquad , \qquad \widehat{\mathcal{T}}^{(t)\dagger}_{\mathcal{C}} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{C}}\ . \tag{4.B.18}$$

Lemma 2 (Support Preservation of Unital Ensemble Twirls)**.** *Given an orthogonal with respect to support basis $\mathcal{P}_d^t$ for $\mathcal{B}[\mathcal{H}^{\otimes t}]$, unital ensemble twirls $\mathcal{C}_\mathcal{U}$ are support-non-increasing and unital projector ensemble twirls $\mathcal{C}_\mathcal{U}$ are support-preserving,*

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}_\mathcal{U}} = \frac{1}{d^t} \sum_{\substack{P,S\in\mathcal{P}_d^t\\ \Gamma_P\subseteq\Gamma_S}} \tau^{(|S|)}_{\mathcal{C}_\mathcal{U}}(P,S)\ |P\rangle\!\rangle\langle\!\langle S| \qquad \leftrightarrow \qquad \tau^{(t)}_{\mathcal{C}_\mathcal{U}}(P,S) = \tau^{(|S|)}_{\mathcal{C}_\mathcal{U}}(P,S)\ \delta_{\Gamma_P\subseteq\Gamma_S} \tag{4.B.19}$$

$$\begin{gathered}
\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}_\mathcal{U}}\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}_\mathcal{U}} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{C}_\mathcal{U}} \quad , \quad \widehat{\mathcal{T}}^{(t)\dagger}_{\mathcal{C}_\mathcal{U}} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{C}_\mathcal{U}}\\
\rightarrow\\
\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}_\mathcal{U}} = \sum_{\Gamma\subseteq[t]} \widehat{\mathcal{T}}^{(\Gamma)}_{\mathcal{C}_\mathcal{U}} \quad : \quad \widehat{\mathcal{T}}^{(\Gamma)}_{\mathcal{C}_\mathcal{U}} = \frac{1}{d^t} \sum_{\substack{P,S\in\mathcal{P}_d^t\\ \Gamma_P=\Gamma_S=\Gamma}} \tau^{(|S|)}_{\mathcal{C}_\mathcal{U}}(P,S)\ |P\rangle\!\rangle\langle\!\langle S|\ .
\end{gathered} \tag{4.B.20}$$

Proof. Let us consider consider the action of unital ensemble $\mathcal{C}_{\mathcal{U}}$ twirls on $|X|$-local operators $X = \otimes_{i\in\Gamma_X} X_i$, given identity operators are invariant under unital operators,

$$\mathcal{T}^{(t)}_{\mathcal{C}_{\mathcal{U}}}(X) = \int_{\mathcal{C}_{\mathcal{U}}} d\Lambda\ \Lambda^{\otimes t}(X)\ = \int_{\mathcal{C}_{\mathcal{U}}} d\Lambda\ \otimes_{i\in\Gamma_X} \Lambda_i(X) \tag{4.B.21}$$

$$= \sum_{\substack{P,S\in\mathcal{P}^{|X|}_d \\ \Gamma_P\subseteq\Gamma_S=\Gamma_X}} \tau^{(|X|)}_{\mathcal{C}_{\mathcal{U}}}(P,S)\ \chi^{(|X|)}_d(S,X)\ P\ = \mathcal{T}^{(|X|)}_{\mathcal{C}_{\mathcal{U}}}(X)\ . \tag{4.B.22}$$

Unital t-order transfer matrices thus reduce to $|S|$-order, that are non-zero $\tau^{(t)}_{\mathcal{C}_{\mathcal{U}}}(P,S) = \tau^{(|S|)}_{\mathcal{C}_{\mathcal{U}}}(P,S) \sim \delta_{\Gamma_P\subseteq\Gamma_S}$ for $P,S\in\mathcal{P}^t_d$ when mapping between basis operators of strictly common support. In the case of projector ensembles $\mathcal{C}_{\mathcal{U}}$, moment operators are self-adjoint, $\widehat{\mathcal{T}}^{(t)\dagger}_{\mathcal{C}_{\mathcal{U}}} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{C}_{\mathcal{U}}}$, constraining such twirls to map between operators of identical support. Projector ensembles can thus be written as a sum of projectors onto $\Gamma\subseteq[t]$-support spaces. □

Further, we note that via Stinespring dilations, an ensemble of unitaries induces an ensemble of quantum channels.

Definition 2 (Unitary Induced Quantum Channel Ensembles)**.** *Unitary ensembles $\mathcal{U}$ over a composite space $\mathcal{H}\otimes\mathcal{E}$, and an environment state $\nu_{\mathcal{E}}\in\mathcal{B}[\mathcal{E}]$, induce a quantum channel ensemble $\mathcal{C}$ over the system space $\mathcal{H}$, with moment operators,*

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}} = I^{\otimes t}_d \otimes \langle\!\langle I^{\otimes t}_s|\ \ \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}}\ \ I^{\otimes t}_d \otimes |\nu^{\otimes t}_{\mathcal{E}}\rangle\!\rangle\ . \tag{4.B.23}$$

Finally, moment operators of channel ensembles induced by projector unitary ensembles are support-non-decreasing.

Theorem 8 (Support Preservation of Quantum Channel Ensembles Induced by Unitary Ensembles)**.** *Moment operators of channel ensembles $\mathcal{C}$ over a d-dimensional space $\mathcal{H}$, induced via Stinespring dilations by projector unitary ensembles $\mathcal{U}$ over a composite ds-dimensional space $\mathcal{H}\otimes\mathcal{E}$, with an environment state $\nu_{\mathcal{E}}\in\mathcal{B}[\mathcal{E}]$, are support-non-decreasing,*

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}} = \frac{1}{d^t}\sum_{\substack{P,S\in\mathcal{P}^t_d \\ \Gamma_P\supseteq\Gamma_S}} \tau^{(t)}_{\mathcal{C}}(P,S)\ |P\rangle\!\rangle\langle\!\langle S| \tag{4.B.24}$$

$$\tau^{(t)}_{\mathcal{C}}(P,S) = \sum_{\substack{T_{\mathcal{E}}\in\mathcal{P}^t_s \\ \Gamma_{P_{\mathcal{H}}}=\Gamma_{S_{\mathcal{H}}}\cup\Gamma_{T_{\mathcal{E}}}}} \langle\!\langle T_{\mathcal{E}}|\nu^{\otimes t}_{\mathcal{E}}\rangle\!\rangle\ \tau^{(t)}_{\mathcal{U}}(P_{\mathcal{H}}\otimes I^{\otimes t}_s, S_{\mathcal{H}}\otimes T_{\mathcal{E}})\ , \tag{4.B.25}$$

given orthogonal with respect to support bases $\mathcal{P}^t_d, \mathcal{P}^t_s$ for $\mathcal{H}$ and $\mathcal{E}$.

Proof. The induced channel ensemble moment operators follow from their definition. □

4.B.2 Properties of Haar and cHaar Twirls

Here, we derive expressions and properties for moment operators for the Haar, cHaar, and Depolarize ensembles. Throughout this section, we will use the fact that the t-order moment operator over the representation of a group projects onto its t-order commutant. We begin with the following definitions of the relevant ensembles.

Definition 3 (Haar ensemble)**.** *The t-order twirl over the Haar ensemble $\mathcal{U}(d)$ is defined as the twirl over t copies of unitaries sampled from the group $\mathcal{U}(d)$ according to the Haar measure. The associated twirl and moment operator are*

$$\mathcal{T}^{(t)}_{\mathcal{U}(d)}(\cdot) \;=\; \int_{\mathcal{U}(d)} dU\; U^{\otimes t} \;\cdot\; U^{\otimes t\dagger} \qquad \rightarrow \qquad \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)} \;=\; \int_{\mathcal{U}(d)} dU\; U^{\otimes t} \otimes U^{\otimes t *}\;. \tag{4.B.26}$$

Since the t-order commutant of the unitary group corresponds to the system-permuting representation of the Symmetric group we can always express the moment operator as

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)} = \frac{1}{d^t}\sum_{\sigma,\pi\in\mathcal{S}_t} \tau^{(t)}_{\mathcal{U}(d)}(\sigma,\pi)\; |\sigma\rangle\!\rangle\langle\!\langle\pi| \qquad : \qquad \tau^{(t)}_{\mathcal{U}(d)}(\sigma,\pi) = \chi^{(t)-1}_{d}(\sigma,\pi)\;. \tag{4.B.27}$$

Definition 4 (cHaar ensemble)**.** *The t-order twirl over the channel-Haar (cHaar) ensemble $\mathcal{C}(d,s)$ is defined as a measure over t copies of random channels, and is a uniform, Lebesgue measure when $s = d^2$. Using the Stinespring formalism, this average is a twirl with respect to the Haar ensemble of unitaries acting on t copies of a composite space, of a system and environment $\mathcal{H}\otimes\mathcal{E}$ of dimensions d, s, and a local pure environment state $\nu\in\mathcal{B}[\mathcal{E}]$,*

$$\mathcal{T}^{(t)}_{\mathcal{C}(d,s)}(\cdot) = \mathrm{tr}_{\mathcal{E}^{\otimes t}}\Big(\mathcal{T}^{(t)}_{\mathcal{U}(ds)}(\cdot\;\otimes\;\nu^{\otimes t})\Big) \;\rightarrow\; \widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)} = I_d^{\otimes t}\otimes\langle\!\langle I_s^{\otimes t}|\;\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(ds)}\; I_d^{\otimes t}\otimes|\nu^{\otimes t}\rangle\!\rangle\;. \tag{4.B.28}$$

Since the cHaar ensemble is generated by the unitary Haar ensemble over $\mathcal{U}(ds)$, $\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)}$ can be obtained by projecting onto the commutant of $\mathcal{U}(ds)$, acting on the environment state ν, and then tracing out the environment,

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)} = \frac{1}{d^t}\sum_{\sigma,\pi\in\mathcal{S}_t} \tau^{(t)}_{\mathcal{C}(d,s)}(\sigma,\pi)\; |\sigma\rangle\!\rangle\langle\!\langle\pi| \qquad : \qquad \tau^{(t)}_{\mathcal{C}(d,s)}(\sigma,\pi) = \chi^{(t)}_{s}(\sigma)\;\tau^{(t)}_{\mathcal{U}(ds)}(\sigma,\pi)\;. \tag{4.B.29}$$

Definition 5 (Depolarize ensemble)**.** *The t-order twirl over the depolarizing (Depolarize) ensemble $\mathcal{D}(d)$ is defined via a single maximally depolarizing channel $\mathcal{D}_d(\cdot)$, with unit probability. The associated twirl and moment operator are*

$$\mathcal{T}^{(t)}_{\mathcal{D}(d)}(\cdot) = \mathcal{D}^{\otimes t}_{d}(\cdot) = \frac{\mathrm{tr}(\cdot)}{d^t} I_d^{\otimes t} \qquad \rightarrow \qquad \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} = \widehat{\mathcal{D}}^{\otimes t}_{d} = \frac{1}{d^t}|I_d^{\otimes t}\rangle\!\rangle\langle\!\langle I_d^{\otimes t}|\;. \tag{4.B.30}$$

The Depolarize moment operator projects onto an invariant, trace-preserving basis of $\{I_d^{\otimes t}\}$.

We now derive explicit forms of the Haar, cHaar, and Depolarize moment operators.

Theorem 9 (Haar Moment Operators are Support-Preserving and Block-Diagonal in Localized Basis)**.** *The Haar moment operator is block-diagonal in the localized permutation basis, as per its support,*

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)} = \begin{bmatrix} \widehat{\mathcal{T}}^{(\{\})}_{\mathcal{U}(d)} & & & \\ & \widehat{\mathcal{T}}^{(\{i,j\})}_{\mathcal{U}(d)} & & \\ & & \ddots & \\ & & & \widehat{\mathcal{T}}^{([t])}_{\mathcal{U}(d)} \end{bmatrix} \tag{4.B.31}$$

$$\widehat{\mathcal{T}}^{(\Gamma)}_{\mathcal{U}(d)} = \frac{1}{d^t} \sum_{\substack{\sigma,\pi\in\mathcal{S}_t \\ \Gamma_\sigma=\Gamma_\pi=\Gamma \\ l=|\Gamma|}} \tau^{(l)}_{\mathcal{U}(d)}([\hat{\sigma}],[\hat{\pi}]) \; |[\hat{\sigma}]\rangle\!\rangle\langle\!\langle[\hat{\pi}]| \tag{4.B.32}$$

$$\tau^{(l)}_{\mathcal{U}(d)}([\hat{\sigma}],[\hat{\pi}]) = \sum_{\substack{\sigma\subseteq\eta\in\mathcal{S}_l \\ \pi\subseteq\kappa\in\mathcal{S}_l}} \chi^{(l)}_d(\eta)\; \chi^{(l)-1}_d(\eta,\kappa)\; \chi^{(l)}_d(\kappa)\ , \tag{4.B.33}$$

with orthogonal support-dependent block moment operators $\widehat{\mathcal{T}}^{(\Gamma)}_{\mathcal{U}(d)}$*, spanned by localized permutations* $[\hat{\sigma}],[\hat{\pi}]\in[\hat{\mathcal{S}}_t]$ *with support* $\Gamma_{[\hat{\sigma}]}=\Gamma_{[\hat{\pi}]}=\Gamma \;:\; \Gamma\subseteq[t]$.

Theorem 10 (cHaar Moment Operators are Support-Non-Decreasing and Block-Lower–Triangular in Localized Basis)**.** *The cHaar moment operator is block-lower-triangular in the localized permutation basis, as per its support,*

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)} = \begin{bmatrix} \widehat{\mathcal{T}}^{(\{\})}_{\mathcal{C}(d,s)} & & & & \\ \widehat{\mathcal{T}}^{(\{\},\{i,j\})}_{\mathcal{C}(d,s)} & \widehat{\mathcal{T}}^{(\{i,j\})}_{\mathcal{C}(d,s)} & & & \\ \widehat{\mathcal{T}}^{(\{\},\{i,j,k\})}_{\mathcal{C}(d,s)} & \cdots & \widehat{\mathcal{T}}^{(\{i,j,k\})}_{\mathcal{C}(d,s)} & & \\ \vdots & \ddots & \widehat{\mathcal{T}}^{(\Gamma,\Gamma')}_{\mathcal{C}(d,s)} & \ddots & \\ \widehat{\mathcal{T}}^{(\{\},[t])}_{\mathcal{C}(d,s)} & \cdots & \cdots & \cdots & \widehat{\mathcal{T}}^{([t])}_{\mathcal{C}(d,s)} \end{bmatrix} \tag{4.B.34}$$

$$\widehat{\mathcal{T}}^{(\Gamma,\Gamma')}_{\mathcal{C}(d,s)} = \frac{1}{d^t} \sum_{\substack{\sigma,\pi\in\mathcal{S}_t \\ \Gamma_\sigma=\Gamma\supseteq\Gamma_\pi=\Gamma' \\ l=|\Gamma|\geq l'=|\Gamma'|}} \tau^{(l)}_{\mathcal{C}(d,s)}([\hat{\sigma}],[\hat{\pi}]) \; |[\hat{\sigma}]\rangle\!\rangle\langle\!\langle[\hat{\pi}]| \tag{4.B.35}$$

$$\tau^{(l)}_{\mathcal{C}(d,s)}([\hat{\sigma}],[\hat{\pi}]) = \sum_{\substack{\sigma\subseteq\eta\in\mathcal{S}_l \\ \pi\subseteq\kappa\in\mathcal{S}_l}} \chi^{(l)}_{ds}(\eta)\; \chi^{(l)-1}_{ds}(\eta,\kappa)\; \chi^{(l)}_d(\kappa)\ , \tag{4.B.36}$$

with orthogonal support-dependent block moment operators $\widehat{\mathcal{T}}^{(\Gamma,\Gamma')}_{\mathcal{C}(d,s)}$*, spanned by localized permutations* $[\hat{\sigma}],[\hat{\pi}]\in[\hat{\mathcal{S}}_t]$ *with support* $\Gamma_{[\hat{\sigma}]}=\Gamma\supseteq\Gamma'=\Gamma_{[\hat{\pi}]} \;:\; \Gamma,\Gamma'\subseteq[t]$.

The Haar and cHaar moment operators have exact identity and transposition coefficients,

$$\tau^{(t)}_{\mathcal{U}(d)}([\hat{\sigma}],[\hat{e}]) = \delta_{\sigma e} \qquad , \qquad \tau^{(t)}_{\mathcal{U}(d)}([\hat{\tau}],[\hat{\tau}']) = \frac{1}{d^2-1}\delta_{\tau\tau'} \ ,$$

$$\tau^{(t)}_{\mathcal{C}(d,s)}([\hat{e}],[\hat{e}]) = 1 \quad , \quad \tau^{(t)}_{\mathcal{C}(d,s)}([\hat{\tau}],[\hat{e}]) = \frac{s-1}{d^2s^2-1} \quad , \quad \tau^{(t)}_{\mathcal{C}(d,s)}([\hat{\tau}],[\hat{\tau}']) = \frac{s}{d^2s^2-1}\delta_{\tau\tau'} \ . \tag{4.B.37}$$

Proof. Let us express the moment operators in terms of the orthogonal with respect to support localized permutation basis $[\hat{\mathcal{S}}_t]$. In particular, cHaar ensemble moment operators are generated by support-preserving Haar ensemble moment operators in the composite system and environment space,

$$\begin{aligned}\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(ds)} &= \frac{1}{d^ts^t}\sum_{\sigma,\pi\in\mathcal{S}_t}\tau^{(t)}_{\mathcal{U}(ds)}(\sigma,\pi)\ |\sigma_{\mathcal{H}}\sigma_{\mathcal{E}}\rangle\!\rangle\langle\!\langle\pi_{\mathcal{H}}\pi_{\mathcal{E}}| \\ &= \frac{1}{d^ts^t}\sum_{\substack{\eta,\eta'\in\mathcal{S}_t\\ \kappa,\kappa'\in\mathcal{S}_t\\ \Gamma_\eta\cup\Gamma_{\eta'}=\Gamma_\kappa\cup\Gamma_{\kappa'}\\ \Gamma=\Gamma_\eta\cup\Gamma_{\eta'}\ ,\ l=|\Gamma|}}\ \sum_{\substack{\eta,\eta'\subseteq\sigma\in\mathcal{S}_l\\ \kappa,\kappa'\subseteq\pi\in\mathcal{S}_l}}\chi^{(l)}_{ds}(\sigma)\ \tau^{(l)}_{\mathcal{U}(ds)}(\sigma,\pi)\ \chi^{(l)}_{ds}(\pi)\ |[\hat{\eta}_{\mathcal{H}}][\hat{\eta}'_{\mathcal{E}}]\rangle\!\rangle\langle\!\langle[\hat{\kappa}_{\mathcal{H}}][\hat{\kappa}'_{\mathcal{E}}]|\,.\end{aligned} \tag{4.B.38}$$

In particular, we note that the support of the input permutations κ,κ' can be changed from $\Gamma_\eta\cup\Gamma_{\eta'}=\Gamma_\kappa\cup\Gamma_{\kappa'} \ \rightarrow\ \Gamma_\eta\cup\Gamma_{\eta'}\supseteq\Gamma_\kappa\cup\Gamma_{\kappa'}$, as any coefficients arising from summations over $\pi\supseteq\kappa,\kappa'$ will cancel to zero for $\Gamma_\kappa\cup\Gamma_{\kappa'}\subset\Gamma_\eta\cup\Gamma_{\eta'}$. Hence, we can write the Haar ensemble moment operator in the composite system and environment space as,

$$\begin{aligned}\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(ds)} &= \frac{1}{d^ts^t}\sum_{\substack{\eta,\eta'\in\mathcal{S}_t\\ \eta,\eta'\subseteq\sigma\in\mathcal{S}_l\\ \Gamma=\Gamma_\eta\cup\Gamma_{\eta'}\ ,\ l=|\Gamma|}}\ \sum_{\substack{\kappa,\kappa'\in\mathcal{S}_l\\ \kappa,\kappa'\subseteq\pi\in\mathcal{S}_l\\ ,\ \Gamma_\eta\cup\Gamma_{\eta'}\supseteq\Gamma_\kappa\cup\Gamma_{\kappa'}}}\chi^{(l)}_{ds}(\sigma)\ \tau^{(l)}_{\mathcal{U}(ds)}(\sigma,\pi)\ \chi^{(l)}_{ds}(\pi)\ |[\hat{\eta}_{\mathcal{H}}][\hat{\eta}'_{\mathcal{E}}]\rangle\!\rangle\langle\!\langle[\hat{\kappa}_{\mathcal{H}}][\hat{\kappa}'_{\mathcal{E}}]| \\ &= \frac{1}{d^ts^t}\sum_{\substack{\eta,\eta'\in\mathcal{S}_t\\ \eta,\eta'\subseteq\sigma\in\mathcal{S}_l\\ \Gamma=\Gamma_\eta\cup\Gamma_{\eta'}=\Gamma_\sigma\ ,\ l=|\Gamma|}}\ \sum_{\substack{\pi\in\mathcal{S}_l\\ \Gamma_\pi\subseteq\Gamma}}\chi^{(l)}_{ds}(\sigma)\ \tau^{(l)}_{\mathcal{U}(ds)}(\sigma,\pi)\ |[\hat{\eta}_{\mathcal{H}}][\hat{\eta}'_{\mathcal{E}}]\rangle\!\rangle\langle\!\langle\pi_{\mathcal{H}}\pi_{\mathcal{E}}|\ .\end{aligned} \tag{4.B.39}$$

From here, we can show that the cHaar ensemble moment is support-non-decreasing

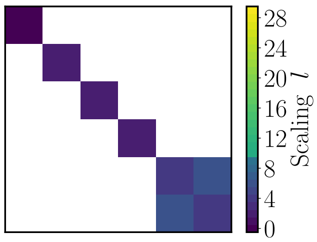
$$
\begin{aligned}
\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)} &= I_d^{\otimes t} \otimes \langle\!\langle I_s^{\otimes t}| \;\; \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(ds)} \;\; I_d^{\otimes t} \otimes |\nu^{\otimes t}\rangle\!\rangle && (4.B.40)\\
&= \frac{1}{d^t s^t} \sum_{\substack{\eta,\eta'\in\mathcal{S}_t\\ \eta,\eta'\subseteq\sigma\in\mathcal{S}_l\\ \Gamma=\Gamma_\eta\cup\Gamma_{\eta'}\ ,\ l=|\Gamma|}} \sum_{\substack{\pi\in\mathcal{S}_l\\ \Gamma_\pi\subseteq\Gamma}} \chi^{(l)}_{ds}(\sigma)\ \tau^{(l)}_{\mathcal{U}(ds)}(\sigma,\pi)\ \langle\!\langle I_s^{\otimes t}|[\hat{\eta}'_{\mathcal{E}}]\rangle\!\rangle\ \langle\!\langle \pi_{\mathcal{E}}|\nu^{\otimes t}\rangle\!\rangle\ |[\hat{\eta}_{\mathcal{H}}]\rangle\!\rangle\langle\!\langle \pi_{\mathcal{H}}| && (4.B.41)\\
&= \frac{1}{d^t} \sum_{\substack{\eta\in\mathcal{S}_t\\ \eta\subseteq\sigma\in\mathcal{S}_l\\ \Gamma=\Gamma_\eta\ ,\ l=|\Gamma|}} \sum_{\substack{\pi\in\mathcal{S}_l\\ \Gamma_\pi\subseteq\Gamma}} \chi^{(l)}_{ds}(\sigma)\ \tau^{(l)}_{\mathcal{U}(ds)}(\sigma,\pi)\ |[\hat{\eta}_{\mathcal{H}}]\rangle\!\rangle\langle\!\langle \pi_{\mathcal{H}}| && (4.B.42)\\
&= \frac{1}{d^t} \sum_{\substack{\eta\in\mathcal{S}_t\\ \eta\subseteq\sigma\in\mathcal{S}_l\\ \Gamma=\Gamma_\eta\ ,\ l=|\Gamma|}} \sum_{\substack{\pi\in\mathcal{S}_l\\ \kappa\subseteq\pi\in\mathcal{S}_l\\ \Gamma_\eta\subseteq\Gamma}} \chi^{(l)}_{ds}(\sigma)\ \tau^{(l)}_{\mathcal{U}(ds)}(\sigma,\pi)\ \chi^{(l)}_{d}(\pi)\ |[\hat{\eta}_{\mathcal{H}}]\rangle\!\rangle\langle\!\langle [\hat{\kappa}_{\mathcal{H}}]| && (4.B.43)\\
&= \frac{1}{d^t} \sum_{\substack{\sigma,\pi\in\mathcal{S}_t\\ \Gamma=\Gamma_\sigma\supseteq\Gamma_\pi\ ,\ l=|\Gamma|}} \tau^{(l)}_{\mathcal{C}(d,s)}([\hat{\sigma}],[\hat{\pi}])\ |[\hat{\sigma}]\rangle\!\rangle\langle\!\langle [\hat{\pi}]| && (4.B.44)
\end{aligned}
$$

$$
\tau^{(l)}_{\mathcal{C}(d,s)}([\hat{\sigma}],[\hat{\pi}]) = \sum_{\substack{\sigma\subseteq\eta\in\mathcal{S}_l\\ \pi\subseteq\kappa\in\mathcal{S}_l}} \chi^{(l)}_{ds}(\eta)\ \tau^{(l)}_{\mathcal{U}(ds)}(\eta,\kappa)\ \chi^{(l)}_{d}(\kappa)\ . \qquad (4.B.45)
$$

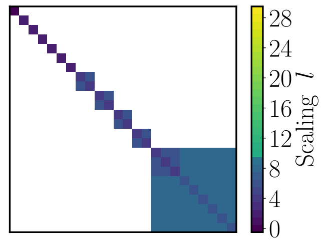
□

To illustrate these properties, we compute the exact $t = 3, 4, 5$-order Haar and cHaar transfer matrices, for $s = d^2$ [332], as shown in Fig. 4.B.4. Here, we can see that a sparse, support-dependent block-diagonal or block-lower-triangular structure occurs. Within the transfer matrices, many of the $\binom{t}{l}$ locality-l blocks, of size $!l$, are identical, and correspond to $l \leq t$-order twirl blocks, due to isomorphisms between permutations. Such structure is primarily attributed to the localized permutation basis being orthogonal with respect to support. For the Haar ensemble, the Haar invariance enforces the support-preserving behaviour, and for the cHaar ensemble, the environment space partial trace enforces the support increasing behaviour. From these results, and inspired by similar discussions in [117], we provide intriguing identities for permutations $\sigma, \pi \in \mathcal{S}_t$ of support $\Gamma = \Gamma_\sigma \supseteq \Gamma_\pi$ and locality $l = |\Gamma|$, and general dimensions d, s,

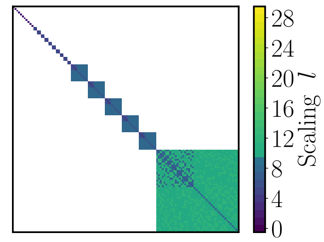
$$
\sum_{\substack{\eta,\kappa\in\mathcal{S}_t\\ \eta\supseteq\sigma,\ \kappa\supseteq\pi}} \chi^{(t)}_{ds}(\eta)\ \chi^{(t)-1}_{ds}(\eta,\kappa)\ \chi^{(t)}_{d}(\kappa) = \sum_{\substack{\eta,\kappa\in\mathcal{S}_l\\ \eta\supseteq\sigma,\ \kappa\supseteq\pi}} \chi^{(l)}_{ds}(\eta)\ \chi^{(l)-1}_{ds}(\eta,\kappa)\ \chi^{(l)}_{d}(\kappa)\ \delta_{\substack{\Gamma_\sigma\supseteq\Gamma_\pi,s>1\\ \text{or}\\ \Gamma_\sigma=\Gamma_\pi,s=1}}\ . \qquad (4.B.46)
$$



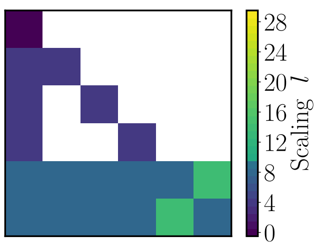
(a) $\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)}$ for $t = 3$



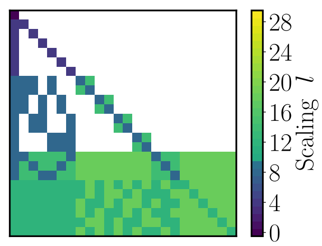
(b) $\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)}$ for $t = 4$



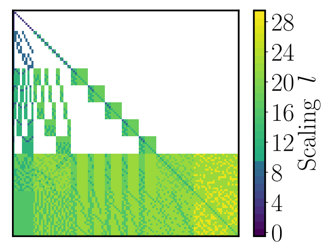
(c) $\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)}$ for $t = 5$



(d) $\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)}$ for $t = 3$



(e) $\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)}$ for $t = 4$



(f) $\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)}$ for $t = 5$

Figure 4.B.4: **Haar and cHaar transfer matrices**. Scaling of t-order moment operator transfer matrix elements for the Haar (top) and cHaar (bottom) ensembles in the localized permutation basis, for $t = 3, 4, 5$ and $s = d^2$. Depicted are matrix elements corresponding to all pairs of $t!$ permutations $\sigma, \pi \in \mathcal{S}_t$, sorted by their support, from the smallest identity (top-left corner), to largest cycles (bottom-right corner), with leading-order scaling l (coloured gradient), such that, $\tau^{(t)}_{\mathcal{U}(d)}([\hat{\sigma}], [\hat{\pi}]), \tau^{(t)}_{\mathcal{C}(d,s)}([\hat{\sigma}], [\hat{\pi}]) \sim \mathcal{O}(1/d^l)$. Sparse, support-dependent block-diagonal (Haar) or block-lower-triangular (cHaar) coefficients occur, and transposition coefficients (light-purple top-left diagonal) are uniquely diagonal $1/(d^2 - 1)$ (Haar), or diagonal and non-unital $(s - \delta_{\pi e})/(d^2 s^2 - 1)$ (cHaar).

Next, we turn our attention to computing the exact $t = 1, 2$-order moment operator for the k-concatenated cHaar ensemble. In particular, $\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)}$ is not a projector, $\widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)} \neq \widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)}$, and we are interested in its convergence properties with k. A straightforward calculation of the exact moment operators and their transfer matrices leads to the following lemma.

Lemma 3 (Exact $t = 1, 2$-order cHaar Moment Operators)**.** *k-concatenated $t = 1, 2$-order cHaar moment operators, for all dimensions d, s, are,*

$$\widehat{\mathcal{T}}^{(1)k}_{\mathcal{C}(d,s)} = \frac{1}{d}|[\hat{e}]\rangle\!\rangle\langle\!\langle[\hat{e}]| \tag{4.B.47}$$

$$\begin{aligned}\widehat{\mathcal{T}}^{(2)k}_{\mathcal{C}(d,s)} =\ & \frac{1}{d^2}|[\hat{e}]\rangle\!\rangle\langle\!\langle[\hat{e}]| \\ & + \frac{1}{d^2}\frac{s-1}{d^2s^2-1}\left[1+\sum_{j>0}^{k-1}\left[s\frac{d^2-1}{d^2s^2-1}\right]^j\right]|[\hat{\tau}]\rangle\!\rangle\langle\!\langle[\hat{e}]| \\ & + \frac{1}{d^2}\frac{1}{d^2-1}\left[s\frac{d^2-1}{d^2s^2-1}\right]^k|[\hat{\tau}]\rangle\!\rangle\langle\!\langle[\hat{\tau}]|\ ,\end{aligned} \tag{4.B.48}$$

where $[\hat{e}] = I_d^{\otimes t}$ is the identity operator, and $[\hat{\tau}] = dS_d - I_d^{\otimes 2} = \sum_{P\in\mathcal{P}_d\setminus\{I_d\}} P \otimes P^\dagger$ is the localized transposition operator.

From the previous results for individual transfer matrix elements, and overlaps between basis operators, we can bound the transfer matrix elements for general t-order moment operators for the k-concatenated cHaar ensemble.

Theorem 11 (Asymptotic t-order cHaar Moment Operators)**.** *k-concatenated t-order cHaar moment operators, in the asymptotic limit $ds \to \infty$, have transfer matrix elements upper bounded by,*

$$\left|\tau^{(l,k)}_{\mathcal{C}(d,s)}([\hat{\sigma}],[\hat{\pi}])\right| \ \leq\ \frac{1}{d^{|\sigma|+|\pi|}s^{|\sigma|+(k-1)|\Gamma_\pi|/2}}\ C^{(l,k)}_{\tau}\ \delta_{\Gamma_\sigma\supseteq\Gamma_\pi}\ , \tag{4.B.49}$$

for $\sigma, \pi \in \mathcal{S}_l$, $l \in [t+1]$, and t, k-dependent constants $C^{(l,k)}_{\tau} = \frac{1}{l!^2\ c_l^4}\ C^{(l)k}_{\lambda}$, $C^{(l)}_{\lambda} = l!^2\ \chi_l\ c_l^7$.

Proof. By definition, for $\sigma,\pi \in \mathcal{S}_l$ we have bounds for the individual transfer matrix elements and overlaps of,

$$\tau^{(l)}_{\mathcal{C}(d,s)}([\hat\sigma],[\hat\pi]) \;=\; \sum_{\sigma\subseteq\eta\;,\;\pi\subseteq\kappa} \chi^{(l)}_{ds}(\eta)\;\chi^{(l)-1}_{ds}(\eta,\kappa)\;\chi^{(l)}_{d}(\kappa)\;\delta_{\Gamma_\sigma\supseteq\Gamma_\pi} \;\;\leq\;\; \frac{1}{d^{|\sigma|+|\pi|}s^{|\sigma|}}\;C^{(l)}_\tau\;\delta_{\Gamma_\sigma\supseteq\Gamma_\pi} \tag{4.B.50}$$

$$\chi^{(l)}_{d}([\hat\sigma],[\hat\pi]) \;=\; \sum_{\eta\subseteq\sigma\;,\;\kappa\subseteq\pi} c(\eta^{-1}\sigma)\;c(\kappa^{-1}\pi)\;\frac{\chi^{(l)}_{d}(\eta,\kappa)}{\chi^{(l)}_{d}(\eta)\chi^{(l)}_{d}(\kappa)}\;\delta_{\Gamma_\sigma\Gamma_\pi} \;\;\leq\;\; d^{|\sigma|+|\pi|}\;C^{(l)}_\chi\;\delta_{\Gamma_\sigma\Gamma_\pi}$$

$$C^{(l)}_\tau \;=\; \chi_l\;c_l^3 \qquad,\qquad C^{(l)}_\chi \;=\; c_\sigma^2\;c_\pi^2 \;\;, \tag{4.B.51}$$

up to l-dependent constants $\chi_l > 0$ [117]. We can also bound the Catalan, Möbius, and Weingarten functions as,

$$c(\sigma) \;\leq\; |c(\sigma)| \;=\; \prod_{\lambda\in\sigma} c_{|\lambda|} \;\leq\; c_\sigma \;=\; \prod_{\lambda\in\sigma} c_{|\lambda|+1} \;\leq\; c_l \;\leq\; e^{2l} \tag{4.B.52}$$

$$\chi^{(l)-1}_{d}(\sigma) \;\leq\; \chi_l\;c(\sigma)\;\frac{1}{d^{|\sigma|}} \;\leq\; \chi_l\;c_\sigma\;\frac{1}{d^{|\sigma|}} \tag{4.B.53}$$

$$\big|\sigma\big| + \big|\pi\big| \pm \big|\sigma^{-1}\pi\big| \;\substack{\geq\\ \leq}\; \big|\sigma\big| + \big|\pi\big|\;,\; \big|\sigma\big| + \big|\sigma^{-1}\pi\big| \;\geq\; \big|\sigma\big|\;,\; \lceil l/2\rceil \;\leq\; \big|\sigma\big| \;\leq\; l-1 \tag{4.B.54}$$

$$\big|\{\sigma\supseteq\pi\in\mathcal{S}_l\}\big| \;=\; c_\sigma \;\leq\; c_l \quad,\quad \big|\{\sigma\subseteq\pi\in\mathcal{S}_l\}\big| \;\leq\; c_{l-|\sigma|} \;\leq\; c_l\;. \tag{4.B.55}$$

Then, k-concatenations of $l \leq t$-order cHaar moment operators have transfer matrix elements for $\sigma,\pi \in \mathcal{S}_l$ of the form,

$$\tau^{(l,k)}_{\mathcal{C}(d,s)}([\hat\sigma],[\hat\pi]) \;=\; \sum_{\substack{\{\sigma_i,\pi_i\in\mathcal{S}_{l_i}\}_{i\in[k]}\\ \Gamma_{\pi_{i-1}}=\Gamma_{\pi_i}\subseteq\Gamma_{\sigma_i}=\Gamma_{\sigma_{i+1}}\\ \sigma_{k-1}=\sigma\;,\;\pi_0=\pi\;,\;l_i=\left|\Gamma_{\sigma_i}\right|}} \;\prod_{i\in[k]} \tau^{(l_i)}_{\mathcal{C}(d,s)}([\hat\sigma_i],[\hat\pi_i]) \prod_{i\in[k-1]} \chi^{(l_i)-1}_{ds}([\hat\pi_{i+1}],[\hat\sigma_i])\;. \tag{4.B.56}$$

Here, we note the support constraints $\{\} \subseteq \Gamma_\pi \subseteq \cdots \subseteq \Gamma_{\pi_i} \subseteq \Gamma_{\sigma_i} \subseteq \cdots \subseteq \Gamma_\sigma \subseteq [l]$ of the propagated localized permutations, resulting in telescoping of the system dimension scalings cancelling from identical scaling of the transfer matrices $\sim d^{|\sigma_i|+|\pi_i|}$ and overlaps $\sim 1/d^{|\sigma_i|+|\pi_{i+1}|}$, and the environment dimension scalings $\sim 1/s^{|\sigma_i|}$ being bounded by the at least $|\Gamma_\pi|$-locality minimal permutations $\{\sigma_i \;:\; \Gamma_{\sigma_i} = \Gamma_{\pi_i} = \Gamma_\pi\;,\; |\sigma_i| \geq |\Gamma_\pi|/2 \geq |\Gamma_\pi|/2\}_{i\in[k-1]}$. Such scalings propagate through concatenations until the $k-1$-th moment operator, where the k-concatenated $l \leq t$-order transfer matrix elements are bounded by,

$$\left|\tau^{(l,k)}_{\mathcal{C}(d,s)}([\hat\sigma],[\hat\pi])\right| \;\leq\; \frac{1}{d^{|\sigma|+|\pi|}s^{|\sigma|+(k-1)|\Gamma_\pi|/2}}\;C^{(l,k)}_\tau\;\delta_{\Gamma_\sigma\supseteq\Gamma_\pi} \tag{4.B.57}$$

$$C_\tau^{(l,k)} \;=\; l!^{2(k-1)}\ \chi_l^k\ c_l^{3k+4(k-1)} \;=\; \frac{1}{l!^2\ c_l^4}\ \left(l!^2\ \chi_l\ c_l^7\right)^k \;=\; \frac{1}{l!^2\ c_l^4}\ C_\lambda^{(l)k} \tag{4.B.58}$$

$$C_\lambda^{(l)} = l!^2\ \chi_l\ c_l^7\ . \tag{4.B.59}$$

Finally, we note that the simplest non-trivial transpositions are orthogonal, with exact transfer matrices and overlaps of,

$$\tau_{\mathcal{C}(d,s)}^{(l,k)}([\hat{\tau}],[\hat{e}]) = \frac{s-1}{d^2s^2-1}\left[1+\sum_{j>0}^{k-1}\left[s\frac{d^2-1}{d^2s^2-1}\right]^j\right] \tag{4.B.60}$$

$$\tau_{\mathcal{C}(d,s)}^{(l,k)}([\hat{\tau}],[\hat{\tau}']) = \frac{1}{d^2-1}\left[s\frac{d^2-1}{d^2s^2-1}\right]^k\ \delta_{\tau\tau'} \tag{4.B.61}$$

$$\chi_d^{(l)}([\hat{\tau}],[\hat{\tau}']) = \left(d^2-1\right)\ \delta_{\tau\tau'}\ . \tag{4.B.62}$$

□

4.B.3 Spectral Properties and Hierarchies of Moment Operators

In this section we derive spectral properties for the moment operators of the reference ensembles. These will allow us to derive a hierarchy between the Haar, cHaar, and Depolarize ensembles via their moment operator norms.

First, we investigate the leading eigenvectors of the cHaar moment operator.

Theorem 12 (Leading Eigenvectors of cHaar Moment Operators)**.** *The t-order k-concatenated cHaar moment operator has a unit leading eigenvalue $\lambda = 1$, a leading right-eigenvector $|\psi\rangle\rangle$ of a uniform combination of permutations, weighted by their characters, and a leading left-eigenvector $|\varphi\rangle\rangle$ reflecting trace-preservation. That is, the cHaar moment operator*

$$\widehat{\mathcal{T}}_{\mathcal{C}(d,s)}^{(t)k} = \frac{|\psi\rangle\rangle\langle\langle\varphi|}{\langle\langle\varphi|\psi\rangle\rangle}\ +\ \cdots\ , \qquad : \qquad |\psi\rangle\rangle = \sum_{\sigma\in\mathcal{S}_t}\chi_{ds}^{(t)}(\sigma)\ |\sigma\rangle\rangle \qquad , \qquad |\varphi\rangle\rangle = |e\rangle\rangle \tag{4.B.63}$$

has a leading-order, concatenation-invariant, non-unital, and trace preserving contribution.

Proof. To confirm that such vectors $|\psi\rangle\!\rangle, |\varphi\rangle\!\rangle$ are eigenvectors of the cHaar moment operator, our goal is to show that, $\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)}|\psi\rangle\!\rangle = |\psi\rangle\!\rangle$, and $\langle\!\langle\varphi|\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)} = \langle\!\langle\varphi|$. Given the orthogonality of the characters and Weingarten functions, the invariance of the characters under inverses of permutations, $\chi^{(t)}_d(\pi,\varsigma) = \chi^{(t)}_d(\pi^{-1}\varsigma) = \chi^{(t)}_d(\varsigma^{-1}\pi)$, and letting $\varsigma \to \pi^{-1}\varsigma$,

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)}|\psi\rangle\!\rangle = \sum_{\sigma,\pi,\varsigma\in\mathcal{S}_t} \chi^{(t)}_s(\sigma)\ \chi^{(t)-1}_{ds}(\sigma,\pi)\ \chi^{(t)}_d(\pi,\varsigma)\ \chi^{(t)}_{ds}(\varsigma)\ |\sigma\rangle\!\rangle \tag{4.B.64}$$

$$= \sum_{\sigma,\pi,\varsigma\in\mathcal{S}_t} \chi^{(t)}_s(\sigma)\ \chi^{(t)-1}_{ds}(\sigma,\pi)\ \chi^{(t)}_{ds}(\pi,\varsigma^{-1})\ \chi^{(t)}_d(\varsigma)\ |\sigma\rangle\!\rangle \tag{4.B.65}$$

$$= |\psi\rangle\!\rangle \tag{4.B.66}$$

$$\langle\!\langle\varphi|\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)} = \sum_{\sigma,\pi\in\mathcal{S}_t} \chi^{(t)}_d(\sigma)\ \chi^{(t)}_s(\sigma)\ \chi^{(t)-1}_{ds}(\sigma,\pi)\ \langle\!\langle\pi| \tag{4.B.67}$$

$$= \sum_{\sigma,\pi\in\mathcal{S}_t} \chi^{(t)}_{ds}(\sigma)\ \chi^{(t)-1}_{ds}(\sigma,\pi)\ \langle\!\langle\pi| \tag{4.B.68}$$

$$= \sum_{\pi\in\mathcal{S}_t} \delta_{\pi e}\ \langle\!\langle\pi| = \langle\!\langle\varphi|\ . \tag{4.B.69}$$

Given the Jucys-Murphy character sums [250], $\sum_{\sigma\in\mathcal{S}_t}\chi^{(t)}_d(\sigma) = \binom{d+t-1}{t}\frac{t!}{d^t}$, the eigenvectors have overlaps in terms of the symmetric subspace dimension $\binom{d^2s+t+1}{t}$,

$$\chi^{(t)}_d(\varphi,\psi) = \chi^{(t)}_d(\psi) = \frac{1}{d^t}\langle\!\langle\varphi|\psi\rangle\!\rangle = \sum_{\sigma\in\mathcal{S}_t}\chi^{(t)}_{d^2s}(\sigma) = \binom{d^2s+t-1}{t}\frac{t!}{d^{2t}s^t} \tag{4.B.70}$$

$$= 1 + \binom{t}{2}\frac{1}{d^2s} + \mathcal{O}\left(\frac{1}{d^4s^2}\right)\ . \tag{4.B.71}$$

□

In fact, we can bound all of the eigenvalues, the norm, and the trace of the Haar, cHaar, and Depolarize ensembles, by making use of the following lemma.

Lemma 4 (Spectrum with respect to Localized Permutation Basis)**.** *The spectrum of the cHaar moment operator in the linearly-independent, but not strictly orthogonal localized permutation basis is,*

$$\text{Spectrum of } \widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)} = \text{ Spectrum of } \left[\tilde{\tau}^{(t,k)}_{\mathcal{C}(d,s)}(\sigma,\pi)\right]_{\sigma,\pi\in\mathcal{S}_t} \tag{4.B.72}$$

$$\tilde{\tau}^{(t,k)}_{\mathcal{C}(d,s)}(\sigma,\pi) = \sum_{\varsigma\in\mathcal{S}_t} \tau^{(t,k)}_{\mathcal{C}(d,s)}([\hat{\sigma}],[\hat{\varsigma}])\ \chi^{(t)}_d([\hat{\varsigma}],[\hat{\pi}]) \tag{4.B.73}$$

$$\tilde{\tau}^{(t,k)\prime}_{\mathcal{C}(d,s)}(\sigma,\pi) = \sum_{\varsigma\in\mathcal{S}_t} \chi^{(t)}_d([\hat{\sigma}],[\hat{\varsigma}])\ \tau^{(t,k)}_{\mathcal{C}(d,s)}([\hat{\varsigma}],[\hat{\pi}])\ , \tag{4.B.74}$$

expressed as the spectrum of the modified transfer matrices $\tilde{\tau}^{(t,k)}_{\mathcal{C}(d,s)}$, $\tilde{\tau}^{(t,k)\prime}_{\mathcal{C}(d,s)}$ in the localized permutation basis, with bounded elements,

$$\tilde{\tau}^{(l,k)}_{\mathcal{C}(d,s)}(\sigma,\pi)\ \leq\ \frac{1}{d^{|\sigma|-|\pi|}}\frac{1}{s^{|\sigma|+(k-1)l'/2}}\ C^{(l,l',k)}_{\tilde{\tau}}\ \delta_{\Gamma_\sigma\supseteq\Gamma_\pi}\ :\ C^{(l,l',k)}_{\tilde{\tau}} = l!^{2(k-1)}\ l'!\ \chi^k_l\ c^{3k+4(k-1)}_l\ c^4_{l'} \tag{4.B.75}$$

$$\tilde{\tau}^{(l,k)\prime}_{\mathcal{C}(d,s)}(\sigma,\pi)\ \leq\ d^{|\sigma|-|\pi|}\frac{1}{s^{l/2+(k-1)l'/2}}\ C^{(l,l',k)}_{\tilde{\tau}'}\ \delta_{\Gamma_\sigma\supseteq\Gamma_\pi}\ :\ C^{(l,l',k)}_{\tilde{\tau}'} = l!^{2(k-1)}\ l!\ \chi^k_l\ c^{3k+4(k-1)}_l\ c^4_l\ ,$$

given l,k-dependent constants $C^{(l,l',k)}_{\tilde{\tau}}$, $C^{(l,l',k)}_{\tilde{\tau}'}$.

Proof. Let $\widehat{\mathcal{T}} = \sum_{\alpha,\beta\in\mathcal{S}}\tau(\alpha,\beta)|\alpha\rangle\!\rangle\langle\!\langle\beta|$ be a matrix, expressed in terms of a linearly independent basis $\mathcal{S}\ni\alpha,\beta$, with overlaps $\chi(\alpha,\beta) = \langle\!\langle\alpha|\beta\rangle\!\rangle$. Its eigenvalue equation, given an eigenvector $|\nu\rangle\!\rangle = \sum_{\alpha\in\mathcal{S}}\nu(\alpha)\ |\alpha\rangle\!\rangle$, and eigenvalue λ, is,

$$\widehat{\mathcal{T}}|\nu\rangle\!\rangle = \lambda\ |\nu\rangle\!\rangle\quad\rightarrow\quad\sum_{\xi,\beta\in\mathcal{S}}\tau(\alpha,\xi)\ \chi(\xi,\beta)\ \nu(\beta) = \lambda\ \nu(\alpha)\quad\forall\ \alpha \tag{4.B.76}$$

$$\rightarrow \tag{4.B.77}$$

$$\tilde{\tau}\ \nu = \lambda\ \nu\quad:\quad\tilde{\tau}(\alpha,\beta) = \sum_{\xi\in\mathcal{S}}\tau(\alpha,\xi)\ \chi(\xi,\beta)\ . \tag{4.B.78}$$

Therefore the eigenvectors and eigenvalues of $\widehat{\mathcal{T}}$ are equivalent to the eigenvectors of components $\{\nu(\alpha)\}_{\alpha\in\mathcal{S}}$ and eigenvalues λ of the matrix $\tilde{\tau}$, with respect to the basis $\mathcal{S}$.

To bound the cHaar transfer matrix elements, we rely on the block-lower-triangular structure of the cHaar moment operator in this localized basis, The cHaar moment operator can be partitioned into blocks labelled by supports Γ,Γ', with associated localities $l = |\Gamma|, l' = |\Gamma'|$, spanned by permutations $\sigma\in\mathcal{S}_l$ and $\pi\in\mathcal{S}_{l'}$ such that $\Gamma_\sigma = \Gamma\supseteq\Gamma' = \Gamma_\pi$. We also recall that the transfer matrix and overlaps for these blocks have bounds of,

$$\left|\tau^{(l,k)}_{\mathcal{C}(d,s)}([\hat{\sigma}],[\hat{\pi}])\right|\ \leq\ \frac{1}{d^{|\sigma|+|\pi|}s^{|\sigma|+(k-1)l/2}}\ C^{(l,k)}_{\tau}\ \delta_{\Gamma_\sigma\supseteq\Gamma_\pi}\qquad: C^{(l,k)}_{\tau}\ =\ l!^{2(k-1)}\ \chi^k_l\ c^{3k+4(k-1)}_l \tag{4.B.79}$$

$$\left|\chi^{(l)}_d([\hat{\sigma}],[\hat{\pi}])\right|\ \leq\ d^{|\sigma|+|\pi|}\ C^{(l)}_{\chi}\ \delta_{\Gamma_\sigma\Gamma_\pi}\qquad: C^{(l)}_{\chi}\ =\ c^4_l\ .$$

The modified transfer matrix elements $\tilde{\tau}^{(l,k)}_{\mathcal{C}(d,s)}(\sigma,\pi)$, as sums over permutations $\varsigma \in \mathcal{S}_l$ can thus be bounded using our expressions for bounds on the original transfer matrix elements $\tau^{(l,k)}_{\mathcal{C}(d,s)}([\hat{\sigma}],[\hat{\pi}])$ and overlaps $\chi^{(l)}_d([\hat{\sigma}],[\hat{\pi}])$,

$$\begin{aligned}
\tilde{\tau}^{(l,k)}_{\mathcal{C}(d,s)}(\sigma,\pi) &= \sum_{\substack{\varsigma\in\mathcal{S}_l \\ \Gamma_\sigma=\Gamma\supseteq\Gamma'=\Gamma_\varsigma=\Gamma_\pi}} \tau^{(l,k)}_{\mathcal{C}(d,s)}([\hat{\sigma}],[\hat{\varsigma}])\ \chi^{(l')}_d([\hat{\varsigma}],[\hat{\pi}]) && (4.B.80)\\
&\le \max_{\substack{\sigma,\varsigma,\pi\in\mathcal{S}_l \\ \Gamma_\sigma=\Gamma\supseteq\Gamma'=\Gamma_\varsigma=\Gamma_\pi}} !l' \left|\tau^{(l,k)}_{\mathcal{C}(d,s)}([\hat{\sigma}],[\hat{\varsigma}])\right| \left|\chi^{(l')}_d([\hat{\varsigma}],[\hat{\pi}])\right| && (4.B.81)\\
&\le \frac{1}{d^{|\sigma|-|\pi|}}\frac{1}{s^{|\sigma|+(k-1)l'/2}}\ C^{(l,l',k)}_{\tilde{\tau}}\ \delta_{\Gamma_\sigma\supseteq\Gamma_\pi} && (4.B.82)\\
&\le C^{(l,l',k)}_{\tilde{\tau}}\ \delta_{\Gamma_\sigma\supseteq\Gamma_\pi} \begin{cases} \left(\frac{1}{ds}\right)^{l/2} & {}^{l>0}_{l'=0} \\ \frac{1}{s^{k+(k+1)(l-l')/2}}\left(\frac{d}{s^k}\right)^{((l'-2)-(l-l'))/2} & {}^{l>0}_{l'>0} \end{cases} && (4.B.83)\\
C^{(l,l',k)}_{\tilde{\tau}} &= l'!\ C^{(l,k)}_{\tau}\ C^{(l')}_{\chi} = l!^{2(k-1)}\ l'!\ \chi^k_l\ c_l^{3k+4(k-1)}\ c^4_{l'}\ , && (4.B.84)
\end{aligned}$$

$$\begin{aligned}
\tilde{\tau}^{(l,k)\prime}_{\mathcal{C}(d,s)}(\sigma,\pi) &= \sum_{\substack{\varsigma\in\mathcal{S}_l \\ \Gamma_\sigma=\Gamma_\varsigma=\Gamma\supseteq\Gamma'=\Gamma_\pi}} \chi^{(l)}_d([\hat{\sigma}],[\hat{\varsigma}])\ \tau^{(l,k)}_{\mathcal{C}(d,s)}([\hat{\varsigma}],[\hat{\pi}]) && (4.B.85)\\
&\le \max_{\substack{\sigma,\varsigma,\pi\in\mathcal{S}_l \\ \Gamma_\sigma=\Gamma_\varsigma=\Gamma\supseteq\Gamma'=\Gamma_\pi}} !l \left|\chi^{(l)}_d([\hat{\sigma}],[\hat{\varsigma}])\right| \left|\tau^{(l,k)}_{\mathcal{C}(d,s)}([\hat{\varsigma}],[\hat{\pi}])\right| && (4.B.86)\\
&\le d^{|\sigma|-|\pi|}\frac{1}{s^{l/2+(k-1)l'/2}}\ C^{(l,l',k)}_{\tilde{\tau}'}\ \delta_{\Gamma_\sigma\supseteq\Gamma_\pi} && (4.B.87)\\
&\le\ C^{(l,l',k)}_{\tilde{\tau}'}\ \delta_{\Gamma_\sigma\supseteq\Gamma_\pi} \begin{cases} s^{l/2-1}\left(\frac{d}{s}\right)^{l-1} & {}^{l>0}_{l'=0} \\ \frac{1}{s^{k+(1-2k)(l-l')/2}}\left(\frac{d}{s^k}\right)^{((l-2)+(l-l'))/2} & {}^{l>0}_{l'>0} \end{cases} && (4.B.88)\\
C^{(l,l',k)}_{\tilde{\tau}'} &= l!\ C^{(l,k)}_{\tau}\ C^{(l)}_{\chi} = l!^{2(k-1)}\ l!\ \chi^k_l\ c_l^{3k+4(k-1)}\ c^4_{l}\ , && (4.B.89)
\end{aligned}$$

where the final bounds follow from considering scaling separately for each case of $l' = 0, l' > 0$, and from the fact that $l > 0$-locality permutations $\sigma \in \mathcal{S}_l$ have sizes bounded by $l/2 \le \lceil l/2\rceil \le |\sigma| \le l-1$. □

We now will make use of the derived bounds for the modified k-concatenated, t-order cHaar moment operator transfer matrix elements to bound its spectra, norm, and trace, and assess its overall spectral properties.

Theorem 13 (Spectrum of cHaar Moment Operator)**.** *The t-order cHaar moment operator has a single leading eigenvalue $\lambda = 1$, and in the asymptotic limit $ds \to \infty$, for fixed t, all non-leading eigenvalues are upper bounded by,*

$$0 \;\leq\; |\lambda| \;\leq\; \frac{1}{s^{l/2}}\; C_\lambda^{(l)} \;<\; 1 \;, \tag{4.B.90}$$

and therefore $|\lambda| < 1$ for sufficiently large s, and t-dependent constants $C_\lambda^{(l)} = l!^2\;\chi_l\;c_l^7$, and some index $2 \leq l \leq t$.

Proof. Here, we use Gershgorin's circle theorem [441], in the non-orthogonal, block matrix case. Eigenvalues λ of the Γ-support, $l \geq 2$-locality, $!l$-sized diagonal blocks of the cHaar moment operator, satisfy

$$|\lambda| = \left|\lambda^k\right|^{\frac{1}{k}} \leq \left[!l \max_{\substack{\sigma,\pi\in\mathcal{S}_l\\ \Gamma_\sigma=\Gamma_\pi=\Gamma}} \left|\tilde{\tau}^{(l,k)}_{\mathcal{C}(d,s)}(\sigma,\pi)\right| \right]^{\frac{1}{k}} \leq \left[\frac{1}{s^k}\left(\frac{d}{s^k}\right)^{\frac{l-2}{2}} C_\lambda^{(l,k)} \right]^{\frac{1}{k}} \overset{k\to\infty}{\leq} \frac{1}{s^{l/2}}\; C_\lambda^{(l)} \tag{4.B.91}$$

$$C_\lambda^{(l,k)} \;=\; l!\; C_{\tilde{\tau}}^{(l,l,k)} \;=\; l!^{2k}\;\chi_l^k\;c_l^{7k} \;=\; C_\lambda^{(l)k} \qquad , \qquad C_\lambda^{(l)} \;=\; l!^2\;\chi_l\;c_l^7 \;, \tag{4.B.92}$$

given eigenvalues of k-concatenations of a matrix are equal to k powers of the eigenvalues, given the bounds on the modified k-concatenated transfer matrix elements $\tilde{\tau}^{(l,k)}_{\mathcal{C}(d,s)}$, and given we can take the $k \to \infty$ limit for fixed t, d, s. □

Theorem 14 (Norm and Trace of Moment Operators)**.** *The k-concatenated Haar, cHaar, and Depolarize moment operators have norms and traces, in the asymptotic limit $ds \to \infty$ for fixed t, k, upper bounded by,*

$$\left\|\widehat{\mathcal{T}}^{(t)k}_{\mathcal{U}(d)}\right\|^2 = \mathrm{tr}\left(\widehat{\mathcal{T}}^{(t)k}_{\mathcal{U}(d)}\right) = t! \qquad , \qquad \left\|\widehat{\mathcal{T}}^{(t)k}_{\mathcal{D}(d)}\right\|^2 = \mathrm{tr}\left(\widehat{\mathcal{T}}^{(t)k}_{\mathcal{D}(d)}\right) = 1 \tag{4.B.93}$$

$$\left\|\widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)}\right\|^2 \leq 1 \;+\; \frac{1}{s^2}\; C^{(t,k)}_{\|\tau\|} \qquad , \qquad \mathrm{tr}\left(\widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)}\right) \leq 1 \;+\; \frac{1}{s^k}\; C^{(t,k)}_{\mathrm{Tr}[\tau]} \;, \tag{4.B.94}$$

given t, k-dependent constants $C^{(t,k)}_{\|\tau\|} = \frac{t!^2-1}{t!^2}\; C_\lambda^{(t)2k}$, $C^{(t,k)}_{\mathrm{Tr}[\tau]} = (t!^2-1)\; C_\lambda^{(t)k}$, $C_\lambda^{(t)} = t!^2\;\chi_t\;c_t^7$.

Proof. The exact norm and trace of the k-concatenated t-order Haar and Depolarize moment operators are given by

$$\left\|\widehat{\mathcal{T}}^{(t)k}_{\mathcal{U}(d)}\right\|^2 = \left\|\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)}\right\|^2 = \mathrm{tr}\left(\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)}\right) = \sum_{\sigma,\pi\in\mathcal{S}_t} \chi^{(t)-1}_d(\sigma,\pi)\ \chi^{(t)}_d(\pi,\sigma) = |\mathcal{S}_t| \tag{4.B.95}$$

$$\left\|\widehat{\mathcal{T}}^{(t)k}_{\mathcal{D}(d)}\right\|^2 = \left\|\widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}\right\|^2 = \mathrm{tr}\left(\widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}\right) = 1\ . \tag{4.B.96}$$

The exact norm and trace of t-order cHaar moment operator do not appear to simplify, however can be bounded as,

$$\left\|\widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)}\right\|^2 = \sum_{\substack{\Gamma'\subseteq\Gamma\subseteq[t]\\|\Gamma|=l\geq l'=|\Gamma'|}} \sum_{\substack{\sigma,\pi\in\mathcal{S}_l\\ \Gamma_\sigma=\Gamma\supseteq\Gamma'=\Gamma_\pi}} \tilde{\tau}^{(l,k)\prime}_{\mathcal{C}(d,s)}(\sigma,\pi)\ \tilde{\tau}^{(l,k)}_{\mathcal{C}(d,s)}(\sigma,\pi) \tag{4.B.97}$$

$$\leq 1\ +\ (t!^2-1)\max_{\substack{l'\leq l\leq t\\ l\leq 2}} \frac{1}{s^{l+(k-1)l'}}\ C^{(l,l',k)}_{\tilde{\tau}}\ C^{(l,l',k)}_{\tilde{\tau}'}\ \leq\ 1\ +\ \frac{1}{s^2}C^{(t,k)}_{\|\tau\|} \tag{4.B.98}$$

$$\mathrm{tr}\left(\widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)}\right) = \sum_{s\in[t!]} \lambda_s^k\ \leq\ 1\ +\ (t!^2-1)\max_{2\leq l\leq t} \frac{1}{s^{kl/2}}\ C^{(l)k}_{\lambda}\ \leq\ 1\ +\ \frac{1}{s^k}C^{(t,k)}_{\mathrm{Tr}[\tau]} \tag{4.B.99}$$

$$C^{(t,k)}_{\|\tau\|} = (t!^2-1)\ C^{(t,k)2}_{\tilde{\tau}} = \frac{t!^2-1}{t!^2}\ C^{(t)2k}_{\lambda} \tag{4.B.100}$$

$$C^{(t,k)}_{\mathrm{Tr}[\tau]} = (t!^2-1)\ C^{(t)k}_{\lambda} \tag{4.B.101}$$

$$C^{(t)}_{\lambda} = t!^2\ \chi_t\ c_t^7\ , \tag{4.B.102}$$

given the bounds on the modified k-concatenated transfer matrix elements $\tilde{\tau}^{(l,k)}_{\mathcal{C}(d,s)}, \tilde{\tau}^{(l,k)\prime}_{\mathcal{C}(d,s)}$. $\square$

For sufficiently large s, all non-leading cHaar moment operator eigenvalues are $|\lambda| < 1$, which agrees with the fact that all channels have $|\lambda| \leq 1$ [442]. For $l = 2$, we have exactly,

$$\lambda = \frac{d^2-1}{d^2s^2-1}s\ . \tag{4.B.103}$$

Next, we note that all trace-preserving moment operators over ensembles $\mathcal{C}$ can be expressed as the Depolarize moment operator $\widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}$, plus deviations $\widehat{\Delta}^{(t)}_{\mathcal{C}}$, and the cHaar $\mathcal{C}(d,s)$ ensemble is invariant under the Haar $\mathcal{U}(d)$ ensemble,

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} \;+\; \widehat{\Delta}^{(t)}_{\mathcal{C}} \tag{4.B.104}$$

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)} = I_d^{\otimes 2t} \otimes \langle\!\langle I_d^{\otimes t}| \;\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(ds)}\; I_d^{\otimes 2t} \otimes |\nu^{\otimes t}\rangle\!\rangle \tag{4.B.105}$$

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}} = \widehat{\mathcal{T}}^{(t)\dagger}_{\mathcal{C}}\widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} \quad , \quad \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)}\widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)}\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{C}(d,s)} \tag{4.B.106}$$

$$\widehat{\mathcal{T}}^{(t)k}_{\mathcal{U}(d)} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)} \quad , \quad \widehat{\mathcal{T}}^{(t)\dagger}_{\mathcal{U}(d)} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)} \;. \tag{4.B.107}$$

From these relationships, and the exact forms of the moment operators obtained above, the following relationships hold:

Corollary 1 (cHaar Moment Operator Limits)**.** *The k-concatenated t-order cHaar moment operator satisfies,*

$$\lim_{ds\to d} \widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)} \qquad , \qquad \lim_{ds\to\infty} \widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} \;. \tag{4.B.108}$$

Corollary 2 (cHaar Moment Operator Converges to Depolarizing and Non-Unital Channel)**.** *The k-concatenated t-order cHaar moment operator is inherently depolarizing, with next-leading-order concatenation-independent non-unital behaviour, and concatenation- dependent unital terms. For all dimensions, given traceless operators $T_d^{(t,k)}, T_d^{(t,k)\prime} \in \mathcal{B}[\mathcal{H}^{\otimes t}]$,*

$$\begin{aligned}\lim_{ds\to\infty} \widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)} = {} & \frac{1}{d^t}|I_d^{\otimes t}\rangle\!\rangle\langle\!\langle I_d^{\otimes t}| \\ & + \;\frac{1}{d^t}\mathcal{O}\left(\frac{1}{ds}\right)\; |T_d^{(t,k)}\rangle\!\rangle\langle\!\langle I_d^{\otimes t}| \\ & + \;\frac{1}{d^t}\mathcal{O}\left(\frac{1}{d^2 s^k}\right) |T_d^{(t,k)}\rangle\!\rangle\langle\!\langle T_d^{(t,k)\prime}| \;.\end{aligned} \tag{4.B.109}$$

Indeed, in the localized permutation basis, the projector Haar moment operators have an invariant block-diagonal structure, and the non-projector cHaar moment operators have a block-lower-triangular structure, whose elements decay with dimension and concatenations. The average behaviour of random quantum channels is thus a useful pedagogical tool, given it interpolates with dimension, between trivial environments and unitarity, and large dimensions and depolarization. In Fig. 4.B.5, we numerically illustrate this bounded behaviour for moment operator norms, for different t, k, and s.

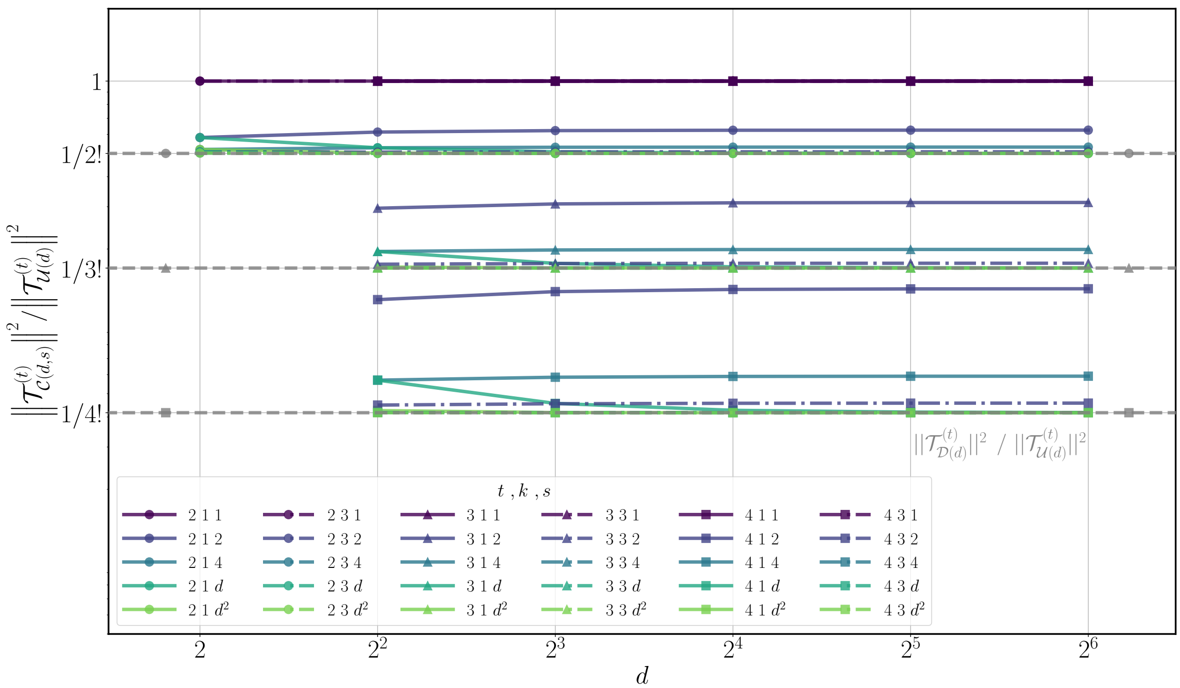


Figure 4.B.5: **cHaar moment operator norm scaling**. Norm of the k-concatenated t-order cHaar moment operator $\|\widehat{\mathcal{T}}^{(t,k)}_{\mathcal{C}(d,s)}\|^2$ as a function of the system dimension d, for environment dimensions s, concatenations k, and orders t. Norms are computed exactly using the derived closed form expressions for $\widehat{\mathcal{T}}^{(t,k)}_{\mathcal{C}(d,s)}$, and symbolic methods [332]. Norms converge to their leading-order, system dimension d-independent term, and decay monotonically with environment dimension s or with concatenation k, for all t, from $\|\widehat{\mathcal{T}}^{(t,k)}_{\mathcal{U}(d)}\|^2 = t!$ (1 on scaled axis) to $\|\widehat{\mathcal{T}}^{(t,k)}_{\mathcal{D}(d)}\|^2 = 1$ ($1/t!$ grey dashed lines on scaled axis). The depicted scalings confirm the bounded behaviour, even for constant environment dimension $s < d$, of the cHaar moment operator as it interpolates between unitary and depolarizing behaviour. Further, such results suggest that the conjecture $\|\widehat{\mathcal{T}}^{(t,k)}_{\mathcal{C}(d,s)}\|^2 \leq \|\widehat{\mathcal{T}}^{(t,k)}_{\mathcal{U}(d)}\|^2$ holds absolutely for all variables t, k, d, s.

The previous results indicate that as we concatenate cHaar random channels k times, the resulting composite transfer matrix elements are suppressed, which intuitively means the moment operator norm should also decrease until it converges to the norm of a single projector with a unit eigenvalue. In the large $ds \to \infty$ limit, this intuition is verified, but given that the cHaar moment operator has no closed form, we instead postulate that the following conjecture should hold.

Conjecture 1 (Hierarchy of Haar, cHaar, Depolarize Ensembles)**.** *Given there exists an exact hierarchy between the k-concatenated, t-order Haar, cHaar, and Depolarize moment operators in the asymptotic limit $ds \to \infty$,*

$$1 = \left\| \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} \right\|^2 \leq \left\| \widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)} \right\|^2 \leq \left\| \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)} \right\|^2 = t! \ , \tag{4.B.110}$$

it is conjectured that such a hierarchy holds for all concatenations k, orders t, and dimensions d, s. Further, it is conjectured that the cHaar norm is monotonically non-increasing with environment dimension and concatenations,

$$\left\| \widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s')} \right\|^2 \overset{?}{\geq} \left\| \widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)} \right\|^2 \qquad \forall\ s' \geq s \quad , \qquad \left\| \widehat{\mathcal{T}}^{(t)k'}_{\mathcal{C}(d,s)} \right\|^2 \overset{?}{\leq} \left\| \widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)} \right\|^2 \qquad \forall\ k' \geq k \ . \tag{4.B.111}$$

Given the relationship between trace-preserving moment operators and the invariant Depolarize moment operator,

$$\left\| \widehat{\mathcal{T}}^{(t)}_{\mathcal{C}} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} \right\|^2 = \left\| \widehat{\mathcal{T}}^{(t)}_{\mathcal{C}} \right\|^2 - \left\| \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} \right\|^2 \geq 0 \qquad \to \qquad \left\| \widehat{\mathcal{T}}^{(t)}_{\mathcal{C}} \right\|^2 \geq \left\| \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} \right\|^2 , \tag{4.B.112}$$

and given the partial invariance of the Haar and cHaar ensembles, the conjecture is true if

$$\left\| \widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)} \right\|^2 \geq 0 \quad \to \tag{4.B.113}$$

$$\left\| \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}(d)} \right\|^2 - \left\| \widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)} \right\|^2 \geq 2 \left(\mathrm{tr}\left(\widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)} \right) - \left\| \widehat{\mathcal{T}}^{(t)k}_{\mathcal{C}(d,s)} \right\|^2 \right) \overset{?}{\geq} 0 \ . \tag{4.B.114}$$

This conjecture, and this bound on the difference of Haar and cHaar moment operator norms, holds in the asymptotic limit $ds \to \infty$, where both norms converge to 1. However, our analysis is limited in part by the lack of closed forms for the Weingarten functions, and from our approaches of (possibly loosely) bounding quantities in terms of maximum cHaar transfer matrix elements. Different, possibly representation-theoretic based approaches are to necessary to prove this non-trivial, but important hierarchy conjecture for general d, s.

4.C Moment Operators of Noisy Ensembles

In these appendices, we calculate moment operators for ensembles of noisy unitary channels, and investigate their scaling with respect to concatenations, dimension, and noise parameters.

4.C.1 Composite Noise Channels

Here, we seek to calculate moment operators for composite ensembles of channels $\mathcal{N} \circ U$, where the unitary component is sampled as $U \sim \mathcal{U}$, and where the noise component is a fixed noise channel $\mathcal{N}$. The t-order composite moment operator, $\widehat{\mathcal{T}}^{(t)}_{\mathcal{UN}} = \mathcal{N} \circ \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}}$, then may be concatenated k times,

$$\widehat{\mathcal{T}}^{(t)k}_{\mathcal{UN}} = \prod_{i\in[k]} \widehat{\mathcal{N}^{\otimes t}} \circ \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}} \;=\; \widehat{\mathcal{D}}^{\otimes t}_{d} \;+\; \widehat{\Delta}^{(t,k)}_{\mathcal{UN}} \tag{4.C.1}$$

with trace-preserving depolarization $\widehat{\mathcal{D}}^{\otimes t}_{d}$ and deviation $\widehat{\Delta}^{(t,k)}_{\mathcal{UN}}$ terms of

$$\widehat{\mathcal{D}}^{\otimes t}_{d} = \frac{1}{d^t} |I^{\otimes t}_d\rangle\!\rangle\langle\!\langle I^{\otimes t}_d| \quad \text{and} \quad \widehat{\Delta}^{(t,k)}_{\mathcal{UN}} = \frac{1}{d^t} \sum_{P\in\mathcal{S}^{(t)}_{\mathcal{N}}\setminus\{I^{\otimes t}_d\},\; S\in\mathcal{S}^{(t)}_{\mathcal{U}}} \tau^{(t,k)}_{\mathcal{UN}}(P,S)\; |P\rangle\!\rangle\langle\!\langle S| \;. \tag{4.C.2}$$

The concatenated composite transfer matrices $\tau^{(t,k)}_{\mathcal{UN}}$ can be expressed in terms of the component transfer matrices $\tau^{(t)}_{\mathcal{N}}, \tau^{(t)}_{\mathcal{U}}$, with respective bases $\mathcal{S}^{(t)}_{\mathcal{U}}, \mathcal{S}^{(t)}_{\mathcal{N}}$ for each component

$$\tau^{(t)}_{\mathcal{UN}}(P,S) = \sum_{Q\in\mathcal{S}^{(t)}_{\mathcal{N}},\; T\in\mathcal{S}^{(t)}_{\mathcal{U}}} \tau^{(t)}_{\mathcal{N}}(P,Q)\; \chi^{(t)}_{d}(Q,T)\; \tau^{(t)}_{\mathcal{U}}(T,S) \tag{4.C.3}$$

$$\tau^{(t,k)}_{\mathcal{UN}}(P,S) = \sum_{Q\in\mathcal{S}^{(t)}_{\mathcal{U}},\; T\in\mathcal{S}^{(t)}_{\mathcal{N}}} \tau^{(t)}_{\mathcal{UN}}(P,Q)\; \chi^{(t)}_{d}(Q,T)\; \tau^{(t,k-1)}_{\mathcal{UN}}(T,S) \;. \tag{4.C.4}$$

The most appropriate basis for each component, and whether $\mathcal{S}^{(t)}_{\mathcal{U}} \neq \mathcal{S}^{(t)}_{\mathcal{N}}$, greatly affects their interplay, and the ease of analysis of the scaling of moment operator norms $\left\|\widehat{\Delta}^{(t,k)}_{\mathcal{UN}}\right\| = \left\|\widehat{\mathcal{T}}^{(t)k}_{\mathcal{UN}} - \widehat{\mathcal{D}}^{\otimes t}_{d}\right\|$, with variables t, k, d.

4.C.2 Unital and Non-unital Noise and Haar Random Unitary Channels

Here, we consider random unitary channels from any Haar distributed unitary group ensembles $\mathcal{U}$, followed by fixed diagonal unital or non-unital noise $\mathcal{N} = \mathcal{N}_\gamma, \mathcal{N}_{\gamma\eta}$, and derive the scaling of their k-concatenated t-order composite moment operators with t, k, d, γ, η.

For the unitary component, t-order moment operators are, in an orthogonal basis $\mathcal{S}_{\mathcal{U}}^{(t)} = \mathcal{P}_d^t$,

$$\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)} = \int_{\mathcal{U}} dU \; U^{\otimes t} \otimes U^{\otimes t\,*} = \frac{1}{d^t} \sum_{P,S \in \mathcal{P}_d^t} \tau_{\mathcal{U}}^{(t)}(P,S) \; |P\rangle\!\rangle\langle\!\langle S| \quad : \quad \tau_{\mathcal{U}}^{(t)}(P,S) \; \propto \; \delta_{\Gamma_P \Gamma_S} \; . \tag{4.C.5}$$

This expansion in an orthogonal basis can be derived, under the assumption the ensemble $\mathcal{U}$, is a Haar distributed group, and has projector moment operators that are support-preserving. The orthogonal basis transfer matrix coefficients $\tau_{\mathcal{U}}^{(t)}(P,S)$ are thus also support-dependent and concatenation-invariant,

$$\sum_{Q \in \mathcal{S}_{\mathcal{U}}^{(t)}} \tau_{\mathcal{U}}^{(t)}(P,Q) \; \tau_{\mathcal{U}}^{(t)}(Q,S) \; = \; \tau_{\mathcal{U}}^{(t)}(P,S) \; \propto \; \delta_{\Gamma_P \Gamma_S} \; . \tag{4.C.6}$$

For the noise component, the noise model $\mathcal{N} = \mathcal{N}_{\gamma\eta}$ moment operator has the special form of diagonal unital noise and non-unital noise components $\gamma_d(P) \geq \gamma$, $\eta_d(P) \leq \eta$ for $P \in \mathcal{S}_{\mathcal{N}_{\gamma\eta}}$, in an orthogonal basis $\mathcal{S}_{\mathcal{N}_{\gamma\eta}} = \mathcal{P}_d$,

$$\widehat{\mathcal{N}}_{\gamma\eta} = \frac{1}{d} \begin{bmatrix} 1 & 0 & \cdots & 0 \\ \eta_d(S) & 1-\gamma_d(S) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \eta_d(P) & 0 & \cdots & 1-\gamma_d(P) \end{bmatrix}_{P,S \in \mathcal{P}_d \setminus \{I_d\}} \tag{4.C.7}$$

$$\widehat{\mathcal{N}_{\gamma\eta}^{\otimes t}} = \mathcal{D}_d^{\otimes t} \; + \; \widehat{\mathcal{N}}_{\eta}^{(t)} \; + \; \widehat{\mathcal{N}}_{\gamma\eta}^{(t)} \; = \; \frac{1}{d^t} \sum_{P,S \in \mathcal{S}_{\mathcal{N}_{\gamma\eta}}^{(t)}} \tau_{\mathcal{N}_{\gamma\eta}}^{(t)}(P,S) \; |P\rangle\!\rangle\langle\!\langle S| \; , \tag{4.C.8}$$

where we have $\widehat{\mathcal{N}}^{\otimes t}$ into depolarizing $\mathcal{D}_d^{\otimes t}$, non-unital $\widehat{\mathcal{N}}_{\eta}^{(t)}$, and unital $\widehat{\mathcal{N}}_{\gamma\eta}^{(t)}$ components, with appropriate additional constraints on γ_d, η_d such that $\mathcal{N}$ is completely positive [335]. The t-order noise component transfer matrix $\tau_{\mathcal{N}_{\gamma\eta}}^{(t)}$, for basis operators $P, S \in \mathcal{S}_{\mathcal{N}_{\gamma\eta}}^{(t)} = \mathcal{P}_d^t$, is

$$\tau_{\mathcal{N}_{\gamma\eta}}^{(t)}(P,S) = \prod_{i \in \Gamma_S} (1-\gamma_d(P_i)) \prod_{i \in \Gamma_P \setminus \Gamma_S} \eta_d(P_i) \; \delta_{P_{\Gamma_S} S_{\Gamma_S}} \in \mathcal{O}\big((1-\gamma)^{|S|} \; \eta^{|P|-|S|}\big) \; \delta_{P_{\Gamma_S} S_{\Gamma_S}} . \tag{4.C.9}$$

We restrict to this special form (a diagonal component arising from Pauli-like-noise, and a dense first column arising from a non-unital component) as we seek to understand exact theoretical properties. More general noise models with other non-diagonal elements, even if initially sparse, become dense and intractable to analyse upon concatenations.

Finally, let us assess the leading-order behaviour of the concatenated transfer matrices,

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}\mathcal{N}_{\gamma\eta}} = \widehat{\mathcal{N}^{\otimes t}_{\gamma\eta}} \circ \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}} = \frac{1}{d^t} \sum_{\substack{P\in\mathcal{S}^{(t)}_{\mathcal{N}_{\gamma\eta}} \\ S\in\mathcal{S}^{(t)}_{\mathcal{U}}}} \tau^{(t)}_{\mathcal{U}\mathcal{N}_{\gamma\eta}}(P,S)\ |P\rangle\!\rangle\langle\!\langle S| \tag{4.C.10}$$

$$\widehat{\mathcal{T}}^{(t)k}_{\mathcal{U}\mathcal{N}_{\gamma\eta}} = \left(\widehat{\mathcal{N}^{\otimes t}_{\gamma\eta}} \circ \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}}\right)^k = \frac{1}{d^t} \sum_{\substack{P\in\mathcal{S}^{(t)}_{\mathcal{N}_{\gamma\eta}} \\ S\in\mathcal{S}^{(t)}_{\mathcal{U}}}} \tau^{(t,k)}_{\mathcal{U}\mathcal{N}_{\gamma\eta}}(P,S)\ |P\rangle\!\rangle\langle\!\langle S|\ . \tag{4.C.11}$$

In particular, given the forms of the individual component transfer matrices, we seek to understand the competition that arises between the unitary components preserving locality, and non-unital noise components increasing locality.

Lemma 5 (Unital and Non-unital Noise and Haar Random Unitary Channel Moment Operators)**.** *Composite ensembles, consisting of a random unitary component $\mathcal{U}$ from a Haar distributed unitary group, with a basis $S \in \mathcal{S}^{(t)}_{\mathcal{U}} = \mathcal{P}^t_d$, and a noise component $\mathcal{N}_{\gamma\eta}$ with a basis $P \in \mathcal{S}^{(t)}_{\mathcal{N}_{\gamma\eta}} = \mathcal{P}^t_d$, and with diagonal unital, and non-unital noise coefficients, $\gamma_d(P) \geq \gamma$, $\eta_d(P) \leq \eta$ for $P \in \mathcal{P}^t_d \backslash \{I^{\otimes t}_d\}$, have k-concatenated t order transfer matrices,*

$$\tau^{(t,k)}_{\mathcal{U}\mathcal{N}_{\gamma\eta}}(P,S) = \mathcal{O}\big((1-\gamma)^{k|S|}\eta^{|P|-|S|}\big)\ \ \tau^{(t)}_{\mathcal{U}}(P_{\Gamma_S},S)\ \delta_{\Gamma_P\supseteq\Gamma_S}\ , \tag{4.C.12}$$

where the leading-order $\mathcal{O}(\gamma,\eta)$ behaviour is strictly due to the unspecified noise component in the $\gamma,\eta \to 0$ limit, and uniformly scales the unitary coefficients by their locality, independent of the specific unitary ensemble itself.

Proof. To begin, we note that the unitary component $\mathcal{U}(d)$ is support-preserving, whereas the non-unital noise component $\mathcal{N}_{\gamma\eta}$ is support-non-decreasing, with individual component transfer matrices, for $P, S \in \mathcal{P}^t_d = \mathcal{S}^{(t)}_{\mathcal{U}} = \mathcal{S}^{(t)}_{\mathcal{N}_{\gamma\eta}}$, of

$$\begin{aligned}\tau^{(t)}_{\mathcal{U}\mathcal{N}_{\gamma\eta}}(P,S) &= \prod_{i\in\Gamma_S}(1-\gamma_d(P_i)) \prod_{i\in\Gamma_P\backslash\Gamma_S}\eta_d(P_i)\ \tau^{(t)}_{\mathcal{U}}(P_{\Gamma_S},S)\ \delta_{\Gamma_P\supseteq\Gamma_S} &\text{(4.C.13)}\\ &\in \mathcal{O}\big((1-\gamma)^{|S|}\eta^{|P|-|S|}\big)\,\tau^{(t)}_{\mathcal{U}}(P_{\Gamma_S},S)\ \delta_{\Gamma_P\supseteq\Gamma_S}\,. &\text{(4.C.14)}\end{aligned}$$

From locality preservation of the unitary component, diagonal unital noise scaling always occurs, for all $\Gamma_P \supseteq \Gamma_S$, whereas non-unital noise scaling only occurs when strictly $\Gamma_P \supset \Gamma_S$, and concatenated operators are constrained to be support-increasing $\Gamma_P \supseteq \cdots \supseteq \Gamma_T \supseteq \cdots \supseteq \Gamma_S$. For instance, the $k = 2$ transfer matrix elements are

$$\tau^{(t,2)}_{\mathcal{UN}_{\gamma\eta}}(P,S) = \sum_{\substack{T\in\mathcal{P}_d^t \\ \Gamma_S\subseteq\Gamma_T\subseteq\Gamma_P}} \tau^{(t)}_{\mathcal{UN}_{\gamma\eta}}(P,T)\ \tau^{(t)}_{\mathcal{UN}_{\gamma\eta}}(T,S) \tag{4.C.15}$$

$$= \sum_{\substack{T\in\mathcal{P}_d^t \\ \Gamma_S\subseteq\Gamma_T\subseteq\Gamma_P}} \Bigg[\prod_{i\in\Gamma_T}(1-\gamma_d(P_i)) \prod_{i\in\Gamma_P\setminus\Gamma_T}\eta_d(P_i) \prod_{i\in\Gamma_S}(1-\gamma_d(T_i)) \prod_{i\in\Gamma_T\setminus\Gamma_S}\eta_d(T_i) \tag{4.C.16}$$

$$\tau^{(t)}_{\mathcal{U}}(P_{\Gamma_T},T)\ \tau^{(t)}_{\mathcal{U}}(T_{\Gamma_S},S)\Bigg]$$

$$= \mathcal{O}\big((1-\gamma)^{2|S|}\ \eta^{|P|-|S|}\big) \sum_{\substack{T\in\mathcal{P}_d^t \\ \Gamma_T=\Gamma_S\subseteq\Gamma_P}} \tau^{(t)}_{\mathcal{U}}(P_{\Gamma_S},T)\ \tau^{(t)}_{\mathcal{U}}(T,S) \tag{4.C.17}$$

$$= \mathcal{O}\big((1-\gamma)^{2|S|}\ \eta^{|P|-|S|}\big)\ \tau^{(t)}_{\mathcal{U}}(P_{\Gamma_S},S)\ \delta_{\Gamma_P\supseteq\Gamma_S}\ , \tag{4.C.18}$$

and k-concatenated composite transfer matrices are thus by induction,

$$\tau^{(t,k)}_{\mathcal{UN}_{\gamma\eta}}(P,S) = \mathcal{O}\big((1-\gamma)^{k|S|}\ \eta^{|P|-|S|}\big)\ \tau^{(t)}_{\mathcal{U}}(P_{\Gamma_S},S)\ \delta_{\Gamma_P\supseteq\Gamma_S}\ . \tag{4.C.19}$$

□

Therefore, the leading-order noise scaling is dictated by the concatenation-invariant unitary components being scaled by strictly unital noise until the last concatenation, at which point non-unital noise increases the support from $\Gamma_S \to \Gamma_P$. Intermediate non-unital noise at concatenations $0 < s < k$ also increase the support $\Gamma_S \to \Gamma_T \to \Gamma_P$, however such intermediate noise contribute next-leading-order unital noise scaling $\mathcal{O}((1-\gamma)^{(k-s)|T|})$. Intermediate non-unital noise, in the case it maps unitary component basis operators to larger-support unitary component basis operators, also contribute dimension-dependent scaling when such concatenations occur between unitary components of differing support.

We now derive the leading-order scaling of moment operator norms for concatenations of composite ensembles, in particular their deviation from the Depolarize ensemble.

Theorem 15 (Unital and Non-unital Noise and Haar Random Unitary Channel Moment Operator Norms). *k-concatenations of t-order ensembles of Haar random unitaries, with commutants with locality-$\geq l$ operators, and rank-r moment operators, and diagonal unital and non-unital noise, with noise coefficients $\gamma_d(P) \geq \gamma$, $\eta_d(P) \leq \eta$ for $P \in \mathcal{P}_d \backslash \{I_d\}$, have moment operator norms relative to the Depolarize ensemble of,*

$$\left\| \widehat{\mathcal{T}}^{(t)k}_{\mathcal{UN}_\gamma} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} \right\|_1 = r \ \mathcal{O}\big((1-\gamma)^{lk}\big) \tag{4.C.20}$$

$$\left\| \widehat{\mathcal{T}}^{(t)k}_{\mathcal{UN}_{\gamma\eta}} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} \right\|_2^2 = t \ |\mathcal{P}_d \backslash \{I_d\}| \ \mathcal{O}\big(\eta^2\big) \ \ . \tag{4.C.21}$$

From the previous theorem it follows that as $k \to \infty$, unital noise decreases moment operator norms for $\gamma > 0$, and approaches a Depolarize t-design, whereas non-unital noise increases moment operator norms for $\eta > 0$, and never approaches a Depolarize t-design.

Proof. Moment operator p-norms follow directly from their expansion in an orthogonal basis. Composite unital noise and unitary component moment operators may be decomposed into support $\Gamma \subseteq [t]$-dependent unitary component projectors $\widehat{\mathcal{T}}^{(\Gamma)}_{\mathcal{U}}$, scaled by unital noise $\gamma_d \geq \gamma$, with deviations from the Depolarize term $\mathcal{D}_d^{\otimes t}$, scaled by non-unital noise $\eta_d \leq \eta$,

$$\widehat{\mathcal{T}}^{(t)k}_{\mathcal{UN}_\gamma} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} \ + \sum_{\Gamma \subseteq [t], s=|\Gamma| \geq l} \mathcal{O}\big((1-\gamma)^{sk}\big) \ \widehat{\mathcal{T}}^{(\Gamma)}_{\mathcal{U}} \tag{4.C.22}$$

$$\widehat{\mathcal{T}}^{(t)k}_{\mathcal{UN}_\eta} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} \ + \sum_{\substack{\Gamma \subseteq [t], s=|\Gamma| \geq l \\ P \in \mathcal{P}_d, \Gamma_P = \Gamma}} \mathcal{O}(\eta^s) \ |P\rangle\!\rangle\langle\!\langle I_d^{\otimes t}| \ + \ \cdots \tag{4.C.23}$$

$$\left\| \widehat{\mathcal{T}}^{(t)k}_{\mathcal{UN}_{\gamma\eta}} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} \right\|_p^p = \sum_{l \leq s \leq t} \binom{t}{s} \ r_s^{\frac{1}{p}} \ \mathcal{O}\big((1-\gamma)^{psk}\big) \tag{4.C.24}$$

$$\left\| \widehat{\mathcal{T}}^{(t)k}_{\mathcal{UN}_\eta} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} \right\|_2^2 = \sum_{l \leq s \leq t} \binom{t}{s} \ |\mathcal{P}_d \backslash \{I_d\}|^s \ \mathcal{O}\big(\eta^{2s}\big) \ \ . \tag{4.C.25}$$

There are $t|\mathcal{P}_d \backslash \{I_d\}|$ leading-order 1-locality non-unital terms, and $\binom{t}{l}$ leading-order l-locality unital terms. For example, Haar ensemble moment operator projectors, given they project onto the permutations, have rank $r_l = !l$, and $r = t!$. $\square$

In fact, we can also obtain bounds on norms of generic composite moment operators,

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}\mathcal{N}_{\gamma\eta}} \to \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}\mathcal{N}_{\gamma\eta}} = \widehat{\mathcal{N}}^{(t)}_{\gamma\eta}\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}} = \left[\begin{array}{cc} \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} & 0 \\ \widehat{\mathcal{N}}^{(t)}_{\eta} & \widehat{\mathcal{N}}^{(t)}_{\gamma\eta}\left(\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}\right) \end{array} \right] , \tag{4.C.26}$$

with ϵ-t-design unitary component ensembles $\mathcal{U}$ with respect to reference ensembles $\mathcal{U}'$,

$$\widehat{\mathcal{T}}_{\mathcal{U}} \to \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'} \ + \ \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}} \ - \ \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'} = \left[\begin{array}{cc} \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} & 0 \\ 0 & \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'} + \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} \end{array} \right] , \tag{4.C.27}$$

and with generic noise components $\mathcal{N}^{(t)}_{\gamma\eta}$,

$$\widehat{\mathcal{N}^{\otimes t}_{\gamma\eta}} \to \widehat{\mathcal{N}}^{(t)}_{\gamma\eta} = \left[\begin{array}{cc} \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} & 0 \\ \widehat{\mathcal{N}}^{(t)}_{\eta} & \widehat{\mathcal{N}}^{(t)}_{\gamma\eta} \end{array} \right], \tag{4.C.28}$$

with depolarizing $\widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}$, non-unital $\widehat{\mathcal{N}}^{(t)}_{\eta}$, and unital components $\widehat{\mathcal{N}}^{(t)}_{\gamma\eta}$, across the t copies,

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)} \quad , \quad \widehat{\mathcal{N}}^{(t)}_{\eta}\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}} = \widehat{\mathcal{N}}^{(t)}_{\eta} \quad , \quad \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}\widehat{\mathcal{N}}^{(t)}_{\eta} = \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}\widehat{\mathcal{N}}^{(t)}_{\gamma\eta} = \widehat{\mathcal{N}}^{(t)}_{\eta}\widehat{\mathcal{N}}^{(t)}_{\gamma\eta} = 0 \ . \tag{4.C.29}$$

For example, we may consider composite $d \to q^n$-dimensional spaces, with local noise,

$$\widehat{\mathcal{N}}_{\gamma\eta} \to \widehat{\mathcal{N}^{\otimes n}_{\gamma\eta}} = \left(\mathcal{D}_q + \widehat{\mathcal{N}^{(1)}_{\eta} + \mathcal{N}^{(1)}_{\gamma}}\right)^{\otimes n} = \left[\begin{array}{cc} \widehat{\mathcal{T}}_{\mathcal{D}(q)} & 0 \\ \widehat{\mathcal{N}}^{(1)}_{\eta} & \widehat{\mathcal{N}}^{(1)}_{\gamma} \end{array} \right]^{\otimes n} = \widehat{\mathcal{T}}^{(n)}_{\mathcal{D}(q)} + \widehat{\mathcal{N}}^{(n)}_{\eta} + \widehat{\mathcal{N}}^{(n)}_{\gamma\eta} \ , \tag{4.C.30}$$

with depolarizing $\widehat{\mathcal{T}}^{(n)}_{\mathcal{D}(q)}$, non-unital $\widehat{\mathcal{N}}^{(n)}_{\eta}$ and unital $\widehat{\mathcal{N}}^{(n)}_{\gamma\eta}$ components. For $q = 2$ qubits, local noise has, without loss of generality, non-unital $\widehat{\mathcal{N}}^{(1)}_{\eta}$ and diagonal unital components $\widehat{\mathcal{N}}^{(1)}_{\gamma\eta}$, up to unitaries [131].

Theorem 16 (Unital and Non-unital Noise and Random Unitary Channel Moment Operator Norms)**.** *k-concatenations of t-order moment operators for composite ensembles consisting of, ϵ-t-design unitary ensembles $\mathcal{U}$ with respect to a reference unitary ensemble $\mathcal{U}'$, and generic noise $\widehat{\mathcal{N}}_{\gamma\eta}^{(t)} = \widehat{\mathcal{T}}_{\mathcal{D}(d)}^{(t)} + \widehat{\mathcal{N}}_{\eta}^{(t)} + \widehat{\mathcal{N}}_{\gamma\eta}^{(t)}$ component, have norms of,*

$$\left\|\widehat{\mathcal{T}}_{\mathcal{U}\mathcal{N}_{\gamma\eta}}^{(t)k} - \widehat{\mathcal{T}}_{\mathcal{D}(d)}^{(t)}\right\|_p \leq \left\|\widehat{\mathcal{N}}_{\eta}^{(t)}\right\|_\infty \frac{\left\|\widehat{\mathcal{N}}_{\gamma\eta}^{(t)}\right\|_\infty^k \left(\left\|\widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{D}(d)}^{(t)}\right\|_p + \left\|\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)}\right\|_p\right)^k - 1}{\left\|\widehat{\mathcal{N}}_{\gamma\eta}^{(t)}\right\|_\infty \left(\left\|\widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{D}(d)}^{(t)}\right\|_p + \left\|\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)}\right\|_p\right) - 1} + \left\|\widehat{\mathcal{N}}_{\gamma\eta}^{(t)}\right\|_\infty^k \left(\left\|\widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{D}(d)}^{(t)}\right\|_p + \left\|\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)}\right\|_p\right)^k . \tag{4.C.31}$$

In the case of the reference $\mathcal{U}'$ having rank-r projector moment operators $\widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)}$, with locality-$\geq l$ terms, and strictly diagonal unital and non-unital noise, with bounded norms, $\|\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)}\|_\infty \leq \epsilon$, $\|\widehat{\mathcal{N}}_{\gamma}^{(t)}\|_\infty \leq 1-\gamma$, $\|\widehat{\mathcal{N}}_{\eta}^{(t)}\|_\infty \leq \eta$, we have the bounds,

$$\left\|\widehat{\mathcal{T}}_{\mathcal{U}\mathcal{N}_{\gamma\eta}}^{(t)k} - \widehat{\mathcal{T}}_{\mathcal{D}(d)}^{(t)}\right\|_p \ \leq \ \eta \, \frac{\left((1-\gamma)^l(r-1) + (1-\gamma)\epsilon\right)^k - 1}{((1-\gamma)^l(r-1) + (1-\gamma)\epsilon) - 1} + \ \left((1-\gamma)^l(r-1) + (1-\gamma)\epsilon\right)^k . \tag{4.C.32}$$

Proof. Given generic composite unitary and noise components, and norm inequalities,

$$\left\|\widehat{\mathcal{T}}^{(t)k}_{\mathcal{U}\mathcal{N}_{\gamma\eta}} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}\right\|_p = \left\|\left(\left(\widehat{\mathcal{N}}^{(t)}_{\eta} + \widehat{\mathcal{N}}^{(t)}_{\gamma\eta}\right)\left(\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'} + \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'}\right)\right)^k\right\|_p \tag{4.C.33}$$

$$\leq \left\|\sum_{s\in[k]}\left(\widehat{\mathcal{N}}^{(t)}_{\gamma\eta}\left(\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'} + \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'}\right)\right)^s\widehat{\mathcal{N}}^{(t)}_{\eta}\right\|_p \tag{4.C.34}$$

$$+ \left\|\left(\widehat{\mathcal{N}}^{(t)}_{\gamma\eta}\left(\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'} + \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'}\right)\right)^k\right\|_p$$

$$\leq \left\|\widehat{\mathcal{N}}^{(t)}_{\eta}\right\|_\infty\sum_{s\in[k]}\left\|\widehat{\mathcal{N}}^{(t)}_{\gamma\eta}\left(\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'} + \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'}\right)\right\|_p^s \tag{4.C.35}$$

$$+ \left\|\widehat{\mathcal{N}}^{(t)}_{\gamma\eta}\right\|_\infty^k\left(\left\|\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}\right\|_p + \left\|\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'}\right\|_p\right)^k$$

$$\leq \left\|\widehat{\mathcal{N}}^{(t)}_{\eta}\right\|_\infty\frac{\left\|\widehat{\mathcal{N}}^{(t)}_{\gamma\eta}\right\|_\infty^k\left(\left\|\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}\right\|_p + \left\|\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'}\right\|_p\right)^k - 1}{\left\|\widehat{\mathcal{N}}^{(t)}_{\gamma\eta}\right\|_\infty\left(\left\|\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}\right\|_p + \left\|\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'}\right\|_p\right) - 1} \tag{4.C.36}$$

$$+ \left\|\widehat{\mathcal{N}}^{(t)}_{\gamma\eta}\right\|_\infty^k\left(\left\|\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}\right\|_p + \left\|\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'}\right\|_p\right)^k .$$

Let us consider the case of the reference unitary ensemble $\mathcal{U}'$ having rank-r projector moment operators $\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'} = \sum_{\Gamma\subseteq[t]}\widehat{\mathcal{T}}^{(\Gamma)}_{\mathcal{U}'}$, with locality-$|\Gamma| \geq l$ terms, and there being strictly diagonal unital $\widehat{\mathcal{N}}^{(t)}_{\gamma\eta} = \widehat{\mathcal{N}}^{(t)}_{\gamma}$ and non-unital $\widehat{\mathcal{N}}^{(t)}_{\eta}$ noise, with bounded norms, $\|\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'}\|_\infty \leq \epsilon$, $\|\widehat{\mathcal{N}}^{(t)}_{\gamma}\|_\infty \leq 1-\gamma$, $\|\widehat{\mathcal{N}}^{(t)}_{\eta}\|_\infty \leq \eta$. The diagonal unital noise $\widehat{\mathcal{N}}^{(t)}_{\gamma} = \widehat{\mathcal{N}}^{\otimes t}_{\gamma}$ thus scales each projective moment operator term according to its locality-$|\Gamma| \geq l$,

$$\left\|\left(\widehat{\mathcal{N}}^{(t)}_{\gamma}\left(\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}\right)\right)^s\right\|_p = \left\|\sum_{\substack{\Gamma\subseteq[t]\\|\Gamma|\geq l}}\widehat{\mathcal{N}}^{(\Gamma)s}_{\gamma}\widehat{\mathcal{T}}^{(\Gamma)}_{\mathcal{U}'}\right\|_p \tag{4.C.37}$$

$$\leq \sum_{\substack{\Gamma\subseteq[t]\\|\Gamma|\geq l}}\left\|\widehat{\mathcal{N}}^{(\Gamma)s}_{\gamma}\right\|_\infty\left\|\widehat{\mathcal{T}}^{(\Gamma)}_{\mathcal{U}'}\right\|_p \tag{4.C.38}$$

$$\leq \left\|\widehat{\mathcal{N}}_{\gamma}\right\|_\infty^{ls}\left\|\widehat{\mathcal{T}}^{(t)}_{\mathcal{U}'} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}\right\|_p , \tag{4.C.39}$$

$$\left\|\left(\widehat{\mathcal{N}}_\gamma^{(t)}\left(\widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{D}(d)}^{(t)}\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)}\right)\right)^s\right\|_p \leq \left(\left\|\widehat{\mathcal{N}}_\gamma\right\|_\infty^l \left\|\widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{D}(d)}^{(t)}\right\|_p + \left\|\widehat{\mathcal{N}}_\gamma\right\|_\infty \left\|\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)}\right\|_p\right)^s. \tag{4.C.40}$$

Therefore, the norms are bounded as,

$$\left\|\widehat{\mathcal{T}}_{\mathcal{U}\mathcal{N}_{\gamma\eta}}^{(t)k} - \widehat{\mathcal{T}}_{\mathcal{D}(d)}^{(t)}\right\|_p \leq \left\|\sum_{s\in[k]}\left(\widehat{\mathcal{N}}_\gamma^{(t)}\left(\widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)} + \widehat{\mathcal{T}}_{\mathcal{U}}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)}\right)\right)^s \widehat{\mathcal{N}}_\eta^{(t)}\right\|_p + \left\|\left(\widehat{\mathcal{N}}_\gamma^{(t)}\left(\widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)} + \widehat{\mathcal{T}}_{\mathcal{U}}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)}\right)\right)^k\right\|_p \tag{4.C.41}$$

$$\leq \left\|\widehat{\mathcal{N}}_\eta^{(t)}\right\|_\infty \sum_{s\in[k]}\left(\left\|\widehat{\mathcal{N}}_\gamma\right\|_\infty^l \left\|\widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{D}(d)}^{(t)}\right\|_p + \left\|\widehat{\mathcal{N}}_\gamma\right\|_\infty \left\|\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)}\right\|_p\right)^s + \left(\left\|\widehat{\mathcal{N}}_\gamma\right\|_\infty^l \left\|\widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{D}(d)}^{(t)}\right\|_p + \left\|\widehat{\mathcal{N}}_\gamma\right\|_\infty \left\|\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{U}'}^{(t)}\right\|_p\right)^k \tag{4.C.42}$$

$$\leq \eta\ \frac{\left((1-\gamma)^l(r-1) + (1-\gamma)\epsilon\right)^k - 1}{((1-\gamma)^l(r-1) + (1-\gamma)\epsilon) - 1} + \left((1-\gamma)^l(r-1) + (1-\gamma)\epsilon\right)^k . \tag{4.C.43}$$

If the unitary ensemble is an exact t-design with respect to the reference ensemble, there is the further simplification,

$$\left\|\widehat{\mathcal{T}}_{\mathcal{U}\mathcal{N}_{\gamma\eta}}^{(t)k} - \widehat{\mathcal{T}}_{\mathcal{D}(d)}^{(t)}\right\|_p \leq \left\|\widehat{\mathcal{N}}_\eta^{(t)}\right\|_\infty \sum_{s\in[k]}\left\|\left(\widehat{\mathcal{N}}_\gamma^{(t)}\left(\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{D}(d)}^{(t)}\right)\right)^s\right\|_p + \left\|\left(\widehat{\mathcal{N}}_\gamma^{(t)}\left(\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{D}(d)}^{(t)}\right)\right)^k\right\|_p \tag{4.C.44}$$

$$\leq \left\|\widehat{\mathcal{N}}_\eta^{(t)}\right\|_\infty \sum_{s\in[k]}\left\|\widehat{\mathcal{N}}_\gamma\right\|_\infty^{ls}\left\|\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{D}(d)}^{(t)}\right\|_p + \left\|\widehat{\mathcal{N}}_\gamma\right\|_\infty^{lk}\left\|\widehat{\mathcal{T}}_{\mathcal{U}}^{(t)} - \widehat{\mathcal{T}}_{\mathcal{D}(d)}^{(t)}\right\|_p \tag{4.C.45}$$

$$\leq \eta\ (r-1)\ \frac{(1-\gamma)^{lk} - 1}{(1-\gamma)^l - 1} + (r-1)\ (1-\gamma)^{lk} . \tag{4.C.46}$$

□

4.C.3 Unital Noise and Parametrized Random Unitary Channels

Here, we consider parametrized random unitary channels $\mathcal{U} = \mathcal{G}_\Theta$, and fixed diagonal unital noise $\mathcal{N} = \mathcal{N}_\gamma$, and derive the scaling of their k-concatenated t-order composite moment operators with t, k, d, γ.

For the unitary component $\mathcal{U} = \mathcal{G}_\Theta$, we assume parametrized unitary channels

$$\mathcal{G}_\Theta = \{U_\theta^G = e^{-i\theta G} \ , \ \theta \sim \Theta\} \ , \tag{4.C.47}$$

generated by single hermitian, involutory operators $G \in \mathcal{P}_d$, with a distribution of parameters Θ, and a basis for the generator's commutant of $\mathcal{S}_G \subseteq \mathcal{P}_d$. t copies of the parametrized unitaries for involutory generators G have the form,

$$U_\theta^{G\otimes t} \ = \sum_{l\in[t+1]} u_\theta^{(l)} \ G^{(l)} \quad : \quad G^{(l)} = \sum_{\substack{\Gamma\subseteq[t] \\ |\Gamma|=l}} \otimes_{i\in\Gamma} \ G_i \tag{4.C.48}$$

$$u_\theta^{(l)} = (-i)^l \ \cos^t(\theta) \ \tan^l(\theta) \ , \tag{4.C.49}$$

and the associated t-order moment operators are functions of trigonometric moments of the parameter distribution,

$$\widehat{\mathcal{T}}^{(t)}_{G\mathcal{G}_\Theta} \ = \ \left\langle U_\theta^{G\,\otimes t} \otimes U_\theta^{G\,\otimes t\,*} \right\rangle_{\theta\sim\Theta} \ = \sum_{l,l'\in[t+1]} \left\langle u_\theta^{(l)} u_\theta^{(l')*} \right\rangle_{\theta\sim\Theta} \ G^{(l)} \otimes G^{(l')*} \ . \tag{4.C.50}$$

Crucial to this analysis is an understanding of the t-order commutant of parametrized random unitaries $\mathcal{S}^{(t)}_{\mathcal{G}_\Theta}$, which depends non-trivially on the generator commutant $\mathcal{S}_G$, and can be described by the locality $l \in [t+1]$ of its operators,

$$\mathcal{S}^{(t)}_{\mathcal{G}_\Theta} = \bigcup_{l\in[t+1]} \mathcal{S}^{(t|l)}_{\mathcal{G}_\Theta} \ : \ \mathcal{S}^{(t|l)}_{\mathcal{G}_\Theta} = \{X \in \mathcal{S}^{(t)}_{\mathcal{G}_\Theta} \ : \ |X| = l\} \ . \tag{4.C.51}$$

A complete understanding of the entire commutant is non-trivial, and is left for future studies. However, given the form of the moment operator, and that higher t-order commutants include lower $l \leq t$-locality commutants, therefore the $l = 1$-locality local component of commutants is the generator commutant at each of the t copies of the space,

$$\mathcal{S}^{(t|1)}_{\mathcal{G}_\Theta} = \bigcup_{i\in[t]} \mathcal{S}_{G_i} \ . \tag{4.C.52}$$

We can also evaluate the trigonometric moments involved in this moment operator exactly for specific $t = 1, 2$.

Lemma 6 ($t = 1, 2$ Twirls of parametrized Random Unitaries)**.** *Twirls over $t = 1, 2$-order parametrized random unitary ensembles $e^{-i\theta G} \sim \mathcal{G}_\Theta$, given Hermitian involutory generators G, uniform parameter distributions Θ, and $X \in \mathcal{B}[\mathcal{H}]$, are*

$$\mathcal{T}^{(1)}_{\mathcal{G}_\Theta}(X) = \frac{1}{2}\left[X + GXG\right] \tag{4.C.53}$$

$$\begin{aligned}\mathcal{T}^{(2)}_{\mathcal{G}_\Theta}(X) = \frac{1}{8}\big[&3\left(X + (G\otimes G)\,X\,(G\otimes G)\right) \\ &- \ \{G\otimes G, X\} + (I\otimes G + G\otimes I)\,X\,(I\otimes G + G\otimes I)\big]\ .\end{aligned} \tag{4.C.54}$$

Given these insights into the local component of the commutant, the t-order $\mathcal{G}_\Theta$ ensemble moment operator is

$$\widehat{\mathcal{T}}^{(t)}_{\mathcal{G}_\Theta} = \frac{1}{d^t}\sum_{P\in\mathcal{S}^{(t|1)}_{\mathcal{G}_\Theta}} |P\rangle\!\rangle\langle\!\langle P| \ + \ \frac{1}{d^t}\sum_{P,S\in\mathcal{S}^{(t)}_{\mathcal{G}_\Theta}\setminus\mathcal{S}^{(t|1)}_{\mathcal{G}_\Theta}} \tau^{(t,k)}_{\mathcal{G}_\Theta}(P,S)\ |P\rangle\!\rangle\langle\!\langle S|\ . \tag{4.C.55}$$

For the noise component, the moment operator $\mathcal{N} = \mathcal{N}_\gamma$ has the specific form of diagonal unital noise components $\gamma_d(P) \geq \gamma$ for $P \in \mathcal{S}_{\mathcal{N}_\gamma}$, in an orthogonal basis $\mathcal{S}_{\mathcal{N}_\gamma} = \mathcal{P}_d$,

$$\widehat{\mathcal{N}}_\gamma = \frac{1}{d}\begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1-\gamma_d(S) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1-\gamma_d(P) \end{bmatrix}_{P,S\in\mathcal{P}_d\setminus\{I_d\}} \tag{4.C.56}$$

$$\widehat{\mathcal{N}^{\otimes t}_\gamma} = \frac{1}{d^t}\sum_{P,S\in\mathcal{S}^{(t)}_{\mathcal{N}_\gamma}} \tau^{(t)}_{\mathcal{N}_\gamma}(P,S)\ |P\rangle\!\rangle\langle\!\langle S|\ , \tag{4.C.57}$$

The t-order noise component transfer matrix $\tau^{(t)}_{\mathcal{N}_\gamma}$, for basis operators $P, S \in \mathcal{S}^{(t)}_{\mathcal{N}_\gamma} = \mathcal{P}^t_d$, is

$$\tau^{(t)}_{\mathcal{N}_\gamma}(P,S) = \prod_{i\in\Gamma_S}(1-\gamma_d(S_i))\ \delta_{PS} \ \in \ \mathcal{O}\big((1-\gamma)^{|S|}\big)\ \delta_{PS}\ . \tag{4.C.58}$$

The t-order noise component basis can also be partitioned by the locality of its operators,

$$\mathcal{S}^{(t)}_{\mathcal{N}_\gamma} = \bigcup_{l\in[t+1]} \mathcal{S}^{(t|l)}_{\mathcal{N}_\gamma} \ : \ \mathcal{S}^{(t|l)}_{\mathcal{N}_\gamma} = \{X \in \mathcal{S}^{(t)}_{\mathcal{N}_\gamma} \ : \ |X| = l\}\ . \tag{4.C.59}$$

and the local component is the noise component basis at each of the t copies of the space,

$$\mathcal{S}^{(t|1)}_{\mathcal{N}_\gamma} = \cup_{i\in[t]}\mathcal{S}_{\mathcal{N}_\gamma i}. \tag{4.C.60}$$

Given the transfer matrices for the unitary and noise components, and that their respective bases satisfy $\mathcal{S}_G \subseteq \mathcal{S}_{\mathcal{N}_\gamma} = \mathcal{P}_d \;\; \rightarrow \;\; \mathcal{S}^{(t|1)}_{\mathcal{G}_\Theta} \subseteq \mathcal{S}^{(t|1)}_{\mathcal{N}_\gamma} = \cup_{i\in[t]}\mathcal{P}_{d_i}$, the k-concatenated t-order composite moment operator is

$$\widehat{\mathcal{T}}^{(t)k}_{\mathcal{G}_\Theta\mathcal{N}_\gamma} = \frac{1}{d^t}\sum_{P,S\in\mathcal{S}^{(t|1)}_{\mathcal{G}_\Theta}} \tau^{(t,k)}_{\mathcal{G}_\Theta\mathcal{N}_\gamma}(P,S)\;|P\rangle\!\rangle\langle\!\langle S| \;+\; \frac{1}{d^t}\sum_{\substack{P\in\mathcal{S}^{(t)}_{\mathcal{N}_\gamma}\backslash\mathcal{S}^{(t|1)}_{\mathcal{N}_\gamma}\\ S\in\mathcal{S}^{(t)}_{\mathcal{G}_\Theta}\backslash\mathcal{S}^{(t|1)}_{\mathcal{G}_\Theta}}} \tau^{(t,k)}_{\mathcal{G}_\Theta\mathcal{N}_\gamma}(P,S)\;|P\rangle\!\rangle\langle\!\langle S| \;, \quad (4.C.61)$$

with diagonal k-concatenated t-order composite transfer matrices, for the local $l = 1$ operators $P,S\in\mathcal{S}^{(t|1)}_{\mathcal{G}_\Theta}$, of,

$$\tau^{(t,k)}_{\mathcal{G}_\Theta\mathcal{N}_\gamma}(P,S) = \prod_{i\in\Gamma_S}(1-\gamma_d(S_i))^k\;\delta_{PS} \;\in\; \mathcal{O}\big((1-\gamma)^{k|S|}\big)\;\delta_{PS} \quad : \quad P,S\in\mathcal{S}^{(t|1)}_{\mathcal{G}_\Theta}\;. \quad (4.C.62)$$

Finally, we can assess the leading-order scaling of moment operator norms for k-concatenations of composite ensembles, with respect to the noise scale.

Theorem 17 (Unital Noise and Parametrized Random Unitary Channel Moment Operator Norms)**.** *k-concatenations of composite ensembles, of parametrized random unitaries, generated by involutory operators G with commutants $\mathcal{S}_G$, and diagonal unital noise $\mathcal{N}_\gamma$ with noise coefficients $\gamma_d(P)\geq\gamma$ for $P\in\mathcal{S}_{\mathcal{N}_\gamma}\backslash\{I_d\}$, have moment operator norms of,*

$$\left\|\widehat{\mathcal{T}}^{(t)k}_{\mathcal{G}_\Theta\mathcal{N}_\gamma} - \widehat{\mathcal{T}}^{(t)}_{\mathcal{D}(d)}\right\|^2_2 = t\;|\mathcal{S}_G\backslash\{I_d\}|\;\mathcal{O}\big((1-\gamma)^{2k}\big)\;. \quad (4.C.63)$$

Therefore unital noise decreases moment operator norms for $\gamma > 0$, and as $k\rightarrow\infty$, approaches a Depolarize t-design.

Proof. Leading-order behaviour of the composite moment operators is associated with local operators in the commutant, $\mathcal{S}^{(t|1)}_{\mathcal{G}_\Theta} = \cup_i\mathcal{S}_{G_i}$. Under the assumption of the intersection of the unitary and noise component bases of $\mathcal{S}^{(t|1)}_{\mathcal{G}_\Theta}\subseteq\mathcal{S}^{(t|1)}_{\mathcal{N}_\gamma}$, the local operator components of the associated moment operator transfer matrices are invariant under k-concatenations, up to scaling by diagonal noise coefficients. The moment operator norm is thus,

$$\left\|\widehat{\mathcal{T}}^{(t)k}_{\mathcal{G}_\Theta\mathcal{N}_\gamma} - \widehat{\mathcal{D}}^{\otimes t}_d\right\|_2^2 = \sum_{P\in\mathcal{S}^{(t|1)}_{\mathcal{G}_\Theta}\backslash\{I_d^{\otimes t}\}} \mathcal{O}\big((1-\gamma)^{2k}\big) \tag{4.C.64}$$

$$+ \sum_{\substack{P,T\in\mathcal{S}^{(t)}_{\mathcal{N}_\gamma}\backslash\mathcal{S}^{(t|1)}_{\mathcal{N}_\gamma}\\ S,Q\in\mathcal{S}^{(t)}_{\mathcal{G}_\Theta}\backslash\mathcal{S}^{(t|1)}_{\mathcal{G}_\Theta}}} \chi^{(t)}_d(P,T)\chi^{(t)}_d(S,Q)\tau^{(t,k)}_{\mathcal{G}_\Theta\mathcal{N}_\gamma}(P,Q)\tau^{(t,k)}_{\mathcal{G}_\Theta\mathcal{N}_\gamma}(T,S)$$

$$= t\ |\mathcal{S}_G\backslash\{I_d\}|\ \mathcal{O}\big((1-\gamma)^{2k}\big)\ +\ \mathcal{O}\big((1-\gamma)^{2k+2}\big)\ . \tag{4.C.65}$$

□

Given both Haar random and parametrized unitary ensembles with unital noise approach a Depolarize t-design, this suggests that unital noise may universally impose depolarization on arbitrary unitary ensembles.

4.D Designs and Trainability

In these appendices, we generalize previous results [106], concerning design properties of unitary ansatz, to the case of general quantum channel ansatze. In particular, we analyse the relationship between what is known as expressivity, or the design properties of ensembles, to trainability, or concentration phenomena of ensemble dependent objectives. By using our derived expressions for channel moment operators, we can bound biases and variances of expectation values and their gradients of functions over channel ensembles. Such bounds allow us to study concentration properties of expectation values, leading to notions of channel-design-induced concentration phenomena.

4.D.1 Expectation Values and Gradients

Let us denote inputs $X, \Pi \in \mathcal{B}[\mathcal{H}],\ Z \in \mathcal{B}[\mathcal{H}^{\otimes t}]$, acted on with t-copies of operators $\Lambda \sim \mathcal{C}$ from an ensemble $\mathcal{C}$, as,

$$X_\Lambda^{\otimes t} = \Lambda^{\otimes t}(X^{\otimes t}) \quad , \quad \Pi_{\Lambda^\dagger}^{\otimes t} = \Lambda^{\otimes t \dagger}(\Pi^{\otimes t}) \quad , \quad Z_\Lambda = \Lambda^{\otimes t}(Z) \ . \tag{4.D.1}$$

with corresponding, not necessarily physical, adjoint operators $\Lambda^\dagger$. With respect to the inner product, the adjoint satisfies the relation for either a forward evolved input X, or backward evolved observable Π, such that $\mathrm{tr}(X_\Lambda O) = \mathrm{tr}(X \Pi_{\Lambda^\dagger})$.

Operator-dependent functions $\mathcal{F}_\Lambda$ of t-order inputs in $\mathcal{B}[\mathcal{H}^{\otimes t}]$, have general moments, and specific means and variances, of,

$$\langle \mathcal{F}_\Lambda(\cdot) \rangle_{\mathcal{C}} = \int_{\mathcal{C}} d\Lambda\ \mathcal{F}_\Lambda(\cdot) \quad , \quad \mu_{\mathcal{F}|\mathcal{C}}(\cdot) = \langle \mathcal{F}_\Lambda(\cdot) \rangle_{\mathcal{C}} \quad , \quad \sigma^2_{\mathcal{F}|\mathcal{C}}(\cdot) = \left\langle \mathcal{F}_\Lambda(\cdot)^2 \right\rangle_{\mathcal{C}} - \langle \mathcal{F}_\Lambda(\cdot) \rangle^2_{\mathcal{C}} \ . \tag{4.D.2}$$

Here, we consider the concentration of functions, within the context of the Chebyshev bound for the probability of quantities differing from their mean by an amount ε, being related to their variance,

$$p\left(\left| \mathcal{F}_\Lambda(\cdot) - \mu_{\mathcal{F}|\mathcal{C}}(\cdot) \right| \geq \varepsilon \right) \leq \frac{\sigma^2_{\mathcal{F}|\mathcal{C}}(\cdot)}{\varepsilon^2} \ . \tag{4.D.3}$$

Generally quantities concentrate due to inherent terms related to taking moments over high dimensional spaces, plus the expressivity terms related to the proximity of ensembles $\mathcal{C}$ to reference ensembles $\mathcal{C}'$ as t-designs. Differences to reference ensembles are $\Delta^{(t)}_{\mathcal{C}\mathcal{C}'}(Z) = \langle Z_\Lambda \rangle_{\mathcal{C}} - \langle Z_\Lambda \rangle_{\mathcal{C}'}$, with p-norms $\epsilon^{(t|p)}_{\mathcal{C}\mathcal{C}'}(Z) = \|\Delta^{(t)}_{\mathcal{C}\mathcal{C}'}(Z)\|_p \leq \epsilon^{(t|p)}_{\mathcal{C}\mathcal{C}'} \|Z\|_p$, for $Z \in \mathcal{B}[\mathcal{H}^{\otimes t}]$.

For example, given the identity I_d and swap operators S_d, moments for the cHaar ensemble are, given inputs $X \in \mathcal{B}[\mathcal{H}],\ Z \in \mathcal{B}[\mathcal{H}^{\otimes 2}]$,

$$\langle X_\Lambda \rangle_{\mathcal{C}(d,s)} = \frac{\mathrm{tr}(X)}{d} I_d \tag{4.D.4}$$

$$\langle Z_\Lambda \rangle_{\mathcal{C}(d,s)} = s^2 \frac{ds\,\mathrm{tr}(Z) - \mathrm{tr}(S_d Z)}{ds(d^2 s^2 - 1)} I_d^{\otimes 2} \ + \ s \frac{-\mathrm{tr}(Z) + ds\,\mathrm{tr}(S_d Z)}{ds(d^2 s^2 - 1)} S_d \ . \tag{4.D.5}$$

and moments for the Depolarize ensemble are, given inputs $Z \in \mathcal{B}[\mathcal{H}^{\otimes t}]$,

$$\langle Z_\Lambda \rangle_{\mathcal{D}(d)} = \frac{\mathrm{tr}(Z)}{d^t} I_d^{\otimes t} \ . \tag{4.D.6}$$

Let us assume operators have the form of a concatenation of operators,

$$\Lambda = \ \circ_\mu \Lambda_\mu \quad , \quad \Lambda_\mu = \mathcal{U}_\mu \circ \mathcal{N}_\mu \ : \ U_\mu = e^{-i\theta_\mu G_\mu} \sim \mathcal{G}_\Theta \tag{4.D.7}$$

where each operator consists of variable unitaries $U_\mu \in \mathcal{G}_\Theta$ with parameters $\theta = \{\theta_\mu\} \sim \Theta$ and generators $G = \{G_\mu\}$, and constant non-unitaries $\mathcal{N}_{\gamma_\mu}$ with parameters γ. We also denote sets of concatenated operators $\Lambda_{\mu_{R,L}} \sim \mathcal{C}_{\mu_{R,L}}$ from ensembles $\mathcal{C}_{\mu_{R,L}}$ before (R) or after (L) an index μ, with corresponding evolved inputs and observables,

$$\Lambda_{\mu_R} = \circ_{\nu \leq \mu} \Lambda_\nu \quad , \quad \Lambda_{\mu_L} = \circ_{\nu > \mu} \Lambda_\nu \quad , \quad X_{\mu_R} = X_{\Lambda_{\mu_R}} \quad , \quad \Pi_{\mu_L} = \Pi_{\Lambda^\dagger_{\mu_L}} \tag{4.D.8}$$

Gradients of the operators with respect to the parameters θ_μ therefore take the form

$$\partial_\mu X_\Lambda = \ - i \Lambda_{\mu_L}([G_\mu, X_{\mu_R}]) \ . \tag{4.D.9}$$

Here, we will focus on properties of objective functions, with respect to operators Λ, states $\rho \in \mathcal{B}[\mathcal{H}]$, and observables $\Pi \in \mathcal{B}[\mathcal{O}]$ with support over a subspace $\mathcal{O} \subseteq H$,

$$\mathcal{L}_\Lambda(\rho, \Pi) = \mathrm{tr}(\rho_\Lambda O) \ \ , \tag{4.D.10}$$

and due to the cyclicity of the trace, gradients of such objective functions are,

$$\partial_\mu \mathcal{L}_\Lambda(\rho, \Pi) = \mathrm{tr}\left(\rho^G_{\mu_R} \Pi_{\mu_L}\right) = \mathrm{tr}\left(\rho_{\mu_R} \Pi^G_{\mu_L}\right) \tag{4.D.11}$$

$$\mathcal{L}_\Lambda(\rho, \Pi) \mathcal{L}_\Lambda(\rho, \Pi) = \mathrm{tr}\left(\rho_\Lambda^{\otimes 2} \Pi^{\otimes 2}\right) \tag{4.D.12}$$

$$\partial_\mu \mathcal{L}_\Lambda(\rho, \Pi) \partial_\nu \mathcal{L}_\Lambda(\rho, \Pi) = \mathrm{tr}\left(\rho^G_{\mu_R} \otimes \rho^G_{\nu_R} \Pi_{\mu_L} \otimes \Pi_{\nu_L}\right) \tag{4.D.13}$$

$$\rho^G_{\mu_R} = -i[G_\mu, \rho_{\mu_R}] \quad , \quad \Pi^G_{\mu_L} = +i[G_\mu, \Pi_{\mu_L}] \ . \tag{4.D.14}$$

Here, we study the mean and variance of channel-dependent functions and their gradients, and how they decay with dimension d, s to conclude if concentration occurs. Such expressions are with respect to quantum state inputs $\rho \in \mathcal{B}[\mathcal{H}]$, hermitian observables $\Pi \in \mathcal{B}[\mathcal{O}]$, and involutory generators $G_\mu \in \mathcal{B}[\mathcal{H}] \ : \ G_\mu^2 = I$. $\mathcal{O} \subseteq \mathcal{H}$ is a subspace of dimension $l \leq d$, with complement space $\overline{\mathcal{O}} \subseteq \mathcal{H}$ of dimension d/l, with identity and swap operators I_l, S_l. Such expressions are also with respect to reference ensembles $\mathcal{C}'$, such as the cHaar $\mathcal{C}' = \mathcal{C}(d,s)$ or Depolarize $\mathcal{C}' = \mathcal{D}(d)$ ensembles.

To bound various quantities, we will use Schatten norms $\|A\|_p^p = \mathrm{tr}(|A|^p)\ ,\ p \in [1,\infty]$ for $A, B \in \mathcal{B}[\mathcal{H}^{\otimes t}]$, which satisfy monotonicity $\|A\|_p \leq \|A\|_q\ ,\ q \leq p$, sub-additivity $\|A+B\|_p \leq \|A\|_p + \|B\|_p$, sub-multiplicativity $\|AB\|_p \leq \|A\|_p \|B\|_p$, and Holder's $\|AB\|_r \leq \|A\|_p \|B\|_q\ ,\ 1/p + 1/q = 1/r$, and von-Neumann $|\mathrm{tr}(AB)| \leq \|A\|_p \|B\|_q\ ,\ 1/p + 1/q = 1$ inequalities [177]. Finally, t-order twirls of ensembles $\mathcal{C}$, given $Z \in \mathcal{B}[\mathcal{H}^{\otimes t}]$, can be written in terms of reference ensembles $\mathcal{C}'$, as $\langle Z_\Lambda \rangle_{\mathcal{C}} = \langle Z_\Lambda \rangle_{\mathcal{C}'} + \Delta^{(t)}_{\mathcal{C}\mathcal{C}'}(Z)$, $\langle Z_{\Lambda^\dagger} \rangle_{\mathcal{C}} = \langle Z_{\Lambda^\dagger} \rangle_{\mathcal{C}'} + \Delta^{(t|\dagger)}_{\mathcal{C}\mathcal{C}'}(Z)$, where the deviation has a p-norm of $\epsilon^{(t|p)}_{\mathcal{C}\mathcal{C}'}(Z) = \|\Delta^{(t)}_{\mathcal{C}\mathcal{C}'}(Z)\|_p$, and $\epsilon^{(t|\dagger p)}_{\mathcal{C}\mathcal{C}'}(Z) = \|\Delta^{(t|\dagger)}_{\mathcal{C}\mathcal{C}'}(Z)\|_p$.

4.D.2 Expectation Value Bias

Let us study the mean of expectation values $\mathcal{L}$,

$$\mu_{\mathcal{L}|\mathcal{C}}(\rho,\Pi) = \langle \mathcal{L}_\Lambda(\rho,\Pi) \rangle_{\mathcal{C}} \quad \to \quad \left| \langle |\mathcal{L}_\Lambda - \langle \mathcal{L}_\Lambda(\rho,\Pi) \rangle_{\mathcal{C}'} | \rangle_{\mathcal{C}} \right| , \tag{4.D.15}$$

and in particular, the bias of expectation values from $\mu_{\mathcal{L}|\mathcal{C}'}$.

Theorem 18 (Expectation Value Bias)**.** *Let $\mathcal{C}, \mathcal{C}'$ be ensembles of quantum channels, let ρ be a quantum state, and let Π be an observable that acts non-trivially on a subspace $\mathcal{O} \subseteq \mathcal{H}$ of dimension l. The following bound on the expectation value $\mathcal{L}_\Lambda$ bias holds,*

$$\langle |\mathcal{L}_\Lambda(\rho,\Pi) - \langle \mathcal{L}_\Lambda(\rho,\Pi) \rangle_{\mathcal{C}'} | \rangle_{\mathcal{C}} \ \leq \ \delta\mu_{\mathcal{L}|\mathcal{C}'}(\rho,\Pi) \ + \ \delta\mu_{\mathcal{L}|\mathcal{C}\mathcal{C}'}(\rho,\Pi) \ , \tag{4.D.16}$$

where the inherent $\delta\mu_{\mathcal{L}|\mathcal{C}'}$ and expressivity $\delta\mu_{\mathcal{L}|\mathcal{C}\mathcal{C}'}$ expectation value bias terms are,

$$\delta\mu_{\mathcal{L}|\mathcal{C}'}(\rho,\Pi) = \sqrt{\mathrm{tr}\left(\left(\langle \rho_\Lambda^{\otimes 2} \rangle_{\mathcal{C}'} - \langle \rho_\Lambda \rangle_{\mathcal{C}'}^{\otimes 2} \right) \left(S_l \otimes I_{d/l}^{\otimes 2} \right) \right)} \ \sqrt{l} \ \|\Pi\|_\infty \tag{4.D.17}$$

$$\delta\mu_{\mathcal{L}|\mathcal{C}\mathcal{C}'}(\rho,\Pi) = \sqrt{\mathrm{tr}\left(\left(\begin{array}{l} \Delta^{(2)}_{\mathcal{C}\mathcal{C}'}\ (\rho^{\otimes 2}) \\ \quad -\Delta_{\mathcal{C}\mathcal{C}'}(\rho) \ \otimes \ \langle \rho_\Lambda \rangle_{\mathcal{C}'} \\ \quad - \langle \rho_\Lambda \rangle_{\mathcal{C}'} \otimes \ \Delta_{\mathcal{C}\mathcal{C}'}(\rho) \end{array} \right) \left(S_l \otimes I_{d/l}^{\otimes 2} \right) \right)} \ \sqrt{l} \ \|\Pi\|_\infty \ . \tag{4.D.18}$$

Proof. To simplify our analysis of the bias of expectation values, we will denote the deviation of a transformed input $\rho \to \rho_\Lambda \in \mathcal{B}[\mathcal{H}]$, where $\Lambda \sim \mathcal{C}$, from its mean with respect to a reference ensemble $\mathcal{C}'$, as the following, with $t = 1, 2$-order moments of these deviations, with respect to an ensemble $\mathcal{C}$, of,

$$\tilde{\rho}_{\Lambda|\mathcal{C}'} = \rho_\Lambda - \langle \rho_\Lambda \rangle_{\mathcal{C}'} \tag{4.D.19}$$

$$\left\langle \tilde{\rho}_{\Lambda|\mathcal{C}'} \right\rangle_{\mathcal{C}} = \Delta_{\mathcal{C}\mathcal{C}'}(\rho) \tag{4.D.20}$$

$$\left\langle \tilde{\rho}^{\otimes 2}_{\Lambda|\mathcal{C}'} \right\rangle_{\mathcal{C}} = \left\langle \rho_\Lambda^{\otimes 2} \right\rangle_{\mathcal{C}'} - \langle \rho_\Lambda \rangle_{\mathcal{C}'}^{\otimes 2} + \Delta^{(2)}_{\mathcal{C}\mathcal{C}'}(\rho^{\otimes 2}) - \Delta_{\mathcal{C}\mathcal{C}'}(\rho) \otimes \langle \rho_\Lambda \rangle_{\mathcal{C}'} - \langle \rho_\Lambda \rangle_{\mathcal{C}'} \otimes \Delta_{\mathcal{C}\mathcal{C}'}(\rho)\ . \tag{4.D.21}$$

The average expectation value bias can then be bounded using the norm inequalities,

$$\begin{aligned}
&\langle | \mathcal{L}_\Lambda(\rho, \Pi) - \langle \mathcal{L}_\Lambda(\rho, \Pi) \rangle_{\mathcal{C}'} | \rangle_{\mathcal{C}} \\
=\ & \langle | \mathrm{tr}(\rho_\Lambda \Pi) - \langle \mathrm{tr}(\rho_\Lambda \Pi) \rangle_{\mathcal{C}'} | \rangle_{\mathcal{C}} && (4.D.22)\\
=\ & \left\langle \left| \mathrm{tr}\left(\tilde{\rho}_{\Lambda|\mathcal{C}'} \Pi \right) \right| \right\rangle_{\mathcal{C}} && (4.D.23)\\
=\ & \left\langle \left| \mathrm{tr}_{\mathcal{O}}\left(\mathrm{tr}_{\overline{\mathcal{O}}}\left(\tilde{\rho}_{\Lambda|\mathcal{C}'} \right) O \right) \right| \right\rangle_{\mathcal{C}} && (4.D.24)\\
\leq\ & \left\langle \left\| \mathrm{tr}_{\overline{\mathcal{O}}}\left(\tilde{\rho}_{\Lambda|\mathcal{C}'} \right) \right\|_1 \|\Pi\|_\infty \right\rangle_{\mathcal{C}} && (4.D.25)\\
\leq\ & \left\langle \left\| \mathrm{tr}_{\overline{\mathcal{O}}}\left(\tilde{\rho}_{\Lambda|\mathcal{C}'} \right) \right\|_2 \|I_l\|_2 \|\Pi\|_\infty \right\rangle_{\mathcal{C}} && (4.D.26)\\
=\ & \left\langle \sqrt{\left\| \mathrm{tr}_{\overline{\mathcal{O}}}\left(\tilde{\rho}_{\Lambda|\mathcal{C}'} \right) \right\|_2^2} \right\rangle_{\mathcal{C}} \|I_l\|_2 \|\Pi\|_\infty && (4.D.27)\\
\leq\ & \sqrt{\left\langle \left\| \mathrm{tr}_{\overline{\mathcal{O}}}\left(\tilde{\rho}_{\Lambda|\mathcal{C}'} \right) \right\|_2^2 \right\rangle_{\mathcal{C}}} \|I_l\|_2 \|\Pi\|_\infty && (4.D.28)\\
=\ & \sqrt{\left\langle \mathrm{tr}_{\mathcal{O}}\left(\mathrm{tr}^2_{\overline{\mathcal{O}}}\left(\tilde{\rho}_{\Lambda|\mathcal{C}'} \right) \right) \right\rangle_{\mathcal{C}}} \|I_l\|_2 \|\Pi\|_\infty && (4.D.29)\\
=\ & \sqrt{\left\langle \mathrm{tr}_{\mathcal{O}^{\otimes 2}}\left(\mathrm{tr}_{\overline{\mathcal{O}}}\left(\tilde{\rho}_{\Lambda|\mathcal{C}'} \right)^{\otimes 2} S_l \right) \right\rangle_{\mathcal{C}}} \|I_l\|_2 \|\Pi\|_\infty && (4.D.30)\\
=\ & \sqrt{\left\langle \mathrm{tr}_{(\mathcal{O} \otimes \overline{\mathcal{O}})^{\otimes 2}}\left(\tilde{\rho}^{\otimes 2}_{\Lambda|\mathcal{C}'}\ S_l \otimes I^{\otimes 2}_{d/l} \right) \right\rangle_{\mathcal{C}}} \|I_l\|_2 \|\Pi\|_\infty && (4.D.31)\\
=\ & \sqrt{\mathrm{tr}\left(\left\langle \tilde{\rho}^{\otimes 2}_{\Lambda|\mathcal{C}'} \right\rangle_{\mathcal{C}} \left(S_l \otimes I^{\otimes 2}_{d/l} \right) \right)} \|I_l\|_2 \|\Pi\|_\infty && (4.D.32)\\
=\ & \sqrt{\mathrm{tr}\left(\left(\begin{array}{l} \left\langle \rho_\Lambda^{\otimes 2} \right\rangle_{\mathcal{C}'} \\ \quad - \langle \rho_\Lambda \rangle_{\mathcal{C}'}^{\otimes 2} \\ \quad + \Delta^{(2)}_{\mathcal{C}\mathcal{C}'}(\rho^{\otimes 2}) \\ \quad - \Delta_{\mathcal{C}\mathcal{C}'}(\rho) \otimes \langle \rho_\Lambda \rangle_{\mathcal{C}'} \\ \quad - \langle \rho_\Lambda \rangle_{\mathcal{C}'} \otimes \Delta_{\mathcal{C}\mathcal{C}'}(\rho) \end{array} \right) \left(S_l \otimes I^{\otimes 2}_{d/l} \right) \right)} \|I_l\|_2 \|\Pi\|_\infty && (4.D.33)
\end{aligned}$$

$$\leq \left(\sqrt{\mathrm{tr}\left(\left(\left\langle \rho_\Lambda^{\otimes 2} \right\rangle_{\mathcal{C}'} - \left\langle \rho_\Lambda \right\rangle_{\mathcal{C}'}^{\otimes 2} \right) \left(S_l \otimes I_{d/l}^{\otimes 2} \right) \right)} \right. \tag{4.D.34}$$

$$\left. + \sqrt{\mathrm{tr}\left(\left(\begin{array}{l} \Delta^{(2)}_{\mathcal{C}\mathcal{C}'}(\rho^{\otimes 2}) \\ \quad -\Delta_{\mathcal{C}\mathcal{C}'}(\rho) \otimes \left\langle \rho_\Lambda \right\rangle_{\mathcal{C}'} \\ \quad - \left\langle \rho_\Lambda \right\rangle_{\mathcal{C}'} \otimes \Delta_{\mathcal{C}\mathcal{C}'}(\rho) \end{array} \right) \left(S_l \otimes I_{d/l}^{\otimes 2} \right) \right)} \right) \left\| I_l \right\|_2 \left\| \Pi \right\|_\infty$$

$$= \delta\mu_{\mathcal{L}|\mathcal{C}'}(\rho, \Pi) \;+\; \delta\mu_{\mathcal{L}|\mathcal{C}\mathcal{C}'}(\rho, \Pi) \; . \tag{4.D.35}$$

□

Corollary 3 (Expectation Value Bias for Reference Ensembles)**.** *For specific observables* Π *and reference ensembles* $\mathcal{C}'$*, the expectation value mean is,*

$$\mu_{\mathcal{L}|\mathcal{C}'}(\rho, \Pi) = \frac{\mathrm{tr}(\rho)\,\mathrm{tr}(\Pi)}{d} = \begin{cases} 0 & \mathrm{tr}(\Pi) = 0 \;,\; \mathcal{C}' = \mathcal{C}(d, s) \\ \frac{1}{d} & \mathrm{tr}(\Pi) = 1 \;,\; \mathcal{C}' = \mathcal{C}(d, s) \\ 0 & \mathrm{tr}(\Pi) = 0 \;,\; \mathcal{C}' = \mathcal{D}(d) \\ \frac{1}{d} & \mathrm{tr}(\Pi) = 1 \;,\; \mathcal{C}' = \mathcal{D}(d) \end{cases} \;, \tag{4.D.36}$$

and the inherent $\delta\mu_{\mathcal{L}|\mathcal{C}'}$ *and expressivity* $\delta\mu_{\mathcal{L}|\mathcal{C}\mathcal{C}'}$ *expectation value bias terms are,*

$$\delta\mu_{\mathcal{L}|\mathcal{C}'}(\rho, \Pi) = \begin{cases} \mathcal{O}\left(\sqrt{\frac{l^2}{ds}} \right) \; \|\rho\|_2 \|\Pi\|_\infty & \mathcal{C}' = \mathcal{C}(d, s) \\ 0 & \mathcal{C}' = \mathcal{D}(d) \end{cases} \tag{4.D.37}$$

$$\delta\mu_{\mathcal{L}|\mathcal{C}\mathcal{C}'}(\rho, \Pi) \leq \sqrt{\epsilon^{(2|\diamond)}_{\mathcal{C}\mathcal{C}'} \|\rho\|_1^2 + 2\; \epsilon^{(1|\diamond)}_{\mathcal{C}\mathcal{C}'} \|\rho\|_1} \; \|\Pi\|_\infty \;,$$

under the assumption that the $t = 1$*-order twirl for a reference ensemble* $\mathcal{C}'$ *is proportional to the identity,* $\left\langle \rho_\Lambda \right\rangle_{\mathcal{C}'} = \frac{\mathrm{tr}(\rho)}{d} I_d$*.*

Proof. The expectation value mean follows from its definition, $\mu_{\mathcal{L}|\mathcal{C}'}(\rho, \Pi) = \left\langle \mathcal{L}_\Lambda(\rho, \Pi) \right\rangle_{\mathcal{C}'} = \mathrm{tr}(\left\langle \rho_\Lambda \right\rangle_{\mathcal{C}'} O)$, and substituting in the assumption that the $t = 1$-order twirl for the reference ensemble $\mathcal{C}'$ is $\left\langle \rho_\Lambda \right\rangle_{\mathcal{C}'} = \frac{\mathrm{tr}(\rho)}{d} I_d$.

The expectation value bias inherent term follows from the deviation $\left\langle \rho_\Lambda^{\otimes 2} \right\rangle_{\mathcal{C}'} - \left\langle \rho_\Lambda \right\rangle_{\mathcal{C}'}^{\otimes 2}$ of the $t = 2$ moment for specific reference ensembles $\mathcal{C}'$, given the forms of their respective $t = 1, 2$-order twirls $\left\langle \rho_\Lambda^{\otimes 2} \right\rangle_{\mathcal{C}'}, \left\langle \rho_\Lambda \right\rangle_{\mathcal{C}'}^{\otimes 2}$.

For the cHaar reference ensemble $\mathcal{C}' = \mathcal{C}(d,s)$,

$$\left\langle \rho_\Lambda^{\otimes 2} \right\rangle_{\mathcal{C}(d,s)} - \left\langle \rho_\Lambda \right\rangle_{\mathcal{C}(d,s)}^{\otimes 2} = \frac{1}{d^2} \frac{1}{1-\frac{1}{d^2 s^2}} \left[\left[\frac{1}{d^2 s^2} \mathrm{tr}^2(\rho) - \frac{1}{ds} \mathrm{tr}\left(\rho^2\right) \right] I_d^{\otimes 2} \right. \tag{4.D.38}$$

$$\left. + \ \frac{1}{s} \left[-\frac{1}{ds} \mathrm{tr}^2(\rho) + \mathrm{tr}\left(\rho^2\right) \right] S_d \right]$$

$$= \frac{1}{d^2} \ \|\rho\|_2^2 \ \left[\mathcal{O}\left(\frac{1}{ds} \right) \ I_d^{\otimes 2} \ + \ \mathcal{O}\left(\frac{1}{s} \right) \ S_d \right] \ , \tag{4.D.39}$$

and for the Depolarize reference ensemble $\mathcal{C}' = \mathcal{D}(d)$,

$$\left\langle \rho_\Lambda^{\otimes 2} \right\rangle_{\mathcal{D}(d)} - \left\langle \rho_\Lambda \right\rangle_{\mathcal{D}(d)}^{\otimes 2} = \frac{1}{d^2} \ \mathrm{tr}^2(\rho) \ I_d^{\otimes 2} \ - \ \left(\frac{1}{d} \ \mathrm{tr}(\rho) \ I_d \right)^{\otimes 2} \ = \ 0 \ . \tag{4.D.40}$$

It follows that the expectation value bias inherent term is,

$$\delta\mu_{\mathcal{L}|\mathcal{C}'}(\rho,\Pi) = \sqrt{\mathrm{tr}\left(\left(\left\langle \rho_\Lambda^{\otimes 2} \right\rangle_{\mathcal{C}'} - \left\langle \rho_\Lambda \right\rangle_{\mathcal{C}'}^{\otimes 2} \right) \left(S_l \otimes I_{d/l}^{\otimes 2} \right) \right)} \ \|I_l\|_2 \ \|\Pi\|_\infty \tag{4.D.41}$$

$$= \begin{cases} \sqrt{\begin{array}{l} \mathcal{O}\left(\frac{1}{ds}\right) \ \mathrm{tr}\left(S_l \otimes I_{d/l}^{\otimes 2} \right) \\ \quad + \mathcal{O}\left(\frac{1}{s}\right) \ \mathrm{tr}\left(I_l^{\otimes 2} \otimes S_{d/l} \right) \end{array}} \ \frac{\sqrt{l}}{d} \ \|\rho\|_2 \ \|\Pi\|_\infty & \mathcal{C}' = \mathcal{C}(d,s) \\ 0 & \mathcal{C}' = \mathcal{D}(d) \end{cases} \tag{4.D.42}$$

$$= \begin{cases} \mathcal{O}\left(\sqrt{\frac{l^2}{ds}} \right) \ \|\rho\|_2 \|\Pi\|_\infty & \mathcal{C}' = \mathcal{C}(d,s) \\ 0 & \mathcal{C}' = \mathcal{D}(d) \end{cases} \ . \tag{4.D.43}$$

The expectation value bias expressivity term follows from the norm inequalities,

$$\delta\mu_{\mathcal{L}|\mathcal{C}\mathcal{C}'}(\rho,\Pi) = \sqrt{\mathrm{tr}\left(\left(\begin{array}{l} \Delta_{\mathcal{C}\mathcal{C}'}^{(2)}(\rho^{\otimes 2}) \\ \quad -\Delta_{\mathcal{C}\mathcal{C}'}(\rho) \otimes \left\langle \rho_\Lambda \right\rangle_{\mathcal{C}'} \\ \quad -\left\langle \rho_\Lambda \right\rangle_{\mathcal{C}'} \otimes \Delta_{\mathcal{C}\mathcal{C}'}(\rho) \end{array} \right) \left(S_l \otimes I_{d/l}^{\otimes 2} \right) \right)} \ \|I_l\|_2 \ \|\Pi\|_\infty \tag{4.D.44}$$

$$\leq \min_{\frac{1}{p}+\frac{1}{q}=1} \sqrt{\epsilon_{\mathcal{C}\mathcal{C}'}^{(2|p)} \ \|\rho\|_p^2 \ + \ 2 \ \epsilon_{\mathcal{C}\mathcal{C}'}^{(1|p)} \ \|\langle \rho_\Lambda \rangle_{\mathcal{C}'}\|_p \ \|\rho\|_p \ \|I_l\|_q \ \|I_d\|_q} \ \|\Pi\|_\infty \tag{4.D.45}$$

$$\leq \sqrt{\epsilon_{\mathcal{C}\mathcal{C}'}^{(2|\diamond)} \|\rho\|_1^2 \ + \ 2 \ \epsilon_{\mathcal{C}\mathcal{C}'}^{(1|\diamond)} \|\rho\|_1} \ \|\Pi\|_\infty \, . \tag{4.D.46}$$

□

4.D.3 Expectation Value Variance

Let us study the variance of expectation values $\mathcal{L}$,

$$\sigma^2_{\mathcal{C}}(\rho,\Pi) \;=\; \left\langle \mathcal{L}_\Lambda(\rho,\Pi)^2 \right\rangle_{\mathcal{C}} - \left\langle \mathcal{L}_\Lambda(\rho,\Pi)\right\rangle^2_{\mathcal{C}} \;. \tag{4.D.47}$$

Theorem 19 (Expectation Value Variance)**.** *Let $\mathcal{C},\mathcal{C}'$ be ensembles of quantum channels, ρ be a quantum state, and Π be an observable. The expectation value $\mathcal{L}_\Lambda$ variance is,*

$$\sigma^2_{\mathcal{C}}(\rho,\Pi) \;=\; \sigma^2_{\mathcal{C}'}(\rho,\Pi) \;+\; \delta\sigma^2_{\mathcal{L}|\mathcal{C}'}(\rho,\Pi) \;, \tag{4.D.48}$$

where the inherent $\sigma^2_{\mathcal{C}'}$ and design $\delta\sigma^2_{\mathcal{L}|\mathcal{C}'}$ expectation value variance terms are respectively,

$$\sigma^2_{\mathcal{C}'}(\rho,\Pi) = \mathrm{tr}\left(\left\langle \rho_\Lambda^{\otimes 2}\right\rangle_{\mathcal{C}'} \Pi^{\otimes 2}\right) - \mathrm{tr}\left(\left\langle \rho_\Lambda\right\rangle^{\otimes 2}_{\mathcal{C}'} \Pi^{\otimes 2}\right) \tag{4.D.49}$$

$$\delta\sigma^2_{\mathcal{L}|\mathcal{C}'}(\rho,\Pi) = \mathrm{tr}\left(\Delta^{(2)}_{\mathcal{C}\mathcal{C}'}(\rho^{\otimes 2})\Pi^{\otimes 2}\right) - \mathrm{tr}^2(\Delta_{\mathcal{C}\mathcal{C}'}(\rho)\Pi) - 2\,\mathrm{tr}(\left\langle \rho_\Lambda\right\rangle_{\mathcal{C}'}\Pi)\;\mathrm{tr}(\Delta_{\mathcal{C}\mathcal{C}'}(\rho)\Pi)\;. \tag{4.D.50}$$

Proof. The expectation value variance can be derived from its definition,

$$\begin{aligned}
\sigma^2_{\mathcal{C}}(\rho,\Pi) =&\; \left\langle \mathcal{L}_\Lambda(\rho,\Pi)^2 \right\rangle_{\mathcal{C}} - \left\langle \mathcal{L}_\Lambda(\rho,\Pi)\right\rangle^2_{\mathcal{C}} && (4.D.51)\\
=&\; \mathrm{tr}\left(\left\langle \rho_\Lambda^{\otimes 2}\right\rangle_{\mathcal{C}} \Pi^{\otimes 2}\right) - \mathrm{tr}\left(\left\langle \rho_\Lambda\right\rangle^{\otimes 2}_{\mathcal{C}} \Pi^{\otimes 2}\right) && (4.D.52)\\
=&\; \left(\mathrm{tr}\left(\left\langle \rho_\Lambda^{\otimes 2}\right\rangle_{\mathcal{C}'} \Pi^{\otimes 2}\right) + \mathrm{tr}\left(\Delta^{(2)}_{\mathcal{C}\mathcal{C}'}(\rho^{\otimes 2})\Pi^{\otimes 2}\right)\right) && (4.D.53)\\
&- \;\mathrm{tr}\left(\left(\left\langle \rho_\Lambda\right\rangle_{\mathcal{C}'} + \Delta_{\mathcal{C}\mathcal{C}'}(\rho)\right)^{\otimes 2} \Pi^{\otimes 2}\right)\\
=&\; \left(\mathrm{tr}\left(\left\langle \rho_\Lambda^{\otimes 2}\right\rangle_{\mathcal{C}'} \Pi^{\otimes 2}\right) - \mathrm{tr}\left(\left\langle \rho_\Lambda\right\rangle^{\otimes 2}_{\mathcal{C}'} \Pi^{\otimes 2}\right)\right) && (4.D.54)\\
&+ \;\left(\mathrm{tr}\left(\Delta^{(2)}_{\mathcal{C}\mathcal{C}'}(\rho^{\otimes 2})\Pi^{\otimes 2}\right) - \mathrm{tr}\left(\Delta_{\mathcal{C}\mathcal{C}'}(\rho)^{\otimes 2}\Pi^{\otimes 2}\right)\right)\\
&- \;\left(\mathrm{tr}\left(\left(\left\langle \rho_\Lambda\right\rangle_{\mathcal{C}'} \otimes \Delta_{\mathcal{C}\mathcal{C}'}(\rho)\right) \Pi^{\otimes 2}\right) + \mathrm{tr}\left(\left(\Delta_{\mathcal{C}\mathcal{C}'}(\rho) \otimes \left\langle \rho_\Lambda\right\rangle_{\mathcal{C}'}\right) \Pi^{\otimes 2}\right)\right)\\
=&\; \sigma^2_{\mathcal{C}'}(\rho,\Pi) && (4.D.55)\\
&+ \;\mathrm{tr}\left(\Delta^{(2)}_{\mathcal{C}\mathcal{C}'}(\rho^{\otimes 2})\Pi^{\otimes 2}\right)\\
&- \;\mathrm{tr}^2(\Delta_{\mathcal{C}\mathcal{C}'}(\rho)\Pi)\\
&- \;2\,\mathrm{tr}(\Delta_{\mathcal{C}\mathcal{C}'}(\rho)\Pi)\;\mu_{\mathcal{L}|\mathcal{C}'}(\rho,\Pi)\;,
\end{aligned}$$

given twirls over ensembles $\mathcal{C}$ can be written in terms of reference ensembles $\mathcal{C}'$ as,

$$\left\langle \rho_\Lambda^{\otimes t}\right\rangle_{\mathcal{C}} = \left\langle \rho_\Lambda^{\otimes t}\right\rangle_{\mathcal{C}'} + \Delta^{(t)}_{\mathcal{C}\mathcal{C}'}(\rho^{\otimes t})\;. \tag{4.D.56}$$

$\square$

Corollary 4 (Expectation Value Variance for Reference Ensembles)**.** *For specific observables Π with $|\mathrm{tr}(\Pi)| \leq 1$, and reference ensembles $\mathcal{C}'$, the expectation value variance is*

$$\sigma^2_{\mathcal{C}'}(\rho,\Pi) = \begin{cases} \frac{1}{d^2}\frac{1}{1-\frac{1}{d^2s^2}}\begin{bmatrix} \left[\frac{1}{d^2s^2}\mathrm{tr}^2(\rho) - \frac{1}{ds}\mathrm{tr}(\rho^2)\right]\mathrm{tr}^2(\Pi) \\ +\frac{1}{s}\left[-\frac{1}{ds}\mathrm{tr}^2(\rho) + \mathrm{tr}(\rho^2)\,\mathrm{tr}(\Pi^2)\right] \end{bmatrix} & \mathcal{C}' = \mathcal{C}(d,s) \\ 0 & \mathcal{C}' = \mathcal{D}(d) \end{cases}, \tag{4.D.57}$$

$$= \begin{cases} \mathcal{O}\left(\frac{1}{d^2s}\right)\ \|\rho\|_2^2\ \|\Pi\|_2^2 & \mathcal{C}' = \mathcal{C}(d,s) \\ 0 & \mathcal{C}' = \mathcal{D}(d) \end{cases}. \tag{4.D.58}$$

and the design $\delta\sigma^2_{\mathcal{L}|\mathcal{C}'}$ expectation value variance term is bounded by,

$$\delta\sigma^2_{\mathcal{L}|\mathcal{C}'}(\rho,\Pi) \leq \left|\mathrm{tr}\left(\Delta^{(2)}_{\mathcal{C}\mathcal{C}'}(\rho^{\otimes 2})\Pi^{\otimes 2}\right)\right| \ +\ 2\left|\mathrm{tr}\left(\left\langle\rho_\Lambda\right\rangle_{\mathcal{C}'}\Pi\right)\mathrm{tr}\left(\Delta_{\mathcal{C}\mathcal{C}'}(\rho)\Pi\right)\right| \tag{4.D.59}$$

$$\leq\ \min\left\{\|\Pi\|_\infty\left(\|\rho\|_1^2\|\Pi\|_\infty\epsilon^{(2|\diamond)}_{\mathcal{C}\mathcal{C}'} + 2\frac{1}{d}\epsilon^{(1|\diamond)}_{\mathcal{C}\mathcal{C}'}\right)\ ,\right. \tag{4.D.60}$$

$$\left.\|\Pi\|_2\left(\|\rho\|_2^2\|\Pi\|_2\epsilon^{(2|2)}_{\mathcal{C}\mathcal{C}'} + 2\frac{1}{d}\epsilon^{(1|2)}_{\mathcal{C}\mathcal{C}'}\right)\right\}\ .$$

Proof. The expectation value variance inherent term follows, for specific reference ensembles $\mathcal{C}'$, given the forms of the $t = 1,2$-order twirls $\left\langle\rho_\Lambda^{\otimes t}\right\rangle_{\mathcal{C}'}$, and given $|\mathrm{tr}(\Pi)| \leq 1$,

$$\sigma^2_{\mathcal{C}'}(\rho,\Pi) = \begin{cases} \frac{1}{d^2}\frac{1}{1-\frac{1}{d^2s^2}}\begin{bmatrix} \left[\frac{1}{d^2s^2}\mathrm{tr}^2(\rho) - \frac{1}{ds}\mathrm{tr}(\rho^2)\right]\mathrm{tr}^2(\Pi) \\ +\frac{1}{s}\left[-\frac{1}{ds}\mathrm{tr}^2(\rho) + \mathrm{tr}(\rho^2)\,\mathrm{tr}(\Pi^2)\right] \end{bmatrix} & \mathcal{C}' = \mathcal{C}(d,s) \\ 0 & \mathcal{C}' = \mathcal{D}(d) \end{cases}, \tag{4.D.61}$$

$$= \begin{cases} \mathcal{O}\left(\frac{1}{d^2s}\right)\ \|\rho\|_2^2\ \|\Pi\|_2^2 & \mathcal{C}' = \mathcal{C}(d,s) \\ 0 & \mathcal{C}' = \mathcal{D}(d) \end{cases}. \tag{4.D.62}$$

The expectation value variance design term follows from the following norm inequalities, given the forms of the $t = 1,2$-order twirls $\left\langle\rho_\Lambda^{\otimes t}\right\rangle_{\mathcal{C}'}$, and given $|\mathrm{tr}(\Pi)| \leq 1$, and $\mathcal{C}' \in \{\mathcal{C}(d,s), \mathcal{D}(d)\}$,

$$\delta\sigma^2_{\mathcal{L}|\mathcal{C}'}(\rho,\Pi) = \mathrm{tr}\left(\Delta^{(2)}_{\mathcal{C}\mathcal{C}'}(\rho^{\otimes 2})\Pi^{\otimes 2}\right)\ -\ 2\,\mathrm{tr}(\langle\rho_\Lambda\rangle_{\mathcal{C}'}\Pi)\ \mathrm{tr}(\Delta_{\mathcal{C}\mathcal{C}'}(\rho)\Pi) \tag{4.D.63}$$

$$-\ \mathrm{tr}^2(\Delta_{\mathcal{C}\mathcal{C}'}(\rho)\Pi)$$

$$\leq \left|\mathrm{tr}\left(\Delta^{(2)}_{\mathcal{C}\mathcal{C}'}(\rho^{\otimes 2})\Pi^{\otimes 2}\right)\right| \ +\ 2\left|\mathrm{tr}\left(\left\langle\rho_\Lambda\right\rangle_{\mathcal{C}'}\Pi\right)\mathrm{tr}\left(\Delta_{\mathcal{C}\mathcal{C}'}(\rho)\Pi\right)\right| \tag{4.D.64}$$

$$
\begin{aligned}
&\le \min_{\frac{1}{p}+\frac{1}{q}=1} \epsilon^{(2|p)}_{\mathcal{C}\mathcal{C}'}(\rho^{\otimes 2})\ \|\Pi\|_q^2 && (4.D.65)\\
&\quad + \min_{\frac{1}{p}+\frac{1}{q}=1} 2\left|\mathrm{tr}\Big(\langle\rho_\Lambda\rangle_{\mathcal{C}'}\,\Pi\Big)\right|\ \epsilon^{(1|p)}_{\mathcal{C}\mathcal{C}'}(\rho)\ \|\Pi\|_q\\
&\le \min_{\frac{1}{p}+\frac{1}{q}=1} \|\rho\|_p^2\ \|\Pi\|_q^2\ \epsilon^{(2|p)}_{\mathcal{C}\mathcal{C}'} && (4.D.66)\\
&\quad + \min_{\frac{1}{p}+\frac{1}{q}=1} 2\left|\mathrm{tr}\Big(\langle\rho_\Lambda\rangle_{\mathcal{C}'}\,\Pi\Big)\right|\ \|\rho\|_p\ \|\Pi\|_q\ \epsilon^{(1|p)}_{\mathcal{C}\mathcal{C}'}\\
&\le \min\Big\{ && (4.D.67)\\
&\qquad \|\rho\|_1^2\ \|\Pi\|_\infty^2\ \epsilon^{(2|\diamond)}_{\mathcal{C}\mathcal{C}'}\ +\ 2\ \frac{1}{d}\ |\mathrm{tr}(\rho)|\ |\mathrm{tr}(\Pi)|\ \|\Pi\|_\infty\ \epsilon^{(1|\diamond)}_{\mathcal{C}\mathcal{C}'}\ ,\\
&\qquad \|\rho\|_1^2\ \|\Pi\|_\infty^2\ \epsilon^{(2|\diamond)}_{\mathcal{C}\mathcal{C}'}\ +\ 2\ \frac{1}{d}\ |\mathrm{tr}(\rho)|\ |\mathrm{tr}(\Pi)|\ \|\Pi\|_2\ \epsilon^{(1|2)}_{\mathcal{C}\mathcal{C}'}\ ,\\
&\qquad \|\rho\|_2^2\ \|\Pi\|_2^2\ \epsilon^{(2|2)}_{\mathcal{C}\mathcal{C}'}\ +\ 2\ \frac{1}{d}\ |\mathrm{tr}(\rho)|\ |\mathrm{tr}(\Pi)|\ \|\Pi\|_\infty\ \epsilon^{(1|\diamond)}_{\mathcal{C}\mathcal{C}'}\ ,\\
&\qquad \|\rho\|_2^2\ \|\Pi\|_2^2\ \epsilon^{(2|2)}_{\mathcal{C}\mathcal{C}'}\ +\ 2\ \frac{1}{d}\ |\mathrm{tr}(\rho)|\ |\mathrm{tr}(\Pi)|\ \|\Pi\|_2\ \epsilon^{(1|2)}_{\mathcal{C}\mathcal{C}'},\\
&\qquad \Big\}\\
&\le \min\Big\{ && (4.D.68)\\
&\qquad \|\Pi\|_\infty\left(\|\rho\|_1^2\ \|\Pi\|_\infty \epsilon^{(2|\diamond)}_{\mathcal{C}\mathcal{C}'}\ +\ 2\ \frac{1}{d}\ \epsilon^{(1|\diamond)}_{\mathcal{C}\mathcal{C}'}\right)\ ,\\
&\qquad \|\Pi\|_\infty^2\ \epsilon^{(2|\diamond)}_{\mathcal{C}\mathcal{C}'}\ +\ 2\ \frac{1}{d}\ \|\Pi\|_2\ \epsilon^{(1|2)}_{\mathcal{C}\mathcal{C}'}\ ,\\
&\qquad \|\rho\|_2^2\ \|\Pi\|_2^2\ \epsilon^{(2|2)}_{\mathcal{C}\mathcal{C}'}\ +\ 2\ \frac{1}{d}\ \|\Pi\|_\infty\ \epsilon^{(1|\diamond)}_{\mathcal{C}\mathcal{C}'}\ ,\\
&\qquad \|\Pi\|_2\left(\|\rho\|_2^2\ \|\Pi\|_2\ \epsilon^{(2|2)}_{\mathcal{C}\mathcal{C}'}\ +\ 2\ \frac{1}{d}\ \epsilon^{(1|2)}_{\mathcal{C}\mathcal{C}'}\right),\\
&\qquad \Big\}\ .
\end{aligned}
$$

$\square$

4.D.4 Expectation Value Gradient Mean

Let us study the mean of gradients of expectation values $\mathcal{L}$ at index μ,

$$\mu_{\partial_\mu \mathcal{L}|\mathcal{C}}(\rho,\Pi) = \langle \partial_\mu \mathcal{L}_\Lambda(\rho,\Pi)\rangle_{\mathcal{C}} \ . \tag{4.D.69}$$

For convenience, we denote the following quantities, given $X, G \in \mathcal{B}[\mathcal{H}],\ Z \in \mathcal{B}[\mathcal{H}^{\otimes 2}]$,

$$\tilde{G} \equiv G^{\otimes 2} - \frac{1}{2}\left(G^2 \otimes I + I \otimes G^2\right) \tag{4.D.70}$$

$$Z_G \equiv \{G^{\otimes 2}, Z\} - \big((G\otimes I)Z(I\otimes G) + (I\otimes G)Z(G\otimes I)\big) \tag{4.D.71}$$

$$X^G \equiv -i[G,X] \qquad , \qquad X_G \equiv \left(X^{\otimes 2}\right)_G = -\left(X^G\right)^{\otimes 2} \ . \tag{4.D.72}$$

Given for the identity operator, $I_G = 0$ and the swap operator, $S_G = 2\tilde{G}S_d$, we have,

$$\mathrm{tr}(Z_G) = 0 \quad , \quad \mathrm{tr}(S_d Z_G) = 2\mathrm{tr}\left(S_d\tilde{G}Z\right) \tag{4.D.73}$$

$$\mathrm{tr}\left(\tilde{G}\right) = \mathrm{tr}^2(G) - d\,\mathrm{tr}\left(G^2\right) \quad , \quad \mathrm{tr}\left(S_d\tilde{G}\right) = 0 \tag{4.D.74}$$

$$\mathrm{tr}\left(\tilde{G}X^{\otimes 2}\right) = \mathrm{tr}^2(GW) - \mathrm{tr}\left(G^2X\right)\mathrm{tr}(X) \quad , \quad \mathrm{tr}\left(S_d\tilde{G}X^{\otimes 2}\right) = \frac{1}{2}\mathrm{tr}\left([G,X]^2\right) \ , \tag{4.D.75}$$

further noting that the Z_G is linear in Z, indicating that for $Z \to Z_\Lambda$, $\langle (Z_\Lambda)_G\rangle_{\mathcal{C}} = (\langle Z_\Lambda\rangle_{\mathcal{C}})_G$. These quantities further have norms bounded by,

$$\|\tilde{G}\|_p \leq \|G\|_p^2 + \|I\|_p\ \|G^2\|_p \tag{4.D.76}$$

$$\|Z_G\|_p \leq \|G^{\otimes 2}Z\|_p + \|ZG^{\otimes 2}\|_p + \|I\otimes GZG\otimes I\|_p + \|G\otimes IZI\otimes G\|_p \tag{4.D.77}$$

$$\leq 4\|G\|_\infty^2\|Z\|_p \tag{4.D.78}$$

$$\|\tilde{G}Z\|_p \leq \|\tilde{G}\|_\infty\|Z\|_p \leq (\|G\|_\infty^2 + \|G^2\|_\infty)\|Z\|_p \tag{4.D.79}$$

$$\left|\mathrm{tr}\left(S_d\tilde{G}Z\right)\right| \leq \min_{\frac{1}{p}+\frac{1}{q}=1} \|S_d\tilde{G}\|_p\|Z\|_q \leq \min_{\frac{1}{p}+\frac{1}{q}=1}\|\tilde{G}\|_\infty\|S_d\|_p\|Z\|_q \tag{4.D.80}$$

$$\leq \min_{\frac{1}{p}+\frac{1}{q}=1}(\|G\|_\infty^2 + \|G^2\|_\infty)\|S_d\|_p\|Z\|_q \tag{4.D.81}$$

$$\left|\mathrm{tr}\left(S_d\tilde{G}X^{\otimes 2}\right)\right| \leq \min_{\frac{1}{p}+\frac{1}{q}=1}\left\{\begin{array}{l} \|G\|_\infty\|WGW\|_1 + \|G^2\|_\infty\|X^2\|_1 \\ \|GWG\|_2\|X\|_2 + \|G^2X\|_2\|X\|_2 \\ \|G\|_\infty^2\|X\|_p\|X\|_q + \|G^2\|_\infty\|X\|_p\|X\|_q \end{array}\right\} \tag{4.D.82}$$

$$\leq \min_{\frac{1}{p}+\frac{1}{q}=1}(\|G\|_\infty^2 + \|G^2\|_\infty)\|X\|_p\|X\|_q \tag{4.D.83}$$

Further, for convenience, we define the following modified norm definitions.

Definition 6 (Modified Norm of $t=2$-order Moments of Inputs)**.** *Let a hermitian operator $W \in \mathcal{B}[\mathcal{H}] \to W_\Upsilon$ within a space $\mathcal{H}$ with dimension d be transformed by a not-necessarily physical operator $\Upsilon \in \mathcal{B}[\mathcal{B}[\mathcal{H}]]$. For example, the adjoint of a channel $\Upsilon = \Lambda^\dagger$ is not necessarily physical. Norms of the transformed operator may then be simplified, given physical operators correspond to Stinespring dilation unitaries $V_\Upsilon \in \mathcal{U}(ds)$, with pure ancillary states $\nu = |\nu\rangle\langle\nu| \in \mathcal{B}[\mathcal{E}]$ from an ancillary space $\mathcal{E}$ with dimension s. We therefore define the modified Schatten p-norms $\|\cdot\|_{p^*}$, with asterisk p^* notation, as,*

$$\|W_\Upsilon\|_1^2 \leq \|W\|_1^2 \|\Upsilon\|_1^2 \leq \|W\|_1^2 \|\Upsilon\|_\diamond^2 \tag{4.D.84}$$

$$\equiv \|W_\Upsilon\|_{1^*}^2 \tag{4.D.85}$$

$$\|W_\Upsilon\|_2^2 \overset{\Upsilon_{\text{non-physical}}}{\leq} \|\Upsilon\|_{\text{op}}^2 \|W\|_2^2 \tag{4.D.86}$$

$$\overset{\Upsilon_{\text{physical}}}{\leq} \left| \operatorname{tr}\left(\left(V_\Upsilon W \otimes \nu V_\Upsilon^\dagger \right)^{\otimes 2} S_d \otimes I_s^{\otimes 2} \right) \right| \tag{4.D.87}$$

$$\leq \min \begin{Bmatrix} \min\limits_{\frac{1}{p}+\frac{1}{q}=1} \|I_d \otimes I_s\|_p^2 \; \|W\|_q^2 & \Upsilon_{\text{physical}} \\ \min\limits_{\frac{1}{p}+\frac{1}{q}=1} \|\Upsilon\|_p^2 \; \|W\|_q^2 & \Upsilon_{\text{non-physical}} \end{Bmatrix} \tag{4.D.88}$$

$$\equiv \|W_\Upsilon\|_{2^*}^2 \ . \tag{4.D.89}$$

Finally, given such definitions, let us calculate the $t=2$ moments of these ensemble-dependent quantities.

Lemma 7 ($t=2$-order Moments of Inputs)**.** *For specific reference ensembles and operators, the $t=2$ moments of G-dependent functions of $Z \in \mathcal{B}[\mathcal{H}^{\otimes 2}]$ can be computed as*

$$\langle (Z_\Lambda)_G \rangle_{\mathcal{C}(d,s)} = -2s \frac{ds \operatorname{tr}(S_d Z) - \operatorname{tr}(Z)}{ds(d^2s^2-1)} S_d \tilde{G} \tag{4.D.90}$$

$$\langle (Z_G)_\Lambda \rangle_{\mathcal{C}(d,s)} = -2s^2 \frac{\operatorname{tr}\left(S_d \tilde{G} Z\right)}{ds(d^2s^2-1)} \left(I_d^{\otimes 2} - dS_d\right) \tag{4.D.91}$$

$$\left\langle \left(\langle (Z_\Lambda)_G \rangle_{\mathcal{C}(d,s)} \right)_\Lambda \right\rangle_{\mathcal{C}(d,s)} = -2s^3 \frac{ds \operatorname{tr}(S_d Z) - \operatorname{tr}(Z)}{ds(d^2s^2-1)} \frac{d \operatorname{tr}(G^2) - \operatorname{tr}^2(G)}{ds(d^2s^2-1)} \left(I_d^{\otimes 2} - dS_d\right) \tag{4.D.92}$$

$$\langle (Z_\Lambda)_G \rangle_{\mathcal{D}(d)} = \langle (Z_G)_\Lambda \rangle_{\mathcal{D}(d)} = \left\langle \left(\langle (Z_\Lambda)_G \rangle_{\mathcal{C}} \right)_\Lambda \right\rangle_{\mathcal{D}(d)} = 0 \ , \tag{4.D.93}$$

and these identities can be used for moments of $Z = X^{\otimes 2}$, and $Z_G = X_G = -\left(X^{\otimes 2}\right)^G$, given $\langle (Z_\Lambda)_G \rangle_{\mathcal{C}} = (\langle Z_\Lambda \rangle_{\mathcal{C}})_G$.

As discussed in the following analysis, the scaling with dimension d, s of quantities, and their upper bounds, are highly dependent on the locality and form of operators $X, \Pi, G \in \mathcal{B}[\mathcal{H}]$, form of norms $\|\cdot\|_p$, and the specific ensemble and reference ensembles $\mathcal{C}, \mathcal{C}'$. In particular, whether operators and their adjoints $\Lambda, \Lambda^\dagger \sim \mathcal{C}, \mathcal{C}'$ are physical, allow specific norm identities to simplify any scaling relations. The irreversibility of quantum channels is shown to be an important property preventing identical, potentially exponential scaling of quantities. Any conclusions about the exact form of scaling, either exponential or otherwise of any quantities, are to be determined for each specific ansatz.

The gradient mean at μ, for R, L ensembles relative to reference ensembles, is,

$$\mu_{\partial_\mu \mathcal{L}|\mathcal{C}}(\rho, \Pi) = \langle \partial_\mu \mathcal{L}_\Lambda(\rho, \Pi) \rangle_{\mathcal{C}} \tag{4.D.94}$$

$$= \mathrm{tr}\left(\left\langle \left(\langle \rho_{\mu_R}^{G_\mu} \rangle_{\mathcal{C}_{\mu_R}} \right)_{\mu_L} \right\rangle_{\mathcal{C}_{\mu_L}} \Pi \right) \tag{4.D.95}$$

$$= \mathrm{tr}\left(\langle \rho_{\mu_R}^{G_\mu} \rangle_{\mathcal{C}_{\mu_R}} \langle \Pi_{\mu_L} \rangle_{\mathcal{C}_{\mu_L}} \right) \tag{4.D.96}$$

$$= \mathrm{tr}\left(\langle \rho_{\mu_R} \rangle_{\mathcal{C}_{\mu_R}} \langle \Pi_{\mu_L}^{G_\mu} \rangle_{\mathcal{C}_{\mu_L}} \right) \ . \tag{4.D.97}$$

There exist several cases where the gradient mean is zero, generally due to the cyclicity of the trace objectives and moments being proportional to the identity.

Theorem 20 (Objective Gradient Mean)**.** *For ensembles $\mathcal{C}_{\mu_R}$ with $t = 1$ moments proportional to the identity,*

$$\mu_{\partial_\mu \mathcal{L}|\mathcal{C}_{\mu_R}, \mathcal{C}_{\mu_L}}(\rho, \Pi) = +i\mathrm{tr}\left(\frac{\mathrm{tr}(\rho)}{d} I_d \ [G_\mu, \langle \Pi_{\mu_L} \rangle_{\mathcal{C}_{\mu_L}}] \right) = 0 \ , \tag{4.D.98}$$

or for ensembles $\mathcal{C}_{\mu_L}$ with $t = 1$ moments proportional to the identity,

$$\mu_{\partial_\mu \mathcal{L}|\mathcal{C}_{\mu_R}, \mathcal{C}_{\mu_L}}(\rho, \Pi) = -i\mathrm{tr}\left(\frac{\mathrm{tr}\Big([G_\mu, \langle \rho_{\mu_R} \rangle_{\mathcal{C}_{\mu_R}}] \Big)}{d} \ \Pi \right) = 0 \ . \tag{4.D.99}$$

Similarly, when ensembles $\mathcal{G}_\Theta$ are uniform with parameters $\theta \sim \Theta = \mathrm{U}[0, 2\pi]$, and involutory generators $G_\mu^2 = I$,

$$\begin{aligned} \mu_{\partial_\mu \mathcal{L}|\mathcal{G}_\Theta}(\rho, \Pi) &= \mathrm{tr}\big(\langle (\sin(2\theta_\mu)[G_\mu \rho_{\mu_R} G_\mu - \rho_{\mu_R}] - i\cos(2\theta_\mu)[G_\mu, \rho_{\mu_R}]) \ \mathcal{U}_\mu(\Pi_{\mu_L}) \rangle_{\theta \sim \Theta} \big) \\ &= 0 \ . \end{aligned} \tag{4.D.100}$$

4.D.5 Expectation Value Gradient Variance

Let us study the variance of gradients of expectation values $\mathcal{L}$ at index μ, which can be bounded as,

$$\sigma^2_{\partial_\mu \mathcal{L}|\mathcal{C}}(\rho,\Pi) = \left\langle \partial_\mu \mathcal{L}_\Lambda(\rho,\Pi)^2 \right\rangle_{\mathcal{C}} - \left\langle \partial_\mu \mathcal{L}_\Lambda(\rho,\Pi) \right\rangle^2_{\mathcal{C}} \ \leq \ \left\langle \partial_\mu \mathcal{L}_\Lambda(\rho,\Pi)^2 \right\rangle_{\mathcal{C}} \tag{4.D.101}$$

$$= \mathrm{tr}\left(\left\langle \left(\left\langle \rho_{\mu_R}^{G_\mu \otimes 2} \right\rangle_{\mathcal{C}_{\mu_R}} \right)_{\mu_L} \right\rangle_{\mathcal{C}_{\mu_L}} \Pi^{\otimes 2} \right) . \tag{4.D.102}$$

Let us now consider the cases of whether ensembles before (R), after (L), or before and after (RL) the gradient index μ, form an ϵ-t-design with respect to a reference ensemble.

Gradient Variance - Before/After Epsilon-t-designs

Theorem 21 (Objective Gradient Variance - Before/After Epsilon-t-designs)**.** *If both (RL) ensembles are ϵ-t-designs, the objective gradient variance is given by inherent $\sigma^{2RL}_{\partial_\mu \mathcal{L}|\mathcal{C}'_{\mu_R}\mathcal{C}'_{\mu_L}}$ and expressivity $\delta\sigma^{2RL}_{\partial_\mu \mathcal{L}|\mathcal{C}_{\mu_{RL}}}$ terms,*

$$\sigma^2_{\partial_\mu \mathcal{L}|\mathcal{C}_{\mu_{RL}}}(\rho,\Pi) \ \leq \ \sigma^{2RL}_{\partial_\mu \mathcal{L}|\mathcal{C}'_{\mu_R}\mathcal{C}'_{\mu_L}}(\rho,\Pi) \ + \ \delta\sigma^{2RL}_{\partial_\mu \mathcal{L}|\mathcal{C}_{\mu_{RL}}}(\rho,\Pi) \ . \tag{4.D.103}$$

For arbitrary observables Π, and various reference ensembles $\mathcal{C}' \in \{\mathcal{C}(d,s),\mathcal{D}(d)\}$, the objective gradient variance inherent term is,

$$\sigma^{2RL}_{\partial_\mu \mathcal{L}|\mathcal{C}'_{\mu_R}\mathcal{C}'_{\mu_L}}(\rho,\Pi) \ = \ \begin{cases} \mathcal{O}\big(\frac{1}{d^2 s^2}\big) \ \|\rho\|_2^2 \ \|\Pi\|_2^2 & \mathcal{C}' = \mathcal{C}(d,s) \\ 0 & \mathcal{C}' = \mathcal{D}(d) \end{cases} , \tag{4.D.104}$$

and the objective gradient variance expressivity term is, with the $p = 2$-norm,

$$\delta\sigma^{2RL}_{\partial_\mu \mathcal{L}|\mathcal{C}_{\mu_{RL}}}(\rho,\Pi) \ \leq \ \begin{cases} \begin{bmatrix} \mathcal{O}\big(\frac{1}{ds}\big) \ \epsilon^{(2|2)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}} \\ + \ \mathcal{O}\big(\frac{1}{ds}\big) \ \epsilon^{(2|\dagger 2)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}} \\ + \ 4 \ \epsilon^{(2|2)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}} \ \epsilon^{(2|\dagger 2)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}} \end{bmatrix} \|\rho\|_2^2 \ \|\Pi\|_2^2 & \mathcal{C}' = \mathcal{C}(d,s) \\ \\ 4 \ \|\rho\|_2^2 \ \|\Pi\|_2^2 \ \epsilon^{(2|2)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}} \epsilon^{(2|\dagger 2)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}} & \mathcal{C}' = \mathcal{D}(d) \end{cases} , \tag{4.D.105}$$

or with the $p=1,\diamond$-norm,

$$\delta\sigma^{2RL}_{\partial_\mu\mathcal{L}|\mathcal{C}_{\mu_{RL}}}(\rho,\Pi) \ \leq \ \begin{cases} \begin{bmatrix} \mathcal{O}\left(\frac{1}{d^2 s}\right) \ \|\rho\|_1^2 \ \|\Pi\|_2^2 \ \epsilon^{(2|\diamond)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}} \\ \quad + \mathcal{O}\left(\frac{1}{d^2 s}\right) \ \|\rho\|_2^2 \ \|\Pi\|_1^2 \ \epsilon^{(2|\dagger 1)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}} \\ \quad + 4 \ \|\rho\|_1^2 \ \|\Pi\|_1^2 \ \epsilon^{(2|\diamond)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}} \ \epsilon^{(2|\dagger 1)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}} \end{bmatrix} & \mathcal{C}'=\mathcal{C}(d,s) \\ \\ 4 \ \|\rho\|_1^2 \ \|\Pi\|_1^2 \ \epsilon^{(2|\diamond)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}} \ \epsilon^{(2|\dagger 1)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}} & \mathcal{C}'=\mathcal{D}(d) \end{cases} \quad . \tag{4.D.106}$$

For ϵ-t-cHaar designs, the inherent term decays with system and environment dimensions, and the expressivity terms decay at least with environment dimension. For ϵ-t-Depolarize designs, the inherent term is identically zero, and the expressivity terms depend strictly on the twirl norms. However, the adjoint channels from $\mathcal{C}_{\mu_L}$ acting on Π are not necessarily physical, and their $\diamond$-norms are not necessarily well defined. Such norms have potentially loose scaling with respect to decaying with dimension, meaning objective gradient concentration statements for both (RL) ensembles being ϵ-t-designs, do not always hold.

Proof. We may derive each term in the objective gradient variance,

$$\begin{aligned} \sigma^2_{\partial_\mu\mathcal{L}|\mathcal{C}_{\mu_{RL}}}(\rho,\Pi) \ &\leq \ \sigma^{2RL}_{\partial_\mu\mathcal{L}|\mathcal{C}'_{\mu_R}\mathcal{C}'_{\mu_L}}(\rho,\Pi) + \delta\sigma^{2RL}_{\partial_\mu\mathcal{L}|\mathcal{C}_{\mu_{RL}}}(\rho,\Pi) \\ \delta\sigma^{2RL}_{\partial_\mu\mathcal{L}|\mathcal{C}_{\mu_{RL}}}(\rho,\Pi) \ &= \ \delta\sigma^{2RL}_{\partial_\mu\mathcal{L}|\Delta_{\mu_R}\mathcal{C}'_{\mu_L}}(\rho,\Pi) + \delta\sigma^{2RL}_{\partial_\mu\mathcal{L}|\mathcal{C}'_{\mu_R}\Delta_{\mu_L}}(\rho,\Pi) + \delta\sigma^{2RL}_{\partial_\mu\mathcal{L}|\Delta_{\mu_R}\Delta_{\mu_L}}(\rho,\Pi) \ , \end{aligned} \tag{4.D.107}$$

given the previously derived $t=2$ moments of G_μ-dependent functions of $\rho,\Pi \in \mathcal{B}[\mathcal{H}]$.

For specific reference ensembles and operators, the inherent term is,

$$\begin{aligned} \sigma^{2RL}_{\partial_\mu\mathcal{L}|\mathcal{C}'_{\mu_R}\mathcal{C}'_{\mu_L}}(\rho,\Pi) &= \mathrm{tr}\left(\left\langle\left(\left\langle\rho^{G_\mu\otimes 2}_{\mu_R}\right\rangle_{\mathcal{C}'_{\mu_R}}\right)_{\mu_L}\right\rangle_{\mathcal{C}'_{\mu_L}} \Pi^{\otimes 2}\right) \\ &= \ -\,\mathrm{tr}\left(\left\langle\left(\left(\left\langle\rho^{\otimes 2}_{\mu_R}\right\rangle_{\mathcal{C}'_{\mu_R}}\right)_{G_\mu}\right)_{\mu_L}\right\rangle_{\mathcal{C}'_{\mu_L}} \Pi^{\otimes 2}\right) \\ &= \begin{cases} -2s^3\,\frac{\left(ds\ \mathrm{tr}\left(\rho^2\right)-\mathrm{tr}^2(\rho)\right)\left(d\ \mathrm{tr}\left(G_\mu^2\right)-\mathrm{tr}^2(G_\mu)\right)\left(d\ \mathrm{tr}\left(\Pi^2\right)-\mathrm{tr}^2(\Pi)\right)}{d^2 s^2 (d^2 s^2-1)^2} & \mathcal{C}'=\mathcal{C}(d,s) \\ 0 & \mathcal{C}'=\mathcal{D}(d) \end{cases} \\ &= \begin{cases} \mathcal{O}\left(\frac{1}{d^2 s^2}\right) \ \|\rho\|_2^2 \ \|\Pi\|_2^2 & \mathcal{C}'=\mathcal{C}(d,s) \\ 0 & \mathcal{C}'=\mathcal{D}(d) \end{cases} \ , \end{aligned} \tag{4.D.108}$$

and the expressivity terms are,

$$\begin{aligned}
&\delta\sigma^{2RL}_{\partial_\mu\mathcal{L}|\Delta_{\mu_R}\mathcal{C}'_{\mu_L}}(\rho,\Pi) && (4.D.109)\\
&= \operatorname{tr}\left(\left\langle\left(\Delta^{(2)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}}(\rho^{G_\mu\otimes 2}_{\mu_R})\right)_{\mu_L}\right\rangle_{\mathcal{C}'_{\mu_L}}\Pi^{\otimes 2}\right)\\
&= -\operatorname{tr}\left(\left\langle\left(\left(\Delta^{(2)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}}(\rho^{\otimes 2}_{\mu_R})\right)_{G_\mu}\right)_{\mu_L}\right\rangle_{\mathcal{C}'_{\mu_L}}\Pi^{\otimes 2}\right)\\
&= \begin{cases} -2s^2\frac{\left(\operatorname{tr}\left(S_d\tilde{G}_\mu\Delta^{(2)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}}(\rho^{\otimes 2})\right)\right)\left(d\operatorname{tr}\left(\Pi^2\right)-\operatorname{tr}^2(\Pi)\right)}{ds(d^2s^2-1)} & \mathcal{C}'=\mathcal{C}(d,s)\\ 0 & \mathcal{C}'=\mathcal{D}(d)\end{cases}\\
&\leq \begin{cases} \min\limits_{\frac{1}{p}+\frac{1}{q}=1}\ 2s^2\frac{(\|G_\mu\|^2_\infty+\left\|G^2_\mu\right\|_\infty)(d\operatorname{tr}(\Pi^2)-\operatorname{tr}^2(\Pi))}{ds(d^2s^2-1)}\|S_d\|_p\|\rho\|^2_q\ \epsilon^{(2|q)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}} & \mathcal{C}'=\mathcal{C}(d,s)\\ 0 & \mathcal{C}'=\mathcal{D}(d)\end{cases}\\
&\overset{\substack{p=2\\q=2}}{\leq} \begin{cases} \mathcal{O}\left(\frac{1}{ds}\right)\ \|\rho\|^2_2\ \|\Pi\|^2_2\ \epsilon^{(2|2)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}} & \mathcal{C}'=\mathcal{C}(d,s)\\ 0 & \mathcal{C}'=\mathcal{D}(d)\end{cases}\\
&\overset{\substack{p=\infty\\q=1}}{\leq} \begin{cases} \mathcal{O}\left(\frac{1}{d^2s}\right)\ \|\rho\|^2_1\ \|\Pi\|^2_2\ \epsilon^{(2|\diamond)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}} & \mathcal{C}'=\mathcal{C}(d,s)\\ 0 & \mathcal{C}'=\mathcal{D}(d)\end{cases}
\end{aligned}$$

$$\begin{aligned}
&\delta\sigma^{2RL}_{\partial_\mu\mathcal{L}|\mathcal{C}'_{\mu_R}\Delta_{\mu_L}}(\rho,\Pi) && (4.D.110)\\
&= \operatorname{tr}\left(\left\langle\rho^{G_\mu\otimes 2}_{\mu_R}\right\rangle_{\mathcal{C}'_{\mu_R}}\Delta^{(2|\dagger)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}}(\Pi^{\otimes 2}_{\mu_L})\right)\\
&= -\operatorname{tr}\left(\left(\left\langle\rho^{\otimes 2}_{\mu_R}\right\rangle_{\mathcal{C}'_{\mu_R}}\right)_{G_\mu}\Delta^{(2|\dagger)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}}(\Pi^{\otimes 2}_{\mu_L})\right)\\
&= \begin{cases} 2s\frac{ds\operatorname{tr}\left(\rho^2\right)-\operatorname{tr}^2(\rho)}{ds(d^2s^2-1)}\operatorname{tr}\left(S_d\tilde{G}_\mu\Delta^{(2|\dagger)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}}(\Pi^{\otimes 2})\right) & \mathcal{C}'=\mathcal{C}(d,s)\\ 0 & \mathcal{C}'=\mathcal{D}(d)\end{cases}\\
&\leq \begin{cases} \min\limits_{\frac{1}{p}+\frac{1}{q}=1}\ 2s\frac{(\|G_\mu\|^2_\infty+\left\|G^2_\mu\right\|_\infty)(ds\operatorname{tr}(\rho^2)-\operatorname{tr}^2(\rho))}{ds(d^2s^2-1)}\ \|S_d\|_p\ \|\Pi\|^2_q\ \epsilon^{(2|\dagger q)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}} & \mathcal{C}'=\mathcal{C}(d,s)\\ 0 & \mathcal{C}'=\mathcal{D}(d)\end{cases}
\end{aligned}$$

$$\overset{\substack{p=2\\q=2}}{\leq} \begin{cases} \mathcal{O}\left(\frac{1}{ds}\right)\ \|\rho\|_2^2\ \|\Pi\|_2^2\ \epsilon^{(2|\dagger 2)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}} & \mathcal{C}' = \mathcal{C}(d,s) \\ 0 & \mathcal{C}' = \mathcal{D}(d) \end{cases}$$

$$\overset{\substack{p=\infty\\q=1}}{\leq} \begin{cases} \mathcal{O}\left(\frac{1}{d^2 s}\right)\ \|\rho\|_2^2\ \|\Pi\|_1^2\ \epsilon^{(2|\dagger 1)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}} & \mathcal{C}' = \mathcal{C}(d,s) \\ 0 & \mathcal{C}' = \mathcal{D}(d) \end{cases}$$

$$\begin{aligned} &\delta\sigma^{2RL}_{\partial_\mu\mathcal{L}|\Delta_{\mu_R}\Delta_{\mu_L}}(\rho,\Pi) \\ &= \mathrm{tr}\Big(\Delta^{(2)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}}(\rho^{G_\mu\otimes 2}_{\mu_R})\ \Delta^{(2|\dagger)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}}(\Pi^{\otimes 2}_{\mu_L})\Big) \\ &= -\,\mathrm{tr}\bigg(\Big(\Delta^{(2)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}}(\rho^{\otimes 2}_{\mu_R})\Big)_{G_\mu}\ \Delta^{(2|\dagger)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}}(\Pi^{\otimes 2}_{\mu_L})\bigg) \\ &\leq \min_{\frac{1}{p}+\frac{1}{q}=1}\ \left\|\Big(\Delta^{(2)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}}(\rho^{\otimes 2}_{\mu_R})\Big)_{G_\mu}\right\|_p\ \left\|\Delta^{(2|\dagger)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}}(\Pi^{\otimes 2}_{\mu_L})\right\|_q \\ &\leq \min_{\frac{1}{p}+\frac{1}{q}=1}\ 4\ \|G_\mu\|_\infty^2\ \epsilon^{(2|p)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}}(\rho^{\otimes 2})\ \epsilon^{(2|\dagger q)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}}(\Pi^{\otimes 2}) \\ &\overset{\substack{p=2\\q=2}}{\leq} 4\ \|\rho\|_2^2\ \|\Pi\|_2^2\ \epsilon^{(2|2)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}}\ \epsilon^{(2|\dagger 2)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}} \\ &\overset{\substack{p=\infty\\q=1}}{\leq} 4\ \|\rho\|_1^2\ \|\Pi\|_1^2\ \epsilon^{(2|\diamond)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}}\ \epsilon^{(2|\dagger 1)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}} \end{aligned} \tag{4.D.111}$$

□

Gradient Variance - Before Epsilon-t-designs

Theorem 22 (Objective Gradient Variance - Before Epsilon-t-designs)**.** *If only the before (R) ensemble is an ϵ-t-design, the objective gradient variance is given by inherent $\sigma^{2R}_{\partial_\mu\mathcal{L}|\mathcal{C}'_{\mu_R}\mathcal{C}_{\mu_L}}$ and expressivity $\delta\sigma^{2R}_{\partial_\mu\mathcal{L}|\mathcal{C}\mathcal{C}'_{\mu_R}}$ terms,*

$$\sigma^2_{\partial_\mu\mathcal{L}|\mathcal{C}_{\mu_R}}(\rho,\Pi)\ \leq\ \sigma^{2R}_{\partial_\mu\mathcal{L}|\mathcal{C}'_{\mu_R}\mathcal{C}_{\mu_L}}(\rho,\Pi)\ +\ \delta\sigma^{2R}_{\partial_\mu\mathcal{L}|\mathcal{C}\mathcal{C}'_{\mu_R}}(\rho,\Pi)\ . \tag{4.D.112}$$

For arbitrary observables Π, and various reference ensembles $\mathcal{C}' \in \{\mathcal{C}(d,s),\mathcal{D}(d)\}$, with the $p=2$-norm, the objective gradient variance inherent term is,

$$\sigma^{2R}_{\partial_\mu\mathcal{L}|\mathcal{C}'_{\mu_R}\mathcal{C}_{\mu_L}}(\rho,\Pi)\ \leq\ \begin{cases} \mathcal{O}\left(\frac{1}{ds}\right)\ \|\rho\|_2^2\ \left\|\left\langle\Pi^{\otimes 2}_{\mu_L}\right\rangle_{\mathcal{C}_{\mu_L}}\right\|_{2^*} & \mathcal{C}' = \mathcal{C}(d,s) \\ 0 & \mathcal{C}' = \mathcal{D}(d) \end{cases}\ , \tag{4.D.113}$$

and the objective gradient variance expressivity term is,

$$\delta\sigma^{2R}_{\partial_\mu\mathcal{L}|\mathcal{C}\mathcal{C}'_{\mu_R}}(\rho,\Pi) \;\leq\; 4\;\|\rho\|_2^2\;\;\left\|\left\langle\Pi_{\mu_L}^{\otimes 2}\right\rangle_{\mathcal{C}_{\mu_L}}\right\|_{2^*}\;\;\epsilon^{(2|2)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}}\;, \tag{4.D.114}$$

or with the $p=1,\diamond$-norm, the objective gradient variance inherent term is,

$$\sigma^{2R}_{\partial_\mu\mathcal{L}|\mathcal{C}'_{\mu_R}\mathcal{C}_{\mu_L}}(\rho,\Pi) \;\leq\; \begin{cases} \mathcal{O}\left(\frac{1}{d^2s}\right)\;\|\rho\|_2^2\;\left\|\left\langle\Pi_{\mu_L}^{\otimes 2}\right\rangle_{\mathcal{C}_{\mu_L}}\right\|_{1^*} & \mathcal{C}'=\mathcal{C}(d,s)\\ 0 & \mathcal{C}'=\mathcal{D}(d)\end{cases}\;, \tag{4.D.115}$$

and the objective gradient variance expressivity term is,

$$\delta\sigma^{2R}_{\partial_\mu\mathcal{L}|\mathcal{C}\mathcal{C}'_{\mu_R}}(\rho,\Pi) \;\leq\; 4\;\|\rho\|_1^2\;\;\left\|\left\langle\Pi_{\mu_L}^{\otimes 2}\right\rangle_{\mathcal{C}_{\mu_L}}\right\|_{1^*}\;\;\epsilon^{(2|\diamond)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}}\;. \tag{4.D.116}$$

For ϵ-t-cHaar designs, the inherent term decays with system and environment dimensions, and the expressivity term decays at least with the moment operator norms. For ϵ-t-Depolarize designs, the inherent term is identically zero, and the expressivity term depends strictly on the twirl norms. However, the adjoint channels from $\mathcal{C}_{\mu_L}$ acting on Π are not necessarily physical, their $\diamond$-norms are not necessarily well defined, and modified p^* norms must be used. Such norms have potentially loose scaling with respect to decaying with dimension, meaning objective gradient concentration statements for strictly before (R) ensembles being ϵ-t-designs, do not always hold.

Proof. We may derive the terms in the objective gradient variance,

$$\begin{aligned}\sigma^{2}_{\partial_\mu\mathcal{L}|\mathcal{C}_{\mu_R}}(\rho,\Pi) \;&\leq\; \sigma^{2R}_{\partial_\mu\mathcal{L}|\mathcal{C}'_{\mu_R}\mathcal{C}_{\mu_L}}(\rho,\Pi) \;+\; \delta\sigma^{2R}_{\partial_\mu\mathcal{L}|\mathcal{C}\mathcal{C}'_{\mu_R}}(\rho,\Pi)\\ \delta\sigma^{2R}_{\partial_\mu\mathcal{L}|\mathcal{C}\mathcal{C}'_{\mu_R}}(\rho,\Pi) \;&=\; \delta\sigma^{2R}_{\partial_\mu\mathcal{L}|\Delta_{\mu_R}\mathcal{C}_{\mu_L}}(\rho,\Pi)\;,\end{aligned} \tag{4.D.117}$$

given the previously derived $t=2$ moments of G_μ-dependent functions of $\rho,\Pi\in\mathcal{B}[\mathcal{H}]$.

For specific reference ensembles and operators, the inherent term is,

$$\begin{aligned}&\sigma^{2R}_{\partial_\mu\mathcal{L}|\mathcal{C}'_{\mu_R}\mathcal{C}_{\mu_L}}(\rho,\Pi)\\ &=\mathrm{tr}\left(\left\langle\rho_{\mu_R}^{G_\mu\otimes 2}\right\rangle_{\mathcal{C}'_{\mu_R}}\left\langle\Pi_{\mu_L}^{\otimes 2}\right\rangle_{\mathcal{C}_{\mu_L}}\right)\\ &=\begin{cases}2s\frac{ds\;\mathrm{tr}\left(\rho^2\right)-\mathrm{tr}^2(\rho)}{ds(d^2s^2-1)}\mathrm{tr}\left(S_d\tilde{G}_\mu\left\langle\Pi_{\mu_L}^{\otimes 2}\right\rangle_{\mathcal{C}_{\mu_L}}\right) & \mathcal{C}'=\mathcal{C}(d,s)\\ 0 & \mathcal{C}'=\mathcal{D}(d)\end{cases}\end{aligned} \tag{4.D.118}$$

$$\le \begin{cases} \min\limits_{\frac{1}{p}+\frac{1}{q}=1} \; 2s\dfrac{(\|G_\mu\|_\infty^2 + \|G_\mu^2\|_\infty)(ds\,\mathrm{tr}(\rho^2)-\mathrm{tr}^2(\rho))}{ds(d^2s^2-1)}\|S_d\|_p \; \left\|\left\langle\Pi_{\mu_L}^{\otimes 2}\right\rangle_{\mathcal{C}_{\mu_L}}\right\|_{q^*} & \mathcal{C}'=\mathcal{C}(d,s) \\ 0 & \mathcal{C}'=\mathcal{D}(d) \end{cases}$$
$$\overset{\substack{p=2\\q=2}}{\le} \begin{cases} \mathcal{O}\left(\frac{1}{ds}\right) \; \|\rho\|_2^2 \; \left\|\left\langle\Pi_{\mu_L}^{\otimes 2}\right\rangle_{\mathcal{C}_{\mu_L}}\right\|_{2^*} & \mathcal{C}'=\mathcal{C}(d,s) \\ 0 & \mathcal{C}'=\mathcal{D}(d) \end{cases}$$
$$\overset{\substack{p=\infty\\q=1}}{\le} \begin{cases} \mathcal{O}\left(\frac{1}{d^2s}\right) \; \|\rho\|_2^2 \; \left\|\left\langle\Pi_{\mu_L}^{\otimes 2}\right\rangle_{\mathcal{C}_{\mu_L}}\right\|_{1^*} & \mathcal{C}'=\mathcal{C}(d,s) \\ 0 & \mathcal{C}'=\mathcal{D}(d) \end{cases} ,$$

and the expressivity term is,

$$\begin{aligned} & \delta\sigma^{2R}_{\partial_\mu\mathcal{L}|\Delta_{\mu_R}\mathcal{C}_{\mu_L}}(\rho,\Pi) \\ &= \mathrm{tr}\left(\Delta^{(2)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}}(\rho_{\mu_R}^{G_\mu\otimes 2})\left\langle\Pi_{\mu_L}^{\otimes 2}\right\rangle_{\mathcal{C}_{\mu_L}}\right) \\ &= -\,\mathrm{tr}\left(\left(\Delta^{(2)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}}(\rho_{\mu_R}^{\otimes 2})\right)_{G_\mu}\left\langle\Pi_{\mu_L}^{\otimes 2}\right\rangle_{\mathcal{C}_{\mu_L}}\right) \\ &\le \min_{\frac{1}{p}+\frac{1}{q}=1} \left\|\left(\Delta^{(2)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}}(\rho_{\mu_R}^{\otimes 2})\right)_{G_\mu}\right\|_p \left\|\left\langle\Pi_{\mu_L}^{\otimes 2}\right\rangle_{\mathcal{C}_{\mu_L}}\right\|_q \\ &\le \min_{\frac{1}{p}+\frac{1}{q}=1} \; 4\,\|G_\mu\|_\infty^2 \; \epsilon^{(2|p)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}}(\rho^{\otimes 2}) \; \left\|\left\langle\Pi_{\mu_L}^{\otimes 2}\right\rangle_{\mathcal{C}_{\mu_L}}\right\|_{q^*} \\ &\overset{\substack{p=2\\q=2}}{\le} 4\,\|\rho\|_2^2 \; \left\|\left\langle\Pi_{\mu_L}^{\otimes 2}\right\rangle_{\mathcal{C}_{\mu_L}}\right\|_{2^*} \; \epsilon^{(2|2)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}} \\ &\overset{\substack{p=\infty\\q=1}}{\le} 4\,\|\rho\|_1^2 \; \left\|\left\langle\Pi_{\mu_L}^{\otimes 2}\right\rangle_{\mathcal{C}_{\mu_L}}\right\|_{1^*} \; \epsilon^{(2|\diamond)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}} \; . \end{aligned} \tag{4.D.119}$$

□

Gradient Variance - After Epsilon-t-designs

Theorem 23 (Objective Gradient Variance - After Epsilon-t-designs)**.** *If only the after (L) ensemble is an ϵ-t-design, the objective gradient variance is given by inherent $\sigma^{2L}_{\partial_\mu\mathcal{L}|\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_L}}$ and expressivity $\delta\sigma^{2L}_{\partial_\mu\mathcal{L}|\mathcal{C}\mathcal{C}'_{\mu_L}}$ terms,*

$$\sigma^2_{\partial_\mu\mathcal{L}|\mathcal{C}_{\mu_L}}(\rho,\Pi) \;\le\; \sigma^{2L}_{\partial_\mu\mathcal{L}|\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_L}}(\rho,\Pi) \;+\; \delta\sigma^{2L}_{\partial_\mu\mathcal{L}|\mathcal{C}\mathcal{C}'_{\mu_L}}(\rho,\Pi)\;. \tag{4.D.120}$$

For arbitrary observables Π, and various reference ensembles $\mathcal{C}' \in \{\mathcal{C}(d,s), \mathcal{D}(d)\}$, with the $p = 2$-norm, the objective gradient variance inherent term is,

$$\sigma^{2L}_{\partial_\mu \mathcal{L}|\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_L}}(\rho,\Pi) \;\leq\; \begin{cases} \mathcal{O}\big(\frac{1}{ds}\big) \quad \big\|\big\langle \rho^{\otimes 2}_{\mu_R}\big\rangle_{\mathcal{C}_{\mu_R}}\big\|_{2^*} \quad \|\Pi\|_2^2 & \mathcal{C}' = \mathcal{C}(d,s) \\ 0 & \mathcal{C}' = \mathcal{D}(d) \end{cases} \;, \tag{4.D.121}$$

and the objective gradient variance expressivity term is,

$$\delta\sigma^{2L}_{\partial_\mu \mathcal{L}|\mathcal{C}\mathcal{C}'_{\mu_L}}(\rho,\Pi) \;\leq\; 4\;\epsilon^{(2|\dagger 2)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}}(\Pi^{\otimes 2})\;\big\|\big\langle \rho^{\otimes 2}_{\mu_R}\big\rangle_{\mathcal{C}_{\mu_R}}\big\|_{2^*} \;, \tag{4.D.122}$$

or with the $p = 1$-norm, the objective gradient variance inherent term is,

$$\sigma^{2L}_{\partial_\mu \mathcal{L}|\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_L}}(\rho,\Pi) \;\leq\; \begin{cases} \mathcal{O}\big(\frac{1}{d^2 s}\big) \quad \big\|\big\langle \rho^{\otimes 2}_{\mu_R}\big\rangle_{\mathcal{C}_{\mu_R}}\big\|_{1^*} \quad \|\Pi\|_2^2 & \mathcal{C}' = \mathcal{C}(d,s) \\ 0 & \mathcal{C}' = \mathcal{D}(d) \end{cases} \;, \tag{4.D.123}$$

and the objective gradient variance expressivity term is,

$$\delta\sigma^{2L}_{\partial_\mu \mathcal{L}|\mathcal{C}\mathcal{C}'_{\mu_L}}(\rho,\Pi) \;\leq\; 4\;\epsilon^{(2|\dagger 1)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}}(\Pi^{\otimes 2})\;\big\|\big\langle \rho^{\otimes 2}_{\mu_R}\big\rangle_{\mathcal{C}_{\mu_R}}\big\|_{1^*} \;. \tag{4.D.124}$$

For ϵ-t-cHaar designs, the inherent term decays with system and environment dimensions, and the expressivity term decays at least with the moment operator norms. For ϵ-t-Depolarize designs, the inherent term is identically zero, and the expressivity objective gradient variance term depends strictly on the twirl norms. However, the adjoint channels from $\mathcal{C}_{\mu_L}$ acting on Π are not necessarily physical, the corresponding adjoint moment operator norms are difficult to manipulate and bound, their $\diamond$-norms are not necessarily well defined, and the modified p^* norm must be used. Similarly, the modified p^* norm must be used for the channels from $\mathcal{C}_{\mu_R}$ acting on ρ. Such norms have potentially loose scaling with respect to decaying with dimension, meaning objective gradient concentration statements for strictly after (L) ensembles being ϵ-t-designs, do not always hold.

Proof. We may derive the terms in the objective gradient variance,

$$\begin{aligned} \sigma^{2}_{\partial_\mu \mathcal{L}|\mathcal{C}_{\mu_L}}(\rho,\Pi) \;&\leq\; \sigma^{2L}_{\partial_\mu \mathcal{L}|\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_L}}(\rho,\Pi) \;+\; \delta\sigma^{2L}_{\partial_\mu \mathcal{L}|\mathcal{C}\mathcal{C}'_{\mu_L}}(\rho,\Pi) \\ \delta\sigma^{2L}_{\partial_\mu \mathcal{L}|\mathcal{C}\mathcal{C}'_{\mu_L}}(\rho,\Pi) \;&=\; \delta\sigma^{2L}_{\partial_\mu \mathcal{L}|\mathcal{C}_{\mu_R}\Delta_{\mu_L}}(\rho,\Pi) \;, \end{aligned} \tag{4.D.125}$$

given the previously derived $t = 2$ moments of G_μ-dependent functions of $\rho, \Pi \in \mathcal{B}[\mathcal{H}]$.

For specific reference ensembles and operators, the inherent term is,

$$
\begin{aligned}
&\sigma^{2L}_{\partial_\mu \mathcal{L}|\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_L}}(\rho,\Pi) \qquad (4.D.126)\\
&= \mathrm{tr}\left(\left\langle \left(\Delta^{(2)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}}(\rho^{G_\mu\otimes 2}_{\mu_R})\right)_{\mu_L}\right\rangle_{\mathcal{C}'_{\mu_L}} \Pi^{\otimes 2}\right)\\
&= -\,\mathrm{tr}\left(\left\langle \left(\left(\Delta^{(2)}_{\mathcal{C}_{\mu_R}\mathcal{C}'_{\mu_R}}(\rho^{\otimes 2}_{\mu_R})\right)_{G_\mu}\right)_{\mu_L}\right\rangle_{\mathcal{C}'_{\mu_L}} \Pi^{\otimes 2}\right)\\
&= \begin{cases} -2s^2\frac{d\,\mathrm{tr}\left(\Pi^2\right)-\mathrm{tr}^2(\Pi)}{ds(d^2s^2-1)}\mathrm{tr}\left(S_d\tilde{G}_\mu\left\langle\rho^{\otimes 2}_{\mu_R}\right\rangle_{\mathcal{C}_{\mu_R}}\right) & \mathcal{C}'=\mathcal{C}(d,s)\\ 0 & \mathcal{C}'=\mathcal{D}(d)\end{cases}\\
&\leq \begin{cases} \min\limits_{\frac{1}{p}+\frac{1}{q}=1}\ 2s^2\frac{(\|G_\mu\|^2_\infty+\left\|G^2_\mu\right\|_\infty)(d\,\mathrm{tr}(\Pi^2)-\mathrm{tr}^2(\Pi))}{ds(d^2s^2-1)}\ \|S_d\|_p\ \left\|\left\langle\rho^{\otimes 2}_{\mu_R}\right\rangle_{\mathcal{C}_{\mu_R}}\right\|_{q^*} & \mathcal{C}'=\mathcal{C}(d,s)\\ 0 & \mathcal{C}'=\mathcal{D}(d)\end{cases}\\
&\overset{\substack{p=2\\q=2}}{\leq} \begin{cases} \mathcal{O}\left(\frac{1}{ds}\right) & \left\|\left\langle\rho^{\otimes 2}_{\mu_R}\right\rangle_{\mathcal{C}_{\mu_R}}\right\|_{2^*}\ \ \|\Pi\|_2^2 \quad \{\mathcal{C}'=\mathcal{C}(d,s)\}\\ 0 & \mathcal{C}'=\mathcal{D}(d)\end{cases}\\
&\overset{\substack{p=\infty\\q=1}}{\leq} \begin{cases} \mathcal{O}\left(\frac{1}{d^2s}\right) & \left\|\left\langle\rho^{\otimes 2}_{\mu_R}\right\rangle_{\mathcal{C}_{\mu_R}}\right\|_{1^*}\ \ \|\Pi\|_2^2 \quad \{\mathcal{C}'=\mathcal{C}(d,s)\}\\ 0 & \mathcal{C}'=\mathcal{D}(d)\end{cases},
\end{aligned}
$$

and the expressivity term is,

$$
\begin{aligned}
&\delta\sigma^{2L}_{\partial_\mu \mathcal{L}|\mathcal{C}_{\mu_R}\Delta_{\mu_L}}(\rho,\Pi) \qquad (4.D.127)\\
&= \mathrm{tr}\left(\left\langle\rho^{G_\mu\otimes 2}_{\mu_R}\right\rangle_{\mathcal{C}_{\mu_R}}\Delta^{(2|\dagger)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}}(\Pi^{\otimes 2}_{\mu_L})\right)\\
&= \mathrm{tr}\left(\left(\left\langle\rho^{\otimes 2}_{\mu_R}\right\rangle_{\mathcal{C}_{\mu_R}}\right)_{G_\mu}\Delta^{(2|\dagger)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}}(\Pi^{\otimes 2}_{\mu_L})\right)\\
&\leq \min_{\frac{1}{p}+\frac{1}{q}=1}\ 2(\|G_\mu\|^2_\infty+\left\|G^2_\mu\right\|_\infty)\ \epsilon^{(2|\dagger p)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}}(\Pi^{\otimes 2})\ \left\|\left\langle\rho^{\otimes 2}_{\mu_R}\right\rangle_{\mathcal{C}_{\mu_R}}\right\|_{q^*}\\
&\overset{\substack{p=2\\q=2}}{\leq} 4\ \epsilon^{(2|\dagger 2)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}}(\Pi^{\otimes 2})\ \left\|\left\langle\rho^{\otimes 2}_{\mu_R}\right\rangle_{\mathcal{C}_{\mu_R}}\right\|_{2^*}\\
&\overset{\substack{p=\infty\\q=1}}{\leq} 4\ \epsilon^{(2|\dagger 1)}_{\mathcal{C}_{\mu_L}\mathcal{C}'_{\mu_L}}(\Pi^{\otimes 2})\ \left\|\left\langle\rho^{\otimes 2}_{\mu_R}\right\rangle_{\mathcal{C}_{\mu_R}}\right\|_{1^*}.
\end{aligned}
$$

□

Chapter 5 Appendices

5.A Distributions of Expectation Values of Operators

In these appendices, we seek to understand the analytical behaviour of distributions of expectation values of operators, with respect to randomly distributed quantum states. Here, we extend the derivations of [144] from the case of pure states to the case of mixed states, and provide new geometric-based derivations that simplify analyses.

5.A.1 Expectation Values of Operators

Here, we consider quantum systems described by general d-dimension s-rank mixed states ρ, that are positive, with unit-trace, $\rho \geq 0$, $\mathrm{tr}(\rho) = 1$. Such states can be expressed via purification [16], in terms of ds-dimensional pure states $\psi = |\psi\rangle\langle\psi|$ in a composite d-dimensional system and s-dimensional environment space,

$$\rho = (I \otimes \mathrm{tr}_s)(\psi) \ , \tag{5.A.1}$$

given the d-dimensional identity I, and s-dimensional partial trace tr_s. Such states can represented by $d \times s$ complex parameters φ, via the mapping in terms of elements $\mu \in [d]$, $\nu \in [s]$, where we denote the set of n indices as $[n] = \{0, 1, \dots, n-1\}$,

$$\varphi \ \rightarrow \ |\psi\rangle = \sum_{\mu\in[d]\ ,\ \nu\in[s]} \varphi_{\mu\nu}|\mu\nu\rangle \ \rightarrow \ \rho = \varphi\varphi^\dagger \ , \tag{5.A.2}$$

We also consider l-rank d-dimensional Hermitian operators,

$$\Pi = \sum_\xi \xi \ I_\xi \ , \tag{5.A.3}$$

with # number of distinct real eigenvalues $\{\sigma \le \xi \le \lambda\}$, with associated d_ξ-dimensional eigenspaces, with projectors I_ξ such that $\sum_\xi I_\xi = I$. Expectation values x are thus,

$$x = \tau_\Pi(\rho) = \mathrm{tr}(\Pi\ \rho)\ , \tag{5.A.4}$$

which are bounded by the maximum λ and minimum σ eigenvalues of the operators,

$$\sigma I \le \Pi \le \lambda I \qquad , \qquad \sigma \le x \le \lambda\ . \tag{5.A.5}$$

If we assume states $\rho \sim P_\rho$ are distributed in terms of their parameters $\varphi \sim P_\varphi$ as,

$$P_\rho(\rho) = \int d\varphi\ P_\varphi(\varphi)\ \delta\big(\rho - \varphi\varphi^\dagger\big)\ , \tag{5.A.6}$$

then expectation values $x \sim P_\Pi$ are thus randomly distributed, with distributions,

$$P_\Pi(x) = \int d\rho\ P_\rho(\rho)\ \delta(x - \tau_\Pi(\rho)) = \int d\varphi\ P_\varphi(\varphi)\ \delta(x - \tau_\Pi(\rho))\ . \tag{5.A.7}$$

Here, the complex parameters $\varphi = \alpha + i\beta$ have a measure of,

$$d\varphi = \prod_{\mu\in[d]\ ,\ \nu\in[s]} d\alpha_{\mu\nu}\ d\beta_{\mu\nu} = d\Omega_{2ds}\ d\|\varphi\|\ \|\varphi\|^{2ds-1}\ , \tag{5.A.8}$$

expressed in terms of $2ds$-dimensional real spherical coordinates, radii $\|\varphi\|^2 = \mathrm{tr}\big(\varphi^\dagger\varphi\big)$, areas $\Omega_{2ds} = 2\pi^{ds}/\Gamma_{ds}$, and Gamma functions $\Gamma_d = (d-1)!$. For d, s-dimensional operators Γ, Γ', such that $\varphi \to \Gamma\varphi\Gamma'^\dagger$, the measure transforms as $d\varphi \to |\det(\Gamma)|^{2s}|\det(\Gamma')|^{2d}\ d\varphi$, and is thus invariant $d\varphi \to d\varphi$ under d,s-dimensional unitaries U, V.

We now choose to make some assumptions about the distribution. Given the unitary-invariance of the parameter measure $d\varphi$, we will assume the parameter distribution $P_\varphi = P_\varphi(\|\varphi\|)$ is also unitarily-invariant and depends strictly on the parameter norm $\|\varphi\|$,

$$P_\varphi(\varphi)\ d\varphi \quad = \quad \frac{1}{\frac{1}{2}\Omega_{2ds}}\delta\big(\|\varphi\|^2 - 1\big)\, d\varphi\ . \tag{5.A.9}$$

5.A.2 Moments of Expectation Values of Operators

While studying distributions of expectation values, it is important to first study the underlying distributions of states which generate such distributions. Here, the unitarily-invariant distribution over ds complex parameters,

$$P_\varphi(\varphi)\ d\varphi \quad = \quad \frac{1}{\frac{1}{2}\Omega_{2ds}}\delta\big(\|\varphi\|^2 - 1\big)\, d\varphi \quad \cong \quad d\psi\ , \tag{5.A.10}$$

is isomorphic to the Haar distribution $d\psi$ of uniformly random ds-dimensional pure states $\varphi\varphi^\dagger \to \psi$, with respect to ds-dimensional operators $\Pi \to \Pi \otimes I_s$, up to tracing over the s-dimensional environment.

Such distributions of ds-dimensional states have t-order moments in terms of projectors $\mathcal{T}_{ds}^{(t)}$ onto the $\binom{ds+t-1}{t}$-dimensional symmetric subspace [248], over the t-copies of the space,

$$\psi_t = \int d\psi\ \psi^{\otimes t} = \frac{1}{\binom{ds+t-1}{t}} \mathcal{T}_{ds}^{(t)}\ , \tag{5.A.11}$$

and t-order moments of expectation values $x = \mathrm{tr}(\psi\ \Xi)$ of operators $\Pi \to \Xi = \Pi \otimes I_s$ are,

$$x_t = \int d\psi\ \mathrm{tr}(\psi\ \Xi)^t = \mathrm{tr}\big(\psi_t\ \Xi^{\otimes t}\big)\ \ . \tag{5.A.12}$$

Finally, the symmetric subspace projector $\mathcal{T}_{ds}^{(t)}$ can be expressed in terms of ds-dimensional unitary representations V_{ds} of the t-order permutations $\mathcal{S}_t$, and we will denote un-normalized moments of the ds-dimensional Ξ, and traces of powers of Ξ as,

$$\mathcal{T}_{ds}^{(t)} = \frac{1}{t!} \sum_{\sigma \in \mathcal{S}_t} V_{ds}(\sigma) \tag{5.A.13}$$

$$\xi_t = \mathrm{tr}\Big(\mathcal{T}_{ds}^{(t)}\ \Xi^{\otimes t}\Big) \qquad , \qquad \zeta_t = \mathrm{tr}\big(\Xi^t\big) \qquad , \qquad \vartheta_t = \sum_{\sum_\xi \sum_{l_\xi \in [d_\xi]} t_{l_\xi} = t}\ \prod_{\substack{\xi \\ l_\xi \in [d_\xi]}} \xi^{t_{l_\xi}}\ . \tag{5.A.14}$$

Now, we can derive closed-form expressions for moments ξ_t in terms of the eigenvalues and multiplicities $\{\xi, d_\xi\}$ of arbitrary Ξ. Such moments have been derived [144] for Ξ with strictly $d_\xi = 1$, and are potentially known to the combinatorics community, given several intermediate results for $\xi_t, \zeta_t, \vartheta_t$ are known [380]. Our derivations use generating functions,

$$\sum_t \xi_t\ x^t\ , \tag{5.A.15}$$

to express ξ_t by identifying the t-order terms in this series in x. Compositions $F(G(x))$ can then be written as expansions, by collecting terms in expansions for $F(x)$ and $G(x)$,

$$F(x) = \sum_t \frac{1}{t!}\ f_t\ x^t \quad , \quad G(x) = \sum_t \frac{1}{t}\ g_t\ x^t \tag{5.A.16}$$

$$F(G(x)) = \sum_t \sum_l \sum_{\substack{\sum_k l_k = l \\ \sum_k l_k k = t}} \frac{1}{l!} \binom{l}{\{l_k\}}\ f_l\ \prod_k \left(\frac{g_k}{k}\right)^{l_k}\ x^t \tag{5.A.17}$$

which is known as the *Faà di Bruno formula* [380] in terms of the partitions $\{l_k\}$ of (t,l) that satisfy the constraints of their sum being l and their sum weighted by k being t. Useful examples of such generating functions are,

$$F(x) = e^x = \sum_t \frac{1}{t!}\, x^t \quad \rightarrow \quad f_t = 1 \tag{5.A.18}$$

$$F(x) = \frac{1}{(1-x)^l} = \sum_t \binom{l+t-1}{t}\, x^t \quad \rightarrow \quad f_t = \frac{\Gamma_{l+t}}{\Gamma_l} \tag{5.A.19}$$

$$G(x) = \log\left(\left(\frac{1}{1-x}\right)\right) = \sum_l \frac{1}{t}\, x^t \quad \rightarrow \quad g_t = 1 \tag{5.A.20}$$

$$\binom{l}{\{l_\xi\}} = \frac{l!}{\prod_\xi l_\xi!} \qquad , \qquad \Gamma_l = (l-1)!\ . \tag{5.A.21}$$

Partitions $\{l_k\}$ of (t,l) are particularly relevant to our derivations. In fact, summations over t-order permutations $\mathcal{S}_t$ can be described as summations over partitions, with a partition identically describing the cycle structure $\{l_k\}$ of a permutation, given permutations with $l \leq t$ cycles have l_k number of k-length cycles,

$$\frac{1}{t!}\sum_{\sigma\in\mathcal{S}_t} \ = \ \sum_l \sum_{\substack{\sum_k l_k = l \\ \sum_k l_k k = t}} \frac{1}{l!}\binom{l}{\{l_k\}} \prod_k \frac{1}{k^{l_k}}\ . \tag{5.A.22}$$

There are $\binom{t}{\{l_k k\}}$ ways of choosing such partitions, times $(l_k k)!$ ways of placing the elements within the l_k k-length cycles, less the $l_k!$ ways of ordering the k-length cycles, and less k^{l_k} ways of choosing the representative first element in each k-length cycle. We also can replace the $t!$ factors with $l!$, given $t!$ occurs in the numerator and denominator.

To derive the moments ξ_t, we note that traces of operators with permutations $\sigma \in \mathcal{S}_t$, with cycle structure described by the partition $\{l_k\}$, equals products of powers of traces of k-powers of such operators, denoted by ζ_k, allowing us to relate ξ_t and ζ_t,

$$\mathrm{tr}\big(V_{ds}(\sigma)\ \Xi^{\otimes t}\big) = \prod_k \mathrm{tr}\big(\Xi^k\big)^{l_k} = \prod_k \zeta_k^{l_k} \tag{5.A.23}$$

$$\rightarrow$$

$$\xi_t = \sum_l \sum_{\substack{\sum_k l_k = l \\ \sum_k l_k k = t}} \frac{1}{l!}\binom{l}{\{l_k\}} \prod_k \left(\frac{\zeta_k}{k}\right)^{l_k}\ . \tag{5.A.24}$$

Further, given ζ_k, ϑ_t are functions of the spectra $\{\xi, d_\xi\}$, we have the relationships to distinct generating functions,

$$\zeta_k = \sum_\xi d_\xi \xi^k \tag{5.A.25}$$

$$\rightarrow$$

$$\sum_k \frac{1}{k}\zeta_k \; x^k = \sum_\xi d_\xi \sum_k \frac{1}{k}(\xi x)^k = \sum_\xi \log\left(\left(\frac{1}{1-\xi x}\right)^{d_\xi}\right) \tag{5.A.26}$$

$$\vartheta_t = \sum_{\sum_\xi \sum_{l_\xi \in [d_\xi]} t_{l_\xi} = t} \prod_{\substack{\xi \\ l_\xi \in [d_\xi]}} \xi^{t_{l_\xi}} \tag{5.A.27}$$

$$\rightarrow$$

$$\sum_t \vartheta_t \; x^t = \sum_t \sum_{\sum_\xi \sum_{l_\xi \in [d_\xi]} t_{l_\xi} = t} \prod_\xi (\xi x)^{t_{l_\xi}} = \prod_\xi \left(\frac{1}{1-\xi x}\right)^{d_\xi} \; . \tag{5.A.28}$$

Such expansions of permutations in terms of partitions for ξ_t, are reminiscent of the expansions of compositions of generating functions, namely, the composition of the exponential function with the generating function of ζ_k,

$$\sum_t \xi_t \; x^t = e^{\sum_k \frac{1}{k}\zeta_k \; x^k} = \prod_\xi \left(\frac{1}{1-\xi x}\right)^{d_\xi} = \frac{1}{\det(I - x\Xi)} = \sum_t \vartheta_t \; x^t \, . \tag{5.A.29}$$

Therefore the t-order moments ξ_t of operators Ξ with respect to the symmetric subspace projector $\mathcal{T}_{ds}^{(t)}$ are in fact the *complete homogenous symmetric polynomials* [380] in the spectra $\{\xi, d_\xi\}$, generated by the determinant of $I - x\Xi$,

$$\xi_t \;=\; \vartheta_t \;=\; \sum_{\sum_\xi \sum_{l_\xi \in [d_\xi]} t_{l_\xi} = t} \prod_{\substack{\xi \\ l_\xi \in [d_\xi]}} \xi^{t_{l_\xi}} \;=\; \frac{1}{t!}\,\partial_x^t \,\frac{1}{\det(I - x\Xi)}|_{x=0} \; . \tag{5.A.30}$$

Finally, we can derive a closed-form expression for such moments ξ_t, using partial fraction decompositions,

$$\frac{1}{\prod_\xi (x-\xi)^{d_\xi}} = \sum_\xi \sum_{l_\xi \in [d_\xi]} \frac{f_{l_\xi}}{(x-\xi)^{ds - l_\xi}} \qquad : \qquad f_{l_\xi} = \frac{1}{l_\xi!}\partial_x^{l_\xi} \frac{1}{\prod_{\zeta \neq \xi}(x-\zeta)^{d_\zeta}}|_{x=\xi} \; , \tag{5.A.31}$$

where derivatives of products of functions $\{f_\xi(x)\}$ and of reciprocal polynomials $1/(x-\xi)^k$ can be expanded as,

$$\partial_x^l \prod_\xi f_\xi(x) = \sum_{\sum_\xi l_\xi = l} \binom{l}{\{l_\xi\}} \prod_\xi \partial_x^{l_\xi} f_\xi(x) \tag{5.A.32}$$

$$\partial_x^l \frac{1}{(x-\xi)^k} = (-1)^l \; \frac{\Gamma_{k+l}}{\Gamma_k} \; \frac{1}{(x-\xi)^{k+l}} \; . \tag{5.A.33}$$

Therefore an expression for the determinant of $I - x\Xi$ is,

$$\prod_\xi \frac{1}{(1-\xi x)^{d_\xi}} = \sum_\xi \sum_{l_\xi \in [d_\xi]} \sum_{\sum_{\zeta \neq \xi} l_\zeta = l_\xi} \prod_{\zeta \neq \xi} \binom{d_\zeta + l_\zeta - 1}{l_\zeta} \frac{\zeta^{l_\zeta}}{(\xi - \zeta)^{d_\zeta + l_\zeta}} \; \frac{(-1)^{l_\xi} \; \xi^{ds - d_\xi}}{(1-\xi x)^{d_\xi - l_\xi}} \; , \tag{5.A.34}$$

and expanding the reciprocal x-dependent terms, we can match the t-order term with the t-order moments,

$$\begin{aligned} \xi_t = & \sum_\xi \sum_{l_\xi \in [d_\xi]} (-1)^{l_\xi} \; \binom{d_\xi - l_\xi + t - 1}{t} \; \xi^{ds - d_\xi + t} \\ & \times \sum_{\sum_{\zeta \neq \xi} l_\zeta = l_\xi} \prod_{\zeta \neq \xi} \binom{d_\zeta + l_\zeta - 1}{l_\zeta} \frac{\zeta^{l_\zeta}}{(\xi - \zeta)^{d_\zeta + l_\zeta}} \; . \end{aligned} \tag{5.A.35}$$

By using combinatorics we avoid any contour integration used in [144], simplifying our analysis to derive expressions for the t-order moments of d-dimensional operators Π with spectra $\{\xi, d_\xi\}$ in s-dimensional environments,

$$x_t = \sum_\xi \sum_{l_\xi \in [d_\xi s]} \chi_{l_\xi, t} \; \xi^t \; , \tag{5.A.36}$$

given the spectrum-dependent coefficients,

$$\chi_{l_\xi, t} = (-1)^{l_\xi} \; \frac{\Gamma_{ds}}{\Gamma_{d_\xi s - l_\xi}} \frac{\Gamma_{d_\xi s - l_\xi + t}}{\Gamma_{ds+t}} \sum_{\sum_{\zeta \neq \xi} l_\zeta = l_\xi} \prod_{\zeta \neq \xi} \binom{d_\zeta s + l_\zeta - 1}{l_\zeta} \frac{\xi^{d_\zeta s} \; \zeta^{l_\zeta}}{(\xi - \zeta)^{d_\zeta s + l_\zeta}} \; . \tag{5.A.37}$$

5.A.3 Distributions of Expectation Values of Operators

Here, we consider expectation values $x = \mathrm{tr}(\rho\,\Pi) = \|\Gamma\varphi\|^2$, with respect to l-rank operators $\Pi = \Gamma^\dagger\Gamma$, with $\#$ distinct eigenvalues $\{\sigma \leq \xi \leq \lambda\}$, and s-rank states $\rho = \varphi\varphi^\dagger$. Here the parameters φ are distributed according to the unitarily-invariant distribution of Haar random ds-dimensional pure states,

$$\varphi \sim P_\varphi \propto \delta\big(\|\varphi\|^2 - 1\big) \quad \rightarrow \quad x \sim P_\Pi \propto \int d\varphi\; \delta\big(\|\varphi\|^2 - 1\big)\;\delta\big(\|\Gamma\varphi\|^2 - x\big)\ . \tag{5.A.38}$$

From the unitary-invariance that defines the expectation value distributions, given any scalars ζ, ς such that operators are transformed as $\Pi \rightarrow (\Pi - \varsigma I)/\zeta$, then the distributions exhibit shift and scale invariance,

$$P_\Pi(x) = \frac{1}{|\zeta|} P_{\frac{\Pi - \varsigma I}{\zeta}}\left(\frac{x - \varsigma}{\zeta}\right) \quad : \quad \sigma I \leq \Pi \leq \lambda I \quad , \quad \zeta\sigma + \varsigma \leq x \leq \zeta\lambda + \varsigma\ . \tag{5.A.39}$$

As such, without loss of generality, it is convenient to normalize the operators as,

$$\Pi \rightarrow \frac{\Pi - \sigma I}{\lambda - \sigma} \quad , \quad x \rightarrow \frac{x - \sigma}{\lambda - \sigma}\ , \tag{5.A.40}$$

which are bounded by their eigenvalues,

$$0 \leq \Pi \leq I \quad , \quad 0 \leq x \leq 1\ . \tag{5.A.41}$$

The normalized operators $\Pi \equiv \Gamma^\dagger\Gamma \geq 0$ are thus positive, l-rank, with $\#$ number of distinct non-negative eigenvalues. The image of the normalized Π thus induces $l, d-l$-dimensional subspaces with associated $l, d-l$-rank projectors $I_\Pi, I - I_\Pi$, and it is often convenient to partition the parameters φ into associated $l, d-l$-dimensional components,

$$\varphi \rightarrow \varphi \oplus \bar{\varphi} \quad , \quad d\varphi \rightarrow d\varphi\; d\bar{\varphi}\ . \tag{5.A.42}$$

Such normalizations and partitionings enforce that the positive normalized $l < d$-rank operators Π always have a non-trivial l-dimensional image where Π acts on φ, and a non-trivial $d-l$-dimensional kernel where Π acts $\bar{\varphi}$. Such a bi-partitioned basis thus simplifies analysis by straightforwardly satisfying the two distribution Dirac-delta functions constraints related to trace-preservation $\|\varphi \oplus \bar{\varphi}\|^2 = 1$ and expectation value-preservation $\|\Gamma\varphi\|^2 = x$.

The distribution of $x \sim P_\Pi(x)$ for $\sigma \leq x \leq \lambda$, thus depends on the number of distinct eigenvalues of $\#$ of the operators Π.

For $\# = 1$-eigenvalue operators $\Pi = I$,

$$P_\Pi(x) = \int d\varphi \; P_\varphi(\varphi) \; \delta(\pi_\Pi(\varphi) - x) \tag{5.A.43}$$

$$= \int d\varphi \; P_\varphi(\varphi) \; \delta\big(\|\varphi\|^2 - x\big) \tag{5.A.44}$$

$$= \frac{1}{\frac{1}{2}\Omega_{2ds}} \int d\varphi \; \delta\big(\|\varphi\|^2 - 1\big) \;\; \delta\big(\|\varphi\|^2 - x\big) \tag{5.A.45}$$

$$= \delta(x - 1) \;\; . \tag{5.A.46}$$

For $\# = 2$-eigenvalue operators $\Pi = I_\Pi$,

$$P_\Pi(x) = \int d\varphi \; P_\varphi(\varphi) \; \delta(\pi_\Pi(\varphi) - x) \tag{5.A.47}$$

$$= \int d\varphi \; d\bar{\varphi} \; P_{\varphi\bar{\varphi}}(\varphi, \bar{\varphi}) \; \delta\big(\|\varphi\|^2 - x\big) \tag{5.A.48}$$

$$= \frac{1}{\frac{1}{2}\Omega_{2ds}} \int d\varphi \; d\bar{\varphi} \; \delta\big(\|\varphi\|^2 + \|\bar{\varphi}\|^2 - 1\big) \;\; \delta\big(\|\varphi\|^2 - x\big) \tag{5.A.49}$$

$$= \frac{\frac{1}{2}\Omega_{2ls} \; \frac{1}{2}\Omega_{2(d-l)s}}{\frac{1}{2}\Omega_{2ds}} \; x^{ls-1} \; \frac{1}{\frac{1}{2}\Omega_{2ls}} \int d\varphi \; \delta\big(\|\varphi\|^2 - 1\big) \left(1 - x\|\varphi\|^2\right)^{(d-l)s-1} \tag{5.A.50}$$

$$= \frac{\Gamma_{ds}}{\Gamma_{ls} \; \Gamma_{(d-l)s}} \; x^{ls-1} \;\; (1 - x)^{(d-l)s-1} \;\; . \tag{5.A.51}$$

For $\# > 1$-eigenvalue operators $\Pi = \Gamma^\dagger \Gamma$,

$$
\begin{aligned}
P_\Pi(x) &= \int d\varphi \; P_\varphi(\varphi) \; \delta\big(\|\Gamma\varphi\|^2 - x\big) && (5.A.52)\\
&= \frac{1}{\frac{1}{2}\Omega_{2ds}} \int d\varphi \; \delta\big(\|\varphi\|^2 - 1\big)\, \delta\big(\|\Gamma\varphi\|^2 - x\big) && (5.A.53)\\
&= \frac{1}{\frac{1}{2}\Omega_{2ds}} \int d\varphi \; d\bar{\varphi} \; \delta\big(\|\varphi\|^2 + \|\bar{\varphi}\|^2 - 1\big)\, \delta\big(\|\Gamma\varphi\|^2 - x\big) && (5.A.54)\\
&= \frac{\frac{1}{2}\Omega_{2(d-l)s}}{\frac{1}{2}\Omega_{2ds}} \int d\varphi \;\; \big(1 - \|\varphi\|^2\big)^{(d-l)s-1} \;\; \delta\big(\|\Gamma\varphi\|^2 - x\big) && (5.A.55)\\
&= \frac{\frac{1}{2}\Omega_{2ls}\frac{1}{2}\Omega_{2(d-l)s}}{\frac{1}{2}\Omega_{2ds}} \frac{1}{|\det(\Gamma)|^{2s}} \; x^{ls-1} \; \frac{1}{\frac{1}{2}\Omega_{2ls}} \int d\varphi \; \delta\big(\|\varphi\|^2 - 1\big) && (5.A.56)\\
&\qquad \times \; \Big(1 - x\big\|\Gamma^{-1}\varphi\big\|^2\Big)^{(d-l)s-1} \\
&= \frac{\frac{1}{2}\Omega_{2ls}\frac{1}{2}\Omega_{2(d-l)s}}{\frac{1}{2}\Omega_{2ds}} \frac{1}{\det(\Pi)^{s}} \; x^{ls-1} \; \frac{1}{\frac{1}{2}\Omega_{2ls}} \int d\varphi \; \delta\big(\|\varphi\|^2 - 1\big) && (5.A.57)\\
&\qquad \times \; \mathrm{tr}\Big(\varphi\varphi^{\dagger\, \otimes(d-l)s-1} \;\; \big(I_\Pi - x\Pi^{-1}\big)^{\otimes(d-l)s-1}\Big) \\
&= \frac{\frac{1}{2}\Omega_{2ls}\frac{1}{2}\Omega_{2(d-l)s}}{\frac{1}{2}\Omega_{2ds}} \frac{1}{\det(\Pi)^{s}} \; x^{ls-1} \; \frac{1}{\mathrm{tr}\Big(\mathcal{T}_q^{(t)}\Big)} \; \mathrm{tr}\big(\mathcal{T}_q^{(t)} \; \Delta(x)^{\otimes t}\big) \;\; , && (5.A.58)
\end{aligned}
$$

where $t = (d-l)s - 1$, $q = ls$, $\Delta(x) = (I_\Pi - x\Pi^{-1}) \otimes I$, and $\mathcal{T}_q^{(t)}$ is the projector onto the $\binom{q+t-1}{t}$-dimensional symmetric subspace of the t-copies of the q-dimensional space. Using our expressions for t-order moments,

$$
P_\Pi(x) = \sum_{\xi \neq 0} \sum_{l_\xi \in [d_\xi s]} \pi_{l_\xi}(x) \; \frac{x^{d_\xi s - l_\xi - 1}(\xi - x)^{(d-d_\xi)s-1}}{\xi^{(d-l)s + d_\xi s - l_\xi - 1}} \;\; , \tag{5.A.59}
$$

with spectrum and x-dependent coefficients,

$$
\begin{aligned}
\pi_{l_\xi}(x) = (-1)^{l_\xi} &\frac{\Gamma_{ds}}{\Gamma_{ls}\Gamma_{(d-l)s}} \frac{\Gamma_{ls}}{\Gamma_{d_\xi s - l_\xi}} \frac{\Gamma_{(d-l)s + d_\xi s - l_\xi - 1}}{\Gamma_{ds-1}} && (5.A.60)\\
&\times \sum_{\sum_{\zeta \neq 0,\xi} l_\zeta = l_\xi} \; \prod_{\zeta \neq 0,\xi} \binom{d_\zeta s + l_\zeta - 1}{l_\zeta} \frac{(\zeta - x)^{l_\zeta}}{(\xi - \zeta)^{d_\zeta s + l_\zeta}} \; .
\end{aligned}
$$

Alternatively, we can derive simpler expressions, by extending previous derivations using contour integration [144], to the case of non-trivial environments $s > 1$.

For $\# > 1$-eigenvalue operators $\Pi = \Gamma^\dagger \Gamma$,

$$P_\Pi(x) = \int d\varphi \; P_\varphi(\varphi) \; \delta(\pi_\Pi(\varphi) - x) \tag{5.A.61}$$

$$= \frac{1}{\frac{1}{2}\Omega_{2ds}} \int d\varphi \; \delta\big(\|\varphi\|^2 - 1\big) \;\; \delta\big(\|\Gamma\varphi\|^2 - x\big) \tag{5.A.62}$$

$$= \frac{1}{\frac{1}{2}\Omega_{2ds}} \frac{1}{2\pi 2\pi} \int du \; dv \; e^{-i(xu+v)} \int d\varphi \; e^{i \; \mathrm{tr}\left(\varphi^\dagger (vI + u\Pi)\varphi\right)} \tag{5.A.63}$$

$$= \frac{i^{ds}\pi^{ds}}{\frac{1}{2}\Omega_{2ds}} \frac{1}{2\pi 2\pi} \int du \; e^{-ixu} \int dv \; \frac{e^{-iv}}{\prod_\xi (v + u\xi)^{d_\xi s}} \tag{5.A.64}$$

$$= \sum_\xi \sum_{l_\xi \in [d_\xi s]} \sum_{\sum_{\zeta \neq \xi} l_\zeta = l_\xi} (-1)^{l_\xi} \; \frac{\Gamma_{ds}}{\Gamma_{d_\xi s - l_\xi}} \prod_{\zeta \neq \xi} \binom{d_\zeta s + l_\zeta - 1}{l_\zeta} \frac{1}{(\xi - \zeta)^{d_\zeta s + l_\zeta}} \tag{5.A.65}$$

$$\times \int \frac{du}{2\pi} \; \frac{e^{iu(\xi - x)}}{(iu)^{(d-d_\xi)s + l_\xi}} \tag{5.A.66}$$

$$P_\Pi(x) = \sum_\xi \sum_{l_\xi \in [d_\xi s]} \pi_{l_\xi} \; \mathrm{sign}(\xi - x) \;\; (\xi - x)^{(d-d_\xi)s + l_\xi - 1} \; , \tag{5.A.67}$$

with cumulative distributions,

$$F_\Pi(x) = \sum_\xi \sum_{l_\xi \in [d_\xi s]} \pi_{l_\xi} \; \left(-\mathrm{sign}(\xi - x) \;\; (\xi - x)^{(d-d_\xi)s + l_\xi} \; + \; (\xi - \sigma)^{(d-d_\xi)s + l_\xi}\right) \; , \tag{5.A.68}$$

given the spectrum-dependent coefficients,

$$\pi_{l_\xi} = \frac{1}{2}(-1)^{l_\xi} \; \frac{\Gamma_{ds}}{\Gamma_{d_\xi s - l_\xi} \Gamma_{(d-d_\xi)s + l_\xi}} \sum_{\sum_{\nu \neq \mu} l_\zeta = l_\xi} \prod_{\zeta \neq \xi} \binom{d_\zeta s + l_\zeta - 1}{l_\zeta} \frac{1}{(\xi - \zeta)^{d_\zeta s + l_\zeta}} \; . \tag{5.A.69}$$

The resulting distribution and moment expressions, in particular their coefficients $\pi_{l_\xi}, \chi_{l_\xi, t}$, have remarkable similarities and differences, however do not appear to immediately follow from each other. The forms of such expressions evidently have deep connections to algebraic combinatorics [380], and the distributions appear to potentially have connections to hypergeometric functions [382].

To derive expressions for these distributions, we use several identities, including for Dirac delta functions,

$$\delta(\xi) = \frac{1}{2\pi} \int dx\ e^{i\xi x}\ , \tag{5.A.70}$$

and for ds-dimensional complex Gaussian integrals, given operators Π with spectra $\{\xi, d_\xi\}$,

$$\int d^{2ds}\varphi\ e^{i\ \mathrm{tr}\left(\varphi^\dagger \Pi \varphi\right)} = \frac{(i\pi)^{ds}}{|\det(\Pi)|^s} \tag{5.A.71}$$

$$\det(xI + \Pi) = \prod_\xi (x + \xi)^{d_\xi} \qquad , \qquad \binom{d}{\{d_\xi\}} = \frac{d!}{\prod_\xi d_\xi!}\ . \tag{5.A.72}$$

Integrals can be evaluated via contour integration of products of functions,

$$\int dx\ \frac{1}{\prod_\xi (x-\xi)^{d_\xi}} = i2\pi \sum_\xi \mathrm{sign}(\xi)\ \frac{1}{\Gamma_{d_\xi}}\ \partial_x^{d_\xi - 1}\ \frac{1}{\prod_{\zeta\neq\xi}(x-\zeta)^{d_\zeta}}\Bigg|_{x=\xi}\ , \tag{5.A.73}$$

with the following identities for various derivatives of functions $f, g, \{f_\xi\}$,

$$\partial^l fg = \sum_k^l \binom{l}{k}\ \partial^k f\ \partial^{l-k} g \qquad , \qquad \partial^l \prod_\xi f_\xi = \sum_{\sum_\xi l_\xi = l} \binom{l}{\{l_\xi\}} \prod_\xi \partial^{l_\xi} f_\xi\ , \tag{5.A.74}$$

$$\partial_x^l\ \frac{1}{(x-\xi)^k} = (-1)^l\ \frac{\Gamma_{k+l}}{\Gamma_k}\ \frac{1}{(x-\xi)^{k+l}} \qquad , \qquad \partial_x^l\ e^{\xi x} = \xi^l\ e^{\xi x}\ , \tag{5.A.75}$$

and therefore we have,

$$\begin{aligned}\partial_x^{d_\xi - 1}\ \frac{e^{zx}}{\prod_{\zeta\neq\xi}(x-\zeta)^{d_\zeta}} = &\sum_{l_\xi \in [d_\xi]} \sum_{\sum_{\zeta\neq\xi} l_\zeta = l_\xi} (-1)^{l_\xi}\ z^{d_\xi - l_\xi - 1}\ \frac{\Gamma_{d_\xi}}{\Gamma_{d_\xi - l_\xi}} \\ &\times \prod_{\zeta\neq\xi} \binom{d_\zeta + l_\zeta - 1}{l_\zeta} \frac{1}{(x-\zeta)^{d_\zeta + l_\zeta}}\ e^{zx}\ .\end{aligned} \tag{5.A.76}$$

Finally, we note that integrals of sign functions are,

$$\int_\alpha^\beta dx\ \mathrm{sign}(\xi - x)\,(\xi - x)^{k-1} = \frac{1}{k}\left(-\mathrm{sign}(\xi-\beta)\,(\xi-\beta)^k\ +\ (\xi-\alpha)^k\right)\ . \tag{5.A.77}$$

5.B Simulations of Expectation Values of Operators

In these appendices, we describe implementation details of numerically simulating expectation values of operators. In particular, we discuss the simulated sets of measurement operators, numerical representations of empirical distributions, perform studies of effects of number of samples on these studies, and finally investigate and interpret the properties of the fit effective analytical model parameters. All numerical experiments are performed using a custom Jax-based quantum circuit simulator [6], and data is available at [5].

5.B.1 Sets of Measurement Operators

Here, we consider measurement probabilities p resulting from sets of positive-operator-valued (POVM) measurement operators $\Pi \in \mathcal{P}$.

Local POVM Measurement Operators

Regarding the specific operators studied, for numerical efficiency during simulations, we will consider tensor-products of $n,\ q\ :\ d = q^n$-dimensional local operators $\Pi \to \Pi = \otimes_{i\in[n]}\Pi_i$. Each local $\Pi_i \in \mathcal{P}_i$ will be from the same set of operators $\mathcal{P}_i = \mathcal{P}$, thus $\mathcal{P} \to \mathcal{P}^{\otimes n}$. We will consider both symmetric and non-symmetric sets of operators.

First, we consider the set of $\left|\mathcal{P}^{\mathrm{PVM}}\right| = d$ symmetric, local, orthogonal, and non-informationally-complete Projector PVM measurement operators, in terms of orthogonal basis states $|\mu\rangle$, $\mu \in [d]$,

$$\begin{aligned} \Pi_\mu^{\mathrm{PVM}} &= \ \otimes_{i\in[n]}\, \psi_{\mu_i}^{\mathrm{PVM}} \\ |\psi\rangle_{\mu_i}^{\mathrm{PVM}} &= |\mu_i\rangle\ . \end{aligned} \tag{5.B.1}$$

Second, we consider the set of $\left|\mathcal{P}^{\text{SIC-POVM}}\right| = d^2$ symmetric, local, non-orthogonal, and informationally-complete Tetrad SIC-POVM measurement operators, in terms of uniformly spread out pure states ψ_μ, $\mu \in [d^2]$,

$$\begin{aligned} \Pi_\mu^{\text{SIC-POVM}} &= \lambda \otimes_{i\in[n]} \psi_{\mu_i} \\ |\psi\rangle_{\mu_i}^{\text{SIC-POVM}} &\in \{|0\rangle\ ,\ \cos(\theta_q/2)|0\rangle + e^{i\mu_i\phi_q}\sin(\theta_q/2)|1\rangle\}_{0<\mu_i<q^2} \\ \lambda = \frac{1}{q^{2n}} \quad , &\quad \cos(\theta_q/2) = \sqrt{\frac{1}{q^2-1}} \quad , \quad \phi_q = \frac{2\pi}{q^2-1}\ . \end{aligned} \tag{5.B.2}$$

Third, we consider the set of $\left|\mathcal{P}^{\text{NON-SIC-POVM}}\right| = d^2$ non-symmetric, local, non-orthogonal, and informationally-complete Pauli NON-SIC-POVM measurement operators, in terms of non-uniform mixed states Ψ_μ, $\mu \in [d^2]$,

$$\begin{aligned}
\Pi^{\text{NON-SIC-POVM}}_{\mu} &= \lambda \otimes_{i\in[n]} \Psi^{\text{NON-SIC-POVM}}_{\mu_i} \\
\Psi^{\text{NON-SIC-POVM}}_{\mu_i} &\in \{|0\rangle\langle 0| \ , \ |+\rangle\langle +| \ , \ |+i\rangle\langle +i| \ , \ |1\rangle\langle 1| + |-\rangle\langle -| + |-i\rangle\langle -i|\} \\
\lambda &= \frac{1}{(q^2-1)^n} \ .
\end{aligned} \tag{5.B.3}$$

Total Distributions of Measurement Probabilities

Here, we discuss an important clarifying remark, regarding the total probability over the set of operators,

$$P_{\mathcal{P}}(p) = \sum_{\Pi} P_{\Pi|\mathcal{P}}(\Pi) P_{\Pi}(p) \quad \to \quad \tilde{P}_{\mathcal{P}}(p) = \sum_{\Pi\in\mathcal{P}} P_{\Pi|\mathcal{P}}(\Pi) \tilde{P}_{\Pi}(p) \ , \tag{5.B.4}$$

which corresponds to sampling from the joint distribution of states $\rho \sim P_\rho$ and operators $\Pi \sim P_{\Pi|\mathcal{P}}$. In our simulations, operators are not sampled, but deterministically iterated over all $\Pi \in \mathcal{P}$, weighted by the exact operator probability $P_{\Pi|\mathcal{P}}(\Pi)$, which we choose to be uniform $P_{\Pi|\mathcal{P}}(\Pi) = 1/|\mathcal{P}|$. This partially-deterministic procedure results in solely empirical conditional distributions $\tilde{P}_{\Pi}(p)$, with a total empirical distribution of $\tilde{P}_{\mathcal{P}}(p)$. Previous works [120, 372] have avoided such technical considerations due to considering individual operators, or symmetric operators representative of the whole set. Sampling operators may be necessary for larger dimensions d as exact iterations over $\mathcal{O}(\text{poly}(d))$ operators become infeasible, compounding uncertainty within total empirical distributions.

5.B.2 Numerical Continuous Variable Empirical Distributions

Here, we construct empirical distributions from numerical simulations. We are considering continuous variable distributions, which given their expressions in terms of integrals as opposed to summations, are significantly more difficult to approximate than discrete variable distributions [377, 443]. We must therefore make two approximations for numerical feasibility, regarding both the form of the resulting empirical distributions formed from samples, and the form of the metric used to quantify differences between distributions.

Binned Empirical Distributions

Regarding the empirical distributions and sampling, here we choose to perform the following binning procedure, to avoid storing exponential in system size number of samples. Given m samples of continuous variables x, for numerical tractability when computing empirical distributions and histograms, we bin samples into m' discretized bins,

$$\{x_i\}_{i\in[m]} \to \{x'_i\}_{i\in[m']} \quad : \quad x_i \to x'_{i'} \text{ if } x_i \in [x'_{i'}, x'_{i'+1}] \ , \tag{5.B.5}$$

with bin density, bin size, and cumulative bin density,

$$\rho(x_{i'}) = \frac{\big|\{x_i \ : \ x_i \in [x'_{i'}, x'_{i'+1}]\}_{i\in[m]}\big|}{m} \quad , \quad \eta(x'_i) = \frac{x'_{i'+1} - x'_{i'}}{m'} \tag{5.B.6}$$

$$\omega(x'_{i'}) = \sum_{j'\leq i'} \rho(x_{j'}) \ . \tag{5.B.7}$$

This procedure yields binned empirical distributions,

$$\tilde{P}'(x) = \frac{1}{m'} \sum_{i'\in[m']} \rho(x'_{i'}) \ \frac{1}{\eta(x'_{i'})} \ \delta\big(x'_{i'} \leq x < x'_{i'+1}\big) \tag{5.B.8}$$

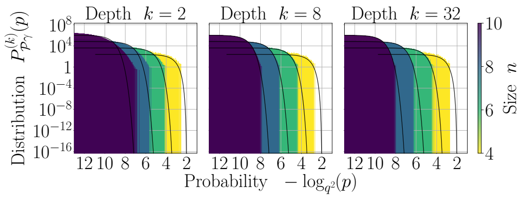
$$\tilde{F}'(x) = \sum_{i'\in[m']} \omega(x'_{i'}) \ \delta(x \geq x'_{i'}) \ , \tag{5.B.9}$$

which may be interpreted in terms of the conditional distributions,

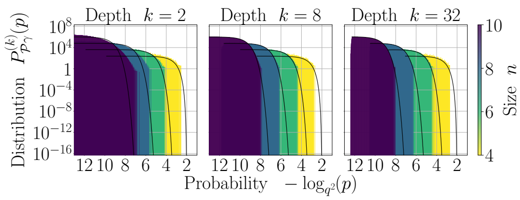
$$\tilde{P}(x|x_i) = \delta(x = x_i) \quad \to \quad \tilde{P}'(x|x'_i) = \rho(x'_{i'}) \ \frac{1}{\eta(x'_{i'})} \ \delta\big(x'_{i'} \leq x < x'_{i'+1}\big) \ . \tag{5.B.10}$$

In these numerical studies of the measurement operator distributions $\mathcal{P}$, for efficiency given the exponentially large number of samples $m \to m|\mathcal{P}|$ required in this continuous variable setting, we use binned empirical distributions. Here, we map our $m|\mathcal{P}| \leq 128|\mathcal{P}|$ samples to fixed $m' = 10^4$ binned samples of equally spaced points on a logarithmic-scale in the range $[10^{-20}, 1]$. Thus, $\tilde{P}(x) \to \tilde{P}'(x)$, and $\tilde{F}(x) \to \tilde{F}'(x)$ are implicitly replaced in any expressions. Such binning introduces bias into the empirical distributions, possibly masks sample complexity effects with m, and ultimately complicates sample complexity analysis [377]. However, given smooth enough distributions, an appropriately chosen logarithmic-scale for binning, which is shown to affect bias less [443], and large enough $m' \gg m$, such binning procedures should not significantly affect any interpretations.

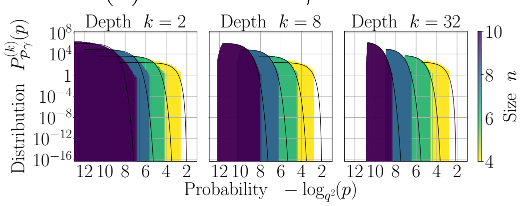
In Figs. 5.B.1 and 5.B.2 we show the resulting respective Tetrad SIC-POVM and Pauli NON-SIC-POVM binned empirical distribution histograms for various system parameters, with comparisons to the fit effective analytical models.



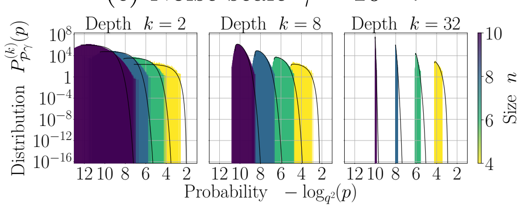
(a) Noise scale $\gamma = 0$.



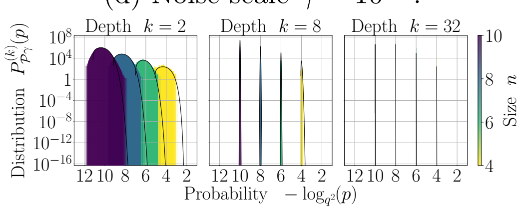
(b) Noise scale $\gamma = 10^{-4}$.



(c) Noise scale $\gamma = 10^{-3}$.

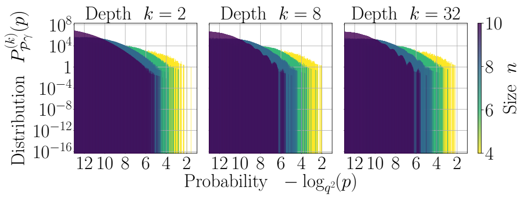


(d) Noise scale $\gamma = 10^{-2}$.

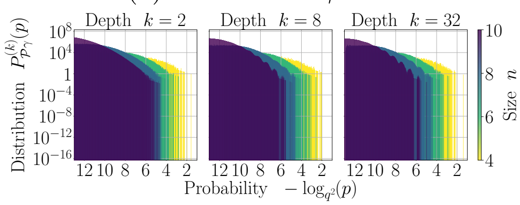


(e) Noise scale $\gamma = 10^{-1}$.

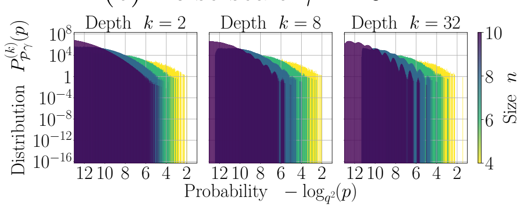
Figure 5.B.1: **SIC-POVM distributions**. Empirical SIC-POVM probability histograms $\tilde{P}^{(k)}_{\mathcal{P}_\gamma}(p)$ for depths k (rows), noise scales γ (a-e), system sizes n (coloured histograms), and $m = 128$ samples, for Haar random brickwork and local depolarization circuits. Solid lines indicate the fit effective analytical distribution $P^{(k)}_{\mathcal{P}_\gamma}$ for each system size. The effective analytical models generally capture the distributions' peaks, particularly as depth or noise scales increase and the distribution peaks sharpen and tails decrease.



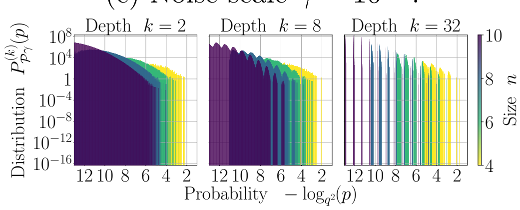
(a) Noise scale $\gamma = 0$.



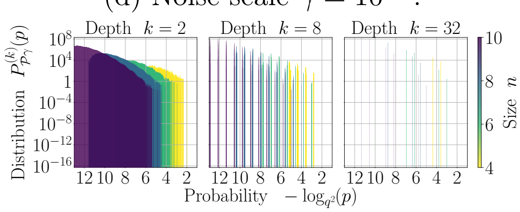
(b) Noise scale $\gamma = 10^{-4}$.



(c) Noise scale $\gamma = 10^{-3}$.



(d) Noise scale $\gamma = 10^{-2}$.



(e) Noise scale $\gamma = 10^{-1}$.

Figure 5.B.2: **NON-SIC POVM distributions**. Empirical NON-SIC-POVM probability histograms $\tilde{P}_{\mathcal{P}_\gamma}^{(k)}(p)$ for depths k (rows), noise scales γ (a-e), system sizes n (coloured histograms), and $m = 128$ samples, for Haar random brickwork and local depolarization circuits. As depth and noise scales increase, the empirical distributions develop multiple peaks, due to being a sum of distributions over all non-symmetric measurement operators, each contributing shifted and peaked conditional distributions to the total distribution.

Empirical Kolmogorov–Smirnov Metrics

Regarding metrics to quantify differences between distributions, here we use an upper bound on the Kolmogorov–Smirnov metric, to avoid maximizations over continuous variable domains. Given the monotonically increasing cumulative distributions, then $F(x_i) \leq F(x_{i+1}) \ \forall \ x_i \leq x_{i+1}$, and similarly given the piecewise-constant empirical distributions, then $\tilde{F}(x) = \tilde{F}(x_i) \ \forall \ x \in [x_i, x_{i+1}]$. From these properties, given m samples $\{x_i\}_{i\in[m]}$, and $x \in [x_i, x_{i+1}]$, then the difference of empirical and analytical distributions is,

$$\left|\tilde{F}(x) - F(x)\right| = \left|(\tilde{F}(x) - F(x_i)) - (F(x) - F(x_i))\right| \tag{5.B.11}$$

$$\leq \left|\tilde{F}(x) - F(x_i)\right| + \left|F(x) - F(x_i)\right| \tag{5.B.12}$$

$$\leq \left|\tilde{F}(x_i) - F(x_i)\right| + \left|F(x_{i+1}) - F(x_i)\right| . \tag{5.B.13}$$

Optimizations over the entire domain x can be replaced by the maximization over the m samples, yielding what we refer to as the empirical Kolmogorov–Smirnov metric $\tilde{\mathcal{L}}$,

$$\mathcal{L} = \max_x \left|\tilde{F}(x) - F(x)\right| \leq \max_{i\in[m]} \ \left|\tilde{F}(x_i) - F(x_i)\right| \ + \ \left|F(x_{i+1}) - F(x_i)\right| \equiv \tilde{\mathcal{L}} \ , \tag{5.B.14}$$

Depending on the smoothness of the distributions, such an upper bound, with the additional difference term $|F(x_{i+1}) - F(x_i)|$, may not necessarily be tight. However, general convergence trends should be demonstrated for sufficiently large m samples and carefully binned m' bins as $\lim_{x_{i+1}\to x_i} F(x_{i+1}) \to F(x_i)$ tightens.

Returning to our measurement probability distributions, to assess the similarity of empirical distributions $\tilde{F}^{(k)}_{\mathcal{P}_\gamma}$ of measurement probabilities p to analytical distributions $F^{(k)}_{\mathcal{P}_\gamma}$, we compute the empirical Kolmogorov–Smirnov metric as,

$$\mathcal{L}^{(k)}_{\mathcal{P}_\gamma} \leq \tilde{\mathcal{L}}^{(k)}_{\mathcal{P}_\gamma} = \max_{i\in[m]} \ \left|\tilde{F}^{(k)}_{\mathcal{P}_\gamma}(p_i) - F^{(k)}_{\mathcal{P}_\gamma}(p_i)\right| \ + \ \left|F^{(k)}_{\mathcal{P}_\gamma}(p_{i+1}) - F^{(k)}_{\mathcal{P}_\gamma}(p_i)\right| \tag{5.B.15}$$

$$\approx \tilde{\mathcal{L}}^{(k)\prime}_{\mathcal{P}_\gamma} = \max_{i\in[m']} \ \left|\tilde{F}^{(k)}_{\mathcal{P}_\gamma}(p'_i) - F^{(k)}_{\mathcal{P}_\gamma}(p'_i)\right| \ + \ \left|F^{(k)}_{\mathcal{P}_\gamma}(p'_{i+1}) - F^{(k)}_{\mathcal{P}_\gamma}(p'_i)\right| , \tag{5.B.16}$$

which we further approximate using our binned distribution $\tilde{\mathcal{L}}^{(k)\prime}_{\mathcal{P}_\gamma}$ with m' samples.

Sample Complexity of Empirical Distributions

It remains to be seen how the number of samples m affects the behaviour of these metrics, in particular given a fixed-size m' binning procedure that is independent of m. In Fig. 5.B.3, we plot the empirical, binned Kolmogorov–Smirnov metric for $m \in \{32, 64, 128\}$ number of samples. Such metrics appear consistent across number of samples, with slightly more stable, and more convergent behaviour for noiseless simulations at larger sample sizes.

The consistency of the empirical Kolmogorov–Smirnov metrics across number of samples is potentially due to masking of sampling effects. First, the exact deterministic operator sampling, which multiplies the number of samples to be $m \to m|\mathcal{P}|$, with $|\mathcal{P}| \gg m$, may mask any m-dependencies. Second, the binning procedure on top of any sampling may predominantly mask any sampling procedures, particularly in parameter regimes where there are highly peaked distribution. Even with fine discretization into $m' \gg m$ uniform logarithmically spaced bins, non-uniform bins around the peaks and tails may be more appropriate to better understand sample complexities.

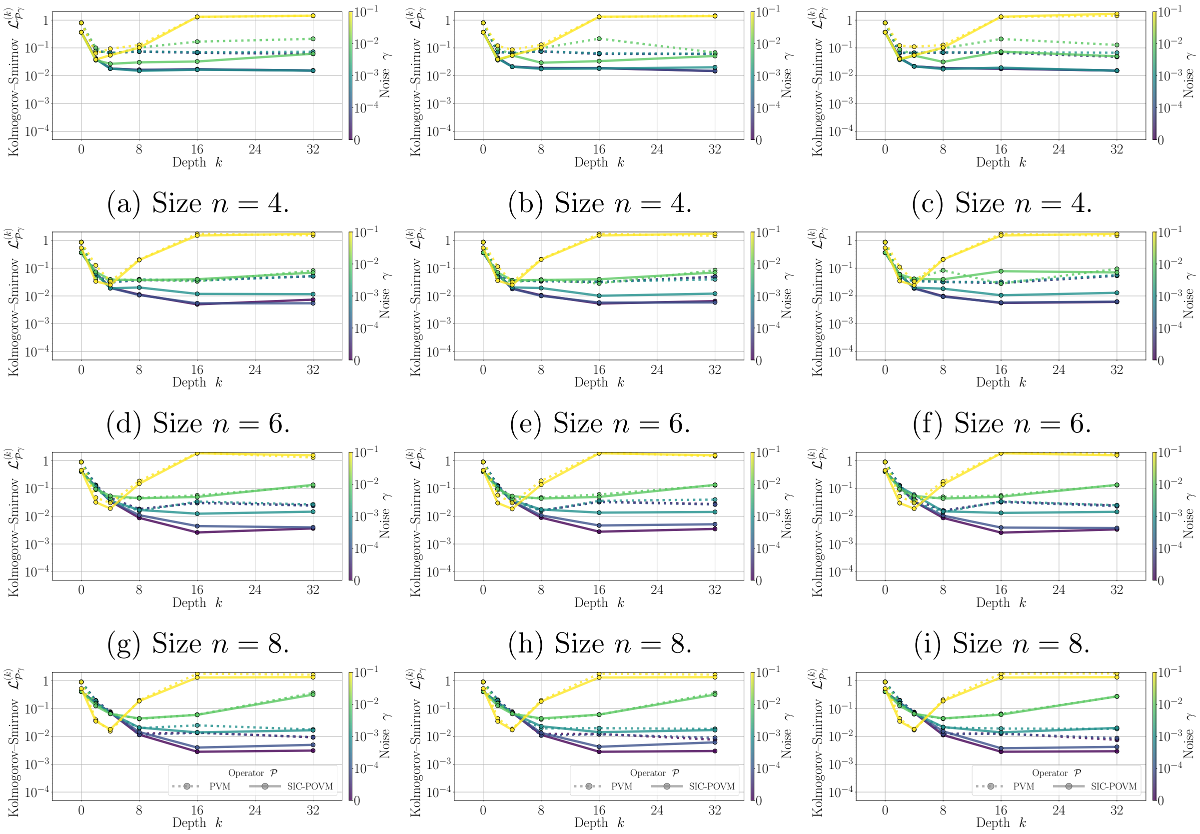
Given these biases, care must be taken in interpreting Chebyshev's inequality of the empirical distribution $\tilde{F}$ differing from its mean, as this mean is not necessarily F, but whichever distribution the empirical distribution converges towards. However, given the variance of the empirical distribution itself is always bounded by a constant independent of the system when $F = 1/2$, the system-independent sample complexity should still hold,

$$m \geq \left(\frac{1}{2\delta\epsilon} \right)^2 , \tag{5.B.17}$$

whose system independence could also partially explain the consistency of our metric across samples m and sizes n.

5.B.3 Effective Analytical Models

Here, given our analytical models of noiseless distributions, we propose an effective analytical model for noisy distributions, and interpret the behaviour of the model's parameters as a function of system parameters.



(a) Size $n = 4$. (b) Size $n = 4$. (c) Size $n = 4$.

(d) Size $n = 6$. (e) Size $n = 6$. (f) Size $n = 6$.

(g) Size $n = 8$. (h) Size $n = 8$. (i) Size $n = 8$.

(j) Size $n = 10$ $m = 32$ samples. (k) Size $n = 10$ $m = 64$ samples. (l) Size $n = 10$ $m = 128$ samples.

Figure 5.B.3: **Kolmogorov–Smirnov metrics**. Sampling-dependence of empirical Kolmogorov–Smirnov metric $\mathcal{L}_{\mathcal{P}_\gamma}$ of upper-bounded maximum difference between empirical and analytical cumulative distributions for SIC-POVM and PVM distributions $\tilde{F}^{(k)}_{\mathcal{P}_\gamma}(p)$, as a function of depth k, for noise scales γ (colours), system sizes n (rows), and m (columns) samples, for Haar random brickwork and local depolarization circuits. As depth increases, the metrics converge towards zero, before plateauing due to inherent biases in the binning procedures, and in the effect models for $F^{(k)}_{\mathcal{P}_\gamma}(p)$.

Selection of Effective Analytical Models

Given our simulated systems with local noise, we must propose an appropriate effective global noise analytical model for the distribution of its resulting expectation values. In fact, such circuits have been studied analytically recently [120, 130], where expressions for moments in asymptotic limits are derived, and distributions are constructed from fitting procedures and truncated series expansions. Here, we desire immediately interpretable closed-form expressions for distributions. We thus take inspiration from results that noisy random quantum circuit probabilities converge to the globally depolarized uniform distribution, within shallow circuit depths scaling logarithmically with system size [136, 143].

Although developing analytical local noise and local Haar random state models appears out of the scope of these studies, we can develop effective global noise and global Haar random state models, which over certain noise scales and circuit depths, approximates the noisy behaviours well. Here, we use our previous insight that isotropic depolarizing-like quantum channels have simple and intuitive shifted and scaled distributions,

$$\Pi \to (1-\tilde{\gamma})\Pi + \tilde{\gamma}\frac{\mathrm{tr}(\Pi)}{d}I \tag{5.B.18}$$

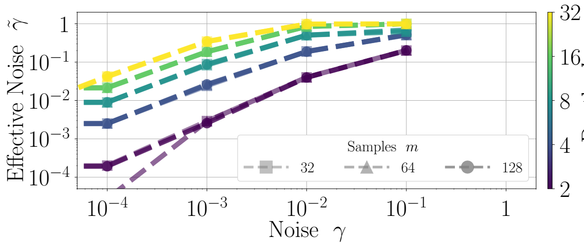
$$P_{\Pi}(p) \to \frac{1}{1-\tilde{\gamma}}P_{\Pi}\left(\frac{p-\tilde{\gamma}\mathrm{tr}(\Pi)/d}{1-\tilde{\gamma}}\right) \;\equiv\; P^{(k)}_{\Pi\gamma}(p) \;\approx\; \tilde{P}^{(k)}_{\Pi\gamma}(p)\ . \tag{5.B.19}$$

We define the arbitrary noise scale $\tilde{\gamma} = \tilde{\gamma}(k,\gamma,n,m)$ and environment dimension $\tilde{s} = \tilde{s}(k,\gamma,n,m)$ as variable effective model parameters, to be fit using empirical data samples $\tilde{P}^{(k)}_{\Pi\gamma}(p_i)$, as functions of system parameters k,γ,n,m.

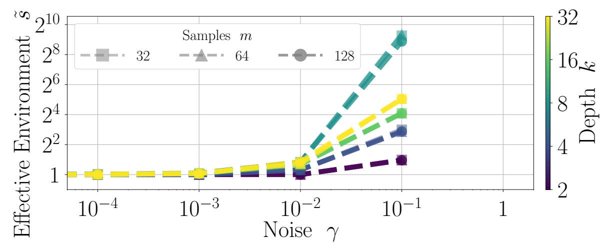
Optimization of Effective Analytical Models

To find the optimal effect model parameters $\tilde{\gamma},\tilde{s}$, we minimize the constrained normalized mean-squared-error between the (binned) empirical distribution and the effective model, evaluated at the $m \to m'$ samples of empirical data,

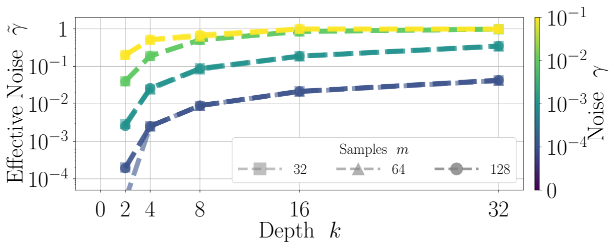
$$\tilde{\gamma},\tilde{s} = \quad \mathrm{argmin}_{\tilde{\gamma},\tilde{s}} \quad \frac{1}{m'}\frac{\sum_{i\in[m']}\left|P^{(k)}_{\mathcal{P}\gamma}(p'_i|\tilde{\gamma},\tilde{s}) - \tilde{P}^{(k)}_{\mathcal{P}\gamma}(p'_i)\right|^2}{\sum_{i\in[m']}\left|\tilde{P}^{(k)}_{\mathcal{P}\gamma}(p'_i)\right|^2} \qquad : \qquad \begin{matrix} 0 \leq \tilde{\gamma} \leq 1 \\ 1 \leq \tilde{s} \leq d^2 \end{matrix} \tag{5.B.20}$$



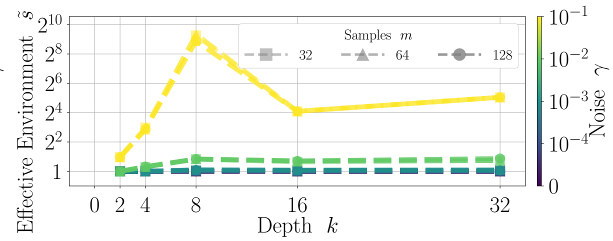
(a) Effective noise $\tilde{\gamma}$
as a function of noise γ.



(b) Effective environment $\tilde{s}$
as a function of noise γ.



(c) Effective noise $\tilde{\gamma}$
as a function of depth k.



(d) Effective environment $\tilde{s}$
as a function of depth k.

Figure 5.B.4: **Effective model parameters**. Optimized parameters of effective noise $\tilde{\gamma}$ and effective environment dimension $\tilde{s}$ of an effective analytical model $P^{(k)}_{\mathcal{P}_\gamma}(p|\tilde{\gamma},\tilde{s})$ for SIC-POVM measurement probability distributions, as a function of system parameters noise scale γ and depth k, for system size $n = 10$. Effective noise scales are shown to be smooth polynomial functions of noise scale, offset by increasing depth, and are consistent with noiseless behaviour, $\lim_{\gamma\to 0}\tilde{\gamma} \to 0$. Effective environment dimensions remain constant at the minimal trivial $\tilde{s} = 1$ environments, consistent with noiseless behaviour, $\lim_{\gamma\to 0}\tilde{s} \to 1$, until sufficient noise causes a potentially exponential jump in environment dimension with noise. Negligible sample size effects occur for number of samples $m \in \{32, 64, 128\}$.

We numerically perform such optimizations using the *scipy.optimize.minimize* optimizer [407], with the options: *method*: *None*, *bounds*: $\tilde{\gamma} \in [0, 0.99999999]$, $\tilde{s} \in [1, \infty]$, *tol*: 10^{-16}, *ftol*: 10^{-16}, *gtol*: 10^{-16}, *eps*: 10^{-8}, *maxiter*: 1000, *maxls*: 64, and did not conduct a full hyper-parameter search. The optimizer reported successful optimizations according to these criteria, however it was not verified that global optima are reached.

Given such optimizations, under the assumption of convergence, we can interpret the effective model parameters. In the noiseless limit, we recover the expected consistent behaviour, indicating the optimization is within a physically consistent local minimum,

$$\lim_{\gamma \to 0} \ \tilde{\gamma} \to 0 \ , \ \tilde{s} \to 1 \ . \tag{5.B.21}$$

Other behaviours to consider include the smoothness of the effective parameters, given they are expected to increase monotonically with noise scales and depth. The effective noise scale appears to have an intuitive form of a monotonic, polynomial function of noise scale. The effective environment dimension, being typically an integer, varies less smoothly with noise scale and depth. At low noise scales, the effective environment dimension remains trivially at one, indicating independent systems and environments in noiseless settings. At noise scales above a threshold, the effective environment dimension increases to scale exponentially with system size, indicative of highly entangled systems and environments in noisy settings. Further, the effective environment is largest at intermediate depths, potentially indicating where the noisy states are maximally Haar random mixed states. In conclusion, although it is not confirmed that such trends are indicative of global optima of the effective model, the smoothly varying behaviours suggest that these effective parameters are physically valid, and representative of the simulated systems.

Chapter 6 Appendices

6.A Numerical Simulations

In these appendices, we define quantities used to assess numerical simulation methods for quantum systems. For simplicity, quantities are indexed by $\mu \in [|\mathcal{P}|]$,

$$|\rho\rangle\!\rangle = p_\mu \ \chi^{-1}_{\mu\nu} \ |\Pi_\nu\rangle\!\rangle \quad : \quad p_\mu = \langle\!\langle \Pi_\mu | \rho \rangle\!\rangle \tag{6.A.1}$$

$$\hat{\Lambda} = L_{\lambda\nu} \ \chi^{-1}_{\mu\lambda} \ |\Pi_\mu\rangle\!\rangle\langle\!\langle \Pi_\nu| \quad : \quad L_{\mu\nu} = \langle\!\langle \Pi_\mu | \Lambda(\Pi_\lambda) \rangle\!\rangle \ \chi^{-1}_{\lambda\nu} \tag{6.A.2}$$

$$p_\mu \to L_{\mu\nu} p_\nu \quad , \quad \chi_{\mu\nu} = \chi_{\mathcal{P}}(\Pi_\mu, \Pi_\nu) \ . \tag{6.A.3}$$

- *Norm*: Normalizations of states ρ and probabilities p
  - Quantum

$$\mathcal{N}^Q_\rho = 1 - \mathrm{tr}(\rho) = 1 - \sum_\mu p_\mu \tag{6.A.4}$$

  - Classical

$$\mathcal{N}^C_\rho = 1 - |p| = 1 - \sum_\mu p_\mu \tag{6.A.5}$$

  - Pure

$$\mathcal{N}^P_\rho = 1 - \sqrt{\mathrm{tr}(\rho^2)} = 1 - \sqrt{p_\mu \chi^{-1}_{\mu\nu} p_\nu} \tag{6.A.6}$$

- *Infidelity:* Similarity of states $\rho \approx \sigma$ or distributions $p \approx s$
  - Quantum
$$\mathcal{L}^{Q}_{\rho\sigma} = 1 - \mathrm{tr}(\sqrt{\rho\sigma}) \tag{6.A.7}$$
  - Classical
$$\mathcal{L}^{C}_{\rho\sigma} = 1 - \sqrt{p} \cdot \sqrt{s} = 1 - \sqrt{p_\mu}\sqrt{s_\mu} \tag{6.A.8}$$
  - Pure
$$\mathcal{L}^{P}_{\rho\sigma} = 1 - \sqrt{\mathrm{tr}(\rho\sigma)} = 1 - \sqrt{p_\mu \chi^{-1}_{\mu\nu} s_\nu} \tag{6.A.9}$$
- *Entanglement:* Entropy of reduced states $\rho_\Gamma = \mathrm{tr}_{\bar{\Gamma}}(\rho)$, $p_\Gamma = \sum_{\bar{\Gamma}} p$
  - Quantum
$$\mathcal{S}^{Q\Gamma}_{\rho} = -\frac{1}{l\log(q)} \mathrm{tr}(\rho_\Gamma \log(\rho_\Gamma)) \tag{6.A.10}$$
  - Classical
$$\mathcal{S}^{C\Gamma}_{\rho} = -\frac{1}{2}\frac{1}{l\log(q)} \mathrm{tr}(p\log(p)) = -\frac{1}{2}\frac{1}{l\log(q)} p_{\mu_\Gamma} \log(p_{\mu_\Gamma}) \tag{6.A.11}$$
  - Renyi
$$\mathcal{S}^{R\Gamma}_{\rho} = 1 - \mathrm{tr}\left(\rho_\Gamma^2\right) = 1 - p_{\mu_\Gamma} \chi^{-1}_{\mu\nu_\Gamma} p_{\nu_\Gamma} \tag{6.A.12}$$
- *Vectorization*: Purification of states $\rho \to |\rho\rangle\rangle$

$$\rho \to |\rho\rangle\rangle = p_\mu \chi^{-1}_{\mu\nu} |\Pi_\nu\rangle\rangle \quad , \quad \langle\langle\rho|\rho\rangle\rangle = \mathrm{tr}\left(\rho^2\right) = p_\mu \chi^{-1}_{\mu\nu} p_\nu \tag{6.A.13}$$
$$\Psi = \frac{|\rho\rangle\rangle\langle\langle\rho|}{\langle\langle\rho|\rho\rangle\rangle} = \frac{1}{p_\mu \chi^{-1}_{\mu\nu} p_\nu} p_\mu p_\sigma \chi^{-1}_{\mu\nu} \chi^{-1}_{\sigma\pi} |\Pi_\nu\rangle\rangle\langle\langle\Pi_\pi| \tag{6.A.14}$$
$$\Psi_\Gamma = \mathrm{tr}_{\bar{\Gamma}}(\Psi) = \frac{1}{p_\mu \chi^{-1}_{\mu\nu} p_\nu} p_{\mu\nu_\Gamma} \chi^{-1}_{\mu\sigma_\Gamma} \chi^{-1}_{\nu\pi_\Gamma} |\Pi_{\sigma_\Gamma}\rangle\rangle\langle\langle\Pi_{\pi_\Gamma}| \tag{6.A.15}$$
$$p_{\mu\nu_\Gamma} = \sum_{\mu_{\bar{\Gamma}}\nu_{\bar{\Gamma}}} p_\mu \chi^{-1}_{\mu\nu_{\bar{\Gamma}}} p_\nu \tag{6.A.16}$$

- *Operator Entanglement:* Entanglement of vectorized states $\Psi_\Gamma = \mathrm{tr}_{\bar{\Gamma}}(|\rho\rangle\!\rangle\langle\!\langle\rho|)$
  - Quantum

$$\mathcal{Q}^{Q\Gamma}_\rho = \; -\frac{1}{2}\frac{1}{l\log(q)}\mathrm{tr}(\Psi_\Gamma \log(\Psi_\Gamma)) \tag{6.A.17}$$

  - Classical

$$\mathcal{Q}^{C\Gamma}_\rho = \; -\frac{1}{2}\frac{1}{l\log(q)}\mathrm{tr}(p_{\mu\nu_\Gamma} \log(p_{\mu\nu_\Gamma})) \tag{6.A.18}$$

  - Renyi

$$\mathcal{Q}^{R\Gamma}_\rho = 1 - \mathrm{tr}\left(\Psi_\Gamma^2\right) = 1 - p_{\mu\nu_\Gamma}\chi^{-1}_{\mu\pi_\Gamma}\chi^{-1}_{\nu\sigma_\Gamma}p_{\sigma\pi_\Gamma} \tag{6.A.19}$$

- *Mutual Information*: Dependencies between partitions $\Gamma, \bar{\Gamma}$
  - Quantum

$$\mathcal{I}^{Q\Gamma}_\rho = \mathcal{S}^{Q\Gamma}_\rho - \mathcal{S}^{Q\Gamma|\bar{\Gamma}}_\rho \tag{6.A.20}$$

$$\mathcal{S}^{Q\Gamma|\bar{\Gamma}}_\rho = \mathcal{S}^{Q}_\rho - \mathcal{S}^{Q}_{\rho_\Gamma} \tag{6.A.21}$$

$$\rho_\Gamma = \mathrm{tr}_{\bar{\Gamma}}(\rho) = p_{\mu_\Gamma}\chi^{-1}_{\mu\nu_\Gamma}\Pi_{\nu_\Gamma} \quad , \quad p_{\mu_\Gamma} = \mathrm{tr}(\Pi_{\mu_\Gamma}\rho) = \sum_{\mu_{\bar{\Gamma}}} p_{\mu_\Gamma\mu_{\bar{\Gamma}}}$$

  - Measurement

$$\mathcal{I}^{Q\Gamma|\mathcal{P}}_\rho = \mathcal{S}^{Q\Gamma}_\rho - \mathcal{S}^{Q\Gamma|\mathcal{P}}_\rho \tag{6.A.22}$$

$$\mathcal{S}^{Q\Gamma|\mathcal{P}}_\rho = \; \sum_{\mu_{\bar{\Gamma}}} p_{\mu_{\bar{\Gamma}}} \; \mathcal{S}^{Q}_{\rho_{\Gamma|\mu_{\bar{\Gamma}}}} \tag{6.A.23}$$

$$\rho_{\Gamma|\mu_{\bar{\Gamma}}} = \frac{1}{p_{\mu_{\bar{\Gamma}}}}\mathrm{tr}_{\bar{\Gamma}}\left(\Pi_{\mu_{\bar{\Gamma}}}\rho\right) = \frac{1}{p_{\mu_{\bar{\Gamma}}}}p_{\mu_\Gamma\mu_{\bar{\Gamma}}}\chi^{-1}_{\mu\nu_\Gamma}\Pi_{\nu_\Gamma} \quad , \quad p_{\mu_{\bar{\Gamma}}} = \mathrm{tr}\left(\Pi_{\mu_{\bar{\Gamma}}}\rho\right) = \sum_{\mu_\Gamma} p_{\mu_\Gamma\mu_{\bar{\Gamma}}}$$

- *Quantum Discord*: Effect of measurements $\mathcal{P}$ on mutual information
  - Quantum

$$\mathcal{D}^{Q\Gamma}_\rho = \mathcal{I}^{Q\Gamma}_\rho - \mathcal{I}^{Q\Gamma|\mathcal{P}}_\rho \tag{6.A.24}$$